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FINAL REPORT OF THE SEATTLE-DENVER INCOME MAINTENANCE EXPERIMENT

VOLUME 1

DESIGN AND RESULTS

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FINAL REPORT OF THE SEATTLE/DENVER INCOME MAINTENANCE EXPERIMENT

VOLUME 1

DESIGN AND RESULTS

SRI International

May 1983

Acknowledgments

This volume presents the final results of one of the country's most ambitious efforts at social experimentation, the Seattle/Denver Income Maintenance Experiment (SIME/DIME). More than 4800 families participated in these experiments. In addition, at the peak of its operation, more than 200 persons were involved in the day-to-day activities of the experiment, including officials in government agencies, payment specialists, interviewers, data managers, data processors, and researchers. The combined efforts of all these people for more than a decade (1969–1981) contributed to the successful completion of the work upon which this book is based. We do not have room here to list all the persons whose contributions are noteworthy, but a few principal individuals deserve mention.

The technical monitors at the United States Department of Health, Education and Welfare have provided guidance, support, and reviews of our research throughout the entire history of the project. Jodie Allen was instrumental in initiating SIME/DIME and providing support during its early years. Joseph Corbett ably guided the project during most of its operational years. David Greenberg, Gary Burtless, and Douglas Wolf provided invaluable services during later years in reviewing and commenting on SIME/DIME-related research efforts for the Department of Health and Human Services. Many persons in the state governments of Washington and Colorado also contributed greatly to the overall success of the project. We wish to give special thanks to the state directors in Washington and Colorado—Mike Linn and Joseph Bell in Seattle, and Jacob Shockley in Denver.

It is very difficult to convey to the reader the complexity of the operations required to carry out simultaneously a functioning welfare program and an experiment. Mathematica Policy Research (MPR), under subcontract to SRI International (which designed and evaluated the experiment), distributed the payments to the families and conducted the household interviews. A special debt of gratitude is expressed to MPR's late president, David Kershaw, whose untimely death saddened all of us who had worked with him on the experiment. Because of Dave's pioneering work in directing the New Jersey Income Maintenance Experiment, and because of his dedication to SIME/DIME, MPR performed ably in carrying out its operational activities for SIME/DIME. Key to MPR's successful handling of the payment operations and interviewing where Alan Brewster and Gary Christophersen in Seattle, and Mary Scowcroft and Robert Williams in Denver. Also playing a crucial role in experimental operations were the community colleges of Seattle and Denver, which conducted the manpower component of the experiment. Special thanks are given to N. John Andersen and Leroy Fails who managed the Seattle Central Community College's participation, and to Ray Urbas and Paul Kludt who managed Denver's community college participation.

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PREFACE

The Seattle/Denver Income Maintenance Experiment (SIME/DIME) was the last in a series of four large-scale income maintenance experiments undertaken in the late 1960s and early 1970s to measure the disincentive effects of cash transfers on the labor force participation of those eligible for them.

The Seattle/Denver Experiment was launched in Seattle, Washington in 1970 and extended in 1972 to a second site in Denver, Colorado. The prime contractors for SIME/DIME were the States of Washington and Colorado, which subcontracted with SRI International for the design, operation, and research evaluation of the experiment. SRI International in turn subcontracted with Mathematica Policy Research (MPR) for the administration of the cash transfer (negative income tax) treatment and all the field data collection from both experimental and control groups. For the administration of the counseling and educational subsidy treatment, SRI subcontracted with the Seattle Central Community College and the Community College of Denver.

The experiment involved almost 5,000 families randomly assigned to four groups: (a) NIT only, (b) counseling/training only, (c) NIT and counseling/training, (d) no treatment.

As in all experiments, more than one version of the negative income tax treatment was tested in order to provide information on how labor response might vary as program generosity varied. SIME/DIME tested three guarantee levels and four tax schedules, combined in such a way as to produce 11 negative income tax plans in all. The three guarantee levels were $3,800, $4,800, $5,600 (1971 dollars), respectively. These correspond to 95%, 120%, and 140% of the official poverty line ($4,000 in 1971 dollars). These percentages were preserved throughout the experimental period by adjusting the dollar guarantee levels regularly according to increases in the Consumer Price Index, as is the poverty line itself. The dollar guarantee levels varied with family size, with larger families qualifying for higher guarantee levels under a given NIT plan. In the three previous income maintenance experiments only constant tax rates were tested, i.e. tax rates that remain the same for every level of family income below the breakeven. SIME/DIME tested two constant tax rates: 50% and 70%. In addition to constant tax rates, however, SIME/DIME also tested two declining tax rate schedules. These rates were 80% and 70%, respectively, for the first $1,000 of nonexperiment income and then declined by 5% for each additional $1,000 of nonexperiment income.

The counseling/training subsidy treatment consisted of three variants: counseling only, counseling combined with a 50% subsidy for approved education or training courses, and counseling combined with a 100% subsidy for approved education or training. Within a family enrolled in the counseling training subsidy programs, every family member aged 16 and older was eligible for the experimental counseling and training subsidies. Important characteristics of the counseling component of the treatment were that it was voluntary, informational, and nondirective. Within this nondirective framework, the counseling had three main features: self-assessment, labor market assessment, and job search assistance.

This volume of the research findings is organized into six parts. Part I discusses the history and design of the experiment. Part II describes how the experiment was implemented and operated. Part III examines the labor supply responses of husbands, wives, single parents and youth in terms of hours worked, wage rates and earnings, and degree of unemployment. Part IV then goes on to look at effects of the counseling and education subsidy programs. Part V examines the relationship between the treatment and the formation and dissolution of families. Finally, Part VI summarizes the findings with respect to health, consumption and social behavior.

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SIME/DIME FINAL REPORT

VOLUME 1, PART I

HISTORY AND DESIGN

by

Robert G. Spiegelman

CHAPTER I

INTRODUCTION AND HISTORY

The NIT as a solution to the welfare problem

Throughout the 1960s there was a growing concern that the mushrooming welfare system was an ill-coordinated and inefficient means for helping the poor. Total income maintenance benefits were estimated to have risen in just 4 years from $28 billion in fiscal year 1965 to $49 billion in fiscal year 1969. Benefits continued to increase to $102 billion in fiscal year 1974. The two major public assistance programs for families—Aid to Families With Dependent Children (AFDC) and Food Stamps—showed the following trends (Lynn, 1977, pp. 85-86, Table 3.1):

	Fiscal year—	
	1965	1969
AFDC:		
Benefits (millions)	$959	$1,704
Recipients (thousands)	4,323	6,076
Food Stamps:		
Benefits (millions)	36	251
Recipients (thousands)	633	3,224

Despite this growth in benefits, there were still 24 million persons living below the poverty line in 1970 (Lynn, 1977, p. 90, Table 3.2). Only one-third of those whose own earnings were unable to lift them over the poverty line were taken out of poverty by transfer programs (p. 91). One reason that so many persons remained in poverty is that the public assistance programs, except Food Stamps, did not provide significant benefits to two-parent families. A RAND study (1979, p. 1) estimated that in 1975, 1.8 million husband-wife families with children were in povety. The AFDC-Unemployed Father program has never been a major factor in the welfare system because of stringent eligibility conditions and low participation by eligibles (Lynn, 1977).

A second concern that motivated interest in an NIT was the belief that high tax rates serve to discourage work effort and perpetuate welfare dependency. The nominal marginal tax rate on the combined AFDC and Food Stamp programs (in the income range below the income level at which benefits cease) is 76 percent. If the family also lives in public housing, the tax rate approaches 100 percent.[1]

For the reasons cited above, interest grew in a system that would give better coverage to two-parent poor families and reduce the tax burdens on those families who wanted to work. These concerns led to a growing support for the Negative Income Tax (NIT) as a solution. A negative income tax is a form of income maintenance that provides a maximum benefit to families with no other income, and reduces the payment according to a specified formula for each dollar of other income received by the family. Income maintenance is a term that encompasses more generally a broad spectrum of income-conditioned programs—AFDC, Food Stamps, publicly supported housing, and SSI, among others. Most of these programs aim at specified groups in the general population, and have eligibility constraints in addition to income (family status, labor force status, age, disability). In spite of the fact that the concept of an NIT as a substitute for the existing potpourri of welfare programs appealed to liberals and conservatives in the late 1960s—the former because of its equity features, the latter because of its apparent efficiency features—it had little chance of being implemented. This was because of the widespread fear that if income maintenance were extended to cover able-bodied heads of low-income intact working families there would be large reductions in the amount they worked. To determine the validity of those fears was the major reason for undertaking the series of income maintenance experiments.

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[1] In a study for SIME/DIME, Harlan Halsey (1978) found that effective tax rates were much lower than nominal rates because of income underreporting. However, the underreporting issue was not well understood at the time NIT experiments were being considered.

The experimental approach and rationale for SIME/DIME

Several negative income tax experiments were carried out to measure their work incentive effects on low-income families, including both welfare families and low-income working families without significant welfare experience.[2] In essence, an experiment is similar to a demonstration, except that there is random assignment to experimental and control status. Random assignment makes it possible to measure the effect of a program directly by comparing the difference in the behavior of persons receiving the treatment (experimentals) and those not receiving the treatment (controls) because random assignment, if successful, creates two groups that are identical except for the presence of the treatment.

For the income maintenance experiments, groups of eligible people were selected based on location, family structure, and incomes, and randomly assigned within designated strata to an income maintenance program or to control status. Those enrolled in an income maintenance treatment were told that for a fixed period of time they could be eligible to receive benefits in accordance with a specified formula. Experimental families were required to submit monthly income reports on the basis of which the grants were calculated and payment checks issued. Periodically, both experimental families and control families were interviewed and detailed behavioral data collected. The evaluation of the effects of the treatment occurred by comparing the information from the experimental group with information from the control group. Differences in hours of work, employment, and other variables were then measured as a function of the treatment. Although an experimental design does not require the testing of more than one program variant, each of the income maintenance experiments tested several NIT programs that differ in generosity. Comparisons, therefore, were possible both between those receiving some treatment and those receiving none, and among those receiving different variants of the program.

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[2] Paralleling the experiments was a series of legislative proposals, beginning with the Family Assistance Plan (FAP) submitted to the Congress by President Nixon in 1969. This program, which would have substituted for the existing AFDC program but retained Food Stamps and required state supplementation in high welfare states, passed the House of Representatives as H.R. 1, but foundered in the Senate. A revised FAP with a jobs component was resubmitted in 1970 and again was defeated in the Senate. No other proposal for a comprehensive NIT has passed either house of Congress. However, several have been proposed. President Ford presented the Income Supplement Program (ISP), which would have substituted an NIT for both the AFDC system and Food Stamps and imposed annual accounting. One of President Carter's first acts in 1977 was to present the Program for Better Jobs and Income (PBJI) to Congress. PBJI introduced a two-tier system, one for work eligible family heads which contained a work requirement and job guarantees, and a second for families with heads not expected to work. However, budgetary problems doomed this proposal from the start (see Bowler, 1974; Congressional Budget Office, 1977 and 1978).

A substantial body of nonexperimental literature on the labor supply effects of unearned income already existed when the experiments were being considered. Such work, however, generally suffered from four drawbacks:[3]

1. The data used did not necessarily encompass the population groups that would be eligible for a new program and thereby necessitated the extrapolation of findings to a different population from the one studied. For example, only 1 of the 12 studies reviewed by Cain and Watts (1973, p. 332, Table 9.1) included wives, and the sample in that study only included nonworking wives. An experiment can collect data on exactly the population of interest.
2. When viewed together, these studies displayed a very wide range of estimates, not only among studies but sometimes within the same study, which could provide only minimal policy guidance. For instance, the nonexperimental labor supply studies collected in the Cain and Watts volume (1973) produced estimates of male labor supply response to a typical NIT ranging from near zero to a reduction of approximately 40 percent. This wide range of estimated responses underlined the importance of getting direct estimates from experimentation.
3. Persons or families on existing programs may be a biased sample of those eligible for a new program because they already have made labor supply decisions that make them eligible for these programs. In nonexperimental studies, for example, differences in the labor supply between persons on welfare and persons off welfare cannot be unequivocally attributed to welfare, because persons with a low taste for work have already selected a labor supply (perhaps zero) that makes them eligible for welfare. In the experimental approach, all nonexperimentally induced influences on labor supply apply to both the experimental group and the control group.
4. Trying to measure causation using nonexperimental data is inevitably questionable because those data are the result of a multitude of factors changing simultaneously, making it impossible to sort out their separate effects with statistical means.
5. The scale of the changes from which labor supply must be deduced in nonexperimental data is small. Experiments, in contrast, can introduce large incentive changes. The SIME/DIME design provided particular richness in this dimension, by including at the top of the experimental benefit range more generous benefit schedules than those likely to be implemented in a national program. This enabled labor supply response estimates to be calculated over a wide range of treatments.

Advantages of the experimental mode are thus considerable; but any social experiment is complex and SIME/DIME, because of its ambition to test many treatment alternatives on a diverse population, was particularly complex. In later chapters of Part II and subsequent parts of this book, we discuss these complexities, problems they created for analysis, and solutions invoked by the research team.

SIME/DIME and its predecessors

The last of four NIT experiments conducted in the United States, SIME/DIME was also the largest and most comprehensive. It was preceded by the New Jersey Experiment, the Rural Experiment in North Carolina and Iowa, and the Gary Experiment in Gary, Indiana. Table 1.1 outlines the main features of the four NIT experiments. A brief history of the experiments will help explain the differences among them.[4]

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[3] See Cain and Watts (1973), especially pp. 328-367.

[4] For more comprehensive descriptions of the other three experiments, see USDHEW (1973, 1976), Kehrer (1977).

The Office of Economic Opportunity (OEO) funded two of the four experiments. The New Jersey Income Maintenance Experiment was launched in 1968, with a sample of 1,375 dual-headed families. It was conducted in three urban sites in New Jersey and one in Pennsylvania. Population groups selected for this experiment were those not eligible for assistance under the welfare system in New Jersey[5] but would be eligible under a universal national NIT program. Following this criterion, eligibility was restricted to families with incomes less than 150 percent of the poverty line (which in 1968 was approximately $3,800). Since few families in which both spouses were employed had incomes below this level, the stringent limitation on family income meant that most families in the sample had only one family head employed (usually the husband). The experimental design had support levels at 75 percent, 100 percent, and 125 percent of the poverty line, and tax rates of 30 percent, 50 percent, and 70 percent. After the New Jersey Experiment was launched, OEO decided to expand the experimental population base to include rural families. None were in the New Jersey Experiment, yet the rural poor could be an important part of a negative income tax program. Two counties, one in rural Iowa and one in rural North Carolina—the former almost entirely white and the latter almost entirely black—were selected in 1969 for a second experiment. The Rural Income Maintenance Experiment, run by the Institute for Research on Poverty, was similar to the New Jersey Experiment in income eligibility levels and tested only a subset of the plans tested in New Jersey. The sample included both two-parent working-age families and subsamples of households headed by single women and by aged persons. Further, the small sample of 800 families was subdivided into samples of farm families and rural nonfarm families.

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[5] When the experiment began there was no AFDC–UF program in New Jersey—a major reason for the selection of that site. This decision lost some of its relevance when a fourth site had to be added in Pennsylvania (*which* did have AFDC–UF) and the rest of its relevance when New Jersey instituted AFDC–UF part way through the experiment.

TABLE 1.1.—FEATURES OF THE NEGATIVE INCOME TAX EXPERIMENTS

Experiment		Sample size (families)	Population groups covered		Guarantee levels[1] (family of 4)	[illegible] treatments		Other treatment
Title	Years		Family types	Races		Tax rates (average)	Years on program	
Seattle/Denver	1970–78	2,042 (Seattle) 2,758 (Denver)	Dual- and single-headed families.	White, black, Chicano.	$3,800 4,800 5,600 (1971)	0.5, .7, .8 with average rates of decline of zero and .025/$1,000.	3, 5, 20	Counseling; education and training subsidies of 50 percent, 100 percent.
New Jersey	1968–72	1,357	Predominantly dual-headed families.	White, black, Spanish-speaking.	1,650 2,457 3,300 4,125 (1968)	.3, .5, .7	3	
Rural	1969–73	809	Dual- and single-headed families.	White, black	1,741 2,612 3,482 (1969)	.3, .5, .7	3	
Gary	1971–75	1,780	Dual- and single-headed families.	Black	3,300 4,300 (1971)	.4, .6	3	Day care subsidies of 35 percent, 60 percent, 80 percent, 100 percent.

[1] The support levels may be adjusted to a common year base relative to either the poverty level or the CPI by using the following indexes. The poverty cutoff for a family of four persons (nonfarm) in 1968 was $3,553. The indexes for 1969, 1970, and 1971 were 105.3, 111.7, and 116.4 (USDHEW, 1976, p. 72, Table 7). Using 1968 as the base year, the CPI indexes for 1969, 1970, and 1971 were 105.4, 111.6, and 116.4 (p. 93, Table 14).

Source: Spiegelman and Yaeger (1980), p. 466, Table 1.

In 1969, the U.S. Department of Health, Education and Welfare (HEW) set in motion two additional negative income tax experiments. The Gary Experiment, run by the University of Indiana, consisted of 1,800 two-parent and one-parent black families. It differed from the New Jersey Experiment by using a higher family income cut-off. In addition, there was a social service component in which randomly assigned people were provided a variety of social services, with particular emphasis placed on child care.

The Seattle-Denver (SIME/DIME) Income Maintenance Experiment was launched in Seattle in 1970. At that time, changes in Seattle's major industry (aerospace) were causing severe unemployment, so the Seattle experiment was extended to include a second, more representative western city, Denver. SIME/DIME included white, black, and Chicano families, both two-parent and one-parent. Its sample of 4,800 families was greater then the 3,900 families enrolled in all the other income maintenance experiments combined. In addition to support levels that ran from 75% to 140% of the poverty level, it featured more complex tax systems with declining marginal tax rates (i.e., as incomes rose, tax rates on each additional $1,000 of income were less than the rates on the previous $1,000 of income). Furthermore, it provided a major test of the combined effects of a negative income tax and a labor market assistance treatment consisting of counseling and subsidized education and training.

The rationale for conducting more than one experiment was threefold: (1) to assure reliability of the results through replication, (2) to ascertain that results were not site-specific, and (3) to increase the scope of the experiments by adding certain complementary programs to the NIT—namely, day care subsidies (the Gary Experiment) and counseling/training treatments (the Seattle/Denver Experiment).

Role of government in experimental operations

The New Jersey Experiment was designed in 1967 and launched in 1968; the Rural Experiment followed a year later. At the same time, the design work for the Gary Experiment and SIME/DIME was started. In the design efforts for SIME/DIME, we were anxious to capitalize on the innovations of the previous experiments without repeating their mistakes. One of the important operational lessons from the New Jersey Experiment was that a high degree of cooperation between the federal government, state welfare agencies, and the contractors carrying out the experiment would greatly facilitate the operation of the experiment and help gain access to necessary public program records on sample families. The New Jersey Experiment used a direct contract between the U.S. Office of Economic Opportunity and the Poverty Institute (which then entered into a subcontract with Mathematica for field operations and part of the research). The state governments of New Jersey and Pennsylvania were not involved. This turned out to create numerous problems for the families and staff of the experiment. State involvement in SIME/DIME avoided some of the problems and brought with it several ancillary benefits.

The most important advantage of the explicit participation of the state welfare departments of Washington and Colorado was that it secured the cooperation of other state agencies in allowing the experiment to be carried out without undue interference (e.g., the New Jersey experiment was continually faced with threatened investigations of sample families for fraud). A second benefit was the agreement reached between the states and the Social Security Administration to allow the federal share of AFDC benefits saved by the switch-over of AFDC recipients to SIME/DIME to be transferred to the SIME/DIME accounts, thereby reducing the budgeted dollars necessary to operate the experiment.

Rules of operation for the experiment provided that families transferring from AFDC to NIT did not lose service benefits available to welfare recipients, the most important of these being Medicaid. Since the intent of SIME/DIME was to measure the effects of changes in cash transfers and labor market assistance, we did not wish to confound the measurement problem by any other changes in the service bundle. Thus, a third benefit was an arrangement with the states for AFDC recipients to continue to receive all their supplementary benefits if they elected to accept the experimental NIT program. This required that the states keep lists of AFDC families transferring to SIME/DIME, and that these lists be periodically reviewed to ensure that families were not receiving double payments. In fact, the entire roster of SIME/DIME families was scrutinized in this way. Furthermore, because of the gap between the income necessary to become eligible for AFDC and the income at which one would lose all AFDC benefits once in the program, there was a range of incomes that would make a family ineligible for enrollment on AFDC but eligible to stay on AFDC once in the program. An agreement was reached that at the conclusion of the NIT program people in that range could transfer to the AFDC program whether previously in the program or not. All these agreements, plus the important agreement of confidentiality, required so much state cooperation that HEW wisely put the state into the contract loop, and essentially assigned to the states the role of managing the SIME/DIME experiments.

The experiment was designed to determine the effect of NITs with specified support levels and tax rates. Since a family faces a multitude of income taxes in the real world, the ability to create a "treatment" defined as a specific tax on income depends on the ability to substitute the experimental rate for all other taxes on income. Enabling this to happen required cooperation between SIME/DIME and the federal government. First, SIME/DIME arranged with the IRS to exempt the NIT payments from positive (regular) income tax liability. The IRS issued a ruling that placed NIT payments in the same category as other welfare payments for tax purposes, thereby exempting them from federal and state income taxes.[6] Second, the experiment reimbursed positive taxes paid by experimental families receiving NIT benefits. The reimbursements were necessary to avoid double taxation. Without reimbursement, the family would face a tax rate equal to the sum of the NIT-imposed rate on outside income and the regular federal and state tax rates.

Reimbursing positive taxes in a way that would not distort the experimental incentives posed a problem since tax liability is determined only once a year, whereas paychecks are received at much more frequent intervals. If taxes were reimbursed only when tax liability is finally determined in April, the payment might be disassociated from the earnings in the minds of the family, in which case the effect would be that of a lump-sum windfall. Thus, we devised a system of reimbursing tax withholding on a monthly basis, with an annual reconciliation to correct for under- or overreimbursement. The need for this somewhat cumbersome system arose because we could not persuade the IRS to alter the reimbursement system with regard to taxes on regular income of experimental families.

SIME/DIME was operated by the states of Washington and Colorado under contract to HEW. The states subcontracted with SRI International for the design, operation, and evaluation of the experiment. SRI subcontracted with Mathematica Policy Research (MPR) to operate the NIT payment system and the household surveys; SRI also subcontracted with the Seattle Central Community College and the Community College of Denver to conduct the counseling/training component of the experiment.

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[6] See Rulings by IRS provided in letters from IRS to SRI, reference T:I:I:2:1 dated March 12, 1971 and April 10, 1972 in which grant disbursements from the general welfare fund were ruled as not to be includible in gross income for tax purposes.

CHAPTER II

EXPERIMENTAL TREATMENT AND EXPECTED RESPONSES TO THEM

The overall purpose of the experiment was to measure the labor supply response to a negative income tax. Since the main behavioral response expected was a reduction in the amount of labor supplied, HEW was interested in testing whether certain supply side treatments might mitigate this negative response. It was decided therefore, to test a second treatment as well—a labor market counseling and education training option. With respect to this treatment also, the primary interest was in the labor supply response. The principal design concern, therefore, was to optimize the ability of the experiment to measure changes in labor supply. Within this constraint, other responses were also of interest, and the SIME/DIME design accommodated as many of these as possible, including marital status as a particular focus of attention.

This chapter first discusses the labor supply responses that were expected from the negative income tax and counseling/training treatments. It then discusses other behavioral responses of interest, including marital status, child-bearing, migration, human capital investment, durable goods acquisition, and housing consumption. This is followed by a detailed description of the negative income tax treatment and the counseling/training treatment. The chapter ends with a discussion of other major factors in the environment facing those in the treatment group that can be expected to influence their response.

Labor supply theory and the NIT

The intense interest in the impact of a universal negative income tax on labor supply behavior arose from the confluence of several features of such a program. At the heart of the issue was the fact that a universal NIT program would provide income supplements to the so-called working poor: families with a male and female head for whom combinations of low earnings capacity and/or large family size can result in low standards of living despite substantial work effort. Thus, such an NIT would result in a more equitable distribution of assistance to the low-income population than do welfare programs that provide benefits primarily to families without a working-age male head. However, the provision of benefits to low-income workers under an NIT alters work incentives in such a way that economic theory predicts a reduction in hours of work among recipient workers. The extent of the expected work reduction is important both to cost projections for a national program and to the political feasibility of an NIT.

Economic theory is unambiguous in predicting the direction of workers' labor supply responses. But the types of incentive changes that would be introduced by a universal NIT are rarely observed in the nonexperimental labor market, so that comparatively little was known before the income maintenance experiments about the critical issue of the size of such responses. Here we review the basic economic theory of labor supply response and the next section discusses the incorporation of that theory into the design of SIME/DIME.

Economists view an individual's choices about labor supply in terms of a tradeoff between the satisfaction derived from money income through purchases of market goods and services, and satisfaction derived from nonmarket time. The latter is often referred to as leisure, though it typically includes such productive activities as child care and home maintenance in addition to strict leisure time. In effect a worker sells his or her leisure time in return for market income at a price given by the net wage rate (i.e., wage rate after taxes). As hours of market work rise, the relative value of remaining leisure rises and the worker chooses an optimal level of work at a point where the last hour's earnings just compensate for the last hour of leisure foregone.

Changes in market wage rates and/or income typically change an individual's optimal level of labor supply. Incentives to change labor supply include some combination of two conceptually distinct effects, the income effect and the substitution effect. The income effect says that an increase in income will reduce the number of hours of market work that an individual wishes to provide. In other words, the individual will prefer to take some (and perhaps all) of the potential increase in income in the form of more leisure, rather than more command over market goods. A decrease in income will have the reverse effect. The substitution effect says that a decline in the marginal net wage rate (due, for example, to a tax rate increase) also causes the individual to want to provide fewer hours of market work because the relative price of an hour of leisure has been reduced. Again an increase in the net wage rate will have the reverse effect. The substitution effect is a somewhat nonintuitive concept because it technically measures the price effect, holding the income effect constant.

Incentive changes that are most often observed in the labor market are composed of income and substitution effects working in opposite directions. Typical increases in wage rates improve the marginal payoff to work, but also increase income by changing earnings at the original labor supply position. Most observed increases in market wage rates result in a positive substitution effect on labor supply together with a negative income effect stemming from the accompanying increase in real income. The observed net response may thus be positive or negative, depending on which effect is the stronger. In any case, the changes observed are usually small enough to induce rather small effects.

Incentive changes introduced by an NIT are markedly different from those usually observed in the labor market. The income maintenance payments increase recipients' real income at the same time that the tax rate by which the guarantee is reduced as other earnings rise reduces the net marginal wage rate. Thus, the income and substitution effects operate in the same direction for a worker on an NIT—both contributing to labor supply reductions—and so the direction of the expected change in labor supply is unambiguously negative. The magnitude of the response depends on the strengths of the income and substitution effects. Measurement of these magnitudes is the primary objective of the financial treatments in SIME/DIME. They are measured directly in the SIME/DIME labor supply analysis by including in the model a variable measuring the change in income caused by the NIT treatment, and a variable measuring the change in the marginal net wage rate caused by the benefit reduction formula, both variables measured at the preexperimental level of labor supply.

NIT treatments tested in SIME/DIME

The financial treatments in SIME/DIME involved a range of guarantee (maximum benefits) levels and tax rates. To ensure that families would volunteer for the experiment, it was necessary to offer guarantee levels at least as high as the available alternatives. The lowest guarantee level in SIME/DIME was 95% of the official poverty level, or $3,800 per year (in 1971 dollars) for a family of four.[1] All guarantee levels were adjusted for family size and increased annually according to the Consumer Price Index. This level approximated existing welfare payments from AFDC plus the Food Stamp program.[2] The next highest support level was 120 percent of the poverty level ($4,800) and was intended to incorporate a differential ($1,000) sufficiently large to produce measurable changes in labor supply. This guarantee level was also equivalent to the highest guarantee in the New Jersey Experiment, aiding interexperimental comparison. The third support level was 140 percent of the poverty level ($5,600) and was included to extend the range of response and test the highest support level suggested as a feasible policy option.

SIME/DIME tested a range of tax rates—that is, the rates at which benefit payments are reduced as family earnings increase. Two different tax systems were tested. One was a system of constant tax rates. The constant tax rate system (which was used in all other NIT experiments) has the advantage of being simple to understand, administer, and analyze. Constant tax rates of 50 percent and 70 percent were tested. The other system was a declining tax rate system. A declining marginal tax system is the reverse of the progressive schedule in the positive tax system. In the positive tax system, at each step in the schedule, the marginal tax rate on that increment is greater than the tax on the previous increment. In the declining tax system imposed in SIME/DIME, each increment (in $1,000 steps) was taxed at a lower rate than the previous increment, until the SIME/DIME payments dropped to zero. In the SIME/DIME program, a single rate of decline was used, providing for a marginal tax rate that declined at a rate of 5 percent for each additional $1,000 per year of earned income. Thus, the marginal tax rate was 5 percent less on each additional $1,000 of income. Two declining rate systems were used, starting at either 80 percent or 70 percent tax on the first $1,000, and declining until payments were exhausted or the marginal tax rate reached zero.[3] Table 1.2 displays the marginal and average tax schedule, starting at 80 percent on the first $1,000 of income.

TABLE 1.2.—DECLINING TAX RATES [1]

Income	Marginal tax rate on additional $1,000 of income at Y_i (percent)	Average tax rate on income Y_i (percent)
$0	80	80
$1,000	75	77.5
$2,000	70	75

---

[1] The official poverty level is defined as the income necessary to provide the family with three times the income necessary to afford the nutritional goals for the lowest of the four Department of Agriculture family food plans (see USDHEW, 1976, p. 8).

[2] The issue of AFDC dominance is an important one in determining the extent to which the nominal NIT treatment is a "real" treatment, especially for single-headed families. A careful evaluation of the guarantee level and breakeven (zero grant) level of monthly income required for dominance of AFDC and Food Stamp benefits shows that at the start of the experiment there were certain cases where dominance was in doubt. These cases involved only the lowest support level, and are discussed below:

(1) In Seattle, 1971, for families larger than five members, the AFDC/Food Stamp guarantee level dominated the $3,800 guarantee in Seattle. However, in that year, there was a maximum grant level which reduced the AFDC/Food Stamp support below that. In Denver, the low support strongly dominated AFDC/Food Stamp for one-parent families with less than four members, and roughly equaled the level for all other family types and sizes.

(2) The breakeven levels were compared for SIME/DIME's least generous plan and AFDC/Food Stamp, using the legal formulas for calculating AFDC/Food Stamp grants. In this comparison, the AFDC/Food Stamp breakeven was higher in Seattle than the least generous experimental plan by about 10 percent for single-headed families of four or five persons and somewhat more for other sizes and for dual-headed families, indicating that working single-headed families already on welfare could be expected to prefer that status to the least generous NIT. In Denver, the least generous plan approximately matched that of the AFDC/Food Stamp combinations in terms of breakeven (see Hall, 1976).

[3] If this occurred first, then the marginal rate was set equal to the average rate, which declines only half as fast as the marginal rate until the payment was exhausted. The need to impose a nonnegativity constraint on the marginal tax rate would have occurred in cases where income exceeded $16,000.

TABLE 1.2.—DECLINING TAX RATES[1]—Continued

Income	Marginal tax rate on additional \$1,000 of income at Y_1 (percent)	Average tax rate on income Y_1 (percent)
\$3,000	65	72.5
\$4,000	60	70
\$5,000	55	67.5
\$6,000	50	65
\$7,000	45	62.5
\$8,000	40	60
\$9,000	35	57.5
\$10,000	30	55
\$11,000	25	52.5
\$12,000	20	50
\$13,000	15	47.5
\$14,000	10	45
\$15,000	5	42.5
\$16,000	0	40

[1] $G = S - Y[t - rY]$,

when

S = support in dollars per year.
Y = income per year.
t = initial tax rate on first \$1,000 of income.
r = average rate of decline per \$1,000 income.
$2r$ = marginal rate of decline per \$1,000.

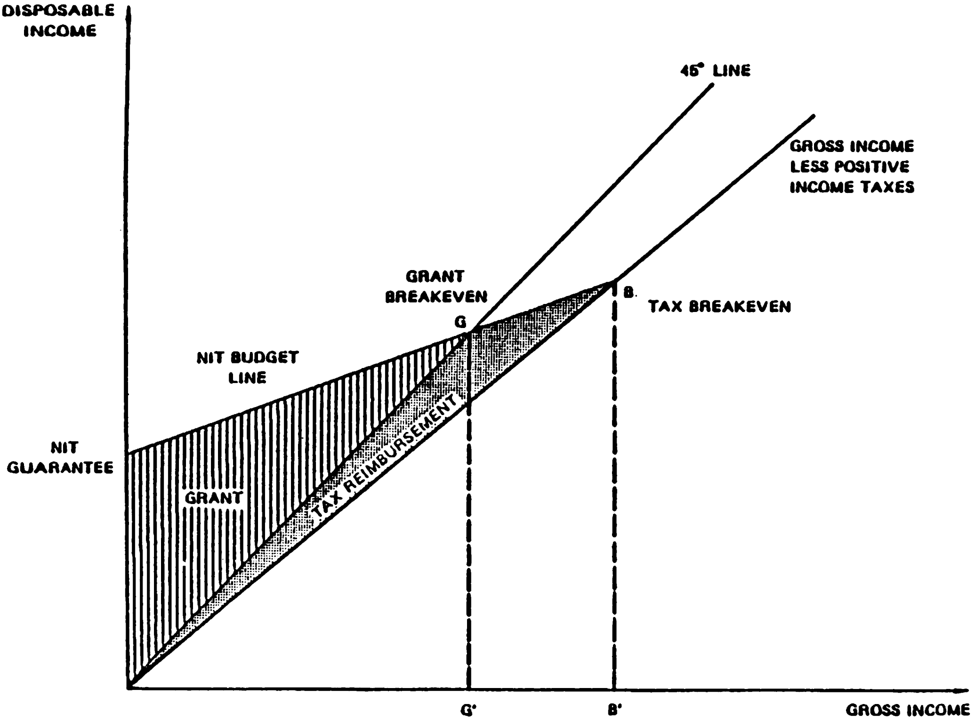
Thus,

$t - ry_1$ = average tax on income Y_1
$t - 2rY_1$ = marginal tax rate on additional dollar was of income at Y_1

Source: Kurz, Spiegelman, and Brewster (1972), SRI Research Memorandum 19, p. 3.

The declining tax system was introduced, despite its being regressive, because a declining tax rate system when compared to a constant tax rate system with the same support level and breakeven point, provides less work disincentive and costs less. Moreover, it can be integrated into the positive tax system without discontinuity in the marginal tax rate. However, what appeared to be an advantage of the declining tax system—the wide range of observed tax rates—turned into a significant problem in the analysis. Because individuals on a declining tax system simultaneously select their level of work and their tax rate, the declining tax rate is endogenous and cannot be used directly in the analysis as a treatment variable. This is corrected in the analysis models by estimating an equivalent constant tax treatment on the assumption that the average and marginal tax of this equivalent treatment equal the marginal tax in the declining tax schedule at the preexperimental level of earnings. (This also requires an adjustment in the support level. See discussion in Part III, Chapter 1.)

In order to control the tax rate for those in the experimental group, SIME/DIME taxed all other public transfers at 100 percent and reimbursed positive income taxes. Figure 1.1 depicts the relationships between the positive income tax system and an NIT program with a constant tax rate. The horizontal axis shows gross income (income before taxes and transfers), and the vertical axis shows disposable income (income after taxes and transfers). Point G is the grant breakeven level. Families with gross incomes less than G′ are eligible to receive positive grants from SIME/DIME. They also receive 100 percent reimbursement of positive income taxes. In order to prevent a drop in disposable income for families who earned slightly more than the grant breakeven level G′, the experiment reimbursed a fraction of positive taxes for incomes up to point B′. Point B is thus the tax breakeven level. Table 1.3 lists the details of guarantee levels, tax rates, and breakeven levels for the 11 financial programs tested.


DISPOSABLE INCOME
45° LINE
GROSS INCOME LESS POSITIVE INCOME TAXES
GRANT BREAKEVEN
G
B
TAX BREAKEVEN
NIT BUDGET LINE
TAX REIMBURSEMENT
NIT GUARANTEE
GRANT
G'
B'
GROSS INCOME

NOTE: Figure assumes no income outside of earnings and a linear positive income tax system.

FIGURE 1.1.—A negative income tax program with positive tax reimbursement.

TABLE 1.3.—SIME/DIME BREAKEVEN LEVELS

[1971 dollars]

Plan	Grant breakeven level	Tax breakeven level
F1 ($S=3800$, $t_e=.5$, $r=0$)	7,600	10,250
F2 ($S=3800$, $t_e=.7$, $r=0$)	5,429	6,350
F3 ($S=3800$, $t_e=.7$, $r=.025$)	7,367	10,850
F4 ($S=3800$, $t_e=.8$, $r=.025$)	5,802	7,800
F5 ($S=4800$, $t_e=.5$, $r=0$)	9,600	13,150
F6 ($S=4800$, $t_e=.7$, $r=0$)	6,867	8,520
F7 ($S=4800$, $t_e=.7$, $r=.025$)	12,000	19,700
F8 ($S=4800$, $t_e=.8$, $r=.025$)	8,000	11,510
F9 ($S=5600$, $t_e=.5$, $r=0$)	11,200	15,700
F10 ($S=5600$, $t_e=.7$, $r=0$)	8,000	9,780
F11 ($S=5600$, $t_e=.8$, $r=.025$)	10,360	16,230

Note: These figures are for a family of four with only one earner and no income outside of earnings. Positive tax reimbursements include the federal income tax and social security taxes. The federal income tax assumes the family takes the standard deduction. State income taxes, which are relevant only for the Denver experiment (there is no state income tax in Washington), were levied at graduated rates up to a maximum of four. We ignore these state income taxes in calculating the tax breakeven levels. The tax breakeven levels are thus slightly higher for the Denver experiment.

Key: S = NIT annual guarantee level.
t_e = initial NIT tax rate.
r = rate of decline of the average NIT tax rate per thousand dollars of income.

Rationale for the counseling/training treatment

The main behavioral response expected from the NIT treatment in SIME/DIME was a reduction in labor supply. HEW was interested in testing certain other treatments whose aim would be to counter this response. The general conceptual approach underlying the choice of treatments to test was the human capital theory that labor market participants choose to increase their investments in human capital to the extent that so doing will improve their economic position (net of cost). It was, thus, expected that the experimental group exposed to subsidized opportunities to increase their human capital would take advantage of those opportunities in the expectation of eventually increasing their gross wage rate—the counseling/training subsidy was expected to be particularly beneficial to those on the NIT treatment because the NIT reduced the after-tax wage rate in the short run (the period of the experiment), thereby reducing the opportunity cost (lost earnings) of reducing work hours to take counseling or training.

Because of this human capital approach, the treatment was designed to minimize intervention in the experimental group's decision-making and to manipulate only the price of the human capital investment, not the content of education or training option available. This led to a voucher-like treatment to subsidize job-related (i.e., human capital enhancing) training and education.

It was recognized that the effect of the voucher on wage rates depends first on the decision to use the voucher, second on the decision of how to use it (i.e., what to buy), and third on the effectiveness of the training or education program taken by the user. The first and second of these depend in part on the level of knowledge possessed by the subject as to what training and education is available and how it might help him/her improve labor market performance. Thus, a counseling (i.e., information) component was made an integral part of the human capital treatment in order to permit all treatment subjects to make presumably optimal decisions as to the use of the voucher (see further discussion in Kurz and Spiegelman, 1972, pp. 17–23). The counseling component was thus included to ensure that individuals had sufficient information about their capacities and opportunities in the labor market to accurately evaluate the costs and benefits of human capital investment. The cost reimbursement component was included to allow for measurement of the price elasticity of demand for vocational training, as well as for the effect of such investment on work effort.

SIME/DIME tested three different counseling and subsidy plans: one provided counseling alone, free of charge; another provided free counseling plus 50 percent training cost reimbursement; and a third provided free counseling plus 100 percent training cost reimbursement. Reimbursable costs included all direct expenses associated with the chosen education or training program, such as tuition, books, commuting costs, and special clothing and child care if necessary. Consumption allowances were not included, and tuition reimbursements were limited to the costs of the lowest-priced institution offering the course. The training subsidy program operated like a voucher system in that an individual was free to choose both the course and place of study. The training or educational institution was directly reimbursed for that proportion of the cost indicated by the participant's subsidy plan.

Other relevant features of the experimental environment [4]

Other financial aspects of SIME/DIME that can be expected to influence the behavior of participants should be mentioned. First, SIME/DIME included a provision that accounted for asset holdings in the determination of benefits. This was done by imputing a 5.75 percent rate of return to certain components of net worth (stocks, savings accounts, and equity in owner-occupied housing and other real property), and taxing the imputed income as regular income. The experiment also provided a subsidy for employment-related child care expenses in order to duplicate the benefits available under AFDC programs and broaden the labor force participation options for mothers in two-parent households.

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[4] The rules and regulations governing administration of the treatment and responsibilities of the sample families are described in part II of this volume. They are discussed in great detail and reprinted in their entirety in volume II.

Second, the experimental design allowed couples who became divorced or separated to remain eligible for program payments as individuals or as new single heads of families. Where marriages occurred during the experiment, new spouses were eligible to participate and receive payments. Young people at least 18 years old who formed new households of their own were also eligible for payments. The maximum benefit available to unmarried individuals without dependents was $1,000 per year. Finally, SIME/DIME allowed families who moved anywhere in the continental United States to remain eligible for program payments.

* * *

As has been mentioned, although the experimental design only considered the labor supply response, data were collected and analyzed on many other behavioral reponses of potential research and policy interest. The rest of this chapter discusses the expected responses in the other major areas of behavior that were included.

Impact of the NIT on marital status

A major issue of concern was the impact of an NIT on the stability of families. In the late sixties and early seventies, it was generally thought that the existing welfare system contained incentives for families to break up because most families were only eligible for public assistance if there was no male head of household present. Although the situation was changed in principle by the introduction in the early 1970s of the AFDC–U program, in which a two-parent family could receive AFDC benefits in the event both heads were unemployed and had exhausted unemployment benefits, the eligibility and administrative structures of this program were such that only a small proportion of poor two-parent families ended up being covered.[5] It was conjectured that an NIT program based only on total family income would eliminate this artificially induced incentive for an unemployed (or very low wage) husband to depart and might therefore have a stabilizing effect.

Although the detailed cell structure of SIME/DIME did not provide for the fine tuning by age or family structure that would have been optimal for a study of family stability, the design did provide the basic two-cell structure in which families of similar income were divided into those headed by a couple and those headed by a single individual, so that key family changes could be measured in the context of the program. At the time the experiment was designed, however, there was no generally accepted theory as to how an NIT should affect family composition. In fact, we were so unaware of the possibility of a complex relationship between income support programs and family stability that in the baseline data collection effort little consideration was given to collecting marital histories of the families. Only later in the experiment did we evolve more complete notions of the data required to measure the NIT impact on family structure. Attempts were then made to recapture marital histories.

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[5] In 1975, RAND reported the AFDC–U caseloads as 1,851 in Colorado and 4,129 in Washington. These numbers represented one-half of 1 percent of presumably two-parent nonaged families in Colorado and 0.9 percent of such families in Washington (Hosek, 1979, p. 41, Table 9). If 6 percent of such families were in poverty, as implied by national statistics (see USDHEW, 1976, p. 113, Table 19), then a very low participation rate of eligibles is implied.

It was not until the first findings regarding the NIT impact were observed and the unexpected conclusion reached that SIME/DIME was apparently having a negative impact on family structure (i.e., the dissolution of two-parent families was greater in the experimental group than in the control group) that we turned serious attention to developing an explanation for the observed relationship between SIME/DIME treatments and family structure. Fundamentally, the explanation that was developed rests on the premise that an income maintenance program does not have a single effect. Rather, there are two separate effects working in opposite directions, with the net effect a question for empirical determination. In this two-dimensional model, the increased income from the NIT provides increased income for the family which increases its stability. At the same time, the increased income available to a nonworking spouse provides an alternative to economic support from the husband and the resulting economic independence weakens the marital bond. This theory, its implications, and some alternatives to it are discussed along with our empirical findings in part V of this volume.

NIT effects on human capital formation

Principally, an NIT guarantee subsidizes the cost of human capital investment by substituting in part for wages lost because of the training. In other words, if an NIT program has a $4,000 guarantee, then an individual with an $8,000 income would lose not $8,000 but $4,000 by quitting work and going to school. Thus, the total cost of making an investment in human capital that requires a reduction in work has been reduced by the NIT program. As a result, an NIT recipient would be expected to increase his or her investment in schooling, compared with his/her control counterpart. An NIT would also reduce the direct costs of education or training by increasing the income at a given earnings level available to purchase tuition and supplies.

Other effects

There are many other behaviors of interest in any discussion of an NIT. However, these are generally regarded as being of less importance than those discussed. The major ones analyzed using SIME/DIME data are described below.

Fertility

An issue of considerable importance in terms of the long-run effects of an NIT is its impact on fertility. Since the decision to have a child represents a commitment over a long period of time, however, a short-term program can be expected to have far less impact than a long-term program. It is unlikely, indeed, to have an effect on the total number of children a family plans to have, although it might change the spacing of a given number of children.

Over the long run, an NIT can be expected to influence child rearing because some part of the increased income could be expected to be spent on children. However, this income could be spent either on increasing numbers of children or on increasing quality per child. Thus, the net effect of increased income on fertility is an empirical question (see Keeley, 1979). An NIT can also be expected to have a direct positive effect on fertility because the family size index provides an implicit subsidy for additional children, and because the relatively high tax rates of an NIT by reducing labor supply make more time available for child rearing.

Migration

Another measure of importance is the impact of the NIT on patterns of migration. The variation in AFDC support levels across states has always given rise to concern about the attraction for welfare recipients of states with generous AFDC programs. It may be assumed that an NIT would affect migration at two levels: first, the probability of moving, second, the conditional probability of selecting a destination. There were certain design features of the experiment that were meant to capture the migratory impacts of an NIT. SIME/DIME benefits were transportable to any place in the United States. This feature of the experiment replicates that of a national program in which support and tax levels would be the same throughout the country. Great expense and effort was exerted to keep track of, follow, and continue to interview families that moved.

Since adjustments were not made to allow for different living costs in different areas of the country, however, it might be assumed that SIME/DIME would cause people to move to low cost areas where the real value of their NIT benefits would increase. It might also be assumed that the NIT would facilitate moves to places that provide higher nonmonetary utility; that is, due to the tax on wages, there would be less tendency to move to high wage areas and more tendency to move to places where the benefits are not taxed, e.g., better weather.

Durable goods purchases

Although the experiment was not designed to permit measurement of total consumption expenditure or savings, an important aspect of savings was measured—the accumulation of assets and the purchase of durable consumption goods. The principal reason for concentrating on these measures was to provide one test of the permanent income hypothesis. The permanent income hypothesis, as presented by Milton Friedman and others, essentially states that transitory income will go primarily into some form of savings; thus, if recipients of the income maintenance grants regarded them as transitory, due either to the short life of the experiment or to its lack of credibility, then we should find most of the additional income showing up as an increase in the assets position of the family.

Other effects

The experimental design gave rise to the possibility of measuring a wide variety of effects not explicitly incorporated into the cell structure of the experiment. Since the experimental and control groups differ primarily with respect to the treatment, measurement of behavioral responses can be carried out through use of the interview process to collect information from both experimental and control families. Thus, in Part VI of this volume, there is a discussion of other effects, such as stressful situations in the household, the use of child care to increase the labor supply of spouses, the performance of children in school, expenditures on health and health characteristics, and even family roles and conflicts.

CHAPTER III

SAMPLE SELECTION AND DISTRIBUTION

In any social experiment, a critical question arises as to the precise populations to be studied and how many members of those populations are to be selected for inclusion in the study sample. In this chapter, we describe the criteria for inclusion in the SIME/DIME sample, the method employed to allocate the sample to treatments and, finally, the characteristics of the sample that was actually enrolled in the experiment.

Sample selection criteria

Eligibility for financial treatment or control status in the experiment was limited at the outset to a particular subset of low-income families: married couples (not necessarily with children) and those headed by one or more (not necessarily married) adults with at least one dependent child. Husbands or single female heads of families had to be physically capable of gainful employment to be enrolled in the experiment. The disabled were omitted, first, because it was believed that their labor supply responses should not be confounded with those of the able-bodied population and, second, because some of the disabled are eligible for a separate set of program options, such as Supplemental Security Income. Disability was defined for enrollment purposes as any physical condition preventing an individual from being able to work at the time of screening and during the next 3 years. Husbands or single female heads had to be between the ages of 18 and 58 at enrollment, because it was felt that the experimental impact on retiring workers and the aged would be substantively different from the impact on younger workers. A further condition guaranteeing the inclusion of minimum numbers of black, white, or Chicano family heads was imposed in order to ensure a sufficiently large number in each racial-ethnic group for reliable estimation of racial effects.[1]

Eligibility was further limited to families earning less than $9,000 if one head was employed or less than $11,000 if both heads were employed (based on income for a family of four in 1971). Income limits were imposed because families with higher incomes were not expected to exhibit measurable labor supply responses to the experiment or to a similar nationwide program.

The process that led to the eventual enrollment of 4,800 families in Seattle and Denver was accomplished in several steps. Details of the enrollment procedure have been discussed in previous work (for example, Murarka and Spiegelman, 1978) and can be found in part II of this volume and in additional depth in Volume II. In all, 2,042 families were enrolled in 1970 and 1971 in Seattle, and 2,758 families were enrolled in 1971 and 1972 in Denver. About 60 percent of the sample familes were eligible for an NIT plan; the remaining 40% were assigned to be NIT controls (some of these families were eligible for the counseling/training treatments).

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[1] A race variable was assigned to the family at the preenrollment interview by interviewer observation of the male or only head of the family unit. These observationally determined racial identifiers were used to classify the families into cells for assignment to treatment, and were used in almost all the analyses as the race variable. In the second periodic interview, a set of two questions on race and national origin were asked of each family head to verify correctness of the observational classification, and two-way tables of the race and national origin variables against racial assignment were constructed for one- and two-parent families. We found only .8% of single-parent families and 1.2 percent of dual-headed families clearly misclassified, with another 5–7 percent whose status was ambiguous (see Spiegelman and Miller, 1974).

Distribution of sample to treatments [2]

In the Seattle and Denver Income Maintenance Experiments distribution of sample observations to financial and counseling/training treatments was accomplished by using a variant of the Conlisk-Watts assignment model—the same basic model as was used in the New Jersey, Gary, and Rural Income Maintenance Experiments, and considered for use in other social experiments.[3]

The intent of this model was to assign the treatments in such a way as to maximize the reliability of the estimates of the experimental effects. The process of distribution of the sample to different treatment cells involved stratification of the sample by the two family types, three races, six normal income categories, and two sites. Because the cost of a given negative income tax treatment depends strongly on normal income and may depend on other variables used to stratify the sample, the Conlisk-Watts model was used to assign treatments according to the level of normal family income, such that the probability of assignment to a given treatment depended on the sample stratification cell. Thus, unlike a simple random assignment, experimental treatment and household characteristics are correlated. However, within a given sample stratification cell, treatments were assigned purely randomly. Because of the correlation between household characteristics and treatment, special care must be taken in properly estimating and interpreting the response to experimental treatments (see Aaron, 1975; Metcalf, 1977; Keeley and Robins, 1980, 1981).

In the remainder of this chapter, the theory of the assignment model and its application to SIME/DIME are reviewed, followed by a discussion of the implications of the model for proper measurement and interpretation of responses to the experiment.

The theory of the assignment model

The procedure used in SIME/DIME to select and distribute the sample had four stages. First, a set of experimental treatments (called the design space) was specified; second, a budget was established for the experiment; third, sample requirements for each treatment were established by a run of the assignment model; fourth, the sample was located and enrolled (see Conlisk and Kurz, 1972; Murarka and Spiegelman, 1978). Although these four stages are highly interdependent, in the remainder of this section we take the treatment specification and sample selection process as given, and focus the discussion on the assignment process.

In principle, with a given budget, the reliability of the estimates of experimental response can be increased by accounting for the different costs of different sample observations. For example, consider the case of one treatment. Statistical considerations indicate that the ratio of the number of treatment to control observations should equal the square root of the ratio of control to treatment costs. If a treatment observation costs the same as a control observation, then the sample should be equally divided between controls and experimentals in order to obtain the most reliable estimates of response. Alternatively, if a treatment observation is four times as expensive as a control, two-thirds of the sample should be assigned to control status and one-third to the experimental treatment. This basic idea is generalized in the Conlisk-Watts model to the case of many treatments and to the case in which treatment costs depend on family characteristics.

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[2] This section draws from previous work by Keeley and Robins (1980, 1981).

[3] The Conlisk-Watts model used in the New Jersey Negative Income Tax Experiment is described in Conlisk and Watts (1969), Metcalf (1977), Rossi and Lyall (1976), and Kershaw and Fair (1976). A more general discussion of the model is in Conlisk (1980, 1981). The SIME/DIME assignment model is described in Conlisk and Kurz (1972) and Keeley and Robins (1980, 1981). Initially, there were plans to use the Conlisk-Watts model in the Health Insurance Experiment (see Newhouse, 1972). For several reasons, which are discussed in Morris et al. (1977), ultimately it was not used.

The objective of the Conlisk-Watts model is to minimize a weighted sum of the variances of the estimated experimental responses to the various treatments, subject to a cost constraint that takes into account the fact that different observations have different costs. The cost of a sample observation is assumed to depend on both the specific treatment to which it is assigned and the characteristics of the household, primarily its normal income.

Results of the assignment model

Because a given treatment is less costly in terms of expected NIT payments the higher the income of the family, low-income families are more likely to be assigned to programs with low guarantee levels (or low breakeven levels), and high-income families are more likely to be assigned to programs with high guarantee levels (or high breakeven levels). This assignment process, therefore, leads to a lower expected payment per treatment observation than an assignment process that ignores the relationship between the cost of an observation and family income. Because expected average payments are lower, a given budget yields more observations. An additional consequence of this procedure, however, is that the distribution of income differs in different treatment categories (including the controls).

Tables 1.4 and 1.5 present distributions of financial treatment by normal income category (referred to as E-level) for the two major family groups: husband-wife families and single female-headed families. The sample includes all originally enrolled families with E-levels 1 through 6.[4] Treatments are aggregated by generosity of the plan (measured by the grant breakeven level.[5] For each individual E-level (row), the percentages assigned to each treatment are presented.[6] The penultimate row in each table gives the distribution of the total sample (all E-levels in the various treatment categories) and the last row gives the mean E-level category for each treatment. In parentheses at the top and left of each table are the numbers of families in each cell.

For husband-wife families, table 1.4 shows that low E-levels (E1–E4) are more likely to be assigned to programs with low grant breakeven levels, and high E-levels (E5 and E6) are more likely to be assigned to programs with high grant breakeven levels, which is consistent with the intent of the model as previously described. The χ^2 tests (with 4 degrees of freedom) indicate that, except for E1, the distribution of treatments within each E-level is significantly different from the overall distribution for all E-levels.

The results for assignment to control status indicate that low E-levels (E1–E4) are less likely and high E-levels (E5 and E6) more likely to be assigned control status, because the relative weights in the Conlisk-Watts objective function for control treatments were increased for E5s and E6s.[7] In table 1.5, the same

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[4] In the actual assignment, E7s (normal income of \$11,000 to \$13,000), and E0s (normal income unknown) are also included. E7s are excluded from the figures in Tables 1.4 and 1.5 because they are all controls, and because, as is the case with E0s, they were not formally assigned by the assignment model. In many of the analyses reported in the subsequent parts of this volume, E7s are included in the analysis samples. This poses no special estimation problems as long as a dummy variable for being in the E7 category can appropriately be included in the control set of the equation.

[5] For the constant tax rate programs in SIME/DIME, the grant breakeven level is given by S/t_e. For the declining tax rate programs, the grant breakeven level is given by $(t_e-\sqrt{t^2_e-4rS})/2r)$, where r (=.025) is the rate of decline of the average tax rate per thousand dollars of income.

[6] The results presented in tables 1.4 and 1.5 are based on the actual assignments, which differed somewhat from the theoretical assignment produced by the assignment model.

[7] The actual assignments in SIME/DIME that resulted from the implementation of the model were influenced by a set of weights that were assigned to each E-level, each racial group, each family composition alternative, and each treatment. These so-called policy weights were intended to give increased relative weight to those treatments

Continued

general pattern holds for female-headed families. High E-levels are more likely to be controls and low E-levels are more likely to be assigned to low grant breakeven level programs.[8]

Problems with the assignment model and use of assignment variables

As discussed in Keeley and Robins (1980, 1981), the characteristics of the assignment model lead to several potential problems that must be kept in mind if the data are to be appropriately analyzed and interpreted.

TABLE 1.4.—DISTRIBUTIONS OF FINANCIAL TREATMENT BY NORMAL INCOME CATEGORY FOR HUSBAND-WIFE FAMILIES

Normal income	Category	Proportion Category		Proportion of financial category by grant breakeven level of financial plans[1]			χ^2
		Controls (N = 1,138)	Financials (N = 1,114)	Low $5,000 to $7,500 (N = 465)	Medium: $7,500 to $10,000 (N = 574)	High: $10,000 to $12,000 (N = 402)	
$0 to $1,000..........	E1 (N = 54)	0.407	0.593	0.204	0.315	0.074	4.67
$1,000 to $3,000.....	E2 (N = 179)	.430	.570	.302	.190	.078	[2] 22.59
$3,000 to $5,000.....	E3 (N = 459)	.362	.638	.333	.211	.094	[2] 78.21
$5,000 to $7,000.....	E4 (N = 738)	.364	.636	.248	.285	.103	[2] 54.57
$7,000 to $9,000.....	E5 (N = 698)	.519	.481	.076	.235	.170	[2] 52.92
$9,000 to $11,000...	E6 (N = 451)	.537	.463	.024	.115	.324	[2] 174.79
	All (N = 2,579)	.441	.559	.180	.223	.156	....
Mean E = level category............	4.24	4.41	4.11	3.53	4.09	4.81	....

[1] The low breakeven plans include F2, F3, F4, F6, the medium breakeven plans include F1, F5, F8, F10; the high breakeven plans include F7, F9, F11

[2] Significant at the 1 percent level

The assumed response function is an analysis-of-variance model in which family composition and race are interactive with the other elements of the experiment. Thus, the assignment model assumes that, in principle, separate response equations could be estimated for each of two family types (families headed by a single adult and families headed by a couple) within each of three racial groups (black, Mexican-American, and other whites). Within each race-family structure group the response model assumes complete interaction between the guarantee level (S_i), the tax rate, (T_i), the counseling and training treatment (M_i), and the normal earnings category (E_i), with the simplifying assumptions that there are no third order interactions and no interactions between the tax rate and either normal earnings category or counseling and training programs. The model is given by

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and population groups that were most important for policy purposes. Thus, the less generous plans were weighted more heavily than the more generous ones, and those combinations of high E-levels and low treatment levels that essentially would cause most of the families to be above the breakeven level were set to zero. However, there was no change in the weight for the control "treatments," and therefore the relative control weights were higher for cells E5 and E6. These factors led to a higher probability of assignment to control status for E5s and E6s and consequently a higher mean income for the control group.

[8] The results presented in tables 1.4 and 1.5 are based on the actual assignment, which differed from the theoretical assignment produced by the allocation model because of difficulties in filling some of the cells (see Murarka and Spiegelman, 1978, and Volume II). However, the overall actual allocation of the sample seems to be reasonably close to that dictated by the allocation model and there is no systematic pattern to the differences.

TABLE 1.5.—DISTRIBUTIONS OF FINANCIAL TREATMENT BY NORMAL INCOME CATEGORY SINGLE FEMALE-HEADED FAMILIES

[Ratio of actual to theoretical assignment in parentheses]

Normal income	Category	Proportion of category		Proportion of financial category by grant breakeven level of financial plans [3]			χ^2
		Controls (N=734)	Financials (N=1,098)	Low: $5,000 to $7,500 (N=451)	Medium: $7,500 to $10,000 (N=454)	High: $10,000 to $12,000 (N=193)	
$0 to $1,000........	E1 (N=316)	0.430	0.570	0.275	0.215	0.080	5.201
$1,000 to $3,000.	E2 (N=455)	.321	.679	.277	.284	.119	[1] 12.208
$3,000 to $5,000.	E3 (N=456)	.301	.699	.311	.266	.122	[2] 21.785
$5,000 to $7,000.	E4 (N=346)	.460	.540	.208	.269	.064	[1] 11.231
$7,000 to $9,000.	E5 (N=220)	.596	.404	.086	.159	.159	[2] 56.677
$9,000 to $11,000...........	E6 (N=26)	.808	.192	.038	.154	....	[2] 18.93
	All (N=1,832)	.401	.599	.246	.248	.105	....
Mean E=level category..........	2.88	3.09	2.73	2.58	2.80	2.95	....

[1] Significant at the 10 percent level.
[2] Significant at the 1 percent level.
[3] The low breakeven plans include F2, F3, F4, F6; the medium breakeven plans include F1, F5, F8, F10; the high breakeven plans include F7, F9, F11.

$$y = \sum_{i=1}^{6} a_iE_i + \sum_{j=1}^{4} \sum_{i=1}^{3} c_{i,j}S_iT_j + \sum_{i=2}^{6} \sum_{j=1}^{3} d_{i,j}E_iS_j$$

$$+ \sum_{i=1}^{6} \sum_{j=1}^{3} e_{i,j}E_iM_j + \sum_{i=1}^{3} \sum_{j=1}^{3} f_{i,j}S_iM_j + u \qquad (1)$$

where y is the variable whose response to the experiment is being studied (e.g., hours of work). The model has a total of 61 coefficients to be estimated for each subgroup as opposed to a possible 288 in a completely interactive model.[9] In a second iteration of the model, duration was introduced with two alternative values—3 years and 5 years. This modification doubled the number of coefficients.

The efficiency criterion function used was a weighted sum of the error variances for predictions of the dependent variable for each of the possible combinations of the right-hand side variables. The assignment model minimizes this criterion function within each race and family structure group subject to the constraint that the cost function equals a preassigned budget.

But this procedure failed to ask a prior question: "Is the budget adequate to obtain statistically reliable measures of the treatment effects?" The budget was given and the assignment model, therefore, simply addressed the question of the optimum allocation of the given budget. In fact, because of the large number of coefficients (or cells) the response function used in the SIME/DIME sample allocation model did not succeed in identifying statistically significant responses for individual treatment cells. For example, in the model for white husbands the re-

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[9] For example, the coefficient for the S=$5,600, T=50 percent, E=$7,000–$9,000, and M–0 treatment cell has a standard error of 5.0 hours. For a 4 hour per week response to be significant, the standard error would have to be 2.04 hours. Since standard error$\sim 1/\sqrt{n}$, where n is the sample size, a reduction of the standard error to 2.04 would require

$$n^* = n \cdot \left(\frac{5.0}{2.04}\right)^2 \approx 6\,n.$$

gression on hours of work in the eighth experimental quarter for a sample of 1,015 white husbands is significant at the 1% level (F=2.88); however, none of the individual treatment coefficients is significant even at the 5% level.

The lack of significance of the individual treatments in a model with 120 treatment variables is not surprising. The evidence from this regression suggests that a sample perhaps six times as large as the one actually used in SIME/DIME might be necessary to find statistically significant results using a response model which fully exploits the separate treatment cells.[9] In this case, the SIME/DIME sample of approximately 5,000 families would have had to be increased to 30,000 to find significant results using a complete analysis-of-variance model (without considering duration).

Thus, the models used to estimate the effects of the experiment in SIME/DIME reported in this volume are either group plans, or use parameterizations that impose "theory" on the behavioral response models. However, it should always be kept in mind that, due to the assignment process, simple mean differences between experimentals and controls or between experimentals on different treatments are not unbiased estimates of program effect. To avoid bias in the measurement of response, all variables used to determine treatment must be included in any estimation model as explanatory variables.

If the assignment variables only affected the mean differences in behavior, then the problem would be easily resolved by including these variables along with treatment variables. However, if the assignment variables affect response (e.g., lower E-level [income] families respond more to a given treatment than higher E-level [income] families), then the assignment variables must also be interacted with the treatment variables (see detailed discussion and presentation of the theory in Keeley and Robins, 1980, and in Robins and West, Part III of this volume).

Sample characteristics at enrollment

The original experimental sample totaled 4,800 family units, with 2,042 families enrolled in Seattle and 2,758 enrolled in Denver. The larger enrollment in Denver resulted from the decision to include a set of Hispanic families in the Denver sample only. Hispancs were not a large enough segment of the Seattle population to be enrolled in that city, but a sufficient number were available in Denver for that purpose and it was a racial group not originally included in the sampling design. The sample comprised 2,031 families headed by a single adult and 2,769 families headed by a couple (some of whom were childless). The larger number of couple-headed families was a result of the application of policy weights in the assignment model (see footnote 7 of this chapter) that favored couples because of the greater policy interest in their response, and the statistical requirements of the model that weighted toward couple-headed families to compensate for the greater tendency for couples to change family status. In general, the distribution of the SIME/DIME sample characteristics resulted from the requirements of the assignment model. To estimate the impact of an NIT on any other population, it is necessary to predict the behavioral response for each population segment (race, family status, etc.) and assign appropriate weights to determine the mean response for the population.

The sample was distributed among four major treatment groups. (Since SIME/DIME essentially consists of two experiments conducted simultaneously, a family may be an experimental in one treatment and a control in the other.) All four cells are occupied in order to test both the separate and combined effects of the counseling/training and financial components of the experiment. With regard to the financial treatment, the sample is divided such that 57% of the original families are assigned to a financial treatment and 43% to a control status.

	NIT controls	NIT experimentals	Total counseling/ training
Counseling/training controls	1,041	946	1,987
Counseling/training experimentals	1,012	1,801	2,813
Total financial	2,053	2,747	4,800

Table 1.6 shows the distribution of the sample by the income categories used as cells in the assignment process. Income includes earnings from work and returns on capital, and excludes transfer income. Although SIME/DIME was a welfare-type program, only a third of the families selected were at the time enrolled in the major welfare program, AFDC. Most of the families were working families who may have been receiving payments under some other transfer programs, such as public housing, Food Stamps, or unemployment insurance, but were not generally considered as welfare families. The selection criterion expanded the income base well beyond the existing welfare set because the principal question of concern in the experiment was "How would nonwelfare, low-income families respond to an NIT?"

TABLE 1.6.—ENROLLMENT BY INCOME CATEGORY

	SIME	DIME	Total
No E[1]	65	101	166
$0 to $1,000	203	179	382
$1,001 to $3,000	291	362	653
$3,001 to $5,000	385	570	955
$5,001 to $7,000	467	670	1,137
$7,001 to $9,000	425	546	971
$9,001 to $11,000	196	307	503
$11,001 to $13,000[2]	10	23	33
Total	2,042	2,758	4,800

[1] Mostly secondary families discovered at time of enrollment and not part of assignment.
[2] Control families only.

Source: Murarka and Spiegelman (1978), pp. 47 and 54.

As noted above, 43 percent of the originally enrolled families were in an NIT control group. Those families also not in the counseling/training treatments were considered as null controls in that they receive no treatment. They were, however, interviewed on the same interview schedule as experimental families.[10]

Since the essence of the experimental design is the existence of the control group, it is important to understand the role of a control group in a social experiment. A control group differs from the experimental group only in that controls receive no treatment. This, however, is not as straightforward or as obvious in terms of measurement issues as appears at first sight. Comparing the mean value of a response variable (e.g., hours of work per year) of any single NIT treatment

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[10] They received a payment of $8 per month for sending a postcard to the Mathematica office indicating their current address. This was done to avoid losing families that moved. To determine if there was an effect on labor market behavior resulting from the submittal of income reports by experimental families, 50 percent of control families in Denver submitted income reports identical to those submitted by families receiving NIT payments. A test of the significance of reporting did not indicate that submitting forms (IRFs) had any effect on labor supply response. The IRF-control variable was included in most of the labor supply models, but was never statistically significant. This is important because the reverse finding would indicate that our estimates of experimental effects based on comparing reported behavior of experimental and control families are contaminated by differences in the reporting system (see part III, chapter 1.)

(e.g., $3,800 support level and 50 percent tax) against the mean value of the variable for the controls will show the impact of the treatment. But, if we wish to interpret the results in terms of a normalized response to a unit of support and tax change, we must know more about the control group than that they did not receive a treatment. To measure tax effects, we must know the tax rate faced by controls. It is certainly not zero, because the controls are ordinary families in society who face tax systems that exist outside of the treatment environment, including the positive income tax and those taxes imposed in other income-conditioned programs that a control family may participate in. The measurement of the NIT effect requires knowledge of transfers received by control families. Since one-third of control families are on AFDC and others receive a mixture of programs that exist in the normal environment, control families on average also have positive transfer payments. It is essential to be able to measure the taxes and transfers in the control environment to know what are the real differences being created by the treatments. Otherwise, the experimental-control differences (i.e, the experimental response) cannot be extrapolated to determine the effects on any other population, or any mix of experimental treatments other than that tested in the experiment. Table 1.7 shows some of the characteristics of the controls at the time of enrollment (i.e., their E-level distributions, percentage on AFDC, mean incomes, and other characteristics) that may be important for comparison with experimentals.

TABLE 1.7.—MONTHLY MEANS OF PUBLIC TRANSFERS AND OF AFDC AMONG ONE- AND TWO-PARENT CONTROL FAMILIES ACROSS NORMAL INCOME GROUPS

Normal income group	AFDC		Other public transfers	
	One parent (N = 801)	Two parents (N = 1,220)	One parent (N = 801)	Two parents (N = 1,220)
Unknown	$152.11	$0.00	$12.72	$44.48
$0 to $1,000	166.14	142.75	67.95	78.21
$1,001 to $3,000	127.90	96.45	56.09	38.38
$3,001 to $5,000	82.75	35.65	49.23	46.13
$5,001 to $7,000	40.90	10.40	38.03	24.48
$7,001 to $9,000	12.32	6.60	31.53	15.59
$9,001 to $11,000	12.78	1.80	16.60	12.06
$11,001 to $13,000	52.42	1.10	0.00	9.64
Overall	82.68	19.54	46.95	24.17

	Number of control families					
	AFDC		Overall	Other public transfers		
	One parent	Two parents		One parent	Two parents	Overall
Missing	11	19	30	11	19	30
Zero transfers	412	1,078	1,490	331	871	1,202
Positive transfers	389	142	531	470	349	819

CHAPTER IV

ISSUES IN THE MEASUREMENT OF EXPERIMENTAL EFFECTS

Most of this volume is concerned with the use of the data generated in SIME/DIME to measure experimental effects. It is important, therefore, to be aware of problems that can arise in using experimental data and how they were handled in SIME/DIME.

Duration of treatment effects

A family's response to a program of limited duration may differ from its response to a permanent program. First, the income effect is weakened (biased downward) because the benefits available in a finite experiment are available for less time and, therefore, have a lower capitalized present value than benefits available indefinitely. Second, the tax rate is imposed on the recipient's wage rate only during the experiment. Thus, the wage rate net of taxes will return to its preexperimental level once the experiment ends. In other words, during the experiment, there is a "sale" on leisure, which means that the substitution effect in a finite experiment is biased upward. SIME/DIME dealt with this problem by instituting experiments of different durations. Families were enrolled in 3-, 5-, or 20-year programs.[1] Differences in response among programs that are identical except for duration provide a way to measure duration effects.

Time horizon effects

In addition to the effect of duration on the equilibrium degree of response, there is the issue of time required to adjust to any equilibrium. Changes in hours of work or in weeks worked per year cannot be made immediately due to obvious institutional constraints (e.g., a job may require 40 hours per week of work; thus, change of hours of work may necessitate a change of jobs, which takes time). The adjustment process is discussed in Part III, Chapter 1. That discussion shows that the adjustment process is lengthy, and that an equilibrium level of labor supply (i.e., the amount of work that a person of given characteristics wants to provide under a specific set of circumstances) will be reached only after considerable time has elapsed (i.e., about 2½ years for husbands, 3½ years for wives, and 4½ years for female heads [see Robins and West, 1978]). This means that the early measures of the labor supply response are underestimates of the longer-run responses.

Site differences

Since the goal of the NIT experiments was to provide guidance for national policy, it is obvious that a national sample, rather than site-specific samples, would have been preferable, other things equal. A national sample was not selected because of high administrative and data collection costs and because of the difficulty in providing information services, job counseling, and training subsidies to a dispersed sample. However, the income maintenance experiments overall were located in six different sites (counting the four cities of the New Jersey Experiment as a single site), providing a test of the hypothesis that site is not important in the response to a financial treatment. Comparisons among the four experiments and between Seattle and Denver lend weight to the argument that the site-specific results are reasonably generalizable to a national population.

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[1] Approximately 200 families were switched from control status or from 3-year treatment groups to the 20-year program in the third year of the experiment. Because of the lateness of this shift and the subsequent finding of negligible marginal effects on 20-year families, almost all the analyses in which duration was a variable consist of comparisons between 3- and 5-year durations.

SIME/DIME was originally scheduled only for Seattle. Previous experiments had been located in eastern or midwestern sites. But when changes in Seattle's major industry (aerospace) brought the Seattle unemployment rate to over 10% in 1970, the Seattle area economy could no longer be considered representative of a significant sector of the nation. Adding the Denver site, with its low unemployment rate, made possible a comparison of effects in a normal setting with those in a setting of economic dislocation.

Attrition

The SIME/DIME experiment was designed explicitly with the aim of making inferences about work responses to a negative income tax program. One of the key features of an experiment is the ability to compare the behavior of randomly selected experimental and control groups. The validity of these inferences depends critically, however, on maintaining the integrity of the sample. One major threat to the integrity of the sample is sample attrition. Sample attrition occurs when a family either cannot be found by an interviewer or refuses to be reinterviewed. Failure to reinterview families in the originally selected sample requires that such families be excluded from analyses that pertain to periods after the time when attrition occurs. In the administration of SIME/DIME, a great deal of attention was paid to developing ways to counter attrition, and substantial resources were devoted to minimizing its occurrence. (These efforts are described in Part II of this volume and discussed in detail in Volume II.)

Sample attrition may reduce the degree of comparability between experimentals and controls that was originally engendered by the random selection process. This reduction in comparability would not occur if the attrition process were purely random, or if the only nonrandom component of the attrition process were a relationship with observable predetermined variables, in which case comparability could be restored by controlling for the effects of these variables. But, if attrition is related to the values of the outcome variables being studied, as it almost certainly is, biased inferences may result from comparisons between the experimental and control groups.

Although similar attrition biases plague both experimental and nonexperimental evaluations, the magnitude of these biases may be increased in an experimental situation because the process generating attrition may be different for experimentals and controls. For instance, in an income maintenance experiment like SIME/DIME, the experimental NIT payments are an added incentive for experimental families to remain in the sample. In addition, families with higher payments are more likely to remain in the sample than families with lower payments. Because the NIT payment depends on earnings which in turn depends on hours of work and wage rates, attrition among experimentals can be expected to be related to hours of work in the experimental period—an outcome variable of key interest. The relationship between attrition and hours of work will undoubtedly be different among controls. This situation can lead to a biased estimate of the effects of the experimental NIT payments on hours of work. These biases are discussed in more detail in Part III, Chapter 1 of this volume, in Robins and West (1978), and in Pencavel and West (1978).

Extent of attrition in SIME/DIME

Because the problem of sample attrition was recognized by the designers of the SIME/DIME experiments, intensive efforts were made to minimize its extent. Families were required to submit monthly address reports so that they would be easy to find at the time of interview. Families were followed and kept in the sample as long as they remained in the United States. Also, families were paid a fee to complete interviews to compensate them for the time and effort.

In spite of intensive efforts throughout the experiment to minimize attrition, there was a substantial amount of attrition during SIME/DIME. During the first 2½ years of the experiment, 20% of husbands, 15% of wives, and 15% of single female heads of families dropped out of the experiment,[2] a rate roughly equivalent to that experienced in the New Jersey Income Maintenance Experiment. These attrition rates are different for husbands and wives because of differential attrition after family separations.

Table 1.8 presents a breakdown of the attrition rate in SIME/DIME by site, race, and experimental control status. For husbands and wives, the attrition rates did not differ between Seattle and Denver, but did differ among ethnic groups, being somewhat higher among blacks and Chicanos than among whites. In contrast for female heads of families the difference between the sites was significant and the ethnic group differences were not. For them, the attrition rate was somewhat higher in Denver than in Seattle. For all three groups, the difference in attrition rates between experimentals and controls was significant. As was expected, experimentals were less likely to drop out of the experiment than were controls. Also, within the experimental population itself, the type of program to which an individual was assigned revealed some interesting contrasts. Table 1.9 presents attrition rates by assigned NIT program and a simple χ^2 test indicates that the differences among the programs are significant for experimentals. This table suggests that attrition rates are associated with the generosity of each program, where generosity is defined in terms of the breakeven level for a family of four. For instance, for wives the program with the highest attrition rate also has the lowest breakeven level (program F2), while the lowest attrition rate is experienced by the program with the second highest breakeven level (program F11).

TABLE 1.8.—ATTRITION RATES BY EXPERIMENTAL STATUS, SITE, AND RACE

	Husbands	Wives	Female heads
Controls	.217	.180	.200
Experimentals	.181	.125	.109
Seattle	.188	.156	.119
Denver	.204	.144	.156
Black	.250	.191	.144
Anglo	.150	.119	.150
Chicano	.215	.149	.137

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[2] This attrition rate is with reference to the families enrolled in the experiment. It does not include 372 families (about 6%) who were selected but refused enrollment and 512 families (about 9%) who were selected but had moved out of the area or could not be found. Nor do the attrition statistics include larger numbers who refused to take the screening or preexperimental interviews. Since, however, these interviews were administered without any reference to an impending experiment, there is no reason to expect any response bias from these refusals (for more details, see Murarka and Spiegelman, 1978).

TABLE 1.9.—ATTRITION RATES BY FINANCIAL PROGRAM

Program	Dollar tax break-even level	Husbands	Wives	Female heads
F1	10,250	.206	.082	.124
F2	6,350	.265	.224	.139
F3	10,850	.268	.214	.134
F4	7,800	.213	.160	.103
F5	13,150	.121	.096	.043
F6	8,520	.244	.181	.120
F7	19,700	.144	.103	.097
F8	11,510	.184	.112	.094
F9	15,700	.088	.075	.083
F10	9,780	.138	.114	.111
F11	16,230	.134	.073	.095

The probability of attrition

Pencavel and West (1978) estimated a model of the probability of attrition. They used the same variables to model the effects of the experimental treatments as are used in the analysis of labor supply effects of the experiment presented in Part III, Chapter 1 of this volume. They found that the change in disposable income induced by the experimental treatments significantly reduced the probability of attrition for husbands and wives and single female heads of families. However, the change in the net wage rate induced by the experiment had a significant effect only for single female heads, for whom increasing the net wage rate increased the probability of attrition. They also found that persons in families which had experienced marital changes were more likely to drop out, that persons and families initially living in subsidized housing were less likely to drop out, and that female heads who had given birth during the experiment were less likely to drop out. Their results also confirmed the ethnic group differences found in table 1.8—blacks and Chicanos dropped out of the experiment at a higher rate than whites. Family size and composition had no effect on the rate of attrition. The counsel/training treatments tended to reduce the probability of attrition. However, the effects were rarely significant.

Reporting error

In social experiments the subjects themselves report most of the data. Reporting behavior is subject to random error and to incentives of the experiment itself and of society at large. Reporting inaccuracies add an additional element to other sources of experimental error, and they affect the analytical methods used. In addition, the experimental effect on reporting error is itself of interest.

Since the experiment obviously operated under a budget constraint, it was necessary to make choices about where to direct resources according to the relative importance of data areas, the relative potential for inaccuracy, and the relative incentives to underreport. For most types of data, there is either little incentive to misreport (the age of family members is an example of this kind of information), or the reports are easily checked (the number of children in the family is an example). Income, however, is different. Income is usually heavily taxed and is quite easily concealed. Therefore, most of the SIME/DIME emphasis on the analysis of data accuracy was placed on income data.

To minimize error in reported income, several precautions were taken in the operational and data acquisition systems, described in Part II of this volume and discussed in detail in volume II. Interviews, upon which the analysis of experimental-control differences was based, were designed, through the inclusion of detailed cross-referencing questions, to make it difficult to misreport. Second, payment analysts checked on any peculiarities or sudden changes in the income reports. They also continually checked welfare records for instances of nonreported welfare payment. Income was divided into three general classifications: wage income, public source nonwage income, and private source nonwage income. Since the high taxes on wage income provided a considerable incentive to underreport, wage income data were collected not only as monthly aggregates but also in the form of the wage rate and hours worked for each individual job. Ideally, all nonwage income would have been taxed at 100 percent to insure that changes in nonwage income represented experimental treatments. Public-source nonwage income was indeed taxed at 100 percent, but private-source nonwage income (which is more easily concealed) was taxed at only 50 percent to lessen the disincentive for families to conceal it, and to reduce the incentive to forgo this source of family income.

To measure error in reported income and determine whether there were discrepancies between SIME/DIME data and other data sources, a data validation study was undertaken (Halsey, Murarka, Spiegelman, 1976) for 10 percent of our sample, focusing on individual income sources. We expected that income streams with infrequent or variable-sized payments would be reported inaccurately, relative to income streams whose payments were current, equal, and closely spaced. Since many nonwage income streams have payments that are infrequent or variable in size, while most wage income is more regular, we expected wage income to be more accurately reported than nonwage income. The results of the validation study tended to support this surmise, although the frequency of some types of nonwage income in our sample was too low to allow reliable statistical inferences.

A large sample wage income study comparing income reported to SIME/DIME with income reported to the Internal Revenue Service (IRS) on forms 1040 and 1040A was undertaken, which showed that underreporting to SIME/DIME was statistically significant but of small magnitude (Halsey, 1980, pp. 42–45). SIME/DIME participants on average reported between $100 and $300 less per year to the experiment than to the Internal Revenue Service—less than 5% of mean income. The variance in the amount underreported to SIME/DIME is on the order of $1,000, or about one-fifth of mean income. Surprisingly, this variance appears to be roughly constant, rather than proportional to income, and is large enough to cause noticeable errors in ordinary least squares parameter estimates.

Underreporting was not universal, however. We observed that almost as many people overreported their income as underreported it. Two reasons might explain the apparent overreporting of income to SIME/DIME. The first is simply that apparent overreports to SIME/DIME may really be underreports to the IRS. Income from odd jobs (for which no W-2 form is ordinarily generated) is a good candidate for this type of underreporting to the IRS. The other reason is that SIME/DIME participants were asked at each interview to recall data for the 4 previous months. In some cases, high current income levels may have been incorrectly attributed to previous months of low income. Further study of individual cases is necessary to resolve these issues. A control might overreport income for reasons of pride on interviews but would not on his/her tax form, where it would be costly to do so; but this is unlikely for experimentals, who would pay for such behavior in lower benefits.

In a third validation study, Greenberg and Halsey (1980) attempted to measure directly the effects of underreporting on the labor supply responses variables. Using unemployment insurance earnings records and social security earnings (grouped to avoid disclosure), the authors reestimated the labor supply models, and concluded that underreporting by experimentals did tend to bias upward the estimation of treatment effects on labor supply.

A validation study should have been conducted concurrently with the survey, because the accuracy, ease of access, and even the availability of corroborative sources deteriorates over time. But the lag between the survey and validation study was at times as long as 4 years, a factor that may weaken the reliability of our validation results.

Comprehension of the experiment on the part of participants

It is recognized that responses of participants are conditional not only on the treatment, but on their perception of the treatment. True perception probably cannot be determined without exhaustive survey techniques that were beyond our abilities and certainly our intent. However, efforts were made to foster good communication with the participants and to develop some knowledge of at least their verbal comprehension. The difficulty with the latter measure, of course, is that verbal comprehension does not necessarily represent true understanding. In the first place, to the extent that people know what they need to know, their lack of knowledge on any point at a particular time may simply reflect their lack of a need for the knowledge, and that they would find out if the need arose. More important, true knowledge is conveyed in behavior and the correlation between verbal understanding and behavior is considerably less than one. Even more strongly, verbal skills of our low-income sample may not have been fully adequate to cope with the questionnaire. For example, Chicanos did on the average considerably worse than whites. We cannot be sure if this is indicative of less comprehension by Chicanos or less ability to deal with questionnaires in English.

Procedures for educating the participants were essentially of the following kind. First, upon enrollment into the experiment, the participant family was visited by a trained enroller who spent more than an hour describing the experiment, the rules of operation, and the tables that the family could use to calculate its grant from any known amount of income. A table was created for each of the NIT plans, such that the family did not need to calculate any formulas in order to arrive at an estimate of the benefit due. Second, permanent field offices were established in several locations in the cities of Seattle and Denver and the families were informed that representatives in these field offices were there for the purpose of responding to their inquiries and questions regarding the program. Third, each month, a detailed statement was sent to the family describing the grant calculation and the procedures. Fourth, one year after the family was enrolled into the program, the family was visited by a representative of Mathematica. At that meeting, the program and grant calculation procedures were again described to the family. In addition, all changes in the rules of operation since the enrollment were discussed, and the family was encouraged to ask questions.

At that time—immediately prior to the discussion—a brief survey was conducted to determine the extent to which the families understood the nature of the program.[3] Survey questions fell into five categories: (1) program eligibility, (2) work effort, (3) family structure effects, (4) income sources, and (5) grant calculation.

The analysis of the response to the so-called "reeducation" survey indicated a moderate verbal understanding of the overall program by both Seattle and Denver families. Seattle families displayed a mean score of 57 percent correct responses on the entire survey, while their counterparts in Denver obtained a comprehension score of 51 percent. A very important conclusion from the survey results was that the percentage of correct answers was indeed changed by prior association with the events being asked about. For example, questions on the extent to which child care services could be reimbursed for work under SIME/DIME were answered more correctly if the respondent had child care expenses prior to that time. In the same manner, questions regarding the effects of being unemployed or refusing to accept work if unemployed were answered better if the person had previous involuntary unemployment experience. We found a striking difference in the ability to use the annual benefit tables that describe the relationship between income and benefit if the respondent had some annual earnings than if not, and so on. Thus, we conclude that the mean responses underestimate the amount of real comprehension in that they do not account for the fact that people will find out about the effects of different behaviors when those behaviors or activities become relevant.

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[3] For a detailed description of the survey, see the study by Halsey, Murarka, and Spiegelman (1979).

The survey was also used by Halsey, Murarka, and Spiegelman (1979) to rerun one of the labor supply models most commonly used to describe the impact of the experimental treatment on work effort with the inclusion of comprehension scores. This variable had no effect on the regression, implying that the comprehension score did not change the measurement of treatment impact on labor supply behavior. Thus, we found no evidence that an inability to communicate an understanding of the program was associated with the labor supply response of the participants to the experiment.

Other possible biases

Other potential problems arising from the experimental nature of the study include Hawthorne effects, dowry effects, and community effects. Hawthorne effects would exist if the act of providing monthly income records affected the data that families reported. The extent of Hawthorne effects was investigated by requiring half the control families in Denver to submit the same monthly income reports as the experimental families. Results show no significant differences in the quality of income reporting in interviews between control families submitting such reports and those not submitting them.

Dowry effects can occur in an experiment if persons eligible for payments can extend eligibility to new spouses. Such a provision in SIME/DIME creates a financial incentive for marriage. The potential bias in the measured effect of the NIT treatment on marriage is addressed in Part V of this volume.

Community effects, or changes in social attitudes and tastes occurring in a communitywide or nationwide program, are not replicable in a short-term study using a dispersed sample. Community effects were, therefore, addressed explicitly in SIME/DIME. A saturation experiment (including all eligible families within a given area) would give more information about these effects, but such a design would pose other severe—possibly insurmountable—difficulties in controlling for changes in the external environment.

CHAPTER V

THE PAYMENT SYSTEM IN SIME/DIME

The payment formula and its complexities [1]

Financial treatments in SIME/DIME provided income subsidies to families in accordance with the principles of a negative income tax (NIT). The NIT is simple in structure, being defined by the following two equations:

$$G(Y) = S - F(Y) \tag{1}$$

$$Y^* = Y + G(Y), \tag{2}$$

where

$G(Y)$ = the NIT grant, which is a function of income
S = the support level, which is the income received by the family from the program if it has no other sources of income
$F(Y)$ = is the NIT tax function, or the benefit reduction formula
Y^* = is the family's total income
Y = is that income received by the family from sources other than the NIT program.

Although the basic NIT formula is simple, there were many complex issues to be resolved in order to implement an NIT experiment. Some of these issues derived from the fact that we were conducting an experiment, which required definitional precision and functional continuity to a degree unnecessary in an operating program. Other issues are complex because the very nature of a program sets a requirement for detailed rules of operation. In this chapter, we briefly discuss the tax formula, the accounting period, indexing for family size, and changes in cost of living.

The tax formula

In all income maintenance experiments except SIME/DIME, the tax formula, $F(Y)$, was represented by a constant proportional tax on income, so that the formula read $f(Y) = TY$, where T is some constant proportion, 30 percent, 50 percent or 70 percent, applied to all incomes eligible for taxation. SIME/DIME, as has already been mentioned, had two different forms of the tax function. One was a constant proportional tax rate and the other was a schedule of declining marginal tax rate. A declining marginal tax rate is one in which each additional unit of income is taxed at a lower rate than the previous unit of income. In SIME/DIME, a fairly simple linear formula was applied, in which the marginal tax rate declined by 5 percent for each $1,000 of income. For example, one program in which the initial $1,000 of income was taxed at 70 percent had a formula in which the second $1,000 was taxed at 65 percent, the third $1,000 at 60 percent, etc. This formula is equivalent to a tax function with the average tax rate declining at 2.5 percent per $1,000 of income. That is, a family with $1,000 of income pays 70 percent on total income, a family with $2,000 of income pays

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[1] All this material is covered in much greater detail in Volume II.

67.5 percent and a family with $3,000 of income pays 65 percent tax on total income. In this system, however, the marginal tax rate could turn negative before the payment reaches zero. To avoid the marginal tax rate becoming negative, resulting in an increase in the grant after declining to a minimum, a constraint was added to the formula setting the average tax rate to one-half the initial tax rate at the point where the marginal tax rate reaches zero. The NIT formulas used in SIME/DIME are set forth below:

$$F(Y) = tY \qquad \text{for constant rate treatments} \qquad (3)$$

$$F(Y) = \text{Max}(t_0 - 0.025Y,\ t_0/2)\ Y \qquad \text{for declining rate treatments,} \qquad (4)$$

where

$$t = 0.50,\ 0.70$$
$$t_0 = 0.70,\ 0.80.$$

The accounting period

An essential feature of the payment system in SIME/DIME is the annual accounting period, which derives from the intent of the income maintenance program to provide a defined level of support over a year. Since payments were made once a month, the system was designed so that every month represented the end of a 12-month year.[2] Each monthly payment was calculated so that the total payments over any 12-month period would be the same as if the payment had been made once at the end of that period; that is, every month was the end of a payment year. More precisely, the grant in any given month was equal to the support available to the family over a year, minus the tax function applied to income over the previous 12 months, minus the payments that the family had already received over the previous 11 months. This formula provides the marginal annual payment, and the one that would be correct in any given month. The annual accounting period did create some problems, mainly resulting from the fact that many families had incomes in some months that were above the breakeven level (i.e., above the level at which payments under the system would be zero). Thus, a situation would often exist in which a family would receive payments for several months because its income was below the breakeven level, then the family's income would rise above the breakeven level, creating a negative entitlement, with the family obligated to pay back some of the grant money it had previously received. We recognized in the experiment that any requirement for a family to pay back money could cause the family to drop out of the experiment, an incentive that probably would not exist in a real world program. To reduce the likelihood of overpayment, a constraint on payments in any one month was introduced, resulting in a modified annual entitlement system with both minimum and maximum monthly payments.

Monthly payments to a family under the modified annual entitlement system were set equal to the marginal annual entitlement subject to two constraints:

1. Payments were not less than a $20 minimum.
2. Payments did not exceed the "monthly marginal payment" which is one-twelfth of the annual entitlement calculated as if current monthly income had been stable for 12 months.

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[2] In Seattle for the first 2 years of the experiment, families had an alternative of being paid once a month or once every 4 weeks on the grounds that most earnings in low-income families are weekly events and a 4-week accounting period would represent a smoother payment system than a monthly period. Midway in the experiment, the 4-week option was eliminated in Seattle—it had never been instituted in Denver—for three reasons: (1) operating a system with dual payment periods was administratively cumbersome, (2) two-thirds of the families opted for the monthly system from the outset, and (3) most families had monthly income of some kind, which created an unnecessary change in savings in grant entitlement under the 4-week system.

The constraints on the payment system had comparatively minor consequences if annualized monthly income was always below breakeven, because the marginal annual entitlement and the monthly marginal payment are approximately equivalent in most such cases. The constraints had more important effects for income movements around the breakeven point. The monthly marginal ceiling on payments served to dampen large variations in payments that could occur with unrestricted marginal annual entitlement calculations, and in some cases helped to limit cumulative overpayments that could occur when income moved from below to above breakeven. The monthly marginal ceiling also introduced a cumulative underpayment when income moved from above to below breakeven, which was roughly symmetrical with overpayments for the opposite income changes. If understood by recipients this would reduce incentives to voluntarily lower labor supply in order to qualify for payments.

The family size index

A family size index was used to adjust payments for families of different sizes in order to compensate for any differences in the incentive to work of family members that are caused by differences in the cost of living. The family size index shown in table 1.10 is positively related to family size; however, it increases less than in proportion to the number of family members. The index used in SIME/DIME was identical to that used in the New Jersey Experiment, and close to the family size index used in the state of Washington's AFDC program. This index reflects the concept that there are substantial economies of scale, especially in housing, that make it unnecessary for income to increase in proportion to the number of people in order to compensate for cost of living differences.

As shown in table 1.10, an additional index was applied to the grant formula for a "secondary" family—that is, a family that lived with a "primary" family and received some of its support in the form of free housing from the primary family. The primary family was the unit that owned the dwelling or was responsible for the rent. The benefit available to the secondary unit was reduced to account for the transfer payment implicit in the rent subsidy. In SIME/DIME, only 5 percent of family units were secondary units.

TABLE 1.10.—SUPPORT GUARANTEE BY PROGRAM AND FAMILY SIZE

Number of family members	Primary family index	$3,800 program		$4,800 program		$5,600 program	
		Primary family	Secondary family	Primary family	Secondary family	Primary family	Secondary family
2	0.62	$2,356	$1,984	$2,976	$2,604	$3,472	$3,100
3	.83	3,154	2,656	3,984	3,486	4,648	4,150
4	1.00	3,800	3,200	4,800	4,200	5,600	5,000
5	1.12	4,256	3,584	5,376	4,704	6,272	5,600
6	1.23	4,674	3,936	5,904	5,166	6,888	6,150
7	1.32	5,016	4,224	6,336	5,544	7,392	6,600
8	1.38	5,244	4,416	6,624	5,796	7,728	6,900

Adjustment for cost of living changes

One of the basic assumptions of all the experiments was that a constant real benefit level should be provided over the life of the experiment. This assumption is fully consistent with the general concern of welfare programs with a family's standard of living rather than its nominal income: if the support level had remained constant in nominal dollars in the face of inflation, the standard of living would have continually declined over the life of the experiment for those families dependent on SIME/DIME payments. Furthermore, declining real benefits would have created serious analytic difficulties, since behavioral responses are hypothesized to be related to real and not nominal changes in income. If real benefits changed over time simply due to inflation, then observed changes in behavior would be confounded by these changes as well as those induced by the intended treatment. It was decided that adjustments for changes in the cost of living would be made quarterly to avoid the discontinuities which might arise from annual adjustments. Thus, every quarter, the cost of living indexes in the two sites were adjusted to reflect changes in the cost of living in Seattle and in Denver. Calculations made after the experiment indicated that in fact the quarterly adjustment method led to some slight deterioration in real value of the program over time due to the rapid escalation in inflation over the life of the experiment.

Income from work effort versus other income

The purpose of the negative income tax experiment was to measure the effects of the negative income tax parameters on work effort. For this reason, the NIT tax rate or function should be applied to all income resulting directly or indirectly from work effort, and nonwork related income should be deducted dollar-for-dollar from the benefit (that is, taxed at 100 percent). With one exception, this principle was followed in SIME/DIME.

Work-related income was defined to include earnings from all jobs, projects from business, and returns from physical wealth (including owner-occupied housing). Returns from physical wealth were included on several arguments. First, past physical wealth is work related since it represents the embodiment of past labor effort at least on someone's part. Second, future wealth is work related and, in the long run at least, taxes on wealth represent a disincentive to work for the purpose of saving and asset accumulation. Third, equity considerations dictate the inclusion of wealth in some appropriate way. Since a family's wealth as such did not exclude a family from eligibility in SIME/DIME, imputed income from wealth had to be taken into consideration.

The way wealth was included was to ignore cash flow entirely and assume that families always had the option of converting any asset into cash and placing that cash in a risk-free bank account. Using this procedure, the equity in homes, other real estate, non-dividend-paying stocks, interest-bearing bonds, and savings accounts were all treated the same.

The rate of return was set at 5½ percent to 5¾ percent (equal to the savings and loan rates on one year's deposits). Neither realized nor unrealized capital gains were taxed, of course, although the return from the wealth that resulted from capital gains was taxed. Taxing wealth at the rate equal to the return from savings accounts allowed families to retain the risk premium associated with holding higher risk assets. The tax was applied to net wealth—which equaled gross wealth minus all liabilities, such as mortgages on real estate. Consumption items, such as household furnishings and autombiles not used for business, were exempt.

All government transfers and public program benefits were taxed at 100 percent. These government transfers include payments from welfare programs such as AFDC, Food Stamps, SSI,[3] and payments under social insurance programs such as unemployment insurance benefits, workers' compensation, and all social security payments. Social security benefits could conceptually be viewed as similar to private pensions or annuities, which were taxed as regular income; however, since there is a substantial transfer component for low wage earners in social security, the decision was made to regard social security payments as transfers rather than as insurance benefits. Since social security benefits were taxed at 100 percent, social security taxes were reimbursed at 100 percent. In a permanent NIT program, with an actuarially sound social security system, none of this would matter since the tax effects of the NIT would be neutral over the life of the recipient.

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[3] Administratively, AFDC recipients received special handling, being required to accept either AFDC or SIME/DIME payments, but not both in a given month.

The single exception to the 100 percent taxation principle for non-work-related income was private transfers. Although these transfers should have been treated the same way as public transfers, they were in fact taxed at 50 percent for two reasons: (1) we did not want to discourage private support payments for program cost reasons, and (2) we did not want to provide an undue incentive to hide such support, which is relatively easy to do. Since this tax rate was constant and the same across all programs, it did not affect the net marginal tax rate. Private transfers from SIME/DIME participants were correspondingly reimbursed at 50 percent.

Tax reimbursement and reconciliation

Since the intent of the NIT experiment was to determine the effects of a specified NIT tax rate on the behavior of families, it was necessary to create an environment in which the experimental tax rate was the only tax rate on work-related income that experimental families faced. Thus, it was necessary to devise a system under which other taxes were either eliminated or restructured in such a way as to negate their effect on the marginal tax rate.

General method of approach

Taxing of other income-conditioned transfer programs at 100 percent accomplished this goal with regard to programs such as AFDC and Food Stamps, as did reimbursement of social security taxes. The IRS, however, was a more complicated matter.

We attempted, but failed, to arrange for a waiver of the positive income taxes paid to the IRS. Instead, the experiment reimbursed these taxes at a rate of 100 percent for all families whose income was below the NIT breakeven level (i.e., receiving positive benefits). Operationally, this was not a simple task because of the withholding process. If there were a one-to-one relationship between income taxes withheld and income tax liability to the IRS, there would be no problem; we could have simply reimbursed the taxes withheld. Since that is not the case, we had to devise another system. One possibility would have been to wait until the families filed their 1040's at the end of the tax year. This was not satisfactory because such reimbursement would take the form of a lump-sum payment once a year, whereas family spending patterns are based on cash flow of income. Annual reimbursement, therefore, could distort the labor supply decisions of families to the extent that they treated it as a capital transfer or windfall and not part of their normal income flow. On the other hand, reimbursement of withheld taxes on a month-to-month basis would create overpayments to the extent that people overwithhold as a forced savings device. Such overpayments in turn would mean that SIME/DIME would have to recover the excess from the family, thus increasing the danger of attrition. This is analogous to the problem of calculating benefits on a monthly basis without regard to the annual total. To mimimize the amount of such recovery after the fact, a system was designed in which withheld taxes were fully reimbursed on a monthly basis as long as the withheld amount did not exceed the calculated value of the family's true tax liability. In cases where the amount withheld was greater than the calculated tax liability, the latter amount was reimbursed. At the end of the tax year, a copy of the family's 1040 was submitted to SIME/DIME and the tax reimbursements and liabilities were reconciled. In cases where there was a discrepancy, this was corrected by making further reimbursements or recovering overpayments through reductions in grant payments, as appropriate.

Payments above the grant breakeven

As noted above and shown in Figure 1.1., there are two breakeven points in the SIME/DIME system. The first, known as the grant breakeven, is the point at which the NIT benefit becomes zero. The second, at the payment or tax breakeven, is the point at which tax reimbursement falls to zero. This system was designed to retain continuity in the area of the grant breakeven. If tax reimbursement had been terminated at the point of the grant breakeven, there would have been great discontinuity in that region and the result would have been a major disincentive to engage in work that would take participants over the grant breakeven. To retain the continuity, therefore, tax reimbursement was continued above the grant breakeven point. As can be seen in figure 1.1, tax reimbursement was phased out by allowing the NIT tax rate to be charged against income above the grant breakeven until all tax reimbursements had been eliminated. Such a system, if unlimited, would have created very high tax breakevens in some cases. Thus, an arbitrary point was established at which the effective marginal tax rate was adjusted upwards to hasten the reduction of payments. We chose 1.3 times the grant breakeven, on the grounds that above this income level the behavioral response would be minimal. This adjustment, referred to as the lambda (λ) function, was intended to increase the marginal tax rate to 70 percent until benefits were reduced to the minimum payment of $20 per month.[4]

The startup procedure

In order to have a payment system that generated a payment each month that would be correct in terms of the annual accounting period, it was necessary to design a startup procedure that would meet this criterion. Of course, at the start there are no payment histories. One possibility would have been to start with month one as if it were one-twelfth of a year; months one and two would be two-twelfths of a year, and so on, until the 12th month payment would be based on a full year of real earnings history. The problem with this approach is that the first year's payment stream would not be an accurate reflection of the annual income of the family until the 12th month. From an analytical point of view, this would greatly reduce the usefulness of the data during the first 11 months of the experiment. From an operational point of view, such a procedure would not likely be included in an ongoing national program. To make payments that simulate an actual annual system, income histories of the families were collected in the enrollment interviews, and payment calculated equal to what the family would have received if it had been on the system for the year ending in that month. Thus, payments from the first month on would be the same as in an ongoing program, and every month in the experimental period would be based on the 12 previous months of income.

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[4] Due to a programming error, an increment of 70% was actually added to the marginal tax rate, which in some cases yielded tax rates in excess of 100% over a narrow income range. The payment formula, including the lambda adjustment (as implemented), is given in the following equation:

$$P(y) = \max \begin{cases} \$240 \\ S - F(y) + T - SR100 - 0.7\,(y - 1.3\beta_o)\,\delta\lambda, \end{cases}$$

where

$P(y)$ = annual NIT payments to an eligible family
T = federal, state, and FICA tax liabilities
$SR100$ = nonexperimental transfer income deducted from payments
β_o = gross income at the grant breakeven
$\lambda = 1$ if $Y > 1.3\beta_o$, 0 otherwise.

CHAPTER VI

THE INFORMATION SYSTEM

The household interview

As with the other experiments, personal interviews administered to experimentals and controls alike were the primary source of data used to analyze the impacts of SIME/DIME. Thus, the focus of the interview was to collect data that would enable us to answer questions about participants' labor supply behavior, family stability, investments in human capital, and, to a lesser extent, about their behavior with respect to durable consumption, housing, migration, and other socioeconomic factors.

A unique feature of SIME/DIME interviews was that much of the economic data for individuals were collected in a continuous-time mode so that each change, for example, in employment status, hours of work per week, wage rates, marital status, and enrollment in training or education could be ascertained. This continuous-time mode of data collection was accomplished by recording each event that occurred during the period covered by each interview and then linking the interview data through time—allowing a complete picture of the major activities of each adult member to be determined at any point in time throughout the experiment. Continuous-time data enabled instantaneous rate models to be estimated in the study of flows into and out of various statuses, and enabled analysis to be conducted over any time period. Other data, less important or inherently sporadic, were collected intermittently in the interests of time and economy.

Interviewing

The SIME/DIME interviewing system was based on a triannual cycle of periodic interviews and covered three distinct periods: the preexperimental period, the experimental period, and the postexperimental period. Figure 1.2 presents a diagram of these interviewing periods. The time spanned for both experimental and control families could be split into these periods, since each family had an enrollment date, a specified length of treatment, and a disenrollment date.

Four interviews covering preexperimental time were fielded: the screening interview, the preenrollment interview, the enrollment interview, and the postenrollment interview (in Denver only). The screening interview was used to determine the basic eligibility of the population residing in the low-income areas initially canvassed in each city and collected for the most part only those data necessary to establish the basic eligibility of the surveyed population. These data were keypunched but never encoded into the SIME/DIME data base or used for any subsequent analytic purpose. Households were eliminated at this point if they did not meet certain eligibility criteria. Then, a selected sample of eligible households was administered the preenrollment interview, designed to gather data on household characteristics for the assignment: race, family structure (number of heads of household), family earnings, nonwage income, work history of the heads of household, and family asset position.

In addition, the preenrollment interviews collected preprogram measures of the dependent variables needed to estimate models that compared the changes in

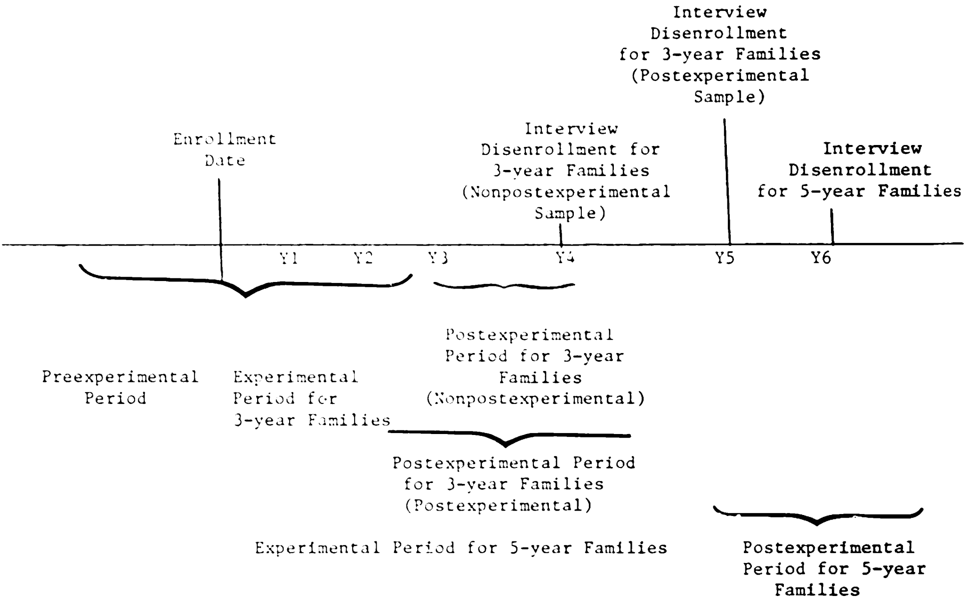
Interview
Disenrollment
for 3-year Families
(Postexperimental
Sample)
Enrollment
Date
Interview
Disenrollment for
3-year Families
(Nonpostexperimental
Sample)
Interview
Disenrollment
for 5-year Families
Y1
Y2
Y3
Y4
Y5
Y6
Postexperimental
Period for 3-year
Families
(Nonpostexperimental)
Preexperimental
Period
Experimental
Period for
3-year Families
Postexperimental Period
for 3-year Families
(Postexperimental)
Experimental Period for 5-year Families
Postexperimental
Period for 5-year
Families

FIGURE 1.2.—Interviewing periods.

dependent variables between experimental and control families and preprogram measures of characteristics thought to affect response.

The enrollment sample was selected from this group and was assigned to an experimental treatment according to the assignment model. The families selected were then enrolled into the experiment and administered the enrollment interview on the same day. The enrollment interview concentrated on collecting economic data. Continuous-time detailed data on employment, unemployment, and nonwage income were reported covering the time period from January 1, 1970, in Seattle or January 1, 1971, in Denver to the date of the interview. (Enrollment took place in Seattle from October 1970 through November 1971 and in Denver from October 1971 through August 1972.)

The postenrollment interview, given only in Denver, collected mostly attitudinal data on such topics as job satisfaction, opinions on wives working, aspirations for children, and use of credit. The data in this interview could have been considered to be preprogram data, since it was administered within 6 weeks of enrollment. They were never used in this manner, though, because of a lack of comparable Seattle data, although they were used in some of the analyses as the first experimental occurrence of these data.

The experimental and postexperimental period data collection consisted of a series of "periodic" interviews administered on an approximately 4- to 5-month cycle. There were 17 periodic interviews in Seattle and 18 periodic interviews in Denver, covering both the experimental and postexperimental periods. Five-year families who completed the experiment were interviewed for at least 6 years after enrollment, including 1 postexperimental year. Three-year families were interviewed for at least 4 years after enrollment. In addition, approximately half of the 3-year families were selected to take a special postexperimental interview that provided a fifth year of data. The 20-year families (Denver only) were interviewed for at least 4 years after their enrollment into the 20-year program. Thus, all families had nearly 2 years of preexperimental data; 3-year families had 3 years of experimental and one-half year of postexperimental data; 5-year families had 5 years of experimental and 1 year of postexperimental data; and 20-year families had 4 years of experimental data.

These regular periodic interviews administered during the experimental and postexperimental periods continuously updated the longitudinal employment, unemployment, and nonwage income data that began in the enrollment interviews. In addition, other topics of research interest were introduced and updated at various points in the interviewing cycle. These data suffered in that adequate preexperimental measures were generally not available, nevertheless studies using simple experimental-control comparisons were done on many of the topics. (Details of the subject matter covered and the construction and administration of these interviews appear in part II of this volume.)

Certain of the topics introduced in the interviews after enrollment were historical in nature and thus, to some extent, allowed for preexperimental measures. This was particularly true for subjects such as marital history, fertility history, training and education history, and mobility history. It was somewhat advantageous to put these questions in later interviews to avoid overburdening the preenrollment and enrollment interviews, and it was reasonable to expect that the historical answers to these questions would not be unduly colored by program participation. However, these adavantages were somewhat balanced by the loss of data for some individuals as a result of attrition.

In addition to the regular periodic interviews, certain special interviews were used during the experimental and postexperimental periods. The postmove/residual series of interviews were designed to collect information from families that were absent for one or more of the periodics or had moved outside the Seattle or Denver areas. There were six cycles of postmove interviews in Seattle and seven cycles in Denver. The postmove interviews were designed to fill in the gaps in interviewing caused by families who had missed one or more regular periodic interviews or who were fairly inaccessible because they had moved from the Denver or Seattle areas. Therefore, these interviews concentrated on primary, continuous information, eliminating many of the point-in-time and attitudinal questions. Employment, benefit and expense, education, housing, real property, and asset data were collected in these interviews. In addition, background modules such as marital history, fertility history, and ethnicity were included. The postmove interviews also included the postmove module, which was asked of families that had moved from the site of their previous interview. The purpose of the postmove module was to measure the impact of the NIT on factors affecting the decision to move and on people's ability to seek out better living situations.

New member interviews were special interviews given to individuals 16 years of age or older upon joining families that were already participating in the experiment. These interviews were designed to collect background data on new individuals to match data that had previously been collected for originally enrolled individuals. The data collected included training and education, ownership of real property and assets, ethnicity, employment history, fertility history, and perceived wage rate of nonworkers. Thereafter, the new members were incorporated into the regular periodic interviews as of their entrance date.

In the first year after the termination of payments (or equivalent time for controls), regular periodic interviews were administered to all families. Then, the postexperimental interview was administered to half of the 3-year sample 5 years after they were enrolled into the experiment. For NIT families, this was 2 years after the end of their benefit period. The interview was intended to measure the longer run experimental effects of both the NIT and the counseling/training program and to capture the permanent effects of the program. Thus, questions about topics that were expected to show these long-run or permanent effects, such as employment, training and education, net worth, housing, health, and educational status of children, were included.

Other data

SIME/DIME made use of data on the participating families other than that collected in the interview. Primarily, these data were collected for use in the operations of various aspects of the experiment—that is, data collected for and used in the counseling program, the (NIT) payments system, and the sites' sample maintenance operation. However, certain of these data were used in the analytic studies as well. In addition, measures of income were collected from various public agencies and housing value data were supplied by independent appraisers and county assessors to support some of the analysis.

Counseling and training subsidy treatment data

Data for the counseling and training subsidy components covered individual and family demographic information and treatment variables, initial contact reports, summaries of counseling sessions, written plans of action, recontact information, and financial payments information. Data were recorded by counseling center counselors and clerks on forms filed in folders prepared for each eligible individual. The folders also contained all correspondence and notes regarding the individual's participation in the program and operational matters. Coders working in the counseling centers reviewed each counseling folder and completed a booklet that summarized the data contained in the folder for use in the analysis of the program. Counseling records were periodically reviewed by counseling center staff to ensure completeness and consistency of coding. In addition, the completed booklets were reviewed by a quality control person to ensure that the counseling folder data had been accurately recorded.

Demographic and treatment data were obtained from SIME/DIME enrollment information at the time the counseling centers received notification of program eligibility. Contact and enrollment data were collected by counselors during in-home visits to the eligible families. During the contact visit, the program was explained and the following information for each eligible family member was collected: occupation, present labor force status, and program response status (accepts, rejects, or defers program participation).

Summary data for all formal group or individual counseling sessions were recorded, including date, type and purpose of session, program response status, type of plan of action, time in counseling session, and counselor ID. The plan of action summarized information pertinent to the objectives of the counseling process, which were to determine an occupational goal and the means to achieve the goal. The counselor worked with the participant to identify a goal and a plan to achieve it, focusing on training, job search, or job retention. Training degree objective and institution were specified in the plan, if applicable.

Counselors made in-home recontact visits after 1 year of eligibility or a lapse of 6 months of activity in the program. Information collected at that time included employment status and evaluation of the counseling program.

Finally, the financial record was used to record the amounts of subsidies paid for tuition, books and supplies, transportation, and child care for each academic quarter in which the recipient was enrolled.

Payments data

Data that generated the monthly NIT benefit calculations were completely separate from the interview data. Payments data were collected on Income Report Forms (IRFs) which were sent to all financial families and selected Denver control families each month. On the forms, the families retrospectively reported family composition, the employment status, earnings and hours of work of all family members, the dollar value of all other income received by the family, and certain expenditures made during the previous month. The forms itemized income and expenditures by type, and explained how each was treated in the grant calculation (i.e., "added to basic income," "100% reduction from support," etc.). These data were used in the studies of income reporting and the payments calculated directly from IRF data were used as a measure of program participation. The behavioral models of labor supply response did not use these measures of income and hours of work, of course, because there were no comparable data for controls.

Reeducation survey data

In order to ensure that the program was properly understood by participating families, a reeducation project was undertaken approximately one year after enrollment. As part of this project, a questionnaire was developed to measure program comprehension and to inform the payments system field personnel about the aspects of the program that were not understood. From February through April 1972, field personnel visted each experimental family in Seattle and left the questionnaire to be filled out and returned by mail. In February 1973 a similar questionnaire was administered personally to experimental families in Denver. Answers to these questions were then used as a basis for further reeducation about aspects of the program that the family appeared not to understand. The survey consisted of questions about program eligibility with regard to family compositon changes, employment status, and AFDC participation, about changes in the grant amount resulting from various types of changes in income to test the families' understanding of the support level and various tax rates, and about the length of time they were eligible to continue receiving payments. Results of this survey were used in a study to measure the impact of program comprehension on response. A second reeducation project was conducted in Seattle in 1974 and in Denver in 1975, the data from which have not been analyzed to date.

Public agency data

The validity of the interview data collected was of crucial importance to the research, because differential misreporting of income and hours of work on the part of either experimental or control families could cause labor supply estimates to be biased. Therefore, income data were collected from a variety of outside sources as a means of measuring the validity of the experimental effects on work estimated from the interview data. These validation studies utilized data collected both from a limited 10 percent sample during the early years of the experiment and from later attempts at more universal data collection.

The 10 percent sample collected both income and benefit data from AFDC, Food Stamps, Department of Employment Security (unemployment insurance data), and public housing for a sample of SIME/DIME individuals covering the years 1971 and 1972 in Seattle and 1972 in Denver. These data were used in early attempts to study the accuracy of both IRF and interview data on earnings, nonwage income, and public transfers.

Attempts at collecting universal validation data resulted in a more diverse set of data. Internal Revenue 1040 forms were collected for virtually all participants for the years 1971–1973 in Seattle and 1972–1973 in Denver. Welfare records for all SIME/DIME families that participated in AFDC during 1970–1971 were collected and used for the AFDC income reporting accuracy study. At a later date, AFDC records for Seattle families were collected through 1976, but only 1975 data were collected for Denver families.

Of principal interest to the later validation studies were employer-reported income data. Thus, unemployment insurance (DES) and social security data were collected for large portions of the SIME/DIME sample. The social security data consisted of yearly earnings data aggregated into five-person cells for all SIME/DIME participants from 1970–1977. From DES we collected quarterly earnings data covering all individuals from 1970–1977, but in Seattle, the data from 1973 to 1977 covered only 50 percent of family heads and 25 percent of persons who were not family heads. The differences in the DES and AFDC data collected at the two sites resulted from differences in record-keeping methods and consequent budgetary considerations.

Housing value data

Outside information was collected to analyze the accuracy of owner estimates of the market value of housing. Owner estimates of housing value were used in SIME/DIME to determine house equity. Estimates of market value were collected from a private appraisal firm and from county assessor records for a random sample of 138 Seattle home-owner families in 1971 and 177 Denver families in 1972. These three estimates of housing market value were combined in a model that predicted the "true" value of housing.[1] From the value of property, the debts were subtracted to obtain an estimate of owner equity. An income was then imputed to that equity and taxed at the NIT rate.

Summary

This report has summarized, but by no means exhausted, the discussion of the design and history of the Seattle and Denver Income Maintenance Experiment (essentially two experiments being conducted simultaneously: the negative income tax, and the counseling/training subsidies). We have discussed the theoretical underpinnings for the experiment, which provided the link between the experimental design and the outcomes that were to be measured, and described the treatments and sample selection procedures and results. We have addressed theoretical and practical issues that had to be considered in the design, and some of the problems that must be overcome in interpreting the experimental results. Lastly, we have discussed the two operational systems for the NIT experiment: the payment system and the information collection system. The former is essentially the operating system of the NIT program that we implemented, and the latter provides the information needed to measure the effects of the experiment.

The rest of the volume

Part II of this volume describes the implementation of the experiment, covering in more detail the operational issues with respect to program implementation that have been alluded to or at best summarized here. Part III presents the analysis of the labor supply response to SIME/DIME. Part IV presents the labor supply effects of the training/subsidy treatment. Part V presents the marital stability results. Part VI summarizes the major analyses of other effects to the experiment.

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[1] The technique for using several approximate measures for estimating "true" measure was developed by Karl Jöreskog and Arthur Goldberger (see Robins and West, 1977).

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SIME/DIME FINAL REPORT

VOLUME 1, PART II

IMPLEMENTATION

by

Gary Christophersen

CHAPTER I

INTRODUCTION AND OVERVIEW OF ADMINISTRATIVE FRAMEWORK

This part of Volume I describes the implementation of the experimental design from its conception through the conclusion of the data collection effort. Issues in the design of the experiment, which have been addressed in Part I, are discussed here only to the extent necessary to provide the foundation on which the administrative structure was built.[1]

The administrative structure of SIME/DIME evolved from that used in the New Jersey Experiment, but was much more elaborate due to the size of the sample, complexity of the experimental and research design, and the operation in two cities distant from each other. Although each site became relatively autonomous in the detail of its operation, the amount of coordination and interdependence programmed into procedures was substantial and in keeping with the need to operate the two sites as one experiment.

Rules of operation

The administration of the NIT payments in SIME/DIME was designed to simulate a small-scale public program. To ensure equitable administration of the treatment, operating regulations and procedures needed to be developed. Further, the essence of the experimental method demanded that treatment and control groups be handled identically in all respects except the provision of the experimental treatment. Family composition changes were a particular problem. If the same set of regulations were not adhered to in defining control group families as used for the experimental families, the integrity and utility of the experiment would have been threatened. To accomplish these ends, the rules of operation were developed to specify operating regulations.

The rules can be divided into two basic sections. The first, which includes family unit definitions and the rules governing eligibility, applied to both treatment and control groups. From the point of establishing the sample frame on throughout the experiment, all eligibility decisions were enforced for both groups according to the same set of rules.

The family unit as defined for enrollment purposes consisted of the eligible head(s) of household (HOH) and all dependents living in the household. To insure that participants would provide a viable observation for the measurement of labor market participation, each family unit had to contain at least one HOH between 18 and 58 years of age who was mentally and physically capable of employment. Individuals who were in the armed forces were systematically excluded because they were constrained in their labor market participation decisions.

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[1] A more thorough treatment of the implementation can be found in Volume II of the SIME/DIME final report, entitled "The Administration of the Seattle/Denver Income Maintenance Experiment." The emphasis in this chapter is to provide background for readers most interested in the research results. Volume II describes in detail how the experiment was actually carried out.

A dependent in the unit was defined as a related individual under 18. The initial eligibility of a household could be established either by the presence of a husband and wife, or by an eligible head of household with a dependent. Once this initial eligibility had been established, all related individuals living in the household were included in the family and fully eligible to participate in the experiment.

In order to observe family unit changes under national program conditions, individuals who were originally enrolled in the experiment were allowed to retain their eligibility if they left the original family unit. Furthermore, if these individuals subsequently formed other family units with nonparticipants, they were allowed (after a waiting period) to transfer eligibility to certain members of their new family. Generally, transference of eligibility occurred when participants married or decided to cohabit on a continuing basis with a new partner. To become eligible, the other individuals joining the household had to be under 18 years of age and a natural child or stepchild of a member of the family unit.

The second part of the rules, governing administration of the NIT treatment, only applied to those families assigned to one of the 11 NIT treatment plans delineated in the experimental design.

The definition of income and expenses considered when calculating income maintenance grant payments was a major component of this portion of the rules. Income and expenses in the grant calculation are summarized in table II.4 below.

In addition, as discussed in Part I, it was necessary to control income-conditioned taxes the families faced outside of the SIME/DIME treatment, making it important to have rules governing nonexperimental taxes due. These rules provided that families would be reimbursed for their entire federal and state income and social security taxes as long as their income was less than 130 percent of their breakeven level of income. Incomes above that level, which is already well beyond the level at which the family would receive any income maintenance benefit, were reimbursed at a declining and ultimately disappearing rate. This segment of the rules also stipulated requirements for filing income report forms (IRFs) and the schedules that families had to adhere to in order to receive timely benefit payments (see figure 2.1).

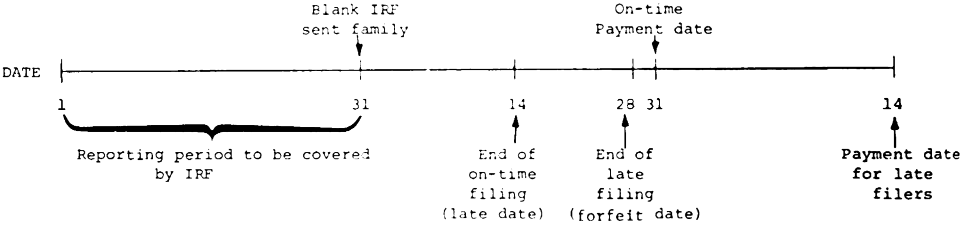
Blank IRF sent family
On-time Payment date
DATE
1
31
14
28
31
14
Reporting period to be covered by IRF
End of on-time filing (late date)
End of late filing (forfeit date)
Payment date for late filers

FIGURE 2.1.—IRF filing schedules.

Every effort was made to administer the rules impartially and equitably. When problems arose, the rules explicitly provided for a mechanism through which a family could contest decisions and actions taken by SIME/DIME. The first step in the process involved an informal internal review directed by the project manager at the request of an aggrieved family. Disputes were usually settled at this level. Throughout the experiment only six cases went to the next and final level of appeal, which was before a state review board. The findings of this board were final.

Although the rules of operation were intended as the overall administrative guide, the size of the sample and the complexity of the situations encountered in the field led to a number of cases in which rules needed clarification or interpretation. Policy decisions were then incorporated into the rules and became the tools used to clarify and interpret the original rule. Policy decisions were used over 250 times during the experiment on payments-related matters covering all aspects of the payments operation—erroneous payments, allowable expenses, equity calculations, transfer payments, and eligibility.

In discussing the New Jersey Experiment, Kershaw and Fair (1976, p. 94) warned that:

> It has become clear that the regulations themselves constitute an important part of the experimental environment, a part that may well interact with the tax rate and guarantee treatments, producing differences in subject responses that are caused by experimentally induced stimuli other than the primary treatments.

In order to minimize the danger of misinterpretation of the experimental effects, considerable care was taken when designing the rules for SIME/DIME. Yet there has still been speculation that differences among the rules of operation of the four income maintenance experiments account for at least some of the differences in the research findings—most notably the marital stability findings (see Christophersen and Kershaw, 1979).

The evolution of the rules over the course of the four income maintenance experiments resulted in a number of differences in family unit and income definitions. Whether or not behavorial effects of the various experiments are related to these differences in the administrative structure of the experiments, it is clear that these types of differences have significant impacts on the level of participation and, therefore, program costs of a national program.

Coordination with public programs

A challenge in implementing an experiment outside a laboratory setting is the necessity to control for all conditions that could potentially affect the experimental treatments or the relationship between experimental and control groups. SIME/DIME was not introduced into a void, but rather into a functioning society with a set of existing public programs involving some of the families selected for enrollment. A number of programs existed, but the most relevant problems for the experiment were aid to families with dependent children (AFDC) and public housing.

The problem of coordination with AFDC proved to be threefold. First, early in the experiment it was noted that the rate at which families were switching from AFDC to the experiment was much slower than anticipated given that the NIT benefits levels dominated AFDC. It became more understandable, however, when it became clear that families switching to the experiment would lose their medical payment eligibility and possibly other categorical services as well. To counter this, waivers providing that SIME/DIME benefits would not be counted in determining eligibility for medical and day care services were filed for and granted by the Department of Health, Education, and Welfare. The second problem of coordination involved the AFDC subsidy of day care, which was a sliding scale according to income. Left unchanged, the sliding scale would have changed tax rates faced by participating families in the same manner (although to a much lesser degree) as the Federal and State income and social security taxes. Waivers were again filed for and granted.

The third problem was the coordination of payments when families switched from AFDC to SIME/DIME or vice versa (as they were free to do throughout the experiment). The rule was that a family could not receive both AFDC and SIME/DIME grant payments in the same month. However, since SIME/DIME had a different income reporting period than AFDC, such a rule was not sufficient to prevent overlapping benefits. AFDC benefits were calculated to meet the families' anticipated need in the upcoming month (a prospective system). In that system the family was sent a check the first of each month designed to meet the family needs during that month. SIME/DIME benefits, in contrast, were based on a retrospective reporting of family income. A SIME/DIME check was sent out the first of each month as in AFDC, but the SIME/DIME benefit was the result of a calculation involving the income and expenses the family had received in earlier months.

Numerous SIME/DIME families lived in public housing. Rents were income-conditioned and the housing authorities are legally obligated to adjust rents for changes in family income. This effectively created a tax on income outside the experimental tax rate as long as SIME/DIME benefits were counted as income. Procedures were therefore developed that provided reimbursement to families for changes in rent caused by SIME/DIME payments, thereby eliminating the implicit public housing tax rate.

The remaining chapters of Part II discuss aspects of the administration of the experiment in more detail. First is a discussion of locating and enrolling the sample.

CHAPTER II

LOCATING, ENROLLING, AND MAINTAINING THE SAMPLE

Sample selection

Sample selection in SIME/DIME was quite different from the more common sampling procedures used in many survey research projects, where simple random samples are typically drawn from a particular population. In SIME/DIME, an attempt was made to find specific numbers of families with specific characteristics in order to meet the requirements of the assignment model described in Part I. The first step in the sample selection process was to identify target neighborhoods within each city with the greatest numbers of families who met the eligibility criteria. The basic criteria were as follows:

- Family income: below $9,000 per year adjusted for a family size of four
- Family structure: participation limited to nuclear families defined to consist of either a married couple (or couple living together as husband and wife) or a parent and minor dependent child
- Race/ethnicity: family head to be either black or white in Seattle and black, white, or Chicano in Denver
- Characteristics of the head of the family: age between 18 and 58 at time of screening interview; physically and mentally capable of employment (not disabled); not in the armed services.

Census data, other available published demographic data, and conversations with persons knowledgeable about the demography of each city produced a list of neighborhoods that were then visually inspected to assess their likely yield of families with the desired characteristics. This process resulted in the designation of 14 Seattle and 19 Denver target areas.

Each designated target area was then canvassed to develop a listing of every household in the neighborhood. Using systematic procedures to ensure that no households were missed, staff listed the neighborhoods by block to establish the sample frame to be used for subsequent interviewing. Once the listing was completed for a particular block, dwelling unit cards were prepared for each household. These cards were used for subsequent sample tracking by recording interview attempts, etc.

The first interview was the screening interview, eventually administered to virtually all households in the sample frame. The purpose of the interview was to identify potentially eligible families with a short questionnaire. Even with the selection of target areas, the proportion of ineligible families was anticipated to be high. Therefore, the screening process was designed to eliminate quickly from further consideration those families which were clearly ineligible. The screening interview thus contained questions on family composition, ages of heads of household, income from various sources over the past year, and race of the heads of household (as recorded by the interviewer from observation).

In both sites, initial screening efforts resulted in unacceptably low numbers of potentially eligible families. As a result a variety of additional steps were taken, involving the addition of new target areas and reworking of the group who had refused or not been reached for an interview.

Table 2.1 reports the results of the screening effort. As can be seen, more emphasis was given to reworking the sample in Denver than in Seattle, resulting in a lower refusal rate (only 10 percent compared to 16 percent in Seattle). Also, the proportion of dwelling units for which no screening interview was ever completed was somewhat lower in Denver than in Seattle (8 percent compared to 11 percent).

TABLE 2.1.—SAMPLE SELECTION IN THE SEATTLE AND DENVER INCOME MAINTENANCE EXPERIMENTS

	Seattle			Denver			Combined total
	Initial listing	Supplementary listing	Total	Initial listing	Supplementary listing	Total	
Housing units listed	24,168	11,856	36,024	50,934	6,893	57,827	93,851
Housing units attempted	24,168	11,856	36,024	40,774	12,807	53,581	89,605
Vacant units	2,383	1,129	3,512	4,811	3,811	5,192	8,704
Occupied housing units	21,785	10,727	32,512	35,963	12,426	48,389	80,901
Refused	3,076	2,276	5,352	2,416	2,226	4,462	9,994
Not completed	2,042	1,560	3,602	2,226	1,853	4,079	7,681
Terminated	8,224	5,208	13,432				13,432
Completed screening interviews	8,443	1,683	10,126	31,321	8,347	39,668	49,794
Ineligible	3,145		3,145	19,559	6,907	26,466	29,611
Eligible for preenrollment	5,298	1,683	6,981	11,672	1,440	13,202	20,183
Selected for preenrollment interviewing	5,298	1,683	6,981	5,911	1,440	7,350	14,331
Moved	521		521			486	1,007
Refused	443		443			1,258	1,701
Ineligible	789		789			591	1,380
Not completed	413		413			332	745
Completed preenrollment interviews	3,132	1,683	4,815			4,683	9,498
Assigned to enrollment			2,542			3,361	5,903
Moved			244			111	355
Refused			125			246	372
Ineligible			84			83	167
Not completed			46			163	209
Total enrolled			2,042			2,758	4,800

Note.—This table is reconstructed from tables 1 and 3 in Murarka and Spiegelman (1978).

Based on the data collected in the screening interview, families were determined to be eligible for the enrollment interview if they met the basic eligibility rules and their income appeared to be below specified levels.[1] In Seattle, all families appearing to be eligible were assigned preenrollment interviews. In Denver, the initial screening process yielded a great many more eligible families in some sample groups than were required for the experiment. In such cases, a random selection of families from the group received a preenrollment interview.

The principal purpose of the preenrollment interview was to improve the measurement of family income. This was accomplished by more detailed questioning, eventually evolving to the methodology used throughout the experiment of developing a continuous history of income and labor force participation for each family member at least 16 years of age. The interview was also used to check and refine the definition of family units when multiple families shared the same dwelling, and to collect other baseline data considered essential for subsequent analyses. Finally, the preenrollment interview checked for any changes in family composition or status of the head of household which would make the family ineligible for SIME/DIME.

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[1] These levels were substantially above those specified in eligibility criteria to allow for measurement error due to lack of precision in the screening interview.

The results of the preenrollment process are also shown in table 2.1. The higher completion rate in Seattle (69 percent) compared to Denver (64 percent) is due entirely to the much lower refusal rate. This difference in the refusal rate simply seems to be the reverse of the screening process, during which heavier sample reworking in Denver reduced the refusal rate on the screening, only to have many of those eventually refuse the preenrollment interview anyway.

Enrollment

The enrollment of families into the experiment was the culminating step in the sample selection process. After the data from the preenrollment interview were processed, determinations of eligibility for enrollment were made. A stratified sample of eligible families was selected to be enrolled at each site and assigned to experimental treatment or control status. The next step was to convince each family to participate and provide the information necessary to conduct the analysis. Successful enrollment of families was critical to the success of the experiment, in order to minimize potential bias from refusals.

The first step in preparing to assign families to the experimental cells was the determination of their "normal income," defined as the expected family income for the year prior to the start of the experiment, and was derived from the preenrollment interview data. The sample frame, from which assignment of families to the experimental cells was made, was established using this normal income designation together with the race and head of household category (single- or two-person) for each family. All families who had the same income class, race, and family type constituted a group. The assignment model then specified the number from each group to be enrolled in each of the experimental treatments and the control group. A random draw from within each group then determined which families would be enrolled and to which treatments.

For those families selected for the control group, enrollment procedures were similar to conducting the preenrollment interview. The enrollment interview was less complicated than the preenrollment interview, and the only additional task for the enroller at this stage was to convince the families to send in monthly address postcards (for which they would be paid $8 a month) and to participate in the interview series.

For the families selected for the experimental group, the enrollment process was far more complex. The interviewer was charged with not only convincing the families to participate in the experiment, but also conducting the enrollment interview, teaching the family to complete the income report form correctly, and providing them with a basic understanding of how their benefit was calculated and how earnings and family composition changes would change the amount received. In order to perform these functions, interviewers selected to enroll experimental families had to be rigorously schooled in the concepts of the negative income tax (a concept not widely understood at that time), the specifics of the rules of operation, and the details of the SIME/DIME benefit calculations.

All enrolled families signed an enrollment agreement which outlined their obligations, eligibility for benefits, time span of enrollment, the experiment's guarantee of the confidentiality of the information they provided, and the assurance that such information would only be used for the operation and evaluation of the experiment.

Table 2.1 also summarizes the results of the enrollment process. Both sites failed to enroll all the families required by the assignment model. The reasons for the shortfall included: refusal of the families to participate, termination of the family at enrollment because of a family structure change that made them ineligible, a move out of the labor market between preenrollment and enrollment, and a failure to locate families that had been assigned. Additionally, the assignment model called for some group sizes which could not be filled from the sample frame of eligible families. Of those families that were assigned, were still eligible for enrollment, and had not moved, 92 percent were enrolled in Seattle and 87 percent in Denver. Explicit refusals to participate were not frequent. In Seattle 6 percent and in Denver 8 percent of those families offered enrollment turned down the offer.

Maintenance of the sample

Once enrolled, the sample required continual effort to minimize attrition, and to ensure both that the assigned experimental treatment and periodic interviews were administered on schedule, and that family composition changes were processed promptly and consistently. The sample maintenance department at each site attended to these issues.

One of the primary responsibilities of this department was to monitor changes over time in the structure of participating families.

The master file consisted of a comprehensive computer file of individual family records. There was a separate record for every family either originally enrolled in the experiment or joining subsequently as a result of families splitting and re-forming. The file was of enormous importance in monitoring family composition changes. As new families were formed, a new family record would be created, and all original and new family members tracked. Thus, the file could link every family and individual back to the original family unit, providing a historical record of all family changes that had transpired during the course of SIME/DIME.

Since attrition is a major source of potential bias in a longitudinal panel study,[2] constant attention was given to countering potential attrition in SIME/DIME. Attrition occurred most often when families moved. Interview refusals also accounted for some attrition, but since the families had been interviewed several times prior to accepting enrollment, those that decided to enroll already had a reasonably good idea of what to expect.

The families in the NIT group had a greater incentive to report current address information on a regular basis than controls since it was one of the requirements that had to be met in order to receive their monthly payment, and was reported on the Income Report Form (from which it was transferred to the sample maintenance department). The control group, in contrast, had less of an incentive to continue participation in the study and, therefore, to report address changes. The control group was only contacted by survey personnel during the periodic interview cycle, which typically took place at 4-month intervals. In order to track control families between these intervals and to offer an incentive to continue participation in the program, families were asked to complete and return a monthly postcard with address and family composition information (for which families were paid $8 per month). The postcards were processed and checks mailed at the end of the month just as for the NIT treatment families. A large percentage of the control sample submitted the postcards each month, an overall average of 85–95 percent.

In Denver, 50 percent of the control group was required to submit monthly income and expenses reports (IRFs), rather than simply submitting postcards, to enable researchers to measure any differences in the respondents' reporting of income and expenses on the periodic interview that might result from the "training" experience the family received from completing a monthly IRF. Control families were paid $15 each month for submitting these IRFs to the payments department.

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[2] Attrition results are discussed in Part I.

Respondent mobility rates at both sites were relatively high and losing track of moved families was potentially a major source of attrition. Even with the implementation of address reporting procedures detailed above, families often moved without reporting their new address. Interviewers made an initial effort to locate these families by contacting neighbors and checking with office personnel for updated address information. Questionnaires belonging to respondents who could not be located by the interviewers were then turned over to a professional searcher in the sample maintenance department. This person was hired for his/her knowledge of the Seattle or Denver area, plus a tenacious and imaginative approach to searching.

A computer disk file, accessible using a visual display terminal, was used to maintain names and addresses of all active families. This file was updated monthly for use in both the control and experiment payments mailings, and also for reference by interviewing personnel.

As a result of family composition changes, attrition, and efforts to counter attrition, the sample of controlled families changed markedly over the course of the experiment. Details of the changes in the Seattle and Denver samples over the course of the experiment are presented in tables 2.2 and 2.3. As can be seen, interview completion rates were very high at both sites, although a large number of sample families failed to fill out appropriate forms at some time during the experiment. The attrition rate was much lower than these figures would indicate, however, because 60 percent of those were only temporary drops from the experiment. The 9-10 percent of the sample who moved out of the Seattle or Denver area were interviewed once per year with a specially designed composite interview.

TABLE 2.2.—SAMPLE DYNAMICS AND ATTRITION IN SEATTLE

[Families of two or more persons]

Periodic interview	1	2	3	4	5	6	7	8	9	10
	Active families from previous interview $(8_p - 10_p)$	Net changes due to family splits, marriage, et cetera	Dropped from experiment	Former drops now restored	Moved out of area [1]	Interview sample (1 + 2 + 4 − 3 − 5)	Interviews not completed	Completed interviews (6 − 7)	Completion rate (8 ÷ 6) (percent)	Completing interview series [2]
1	2,038	23	51	0	47	1,968	4	1,964	99.8	0
2	1,964	14	43	3	30	1,905	8	1,897	99.6	0
3	1,897	28	29	2	41	1,857	19	1,820	98.0	0
4	1,820	18	17	10	13	1,818	19	1,799	99.0	0
5	1,799	9	68	6	40	1,798	78	1,720	95.7	0
6	1,720	11	16	31	22	1,724	35	1,689	98.0	0
7	1,689	−42	9	2	16	1,624	34	1,590	97.9	
8	1,590	−18	17	43	12	1,557	20	1,537	98.7	
9	1,537	−2	35	51	6	1,605	3	1,602	99.8	
10	1,602	17	40	81	20	1,668	7	1,661	99.6	0
11	1,661	26	42	50	13	1,637	6	1,625	99.3	289
12	1,336	20	17	29	7	1,361	6	1,355	99.6	460
13	895	16	15	20	105	906	1	905	99.9	326
14	580	7	5	4	77	579	11	568	98.1	5
15	562	4	9	5	2	561	16	545	97.1	0
16	545	16	14	5	7	545	5	540	99.1	140
17	400	0	20	5	4	381	6	375	98.4	375
Total		152	385	256	261	23,494	302	23,192	98.7	

---

[1] Still in experiment; interviewed annually as part of moved out of area project.

[2] Families were administered periodic interviews for their entire period of enrollment plus 1 year.

TABLE 2.3.—SAMPLE DYNAMICS AND ATTRITION IN DENVER

[Families of two or more persons]

Periodic interview	1 Active families from previous interview (8_p–10_p)	2 Net changes due to family splits, marriage, et cetera	3 Dropped from experiment	4 Former drops now restored	5 Moved out of area [1]	6 Interview sample (1+2+4−3−5)	7 Interviews not completed	8 Completed interviews (6–7)	9 Completion rate (8÷6) (percent)	10 Completing interview series [2]
1	2,699	20	18	0	26	2,675	15	2,660	99.6	0
2	2,660	11	29	4	45	2,601	13	2,588	99.5	0
3	2,588	25	28	4	28	2,561	20	2,541	99.2	0
4	2,541	26	44	19	33	2,499	15	2,483	99.4	0
5	2,483	3	28	16	20	2,454	29	2,425	98.8	0
6	2,425	30	31	33	16	2,441	29	2,212	98.8	0
7	2,419	11	24	29	17	2,411	13	2,398	99.5	0
8	2,398	29	25	13	8	2,407	13	2,394	99.5	0
9	2,394	30	26	12	10	2,400	25	2,375	99.0	0
10	2,375	13	24	14	27	2,351	24	2,327	99.0	0
11	2,327	29	30	20	35	2,311	23	2,288	99.0	106
12	2,186	34	30	23	20	2,189	33	2,156	98.5	725
13	1,431	13	15	22	−15	1,471	23	1,448	98.4	372
14	1,076	20	9	10	−7	1,104	11	1,087	98.5	54
15	1,033	17	22	12	9	1,031	17	1,003	97.3	5
16	998	7	10	15	4	1,006	28	979	97.0	0
17	976	15	6	11	6	990	30	957	96.7	478
18	479	6	27	17	5	470	33	470	100.0	470
Total	....	344	426	264	285	35,372	384	34,988	98.9	....

[1] Still in experiment; interviewed annually as part of moved out of area project.

[2] Families were administered periodic interviews for their entire period of enrollment plus 1 year.

CHAPTER III

ADMINISTERING THE PAYMENTS

To calculate each monthly payment a 12-month historical data set was needed plus a record of income and expenses for the current month. At the start of the experiment the historical (or "startup") data set was collected from each family by the enrollment interviewer. Whenever new persons joined the experiment, as per the provisions of the rules of operation, historical data were also collected from the new member to merge with the financial information of the already enrolled family members. This was done by the field representatives using the same type of income questionnaire booklet as was used by the initial enroller.

The source of information for the monthly data used to compute payment was the income and expense report form (IRF). This form was completed at the end of each month by each family, and reported all receipts and certain allowable expenses which occurred during that month.

The initial IRF used during SIME/DIME was derived from the revised Income Report Form used in the New Jersey Experiment. The experience gained from that experiment plus the additional data requirements of the SIME/DIME design led to the addition of more detailed questions on the SIME/DIME IRF. More detail on family composition changes, monthly reporting of employment status of all family members over 16 years of age, sources of receipts and expenses for the self-employed, and the sale of capital assets were all incorporated.

The amounts included in the SIME/DIME definition of "basic income" was subject to the tax rate to which the particular family was assigned. Other sources of family income, such as government or private transfers, were taxed at either 50 percent or 100 percent. Table 2.4 outlines the items included in each category. An IRF filing schedule was adopted, as described earlier, to both allow adequate time for families to complete and submit the IRF and also provide sufficient time for the IRF to be processed and the payment calculated. A blank IRF was sent to each family, along with their grant check, on the last working day of the month. This IRF was stamped with the dates of the period covered by the form and the date it was due to be returned.The completed IRF was due 14 days after it was sent out. If it was submitted on time the family would receive payment on the regularly scheduled payment date, the last business day of the month. If it was submitted late, as long as it was filed within the 2-week period after the end of the on-time filing period, the family would receive payment 2 weeks after the regularly scheduled payment date.

To promote IRF filing, a $20 minimum payment was made in return for all IRFs that were filed on time or within the next two weeks. For families who were to receive an NIT payment anyway, this minimum was exceeded by the benefit and enough incentive to file the IRF already existed. For families who were on AFDC, in contrast, or whose income level was too high to qualify for a payment, the $20 minimum provided a needed incentive to file. Not surprisingly, the filing rate for the $20 families was regularly slightly lower than for those families receiving the more sizable benefit payments. IRFs that were not filed by the end of the late filing period were subject to a penalty. The initial rules of

TABLE 2.4 ITEMS INCLUDED IN INCOME AND SUPPORT

Taxed at experimental assigned rate		Transfer payment income, taxed at 100 percent	Support payments received, taxed at 50 percent	Support payments paid out, increased NIT by 50 percent
[illegible] basic income	Deductions from basic income			
1. Wages, tips, salaries, commissions	1. Nonreimbursed day care for job related reasons	1. Welfare (old age assistance, disability assistance, general assistance)	1. Alimony received.	1. Alimony paid.
2. Asset imputation	2. Care for aged or invalid family members for job-related reasons	2. Unemployment compensation	2. Child support or support of other relation received.	2. Child support or support of other relative paid out.
3. Life insurance benefits (over $1,000)	3. Nonreimbursed medical expenses (including dental care and prescription medicines)	3. Veterans benefits (including disability)		
4. Rent from income property	4. Capital losses	4. Training stipends (excluding tuition and fees)		
5. Rent from boarders in excess of $150 per month.	5. Business or self-employment losses	5. Social security benefits		
6. Pensions or annuities		6. Food stamps		
7. Fellowships and scholarships.		7. Workmen's compensation.		
8. Short-term capital gains.				
9. Business or self-employment income.				

operation called for a forfeiture of any payment due if the IRF was not filed before this penalty date. After 2 years of operation, however, this rule was relaxed so that a penalty of only $20 was exacted for any IRF submitted within 1 year of its original due date. The rule again reverted to its original form, however, so that over most of the experiment payments were fully forfeited for any period for which the IRF was not received by the penalty date.

This schedule did not prove overly onerous for participating families, and over the course of the experiment 88 percent of the Seattle IRFs and 84 percent of the Denver IRFs were filed on time.

Processing the income report form (IRF)

An overview of the monthly flow of operation within the council for grants to families appears as figure 2.2. This exhibits, in simplified form, the steps involved in producing a monthly payment from each IRF submitted.

The manual process of preparing the IRF for data processing was primarily a quality control function. This involved an item-by-item review of the IRF to examine internal consistency, completeness, legibility, consistency with the historical sequence of IRFs in the family file, and to check for documentation of any changes. This rather rigorous review took from 10 to 25 minutes for each IRF.

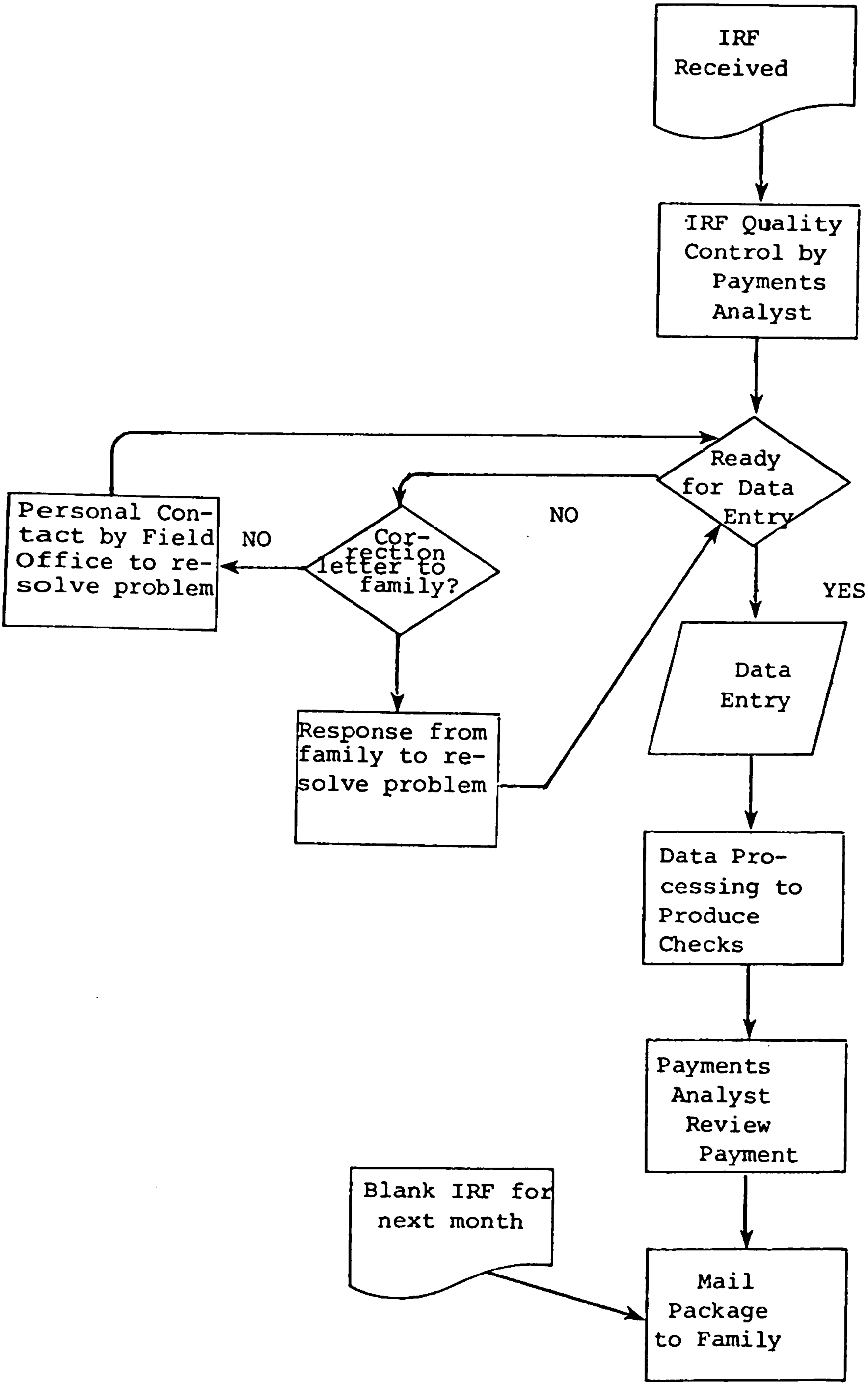

IRF Received
IRF Quality Control by Payments Analyst
Ready for Data Entry
NO
Correction letter to family?
NO
Personal Contact by Field Office to resolve problem
Response from family to resolve problem
YES
Data Entry
Data Processing to Produce Checks
Payments Analyst Review Payment
Blank IRF for next month
Mail Package to Family

FIGURE 2.2.—Overview of operational flow

The review would result either in an IRF that was acceptable for further processing, or in a contact with the family to correct any problems. The problems which delayed payments most often were

- missing signatures
- unclear family composition
- suspected unreported or misreported earnings
- missing current welfare statement.

When these problems arose, a contact letter was sent to the family describing the problem, and the matter referred to a SIME/DIME field office for resolution.

Once the IRF was edited it was prepared for data entry by the insertion of a brief set of codes. The data were then keypunched, and passed through an edit routine which rejected out-of-range values. Once all IRFs had passed the edit routine, the IRF tape was merged with the permanent payments file, which contained the 12-month history for each family. The product of the merger was the data set used by the computer to calculate the grant payments, which then produced the following outputs:

- payment checks
- grant explanation sheets
- check registers
- management information and control reports.

Disbursement of payments to families

Following the automated processing of payments, the materials listed above were sent to the payments analysts, who made one last check on the amount and the accompanying reported data reported on the IRF. The checks and grant explanation sheet (reproduced as figure 2.3) were then sent to the family, according to the schedule mentioned earlier, along with a blank IRF for the next month.

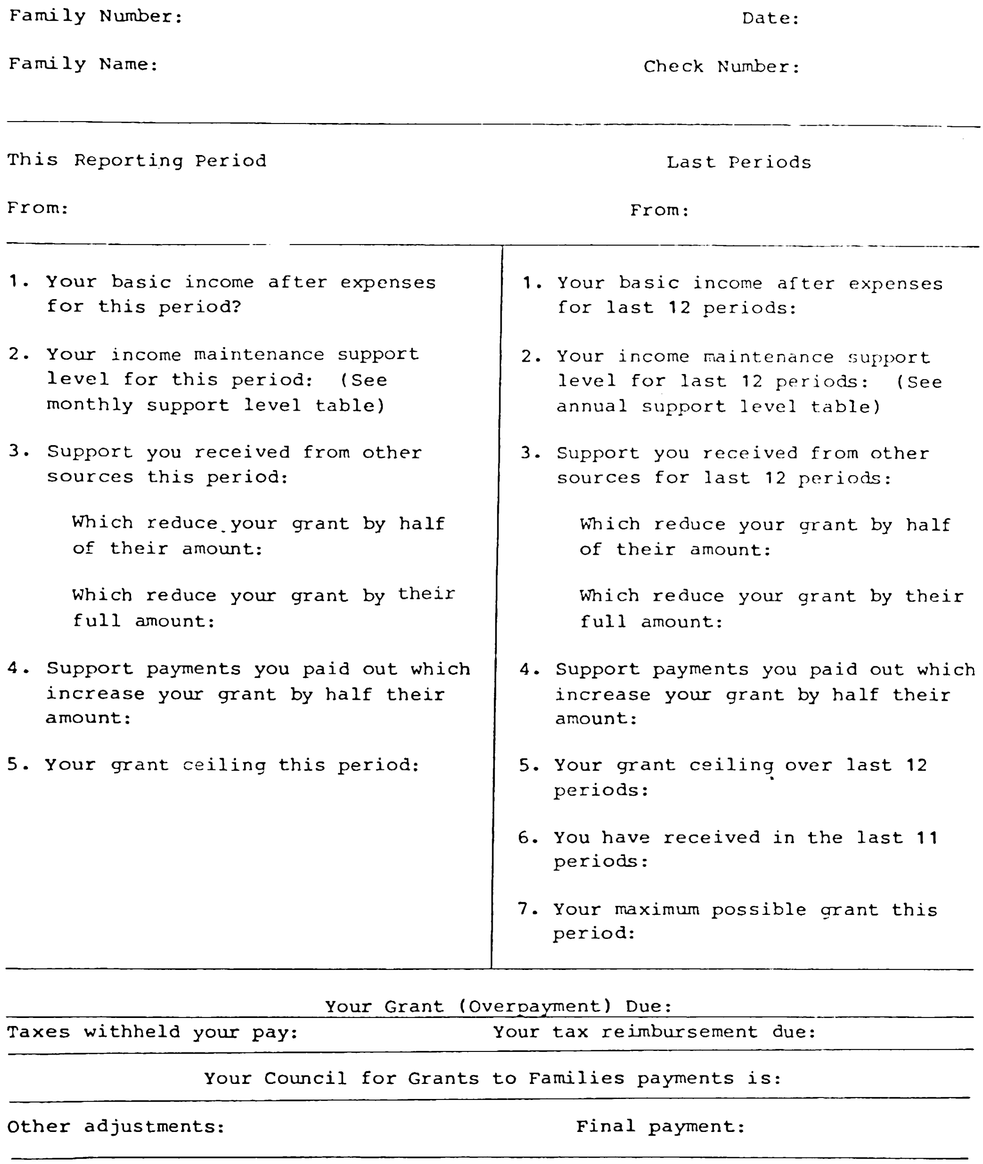
Tax reconciliation

At the end of each calendar year a reconciliation was made of the families' actual Federal and State income taxes and the amount that SIME/DIME had reimbursed the family via the monthly tax reimbursements that were a component of their monthly checks. To facilitate this reconciliation, copies of the 1040 form filed by each family with the IRS were obtained. The amount that should have been reimbursed by SIME/DIME was then calculated from the 1040 form and compared with the sum of the actual reimbursements made over the course of the year. Any discrepancy was either paid to or collected from the family, as appropriate.

Auditing

To ensure the veracity of the data and the credibility of the experiment, audits were built into the operating system. Extensive auditing was done within the day-to-day operation, professional auditing staff performed periodic in-depth audit reviews, and (subject to constraints on the confidentiality of participants) some information was audited against government records. The accounting system used to control disbursements to families was straightforward and eventually tied into the payments system such that only the automated system could issue checks.

The AFDC audit quickly became a regular monthly activity of the SIME/DIME staff. The basic rule regarding AFDC was that no family could receive a



Family Number: Date:

Family Name: Check Number:

This Reporting Period From:	Last Periods From:
1. Your basic income after expenses for this period?	1. Your basic income after expenses for last 12 periods:
2. Your income maintenance support level for this period: (See monthly support level table)	2. Your income maintenance support level for last 12 periods: (See annual support level table)
3. Support you received from other sources this period:	3. Support you received from other sources for last 12 periods:
Which reduce your grant by half of their amount:	Which reduce your grant by half of their amount:
Which reduce your grant by their full amount:	Which reduce your grant by their full amount:
4. Support payments you paid out which increase your grant by half their amount:	4. Support payments you paid out which increase your grant by half their amount:
5. Your grant ceiling this period:	5. Your grant ceiling over last 12 periods:
	6. You have received in the last 11 periods:
	7. Your maximum possible grant this period:

Your Grant (Overpayment) Due:

Taxes withheld your pay: Your tax reimbursement due:

Your Council for Grants to Families payments is:

Other adjustments: Final payment:

FIGURE 2.3.—Grant explanation sheet.

payment from both AFDC and SIME/DIME during any given month.[1] Because of the different income reporting schemes described above, confusion arose from the very beginning—which led to duplicate payments, not only resulting from attempts by families to defraud, but because of imperfect coordination between the experiment and the welfare offices which could potentially serve a participating family. As time passed inadvertent duplicate payments decreased, and fraudulent receipt of duplicate payments became our primary concern. Auditing to detect and prevent these duplicate payments necessitated a matching of the State welfare rolls and the list of families receiving SIME/DIME payments for each month.

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[1] Except for the SIME/DIME $20 monthly filing fee.

SIME/DIME field office

The field offices were responsible for all direct contact with the enrolled families. Their dual role was to provide a service to the families in responding to questions and concerns about the operation of the NIT payment system, and to contact families in response to directives from the payments department concerning specific problems or routine procedures. To provide a deterrent to any possible collusion between enrolled families and the payments analysts, the analysts did not communicate directly with the families. Other than formal notices sent by the analyst, all communications were channeled through the field office and its staff of field representatives.

Most contacts with families were completed over the telephone, although in exceptional cases home visits were made. Occasionally families would come to the field offices with questions. Contacts relating routine information or requiring a written response or signature were handled by mail.

Contacts initiated by the families were usually requests for additional information about the program. Requested most often were

- clarification of the rules of operation
- explanation of the benefit calculation printout
- help in completing the IRF
- clarification of manpower or childcare provisions
- requests for referrals to other agencies.

All contacts with families were recorded to provide a historical record—serving as the data base for research on contacts for national program planning. Table 2.5 shows the contacts generated per 100 IRF'S submitted by reason of

TABLE 2.5.—TYPE OF CONTACT AND REASON FOR CONTACT PER 100 IRFs BY LENGTH OF TIME FAMILIES WERE IN THE PROGRAM

[Number of months in program]

	Seattle					Denver				
	6	12	18	24	Total	6	12	18	24	Total
Contacts	43	38	33	26	34	103	87	71	55	73
Initiated by:										
Families	36	30	25	19	27	84	69	53	37	55
Field offices	7	7	8	8	7	20	19	18	17	18
Disruptive contacts	11	11	10	10	10	26	25	24	23	24
Reason for contact:										
Changes in family composition	6	6	6	5	6	7	7	6	6	6
Explanations of program rules	6	5	5	4	5	16	11	7	3	8
Income and expenses	7	6	5	4	5	21	18	15	11	15
Payments	10	9	8	7	9	29	28	29	29	29
Other IRF problems	8	6	5	3	5	21	15	9	2	9
Changes in addresses, names, etc	6	5	4	3	4	7	7	6	5	6

NOTE: Rates adjusted by regression for differences among the families in income, average payment, number of parents, education of male or only head, family size, number of IRFs filed, number of earners, whether the family contained a self-employed member, whether the IRF was filed before or after orientation sessions for participating families, which season of the year in which the IRF was filed.

contact, site, and length of time in the program. Notably, contacts with families declined in both sites as the experiment progressed and the families became more knowledgeable about program operations.

Reeducation

To measure family comprehension of SIME/DIME, update them as to recent changes, and improve on their understanding when it was found lacking, the field office administered a reeducation program to coincide with the end of the first year of participation in the experiment. The process involved at least one in-person session, usually at the family home. This session included an explanation of the changes to the rules of operation since the family was enrolled, a review of filing procedures (particularly any specific problems that may have been causing the family difficulties), and administration of a short questionnaire to measure family comprehension of the program.

The questionnaires were somewhat different in the two cities, but shared a common core of questions in five subject areas. Indices of comprehension were developed to represent the percentage of correct answers in each subject area, displayed in table 2.6. As can be seen, the overall mean scores were similar for the two cities, although there were striking differences for several of the categories. Further analysis of these questionnaires, as noted in Part I, led to the conclusion that families learned what they needed to know to function within the experiment, and that the average comprehension scores understated behaviorally-related knowledge.

A second, more informal, reeducation project was undertaken at each site for families enrolled for 5 years, timed to coincide with the end of the family's third year of participation. For most families, this effort amounted to no more than a telephone call from the field representative to ask if they had any questions, and to remind them of the field office's availability for assistance if any questions arose.

TABLE 2.6.—REEDUCATION SURVEY INDICES OF COMPREHENSION

	Percentage correct	
	Seattle	Denver
Eligibility rules and requirements	68	56
Relationship between working and income maintenance payments	57	84
The effect of family structure on payments	32	41
Taxation of different types of income	58	37
Calculation of grant payments	38	46
Overall score	57	51

Source: Halsey, Murarka, and Spiegelman (1979).

Disenrollment of families

The disenrollment of the families at the conclusion of the experiment proved far less difficult than originally feared. Considerable prior notice was given to each family along with offers to assist them in the transition to any other programs and assurance of postexperimental referral and counseling through the SIME/DIME field office. The thorough and extensive prior notice were reasons, but not the most important ones, why the disenrollment process went smoothly. The principal reasons for the ease of the disenrollment process seemed first to be the clarity with which the limited duration of the experiment was emphasized during the enrollment process, and secondly, that SIME/DIME was not the Government. None of the families expected to receive income support from a private organization (even though most of them had to be aware of government backing) for 3 or 5 years, and certainly not in perpetuity.

CHAPTER IV

DESIGN AND ADMINISTRATION OF PERIODIC INTERVIEWS

The periodic interview series provided the primary data set for evaluation of the experimental treatments. Interviews were administered three times each year to all experimental and control group families. The focus of the interview was to collect data that would answer questions about participants' labor supply behavior, family stability, investments in human capital and, to a lesser extent, behavior with respect to durable consumption, housing, migration, and other socioeconomic factors. Interviews were conducted in sample members' homes by highly-trained interviewers using bound interview books 325-400 pages in length. It took, on the average, about 50 minutes to administer each interview. A discussion of the design of these questionnaires opens this chapter, followed by the field administration of the interviews.

Design and content

A series of questionnaire modules was developed to cover each topic. The interviews were then constructed from various combinations of these modules, where some were repeated within an interview for more than one person, some were repeated every interview, and some were repeated either yearly or sporadically throughout the experiment. These modules, then, persisted over time with minimal changes to their form and content. A list of each module and its frequency of occurrence appears in table 2.7.

TABLE 2.7.—SCHEDULE OF QUESTIONNAIRE MODULES

Module title	Respondents	Seattle interviews	Denver interviews
Employment and Income Data			
Job core	All members 16+	PRE,[1] ENR [2] 1-17, c[3] PE,[4] PM1, e[5] PM6.	PRE, ENR, 1-18, PE, PM1-PM7.
Benefits and expenses.	Male or only head	PRE, ENR, 1-17, PE, PM1-PM6.	PRE, ENR, 1-18, PE, PM1-PM7.
Perceived wage rate of nonworkers.	All members 16+	2, 5, 8, PM1-PM2, NM1 [6]-NM5.	1, 6, PM1-PM2, NM1-NM5.
Job satisfaction and stability.	Male or only head	2	POST.[7]
Do	Male head only	5, 8, 11, 14, 17	2, 5, 8, 11, 14, 17.
Do	Female head in 2-head family only.	3	
Do	Female head only	6, 9, 12, 15	3, 6, 9, 12, 15.
Female work attitudes.	Female head only	PRE, 4, 8, 14	1, 4, 8, 14, 17.
Husband's attitude toward wife working.	Male head in 2-head family only.	2, 7, 13, 17	POST, 7, 13, 17.
Point-in-time work history.	All members 16+	9, PM3-PM6, NM2-NM5	9, PM3-PM7, NM2-NM5.

TABLE 27.—SCHEDULE OF QUESTIONNAIRE MODULES—Continued

Module	Respondents	Seattle interviews	Denver interviews
Work history for new members	All members 16+	8, PM3-PM6, NM2-NM5	8, PM3-PM7, NM2-NM5.
Employment history	do	11, PM3-PM6, NM3-NM5	9, PM3-PM7, NM3-NM5.
Search for work	Male head only	2, PM1	2, PM1.
	New member male head only	NM1	NM1.
Asset and debt data			
Net worth	Male or only head	1, 3, 5, 7, 10, 13, 16, PM1-PM6	PRE, 1, 3, 5, 7, 10, 13, 16, 18, PM1-PM7.
Do	New member head	NM1-NM5	NM1-NM5.
Real property	Male or only head	PRE, 1, 4, 7, 10, 13, 16, PE, PM1-PM6	PRE, 2, 4, 7, 10, 13, 16, 18, PE, PM1-PM7.
Do	New member head	NM1-NM5	NM1-NM5.
Sale of real property	Male or only head	13, 16, PM3-PM6	10, 13, 16, 18, PM3-PM7.
Attitudes toward use of credit	Male or only head	PRE, 6, 11	POST, 6, 11.
Time horizon and planning	Male or only head	3, 7, 9	POST, 8.
Budget attitudes	Male or only head		PRE.
Education and training data			
Education and training of family members	Male or only head		PRE.
Education history	All new members 16+	1, PM1-PM6	2, PM1-PM7.
Do	All members 16+	NM1-NM5	NM1-NM5.
Job training history	All members 16+	1, PM1-PM6	2, PM1-PM7.
Do	All new members 16+	NM1-NM5	NM1-NM5.
Training and education update.	All members 16+	4, 7, 10, 13, 16, PE, PM1-PM6	4, 7, 10, 13, 16, 18, PE, PM2-PM7.
Attitudes toward training	Male or only head		PRE.
Do	Male head only	PRE	
Do	Female head only	PRE	1.
Family Composition and stability			
Family composition.	Male or only head	PRE, ENR, 1-17, PE, PM1-PM6	PRE, ENR, POST, 1-18, PE, PM1-PM6.
Intraperiodic composition changes.	Female or only head	14-17, PE, PM4-PM6	12-18, PE, PM4-PM7.
Secondary unit identification.	Male or only head	9, 15, PM3-PM6	9, 13, PM3-PM7.
Relationships of family heads and children.	Female or only head	10, PM3-PM6	10, PM3-PM7.
Family opinions	Male head in 2-head family only.		PRE.
Do	Female head in 2-head family only.	1	
Do	Male head only	PRE, 4, 7, 12	4, 7, 12, 17.
Do	Female head only	5, 8, 13	1, 5, 8, 13, 17.
Family roles	Male head in 2-head family only.	1, 9, 15	9, 15.

TABLE 2.7.—SCHEDULE OF QUESTIONNAIRE MODULES—Continued

Module title	Respondents	Seattle interviews	Denver interviews
Family roles	Female head in 2-head family only.	PRE, 8, 14, 17	8, 14, 17.
Family arguments.	Female head in 2-head family only.	PRE, 5, 11	2, 5, 11, 14.
Family leisure time.	Male head only	PRE, 4, 8, 12, 17	4, 8, 12, 17.
Do	Female head only	3, 7, 11, 17	3, 7, 11, 17.
Housing Data			
Subsidized housing.	Male or only head	2-17, PE, PM1-PM6	1-18, PE, PM1-PM7.
Housing	Male or only head	1	
Housing expenditure.	Male or only head		POST.
Housing rent update.	Male or only head	4, 7, 10, 13, 16, PE, PM3-PM6.	3, 7, 10, 13, 16, PE, PM3-PM7.
Respondent Behavior Data			
Postmove	Male or only head	PE, PM1-PM6	PE, PM1-PM7.
Fertility history	Female head only	11, PE	9, PE.
Do	New female head	NM3-NM5	NM3-NM5.
Household chores.	Male and female heads	12	12.
Utilization of child care.	Female or only head	PRE, 4, 8, 15	POST, 9, 15.
Job counseling	All members 16+	3, 6, 9, 12, 15	4, 6, 9, 12, 15.
Childrens' paying jobs 10-15 years.	Female or only head	2, 9, 12, 15	1, 10, 12, 15.
Educational status of children.	Female or only head	3, 6, 10, 13, PE	1, 3, 6, 10, 12, 14, PE.
Quality of Life Data			
Mental health status.	Male or only head	1	
Health mental health status.	Female head in 2-head family only.	2	
Do	Male head only	5, 11, 16, PE	1, 5, 11, 16, PE.
Do	Female head only	6, 12, 16, PE	2, 6, 12, 16, PE.
Nondurable consumption.	Male or only head	3, 6, 9, 12, 15	1, 4, 7, 10, 13, 17.
Community ties	Male or only head	PRE, 1, 6, 14	1, 6, 14.
Do	Female or only head	9	
Do	Female head only	15	9, 15.
Educational and occupational expectations for children.	Female or only head	1, 9, 14	POST, 9, 14.
Postexperiment question.	Male or only head	PE	PE.
Background and Attitudinal Data			
Marital status	Either male or female head		PRE.
Marital status and history.	Male head only	4, 10, PM1-PM6	2, 4, 10, PM1-PM7.
Do	Female head only	4, 10, PM1-PM6	2, 4, 10, PM1-PM7.
Do	New member head	NM1-NM5	NM1-NM5.
Ethnicity	Male head only	2, PM1-PM6	2, PM1-PM7.
Do	Female head only	2, PM1-PM6	2, PM1-PM7.
Do	New member head	NM1-NM5	NM1-NM5.
Mobility history	Male head only	2, PM1	3, PM1.

TABLE 2.7 SCHEDULE OF QUESTIONNAIRE MODULES—Continued

[illegible]	[illegible]	Seattle interviews	Denver interviews
[illegible]	[illegible]	2, PM1	3, PM1.
Do	[illegible]	PM3–PM6, NM1–NM5	PM2–PM7, NM1–NM5.
[illegible]	[illegible]	13	11.
Do	[illegible]	13	11.
[illegible]	[illegible]	4	3
Do	[illegible]	4	3.
Do	[illegible]	10, PM3–PM6, NM3–NM5	9, PM3–PM6, NM3–NM5.
[illegible]	[illegible]	2, 14	PRE, 12.
[illegible]	[illegible]	2, 8, 15	POST, 8, 15.
[illegible]	[illegible]	5	5.
	[illegible]	4	4.

[illegible]

There were multiple advantages to the module system of interview development. Besides providing consistency to the data content over time, it allowed for efficiencies in interview construction and simplified data processing. It also facilitated administration of the interview and perhaps provided better quality responses because of interviewer and respondent familiarity with the questions.

Employment data

The employment data were the most comprehensive continuous-time data collected in SIME/DIME. The data were collected so that each person's employment status could be established on any given day during the course of the experiment or over any given period of time. In addition, hours of work, desired hours of work, wage rates, reasons for job terminations and unemployment, as well as many other employment-related data were collected in this same continuous-time fashion. Thus, dynamics of labor force participation, job turnover, the pattern of earnings could all be studied with these data. Earnings could also be calculated over various time periods, enabling comparisons with other income-reporting sources such as the payments system, IRS or Social Security.

The module in which these data were collected was called "Job Core." Each family member over the age of 16 was asked this section. As younger children turned 16 during the experiment they also began answering Job Core. The Job Core module began with retrospective employment data. Each interview contained this module, collecting data that spanned from the date of the previous interview to the date of the current interview. The module provided complete continuous-time data on all periods during which the respondent was working, as well as periods during which he or she did not work. The data collected were designed to enable the construction of variables that could define, at any point in time, the level of and changes in hours of work, wage rates and earnings, desired hours of work, voluntariness of unemployment, and ability to find optimum jobs. Questions that could be used to define reservation wage rates of nonworkers were originally asked in a separate module, perceived wage rate of nonworkers, but were incorporated into Job Core later in the experiment.

Data for self-employed respondents were collected in a supplement to the Job Core. This supplement collected monthly hours of work and receipts and expenses that could be used to determine the earnings of the business.

Certain other attitudinal modules were administered during the experiment that addressed employment-related issues expected to affect the NIT impact on individuals' work behavior or issues that were expected to be affected by the experiment themselves. Examples of the former type of questions are found in the modules on Female Work Attitudes, Husband's Attitude Toward Wife Working, Social Ethics of Work, and Perception of Discrimination. An example of the latter is Job Satisfaction and Stability.

Nonwage income data

Data on income from sources other than work were also collected in a continuous-time fashion in the Benefits and Expenses module which appeared in each interview. This module covered the value of monthly receipts of 24 categories of nonwage income and certain family expenditures as well. Nonwage income categories covered income from assets, public transfer payments, and private transfers. Monthly expenses for child care, medical care and support given to individuals outside the family were asked because these items were treated as deductions from income in the NIT payments calculation.

The Benefits and Expenses data were used primarily in the calculation of full income needed in order to compute expected grants and the change in disposable income generated by the NIT payments. They were also used in a few of the impact studies as right-hand variables.

Training and education data

Training and education data were collected yearly in, for the most part, a continuous-time fashion. Initially, all individuals over the age of 16 were administered a Job Training History module and an Education History module. These were designed to pick up past and current investments in human capital. Subsequently, the initial data were updated with reports on all programs previously enrolled in and all new programs undertaken since the last report. The complete education and training data set consisted of a measure of cumulative training and education taken before the experiment, and the start and end dates of all programs taken during the experiment, along with characteristics of these programs such as type of program, hours, and costs.

Measures of schooling and training were used as explanatory variables in determining how prior levels of education affected individuals' responses to the NIT program and were also used to study how both the NIT and counseling/training program affected peoples' decision to invest in more schooling. (Data were collected directly from participants in the counseling/training program as well. These data are discussed in part IV of this volume).

Asset and debt data

Data on assets and debts were collected primarily to assess how the program affected participants' decisions about savings and investments and to assess participants' time horizon and planning for the future. These questions were important both in determining the overall impacts of the experiment (particularly whether the details of the payments system created incentives for families to invest in non-income-producing assets), and in determining whether people would behave in a 3- or 5-year program as if it were a lifetime program—i.e., whether they would treat the income from the NIT experiment as a permanent or transitory addition to income.

Asset and debt data were contained in numerous modules in the interview. The Net Worth module collected continuous data on the purchase and disposal of consumer durables and point-in-time data on savings and debts. During the first administration of the Net Worth module, families were asked to list each of their durable items along with cost and date of purchase. In subsequent yearly interviews, families were read this list of items and asked if they still had each of the items; if not, when they ceased to own the item, and if they had acquired any new items. Self-employed individuals were asked a somewhat different set of questions in the Self-Employment Net Worth Supplement to reflect differences in the type of assets owned by a business and to discern which items were used by the business.

The Real Property module collected additional asset data in the form of owned housing and other real property. Each year, families were asked details about their ownership of real property, including data of purchase, mortgage information, and current market value. These data were not continuous, however, and did not enable one to trace the disposition of real property until the Sale of Real Property module was introduced into the interviews somewhat later in the experiment. This module, which was also asked yearly, recorded the date and sales price, among other information, of all property transferred out of the family's ownership during the year.

Family structure data

The primary data source for measures of family stability and changes in marital status was the master file rather than the interview. The interviews were administered to family units, and when families broke up the continuity of family data was broken. Thus, changes in family status could not be followed easily using the interviews. The Master File, however, provided a record of the family status of each individual throughout the experiment, making it the best source of data for studies of marital stability.

The sources used in creating the master file were not only the interviews but also family composition information collected in the monthly forms submitted for NIT payments and monthly forms sent to control families asking about changes in family composition during the preceding month. These monthly data sources were also used to prerecord family composition data in the periodic interviews. In this manner, changes reported in monthly forms were verified in the next interview.

Data on changes in family composition were augmented by modules attempting to determine earlier marital patterns, family roles, the level of family interaction, and family friction as factors that were expected to affect family stability. The Marital Status and History module collected information on respondents' current marital status, the length of time they had been in their current status, and how any former marriages had ended. These historical data were used in the models to explain current marital events. The Family Roles and Family Opinions modules asked who performed various roles within the family and who, ideally, should perform these roles. These measured to what extent respondents held traditional ideas about roles within marriages, to what extent their marriages conformed to these traditional notions, and to what extent there was potential conflict between spouses over their marital expectations. Family Opinions and Family Arguments measured marital satisfaction and the topics and frequency of arguments between spouses. Family Leisure Time measured the amount of time families spent together and in what activities they engaged.

Other attitudinal and behavior modules

To measure the NIT's impact on housing consumption patterns, yearly point-in-time information was collected on purchases and sales of homes and home value. Details on housing characteristics and improvements, and rental and utilities costs were collected in the Housing Rent Update, Real Property, and Sale of Real Property modules. In addition, longitudinal data on utilization of subsidized housing were collected that included the dates of all stays in subsidized housing, the rent or mortgage amounts paid, and each change in these amounts, as well as the number of rooms in the dwelling.

The Utilization of Child Care module collected detailed information throughout the experiment on access to and use of various modes of child care and child care subsidies. These data enabled assessment of the impact of both the NIT and child care subsidy programs on the demand for child care and on women's decisions to work.

The Job Counseling module questioned respondents about their experiences with job counseling services, including those provided by the SIME/DIME counseling training program, and asked them to rate the services they received. These questions were asked as a means for controlling the external job counseling experiences in order to get a more accurate measure of the effects of the SIME/DIME counseling program, and to evaluate the impact of job counseling programs in general.

A number of modules were designed to evaluate possible improvements in the quality of life resulting from participation in SIME/DIME. The NIT could be expected to raise individuals' quality of life through both direct and indirect benefits of increased income. Direct benefits would be those related to higher rates of consumption. Indirect benefits of a higher or more secure income would be things like a lower level of stress, a greater sense of independence, and increased leisure time. Certain modules described earlier measured levels of family stress, interaction, and leisure time activities, which might be expected to show quality of life changes. The Health and Mental Health Status module collected data on frequency and length of illness and data on psychological distress via a series of psychophysiological symptom items. The Postmove module measured people's reasons for moving and the resources they had available to them that made the move possible. Community Ties measured the level of the respondents' social interaction with their family and friends and within their communities. The level of community involvement was measured by the number of memberships in various types of community organizations, the frequency of attendance at meetings, and the number of official positions held. Nondurable Consumption measured the expenditures made in certain months for various forms of entertainment, household items, and clothing. Information collected in these modules indicated how NIT recipients chose to spend their increased income.

Another set of modules measured specific behaviors expected to be altered by the NIT treatment. Household Chores, by asking how much time was spent on a number of household tasks, attempted to measure the extent to which increased time out of the labor market was spent on working at home. Fertility History measured the experiment's impact on decisions to bear children. Childrens' Paying Jobs and Educational Status of Children monitored yearly some of the primary activities of children to measure possible impacts on their behavior.

Lastly, there were a series of modules soliciting background and attitudinal data to be used to control for exogenous factors in the impact analyses. Some of these modules could be considered pure background data in that they collected specific information about respondents' identities and their past. The Ethnicity module asked individuals specific information about their racial, national, and religious backgrounds, and the Occupation and Education of Parents module asked about the status of parents. The Verbal Awareness module was a sentence completion test that measured individual's verbal comprehension and learning. (The test was translated into Spanish for Spanish-language respondents.) The Mobility History module collected information on the type of area the respondent grew up in and the number of moves he or she experienced prior to enrollment. Employment History asked for the total number of years respondents had worked, and Point-In-Time Work History asked for the working and marital status of all respondents on November 22, 1963, the day President Kennedy was killed.

Another group of modules asked questions that were expected to pick up attitudinal differences among the experimental population that could influence their response to the program. Attitudes Towards Welfare asked respondents their opinions about government aid programs and people who participate in them. Social Ethics of Work asked respondents their feelings about the importance of work and whether they felt that people have an obligation to work. These questions were intended to capture differences in program response resulting from varying attitudes about government programs or beliefs in the work ethic. Other questions about individuals' attitudes towards money and credit were found in Attitudes Toward Use of Credit, in which people were asked about their opinions and experiences in borrowing money and using various forms of credit, and in Time Horizon and Planning, which tested peoples' rate of time discount for money. The Perception of Discrimination module and the Self-perception of Socioeconomic Status module were intended to measure differences in respondents' perceptions of the degree to which their circumstances were determined by forces outside of their control. The former asked respondents if they thought that the quality of education and employment opportunities are affected by an individual's race, sex, or ethnic background, and if they felt that they had personally been affected by discrimination. In the latter module, the respondents were asked to classify their socioeconomic status and to express their opinions about the degree of class mobility available to people.

Interview administration

Conducting the periodic interview series was the most expensive task in the administration of SIME/DIME. Over the course of the experiment, more than 65,000 detailed in-person interviews were administered. There were three periodic interview waves administered per year. The task for each wave began with a listing of all families to be given that particular interview. This was followed by the preparation of an interview packet for each family and the administration of the interviews in the family home. Rigorous quality control readings and, finally, entry into the data base completed the process.

Figure 2.4 depicts the flow of a typical periodic interview. The process began with the sample maintenance department assigning the interview so as to make sure that each family was interviewed at 4-month intervals. The 4-month scheduling was not always strictly adhered to, but in no case was the interval allowed to be less than 2 months. Backlogs would often develop at different stages of the process, most commonly in quality control and data entry, so the more usual problem was too long an interval rather than too short a one.

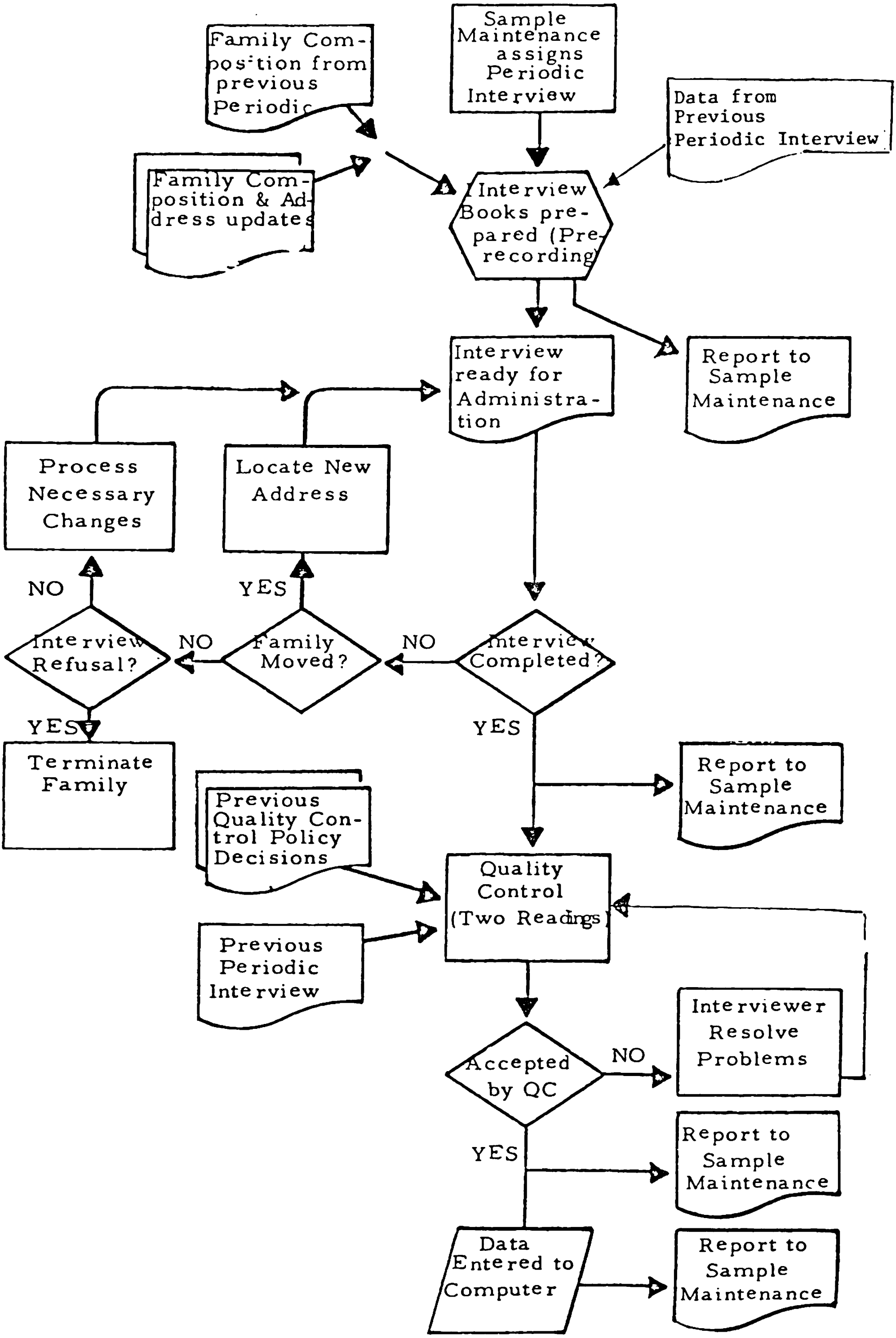
Family Composition from previous Periodic
Sample Maintenance assigns Periodic Interview
Data from Previous Periodic Interview
Family Composition & Address updates
Interview Books prepared (Prerecording)
Interview ready for Administration
Report to Sample Maintenance
Process Necessary Changes
Locate New Address
NO
YES
Interview Refusal?
NO
Family Moved?
NO
Interview Completed?
YES
Terminate Family
YES
Report to Sample Maintenance
Previous Quality Control Policy Decisions
Quality Control (Two Readings)
Previous Periodic Interview
Interviewer Resolve Problems
Accepted by QC
NO
YES
Report to Sample Maintenance
Data Entered to Computer
Report to Sample Maintenance

FIGURE 2.4.—Flow chart of periodic interviews.

Once the periodic was assigned, an interview book for each family had to have certain information prerecorded. This interview book was the printed and bound periodic interview, which varied in length from 325–400 pages depending on the modules included. The prerecording process varied over time as policy changes were made; it also varied in detail depending on the content of a particular periodic. In general, the prerecording process included:

- labeling the interview book with family numbers and designating the family member who should answer each section of the interview
- checking for recent changes and then inserting an updated Master File printout of family composition
- entering certain pieces of factual economic information from the previous periodic interview
- inserting computer printouts containing data from specific modules of earlier periodic interviews (as appropriate).

The prerecorded interview books were then given to the interviewing department. Interviewers were provided up to 10 books a day to work on. They would first attempt to telephone the family to set up an appointment to do the interview. If the family had no phone or could not be reached, the interviewer would have to contact them at their home, at which time they hoped and usually managed to conduct the interview. If the family had moved and could not be traced by the interviewer, the interview was sent to the sample maintenance searcher whose job was to locate moved-not-found families. If the interview was refused by the family it would first be given to another interviewer specially trained in converting refusals. If that effort failed, the interview book was sent to the sample maintenance department, which would terminate the family's formal participation in the experiment.

As already noted, very few families refused periodic interviews. There were a number of reasons for this. First, the periodic interviews included only families who agreed to participate in the study by signing an Enrollment Agreement. Second, by the time they were enrolled they had already willingly participated in three interviews. Refusals did occur during the sample selection phases of the experiment, so having already dropped those families from the periodic interview sample was a factor in the high completion rates. Another factor was the economic incentive to participate. All families receiving experimental treatment, financial and/or training/counseling, were required to complete interviews as requested in order to remain eligible for the treatment. Control families had to continue with interviews to be eligible to receive $8 a month for submitting their address postcard. In addition, all heads of household and dependents over age 16 were required to answer appropriate sections in the questionnaire. For this they received a $7.50 and $5.00 payment, respectively.

The interview length and complexity, and the types of data included required a lot of interviewer training. Training began with at least three days of intensive training on interview techniques and the specifics of each periodic interview. This training included lectures, viewing and discussion of an MPR film "The Art of Interviewing," question-by-question discussion, mock interviews, observation of actual interviews and finally, the administration of actual interviews under the guidance of experienced interviewers. This training was continued longer for any interviewer whose work was not up to quality standards. Each time a new periodic interview was released to the field, all interviewers underwent additional formal training using specially prepared training manuals.

Actual questionnaire administration time varied from 30 minutes to 2½ hours, depending on the length of the specific modules included, the size of the household, and the complexity of their economic situation. A typical questionnaire took around 30–45 minutes to administer. Administration time was reduced as the study progressed and respondents learned to have pay stubs and other required documentation of benefits and expenditures available for the interviewer.

Completed interviews were then submitted to the quality control process. Each interview underwent two rigorous readings by separate reviewers for completeness, internal consistency, consistency with particular data from previous interviews, and conformity with established data quality standards.

The two-reading procedure used during SIME/DIME was certainly not typical of most surveys; nor were the data quality standards that were imposed. Quality control was not allowed to accept inconsistencies within the interview. Redundant questions were deliberately built into the questionnaire and the reconciliation of inconsistencies by the interviewer was required. Indeed, the redundancies were often the basis for further interviewer probes.

The first analytic reading of the period interview by quality control staff was to accomplish the following:

- insure that all factual and opinion questions were answered by the designated respondent
- double check calculations of earnings and other critical economic variables; add totals to be used by the data entry program for cross-checking
- ascertain logical consistency between factual portions of the interview and between certain economic variables in this interview and the preceding one
- insure that the interview was legible
- insure that there was proper notation to explain atypical situations, particularly if reported income for the period did not cover reported expenses
- insure that the correct interview question patterns were followed
- insure that collected data corresponded to research specifications as promulgated in quality control data decisions.

All problems with the interview, identified by page and question number, were noted on problem sheets attached to the interview. The second reader went through essentially the same process as the first, but the length and complexity of the interview were such that the first reader typically missed several problems. The average reading times were about 40 minutes for the first and 30 minutes for the second.

If necessary, the interview was then transferred back to the interviewer who resolved the problems noted on the problem sheet. Due to the length and complexity of the interview, virtually every interview had some problem. Often the resolution necessitated recontacting the family to clarify inconsistencies or collect missing information. When problems were resolved, quality control reviewed the resolution and, if acceptable, passed it along to the precoder. The precoder then entered certain numeric codes (primarily relating to jobs held by family members) to the interview before sending it on to the data entry department.

Data entry involved keying the data from the questionnaire into a computer disk file through a special data entry program. This program prompted the data entry with a shortened version of each interview question and edited with range checks and cross-checks against totals (provided by the quality control department) to insure the correctness of the entered data.

There were continuing efforts throughout the experiment to locate and interview those families in the less than 1 percent who missed an interview. When a periodic cycle was pronounced closed, any remaining families were placed in a residual category. If they were enrolled in experimental treatments, that treatment was discontinued as long as they remained in the residual group. If they were subsequently found, they were administered the then-current periodic interview plus a special interview that collected just the necessary economic information from the date the family members had last been interviewed. Those families therefore were able to have a complete continuous economic data set on all family members through the experiment, just as if they had never missed an interview.

Within the regular periodic interview data collection, several special interviews were designed for special cases. These included the new member interview, self-employment module, and moved-out-of-area (MOA) interview. The new member interview was designed to collect background information on eligible persons over age 16 joining enrolled families during the experiment. The need for a special self-employment module arose from the difficulties of trying to fit earnings data on the self-employed into the highly structured job core module. Both these special interviews were administered as supplements to the regular periodics.

The need for an MOA interview arose because a number of families moved out of the greater Seattle and Denver areas. Continuing eligibility required only that families remain in the continental United States. In order to create the data set necessary to evaluate the experimental treatments, comparable interviews needed to be administered to the families who moved. Each year, therefore, a group of interviewers traveled around the country and administered an interview to the 250–500 MOA families.

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SIME/DIME FINAL REPORT

Volume 1, Part III

LABOR SUPPLY RESPONSE

by

Philip K. Robins and Richard W. West

CHAPTER I

INTRODUCTION AND SUMMARY OF RESULTS

Since the early 1960s, there has been considerable debate about the merits of a universal negative income tax (NIT) as a replacement for the existing welfare system. In principle, a universal NIT would differ from existing transfer programs by extending cash assistance to two-parent families and by removing work requirements as a condition of eligibility. Although the addition of AFDC–U and Food Stamps to the present array of welfare programs has moved the country closer to an NIT, consolidation of all existing programs into a single unrestricted cash transfer program has yet to become a reality.

During the late 1960s and early 1970s, legislative attempts to implement an NIT nationwide were thwarted partly because of a lack of definitive empirical evidence regarding the size of the labor supply impacts of an NIT. Knowledge of the labor supply impacts of an NIT is of enormous policy interest because such impacts are likely to have a major effect on program costs. Moreover, there is great political concern that an NIT would create a sizeable dependent population by inducing large-scale withdrawals from the labor force.

In response to this void in knowledge, a large number of empirical studies were conducted in an attempt to estimate the labor supply effects of an NIT. These studies, which were generally based on nonexperimental cross-sectional data, concluded that labor supply reductions in response to an NIT would not be large enough to cause massive withdrawals from the labor force (see Cain and Watts, 1973). However, the wide variance of these nonexperimental estimates and the methodological problems associated with the various studies that produced them (see the sensitivity study by Da Vanzo, Detray, and Greenberg 1973, and the survey by Keeley, 1981) led to proposals for experimental research in which an NIT is imposed on a random sample of low-income families.

The early NIT experiments, conducted in New Jersey and Pennsylvania from 1968 to 1972, in rural areas of Iowa and North Carolina from 1969 to 1973, and in Gary, Indiana from 1970 to 1974, were beset with a variety of design and operational problems that limited their ability to provide precise estimates of the labor supply effects of an NIT. Probably the most important problem with these early experiments was their small sample sizes. Although these experiments produced estimates suggesting that labor supply would be reduced under an NIT, the precision of these estimates was generally quite low. Hence, the degree of confidence that could be placed in these estimates by policymakers was not great. Nevertheless, the early experiments made the important contribution of putting to rest any fears that an NIT would lead to massive withdrawals from the labor force.

The Seattle and Denver Income Maintenance Experiment (SIME/DIME), conducted from 1971–77 was the last of the NIT experiments,[1] and contained a sample that was larger than all the other experiments combined. The larger sample enabled researchers using SIME/DIME data to obtain more precise estimates of the size of the labor supply response to an NIT than had been possible before. In addition, SIME/DIME researchers developed an empirical labor supply model that could be generalized easily to other NIT programs and to the national population. As a result, policymakers concerned with reform of the present welfare system have made extensive use of the SIME/DIME results.[2]

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[1] Actually, SIME/DIME began about the same time as the Gary Experiment but lasted somewhat longer.

In addition to developing a methodology for estimating the labor supply response to alternative nationwide NIT programs, researchers in SIME/DIME have also had considerable success in identifying the various dimensions of labor supply that are affected by an NIT. Labor supply is a multidimensional concept and a comprehensive analysis of how each of these dimensions is affected is an important ingredient for policymaking. Figure 3.1 shows the relationship between several dimensions of labor supply that may be affected by an NIT and for which data are available in SIME/DIME.[3] Perhaps the most comprehensive measure of labor supply, and the measure used in the SIME/DIME design model, is annual earnings.[4] However, annual earnings, while being directly related to the costs of an NIT program, is not strictly a measure of labor supply because it combines hours worked per year with the average hourly wage rate. An alternative, and perhaps better, measure of labor supply is annual hours of work, which is the measure used almost exclusively in the early evaluation studies of SIME/DIME (Keeley et al., 1978a, 1978b, Robins and West, 1978a, 1978b).

Although annual hours of work is useful for many purposes, it too suffers from several deficiencies. First, it combines effects on the probability of working with the effects on weeks worked and hours worked per week. Since many existing and proposed transfer programs other than an NIT have benefits that depend on either employment status or length of employment, it is useful to determine whether the experiment has different effects on these components of annual hours of work. An additional limitation of annual hours of work is that persons who are temporarily unemployed and looking for work are treated the same as persons who are out of the labor force—that is, whose observed hours of work are zero. At the beginning of the experiment in Seattle, the unemployment rate was quite high relative to the unemployment rates in other urban areas of the United States. In light of this adverse economic situation, SIME/DIME researchers examined an alternative measure of labor supply—desired hours of work—which may be more closely related to long-run hours under a national program. Such a measure associates a hypothetical quantity of work to persons who are unemployed;[5] however, it is based on subjective data and may contain substantial measurement error. In addition, the policy relevance of such a measure is somewhat unclear because it has no observable counterpart in the real world.

Various components of annual hours of work can be further broken down in order to provide answers to additional questions about the nature of the labor supply response to an NIT. For example, do persons without jobs remain out of work longer than they would without the NIT program? Do workers eligible to receive NIT payments leave employment more readily than workers who are not eligible for such payments? Do workers continue to hold jobs but cut back their weekly hours of work by eliminating overtime, leaving secondary jobs, or

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[2] Probably the most important use of SIME/DIME results is the microsimulation study by Betson, Greenberg, and Kasten (1980) which provided the intellectual basis for the design of the Program for Better Jobs and Income (PBJI), the Carter administration's first proposal for welfare reform (also, see Aaron and Todd, 1979). It should be pointed out, however, that there are problems associated with using SIME/DIME results to determine the labor supply effects of welfare reform alternatives like the PBJI, which provide jobs in addition to cash transfers, because SIME/DIME did not impose any work requirements or provide jobs.

[3] One important dimension of labor supply that may be affected by an NIT, but for which data are not available, is intensity of work effort. For a discussion of the importance of this concept, see Pencavel (1977) and Rosen (1981).

[4] For a detailed discussion of the SIME/DIME design model, see Conlisk and Kurz (1972) and Keeley and Robins (1981).

[5] Desired hours of work can also account for underemployment (persons working part time who wish to work full time).

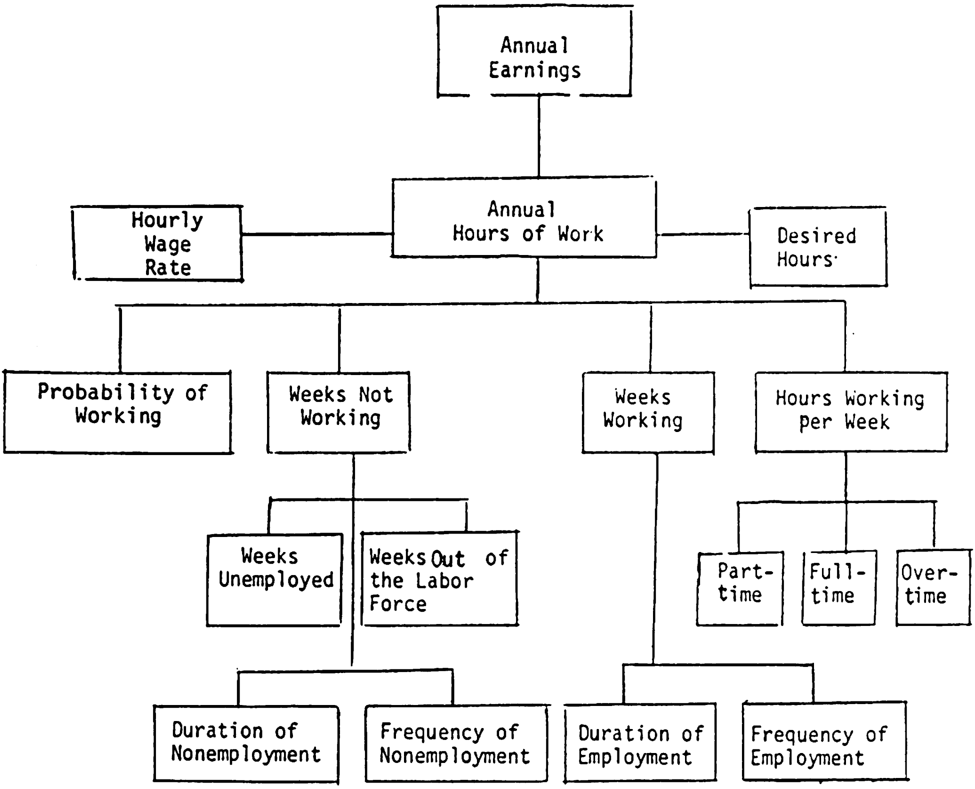
Annual Earnings
Hourly Wage Rate
Annual Hours of Work
Desired Hours
Probability of Working
Weeks Not Working
Weeks Working
Hours Working per Week
Weeks Unemployed
Weeks Out of the Labor Force
Part-time
Full-time
Over-time
Duration of Nonemployment
Frequency of Nonemployment
Duration of Employment
Frequency of Employment

FIGURE 3.1.—Various dimensions of labor supply that may be affected by an NIT.

switching from full-time to part-time work schedules? Does the allocation of market and nonmarket time in a given year with respect to weeks worked, weeks unemployed, and weeks out of the labor force change in response to an NIT?

The answers to these and other related questions have implications for the long-run economic improvement and self-sufficiency of the low-income population, as well as for the design of a national NIT program. If an NIT program significantly lengthens periods spent out of work, it may exacerbate the deterioration of job skills through disuse, thereby inhibiting future job attainment. On the other hand, longer spells out of work could have some benefits if individuals use the time to find more satisfying and higher-paying jobs or to participate in education and training programs. But if an NIT program induces people to leave jobs more readily, these individuals may accumulate less job-specific training and may be less likely to advance beyond low-paying, entry-level jobs. A finding that leaving employment is more prevalent under an NIT program would suggest that more attention be paid to the possible need for (and design of) work requirements for proposed transfer programs. On the other hand, if an NIT program only affects weekly hours worked, it may not affect job skills and advancement opportunities of individual workers.

We attempt to shed light on some of these questions by presenting the findings, based on several years of research, of the labor supply effects of SIME/DIME. Although our main objective is to describe how the families in SIME/DIME responded to the experimental treatments, we also devote considerable attention to generalizing the SIME/DIME findings to the national population, using the technique of microsimulation. Our strategy is to present both simple estimates that do not distinguish the effects of the various treatments and estimates that identify separately the effects of changing guarantee levels and tax rates. Many of the results presented are broken down by race, site, and experimental duration.

Our analysis suggests that, as a replacement for the current welfare system, a universal NIT program without any work requirements in which the mean guarantee level is about 110 percent of the poverty level and the mean tax rate is about 50 percent would lead to significant reductions in virtually every major dimension of labor supply.

Within the SIME/DIME sample, the pattern of the response among the various measures of labor supply is similar for each family group. This pattern is exemplified by the annual hours of work response, which in the second experimental year is −9 percent for husbands, −21 percent for wives, −14 percent for single female heads, −24 percent for male youth, and −18 percent for female youth. It is important to recognize that these estimated impacts in the SIME/DIME sample are averages over the 11 NIT treatments tested. These estimated impacts also combine the responses of 3- and 5-year families and are for a sample truncated on the basis of income. Hence, estimated impacts represent the average responses for persons eligible to receive NIT benefits under a very generous set of temporary programs, for a sample of low- and middle-income families whose incomes did not exceed 3.25 times the poverty level in the preexperimental year.

Although the pattern of response in the SIME/DIME sample among the various measures of labor supply is similar for each family group, there are notable exceptions. For example, male heads of families who cease working generally remain in the labor force whereas female heads do not. For all groups, a major impact of the experiment is on spells of nonemployment which increase by between 40 percent and 70 percent. Because the labor force participation rate of women in the control group increased significantly during the experiment, the implication can be drawn that the experiment delayed entry into the labor force for a sizable number of women eligible to receive NIT benefits. Males in general, rather than withdrawing from the labor force, appeared to have spent more time searching for employment in response to the experiment. All groups reduced full-time employment, but only men reduced overtime employment and only women reduced part-time employment. Youth have the largest percentage responses, although there is some evidence that income was significantly underreported by this group.

Because SIME/DIME tested such generous programs, a major effort was undertaken to generalize the experimental findings to specific programs that represented feasible policy options. This required development of a structural model of labor supply response that identified guarantee and tax effects. Based on the results from this model, estimates of behavioral responses and programmatic outcomes for four nationwide NIT programs with guarantee levels of 75 percent and 100 percent of the poverty level and tax rates of 50 percent and 70 percent were derived from a microsimulation model. One of the most striking findings from the microsimulation analysis is that single female heads are generally made worse off by the less generous NIT programs, highlighting the fact that a very generous welfare system already exists for them in the United States. This is further supported by the finding that from almost one-half to three-quarters of dual-headed families receiving welfare would be made worse off by such programs, even though making an NIT universally available to this group could cost up to an additional $6 billion. In general, because labor supply is significantly reduced under a universal NIT and because program costs depend to a large extent on labor supply reductions, the findings from SIME/DIME suggest that an NIT coupled with a rigorously supervised job search program and strong penalties for underreporting of income, could potentially satisfy three often contradictory goals of welfare reform: adequate minimum benefits, desirable economic incentives, and reasonable cost.

In the remainder of Part III we present details of our analyses. Chapter 2 presents a theoretical model of the labor supply response to SIME/DIME. Potential biases resulting from the limited duration of the experiments are discussed in detail. Chapter 3 discusses types of empirical models that can be estimated using SIME/DIME data, along with the more important methodological problems associated with such models. Chapter 4 describes the SIME/DIME data base with particular attention paid to the employment data. Chapter 5 presents the findings for heads of families, chapter 6 the experimental findings for youths, and chapter 7 the findings for heads of families generalized to specific nationwide NIT programs.

CHAPTER II

A THEORETICAL MODEL OF LABOR SUPPLY RESPONSE[1]

In this chapter, we present a theoretical model of the labor supply response to an NIT program. This model forms the basis of much of the empirical analyses in subsequent sections. We assume a well-behaved utility function in a world of perfect certainty for a family in which there is only one potential earner.[2] To highlight the implications of this model, we first assume that an NIT is implemented permanently. We then relax this assumption and discuss how the implications are altered by a limited duration experiment.

A one-period model[3]

Theory

In a world in which an NIT is implemented permanently, utility is assumed to depend on lifetime consumption of leisure, L, and lifetime consumption of market goods, C:

$$U = U\ (L,\ C), \tag{1}$$

where L = leisure consumption
C = market consumption.

In the absence of taxes and transfers, the budget constraint, which measures the total amount of money available for consumption, is given by

$$C = WH + N, \tag{2}$$

where H = hours of work
W = the market wage rate
N = nonwage income.

Since total time available, T, is the sum of hours of work and leisure ($T = H + L$), we can rewrite (2) as follows

$$F = WT + N = WL + C, \tag{3}$$

where F is full income.

Maximization of the utility function (1) subject to the budget constraint (3) yields a leisure demand function, $L = L(W,\ F)$, that depends on the wage rate and full income. Totally differentiating the leisure demand function gives

$$dL = \frac{\partial L}{\partial W}dW + \frac{\partial L}{\partial F}\ \frac{\partial F}{\partial W}dW + \frac{\partial L_1}{\partial F}\ \frac{\partial F}{\partial N}dN. \tag{4}$$

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[1] Readers not interested in the technical theoretical details that motivated much of the empirical analysis may either skim or skip this chapter. The theoretical models will be briefly reviewed (in a nontechnical fashion) in the empirical chapters that follow.

[2] The implications for two-earner families are discussed below.

[3] This section is based on Keeley et al. (1978a) and Robins, West, and Keeley (1980).

Using the fact that $\partial F/\partial W = T$, $\partial F/\partial N = 1$, and $H = T - L$, and substituting in the Slutsky equation,[4] gives

$$dL = \frac{\partial L}{\partial W}\bigg|_u dW + \frac{\partial L}{\partial F}(Hdw + dN), \tag{5}$$
$$= \alpha dW + \beta dY,$$

where

α = the substitution effect
dW = the total differential of the wage rate
β = the income effect
dY = $HdW + dN$ (the total differential of disposable income, holding constant hours of work, H).

Analyzing the response to an NIT program

Equation (5) states that the effects of shifts in the budget constraint on leisure demand can be decomposed into a substitution effect, α, which depends on the change in the net wage rate, dW, and an income effect, β, which depends on the change in disposable income, dY, at fixed hours of work, H. For a person placed on an experimental NIT program, these differential changes can be approximated by discrete changes resulting from the NIT program. The change in the net wage rate, ΔW, is equal to the gross wage rate times the difference between the preexperimental and experimental tax rates, and the change in disposable income, ΔY, is equal to the payment the person would receive if he or she were to maintain labor supply at the preexperimental level.[5] Referring to Figure 3.2, consider a person who is in equilibrium at point Z before the imposition of an NIT program. The change in the quantity of leisure demanded comprises a substitution effect, $L_S - L_Z$, holding disposable income constant at preprogram labor supply, and an income effect, $L_{[illegible]} - L_S$, holding constant relative prices (i.e., the wage rate relative to the price of goods).[6] Disposable income is held constant by rotating the budget line through the initial equilibrium point Z, where, at the new net wage rate and the new monetary full income, the consumer could still purchase the initial consumption bundle.

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[4] The Slutsky equation decomposes the total effect of a wage change on labor supply into a substitution effect and an income effect:

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial W}\bigg|_u + H\frac{\partial L}{\partial F}$$

[5] These specifications of ΔW and ΔY assume that the gross wage rate is unaffected by the NIT program, that no other changes occur in the parameters of the budget constraint, and that labor supply is in equilibrium prior to the program. In our empirical formulation of this model (see chapter 5), we discuss the implications of these assumptions, various ways of relaxing them to conform to the real world, and test the sensitivity of the estimates to alternative assumptions.

[6] This is the Slutsky, as opposed to the Hicks, decomposition of substitution and income effects. Hence, the substitution effect is not a utility compensated substitution effect but rather an income compensated substitution effect. For a derivation of the utility compensated substitution effect for this model, see Robins, West and Lohrer (1980). Also see the discussion regarding generalization of SIME/DIME structural estimates in chapter 8 of part III.

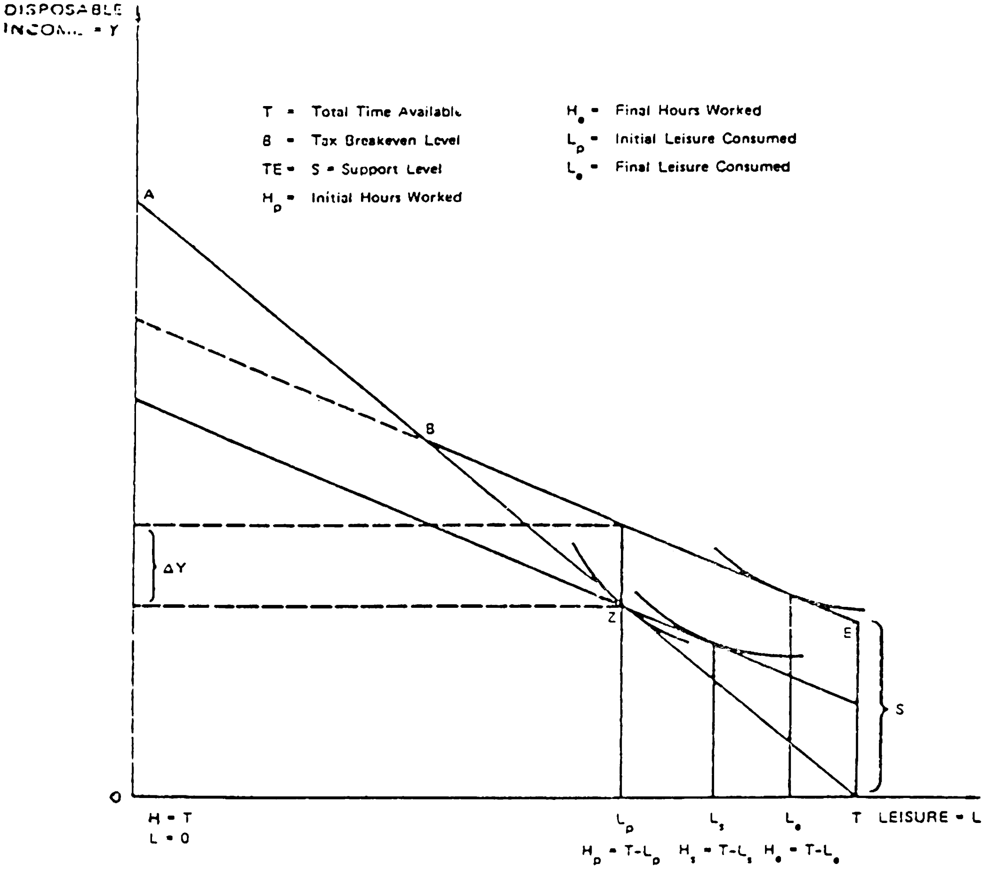

FIGURE 3.2.—Labor supply effects of an NIT program.

Extensions to a Two-Earner Family

When there is more than one earner in a family, the above theory has to be modified to account for cross substitution effects. In the two-earner (husband-wife) case, assuming pooling of resources, the utility function may be rewritten as

$$U = U(L_H, L_W, C), \qquad (6)$$

where L_H = leisure consumption of the husband
L_W = leisure consumption of the wife,

and the budget constraint as

$$F = W_H T_H + W_W T_W + N = W_H L_H + W_W L_W + C, \qquad (7)$$

where W_H = the market wage rate of the husband
W_W = the market wage rate of the wife
T_H = total time of the husband
T_W = total time of the wife.

Maximization of (6) subject to (7) yields a pair of leisure demand equations, $L_i = L(W_H, W_W, F)$ where $i = H, W$, each depending on the wage rate of both spouses and full income. Taking the total differential of the leisure demand equations and substituting in the appropriate Slutsky equations yields

$$dL_i = \left(\frac{\partial L_i}{\partial W_i}\right)_U dW_i + \left(\frac{\partial L_i}{\partial W_j}\right)_U dW_j + \frac{\partial L_i}{\partial F}(H_H dW_H + H_W dW_W + dN), \qquad (8)$$

$i, j = H, W,\ i \neq j$

The difference between equations (8) and (5) is the addition of a cross substitution term representing the effect of a change in one spouse's wage rate on the demand for leisure of the other spouse. In the one-earner case, the effects of an NIT on leisure are unambiguously predicted to be positive. In the two-earner case, as Killingsworth (1976) and others have noted, it is possible for the leisure of one spouse to decrease in response to an NIT if the cross substitution effect is large enough to offset both the own substitution effect and the income effect. The effect of an NIT on both spouses' leisure cannot be negative and total earnings must decrease, however, because of Hicks' composite commodity theorem. Empirically, it is unlikely that the cross substitution effect will be large enough to cause an overall decrease in leisure for either one of the spouses.

A two-period model

The theoretical model just described assumes that the budget constraint is altered permanently by an NIT. In the case of a nationwide NIT this is probably a reasonable assumption.[7] For the NIT experiments, however, it is not. The experiments were of limited duration and families were informed of this fact at enrollment.[8] To illustrate the effects of limited duration on labor supply response, we present a simple extension of the previous one-period model to two periods.[9]

It is assumed that an individual's time horizon can be divided into two period of unequal length. The first period covers the time of the experiment and the second period covers the remainder of the individual's life. Utility depends on the rates of consumption of leisure and market goods in each period:

$$U = U(L_1, L_2, C_1, C_2). \qquad (9)$$

The budget constraint over both periods is given by

$$R_1C_1 + R_2C_2 = R_1W_1H_1 + R_2H_2W_2 + R_1N_1 + R_2N_2, \qquad (10)$$

where

$$R_1 = \int_0^{T_1} e^{-rt}dt$$

$$R_2 = \int_{T_1}^{T_1+T_2} e^{-rt}dt$$

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[7] It can be argued, however, that some individuals would not believe a nationwide program to be permanent and would behave as if the program had a fixed duration. Furthermore, persons who expect to be above the breakeven level in the future may behave as if the program were temporary. On the other hand, even if a program was implemented with a fixed duration, some individuals might behave as if it were permanent under the assumption that, once enacted, such a program would be difficult to terminate.

[8] A small number of families in Denver were enrolled for 20 years in an attempt to help determine potential biases caused by the limited duration. The 20-year program is discussed below.

[9] This discussion is based on Keeley and Wai (1980). For an alternative, but similar, model see Metcalf (1973, 1974).

T_1 = the duration of the experiment
T_2 = the duration of an individual's life after the experiment
r = the interest rate.

Note that T_1 is either 3 years or 5 years in SIME/DIME and T_2 varies according to an individual's age and life expectancy. If we define $R=R_2/R_1$, the budget constraint (10) can be rewritten as

$$C_1+RC_2=W_1H_1+RW_2H_2+N_1+RN_2. \tag{11}$$

Given the fact that $T=H_i+L_i$ and $i=1, 2$, we can rewrite (11) as

$$F=W_1T+N_1+R(W_2T+N_2)=W_1L_1+C_1+R(W_2L_2+C_2). \tag{12}$$

Maximizing equation (9) subject to equation (12) yields a pair of leisure demand equations for periods one and two:

$$L_1=L_1(W_1, RW_2, F), \tag{13a}$$

$$L_2=L_2(W_1, RW_2, F). \tag{13b}$$

Totally differentiating (13a), given that $\partial F/\partial W_1=\partial F/\partial RW_2=T, \partial L_1/\partial RW_2=(1/R)\,\partial L_1/\partial W_2$, $dRW_2=RdW_2$, $\partial F/\partial N_1=1$, $\partial F/\partial N_2=R$, and substituting in the appropriate Slutsky equations, gives

$$\begin{aligned} dL_1 &= \frac{\partial L_1}{\partial W_1}\bigg|_u dW_1 + \frac{\partial L_1}{\partial W_2}\bigg|_u dW_2 + \frac{\partial L_1}{\partial F}(H_1dW_1+dN_1+R\,(H_2dW_2+dN_2)) \\ &= \frac{\partial L_1}{\partial W_1}\bigg|_u dW_1 + \frac{\partial L_1}{\partial W_2}\bigg|_u dW_2 + \frac{\partial L_1}{\partial F}(dY_1 + RdY_2). \end{aligned} \tag{14}$$

Equation (14) can be used to illustrate the differences between the effects of a temporary and a permanent NIT program. Under a temporary program, the net wage rate and nonwage income are only changed in period one.[10] The change in leisure demand during the experimental period is thus given by

$$dL_1 = \frac{\partial L_1}{\partial W_1}\bigg|_u dW_1 + \frac{\partial L_1}{\partial F}dY_1. \tag{15}$$

Under a permanent NIT, the net wage rate and nonwage income are changed in periods one and two. The change in leisure demand during the experimental period is thus given by equation (14). Assuming $dW_1=dW_2=dW$ and $dY_1=dY_2=dY$, the difference in response (during the experimental period) between a permanent and temporary program is given by

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[10] The change we refer to is the change caused by the experiment; nonexperimental changes in the budget constraint are ignored.

$$B = \frac{\partial L_1}{\partial W_2}\bigg|_u dW + \frac{\partial L_1}{\partial F} R\,dY, \tag{16}$$

which represents the bias of a temporary program. Hence the bias in the substitution effect is

$$\frac{\partial L_1}{\partial W_2}\bigg|_u dW$$

(the intertemporal substitution effect) and the bias in the income effect is

$$\frac{\partial L_1}{\partial F} R\,dY$$

(the discounted present value of the income effect in the postexperiment period).[11] If leisure between periods are substitutes, then

$$\frac{\partial L_1}{\partial W_2}\bigg|_u > 0.$$

Since

$$\frac{\partial L_1}{\partial W_1}\bigg|_u < 0.$$

and $dW > 0$, the bias in the substitution effect is positive (that is, the substitution effect will be overestimated in absolute value, in a temporary experiment).[12] Since $\partial L_1/\partial F > 0$, $dY > 0$, and $R > 0$, the income effect will be underestimated (in absolute value) by an amount that depends on the length of the temporary program, the size of the income effect, the generosity of the NIT program, the individual's age and life expectancy, and the interest rate. Overall, the total bias of a temporary program may be positive or negative, depending on which term in equation (16) dominates.

This theory suggests several ways in which potential biases due to the limited duration of the experiment can be identified. In the case of the income effect, separate estimates for 3- and 5-year families can be used to derive an estimate of the interest rate, r, faced by individuals, which in turn can be used to estimate the permanent income effect. Alternatively, differences in response by age can also be used to derive an estimate of r. In the case of the substitution effect, postexperimental data can be used to obtain an estimate of the intertemporal substitution effect.

Another approach is to use data on the length of program participation (movement above and below the breakeven level) for both 3- and 5-year families. Under this approach, natural variation in length of program participation may be used to derive an estimate of the effect of a permanent program.

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[11] Another way of looking at the bias in the income effect is to take the ratio of the income effect under a permanent program to the income effect under a temporary program, or

$$\frac{\partial L_1/\partial F\,(1+R)\,dy}{\partial L_1/\partial F\,dy} = 1+R.$$

[12] If leisures in different periods are complements, the substitution effect is underestimated (in absolute value) in a temporary experiment.

Although the above approaches may yield important information about potential biases due to the limited duration of the experiments, they rely on somewhat restrictive assumptions about behavioral response and impose heavy burdens on the data. An alternative to deriving permanent program labor supply effects from limited duration experimental data is to estimate the response of individuals to a third program of sufficient duration to approximate a permanent program. A longer program duration means smaller differences in the net present value of available program income and the relative price of leisure; therefore, estimates of the permanent program effect will converge on the "true" effect with increasing program duration. Since a small number of families in SIME-DIME participate in a 20-year program, the difference between the estimates and true permanent program labor supply response for these families is likely to be small. Hence, if the labor supply response of individuals in this subsample can be suitably estimated, a measure of the limited duration bias in the 3- or 5-year estimates is available and an estimate of the true response to a permanent NIT program can be derived. As we shall see, the data are not rich enough to provide definitive evidence about the nature of the biases of a temporary program using any of the above approaches, although many of the results are suggestive.

Extensions

The theory developed in this section is based on a comparative static model of family labor supply. The predicted effects of an NIT are stated in terms of hours of work (or earnings) measured over a period of time (for example, a year). However, the theory also has implications for the components of annual hours of work, the probability of working, and weeks worked. Specifically, an NIT should decrease both the probability of working and weeks worked (and hence, annual hours of work). In an elaboration of the standard theory of labor supply, Hanoch (1976) argues that if the utility of workweek leisure differs from that of nonwork leisure, individuals would not be indifferent to working fewer hours per week while employed and having longer periods between episodes of employment, even if total hours worked per year is the same. From a policy perspective, separating these two possible effects is important because, as we mentioned earlier, a number of existing and proposed transfer programs have eligibility requirements that are based, at least in part, on employment status. Although our main focus here is on annual hours of work, we shall also devote considerable attention to several other measures of labor supply.

CHAPTER III

GENERAL EVALUATION ISSUES

The theoretical model just described suggests that labor supply response to an NIT can be evaluated within a framework that identifies income and substitution effects. This structural model approach usually imposes significant constraints on the form of the response. An alternative approach that makes use of dummy variable models entails less constrained specifications. In this chapter, we briefly describe both types of models, noting their advantages and disadvantages. We also discuss several methodological issues associated with estimating such models—including accounting for nonexperimental effects, dealing with family structure changes, choosing the appropriate time period of analysis, and correcting for potential biases caused by sample attrition, income underreporting, and income truncation. Finally, we discuss issues related to generalization of the experimental findings to broader population groups.

Estimating experimental effects

Dummy variable models

The simplest type of response model that can be used to estimate experimental effects is the dummy variable model. We consider two types of dummy variable models in this chapter. The first type represents the entire set of experimental treatments by a single variable that takes on the value of one for persons assigned to an NIT treatment and zero for persons in the control group. We refer to this model as the "single dummy variable model." The coefficient of the dummy variable in this model measures the average effect of all the experimental treatments on labor supply (or the average effect of eligibility for the "typical" treatment). On the basis of the theoretical discussion in the previous chapter, which indicates that higher nonlabor income and higher tax rates lead to a reduction in labor supply, we expect this coefficient to be negative. The single dummy variable model has considerable appeal because it is easy to interpret and provides a useful summary of the overall impact of the experiment.

The second type of dummy variable model we examine exploits the richness of the experimental design. In this version of the model, the experimental treatments are represented by 11 dummy variables, one for each NIT treatment. We refer to this model as the "11 dummy variable model." The coefficients of the 11 dummy variables measure the effects of each of the various combinations of guarantee levels and tax rates. Differences in the coefficients measure effects of changes in guarantee levels and tax rates, which vary according to the set of treatments being compared. The 11 dummy variable model is a useful specification because it provides, in a relatively unconstrained fashion, a general indication of the pattern of response to the various guarantee levels and tax rates tested in the experiments. However, as we indicate below, it is of limited value for generalizing the results of the experiment to other population groups and other programs.

The two versions of the dummy variable model may be written as follows:

Single dummy variable model

$$H_e - H_p = a_0 + a_1E + e, \qquad (1)$$

Eleven dummy variable model

$$H_e - H_p = b_0 + b_1E_1 + \ldots + b_{11}E_{11} + e, \qquad (2)$$

where H_e = labor supply during the experimental period
H_p = labor supply in the preexperimental period
E = a dummy variable for treatment (1 for experimentals, 0 for controls)
$E_1 \ldots E_{11}$ = dummy variables for each treatment (1 for a particular treatment, 0 for controls and for all other treatments)
a_1 = overall effect of the experiment
$b_1 \ldots b_{11}$ = effects of each treatment
e = a random error term.

The dependent variable in both models is in first-difference form, to correct for any permanent differences in labor supply that may exist between experimentals and controls (see Robins and West, 1980b for further discussion). As we indicate below, several additional variables are added to the right-hand side of these equations to control for a variety of factors that may influence labor supply other than the experiment.

Structural models

Another type of response model utilizes economic theory to suggest an empirical formulation of the response function. We call such models structural models. Like the 11 dummy variable model, structural models attempt to isolate the effects of changing guarantee levels and tax rates. However, unlike the 11 dummy variable model, structural models impose constraints on the pattern of response to the NIT parameters. Because of these constraints, structural models can be generalized to NIT programs other than the ones tested in the experiments. Such generalization cannot be performed with the 11 dummy models because responses to NIT programs other than those tested in the experiments cannot be derived from any of the estimated parameters.

Many types of structural models can be estimated with SIME/DIME data. We focus on the structural model developed in Chapter 2. This model replaces the treatment dummy variables in equations (1) and (2) with the change in the net wage rate, ΔW, and the change in disposable income (measured at preexperimental labor supply), ΔY, resulting from the NIT:

$$H_e - H_p = C_0 + \alpha\Delta W + \beta\Delta Y + e. \qquad (3)$$

Hence, in our model, the total experimental response is decomposed into a substitution effect, α, and an income effect, β. Since ΔW and ΔY depend on the NIT guarantee level and tax rate, the estimated income and substitution effects can be used to derive estimates of the labor supply response to specific NIT programs, including programs different from those tested in SIME/DIME.[1]

Although structural models identify guarantee and tax effects, they often rest on a variety of assumptions that may not hold in the real world.[2] Dummy variable models, on the other hand, generally do not impose any constraints suggested by theory and provide reliable estimates of mean labor supply response in the experimental sample. We present estimates from both dummy variable models and structural models in an attempt to provide a more complete picture of the labor supply response to an NIT, although it is only appropriate to use the structural model to generalize the SIME/DIME findings to alternative population groups and programs.

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[1] The structural model given by equation (3) has a problem in that ΔY, because it depends on preexperimental labor supply, H_p, is likely to be correlated with the error term, e. This correlation would lead to biased estimates of α and β. In our empirical implementation of the structural model we describe how this problem is dealt with econometrically.

[2] Some of these assumptions are discussed in Chapter 5 when we examine an alternative version of the structural model.

Accounting for nonexperimental effects

The role of control families

Approximately 40 percent of the families in the SIME/DIME served as controls and were not eligible for NIT payments.[3] These families play an important role in the analysis. Controls are essential for the estimation of dummy variable models since, in these models, the estimated response is the difference in labor supply, adjusted for preexperimental differences, between experimentals and controls. It is possible to estimate dummy variable models without controls by comparing the labor supply of experimentals during the experiment with their labor supply before it. However, such a procedure is based on the assumption that the experiment is the only systematic cause of the change in the labor supply of experimentals between the two periods—an assumption that does not hold for several reasons. First, the business cycle generates changes in labor supply over time that are unrelated to the experiment. Second, there are important secular trends in labor supply, particularly for women, who have been steadily increasing their labor force participation. Third, the exclusion of higher income families from the experiment means that the incomes of experimental families tended to increase over time because some experimental families had incomes that were only temporarily low when they were chosen for inclusion in the experiment.[4] The presence of control families allows us to account for these effects when estimating the average effect of the experiment using dummy variable models.

Control families are not quite as essential when estimating structural models, since these models (properly specified) can be estimated by relying on the differences among the experimental treatments within the group of experimental families and on the constraints imposed on the pattern response. However, the inclusion of control families in the sample increases the efficiency of the estimates at relatively little cost since controls do not receive experimental NIT payments.

The assignment model

As discussed in Part I of this volume, families were assigned to experimental plans and control status on a stratified random basis. The strata used were racial-ethnic group (black, white, Chicano), family type (husband-wife, single-headed), site (Seattle, Denver), and an estimate of the family's normal income category. The probabilities of assignment to control status or the various experimental plans varied among the strata defined by these characteristics. Because of this assignment procedure, families assigned to the various experimental plans are not directly comparable one to another, nor to controls, because the likelihood of being in a particular cell varied by stratum. In particular, high-income families were more likely to be assigned to more generous programs and to control status, and low-income families to less generous programs.

In order to account for these differences, dummy variables for each of the assignment categories are included in all of the models estimated in this chapter (see Keeley and Robins, 1980, for a discussion of the rationale for including the assignment variables in the models). These variables are specified in an additive, rather than interactive, fashion for simplicity.[5] Thus, we include dummy variables for racial-ethnic group, site, and normal income level. Since separate models are estimated for husbands, wives, and single female heads of families, however, family structure type is interacted with all the other assignment variables.

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[3] About one-half of these control families were eligible for the experimental counseling and/or training treatments. The remainder (20 percent of the total sample) were pure controls and eligible for neither NIT payments nor counseling/training.

[4] See the section on income truncation later in this chapter. This phenomenon is often referred to as "regression toward the mean."

Additional control variables

Apart from the assignment variables it is not, in general, necessary to include additional control variables in the estimated equations. If the experimental treatments are represented by dummy variables, the assignment process guarantees that, ex ante, other variables are uncorrelated with the experimental treatments after accounting for (i.e., partialing out) the assignment variables. Thus, it is not necessary to include additional variables to insure that the estimated treatment effects are unbiased. However, the inclusion of additional variables that may be correlated with the dependent variable can reduce the standard errors of the estimators of the treatment effects by reducing the variance of the error term. Also, it is possible that additional variables may correct for accidental (ex post the assignment process) differences between experimentals and controls. Thus, most models estimated in this chapter include additional independent variables. The most important additional independent variable is the preexperimental value of the dependent variable (H_p), which is likely to both reduce variance and account for any unintended permanent differences between experimentals and controls.[6]

Family structure changes

Many families underwent family structure changes during the experimental period. Approximately 20 percent of the husband-wife families in the sample experienced a divorce or dissolution and 22 percent of the single female heads married in the first 3 years of the experiment.[7] Since we are interested in presenting estimates of the total effects of the experiment on labor supply, and not merely the effects conditional on unchanged marital status, we define the groups for analysis (husbands, wives, and single female heads) on the basis of family structure at the time of enrollment into the experiment. Estimates presented thus represent the sum of both the direct effects of the NIT programs on labor supply and the indirect effects that operate through marital status changes. Keeley (1980) presents estimates of the NIT effects conditional on marital status during the experiment and discusses their relationship to the type of estimates presented here, which are conditional on initial marital status.

Time frame of analysis

Data collected in the experiment (for each family) cover the following periods: approximately 1 year prior to the beginning of the experiment, 3 or 5 years during the experiment (depending on the family's assigned experimental duration), and either 2 years (for about half of the 3-year families) or 1 year after the end of the experiment. The response to the experiment is unlikely to be constant throughout the experimental period. For instance, families may adjust only gradually to the changes in their budget constraints induced by the experiments.[8]

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[5] Ignoring interactive effects may be particularly troublesome for dummy variable models because the results may not be easily interpretable. For structural models, interactive effects may be less important if the model is properly specified. See Keeley and Robins (1980) for a discussion of the relative importance of interaction effects in dummy variable models and structural models.

[6] Other variables often included in the models are age, number of family members, number of family members aged 0–4 years, AFDC payments in the preexperimental year and a dummy variable for control families filing income report forms. It should be noted that when H_p is included on the right-hand side of the equation, having H_e as the dependent variable will give the same results as having $H_e - H_p$ as the dependent variable.

[7] Single male heads made up a very small proportion of the experimental population and are excluded from all analyses reported here.

[8] See Robins and West (1980b) for a discussion of the reasons for this gradual adjustment and estimates of the length of time required to adjust to the experiment.

For similar reasons, a person may start to adjust to the known end of the experiment before the experiment actually ends. A third reason for delayed adjustment to the experiment is that it may take some time before the payment guarantees of the experiment are credible enough to warrant adjustment.

For the single dummy variable models estimated, we present estimates for each of the years during and after the experiment, to allow us to examine in an unrestricted fashion the time pattern of response to the experiment, including any potential postexperimental response. For the 11 dummy variable models and the structural model, we concentrate on the second year of the experiment, which allows us to present the results from these models without being overwhelmed by the number of time periods. The second year is chosen because it is most likely to allow sufficient time for much of the adjustment to the experiment to have taken place, while being far enough from the end of third year of the experiment to precede early adjustments by 3-year families to the end of NIT eligibility.

Potential sources of bias in the estimated experimental effects

Income truncation

As discussed in part I, only families with preexperimental incomes below a certain level were included in the experimental sample. Hence, preexperimental incomes in the sample are truncated and the sample is not representative of the population as a whole, even in the two sites where the experiment was conducted.

This income truncation has several implications. First, the sample means of preexperimental earnings and related variables, such as preexperimental hours of work, are likely to differ from the population means. Some variables, like earnings and hours, are likely to have means below the population mean; others, like AFDC benefits, are likely to have means above the population means. To the extent these variables were temporarily low (or high) in the preexperimental period, they will tend to increase (decrease) over time because of the statistically inevitable regression toward the mean.[9]

Income truncation has several implications for the estimation of experimental effects in models that do not allow response to vary with income (such as equations 1 and 2). First, the results from such models are likely to apply only to

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[9] To see this, consider the following model. Suppose that the logarithms of income in periods one and two (y_1 and y_2, respectively) are generated by

$$y_1 = y^p + y^t_1,$$

$$y_2 = y^p + y^t_2,$$

where y^p is the logarithm of permanent income, and y^t_i is the logarithm of transitory income in period i. It is assumed that y^p and y^t are independently and normally distributed random variables. If the sample is chosen so that $y_1 < y^*$, where y^* is a constant representing the logarithm of the income truncation point, then the expected values of y_1 and y_2 in the sample are given by

$$E(y_1 \big| y_1 < y^*) = \bar{y}^p - \frac{\sigma^2_p + \sigma^2_p}{(\sigma^2_p + \sigma^2_t)^{1/2}} \frac{f(y^* - \bar{y}^p)}{F(y^* - \bar{y}^p)},$$

$$E(y_2 \big| y_1 < y^*) = \bar{y}^p - \frac{\sigma^2_p}{(\sigma^2_p + \sigma^2_t)^{1/2}} \frac{f(y^* - \bar{y}^p)}{F(y^* - \bar{y}^p)},$$

where $\bar{y}^p$ is the mean of permanent income, σ^2_p and σ^2_t are the variances of y^p and y^t, respectively, $f(\cdot)$ is the standard normal density function, and $F(\cdot)$ is the standard normal distribution function. It is clear that

$$E(y_1 | y_1 < y^*) < E(y_2 | y_1 < y^*) < y^p.$$

Thus, average income in the truncated sample, which is less than average permanent income, tends to rise over time.

similarly truncated samples. Only if the response to the experiment does not vary with preexperimental income will a dummy variable response be applicable to general populations. Since it is likely that high-income persons (e.g., those above the breakeven level) will have a smaller response to NIT programs than low-income persons, estimates of these types of models are best interpreted as average responses for the sample used in the experiments. In other words, response to an NIT varies with the distribution of income in the sample studied. Because the distribution is different in the experimental sample than in the U.S. population, responses will also be different.

Truncation may not pose problems for response estimated in a structural model or in dummy variable models that allow response to vary with income.[10] If these models are properly specified, the results can be extrapolated outside the sample range. However, given the possibility of misspecification, it is not recommended that the results from these models be extrapolated very far beyond the income range found in the experimental sample.

Sample attrition

Sample attrition is a potentially serious problem for analyses of experimental data. Sample attrition occurs when, for one reason or another, an observational unit (usually a family) either cannot be found by an interviewer or refuses to be reinterviewed. The inability to reinterview families that constitute part of an originally selected sample means that such families must be excluded from analyses that pertain to periods subsequent to their last interview.

The problem of sample attrition was recognized by the designers of the SIME/DIME experiment and efforts were made to minimize its occurrence, as noted in Part I. For example, families were required to submit monthly address reports, were paid a fee to complete interviews, and were followed if they changed residences (see Harper, 1978, for more details). Despite such efforts, attrition averaged approximately 7 percent per year during the first 3 years of the experiment—about the same rate of attrition that occurred during the three years of the New Jersey Income Maintenance Experiment (Kershaw and Fair, 1976).

If, conditional on observed variables, sample attrition is a purely random process, analyses of the sample of nonattriters will not in general pose any special estimation problems, although statistical efficency is reduced because of the smaller sample size. If the attrition process is not random, however, estimated parameters of a behavioral model based on the sample of nonattriters will, in general, be biased. The bias arises because the attriters are systematically different from the nonattriters in a way that is not captured by observable variables. As Heckman (1979, p. 154) puts it, "fitted regression function [on the sample of nonattriters] confound the behavioral parameters of interest with parameters of the function determining the probability of entrance into the sample."

For experimental panel data, such as those collected in SIME/DIME, the problem of sample attrition can be especially serious. In addition to the sample of nonattriters being unrepresentative of the original sample, the attrition process is likely to be different for experimentals and controls, generating biased estimates of the experimental effects. For example, suppose that in the absence of an experimental treatment attrition is random. This means that for control families sample attrition does not pose any special estimation problems. However, for experimental families the NIT payment is an added incentive to remain in the

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[10] The following elaboration of equation (2) is an example of a dummy variable model that allows response to vary with income:

$$H_e - H_p = b_0 + b_1F_1 + \ldots b_{11}F_{11} + b_{12}F_1Y + \ldots b_{22}F_{11}Y,$$

where Y is income. Response in this model for, say, F_1, is equal to $b_1 + b_{12}Y$, which varies with income. Unfortunately, the large number of parameters in this model make it very difficult to estimate with any degree of precision.

sample. In addition, experimental families with higher payments are more likely to remain in the sample than experimental families with lower payments. Because the size of the NIT payment varies inversely with labor supply, attrition can be expected to lead to a general decrease in labor supply of the sample of experimental nonattriters over time. Since, in this example, attrition in the control sample is assumed to be random, the experimental-control differential in average hours of work will become increasingly more negative over time, even in the absence of a true experimental effect on hours of work. Hence, sample attrition can potentially lead to a significant overestimate (in absolute value) of the labor supply response to the experiment.

In a study of the effects of sample attrition on labor supply response, Robins and West (1980d) use data from the Social Security Administration (on both attriters and nonattriters) to determine whether sample attrition generates serious biases in the estimates of labor supply response in SIME/DIME. Using a single dummy variable model in which labor supply is measured by earnings and employment status,[11] they find that estimates corrected for attrition are uniformly larger than estimates not corrected for attrition, which is contrary to theoretical expectations. The biases are slightly larger for earnings than for employment status, but never exceed 2 percent. For example, the estimated earnings effect for husbands is 8 percent uncorrected for attrition and 10 percent corrected for attrition. Although tests of the statistical significance of the biases are not performed, biases are unlikely to be significant at conventional levels.

The estimates presented in this chapter do not account for attrition, primarily because we believe it is not a serious enough problem in SIME/DIME data to warrant elaborate (and expensive) statistical correction procedures. As long as the results of Robins and West (1980d) can be applied to models other than the ones specifically estimated in their study, the results presented here should not be biased in any serious way because of attrition.

Income underreporting

Income underreporting is also a potentially serious problem in the analysis of the effects of the experiment on labor supply. Since the NIT payments experimental families receive are determined on the basis of reported income, there is an incentive to underreport income. While this incentive applies only to income reported on the income report forms (IRFs), it is also likely that there will be underreporting on interviews to the extent that the experimental families want to preclude the possibility that a comparison of the two sources of income will detect a discrepancy. There may also be underreporting of hours of work as well as earnings if subjects also want to minimize inconsistencies between these two types of data.

Control families have no similar incentive to underreport their income. Thus, reported incomes and hours of work may be lower for experimental families than for control families even if actual incomes and hours are the same. Consequently estimates of experimental effects would have a negative bias that again exaggerates the true labor supply response. These biased estimates, however, are appropriate for predicting the costs of nationwide NIT programs if it is assumed that similar underreporting would exist under a nationwide program.

Underreporting was also recognized to be a potentially important problem and care was taken during the administration of SIME/DIME to minimize its extent (see the discussion in part I). For example, experimental families were required to submit copies of paycheck stubs with their income report forms. While this procedure probably eliminated much underreporting, it does not preclude the possibility that a working person might not report having a job at all or might fail to report a second job or income from casual employment.

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[11] Hours of work are not available from Social Security Administration records.

Halsey (1980) conducted a study of the accuracy of SIME/DIME data by comparing interview data to data in public records; he concluded that "the bias in wage and salary data is too small to worry about in almost any analyses" (p. 146). He did, however find large apparently random differences between earnings reported in interviews and earnings recorded on public records. Robins and West (1980d), in contrast, find that estimates of the labor supply response based on interview data imply a larger response than estimates based on social security data. These differences could have occurred because of underreporting or because much of the response was concentrated in jobs not covered by social security. Similar findings by Greenberg and Halsey (1980) using data from another source (unemployment insurance records) further strengthens the possibility of significant underreporting by experimental families (particularly wives and youth). However coverage is extremely low in the Greenberg and Halsey sample. Approximately 50 percent of the husbands, 75 percent of the wives, 65 percent of the single female heads, and 55 percent of the youth are either missing or do not have unemployment insurance records. As Greenberg and Halsey's results indicate, inferences about underreporting biases are sensitive to assumptions made about underreporting for cases with missing data.

Both Greenberg and Halsey and Robins and West conducted their analyses on earnings and employment status (hours of work were not available). In general, when Social Security Administration data are used, the biases tend to be larger for employment status; but when unemployment insurance records are used, the biases tend to be larger for earnings (little bias exists for wives in employment status). For a variety of reasons, both these data sources have limitations that prevent them from providing firm evidence on the extent of an underreporting bias in SIME/DIME data. First, there is the possibility that neither of these data sources contains "true" earnings and employment status for the periods covered in the SIME/DIME data. For example, biases may arise from timing differences in the reporting of earnings information to the various sources. In the Greenberg and Halsey analysis of unemployment insurance records, earnings in a particular quarter during the experiment are examined and there are several cases of overreporting, which is somewhat perplexing. Data aggregated over a longer time period (say, a year or two) are available and should be analyzed to shed possible light on whether timing differences are responsible for a portion of the estimated underreporting bias. Second, the important omission of income from uncovered employment may be a potentially serious problem for both data sources, although Greenberg and Halsey attempt to deal with this problem analytically. Finally, the evidence presented by Greenberg and Halsey and Robins and West is also consistent with an overreporting bias on the part of controls. As pointed out by Burtless (1981), this explanation may be even more plausible than an underreporting bias by experimentals. Controls may have an incentive to overreport earnings because gainful employment is a socially approved activity. Experimentals have no such incentive because it would result in a reduction in their NIT payments. While an overreporting bias still implies that the estimated labor supply reductions of experimentals are overstated, they do not have the same cost implications in the sense that experimental families are fraudulently obtaining NIT payments.

Despite the limitations of the existing studies, it is probably fair to say that the evidence suggests a possible underreporting bias in the SIME/DIME data. Although the precise size of the bias cannot be stated with any degree of accuracy, it may be significant for all groups. Because the estimates presented here do not adjust for underreporting, they should be interpreted as reflecting potential reporting errors as well as true reductions in labor supply.

Interpretation and generalizability of the results

Macroeconomic and labor market issues

The response to the experiment may depend on both the stage of the U.S. business cycle and on the specific labor market conditions in Seattle and Denver. Of these two, the effect of the local labor market is probably the more important. As noted, labor market conditions differed in the two sites. During enrollment into the experiment (1971 in Seattle, 1972 in Denver), the unemployment rate was about 13 percent in Seattle and 4 percent in Denver. In the second year of the experiment, the unemployment rate had fallen to a little less than 8 percent in Seattle, but was still at 4 percent in Denver. Thus, labor market conditions were very different in the two sites during the preexperimental period but tended to converge during the experiment.

Two potential hypotheses link these labor market differences to differences in the response to the experimental NIT programs. First, unemployed persons may be more likely to rely on income maintenance rather than to make the extensive effort that might be necessary to find a job in a high unemployment labor market. This would imply a larger response in Seattle. Alternatively, persons with relatively poor labor market prospects in normal conditions (e.g., low wage rates) may be most likely to respond to the NIT programs. In a high unemployment situation, these persons may be unemployed anyway even without the experiment. This would imply a lower response in Seattle. Later in this report we explore these hypotheses by estimating models separately for each of the sites. Although the estimated differences are generally not statistically significant, they do tend to be consistent with the latter hypothesis because the response tends to be smaller in the site with high unemployment (Seattle).

Another important labor market issue concerns the response of employers in Seattle and Denver. Because SIME/DIME was not a saturation experiment, it is unlikely to have had any labor market impacts (e.g., employer induced changes in wage rates). In general the importance of labor market impacts depends on the slopes of the aggregate demand curves for low-income labor in Seattle and Denver. If the demand curves are flat (i.e., there is elastic demand for low-income labor), there will be no labor market impacts and the estimated responses in SIME/DIME will be an accurate reflection of the true labor supply responses to a universal NIT in the two sites. If the demand curves are downward-sloping, however, it is not possible to assess the effects of a universal NIT because knowledge of the aggregate supply curve of labor and of how this supply curve shifts (the shift is not a parallel shift) is needed. Actual response may be either larger or smaller than the calculated response, depending on the slope of the aggregate supply curve. If aggregate labor supply is upward-sloping, the aggregate labor supply response would be smaller than is calculated assuming perfectly elastic demand. But if aggregate supply is assumed to be backward-bending, actual labor supply response would be larger than that calculated assuming perfectly elastic demand.

Generalizability and sample representativeness

An important issue to consider in evaluating the experimental findings is the generalizability of the estimated labor supply responses to broader population groups.

There are several reasons why the response estimates from dummy variable models are not generalizable to populations other than those from which the experimental samples were drawn. First, it is likely that response to an NIT program in a given area depends on the structure of existing welfare programs in that area. The generosity (guarantee levels) and, to some extent, the implicit tax (or benefit reduction) rates of AFDC (the major existing welfare program) vary substantially among the states. Furthermore, many states do not extend AFDC eligibility to husband-wife families. Thus, the estimated responses from dummy variable models are at best generalizable only to states with welfare programs similar to those in Washington and Colorado. Second, response depends on the distribution of income and wage rates in a given area. Clearly, high-income persons are less likely than low-income persons to respond to an NIT program. Thus, dummy variable models that do not control for differences in response by income are generalizable only to groups with income and wage distributions similar to the experimental samples. (Dummy variable models that do control for differences in response by income are likely to be imprecisely estimated because of the large number of parameters.) Third, SIME/DIME tested 11 different NIT programs. While this is a large number of programs, an actual program may resemble none of those tested.[12] As indicated earlier, results from dummy variable models cannot be easily generalized to programs other than those tested in the experiments because there are no restrictions on the pattern of the response with respect to changing guarantee levels and tax rates. Finally, the structure of the labor supply response may vary over different regions of the country. Since SIME/DIME consisted of only two sites, this could be a potentially serious problem for areas with substantially different economic and demographic characteristics.

The structural model we use here is designed to deal with all but the last problem. That is, it allows a person's response to depend on (1) eligibility and participation in existing income transfer and tax programs, (2) income and wage rates, and (3) guarantee levels and tax rates other than those tested in the experiments. However, structural models are restrictive in how they allow responses to vary (that is, in the functional forms of the response models). If these restrictions are wrong, then generalizability is less valid. Generalization is also only valid to the extent that the structure of the labor supply response is similar in the two sites and in the rest of the country. In Chapter 8, we use results from the structural model to predict the effects of four possible nationwide NIT programs.

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[12] For example, welfare reform proposals presented by the Carter Administration specified an NIT program with a guarantee equal to 65 percent of the poverty level. The lowest guarantee in SIME/DIME was 95 percent of the poverty level.

CHAPTER IV

DATA

Families enrolled in SIME/DIME were interviewed approximately three times a year. The data collected included detailed information on family composition, employment history, nonlabor income, assets and debts including housing, and training and education. The interview questions used to elicit employment data were designed to allow the construction of a continuous history of employment status, weekly hours of work, wage rates and earnings.

At each interview the respondent was reminded of the employer, job title, hours of work, and wage rate he/she reported at the previous interview. The respondent was then asked to report any changes in these items. The dates of all changes were recorded, along with the new value of the item. The employment questions ended by obtaining the current value for each item. Thus, it is possible to calculate hours of work, wage rates and earnings for any interval during the period covered by the interviews. Additional questions ascertained whether or not the respondent was unemployed during periods he/she was not working. From these data we constructed quarterly and annual histories of hours of work, employment status (working, unemployed, out of labor force), and earnings. These work histories provide the data used for the dependent variables in the models we estimate.

As with any panel, observations are lost over time. Table 3.1 shows how the sample size varied from the preexperimental year throught the sixth experimental year for experimental families on the 3- and 5-year programs and for control families. During the first 4 years the sample size was reduced by 30 percent for husbands (7.5 percent per year) and by 25 percent for wives and single female heads (6.3 percent per year). Because attempts were made to interview all families during the first 4 years, the principal cause of loss in sample size during this time period was attrition (that is, individuals either dropped out of the experiment or could not be found by the interviewers). The subgroups (husbands, wives, and single female heads) are defined as of the enrollment date. The sample sizes differ for husbands and wives because of differential attrition after a marital separation.

TABLE 3.1.—SAMPLE SIZE BY EXPERIMENTAL YEAR

	Experimental year						
	Preexperimental	1	2	3	4	5	6
Husbands	2,736	2,464	2,284	2,034	1,911	1,243	647
Experimentals	1,529	1,393	1,294	1,183	1,107	681	311
3-year	1,048	953	879	791	733	341	0
5-year	481	440	415	392	374	340	311
Controls	1,207	1,071	990	851	804	562	336
Wives	2,736	2,497	2,369	2,148	2,043	1,347	705
Experimentals	1,529	1,408	1,343	1,253	1,194	747	341
3-year	1,048	963	916	837	788	375	0
5-year	481	445	427	416	406	372	341
Controls	1,207	1,089	1,026	895	849	600	364

TABLE 3.1 SAMPLE SIZE BY EXPERIMENTAL YEAR—Continued

	Experimental year						
	Preexperiment year	1	2	3	4	5	6
[illegible]	1,941	1,801	1,715	1,559	1,459	951	458
Experimentals	1,179	1,142	1,066	997	934	570	235
5-year	856	798	772	711	656	310	0
3-year	323	344	294	286	278	260	235
Controls	762	659	649	562	525	381	223

As one would expect, experimentals on the 5-year programs had a lower rate of attrition than experimentals on the 3-year programs, and experimentals had a lower rate of attrition than controls. As discussed earlier, attrition tended to be fairly random with respect to labor supply and we do not expect it to lead to any significant biases in the estimated labor supply responses.

In the fifth and sixth experimental years, the 3-year families were phased out of the interviewing process. About half the 3-year families were interviewed in the fifth year, while none of the 3-year families were interviewed in the sixth year. The families dropped from the interviewing process in the fifth experimental year were chosen randomly.

The treatments in SIME/DIME were generous by current program standards, as we have noted, with guarantee levels ranging from 95 percent of the poverty level to 140 percent of the poverty level. To gain some perspective on how generous these plans were relative to transfer payments available in the control environment, table 3.2 presents average NIT payments to experimental families and average transfers received by both experimental and control families for each of the 5 years of the experiment, along with the fraction of families receiving the various benefits. These benefit amounts, like all other income variables used in our analyses, are in real (1971 Denver) dollars.[1]

Averaged over the entire sample (that is, averaged over both recipients and nonrecipients), the SIME/DIME payments to husbands and wives were considerably more generous than benefits received by control families for each of the 5 years of the experiment. They were also more generous for single female heads, although the differences are not nearly as great, particularly in the earlier years.

A significant number of experimental families reported receiving welfare benefits throughout the experiment. SIME/DIME did not allow families to simultaneously receive NIT and AFDC (or AFDC-U), and taxed Food Stamp benefits at the rate of 100 percent. Although families were not allowed to receive both NIT and AFDC benefits in a single payment period, they were free to move from one program to another. In effect, then, AFDC benefits, like Food Stamp benefits, were taxed at the rate of 100 percent by SIME/DIME. Because of this feature of the experiment, the average NIT and welfare benefit amounts for experimental families in table 3.2 can be added to obtain a total representing average benefits received during a given year. Thus, when one considers both NIT and welfare benefits, SIME/DIME experimental families were considerably better off relative to control families.[2] These differences should be kept in mind when interpreting the results for the dummy variable models presented in later chapters.

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[1] Nominal income variables are deflated by the respective consumer price indexes in Seattle and Denver (U.S. Department of Labor, various issues) and an intercity cost of living adjustment is made on the basis of a lower income budget for an urban family of 4 persons (U.S. Department of Labor, 1979).

[2] Because AFDC and Food Stamp benefits were taxed at the rate of 100 percent, there was considerable incentive to underreport these benefits in the interviews (although efforts were made to prevent this from occurring). Thus, experimental families may have been even better off relative to control families than is indicated in table 3.2.

TABLE 3.2.—EXPERIMENTAL PAYMENTS AND TRANSFERS RECEIVED BY SIME/DIME FAMILIES

[1971 dollars]

	Experimental families					Control families transfers received [1]		
	Experimental payments		Transfers received				Fraction receiving	
	Average amount received (over entire sample)	Fraction receiving more than minimum payment ($240 per year) transfers received	Average amount received (over entire sample) [2]	Fraction receiving [3]		Average amount received (over entire sample) [2]	AFDC, AFDC-U	Food stamps
				AFDC, AFDC-U	Food stamps			
Husbands (year):								
1	$1,140	.79	$221	.15	.18	$263	.13	.21
2	1,168	.77	128	.07	.11	220	.11	.17
3	1,163	.71	113	.07	.11	210	.11	.15
4	1,137	.63	96	.05	.12	195	.10	.17
5	899	.59	91	.05	.10	165	.09	.16
Wives (year):								
1	1,199	.80	256	.17	.20	295	.15	.23
2	1,343	.81	181	.10	.14	327	.15	.22
3	1,399	.75	165	.09	.15	383	.18	.22
4	1,476	.70	142	.07	.16	397	.18	.26
5	1,206	.66	117	.06	.13	348	.17	.24
Single female heads (year):								
1	1,270	.77	835	.53	.48	1,171	.57	.53
2	1,713	.77	472	.23	.24	1,089	.50	.49
3	1,630	.71	402	.19	.23	1,014	.47	.46
4	1,647	.71	261	.16	.18	932	.42	.47
5	1,560	.68	216	.10	.15	748	.37	.41

[1] Based on official payments record information.
[2] AFDC, AFDC-U, and bonus value of food stamps reported in interviews.
[3] Excludes transfers received by 3-year families in the fourth and fifth years.

For all groups (except control wives) and for each benefit type, there was a gradual decline over time in both average benefits received and the fraction receiving benefits. This gradual decline reflects primarily the "regression toward the mean" phenomenon resulting from initial truncation of the sample by income. In other words, incomes generally increased over time in the SIME/DIME sample for nonexperimental reasons, causing a decline in eligibility for NIT and welfare benefits.

During the early years of the experiment, average benefits for families receiving AFDC and Food Stamps were somewhat larger than average experimental payments for SIME/DIME recipients. Over time, this gap narrowed somewhat and eventually the level of SIME/DIME payments exceeded the level of welfare benefits. SIME/DIME payments grew relatively faster than welfare benefits primarily because the regular cost of living adjustments applied to them exceeded sporadic adjustments made to welfare benefits.

Most of our analysis focuses on families in the second year of the experiment. Table 3.3 presents a breakdown of the sample in the second year by generosity of treatment, as measured by the grant breakeven level.[3] For husbands and wives, the sample is distributed fairly evenly across the treatments, ranging from about 5 percent to 13 percent of the experimental families in each treatment. For single female heads, the sample is distributed somewhat less evenly; for example, less than 2 percent of the single female heads are in the 5,600/50 treatment and over 17 percent are in the 3,800/50 treatment. About 45 percent of the experimental families were on a declining tax rate treatment. About 57 percent of the husbands and wives were in experimental families while about 62 percent of the single female heads were in experimental families.

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[3] For the constant tax rate programs, the grant breakeven level is given by G/t, where G is the guarantee level and t is the tax rate. For the declining tax rate programs, the grant breakeven level is given by $(t-\sqrt{t^2-4rG})/2r$, where r is the rate of decline of the average tax rate per dollar of income. As discussed in part I of the this volume, the grant breakeven level is to be distinguished from the tax breakeven level, which is the level of income at which the grant plus the positive tax reimbursement equals zero.

TABLE 3.3.—SAMPLE SIZE IN THE SECOND EXPERIMENTAL YEAR BY GENEROSITY OF TREATMENT [1]

Treatment G	Treatment t	Treatment r	Generosity (grant breakeven level)[2]	Sample size Husbands	Sample size Wives	Sample size Female heads
$3,800	0.7	0	(1) $5,429	79	79	105
$3,800	.8	.025	(2) 5,802	82	84	79
$4,800	.7	0	(3) 6,867	132	141	137
$3,800	.7	.025	(6) 7,367	100	99	104
$3,800	.5	0	(5) 7,600	150	161	185
$4,800	.8	.025	(7) 8,000	118	125	99
$5,600	.7	0	(4) 8,000	116	121	72
$4,800	.5	0	(8) 9,600	146	151	86
$5,600	.8	.025	(10) 10,360	163	172	79
$5,600	.5	0	(9) 11,200	72	68	21
$4,800	.7	.025	(11) 12,000	136	142	99
All experimentals				1,294	1,343	1,066
Controls				990	1,026	649
Total sample				2,284	2,369	1,715

[1] G = guarantee level (1971 dollars)
t = initial tax rate
r = rate of decline of the average tax rate per $1,000 of income

[2] Rankings of generosity in terms of the tax breakeven level are presented in parentheses.

CHAPTER V

FINDINGS FOR HEADS OF FAMILIES

Annual hours of work

The first set of results we present is for annual hours of work of heads of families. We take a systematic approach to the presentation of these results by beginning with the simplest specification and following with models of increasing complexity. First we present estimates from the single dummy variable model, which give overall mean effects. The results from the single dummy variable model indicate a substantial disincentive effect of the average of the NIT treatments. We then present estimates from the 11 dummy variable model which give mean effects by treatment. The 11 dummy variable model suggests that a major portion of the response is due to an income (or guarantee) effect. Next, we present estimates from the structural model. The structural model partitions the response into income and substitution effects according to a linear specification. We find that the major portion of the response to the NIT treatments is due to the income effect, corroborating the results from the 11 dummy variable model. Finally, we discuss the robustness of the structural estimates to an alternative specification and compare the structural estimates with the dummy variable estimates.

Single dummy variable model estimates

The single dummy variable models estimated in this section all have the following form:

$$H_t - H_p = a_{0t} + a_{1t}F + a_{2t}E + a_{3t}X + e_t, \qquad (1)$$

where

H_t = hours of work in year t
H_p = hours of work in preexperimental year
F = experimental treatment dummy
E = vector of assignment variables (dummy variables for E-level)
X = vector of preexperimental variables (including H_p)[1]
$a_{0t}, a_{1t}, a_{2t}, a_{3t}$ = parameters to be estimated
e_t = random error term.

Equations are estimated for each of 6 years after enrollment into the experiment. In the analysis of years 1 through 4, the fourth year sample is used; for years 5 and 6, the maximum available sample is used.[2] Equations are also estimated in which the treatment dummy, F, is interacted with dummy variables for race-ethnicity, site, and experimental duration.

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[1] Included in X besides H_p are dummy variables for race-ethnicity (black, white, Chicano), dummy variables for site (Seattle, Denver), age at enrollment, number of family members at enrollment, number of children under 5 years of age at enrollment, AFDC benefits in the preexperimental year, and three dummy variables for the counseling/training component of the experiment.

[2] Appendix A of Robins and West (1980b) reports estimates of the hours of work response during the first 3 years of the experiment when the maximum available sample is used as well as when the fourth year sample is used. The results don't generally indicate any sizable or systematic differences between the two sets of estimates, although the 3- and 5-year differences tend to be somewhat smaller when the maximum available sample is used.

Tables 3.4 through 3.6 present the estimated experimental effect on annual hours of work from the single dummy variable model for husbands, wives, and single female heads of families, respectively.

TABLE 3.4. NIT TREATMENT EFFECTS ON ANNUAL HOURS OF WORK FOR HUSBANDS

[Standard errors in parentheses]

	Experimental year					
	1	2	3	4	5	6
Total	[1] 53.5	164.5	[3] 165.3	...	...	...
	(29.3)	(34.6)	(37.5)	...	...	...
3-year treatment	27.6	[3] 133.1	[3] 128.6	–7.8	–2.9	...
	(31.7)	(37.4)	(40.5)	(43.0)	(61.0)	...
5-year treatment	105.8	227.6	[3] 239.4	[3] –234.3	[3] –214.6	49.2
	(38.1)	(45.0)	(48.8)	(51.8)	(60.2)	(70.5)
F-test for duration differences	[2] 4.56	[2] 4.78	[2] 5.59	[2] 20.77	[3] 10.87	...
Black	67.4	[3] 169.3	[3] 197.3	...	...	...
	(48.1)	(56.7)	(61.5)	...	...	...
3-year	4.5	101.2	[2] 143.0	–27.2	–3.4	...
	(53.7)	(63.4)	(68.7)	(72.9)	(101.7)	...
5-year	171.9	[3] 282.7	[3] 287.9	[3] –290.8	[2] –243.1	11.8
	(62.5)	(73.8)	(80.0)	(84.8)	(98.1)	(113.6)
White	32.0	[3] 143.5	[1] 118.1	...	...	...
	(38.2)	(45.1)	(48.9)	...	...	...
3-year	34.2	[2] 125.5	–75.0	43.0	12.7	...
	(41.6)	(49.2)	(53.3)	(56.5)	(80.7)	...
5-year	30.2	[3] 184.9	[3] 212.7	[3] –223.0	[2] –197.3	60.5
	(52.3)	(61.7)	(66.9)	(71.0)	(81.3)	(91.5)
Chicano	101.4	[3] 230.8	[3] 267.6	...	...	...
	(66.1)	(78.0)	(84.5)	...	...	...
3-year	61.4	[3] 217.4	[3] –269.7	–121.0	–45.7	...
	(70.4)	(83.2)	(90.1)	(95.6)	(131.8)	...
5-year	[2] 190.6	[3] 262.8	[2] 266.4	[1] –199.4	[1] –221.8	91.5
	(85.0)	(100.4)	(108.8)	(115.4)	(132.8)	(155.0)
F-test for ethnic differences (conditional on duration differences)	1.48	.75	1.12	.98	.07	.11
Seattle	21.3	[3] 143.6	[2] –136.0	...	...	...
	(39.3)	(46.4)	(50.3)	...	...	...
3-year	24.8	[2] 123.0	[2] 116.8	0.2	2.4	...
	(42.9)	(50.8)	(55.0)	(58.3)	(84.6)	...
5-year	11.8	[3] 190.7	[2] 179.4	–90.1	–133.4	50.0
	(56.7)	(67.0)	(72.6)	(76.9)	(89.0)	(99.0)
Denver	[2] 92.3	[3] 189.6	[3] –200.7	...	...	...
	(43.0)	(50.8)	(55.1)	...	...	...
3-year	44.1	[3] –149.6	[2] –150.2	–37.3	–22.7	...
	(46.2)	(54.6)	(59.2)	(62.6)	(84.7)	...
5-year	[3] –175.4	[3] –258.7	[3] –288.0	[3] –342.8	[3] –227.1	48.4
	(51.9)	(61.4)	(66.5)	(70.5)	(81.7)	(96.8)
F-test for site differences (conditional on duration differences)	[1] 2.58	.29	.62	[2] 3.27	.78	.00

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Note.—Sample size for years 1 through 4 is 1,911. For year 5 the sample size is 1,243. For year 6 the sample size is 647. Control variables include six dummy variables for normal income categories, preexperimental hours of work, dummy variables for black and Chicano, a dummy variable for Denver, number of family members at enrollment, number of children under 5 years of age at enrollment, AFDC benefits in the preexperimental year, age at enrollment, a dummy variable for a control filing an Income Report Form, three dummy variables for the counseling/training component of the experiment, and a dummy variable for both heads working in the preexperimental year and in normal income category $9,000-$11,000.

TABLE 3.5.—NIT TREATMENT EFFECTS ON ANNUAL HOURS OF WORK FOR WIVES

[Standard errors in parentheses]

	Experimental year					
	1	2	3	4	5	6
Total	−44.2	[3] −128.3	[3] −121.2			
	(27.2)	(33.1)	(36.0)			
3-year treatment	−20.9	[3] −101.4	[3] −102.6	−14.4	[1] 100.4	
	(29.5)	(35.8)	(39.0)	(40.8)	(60.8)	
5-year treatment	[2] −90.0	[3] −181.3	[3] −157.7	[3] −205.5	[3] −192.7	−33.8
	(35.2)	(42.8)	(46.6)	(48.7)	(59.8)	(71.8)
F-test for duration differences	[2] 4.22	[1] 3.81	1.53	[3] 16.85	[3] 21.45	
Black	−67.7	[3] −153.0	[2] −133.1			
	(44.1)	(53.6)	(58.3)			
3-year	[1] −82.3	[2] −152.5	[1] −126.1	−6.5	[2] 195.3	
	(49.0)	(59.7)	(65.0)	(67.9)	(98.8)	
5-year	−42.3	[2] −153.2	[1] −144.7	[3] −253.7	[1] −167.8	98.3
	(57.3)	(69.8)	(76.0)	(79.5)	(96.7)	(113.2)
White	−21.7	[2] −107.2	[2] −105.4			
	(35.9)	(43.6)	(47.5)			
3-year	22.2	−66.9	−81.5	−24.0	112.9	
	(39.1)	(47.6)	(51.9)	(54.2)	(81.6)	
5-year	[2] −113.5	[3] −192.0	[2] −155.8	[3] −183.6	[3] −216.6	−90.8
	(48.7)	(59.3)	(64.6)	(67.5)	(81.1)	(93.7)
Chicano	−70.8	[2] −147.4	[1] −150.3			
	(60.9)	(74.1)	(80.7)			
3-year	−44.6	−117.3	−129.5	7.1	−100.5	
	(64.9)	(79.0)	(86.1)	(90.0)	(130.6)	
5-year	[1] −129.5	[2] −214.5	[1] −196.4	−166.4	[1] −232.9	−143.6
	(77.5)	(94.4)	(102.8)	(107.5)	(132.5)	(156.9)
F-test for ethnic differences (conditional on duration differences)	1.60	.60	.16	.23	1.00	1.22
Seattle	−18.9	[3] −124.2	[2] −98.7			
	(37.0)	(45.0)	(49.0)			
3-year	−8.4	[2] −100.2	−82.4	6.7	[1] 141.0	
	(40.5)	(49.3)	(53.7)	(56.1)	(84.9)	
5-year	−41.1	[3] −177.7	[1] −135.0	[3] −202.8	[1] −164.9	−96.5
	(53.1)	(64.6)	(70.4)	(73.6)	(88.8)	(101.2)
Denver	[1] −73.1	[3] −133.0	[3] −146.9			
	(39.6)	(48.1)	(52.4)			
3-year	−40.6	[2] −103.2	[2] −126.7	−36.6	58.3	
	(42.4)	(51.6)	(56.2)	(58.8)	(84.0)	
5-year	[3] −129.0	[3] −184.3	[3] −181.7	[3] −215.8	[3] −224.7	25.5
	(47.6)	(57.9)	(63.1)	(65.9)	(81.1)	(98.6)
F-test for site differences (conditional on duration differences)	.77	.00	.20	.15	.27	.77

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Note.—Sample size for years 1 through 4 is 2,043. For year 5 the sample size is 1,347. For year 6 the sample size is 705. Control variables include six dummy variables for normal income categories, preexperimental hours of work, dummy variables for black and Chicano, a dummy variable for Denver, number of family members at enrollment, number of children under 5 years of age at enrollment, AFDC benefits in the preexperimental year, age at enrollment, a dummy variable for a control filing an Income Report Form, three dummy variables for the counseling/training component of the experiment, and a dummy variable for both heads working in preexperimental year and in normal income category $9,000-$11,000.

TABLE 3.6. NIT TREATMENT EFFECTS ON ANNUAL HOURS OF WORK FOR SINGLE FEMALE HEADS

[Standard errors in parentheses]

	Experimental year					
	1	2	3	4	5	6
Total	66.6	144.5	³ 231.3	....	....	....
	(36.7)	(43.3)	(47.1)	....	....	....
3-year treatment	8[illegible]	134.1	³ 222.8	¹ 96.7	−87.3	....
	(38.7)	(45.7)	(49.7)	(52.0)	(73.3)	....
5-year treatment	87.[illegible]	170.4	³ 252.1	³ 340.9	³ −405.7	¹ −170.1
	(48.[illegible])	(56.6)	(61.6)	(64.4)	(76.3)	(94.4)
F-test for duration differences	.73	.51	.28	³ 17.61	³ 17.82	....
Black	88.[illegible]	179.4	³ 238.0	....	....	....
	(49.7)	(58.6)	(63.8)	....	....	....
3-year	87.3	196.3	265.4	109.0	−84.6	....
	(52.8)	(62.2)	(67.6)	(70.9)	(98.8)	....
5-year	78.2	134.1	162.4	³ 308.3	³ −459.8	−208.0
	(68.7)	(81.0)	(88.0)	(92.2)	(108.7)	(127.7)
White	44.8	84.6	³ 197.4	....	....	....
	(54.1)	(63.8)	(69.4)	....	....	....
3-year	28.8	60.0	² 175.0	82.7	−130.8	....
	(58.0)	(68.4)	(74.3)	(77.9)	(109.7)	....
5-year	88.5	130.7	³ 245.7	³ 367.4	³ −369.8	¹ −231.1
	(73.1)	(86.1)	(93.7)	(98.1)	(114.3)	(134.6)
Chicano	16.8	² 201.9	³ 316.1	....	....	....
	(87.2)	(102.9)	(111.9)	....	....	....
3-year	15.1	142.4	¹ 232.7	−92.6	38.5	....
	(93.8)	(110.2)	(119.9)	(125.6)	(179.5)	....
5-year	76.9	² 310.5	³ 466.1	² 358.4	320.0	144.4
	(107.1)	(126.2)	(137.3)	(143.8)	(170.0)	(210.8)
F-test for ethnic differences (conditional on duration differences)	.42	1.27	1.62	.12	.34	1.40
Seattle	¹ 84.2	³ 159.0	³ 215.0	....	....	....
	(48.2)	(56.9)	(61.8)	....	....	....
3-year	71.6	³ 158.1	³ 229.5	¹ 129.6	−122.3	....
	(51.1)	(60.3)	(65.6)	(68.6)	(96.9)	....
5-year	124.8	¹ 158.0	¹ 160.4	³ 285.6	³ −382.2	¹ −232.0
	(75.3)	(88.8)	(96.6)	(101.0)	(118.3)	(132.3)
Denver	28.8	¹ 125.0	³ 253.0	....	....	....
	(55.8)	(65.8)	(71.6)	....	....	....
3-year	17.8	105.2	³ 226.1	62.2	−49.5	....
	(58.8)	(69.4)	(75.4)	(78.9)	(107.2)	....
5-year	50.2	² 161.3	³ 301.4	³ 350.9	² −401.8	−108.9
	(65.4)	(77.2)	(83.9)	(87.7)	(103.6)	(131.6)
F-test for site differences (conditional on duration differences)	.35	.23	.84	.65	.22	.45

¹ Significant at the 10-percent level.
² Significant at the 5-percent level.
³ Significant at the 1-percent level.

Note:—Sample size for years 1 through 4 is 1,459. For year 5 the sample size is 951. For year 6 the sample size is 458. Control variables include five dummy variables for normal income categories, preexperimental hours of work, dummy variables for black and Chicano, a dummy variable for Denver, number of family members at enrollment, number of children under 5 years of age at enrollment, AFDC benefits in the preexperimental year, age at enrollment, a dummy variable for a control filing an Income Report Form, and three dummy variables for the counseling/training component of the experiment.

Results for husbands

The results for husbands indicate a statistically significant disincentive effect of the NIT treatments. For 3-year families, the maximum response is −133 hours in the second year or approximately a 7 percent decline.[3] The effects are larger for 5-year families than for 3-year families in every year, and the differences are always statistically significant. The maximum response for annual hours of work for 5-year families is -239 hours in the third year, or approximately a 13 percent decline.

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[3] Percentage responses are calculated using the data in Appendix B of Robins and West (1980b), as $b_i/(H_i-b_i)$, where b_i is the estimated response for the ith group and H_i is the mean hours of work for the ith group.

The finding of a significant difference in the estimated responses of 3- and 5-year husbands is important because of its implications for the effects of a permanent program. Presumably, the larger response of 5-year husbands implies that husbands in a permanent program would have an even larger response. However, it is important to note that the estimated differences between the 3- and 5-year programs are for some average of the programs tested in SIME/DIME. Since the theoretical biases of a short-duration experiment have opposite signs according to whether the response is generated by the substitution effect (which is overestimated) or the income effect (which is underestimated), the total bias of a particular program will vary with the guarantee level and the tax rate of that program. Most programs that might actually be implemented are likely to have lower guarantee levels than the average guarantee level of SIME/DIME. Consequently, the difference between the permanent effects of such a program and predictions from a limited duration experiment may be less than the difference implied by our estimates. We shall have more to say about the experimental duration issue when we discuss the 11 dummy variable model estimates and the structural model estimates.

After the experiment (in the fourth and fifth years for 3-year families and in the sixth year for 5-year families), the responses are not significantly different from zero. This lack of response after the experiment provides strong support for the hypothesis that the responses estimated during the experimental period are real responses and not artifacts of the assignment process. The lack of response in the first year after the experiment suggests that husbands are able to readjust rapidly to the nonexperimental environment.

In addition to differences in responses by length of treatment, there are also differences in response by race-ethnic group and by site, although few of them are statistically significant. There is weak evidence that blacks and Chicanos have larger responses than whites throughout the experimental period; however, the race-ethnic differences for 5-year families are not nearly as great as they are for 3-year families. The lower responses for white husbands, compared to black and Chicano husbands, are somewhat puzzling and we have no convincing explanation for it. In commenting on these findings, Garfinkel (1978) has offered an explanation that appears plausible. He argues that blacks and Chicanos on average have worse labor market opportunities than whites (because of discrimination) and hence respond relatively more to an experimental treatment that increases the attractiveness of not working.

Responses of husbands in Denver appear to be slightly larger than the responses of husbands in Seattle, but most of the site differences occur for 5-year families; 3-year families have very similar responses in both sites. The lower responses in Seattle have a plausible explanation. When the experiment began, the unemployment rate in Seattle was much larger than the unemployment rate in Denver. Average annual hours of work was about 23 percent less for husbands and 25 percent less for single female heads (there was no significant difference in average hours of work for wives). Over time, this difference narrowed somewhat for the control group. Since labor supply response is measured in terms of actual (as opposed to desired) hours of work, it is not surprising to observe a smaller response in Seattle, since unemployed persons (with positive desired hours) cannot reduce actual hours of work. Also, given bleak employment prospects in Seattle, individuals may be less willing to give up a job already held and more willing to accept a job offered.

Results for wives

Results for wives also indicate a statistically significant work disincentive effect during the experimental period. For 3-year families, the maximum response for annual hours of work is −103 hours in the third year, or approximately a 15 percent decline.[4] Like husbands, wives on the 5-year programs have substantially larger responses than wives on the 3-year programs, although the differences are only statistically significant in the first 2 years. The maximum response for 5-year families is −206 hours in the fourth year, or approximately a 27 percent decline.

After the experiment, wives (like husbands) appear to readjust quickly to the nonexperimental environment. In the second year after the experiment, 3-year wives actually had higher annual hours of work than comparable control families. This suggests that the financial treatments may have induced some intertemporal substitution of labor supply.

Like husbands, there are no significant race-ethnic or site differences in response for wives. White wives appear to respond somewhat less than black and Chicano wives, but this difference is present only for 3-year families.

Results for single female heads

Results for single female heads indicate a sizable response that for the most part exceeds the responses of husbands and wives in both absolute and percentage terms.

For each group the response of single female heads grows over time. For 3-year families the maximum response is −223 hours in the third year, or approximately a 22 percent decline. During the first 3 years, the responses of 3- and 5-year single female heads are not signficiantly different (unlike husbands and wives) although the responses of 5-year families exceed the responses of 3-year families by a slight amount. In the fourth and fifth years, the responses of 5-year families continue to grow. The maximum response is −406 hours in the 5th year, or approximately a 32 percent decline. Hence, while there is no evidence of a differential response according to the length of the treatment for single female heads, there is evidence that adjustment to the NIT treatments takes longer than the 5 years the experimental group was observed. This implies that the long-run response to an NIT program may be even larger than the estimates reported here, not because of the limited duration of the experiment, but because adjustment to the NIT treatments appears to be very slow.

After the experiment, single female heads begin to readjust to the nonexperimental environment. Unlike husbands and wives, however, the readjustment is gradual. In the first year after the experiment, the responses of both 3- and 5-year single female heads are statistically significant. Based on evidence both during and after the experiment, it appears that single female heads adjust more slowly to changes in their budget constraints than do husbands and wives. Perhaps the absence of a spouse imposes restrictions on single female heads (both in the labor market and at home) that prevent them from altering their behavior rapidly.

Finally, there is no evidence for single female heads of a significant difference in response according to race-ethnic group or site. Hence, for all the groups, the single dummy variable model estimates suggest that sample stratification by race and site does not pose a problem in generalizing the SIME/DIME findings to other population groups.

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[4] It is important tor remember that these responses are based on interview-reported hours of work. Greenberg and Halsey (1980) find that virtually all of the earnings response of wives (but less than one-third of the employment status response) disappears when data from unemployment insurance records (primarily from Denver) are used, presumably reflecting a differential reporting bias. Such a bias does not appear to exist to any large extent for husbands or for single female heads of families.

Comparison with 20-year families

As indicated in part I of this volume, during the latter half of 1974, 169 families in Denver were enrolled in a 20-year NIT program. In principle, this "experiment within an experiment" should provide definitive evidence about the labor supply response to a permanent NIT. Robins and Stieger (1980) present a detailed analysis of the labor supply response of 20-year families. In table 3.7, we reproduce their findings (which were estimated using a model virtually identical to equation (1) and compare them to the findings for 3-year and 5-year families reported earlier.[5]

Estimates presented in table 3.7 suggest that during the first 3 years of the 20-year program, husbands and wives did not reduce their labor supply relative to comparable control families. This is in direct contrast to evidence presented in tables 3.4 and 3.5 for 3- and 5-year husbands and wives. Moreover, experimental duration seems to matter for 3- and 5-year husbands and wives. Because of the small size of the 20-year sample, it is difficult to estimate precisely the size of their response. Otherwise, we are unable to offer a convincing explanation why the findings for 20-year husbands and wives differ so dramatically from the responses of 3- and 5-year husbands and wives (see Robins and Stieger, 1980, for further discussion).

TABLE 3.7.—COMPARISON OF RESPONSES OF 3-YEAR, 5-YEAR, AND 20-YEAR FAMILIES ON ANNUAL HOURS OF WORK

[Standard errors in parentheses]

	Experimental year		
	1	2	3
Husbands:			
3-year	−27.6	[3]−133.1	[3]−128.6
	(31.7)	(37.4)	(40.5)
5-year	[3]−105.8	[3]−227.6	[3]−239.4
	(38.1)	(45.0)	(48.8)
20-year	79.2	−31.2	51.8
	(99.7)	(117.0)	(118.1)
Wives:			
3-year	−20.9	[3]−101.4	[3]−102.6
	(29.5)	(35.8)	(39.0)
5-year	[2]−90.0	[3]−181.3	[3]−157.7
	(35.2)	(42.8)	(46.6)
20-year	−19.7	66.1	4.9
	(105.6)	(117.7)	(125.0)
Single female heads:			
3-year	−50.0	[3]−134.1	[3]−222.8
	(38.7)	(45.7)	(49.7)
5-year	[1]−87.0	[3]−170.4	[3]−252.1
	(48.0)	(56.6)	(61.6)
20-year	−124.4	[1]−200.2	[1]−252.0
	(116.3)	(116.6)	(135.2)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Note.—The sample sizes for 3- and 5-year families are the same as given in Tables 3.4–3.6. The sample sizes for 20-year families are 232, 259, and 163 for husbands, wives, and single female heads, respectively. The 20-year results are only for families who had previously been control families. For greater details on selection of the 20-year analysis sample see Robins and Stieger (1980).

For single female heads of families on the 20-year program, labor supply is reduced by about the same amount as estimated for 3- and 5-year families. Al-

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[5] It should be noted that the coefficients of the counseling/training, assignment, and other nonexperimental variables for 20-year families, are not the same as the coefficients of these variables for 3- and 5-year families (which are constrained to be the same). This should not appreciably affect the estimates, however. For a discussion of why the 20-year families were analyzed separately, see Robins and Stieger (1980).

[illegible] experimental duration doesn't seem a matter for single female heads, the length of the observation period does, suggesting that this group responds slowly to the NIT. The results in table 3.6 indicate that the response of single female heads to the 5-year program continues to increase up until the end of the experiment, reaching a level of −405 hours per year in the fifth experimental year, or a 32 percent reduction. The implication of this pattern is that it would take [illegible] years to observe the long-run response of 20-year single female [illegible]. Because the interviewing process terminated, however, data are not available for 20-year families beyond their third experimental year.

Interactions with the counseling/training treatments

Because the SIME/DIME experiment was run jointly with a counseling/training experiment, it is of considerable importance to determine whether the effects of the two sets of treatments are additive or whether there is some interaction between them. The results presented in tables 3.4 through 3.6 constrain these effects to be additive. Table 3.8 presents results for a model which allows for interactions. For simplicity, results are presented only for the second experimental year. The model specifies a main effect for the NIT treatments, represented by a single dummy variable, and a main effect for each of the three counseling/training treatments, represented by three dummy variables. In addition, interactions between the NIT treatment variable and each of the three counseling/training treatment variables are also specified. The main counseling/training effects are not statistically significant and are not presented in the table.

TABLE 3.8.—FINANCIAL TREATMENT EFFECTS ON ANNUAL HOURS OF WORK IN THE SECOND EXPERIMENTAL YEAR: INTERACTIONS WITH COUNSELING/TRAINING TREATMENTS

(Standard errors in parentheses)

	Husbands	Wives	Single female heads
Financial treatment effect without interactions[3]	[2] 163.1	[2] −141.9	[2] −150.0
	(32.5)	(30.6)	(39.3)
Financial treatment effect with interactions[4]	[2] 140.2	[2] −140.6	[1] −119.9
	(46.8)	(43.9)	(56.8)
Interaction effects[5]			
Interaction with counseling-only treatment	4.3	−35.8	−84.1
	(81.6)	(76.7)	(96.9)
Interaction with counseling and 50% subsidy treatment	107.2	−5.7	30.9
	(73.9)	(69.7)	(89.9)
Interaction with counseling and 100% subsidy treatment	15.8	43.7	−141.6
	(88.3)	(83.1)	(105.2)
F-test for interactions	.91	.23	1.01

[1] Significant at the 5-percent level.

[2] Significant at the 1-percent level.

[3] These effects differ slightly from those in tables 3.4 through 3.6 because of differences in the sample used. In this table the second year sample is used, whereas in the other tables the fourth year sample is used.

[4] These represent the NIT effects for persons without a counseling/training treatment.

[5] These represent the additional effects for NIT persons of having a counseling/training treatment.

The results in table 3.8 indicate that the estimated main effects of NIT treatments are somewhat smaller for husbands and single female heads when interactions are included in the model. The largest difference occurs for single female heads, for whom the NIT effect drops from −150 to −119. However, the interactions are not jointly significant for any of the three groups; the signs of the interaction terms are also mixed (5 negative, 4 positive), although the negative

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[6] The adjustment problem, which differs from the limited duration problem, has been pointed out by Metcalf (1973), among others.

estimates are somewhat larger in magnitude. Because these interactions are generally small in magnitude and because they are not statistically significant, we conclude that it is possible to ignore interactions between the two sets of treatments in the remainder of our analysis.

Eleven dummy variable model estimates

We now turn to estimates of the 11 dummy variable model, which has the following form:

$$H_2 - H_p = b_0 + b_1 F_1 + \ldots b_{11} F_{11} + b_{12} E + b_{13} x + e_2, \qquad (2)$$

where H_2 = hours of work in the second experimental year
H_p = hours of work in the preexperimental year
$F_1, \ldots F_{11}$ = dummy variables for each experimental treatment
E = a vector of assignment variables
X = a vector of preexperimental variables (including H_p)
b = parameters to be estimated
e_2 = a random error term.

Table 3.9 presents estimates of this model for husbands, wives, and single female heads. Reading down the table gives estimates of response by guarantee level, holding the tax rate constant. Reading across gives estimates of response by tax rate, holding the guarantee level constant.

The results in table 3.8 indicate that response in SIME/DIME varies with the parameters of the NIT programs, although formal tests of the differences from the single dummy variable model are not statistically significant (as given by the F_2 statistics). For each tax rate program, the responses tend to get larger as the guarantee level increases, although there is no conspicuous pattern with respect to the tax rate. For the constant tax rate programs, the responses are generally smaller the higher the tax rate. For the declining tax rate programs, a mixed pattern emerges.

TABLE 3.9.—HOURS OF WORK RESPONSE BY GUARANTEE LEVEL AND TAX RATE

[Second experimental year]

	Tax rate			
	.5	.7	.7d	.8d
(1) Husbands' guarantee level				
$3,800	[1] −114.1	−94.4	[2] −156.1	[1] −143.9
	(63.3)	(83.2)	(75.5)	(81.9)
$4,800	[3] −164.6	−29.1	[3] −278.7	[2] −174.0
	(62.5)	(65.5)	(64.1)	(69.2)
$5,600	[3] −232.7	[3] −210.5	n.t.	[3] −170.0
	(85.6)	(69.4)		(60.3)
$F_1 = 3.32$[3]				
$F_2 = 1.14$				
(2) Wives' guarantee level:				
$3,800	[2] −133.6	−62.1	−9.5	−88.2
	(58.8)	(79.5)	(72.2)	(77.4)
$4,800	[3] −226.8	[3] −154.8	[2] −143.2	[1] −113.5
	(58.8)	(60.9)	(59.9)	(64.5)
$5,600	[3] −227.9	[3] −230.2	n.t.	[1] −97.4
	(83.9)	(65.0)		(56.2)
$F_1 = 3.03$[3]				
$F_2 = 1.18$				
(3) Single female heads' guarantee level:				
$3,800	−66.8	[2] −187.4	−22.2	−130.1
	(61.0)	(76.7)	(76.9)	(85.5)
$4,800	[2] −176.5	[2] −135.1	[3] −206.2	−110.8
	(82.2)	(67.4)	(76.9)	(77.5)

TABLE 3.9 HOURS OF WORK RESPONSE BY GUARANTEE LEVEL AND TAX RATE—Continued

[Second experimental year]

	Tax rate			
	5	.7	.7d	.8d
[illegible]	73.5	³ −306.4	n.t.	³ −300.3
	(156.5)	(88.7)		(85.4)
[illegible]				
[illegible]				

[illegible] SIME DIME

The increasing response to the guarantee level is consistent with the theoretical discussion in chapter II, which indicates that the demand for leisure will increase (or the supply of labor will decrease) as nonwage income increases. The absence of a tax rate effect is also consistent with the theory because, holding the guarantee level constant, increases in the tax rate generate both an income and a substitution effect, which are offsetting. The fact that response tends to decrease with the tax rate suggests that the income effect dominates the substitution effect. As we shall see later, this finding is corroborated by the results from the structural model.

One of the difficulties in drawing inferences from the results in table 3.9 is that the distribution of income differs across the various treatments. Because economic theory indicates the response varies with income, comparison of the responses across the various treatments is not really meaningful because income characteristics of the population are not held constant (the design model tended to allocate higher income families to the more generous treatments and vice versa). Hence, the differences don't measure pure treatment effects for identical populations. This is especially true when comparing the constant tax rate programs with the declining tax rate programs, because the value of the tax rate in the declining tax rate progams depends on income. Attempts to estimate versions of the 11 dummy variable model that interact the treatment variables with normal income give results that are difficult to interpret. Even when the 11 treatment variables are collapsed into 3 treatment variables, measuring generosity of treatment (low, medium, and high breakeven level), the results are not very revealing, although response does tend to be larger for husbands and wives with low normal incomes. For single female heads, the responses were unstable with respect to normal income.

The 11 dummy variable model was also estimated separately for 3- and 5-year families. Generally, the results show larger responses for 5-year families, which is consistent with the findings from the single dummy variable model. The 3- and 5-year differences are not statistically significant, however, for any group. In table 3.10, we present 3- and 5-year coefficients for the model in which the 11 treatment variables are collapsed into 3 treatment variables, measuring generosity of treatment. Effects for the combined 3- and 5-year samples are also presented.

As table 3.10 indicates, a pattern of increasing response with respect to generosity of treatment exists for both 3- and 5-year families. For the combined 3- and 5-year sample, the differences by breakeven level are statistically significant for husbands and female heads. As in the case of the 11 dummy variable model, however, the 3- and 5-year differences are not statistically significant for any group, although the 5-year effects again tend to be larger than the 3-year effects for husbands and wives. The 3- and 5-year differences tend to be smallest for the least generous programs, confirming our earlier speculation that permanent programs less generous than SIME/DIME may not result in as large a response as implied by the more generous 5-year SIME/DIME treatments.

TABLE 3.10.—HOURS OF WORK RESPONSE BY GENEROSITY OF TREATMENT AND EXPERIMENTAL DURATION

[Second experimental year]

	Breakeven level		
	Low ($5,000 to $7,000)	Medium ($7,001 to $8,000)	High ($8,001 to $12,000)
Husbands			
Total ($F_1 = 2.95$ [2]) [5]	−78.3	[3] −160.1	[3] −205.8
	(49.0)	(41.9)	(40.2)
3-year	−63.5	[3] −128.6	[3] −184.2
	(57.3)	(47.1)	(44.8)
5-year	−107.0	[3] −224.8	[3] −255.8
	(72.9)	(61.9)	(61.5)
$F_2 = 1.13$ [6]			
Wives			
Total ($F_1 = 0.41$)	[2] −117.8	[3] −132.4	[3] −160.8
	(46.2)	(39.4)	(37.8)
3-year	[2] −105.6	[3] −120.6	[3] −131.7
	(53.8)	(44.3)	(42.1)
5-year	[2] −139.5	[3] −153.9	[3] −230.1
	(69.1)	(57.8)	(58.4)
$F_2 = 0.95$			
Single female heads			
Total ($F_1 = 2.12$ [1])	[3] −152.8	[2] −105.7	[3] −214.4
	(51.1)	(46.4)	(52.7)
3-year	[3] −157.7	−79.6	[3] −220.2
	(57.8)	(50.9)	(56.9)
5-year	[1] −142.9	[2] −171.3	[1] −189.8
	(76.3)	(70.0)	(97.2)
$F_2 = 0.57$			

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.
[4] Based on treatment description in chapter 4 table 3.3.
[5] F_1 test is for differences from single dummy variable model.
[6] F_2 test is for 3- and 5-year differences.

A structural model [7]

Estimated experimental effects in dummy variable models do not lend themselves easily to generalization, which is a major objective of SIME/DIME. In particular, dummy variable models are of limited use for estimating the effects of NIT programs other than those tested in the experiment and do not model the control environment in terms of welfare participation and benefit amounts.

The structural model presented in this section, which is based on the theory presented in chapter II, attempts to deal with these issues by parameterizing the response in terms of changes in disposable income and net wage rates (income and substitution effects) induced by the experimental NIT programs. Because the changes in disposable income and net wage rates depend on both the characteristics of the specific NIT program and on the person's initial income position, the estimated parameters of this model can be used to predict the effects of alternative NIT programs in general populations. They are used in this way in chapter 8 to predict the nationwide effects of several different NIT programs. In the remainder of this chapter we discuss the model, our empirical specification, the estimation procedure, and the estimation results.

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[7] Much of this section is based on Robins, West, and Keeley (1980).

The model

In its discrete form, the structural model given by equation (5) in Chapter II can be written as

$$\Delta H = \alpha \Delta W + \beta (H_p \Delta W + \Delta N) = \alpha \Delta W + \beta \Delta Y \tag{3}$$

where ΔW is the change in the net wage rate, ΔN is the change in nonlabor income, and ΔY is the change in disposable income of an individual, holding constant his or her preexperimental labor supply, H_p. Thus, as discussed in chapter 2, each person's response depends on both the change in the net wage rate (ΔW) and the change in disposable income (ΔY) he or she experiences as a result of the experimental NIT plans. For a family with more than one potential earner, this equation should include a cross substitution effect (the change in the spouse's net wage rate). We assume that cross substitution effects are zero at the preexperimental position, partly because the net wage changes of the two spouses are highly correlated and their effects difficult to distinguish empirically. This assumption is likely to have little influence on predictions of the effects of NIT programs since actual programs, like the experiment, will change the tax rates and consequently net wage rates of spouses in a similar manner. The assumption does, however, limit the usefulness of these results for making statements about the own elasticity of individual labor supply functions.

The assumptions underlying this model are different from those implicit in most cross sectional studies. Instead of assuming that each person has the same preference structure, it is assumed that differences in taste are reflected in differences in initial labor supply, after controlling for differences in budget constraints. Therefore, no single utility function is consistent with the model, although each person is assumed to maximize a well-behaved utility function.

In estimating this model we assume that α and β are the same for all individuals. Thus, we assume that individuals have equal income and substitution effects at their preexperimental positions.[8] If, in fact, income and substitution effects differ among individuals, the empirical method we use measures average income and substitution effects in the SIME/DIME sample.

The assumption that different individuals have equal substitution and income effects at their preexperimental positions implies that response to a given NIT program depends on the preexperimental position. For example, if a person with low initial income experiences a considerable change in disposable income and net wage rate, a large response is expected. On the other hand, a person initially at the breakeven level experiences a change only in the net wage rate; response for this person consists only of a (Slutsky) substitution effect and is therefore smaller. As another example, consider a person above the breakeven level who experiences changes in disposable income and the net wage rate only if the elasticity of substitution in consumption is sufficiently large that the indifference curve through the initial point intersects the NIT segment of the new budget line. For a person initially above the breakeven level we would expect a very small probability of response. Finally, consider an individual who is not working: this person experiences a considerable change in disposable income and net wage rate, but has zero response.[9] Thus, response depends on preexperimental position.[10]

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[8] Such an assumption does not imply constant substitution and income effects at all levels of the labor supply for a given person. In fact, it is impossible to have a labor supply function with constant income and substitution effects (Dickinson, 1975, p. 31). If we integrate the continuous version of equation (3), we obtain the level version of the model, $H = c + \alpha W + \beta\ (H_p W + N)$. From this equation, the substitution effect is equal to $\alpha + \beta(H - H_p)$. At $H = H_p$, the substitution effect is equal to α. Hence, α can be interpreted as the substitution effect at the preexperimental position.

These differences in response to an NIT program arise, not because persons with different tastes for work have inherently different responses to changes in disposable income or net wage rates, but because individuals with different propensities to work (different initial equilibria on a given budget constraint) are offered different inducements to change their behavior. Those with the smallest propensities to work experience the largest changes in income.

Empirical specification

We estimate equation (3) for the second year of the experiment. The second year is chosen because the estimates of the single dummy variable model indicate a lag of at least 1 year in experimental response. Using data for the third year might introduce a bias because individuals on the 3-year programs may be starting to adjust their labor supplies to the end of the experiment. The response variables, ΔW and ΔY, depend on the treatment to which an individual is assigned and on his or her preexperimental income and budget constraint. Hours of work and income are measured on an annual basis. For reasons described below, several modifications are made to equation (3) regarding nonlinearity of the budget constraint, missing data on wage rates for nonworkers, additional experimental variables, selectivity bias, and errors in variables.[11]

The sample

The second year sample is used to estimate this structural model. The sample includes 2,249 husbands, 2,332 wives, and 1,715 single female heads. This sample for husbands and wives differs slightly from the second year sample used for table 3.1 in chapter 4 because persons in families that are below the tax breakeven level of income, but have incomes greater than 1.3 times the grant breakeven level of income (based on preexperimental income), are excluded from the sample.[12] Families in this earnings region were erroneously given an experimental tax rate that could exceed 100 percent.[13] Thirty-five husbands and 37 wives were excluded from the sample for this reason. There were no single female heads in this income range.

Calculating the change in disposable income

The change in disposable income evaluated at preprogram hours of work is given by

$$\Delta Y = Y_{de} - Y_{dp} , \tag{4}$$

where Y_{de}=disposable income, evaluated at H_p, under the NIT program. For this study ΔY is calculated on the basis of earnings and nonwage income in the year before enrollment into the experiment.[14]

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[9] Response is subject to truncation because, at most, a person can reduce hours to zero. The estimation technique we employ accounts for this problem.

[10] Note that in a model where the treatment variables are the change in nonlabor income (ΔN) and the change in the net wage rate (ΔW), response would not depend on the initial position (below breakeven and ignoring truncation), because the changes in nonwage income and the net wage rate do not depend on the initial position.

[11] As in the dummy variable models presented above, assignment variables and additional control variables are included in the estimated equations.

[12] See Keeley et al. (1978b) for a discussion of the distinction between the tax breakeven level and the grant breakeven level.

[13] See Burtless and Paquette (1978) for a discussion of how this error was made.

[14] $\Delta Y = S - SR100 — .5\ (SR50 — SA50) — (t - r(Y-E))\ (Y-E) + Q$, where S is the support level, t is the initial tax rate, and r is the rate of decline of the average tax rate. The term SR100 represents items taxed at 100 percent: bonus value of food stamps, welfare benefits including AFDC, unemployment insurance, worker's compensation, veteran's survivors and disability benefits, training stipends net of tuition, fees and books, and social security benefits. The term SR50 represents items taxes at 50 percent: alimony and child support received, and other support received. The term SA50 represents items reimbursed at 50 percent: alimony and child support paid and other sup-

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Calculating the change in the net wage rate

The change in the net wage rate is given by:

$$\Delta W = w(1-t_e) - w(1-t_p) = -w(t_e-t_p)\ , \tag{5}$$

where w is the gross wage rate before response. Thus, ΔW depends upon both the change in the tax rate induced by the experimental NIT plans and the person's gross wage rate. In calculating this variable from SIME/DIME data, two problems arise. First, the preexperimental tax function and many of the experimental tax functions are nonlinear. Second, wage rates are not observed for nonworkers. We discuss these problems in the following paragraphs.

Nonlinearity of the tax function—One effect of the declining tax rate plans is that the experimental tax rate, t_e, depends on income and, hence, on labor supply. That is, the tax rate decreases as income increases. We linearize the experimental budget constraint around the preexperimental position and treat the individual as if he or she were on the tangent linear budget constraint. This procedure is deficient in that all final equilibrium points are not on the linearized budget constraint. In addition, since the experimental budget set is not convex for families on the declining tax rate programs, the linearization may not be a reasonable approximation to the true budget set because small changes in nonconvex budget sets may lead to large changes in behavior. To account for the linearization procedure during estimation, we include a dummy variable for persons on the declining tax rate programs who are below the breakeven level.

The preexperimental budget constraint is also nonlinear because of the progressivity of the positive income tax system and the interrelations among tax rates in income-conditioned public transfer programs. The preexperimental tax rates we specify for this model are derived on the basis of preexperimental income and participation in certain income-conditioned tax and transfer programs. The income-conditioned programs we consider include Federal and State income taxes, social security taxes, AFDC, AFDC-U, and food stamps. The tax rates are derived in accordance with the tax laws and the administrative regulations of the public transfer programs.[15]

Wage rates for nonworkers—Because wage rates are not observed for nonworkers, we adopt the approach of estimating a wage equation for workers based on their personal characteristics to predict wage rates for the entire sample. The wage equation we estimate is a simple linear formulation based on the human capital model of Mincer (1974).[16] The change in the net wage, ΔY W, is calcu-

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port paid. The term Y represents items taxed as income: earnings, insurance benefits, pensions and annuities, payments from private disability plans, and a fraction of net worth. The term E represents items subtracted from income: child care expenses, care for the aged, and medical expenses. The term Q represents items reimbursed at 100 percent: federal and state income taxes, and social security taxes. All variables are measured in real terms with an adjustment for cost of living differences between Seattle and Denver (a deflation factor of .918 is applied to all Seattle data). If ΔY > \$240 in Denver or \$220 in Seattle (the minimum annual payment) before deducting AFDC, a family is defined as being below the breakeven level; ΔY is set equal to \$240 in Denver and \$220 in Seattle for families above the breakeven level. See the section on "Additional Experimental Variables" for a discussion of how families above the breakeven level are treated in the empirical model.

[15] See Kurz et al. (1974) for a discussion of the methodology used to calculate the positive tax rates. For transfer programs, the following tax rates, based in part on Maxfield (1977), are used. If the family received AFDC benefits in the preexperimental year and earnings of the head(s) exceeded \$360, the family was assigned an AFDC tax rate of .37 (which reflects the fact that positive taxes and work related expenses are reimbursed under the AFDC program). If the family received food stamps, it was assigned a food stamps tax rate of .27 if earnings of the head(s) were less than \$3,600 and .30 if earnings of the head(s) were greater then \$3,600. The total marginal tax rate for the individual is given by $t_p = t_i + t_f(1-t_i-t_a) + t_a$, where t_i is the positive (federal, state, and social security) tax rate, t_f is the Food Stamp tax rate, and t_a is the AFDC tax rate. This formula takes account of the fact that the Food Stamp program taxes AFDC benefits and reimburses positive taxes.

[16] This approach follows Hall (1973) and Kalachek and Raines (1970). Gronau (1974) and Heckman (1974) demonstrate that the wage equation approach yields biased estimates for nonworkers, and develop alternative estimation procedures. However, we estimate the substitution effects as the coefficient of the change in the net wage rather than the coefficient of the gross wage rate. It is unlikely that small biases in estimating gross wage rates significantly affect the change variable, which depends primarily on the difference between the experimental and

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lated as the product of the predicted wage rate and the difference between the linearized preexperimental and experimental tax rates.

Additional experimental variables

Based on their preexperimental income, many of the enrolled families were above the breakeven level.[17] We measure response for families above the breakeven level by a dummy variable (FABOVE) that takes on the value of one if the family is above the breakdown level, and zero otherwise. The estimated coefficient of this variable measures the combined effect of the probability of moving below the breakeven level and the response conditional on moving below the breakeven level.[18] For these families ΔY is set equal to the minimum payment and ΔW is set equal to zero.

SIME/DIME, as noted, did not allow families to receive NIT and AFDC benefits simultaneously. Hence, families had to choose which program they wished to participate in although they were free to move from one program to another over time. Based on preexperimental income, it is not clear which choice the family will actually make. We suspect that families made better off (in an income sense) by the NIT would choose the NIT program. However, some families not made better off by the NIT might also choose the NIT because of the stigma associated with welfare. In our empirical analysis we choose to treat separately AFDC families not made better off in an income sense by the NIT. We define a dummy variable (DOMINATE) for families that would be below the breakdown level if they were not receiving AFDC benefits in the preexperimental year, but whose preexperimental AFDC benefits exceed their predicted NIT payments. Hence, if the NIT simple taxed AFDC benefits at 100 percent, these families would be above the breakdown level.[19] The calculated values of ΔY and ΔW are set equal to the minimum payment and zero, respectively, for these families.

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preexperimental tax rates, and only slightly on the predicted gross wage rate. In a recent paper, Heckman (1977) finds that in a national sample of white married women (National Longitudinal Survey), the selectivity bias in wage rates is quantitatively small. Predicted wage rates were used for the entire sample, rather than just nonworkers, because using predictions only for a subsample may lead to biased estimates.
The estimated wage equations are

$$
\begin{aligned}
W = &- .146D - .085B - .092C + .119N + .023E + .040X \\
&\quad (.046) \quad (.047) \quad (.062) \quad (.016) \quad (.010) \quad (.008) \\
&- .00069X^2 + .00110T + .0000020T^2 + 2.145 \text{ for husbands} \\
&\quad (.00018) \quad (.00087) \quad (.0000032) \\
W = &- .042D + .053B - .001C + 048N + .091E + .020X \\
&\quad (.048) \quad (.048) \quad (.069) \quad (.018) \quad (.012) \quad (.009) \\
&- .00041X^2 + .0014T - .0000024T^2 + .715 \text{ for wives} \\
&\quad (.00022) \quad (.0012) \quad (.0000024) \\
W = &- .073D + .035B + .021C + .017N + .090E + .046X \\
&\quad (.040) \quad (.041) \quad (.060) \quad (.016) \quad (.011) \quad (.008) \\
&- .00098X^2 + .0017T - .00000015T^2 + .82 \text{ for single female heads,} \\
&\quad (.00018) \quad (.0012) \quad (.00001)
\end{aligned}
$$

Where D is a dummy variable for site (Denver = 1), B and C are dummy variables for race (black = 1 and Chicano = 1, respectively), N is number of family members, E is years of schooling, C is experience (defined as age minus years of schooling minus 5), and T is months of training. Standard errors are in parentheses. The R^2s are .085, .077, and .106, respectively. The sample sizes are 2,404, 1,173, and 1,289, respectively. The dependent variable is earnings divided by hours worked in the preexperimental year.

[17] Based on preexperimental income, about 30 percent of the experimental families above the breakeven level.

[18] Under the assumption that substitution effects are constant, it can be shown that families above the tax breakeven level will respond only if income in excess of the breakeven level is less than one-half of the absolute value of the change in income they would experience if they did respond. This amount of income is likely to be small. See the discussion in Robins and West (1978b, 1980c). This statement is based upon the theory described in chapter 2. Less restrictive theories may imply response over a broader range; for instance, the insurance aspects of an NIT in the presence of uncertainty may induce response even by persons who do not move below the breakeven level.

[19] Based on preexperimental income, 4 percent of the husbands, 5 percent of the wives, and 17 percent of the single female heads are predicted to be better off if they choose to remain on welfare. These percentages are somewhat less than the actual percentages of experimental families who reported receiving welfare during the second experimental year—namely, 7 percent for husbands, 10 for wives, and 23 percent for single female heads.

Errors in variables

As described earlier, the calculation of ΔY is based on earnings and nonwage income in the year before enrollment in the experiments. Thus, ΔY depends on both transitory and permanent components of labor supply in the preexperimental year. If response to the NIT depends only on the permanent component of ΔY, our estimated income effect may be biased because of an errors-in-variables problem.[20]

Using a simplified version of the model actually estimated, it is shown in Robins, West, and Keeley (1980, Appendix A) that if response depends only on the permanent component of ΔY, the bias due to errors in variables contains two parts. The first part is multiplicative and biases the estimated income effect toward zero. The second part is additive and biases the income effect positively. It is further shown that if preexperimental labor supply is included on the right-hand side of the equation, an estimator is obtained with only a multiplicative bias toward zero. This bias, however, is smaller than the first component of the bias using direct estimation of equation (3) from chapter 2. For this reason, H_p is included among the explanatory variables and H_e is the dependent variable.

Selectivity bias

The model of labor supply discussed above can be written as

$$\begin{aligned} H_e = {} & b_0 + b_1 H_p + b_2 C + b_3 M + b_4 \Delta Y\,F + b_5 \Delta W\,F \\ & + b_6\,\text{FABOVE}\,F + b_7\,\text{DOMINATE}\,F + b_8\,\text{DECLINE}\,F + \\ & e, \end{aligned}$$

where H_e = hours of work in the second year of the experiment
H_p = hours of work in the year prior to the experiment
C = a vector of control variables
M = a vector of three counseling/training treatment variables
ΔY = the change in disposable income resulting from the NIT treatment, evaluated at preexperimental income (the minimum payment for persons above the breakeven level and for persons having an NIT treatment that is dominated by welfare)
ΔW = change in the net wage rate resulting from the NIT treatment (zero for persons above the breakeven level and for persons having an NIT treatment that is dominated by welfare)
FABOVE = one if above the breakeven level (based on preexperimental income), zero otherwise
DOMINATE = one if NIT treatment dominated by welfare received in preexperimental year (in the absence of welfare, the family would be below the breakeven level), zero otherwise
DECLINE = one if on declining tax rate program and below the breakeven level, zero otherwise
F = one if experimental family, zero if control family
e = a random error term.

Estimates of this model may suffer from a potentially serious selectivity bias, arising because the treatment variables (ΔY, ΔW, etc.) are complicated functions of the financial treatments and preexperimental income. Since similar functions of income are not allowed to affect hours of work for control families, OLS (or similar) estimates could confound the experimental effects with the direct effects of these functions of income. In essence, this problem arises because we are comparing experimentals to noncomparable controls. Rather than comparing each experimental to controls with similar preexperimental incomes, estimates of equation (11) compare each experimental to all controls.[21]

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[20] The substitution effect would also be biased to the extent that the preexperimental tax rate depends on preexperimental labor supply.

The nature of the problem can be seen by examining a simplified model of response that contains only one treatment variable (B):[22]

$$H_e = BFb + u + v_e \,. \tag{7}$$

For simplicity, we assume that the mean of H_e is zero for controls and that the mean of B is zero for experimentals.

We have partitioned the error term, e, into a permanent component, u, and a purely random component, v_e, in order to allow the error term to be autocorrelated. If we estimate equation (7) by OLS, then the expected value of the estimator of b is given by

$$E(b) = b + \frac{S_{Bu}}{S_{BB}}, \tag{8}$$

where S_{Bu} is the covariance of B and u and S_{BB} is the variance of B. Thus, OLS estimation of b is biased unless $S_{Bu}=0$, which is unlikely to be true because B depends on u through H_e preexperimental incomes and labor supply.

A natural way to correct for the bias is to adopt a procedure similar to the procedure used to correct for the assignment process, namely, to include B in the regression equation. The model to be estimated then becomes

$$H_e = Bb' + BFb + u + v_e, \tag{9}$$

where B is defined for both experimentals and controls. If we regress H_e on BF and B, it can be shown (see Robins, West, and Keeley, 1980) that the expected values of the estimated coefficients of BF and B are given by

$$E\begin{bmatrix} \hat{b} \\ \acute{b} \end{bmatrix} = \begin{bmatrix} b \\ S_{B\mu}/S_{BB} \end{bmatrix}. \tag{10}$$

Hence, inclusion of B in the equation leads to an unbiased estimate of the experimental effect b.

If only one NIT treatment were tested in SIME/DIME, it would be straightforward to calculate B for control families. However, more than one NIT treatment was tested. One way of approximating the procedure when there is more than one treatment is to "hypothetically" assign control families to the same set of treatments as experimental families and then calculate the relevant variables on the basis of this hypothetical assignment. To account for the stratified random assignment, the probability of obtaining a given treatment can be set equal to the proportion of experimentals with that treatment having the same normal income level, ethnic group, and family status.

One drawback to this procedure, however, is that the estimated experimental effect will vary depending on the outcome of the hypothetical assignment of controls. Alternative procedures with more desirable properties include (1) repeated assignment of controls (using a Monte Carlo procedure) or (2) assigning each control to all the treatments and taking the weighted average of the calculated breakeven status variables (where the weights represent the probability of an equivalent experimental being assigned to the particular treatment).

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[21] This selectivity bias is discussed in detail in Robins, West, and Keeley (1980), Robins, West, and Stieger (1980), and Burtless and Greenberg (1980).

[22] The argument generalizes easily to a vector of treatment variables.

In spite of its drawback, however, we adopt the procedure of performing only one assignment, described in detail in Robins, West, and Stieger (1980), for simplicity. We then calculate the simulated treatment variables on the basis of income reported in the preexperimental year with the same formulas used to construct treatment variables for experimental families. These simulated treatment variables (defined for everybody) are then included in the response equation.

Estimation procedure

Based on the preceding discussion, the full empirical model we estimate is given by

$$\begin{aligned} H_e = {} & b_0 + b_1 H_p + b_2 C + b_3 M + b_4 \Delta Y\ F + b_5 \Delta W\ F \\ & + b_6\ FABOVE\ F + b_7\ DOMINATE\ F \\ & + b_8\ DECLINE\ F + b_9 \Delta Y + b_{10} \Delta W + b_{11}\ FABOVE \\ & + b_{12}\ DOMINATE + b_{13}\ DECLINE + e. \end{aligned} \tag{11}$$

Because H_e cannot take on negative values, estimation of equation (11) by OLS will lead to inconsistent coefficient estimates if there are numerous observations for which $H_e = 0$. Furthermore, estimates will be inefficient because the error term is heteroscedastic. To account for these statistical problems, we replace the specification in equation (11) with a tobit model, which is designed to handle cases where the dependent variable is truncated normal.[23] The tobit model may be written as:

$$\begin{aligned} H_e = \mathrm{Max}[& b_0 + b_1 H_p + b_2 C + b_3 M + b_4 \Delta Y\ F + b_5 \Delta W\ F + b_6 FABOVE\ F \\ & + b_7 DOMINATE\ F + b_8\ DECLINE\ F + b_9 \Delta Y + b_{10} \Delta W \\ & + b_{11} FABOVE + b_{12} DOMINATE + b_{13}\ DECLINE + e; 0], \end{aligned} \tag{12}$$

where e is a random error term assumed to be distributed normally with variance σ^2. The b_i are estimated by maximum likelihood using an iterative maximization technique.

Parameters in a tobit model cannot be interpreted in the same way as parameters in a linear model. For a linear model, treatment parameters are interpreted as the average response of the population to the imposition of a negative income tax. In a tobit model, treatment parameters give the average response only of persons whose response is not truncated (hours do not fall to zero). The response of all other persons is somewhat smaller in magnitude because of the lower bound on the dependent variable. Coefficients of ΔY and ΔW, however, can be interpreted as income and substitution effects for persons whose response is not truncated at zero.

Results

Estimation results are presented in table 3.11.[24] There are statistically significant experimental effects on hours worked for husbands, wives, and single female heads. The income effect (the coefficient of ΔY) is significant for all three groups. While the wage effect is not significant for any group, it is positive for all three groups and is substantial in magnitude.[25] Each dollar decrease in the net wage rate has an effect roughly similar to a $1,000 increase in disposable income.

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[23] See Amemiya (1973) or Tobin (1958) for a discussion of the tobit model.

[24] Means and standard deviations of the variables, estimated coefficients of the other variables, and results excluding the simulated treatment variables are presented in Robins, West, and Keeley (1980).

[25] Because the standard errors of the estimated substitution effects are larger than those of the income effects, it may appear that the substitution effects are less precisely estimated. However, the mean change in ΔW (in dollars) is considerably less than the mean change of ΔY (in thousands of dollars). These mean changes are reported in table 3.13 for persons below the breakeven level. If one takes the product of the mean change and the standard error as a measure of precision, the substitution effect is about 30 percent more precise than the income effect for husbands and wives and 15 percent more precise for single female heads. This conclusion is strengthened when one realizes that the experimental mean for ΔY is large relative to that of politically feasible NIT programs, but the experimental mean of ΔW is near the range of feasible programs. The mean change in ΔW is larger for husbands than for wives because they have larger gross wage rates and not because they experience larger tax rate changes.

TABLE 3.11.—ESTIMATED EFFECTS OF NIT ON ANNUAL HOURS OF WORK[1]

[Estimated standard errors in parentheses]

	Husbands	Wives	Single female heads
ΔY ($1,000's)	[2] −95.15	[3] −196.75	[2] −147.12
	(41.59)	(63.75)	(70.28)
ΔW ($1 per hour)	71.33	151.76	176.37
	(56.31)	(136.33)	(119.57)
1 = Above breakeven level	85.46	[2] −201.07	139.30
	(64.22)	(97.71)	(128.86)
1 = Declining tax rate program and below breakeven level	−110.20	113.81	−53.07
	(70.90)	(111.21)	(118.44)
1 = NIT dominated by welfare	−191.48	−317.48	−123.47
	(157.15)	(237.40)	(131.06)
Sample size	2,249	2,332	1,715

[1] Tobit estimates.
[2] Significant at the 5-percent level.
[2] Significant at the 1-percent level.

The other experimental variables are generally not significant. There is, however, a significant decrease in hours worked for wives initially above the breakeven level. However, the above breakeven effects for husbands and single female heads are positive and not significant. Families on the declining tax rate programs (below the breakeven level) do not appear to be responding differently than families on the constant tax rate program (although the results suggest a somewhat greater response for husbands). The coefficients of the domination variable are negative, but not significant, for all three groups.[26]

Mean effects of the experimental treatments

Based on the results in table 3.11, we have calculated estimates of the mean effects of SIME/DIME on hours of work for persons initially below the breakeven level whose treatment is not dominated by welfare. These calculations are presented in table 3.12. When interpreting these mean effects the reader should recall that the average SIME/DIME program is quite generous, with an average support level of 110 percent of the poverty level and an average tax rate of about 50 percent. The expected change due to the experiment is calculated for each person using his or her values of ΔY, ΔW, and DECLINE and accounts for the truncation of response at zero hours of work.[27] Expected changes are then averaged over all persons initially below the breakeven level whose NIT payment is not dominated by welfare. Estimated mean effects indicate a substantial disincen-

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[26] The estimated coefficients of the simulated welfare domination variable (not reported in table 3.11) are similar in magnitude, but of opposite sign. Thus, similar controls increase their labor supply over time, whereas experimentals keep their labor supply roughly constant. This might arise because it is more difficult to maintain eligibility for existing welfare programs than for the experimental programs.

[27] The formula used for the expected change due to the experiment is given by

$$\mathrm{E}(\Delta \text{ hours}) = X\beta\, F\left(\frac{X\beta+T\gamma}{\sigma}\right) - F\left(\frac{X\beta}{\sigma}\right) + T\gamma F\left(\frac{X\beta+T\gamma}{\sigma}\right)$$

$$+ \sigma\, f\left(\frac{X\beta+T\gamma}{\sigma}\right) - f\left(\frac{X\beta}{\sigma}\right),$$

where
X = a vector of all nontreatment variables (including the simulated treatment variables)
β = the vector of estimated coefficients of the nontreatment variables from the hours, weeks, or probability of working equations, as appropriate
T = a vector of treatment variables (ΔY, ΔW, DECLINE)
γ = the vector of estimated coefficients of the treatment variables from the appropriate equation
σ = the estimated standard deviation of the error term in the appropriate equation
$F(\cdot)$ = the cumulative standard normal distribution
$f(\cdot)$ = the standard normal density.

tive effect, particularly for women and are very similar to dummy variables estimated for persons initially below the breakeven level (Robins, West, and Stieger, 1980). Thus, the structural model adequately explains the mean response of persons initially below the breakeven level.

TABLE 3.12. MEAN EFFECTS ON ANNUAL HOURS OF WORK FOR PERSONS INITIALLY BELOW THE BREAKEVEN LEVEL

[Percentage effects in parentheses]

	Husbands	Wives	Single female heads
Mean effects	−227 (−12.5)	−139 (−23.4)	−207 (−20.7)
Means of:			
ΔY ($1,000's)	1.378	1.387	1.258
ΔW ($1 per hour)	−.728	−.447	−.625
Decline	.496	.495	.448

Table 3.13 presents estimated income and substitution effects on hours of work evaluated at the sample means of ΔY and ΔW. These income and substitution effects are for persons below the breakeven level who are working in both the preexperimental and experimental periods. The estimated effects do not take into account either the truncation at zero hours of work or the effect of being on a declining tax rate program. They do, however, allow us to determine the relative importance of income and substitution effects for persons who are able to respond fully to the treatments.

Results in table 3.13 indicate that most of the experimental response is due to the income effect. The percentage of the total effect accounted for by the income effect is 73 percent for husbands, 84 percent for wives, and 67 percent for single female heads. However, it is important to note that these effects are based on the mean changes in disposable income and net wage rates that result from the set of programs tested in SIME/DIME, rather than for any single NIT program. Less generous programs—the more likely policy alternatives—would have smaller overall effects, in which the substitution effect is relatively more important. We examine such programs in chapter 8 where we use these estimates to predict the effects of nationwide NIT alternatives.

TABLE 3.13.—ESTIMATED INCOME AND SUBSTITUTION EFFECTS ON ANNUAL HOURS OF WORK AT THE MEAN [1]

[Estimated standard errors in parentheses]

	Husbands	Wives	Single female heads
Effects at the mean			
Income ($b_4 \overline{\Delta Y}$)	[2] −133.7 (58.4)	[3] −269.4 (87.3)	[2] −175.1 (83.7)
Substitution ($b_5 \overline{\Delta W}$)	−49.7 (39.2)	−49.6 (44.9)	−87.7 (59.4)
Total	[3] −183.4 (62.9)	[3] −319.0 (85.2)	[3] −262.8 (86.3)
Means			
$\overline{\Delta Y}$ ($1,000's)	1.405	1.369	1.190
$\overline{\Delta W}$ ($1 per hour)	−.697	−.329	−.497

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[1] For persons below the breakeven level working in both the experimental and preexperimental periods.
[2] Significant at 5-percent level.
[3] Significant at 1-percent level.

Effects by experimental duration

As the theoretical discussion in chapter 2 indicates, in a finite duration experiment the income effect is likely to be underestimated and the substitution effect overestimated. These biases should be reflected in differences in the estimated income and substitution effects between persons who are eligible for the experimental NIT payments for 3 and 5 years. Table 3.14 presents separate structural model estimates of the experimental effects by experimental duration. The χ^2 tests indicate that the differences by experimental duration are not statistically significant. However, estimated income effects for husbands and wives on the 5-year program are substantially higher than the estimated income effects for those on the 3-year program. There is virtually no difference in the estimated income effects for single female heads. Results for husbands and wives are consistent with the hypothesis that the income effect of a finite duration experiment is less than the income effect of a permanent program because lifetime income is changed less the shorter the program.

TABLE 3.14.—ESTIMATES OF THE STRUCTURAL MODEL WITH DIFFERENT EFFECTS FOR THE 3- AND 5-YEAR PROGRAMS

[Estimated standard errors in parentheses]

	Husbands		Wives		Single female heads	
	3-year	5-year	3-year	5-year	3-year	5-year
Coefficient estimates						
$Y_c(H_p)$	−72.50	[2]−147.17	[2]−172.72	[3]−256.15	[1]−143.79	−152.83
	(44.67)	(57.76)	(69.14)	(89.99)	(74.79)	(98.22)
W	69.31	68.16	221.08	−17.74	171.70	188.22
	(60.58)	(80.35)	(148.07)	(194.55)	(127.38)	(174.17)
1 = Above breakeven level	110.70	43.91	−164.93	[2]−259.82	213.92	2.47
	(71.90)	(86.13)	(109.70)	(132.37)	(142.38)	(171.01)
1 = Declining tax rate program and below breakeven level	−125.54	−72.88	−103.72	126.16	−74.00	12.51
	(76.83)	(98.07)	(121.41)	(155.894)	(125.66)	(174.35)
1 = NIT dominated by welfare	−81.62	[2]−511.23	−392.26	−156.53	−135.56	−92.01
	(167.57)	(230.91)	(261.98)	(329.23)	(140.64)	(186.57)
2 test for	8.06		2.64		1.87	
Calculated variables						
R = 5/3	2.03		1.48		1.06	
r	.01		.13		.90	
p	−781.6		−525.1		−154.3	

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Only for wives is there a substantial difference in substitution effects between the two experimental durations. For wives, the 3-year substitution effect is considerably larger. This is consistent with the notion that the substitution effect in a temporary program will induce greater intertemporal substitution.

Keeley and Wai (1980) used the results reported in table 3.14 to calculate implied rates of discount and permanent income effects. The calculated discount rate is based on the assumption that persons initially below the breakeven level expect to remain below the breakeven level for their entire period of eligibility, or that periods below the breakeven level are evenly distributed throughout the period of eligibility.[28] The implied discount rate is based on the formula

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[28] In fact, this may not be the case since the number of months of participation for 5-year families was only 46 percent (husbands), 55 percent (wives) and 65 percent (single female heads) greater than that for 3-year families, whereas their period of eligibility was 67 percent greater (see Keeley and Wai (1981)).

$$\frac{\beta_5}{\beta_3} = \frac{1-e^{-5r}}{1-e^{-3r}}, \tag{13}$$

where β_5 = the income effect for persons on the 5-year program
β_3 = the income effect for persons on the 3-year program
r = the discount rate.

For husbands, the formula implies a negative value for r since the estimated β_5/β_3 is outside its theoretical range (1 to 1.67). Keeley and Wai chose to assign a small positive value (1 percent) to r for husbands. The calculated discount rate is 13 percent for wives and 90 percent for single female heads (table 3.14). Keeley and Wai then calculated the implied permanent income effects using the formula

$$\beta_p = \frac{1-e^{-Nr}}{1-e^{-5r}}\beta_5, \tag{14}$$

where β_p = the permanent income effect
N = the length of a permanent program (assumed to be 30 years.)[29]

This calculation assumes that persons would participate at the same rate throughout the period covered by the permanent program. The permanent income effects (last row of table 3.14) are substantially greater than the 5-year income effects for husbands (430 percent greater)[30] and wives (87 percent greater) but are about the same for single female heads.[31] These calculated permanent effects imply that the response to a permanent program for husbands and wives might be much larger than the response observed during the experiment. However, in all cases a 90 percent confidence interval around β_5/β_3 contains the entire theoretical range. Consequently, a corresponding confidence interval for r would range from 0 to $+\infty$, since $r = 0$ when $\beta_5/\beta_3 = 1.67$ and $r = +\infty$ when $\beta_5/\beta_3 = 1$. Thus, the empirical estimates yield very little formation on the true magnitude of the permanent effect. A 90 percent confidence interval for $\beta_p$would include both β_5 and β_3 as well as the calculated β_p.

Keeley and Wai (1981) also estimated a model in which the hours worked response during the second year of the experiment is proportional to the number of months of participation during the entire experiment. The model was estimated using predicted months of participation and appears to be consistent with the data for husbands and single female heads, but not for wives. Their estimates indicate a response of about −8 hours per year for each month of participation for husbands and single female heads. Extrapolation of these results to longer periods of eligibility also leads to large estimates of the permanent response. Like the estimates from their other model, these estimates of the permanent effects are quite imprecise.

In addition to being imprecise, the calculated permanent effects in both models are based upon a number of strong assumptions that are probably unrealistic. First, as mentioned above, the participation rate is assumed to be the same throughout the 30 years of the permanent program as it was during the experiment. This is unlikely to occur under a real permanent program for several reasons. Most people are apt to increase their incomes over their life cycles. The families in the experiment have temporarily low incomes because of the way the samples are chosen. Their incomes will increase automatically because of "regression toward the mean" (see the earlier discussion in chapter II). This phenomenon is especially important for the group of people whose incomes initially fell below the breakeven level. Thus, the average participation rate is likely to be lower in a national program than in the experiment.

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[29] The assumption of 30 years is based on the difference between the current average age of the sample (35) and the age at which persons become eligible for full social security retirement benefits (65).

[30] The calculated β_p for husbands would be −636 rather than −782 if based on β_3 and on r=.01. The choice between β_3 and β_5 does not matter for wives and single female heads because their values of r are consistent with β_5/β_3.

[31] Keeley and Wai calculate large permanent effects for female heads when they use the income effect for 3-year families in the second year and the income effect for 5-year families in the third year.

Another assumption behind the calculations is that the differences in the estimated income effect between the 3- and 5-year programs is generated solely by the different changes in lifetime income. However, other causes are also possible. For example, there may be substantial fixed costs associated with labor supply changes. This would be particularly true if the person had to leave his/her current job to accomplish the change; leaving a job may imply not only costs in searching for a new job but also the potential loss of pension benefits and the return to firm-specific human capital. Although such fixed costs would also suggest that longer duration programs would have larger effects, the method of extrapolation from the difference between the 3- and 5-year programs to a permanent program may be vastly different from the methods used in either the permanent income model or the months of participation model.

In addition to the above problems, some of the year-by-year results in Keeley and Wai (1980) are inconsistent with the underlying theory. For example, since only changes in lifetime income matter in the model, experimental NIT payments should have the same income effect after the experiment as during it. In fact, estimated income effects tend to be much lower in the postexperimental period. These differences are not significant, but they are larger than the differences between β_3 and β_5 which are used to calculate the permanent income effect. In addition, postexperimental 5-year program income effects are generally less than third year 3-year program income effects even though the permanent income theory implies the opposite. Finally, estimated mean effects for persons on the 20-year program presented in table 3.7 are not supportive of a substantially larger response for persons on longer programs. Only for single female heads is the 20-year program response larger than the 5-year program response. And even for this group the difference is small.

Given the large standard errors of the calculated permanent effects, the strong assumptions underlying the computations, and the large number of findings inconsistent with theory, the Keeley-Wai results are best viewed as suggestive. Therefore, we conclude that SIME/DIME data are not rich enough to shed much light on the biases inherent in limited duration experiments.

An alternative estimate of the structural model

Estimates of the structural model presented in the previous sections calculate the values of the treatment variables (ΔY, ΔW, FABOVE, DECLINE, and DOMINATE) on the basis of incomes and labor supply during the year prior to the beginning of the experiment. In theory, however, these variables should be measured during the experimental period but before any response to the experiment in order to minimize effects caused by changes in the control environment. Ideally, it would be most desirable to calculate the treatment variables on the basis of what each person's income would have been in the absence of the treatments during the second year of the experiment. Unfortunately, income in the absence of the treatments is not observable during the second year. Thus, estimates presented earlier used the preexperimental situation as a proxy for this "nonexperimental" situation.

However, nonexperimental incomes might differ substantially from preexperimental incomes for several reasons. First, persons were chosen to participate in the experiment (either as experimentals or controls) in part because their incomes fell below a certain level during the period prior to the experiment. To the extent their incomes were only temporarily low, the incomes of sample members will tend to increase over time as income regresses towards the mean. Second, there may have been changes in other tax and transfer programs over time that are not reflected in either preexperimental labor supplies or the algorithm used to calculate the treatment variables. Third, the trend of increasing labor supply for women implies an increase in their incomes over time. Fourth, there may have been changes in incomes due to cyclical changes in labor market conditions.

Other statistical models attempt to account for these changes by using an instrumental variables procedure based upon the incomes of controls during the second year of the experiment. Predicted second year values of ΔY, ΔW, FABOVE, DOMINATE, and DECLINE before response are developed using data from the control group. These predicted variables are then used as treatment variables in the hours of work equation, instead of the treatment variables used earlier based on preexperimental incomes.

We calculate second year values of ΔY, ΔW, FABOVE, DOMINATE and DECLINE for controls using their simulated NIT treatments and their second year incomes and family structure.[32] The formulas used are similar to those used for the preexperimental period except that they account for effects of the cost of living adjustments used in SIME/DIME on real payments, and they use the second year structure of the positive tax system. After calculating current values of the treatment variables, regression equations are estimated in which the dependent variables are the calculated treatment variables and the independent variables are a set of preexperimental and assignment variables. The regressisons are run using the sample of controls. Appendix E in Robins and West (1980a) presents a complete description of these estimated equations. These equations are used to calculate predicted treatment variables for experimental families based on their values of the independent variables, which are then entered into the hours of work equations. As in our other estimates, simulated treatment variables are also included in these equations. The simulated treatment variables are also predicted values, but they are calculated for control families as well as for experimental families.

Estimated income and substitution effects using these predicted treatment variables are presented in table 3.15. The estimates presented earlier are repeated to facilitate comparisons. These new estimates are substantially different from the basic model estimates. It should be noted that the use of predicted treatment variables causes a substantial loss of precision in the estimates (as indicated by the larger standard errors). For husbands, the income effect is slightly larger and the substitution effect is much larger than in the basic model. However, neither the income effect nor the substitution effect is significantly different from zero because of the large standard errors. For wives, both the income effect and the substitution effect are larger than the previous estimates. Only the income effect is significant, however. For single female heads, both income and substitution effects are larger than previous estimates, but neither is significant. For all three groups, a greater portion of the response is due to the substitution effect because the substitution effect is increased relatively more than the income effect.[33] However, the lack of statistical significance and large standard errors of these estimates limit their usefulness for purposes of prediction relative to the basic structural estimates.

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[32] The simulated NIT treatments are identical to those discussed earlier in this chapter in the section "Selectivity Bias."

[33] Dickinson (1980) and Johnson and Pencavel (forthcoming) also find larger substitution effects using somewhat different structural models.

TABLE 3.15.—COMPARISON OF BASIC ESTIMATES OF THE STRUCTURAL MODEL WITH ESTIMATES BASED ON PREDICTED SECOND YEAR TREATMENT VARIABLES

[Standard errors in parentheses]

	ΔY	ΔW	F ABOVE	Decline	Dominate
Husbands:					
Basic model	[1]−95.2	71.3	85.5	−110.2	−191.5
	(41.6)	(56.3)	(64.2)	(70.9)	(157.2)
Predicted treatment variables	−137.3	149.4	57.3	−123.3	−317.5
	(105.3)	(126.2)	(71.7)	(127.2)	(310.8)
Wives:					
Basic model	[1]−196.8	151.8	[1]−210.1	113.8	−317.5
	(63.8)	(136.3)	(97.7)	(111.2)	(237.4)
Predicted treatment variables	[1]−277.7	460.2	−7.9	221.5	−641.4
	(132.2)	(331.3)	(117.5)	(179.1)	(412.4)
Single female heads:					
Basic model	[1]−147.1	176.4	139.3	−53.1	−123.5
	(70.3)	(119.6)	(128.9)	(118.4)	(131.1)
Predicted treatment variables	−247.1	337.3	179.8	123.8	[1]−427.7
	(160.9)	(235.2)	(140.3)	(176.9)	(207.9)

[1] Significant at the 5-percent level.

Labor force status and other measures of labor supply

The first part of this chapter presented a comprehensive analysis of the effects of SIME/DIME on annual hours of work. As indicated in the introduction to this chapter, however, effects on annual hours of work tell only part of the story. Also important from a policy perspective are effects on the various components of annual hours of work, namely the probability of working, weeks worked, and hours worked per week. Furthermore, effects on other measures of labor supply that are either derived from the components of annual hours of work (e.g., duration of employment and nonemployment) or are based on alternative conceptual definitions of labor supply (e.g., desired hours of work) are also of considerable policy interest. Evidence about the size of the experimental impact on these various measures of labor supply can have important implications for the design of a nationwide NIT program. For example, the finding that the probability of working is reduced under an NIT may suggest that more attention be paid in program design to work requirements as a method of reducing work disincentives and program costs.

In the last part of this chapter, we present evidence on the size of the experimental impact on several other measures of labor supply. Table 3.16 shows mean effects of the NIT treatments on various measures of labor supply in the second year of the experiment. Results are based on the single dummy variable model specification which was used for hours of work in "Single Dummy Variable Model Estimates" earlier in the chapter.

TABLE 3.16.—AVERAGE EFFECTS OF THE EXPERIMENT ON ALTERNATIVE MEASURES OF LABOR SUPPLY IN THE SECOND YEAR

[Standard errors in parentheses]

	Husbands	Wives	Single female heads
Labor force status:			
Weeks employed	[2]−2.84	[2]−3.79	[2]−3.91
	(.69)	(.82)	(.99)
Weeks unemployed	[2] 2.16	.14	−.01

TABLE 3.16.—AVERAGE EFFECTS OF THE EXPERIMENT ON ALTERNATIVE MEASURES OF LABOR SUPPLY IN THE SECOND YEAR—Continued

[Standard errors in parentheses]

	Husbands	Wives	Single female heads
	(.54)	(.40)	(.63)
Weeks out of labor force	.68	[1] 3.64	[2] 3.92
	(.55)	(.85)	(1.00)
Worked at all	[2] −.045	[2] −.109	[2] −.073
	(.012)	(.021)	(.023)
Year-end labor market status			
Employed overtime	−.031	−.004	−.008
	(.019)	(.008)	(.012)
Employed fulltime	[1] −.055	[2] −.054	[2] −.087
	(.023)	(.017)	(.024)
Employed parttime	.013	[2] −.039	−.012
	(.011)	(.014)	(.016)
Unemployed	[2] .039	[1] .019	.018
	(.014)	(.011)	(.017)
Out of labor force	[1] .033	[2] .078	[2] .074
	(.014)	(.021)	(.024)
Desired hours of work	[1] −57.6	[1] −139.7	−168.0
	(29.4)	(32.6)	(41.2)
Hours worked	[2] −163.1	[2] −141.8	[2] −150.0
	(32.5)	(30.6)	(39.3)
Earnings	[2] −485.4	[2] −300.7	[2] −426.6
	(134.6)	(75.9)	(104.0)
Log wage rate	−.005	−.005	[2] −.070
	(.018)	(.030)	(.028)
Sample size	2,284	2,369	1,715
Sample size for wage rate	1,972	1,125	1,053

[1] Significant at the 5-percent level

[2] Significant at the 1-percent level

Labor force status

Weeks worked, unemployed, and out of the labor force

The first measures we consider are annual weeks worked, weeks unemployed, and weeks out of the labor force.[34] The principal difference between annual weeks worked and annual hours of work is that the former does not account for variation in hours worked per week. If there are significant effects of the experiment on weeks worked, then the effects on weeks out of the labor force and weeks unemployed allow us to infer whether individuals are withdrawing from the labor market or extending periods of unemployment in response to the experimental treatments.

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[34] The interview data collected for this analysis contain information on spells of employment and nonemployment. A person is defined to be working as long as he or she held a job with positive weekly hours. Periods of paid sick leave and paid vacation are treated as work. Periods of nonemployment, including periods when a person held a job but did not work because of layoff, are classified as either periods out of the labor force or periods of unemployment. A person is classified as out of the labor force during a period if the person did not want to work or did not look for work; a person who both wanted to work and looked for work is classified as unemployed. The interviews did not collect data about spells of nonemployment lasting less than 7 days; persons in such spells are classified as unemployed. Since these three categories of labor force status are mutually exclusive and exhaustive, the sum of the three proportions of time is equal to one. The equations for these variables are estimated by OLS. Since the dependent variables are limited in range (weeks in a work status is always between 0 and 52), OLS is not an entirely appropriate procedure. The OLS predictions do not necessarily satisfy the range restrictions, and the error terms are not homoscedastic, as is necessary for OLS to be efficient. However, since the range of predictions is apt to be fairly narrow and bounded away from 0 and 52, and since OLS predictions satisfy the constraint that the sum of the three categories equals 52, we believe that use of OLS estimates of the experimental effects does not yield misleading estimates, although the standard errors may be biased.

As was the case for hours of work, the results indicate large and significant reductions in weeks worked for all three groups. For husbands, annual weeks worked decreased by 2.8 or approximately 7 percent; this compares to a 9 percent decline in annual hours (Table 3.16). The results for unemployment and being out of the labor force indicate that most of the employment response for husbands is due to a significant increase in unemployment. Although weeks employed decreased by 2.8 weeks, weeks unemployed increased by 2.2 weeks. This represents a 48 percent increase in unemployment. Thus, for husbands, NIT programs may result in a substantial increase in reported unemployment.

This result is difficult to interpret. To the extent that unemployment is involuntary, there is no reason to expect unemployment to be affected by the experiment. However, it is possible that experimental husbands are using the NIT payments to allow them to take the time to search for better jobs than they would accept if they did not have the NIT payments. That is, the NIT may allow them to be more selective about the kinds of jobs they accept. If this were true, we would expect them eventually to find better jobs than are found by similar controls. However, the evidence indicates that experimental husbands did not find jobs with higher wage rates than did controls. On the other hand, it is also possible that experimentals who have reduced their employment because of the experiment are unwilling to admit that their employment reduction is, in fact, voluntary. That is, they may claim to be looking for work when they are actually out of the labor force. There is no financial incentive to report in this manner because the NIT payments do not depend on the labor force status of nonworkers; however, they may not recognize themselves that they are not actually looking for work.

For wives, the weeks worked response is −3.8 weeks or approximately a 19 percent decline. This compares to a 21 percent decline in annual hours of work in the same period. Virtually all of this reduction in weeks worked is reflected in a significant increase of 3.6 in weeks out of the labor force, an 11 percent increase.

Results for single female heads are similar to those for wives. The weeks worked response is −3.9, or a 14 percent decline. This is identical to the 14 percent decline in annual hours of work during the same period. Weeks out of the labor force increase by the same amount as weeks worked decline (3.9), while weeks unemployed were unchanged.

These results indicate that the experimental NIT plans induce a reduction in weeks worked and, hence, in the probability of working at any given time. Therefore, rather than each person reducing weekly hours by approximately the same amount, it appears that some individuals are making large reductions in weekly hours of work (to zero), whereas others are not responding at all. These results do not, however, indicate whether employment reductions are spread evenly through the sample or concentrated among a few individuals. Nor do the results indicate whether the reductions in employment are due to an increased willingness to leave employment or to a reduced likelihood of moving from not employed to employed. We explore these issues next.

Probability of working at all during a year

Table 3.16 also presents estimates of the effects of the NIT treatments on the probability of working at all during the second year. These estimates are large and significant for each group. The probability of working decreases by .045 for husbands, by .109 for wives, and by .073 for single female heads. These represent percentage reductions of 7 percent, 19 percent, and 11 percent, respectively. Similar results persist throughout the experimental period (West and Stieger, 1980). Thus, a substantial number of people are withdrawing from the work force for long periods of time. Whether this withdrawal accounts for a major portion of the total hours worked response is unclear since we do not know how much these persons would have worked if it were not for the NIT.

A stochastic model of work force status

Tuma, Robins, and Smith-Donals (1980) estimated a stochastic model of work force status using a methodological framework developed by Tuma and Robins (1980), and Robins, Tuma, and Yaeger (1980) for analyzing transition rates of entering and leaving employment. This approach enables us to determine whether the employment response discussed above is due to increases in the likelihood of leaving employment or decreases in the likelihood of obtaining employment.

Their approach is based on the following statistical model:

$$\ln r = Xa + Fb, \tag{15}$$

where $\ln r$ = natural logarithm of the instantaneous rate of entering (or leaving) employment
X = vector of nonexperimental (and counseling/training) variables
F = dummy variable for NIT treatment
a, b = parameters to be estimated.

As described in Tuma, Robins, and Smith-Donals (1980), this model can be estimated by the method of maximum likelihood using event history data.[35] Separate models are estimated for entering and leaving employment. In this analysis unemployment and being out of the labor force are both treated as nonemployment. From the estimates, predictions of the probability of employment at various points in time, the duration of employment and nonemployment, and the frequency of employment and nonemployment are easily derived.

Table 3.17 presents estimated effects of the NIT treatments on the (logarithm of the) rates of entering and leaving employment for husbands, wives, and single female heads. The results show a consistent, significant decrease in the rate of entering employment as a result of the NIT treatments. The estimates imply a decrease in the rate of entering employment of 31 percent for husbands, 29 percent for wives, and 41 percent for single female heads of families.[36] These results imply, for example, that nonemployed husbands with an NIT treatment are 31 percent less likely to become employed than are controls during any short period of time. The treatments tend to increase the rate of leaving employment for husbands and single female heads of families (by 16 percent for both groups), but not for wives.[37]

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[35] The statistical model is identical to that used in Part V of this volume to study marital status changes. See Part V for a general discussion of the statistical model, or Tuma and Robins (1980) for a discussion of the model as it is applied to study employment status changes. The model uses information on the length of time a person is in a particular employment state. Hence, more than one observation is possible per individual. The model also accounts for the fact that many spells are not completely observed due to termination of interviewing (including attrition), termination of treatment eligibility, and changes in marital status. Observations are terminated when a marital status change occurs because work by Tuma (1979) suggests that ignoring the occurrence of a rare event like a change in marital status can bias the estimated coefficients in a model explaining more frequent events like employment status changes. Between one-fourth and one-half of all spells are censored.

[36] The percentage effects are calculated as $(r_e - r_c)/r_c$, where r_e is the transition rate when the experimental treatment variable is equal to 1 and r_c is the transition rate when the treatment variable is equal to 0.

This expression can be written as $e^{b_{jk}} - 1$, where b_{jk} is the estimated coefficient of the treatment variable for moving from state j to state k.

[37] It is interesting to note that the direction of the experimental effect on the rate of leaving employment is inconsistent with Spilerman and Miller's finding (1977) of reduced job departures among husbands in the New Jersey NIT experiment. However, the analysis discussed here examines departures from employment, whereas Spilerman and Miller examine departures from jobs (i.e., employers).

TABLE 3.17.—ESTIMATES OF THE FINANCIAL TREATMENT EFFECTS ON THE RATES OF ENTERING AND LEAVING EMPLOYMENT AND PREDICTED FINANCIAL TREATMENT EFFECTS ON VARIOUS MEASURES OF EMPLOYMENT BEHAVIOR

[Estimated standard errors in parentheses]

	Husbands	Wives	Single female heads
Log rate of entering employment	[3] −.370	[3] −.340	[3] −.521
	(.047)	(.047)	(.064)
Log rate of leaving employment	[3] .145	.079	[3] .153
	(.045)	(.050)	(.064)
Predicted Effects			
Probability of working:			
After 1 year	[3] −.054	[3] −.056	[3] −.086
	(.007)	(.009)	(.011)
Steady state	[3] −.063	[3] −.079	[3] −.126
	(.008)	(.013)	(.016)
Duration of nonemployment (percentage)	[3] 42.3	[3] 49.0	[3] 66.6
	(7.7)	(5.7)	(11.0)
Duration of employment (percentage)	[3] −15.8	[2] −12.6	[3] −20.9
	(4.8)	(5.7)	(7.1)
Number of entries into nonemployment within 1 year (percentage)	[2] 14.0	[1] −6.4	−2.1
	(5.9)	(3.4)	(5.5)
Number of entries into employment within 1 year (percentage)	[2] −7.2	[3] −23.3	[3] −26.7
	(3.1)	(3.6)	(3.6)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Tuma, Robins, and Smith-Donals (1980) also consider several possible interactive effects to learn more about the nature of responses. The results generally suggest that the responses tend to be concentrated more among persons who are younger, less educated, less attached to the labor force, and have lower income.

Table 3.17 also presents the predicted effects of the NIT treatments on several outcome measures related to the rates of entering and leaving employment. These predictions are discussed in detail in Robins, Tuma, and Yaeger (1980). After the first year, the NIT is predicted to reduce significantly the probability of employment by 0.05 for husbands, by 0.06 for wives, and by 0.09 for single female heads. In the long run (or steady state), the predicted effects are −.06 for husbands, −.08 for wives, and −.13 for single female heads.[38]

The experiment substantially lengthens periods of nonemployment for all groups. The average length of nonemployment spells increases by 14 weeks (42 percent) for husbands, 69 weeks (49 percent of 2.7 years) for wives, and 68 weeks (67 percent) for single female heads. Employment spells are shortened by 34 weeks (16 percent) for husbands, 11 weeks (13 percent) for wives, and 33 weeks (21 percent) for single female heads.

The final two sets of predictions in table 3.17 show how the financial treatments affect the expected number of entries into employment and nonemployment during the first year of the experiment. The results indicate that, with one exception, the experiment actually reduces the expected number of entries into employment and nonemployment. Because the work disincentive effect of the financial treatments primarily lengthens spells of nonemployment, the treatments lead to less individual movement in the labor market.

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[38] In a study of the interaction between husband and wife employment behavior using the same methodological framework, Tuma and Smith-Donals (1980) find that the NIT treatments decrease the long-run probability of both a husband and wife working and increase the long-run probability of neither one working. However, the study also finds that the long-run probability of only the husband working (i.e., the probability that the husband works and the wife does not work) increases under the NIT treatments.

Other measures of labor supply

Full-time, part-time, and overtime work

We examine next effects of the experimental NIT treatments on a more detailed measure of work status that distinguishes part-time, full-time, and overtime work, as well as being out of the labor force and unemployed. Variables used are defined on the basis of the person's status as of a single day in the year (the 15th of the last month of the second year).[39] This analysis allows us to determine to what extent the work effort response to the experiment is due to shifts from overtime to full-time work, or full-time to part-time, etc. One might expect that most of the NIT response would be concentrated in part-time work if marginal workers are induced to leave the work force. In fact, this is not the case. For husbands, as seen in table 3.16, all the employment response occurs in overtime and full-time work. For single female heads all the response is in full-time work. Only for wives is there a substantial reduction in part-time work. This reduction accounts for about half the employment response of wives. Although it is possible that full-time workers were shifting to part-time work while part-time workers were ceasing to work, the net observed effect is a reduction in full-time and overtime work with a reduction in part-time work occurring only for wives.

Estimated responses for husbands represent a 12 percent decrease in the probability of working overtime, a 11 percent decrease in the probability of working full time and a 24 percent increase in the probability of working part time. Consequently, experimental husbands who worked, tended to work fewer hours per week than controls with comparable characteristics. The responses for wives imply an 11 percent reduction in the probability of working overtime, a 22 percent reduction in the probability of working full time, and a 29 percent reduction in the probability of working part time. Consequently, it appears that experimental wives who continued to work actually work more than comparable controls. Responses for single female heads imply a 16 percent reduction in the probability of working overtime, a 21 percent reduction in the probability of working full time, and a 12 percent reduction in the probability of working part time.

Desired hours of work

It is of considerable interest to assess the impact of the experimental NIT plans on desired hours of work because desired hours may come closest to the theoretical concept of "labor supply." Unemployed individuals are in a state of disequilibrium with respect to labor supply and in many respects are more similar to workers than to persons who are out of the labor force. Given the significant level of unemployment in Seattle at the beginning of the experiment, this may have some implications for evaluating labor supply response in that city. Desired hours of work is a measure of labor supply that considers employment, unemployment, and underemployment as equivalent. We define desired hours of work as the product of weekly desired hours of work while working and weeks worked during the year, plus the product of desired weekly hours of work while unemployed and weeks unemployed.[40]

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[39] A person is classified as working part time if he or she is working at a weekly rate less than 35 hours per week. Full-time work is defined as being between 35 and 40 hours per week. Over 40 hours per week is defined as overtime work. Forty hours per week was chosen as the separation point between full-time and overtime work because federal wage and hours laws require premium pay for work in excess of 40 hours per week. Thirty-five hours per week was chosen as the demarcation point between full-time and part-time work because 35 hours appears to be about the lower boundary for full-time work (five 7-hour days or four 9-hour days) and because there appears to be a decline in the frequency density of hours per week below 35. The estimation procedure used is OLS. Since the dependent variables are binary, of course, OLS is not an entirely appropriate procedure. Appendix B in West and Stieger (1980), however, shows that in these models OLS and logit give similar results.

The experimental desired-hours response may be closer than the experimental actual-hours response to the response to a national NIT program for two basic reasons. First, the response of actual hours is likely to vary with the extent of unemployment in the labor market. Persons who are not working because they have been laid off or can't find a job cannot reduce their hours of work. Consequently, the response of actual hours is likely to be smaller in a region or period of relatively high unemployment than in a region or period of low unemployment. Indeed, the results in "Single Dummy Variable Model Estimates" report smaller actual-hours responses in Seattle, which had a higher unemployment rate, than in Denver. However, unemployment does not place any restrictions on the response of desired hours of work since nonworkers can shift from being unemployed to being out of the labor force. If estimates of the labor supply impact of the desired-hours response to an NIT program are more invariant to differences in unemployment than the actual-hours response, desired-hours responses in both sites may be quite similar. Furthermore, the desired-hours response may be closer than the actual-hours response to the labor supply response to a national program in an environment of low unemployment.

Second, the experimental response of actual hours may differ from the response to a national program because of institutional rigidities in hours of work. Rather than being freely adjustable by employees, hours of work may be determined by employers. In that case, individuals can adjust their labor supply only by changing employers. But desired hours is not subject to employer restrictions. Since only a small proportion of workers in any one work place are subject to the incentive of an experimental NIT program, employers have little incentive to adjust their requirements. Under a national NIT program, however, employers in industries characterized by low-income workers will have a greater incentive to adjust required hours of work. The desired-hours response may therefore be closer to the actual-hours response in a national program.

Effects of the experiment on desired hours of work are substantial for all three groups; the effects range from −58 hours for husbands to −168 hours for single female heads (table 3.16). Effects on actual hours are also presented for comparison.[41] For husbands, the desired-hours effects are substantially (65 percent) smaller than the actual-hours estimates. This perplexing result is due to the experimentally induced increase in unemployment among husbands. For wives the desired-hours response is virtually identical to the actual-hours response; for female heads the desired-hours response is somewhat larger. Thus, the arguments presented earlier that the desired-hours response might be larger than the actual-hours response are not confirmed by the data. Consequently, there is no evidence that the experimental actual-hours response is a downward-biased estimate of the response to a national program.

Earnings and wage rates

Effects of the experiment on earnings are important for several reasons. First, changes in earnings affect the budgetary costs of NIT programs. Second, reductions in earnings decrease the antipoverty effects of the NIT by reducing the after-response disposable income of participants. Third, earnings reductions are a measure of the effect of the NIT or GNP.

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[40] Weekly desired hours of work is derived from a question asking how many hours per week the individual wanted to work. The question was asked for each spell of unemployment and for each period of work with different actual hours (but at least once in every interview).

[41] These actual hours estimates differ somewhat from those presented in the section on Single Dummy Variable Model Estimates because the fourth year sample was used in that section while the second year sample is used here.

Earnings can change due to changes in either hours worked or wage rates. We have already learned that the experimental NIT plans reduced hours worked. There are a number of reasons why one might expect the experimental NIT treatments to affect wage rates. The NIT treatments could affect the tradeoffs eligible persons make between the pecuniary and nonpecuniary returns to work. First, because income is increased and the pecuniary returns to work are taxed while nonpecuniary returns are not taxed, individuals can be expected to slant their job choice more toward nonpecuniary returns. Second, because the experimentally imposed tax rates are temporary, individuals can be expected to shift to jobs with a higher on-the-job training and lower wage rates. Thus, there are several theoretical reasons to expect the experiment to increase the nonpecuniary returns to work and reduce wage rates. On the other hand, the income support provided by the experiment might afford persons an opportunity to search for better jobs and thus lead to an increase in wage rates and an upgrading of occupation.

As seen in table 3.16 the financial treatments had virtually no effect on wage rates for husbands and wives, but did induce a 7 percent decline in the wage rates of single female heads. Thus, single women may have moved to jobs with relatively higher nonpecuniary rewards.

Results for annual earnings indicate substantial and significant reductions for all three groups. Because of the lack of response on wage rates for husbands and wives, most of their earnings response is due to a reduction in hours of work. Percentage reductions are 8 percent for husbands, 20 percent for wives and 16 percent for single female heads. These percentage changes are similar to the percentage changes in hours worked (9 percent for husbands, 21 percent for wives, and 14 percent for single female heads).

CHAPTER VI

FINDINGS FOR YOUTH

In this chapter we examine the effects of the experiment on the labor supply of dependent youth, defined as at least age 16 but under age 21.[1] These youth were exposed to the experimental treatments at a time when they were likely to be entering the labor market for the first time. Any disincentive impact of the experiment on their labor supply during this crucial period could have important long run effects by postponing or reducing their initial labor market experience. National attention has recently focused on the high unemployment rates observed among young persons. Any impact of the experiment on their behavior is, thus, clearly of policy interest.

The manner in which experimental NIT plans can be expected to have an effect on the labor supply of youth depends on the extent to which these youth are integrated into the family decision-making process. For youth living in the parental home, there are two possible extremes. At one extreme, youth are fully integrated into a decision-making process that maximizes a family utility function. In this case, the experimental effects for youth are similar to those discussed for family heads in Chapter 2. We can expect the experimentally induced increase in family disposable income and the increase in the net wage rate to have negative effects on the youth's labor supply. At the other extreme, the youth is an independent decision-maker who benefits from being a member of the family. In this case, the experimental effects can be expected to depend on whether the family is able to induce the youth to reimburse the family for the taxes (benefit reductions) that the family pays on the youth's income. Under the NIT the income of the family (other than the youth) increases, and, if the family is able to collect these taxes from the youth, the youth's income and the net wage rate decrease. The increase in family income and decrease in the youth's net wage rate should reduce his labor supply, while the decrease in his income should increase his labor supply. If, in contrast, the family does not collect the taxes from the youth, family income increases while the youth's income and net wage rate are unchanged. In this case, the increase in family income should unambiguously decrease the youth's labor supply.

For youth leaving home, none of this can be expected to apply. In fact, many of the youth in the SIME/DIME sample moved out of their parents' family and formed their own households. These youth can be expected to respond in a manner similar to other family heads.

Results

Here, we examine the effects of the NIT treatments on the labor supply of youth during the third year of the experiment (experimental quarters 8 to 11), and consider the effects on five different dependent variables—average hours worked per week, the proportion of the year worked, the proportion of the year unemployed, the proportion of the year spent out of the labor force, and average in-school status.[2] The sample used for the analysis comprises male and female youth who at enrollment were between the ages of 16 and 21 and were either children, stepchildren, or grandchildren of the family head.[3] Of 1,177 persons meeting these criteria, the sample analyzed is reduced to 733 because of attrition (378) and missing data (66). It should be noted that the analyses presented in this chapter do not attempt to account for potential biases that might be generated by the large amount of attrition (32 percent).

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[1] This chapter is based primarily on West (1979, 1980).

We present the results for youth in the following manner. First, we present some simple evidence to indicate whether or not the experiment had any effect at all on youth. The possibility of an underreporting bias is also discussed. Second, we explore the extent to which the response to the treatment varied with family status in the third year of the experiment and experimental duration. Third, we examine the interrelation of the effects on labor supply and schooling. Males and females are analyzed separately throughout.

A simple estimate of the effect of the NIT plans

There are good reasons for expecting either some negative effect or no effect at all on the labor supply of youth. Since youth are typically considered as marginal workers, they may be especially responsive to economic incentives. However, youth living in their parental families are not directly affected by the NIT plans. Although families receive NIT payments that depend on the youths' earnings, youths themselves receive nothing. Unless their work behavior is affected by family income or the family is able to transfer the NIT tax on earnings to the youth, as mentioned above, the youth will not respond to the treatment. Youth who have formed their own families, in contrast, are directly affected by the NIT treatments.

In order to address the issue of whether or not there is any financial treatment effect on labor supply, we use the simplest possible representation of the financial treatments: a single dummy variable which takes the value 1 for a person eligible for any of the 11 experimental NIT plans and 0 for controls.[4]

Results presented in the first column of table 3.18 indicate that the NIT plans did have a significant effect on the labor supply of both male and female youth. For males, the effects are substantial. Hours of work per week are reduced by 4.63 hours, or about 24 percent, and the proportion of the year worked is reduced by .106, or 21 percent.[5] These substantial decreases in work effort are reflected mostly in an increase in time unemployed rather than time out of the labor force. The results imply that a national NIT program could substantially raise the measured unemployment rate for young males. The results for schooling status indicate that there is no evidence that male youth reduced labor supply in order to increase school attendance.

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[2] Average hours per week is calculated as total hours worked during the year divided by 365/7. The proportion of the year worked is calculated as the number of days the person held jobs in which he or she was working at a positive weekly rate divided by 365. Thus, for example, it excludes periods when the person held a job but was not working, such as might occur for a school teacher during the summer. Both average hours worked and the proportion of the year worked treat paid vacation and sick leave as work. The proportion of the year unemployed is calculated as the length of periods during which the person did not work, but looked for work, divided by 365. The proportion of the year out of the labor force represents the remainder of the period. Average in-school status is the number of quarters during the year in which the nonhead attended school divided by four.

[3] Persons who were married at enrollment are not considered unless they were separated from their spouse and living with their parents.

[4] Other variables included in the equation are dummy variables for the normal income level of the youth's original family, dummy variables for ethnic group, a dummy variable for being in Denver, a dummy variable indicating that the youth's original family was headed by a single person, 3 dummy variables for the counseling/training treatments and a set of continuous variables consisting of the preexperimental values of average hours worked, the proportions of the year worked and unemployed, average in-school status, education, weeks of training, number of family members, number of family members aged 0 to 5, age (piecewise linear with a change of slope at 18), and family income.

[5] The percentage effects are calculated by dividing the response by the mean of the variable for controls.

Results for females indicate a somewhat smaller and less significant effect. Hours worked per week are reduced by 2.78 hours or 18 percent, and the proportion of the year worked is reduced by .074 or 17 percent. Most of the reduction in employment appears to be due to increased likelihood of being out of the labor force, rather than to an increase in unemployment. As with males, there is no evidence that work effort was reduced in order to increase school attendance.

TABLE 3.18.—ESTIMATED OVERALL AND 3- AND 5-YEAR EFFECTS ON LABOR SUPPLY OF YOUTH

[Standard errors in parentheses]

	Males			
	All	5-year program	3-year program	Difference between 3- and 5-year programs
Hours worked per week	[3] −4.63	[2] −5.78	[2] −4.10	−1.68
	(1.67)	(2.32)	(1.82)	(2.36)
Proportion of year worked	[3] −.106	[2] −.129	[2] −.096	−.034
	(.040)	(.056)	(.044)	(.057)
Proportion of year unemployed	[1] .067	.076	[1] .064	.012
	(.035)	(.048)	(.038)	(.049)
Proportion of year out of labor force	.039	.054	.032	.022
	(.035)	(.048)	(.038)	(.049)
Average in-school status	.035	.038	.034	.004
	(.047)	(.066)	(.052)	(.067)
Sample size				351
Females				
Hours worked per week	[1] −2.78	[2] −5.35	−1.83	−3.52
	(1.62)	(2.37)	(1.74)	(2.37)
Proportion of year worked	[1] −.074	[3] −.163	−.041	[2] −.122
	(.042)	(.061)	(.045)	(.064)
Proportion of year unemployed	.014	−.006	.022	−.028
	(.029)	(.043)	(.031)	(.043)
Proportion of year out of labor force	.060	[3] .169	.019	[2] .150
	(.041)	(.060)	(.044)	(.060)
Average in-school status	−.011	.033	−.027	.060
	(.043)	(.062)	(.046)	(.062)
Sample size				382

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Accounting for income underreporting

As indicated in chapter 3, Greenberg and Halsey (1980) investigate the extent to which estimated experimental labor supply responses might be due to biases generated by differential underreporting of income between experimentals and controls, by matching experimental interview data (Denver only) with unemployment insurance (UI) records data. Since experimentals were taxed on their income by the experiment, they had an incentive to underreport that was not present for controls. The Greenberg and Halsey results for youth are striking. They estimate an employment response for males and females combined of −.18 using interview data, but only −.03 using UI data. According to these estimates, virtually all of the estimated youth labor supply response appears to be due to underreporting rather than to actual reductions in work effort.

Their results, however, are based on data for only a single quarter, using only the Denver half of the sample. Furthermore, their methodology has certain limitations. They adjust the observed work status of youth using the assumption that the likelihood of underreporting to SIME/DIME was equal in sectors both covered and uncovered by unemployment insurance (see Robins, 1981, for a discussion of their methodology). This adjustment is done on the basis of the following probability:

$$\frac{BC}{AD},$$

where A worked in both interview and UI data
B worked only in UI data
C worked only in interview data
D did not work in both interview and UI data.

The probability is calculated based on a multinomial logit model of the probability of being in each of the four categories. However, this probabilistic adjustment is performed on a relatively small number of nonworking youth (84 experimentals and 34 controls). Thus, the number probabilistically assigned to work may differ substantially from its expected value. Indeed, the numbers probabilistically assigned differ substantially from the number expected based on the mean probabilities of being in the various classes for experimentals and controls (40 rather than 25 experimentals, 0 rather than 8 controls). Thus, too many experimentals and too few controls are assumed to be underreporting their work status, leading to a serious overstatement of the bias due to underreporting. An estimate of the experimental effect corrected for underreporting based on the simple proportion in categories A, B, C, and D is −.07 (Robins, 1981). This measure is apt to be an understatement of the true experimental effect, however, since it does not correct for preexperimental differences between experimentals and controls, and controls tended to work more than experimentals in the preexperimental period.

One striking difference between the results reported in table 3.18 and those reported in Greenberg and Halsey is the large experimental effect they find on work using interview data (−.18) compared to about-.08 reported here (average of male and female results) and −.11 for Denver in West (1979). While there is a conceptual difference between the measures of work used (worked at all in the 10th quarter in Greenberg and Halsey vs. proportion of the year worked in this chapter), it is unlikely that these conceptual differences account for the large difference in the estimated impacts.[6]

Separate estimates of 3- and 5-year program effects

Since the experiment was conducted for only a limited period of time, as we have noted, it is not clear that the estimated effects represent the effects of a permanent NIT program. As discussed in chapter 2, in a temporary program the income effect is expected to be smaller and the substitution effect is expected to be larger than in a permanent program. In this section, we present results based on the two different experimental durations.

Separate estimates of the experimental effects for persons on 3- and 5-year NIT plans are also presented in table 3.18. The model is identical to the single dummy variable model except that there are separate dummy variables for the 3- and 5-year programs. The estimated effects of the 5-year program on hours worked and the proportion of the year worked are somewhat larger for males and considerably larger for females than the effects of the 3-year program. However, the difference between the 3- and 5-year effects on the proportion of the year worked is significant only for females. Thus, while the analysis cannot demonstrate conclusively that the work effect response increases with the duration of the NIT plan, there is a distinct possibility that responses to a 5-year program are considerably larger than responses to a 3-year program. Responses to a permanent NIT program may be considerably larger again than responses to a 5-year program.

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[6] There is little difference between the estimated effects on proportion of year worked and worked at all during the year presented in chapter 5, table 3.16, for family heads.

For males the estimated 3- and 5-year effects on being unemployed and out of the labor force are similar to each other and to the overall effect for males. For females, only the 5-year program has a significant effect on being out of the labor force, and there is no effect on unemployment. For neither males nor females is there a significant effect on school attendance.

Variation in response by treatment

The analysis of the single dummy variable model has demonstrated that experimental NIT plans have a considerable effect on the labor supply of youth, especially males. To test for possible variations by guarantee and tax rate, West (1979) estimated a model in which response depends on a dummy variable for being on an NIT plan, the guarantee level minus $3,800, and the tax rate −.5. For males, all the coefficients have the expected sign (negative), but none are significantly different from zero. The coefficients are also all negative for females and one coefficient is significant—the coefficient of the dummy variable (which represents the effect of the $3,800 guarantee level, 50 percent tax rate treatment).

Variation in response by family status

Many persons who were dependent youth at enrollment into the experiment became either heads of families or unrelated individuals during the experimental period. By the 11th experimental quarter, only 48.4 percent of male youth were still living with their parental family; 31.1 percent had become unrelated individuals and 20.5 percent had become heads of their own families. The figures for females are similar except that females were less likely to have become unrelated individuals (22.3 percent) and more likely to have become family heads (40.8 percent). These large changes in family status suggest that it may be important to account for changes in family status in the analysis of the effects of the experiment on the youth's labor supply. The NIT plans provide very different incentives to youths in different family statuses during the experimental period. For instance, the guarantee level for an unrelated individual is $1,000, which is less than half the guarantee level for a two-person family in the $3,800 programs (the two-person guarantee level is $2,356). Also, a youth who forms his or her own family may be more directly affected by a given program than a youth who remains in his or her parental family, although the guarantee level is the same in each case (for a given sized family). Thus, it is likely that an individual's response will depend upon his or her status during the experimental period.

We present separate estimates of response for youths in each of the three family statuses, defined on the basis of family status in experimental quarter 11.[7] This measure of family status is not exogenous with respect to the NIT treatments since family status itself could have been affected by the NIT plans. Consequently, the experimental impacts disaggregated by family status cannot be averaged to obtain the total experimental effect on labor supply, since they do not incorporate effects that operate through changes in family status. The estimates do, however, indicate how the experimental response varies with family status during the experimental period.

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[7] West (1979) also presents estimates based on family status in experimental quarter 7. The quarter 7 definition has the advantage of being determined prior to the dependent variables, which represent averages for quarters 8 to 11. The 11th quarter defintion has the advantage of capturing many more status changes. The results turn out to be similar.

The estimates are presented in table 3.19. The model contains three dummy variables representing persons eligible for an NIT treatment in each of the three family status categories. The model also includes dummy variables for two of the family status categories among the control variables in order to account for the direct effects of family status on the dependent variables. The results for males indicate that there is a large and significant reduction in work effort for youths who remain dependents. The reduced work effort is reflected about equally in increased time unemployed and increased time out of the labor force. For males who become heads of their own families, there is also a substantial reduction in work effort. The reduction for new heads is primarily reflected in an increase in unemployment. The response is −7.37 hours (−43 percent) for 11th quarter nonheads and −10.36 hours (−33 percent) for 11th quarter husbands. These responses are much larger in magnitude than the response estimated for originally enrolled husbands. There is no significant effect on work effort for persons who become unrelated individuals.

TABLE 3.19.—VARIATION OF YOUTH RESPONSE BY FAMILY STATUS

[illegible]

Dependent variable	Males			Females		
	Dependent	Head	Unrelated individual	Dependent	Head	Unrelated individual
Hours worked per week	7.37	10.36	2.16	6.77	2.12	1.90
	(2.23)	(3.38)	(2.80)	(2.58)	(2.43)	(3.26)
Proportion of year worked	.157	.286	.067	.195	.034	.036
	(.054)	(.082)	(.068)	(.067)	(.063)	(.085)
Proportion of year unemployed	.076	.204	.011	.051	.025	−.055
	(.047)	(.071)	(.059)	(.046)	(.044)	(.059)
Proportion of year out of the labor force	[1] .081	.082	.078	[2] .144	.009	.019
	(.047)	(.071)	(.059)	(.065)	(.061)	(.082)
Average in-school status	.035	.128	.028	.007	.047	.059
	(.065)	(.099)	(.082)	(.067)	(.063)	(.085)
Number in group	170	72	109	141	156	85
Controls	82	29	42	62	68	37
Experimentals	88	43	67	79	88	48
Sample size		351			382	

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Results for females show a significant response only for youth who remain dependents. The response is −6.77 hours (42 percent) for 11th quarter dependent youth. The reduction in work effort corresponds primarily to an increase in the amount of time spent out of the labor force. As in the previous results, the effect on school attendance is not significant for either males or females.

Thus, the breakdown of response by family status reveals two important results. The experimental NIT plans induce a reduction in the work effort of male youth who either remain dependents or become heads of new families; there is no reduction for males who become unrelated individuals. There is a reduction in work effort induced by the experimental NIT plans for females who remain dependents, but not for females who become wives or unrelated individuals.[8] This result contrasts with the substantial response by women who were married at the beginning of the experiment (see Chapter 5).

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[8] Similar analyses in West (1979) indicate the response of youth did not vary by age, race, or site.

Interrelation of labor supply and schooling

Having determined that youth are indeed working less in response to the experimental NIT plans, it is important to know what they are doing during the additional time they are not working—in particular, whether or not they are decreasing their labor supply in order to increase school attendance. Although estimation results using the single dummy variable model for average in-school status showed no significant impact of the NIT plans on school attendance, it seems desirable to explore the schooling question in somewhat more detail. We define four dummy variables that measure combined work and school status: working and attending school, working and not attending school, not working and attending school, and not working and not attending school. Each variable is defined for each quarter from 8 to 11 and then averaged over the quarters where information is available. Since the states defined by the four variables are mutually exclusive and exhaustive, the four variables sum to one. In table 3.20 we present estimation results using these variables as dependent variables both with a model that allows responses to vary by 11th quarter family status and also with the single dummy variable model. Results for the single dummy variable model are presented in the rows marked "All;" the responses by family status are presented in the remaining rows. The estimates are calculated by OLS.

TABLE 3.20.—ESTIMATED EFFECTS ON COMBINED WORK AND SCHOOL STATUS BY 11TH QUARTER FAMILY STATUS

[Standard errors in parentheses]

11th quarter family status	Work, school	Work, no school	No work, school	No work, no school
Males:				
Dependent	−.057	[3] −.162	[1].092	[2].126
	(.049)	(.062)	(.049)	(.052)
Head	−.014	[3] −.266	[1].143	[1].137
	(.075)	(.093)	(.074)	(.078)
Unrelated individual	−.046	.038	.018	−.010
	(.062)	(.077)	(.061)	(.065)
All	−.046	[1] −.113	[2].082	[2].078
	(.036)	(.046)	(.035)	(.037)
Mean of dependent variable	.171	.411	.180	.238
Females:				
Dependent	[2] −.112	[2] −.170	[1].105	[3].177
	(.048)	(.068)	(.054)	(.066)
Head	−.007	−.038	−.040	.085
	(.046)	(.064)	(.051)	(.062)
Unrelated individual	.011	.006	.049	--.065
	(.061)	(.085)	(.068)	(.083)
All	−.041	[1] −.074	.030	[2].084
	(.030)	(.042)	(.034)	(.042)
Mean of dependent variable	.135	.354	.180	.331

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

For males, results indicate that the response occurs only among those who remain dependents or become heads of families; there is no response for male youth who become unrelated individuals. This result is consistent with the results reported earlier. For males who either remain dependents or become heads, the results indicate that the experimental NIT plans decrease the probability of working and not going to school, and increase both the probability of not working and going to school and the probability of not working and not going to school. Responses are somewhat larger for males who become husbands. Thus, it is clear that many males are decreasing their work effort even though they do not go to school. However, the increase in the probability of not working and going to school indicates that some of the reduction in work effort occurs among persons who choose to concentrate on schooling. The net effect on school attendance is, however, not significant.

As in our other estimates, only those females who remain dependents exhibit a response to the experimental NIT plans. Results indicate a reduction in the probability of both working and going to school and an almost equal increase in the probability of not working and going to school. Thus, a substantial portion of the work effort response occurs among females who choose to concentrate on schooling. There is, however, no net effect on school attendance. The major part of the work effort response is associated with a reduction in the probability of working and not attending school and an increase in the probability of not working and not attending school. Thus, for both males and females, there is a significant decrease in work effort that is not accompanied by school attendance.

We also calculate separate estimates of the NIT treatment effect on hours worked conditional on schooling status in the experimental periods. In these estimated equations, independent variables are the same as those used in the model reported in table 3.19 except that average in-school status is also included.[9] Results, presented in table 3.21, indicate that the response tends to be largest for persons not in school. Although the differences in the response of persons in school and persons not in school are substantial, they are not statistically significant. These results confirm earlier results which indicate that most of the reduction in work effort occurs among nonstudents.

Summary and conclusions

Experimental NIT plans definitely induce a reduction in work effort among youth. Both hours of work per week and the proportion of time worked are reduced. The analysis also demonstrates that this reduction in work effort is concentrated in particular groups of youth. The response is about −7.7 hours per week (−43 percent) for males who remain dependents and −10.36 hours per week (−33 percent) for males who become husbands. Among females only those who remain dependents responded. Their response is −6.8 hours per week (−42 percent). There is no response for males and females who become unrelated individuals or females who become wives.

There is no evidence that the work effort reduction is accompanied by an increase in school attendance. Somewhat less than half the response appears to occur among persons who would have worked while attending school in the absence of the experiment, but who don't work under the influence of the experiment although continuing to attend school. Separate estimates of response for the 3- and 5-year programs indicate that there is a possibility that response increases with experimental duration.

Estimated responses are large in percentage terms and should not be ignored when considering the possible effects of national NIT plans, although a possible underreporting bias may exist for this group. Furthermore, the response to a national plan could not be reduced by refusing eligibility to unrelated individuals since the estimated response comes from other groups that could not be excluded so easily from eligibility.

The response by male and female youth who remain dependents, although large in percentage terms, may not be of great importance since these nonheads are generally secondary earners in their families. The major danger would be if a

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[9] The experimental treatment variables are FGTO ×INSCH and FGTO×(1-INSCH), where FGTO is a dummy variable for having an NIT treatment and INSCH is average in-school status in the experimental period (with a range from 0 to 1). INSCH is endogenous with respect to the treatments since in-school status could have been affected by the treatments. Thus, these estimates represent only the partial effects of the experiment on hours worked other than effects that operate through schooling status.

TABLE 3.21.—EXPERIMENTAL EFFECTS ON HOURS WORKED CONDITIONAL ON EXPERIMENTAL SCHOOLING STATUS

[Standard errors in parentheses]

	Males	Females
Not in school	[2]−5.72 (2.06)	[1]−3.44 (2.01)
In school	−1.84 (2.76)	−1.58 (3.05)
Difference	−3.88 (3.52)	−1.86 (3.81)

[1] Significant at the 10-percent level.
[2] Significant at the 1-percent level.

reduction in work effort during this portion of their life cycles has continuing effects by reducing their labor market experience and thus possibly reducing their ability to obtain good jobs in the future. Since we have no long term data, of course, such an effect must remain purely speculative.

The reduction in work effort by male youth who become husbands is more important because these males are reducing their work effort just at the time when they are undertaking family responsibilities. And in addition to their response in the current period, long term effects on their labor supply behavior are, as above, also a possibility.

CHAPTER VII

COMPARISON OF LABOR SUPPLY FINDINGS ACROSS EXPERIMENTS

As indicated earlier, SIME/DIME is one of four NIT experiments conducted in the United States.[1] In this chapter, we present a rough comparison of the labor supply findings across the experiments. More detailed comparisons are presented in two excellent surveys by Keeley (1981) and Moffitt and Kehrer (forthcoming).

Descriptions of the experiments

The four NIT experiments were geographically dispersed across the country. Their main features are described in table 3.22. The New Jersey Experiment, conducted in urban areas of New Jersey and Pennsylvania from 1968 to 1972, although the smallest of the three urban experiments, is generally the best known, probably because of its distinction as the first major social experiment. The New Jersey Experiment enrolled only dual-headed families for a period of three years. Three racial groups (black, white, and Hispanic) were assigned to eight different treatments, with guarantee levels ranging from 50 percent to 125 percent of the poverty level, and constant tax rates ranging from 30 percent to 70 percent. Breakeven levels ranged from 100 percent to 300 percent of the poverty level. New Jersey (along with the Rural experiment) had the most stringent income selection criterion, limiting enrollment to families with incomes equal to 150 percent of the poverty level or less. Although New Jersey enrolled 1,357 families, only the 60 percent of families that did not change marital status during the experiment were included in the labor supply analysis.

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[1] An NIT experiment was also conducted in Canada, but results from this experiment are not yet available.

TABLE 3.22.—MAIN FEATURES OF THE FOUR NIT EXPERIMENTS

Experiment	Site(s)	Length of treatments	Number of families enrolled	Type of families enrolled	Racial composition of sample	Income cutoff[2]	NIT Plans Tested[1]		
							Guarantee level[2]	Tax rate[3]	Grant breakeven level[2]
New Jersey (1968–72).	Trenton, N.J. (12%).	3 years	1,357	Dual-headed	Black (37%)	1.5	0.5	.3, .5	1.7, 1.0.
	Paterson-Passaic, N.J. (36%).				White (32%)		.75	.3, .5, .7	2.5, 1.5, 1.1.
					Hispanic-American (31%).		1.00	.5, .7	2.0, 1.4.
	Jersey City, N.J. (29%).						1.25	.5	2.5.
	Scranton, Pa. (23%).						(.81)	(.5)	(1.7).
Rural (1969–73)	Rural Iowa (38%).	3 years	809	Dual-headed, prime age (73%).	Black (35%)	1.5		.5	1.0.
	Rural N.C. (62%).				White (65%)		.75	.3, .5, .7	2.5, 1.5, 1.1.
				Single-headed, prime age (13%).			1.00	.5	
				Dual or single, head over 59 (14%).			(.85)	(.5)	(1.6).
Gary (1971–74)	Gary, Ind	3 years	1,780	Dual-headed (41%)	Black	[4]2.4	.75	.4, .6	1.9, 1.3.
				Single-headed (54%)			1.00	.4, .6	2.5, 1.7.
							(.88)	(.5)	(1.9).
Seattle-Denver (1971-present).	Seattle, Wash. (43%).	3 years (71%)	4,800	Dual-headed (61%)	Black (43%)	3.25	.95	.5, .7, .7–.025Y, .8–.025Y.	1.9, 1.4, 1.8.
	Denver, Colo. (57%).	5 years (25%)		Single-headed (39%)	White (39%)				1.5.
		20 years (4%)			Chicano (18%)		1.20	.5, .7, .7–.025Y, .8–.025Y.	2.4, 1.7, 3.0, 2.0.
							1.40	.5, .7, .8–.025Y	2.8, 2.0, 2.6.
							(1.16)	(.5)	(2.1).

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[1] Approximate average values of the guarantee level, the tax rate, and the breakeven level are presented in parentheses.

[2] Expressed as a percentage of the poverty level for a family of four in the year of enrollment.

[3] Y is annual income in thousands of 1970 dollars.

[4] 125 additional families with income above this amount were added after enrollment.

The Rural Experiment, conducted in Iowa and North Carolina, was the only experiment that focused on families in nonurban areas. The smallest of the NIT experiments, enrolling only 809 families, it also yielded the least conclusive findings (see Palmer and Pechman, 1978). The design of the Rural Experiment followed the New Jersey design closely, using a subset of the same treatments and the same sample selection criteria. The analysis sample for the Rural Experiment was broken down into two categories, wage earners and farm operators. The difficulties of analyzing the latter group reduced the effective sample size for analysis from its already low level and probably was the main factor responsible for the lack of precision of the results. Our comparisons focus only on the wage earner sample in the Rural Experiment.

The Gary Experiment was the second urban experiment, conducted in Gary, Indiana, from 1971–1974. Gary had a higher income cutoff than either the New Jersey or Rural Experiments, but focused only on blacks. Gary had the second largest sample size, although still too small to yield very precise findings. The treatments in Gary were somewhat different from the treatments in the New Jersey and Rural Experiments, particularly in the tax rates tested.

SIME/DIME was the last of the NIT experiments and differed from the other experiments in several important respects. First, the sample size was considerably larger than the combined sample sizes of the other experiments, enabling more precise estimates of behavioral response. Second, the treatments had considerably more generous guarantee levels. The least generous treatment, with a guarantee level of 95 percent of the poverty level, was approximately equivalent to the most generous treatment in the other experiments. SIME/DIME was also the only experiment that tested treatments of differing length (3, 5, and 20 years) and nonconstant tax rates.

In comparing the labor supply findings across experiments, it is important to keep in mind that the differences in design, operations, and methods of analysis can be expected to yield different findings. However, each experiment tested basically the same type of welfare program—namely, a negative income tax—and for this reason should provide relatively consistent estimates of labor supply response.

In the remainder of this chapter we compare labor supply findings across the four experiments from the perspective of both single dummy variable models and structural models. We expect the single dummy variable estimates to be different across experiments primarily because of differences in the generosity of treatments (recall that our earlier theory indicates that the size of the response depends on the generosity of treatment). Structural estimates, in contrast, are supposed to account for the variation in treatment generosity; any differences in the estimates, therefore, should be due primarily to differences in the methods of analysis used.

Single dummy variable estimates

Each experiment has produced several estimates of overall treatment effects. Because of the similarities in these results, we report only one set of estimates from each experiment. We have attempted to select those estimates that are based on the most similar sample selection critera, time period, and variable specification.[2] For example, each set of estimates is based on a sample that excludes all families in which a marital dissolution occurred. Hence, although all previously reported results for SIME/DIME include such families, the results reported here do not.[3] Similarly, whereas SIME/DIME was the only experiment with a 5-year treatment, we report results that both include and exclude such families. Before discussing the results, we note again that the SIME/DIME estimates can be expected to be more precise and larger than the estimates for the other experiments because the SIME/DIME sample sizes are much larger and the treatments are more generous.

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[2] A file was produced for the purpose of performing a cross-experimental analysis (see Citro, 1980) but was not used to produce the estimates reported here. While stricter uniformity in the estimation methodology could be ensured using the cross-experimental file, we do not believe such uniformity would produce dummy variable estimates that would differ significantly from the ones reported here, because the results from dummy variable models tend to be fairly insensitive to slight variations in sample selection and variable specification.

Tables 3.23 through 3.26 present the single dummy variable estimates of annual hours of work and the employment rate for husbands, wives, single female heads, and youth, respectively. Two things stand out. First, in each experiment, there is an unambiguous decrease in labor supply as a result of the NIT treatments. Second, although there is substantial variation in the size of the estimates, they do exhibit a fairly consistent pattern for each group. For example, blacks and Chicanos generally have larger responses than whites, husbands generally have smaller responses than the other groups, and in no case is there evidence of a massive withdrawal from the labor force. Hours reductions generally range from about 1 week to about 5 weeks of full-time work, which translate into percentage responses that vary between 5 percent and 25 percent. As ex-

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[3] Because of this difference in sample selection criteria, the SIME/DIME single dummy variable estimates reported here differ slightly from the single dummy variable estimates reported earlier.

TABLE 3.23.—SINGLE DUMMY VARIABLE ESTIMATES OF LABOR SUPPLY RESPONSE OF HUSBANDS

[Estimated standard errors in parentheses]

Total sample	Annual hours of work		Employment rate	
New Jersey [1] [2]	34.2	....	.01	....
Rural [1] [2] [3]	56.0	....	−.003	....
Gary	35.4	(65.1)	−.01	(.03)
Seattle-Denver	[6] 128.6	(30.1)	[6] .04	(.01)
3-year only	[6] 112.8	(33.1)	[6] −.04	(.01)
Blacks				
New Jersey	+36.5	....	+.01	....
Rural	[5] 152.0	....	−.01	....
Gary	35.4	(65.1)	−.01	(.03)
Seattle-Denver	[4] 95.5	(50.9)	[4] −.04	(.02)
3-year only	60.5	(56.5)	−.03	(.02)
Whites				
New Jersey	99.1	....	−.02	....
Rural [3]	+80.0	....	−.001	....
Seattle-Denver	[4] 85.1	(43.6)	−.01	(.02)
3-year only	65.2	(48.0)	−.01	(.02)
Hispanic = Chicano				
New Jersey (Hispanic)	10.4	....	−.02	....
Seattle-Denver (Chicano)	[6] 296.1	(67.7)	[6] −.11	(.03)
3-year only	[6] −308.2	(72.3)	[6] −.12	(.03)

[1] Results for the total sample are weighted averages of the separated responses by race. Standard errors are not available for any of the groups.

[2] Significance levels not available.

[3] Results are for the wage earner sample only.

[4] Significant at the 10-percent level.

[5] Significant at the 5-percent level.

[6] Significant at the 1-percent level.

Note.—Estimates are for a continuous sample in quarters 3–10 of the experiments (quarters 1–12 for the Rural Experiment). For the Gary and Seattle-Denver results, the model specifications and sample selection procedures are identical and are described in Moffitt (1979a); full results are presented in Moffitt (1979b) and Robins and McNicoll (1979, 1981). The New Jersey results are from Watts and Cain (1973) and the Rural results are from USDHEW (1976).

TABLE 3.24.—SINGLE DUMMY VARIABLE ESTIMATES OF LABOR SUPPLY RESPONSE OF WIVES

[Estimated standard errors in parentheses]

Total sample	Annual hours of work		Employment rate	
New Jersey [1]	−55.7	....	−.04	....
Rural [1] [2] [3]	−178.0	....	−.16	....
Gary	−57.6	(62.7)	−.03	(.04)
Seattle-Denver	[5]−145.1	(31.4)	[5]−.08	(.02)
3-year only	[5]−141.2	(34.5)	[5]−.08	(.02)
Blacks				
New Jersey	−5.2	....	−.003	....
Rural [2]	[4]−268.0	....	[4]−.18	....
Gary	−57.6	(62.7)	−.03	(.04)
Seattle-Denver	[5]−176.2	(53.2)	[5]−.09	(.03)
3-year only	[5]−165.6	(59.1)	[5]−.10	(.03)
Whites				
New Jersey	−73.0	....	−.06	....
Rural [2] [3]	+88.0	....	−.03	....
Seattle-Denver	[4]−111.8	(45.6)	−.05	(.02)
3-year only	[4]−111.8	(50.3)	[4]−.05	(.03)
Hispanic-Chicano				
New Jersey (Hispanic)	−99.1	....	−.05	....
Seattle-Denver (Chicano)	[4]−170.7	(70.8)	[5]−.11	(.04)
3-year only	[4]−168.7	(75.7)	[5]−.11	(.04)

[1] Results for the total sample are weighted averages of the separated responses by race. Standard errors are not available for any of the groups.

[2] Results are for the wage earner sample only.

[3] Significance levels not available.

[4] Significant at the 5-percent level.

[5] Significant at the 1-percent level.

NOTE.—Estimates are for a continuous sample in quarters 3-10 of the experiments (quarters 1-12 for the Rural Experiment). For the Gary and Seattle-Denver results, the model specifications and sample selection prodedures are identical and are described in Moffitt (1979a); full results are presented in Moffitt (1979b) and Robins and McNicoll (1979, 1981). The New Jersey results are from Watts and Cain (1973) and the Rural results are from USDHEW (1976).

TABLE 3.25.—SINGLE DUMMY VARIABLE ESTIMATES OF LABOR SUPPLY RESPONSE OF SINGLE FEMALE HEADS

[Estimated standard errors in parentheses]

Total sample	Annual hours of work		Employment rate	
Gary	−37.1	(68.3)	−.04	(.03)
Seattle-Denver	[3]−154.9	(36.6)	[3]+.08	(.02)
3-year only	[3]−163.3	(39.4)	[3]−.08	(.02)
Blacks				
Gary	−37.1	(68.3)	−.04	(.03)
Seattle-Denver	[3]−205.9	(51.5)	[3]−.10	(.02)
3-year only	[3]−232.3	(54.9)	[3]−.11	(.03)
Whites				
Seattle-Denver	−59.5	(60.5)	−.03	(.03)
3-year only	−53.4	(65.4)	−.04	(.03)
Chicanos				
Seattle-Denver	[2]−207.1	(88.1)	[2]−.10	(.04)
3-year only	[2]−189.1	(95.0)	[1]−.09	(.05)

[1] Significant at the 10-percent level.

[2] Significant at the 5-percent level.

[3] Significant at the 1-percent level.

NOTE.—Estimates are for a continuous sample in quarters 3-10 of ther experiments. The model specifications and sample selection prodedures are identical and are described in Moffitt (1979a). The full results are presented in Moffitt (1979b) and Robins and McNicoll (1979, 1981).

TABLE 3.26.—SINGLE DUMMY VARIABLE ESTIMATES OF LABOR SUPPLY RESPONSE OF YOUTH

[Estimated standard errors in parentheses]

Total sample	Annual hours of work		Employment rate	
New Jersey [1]	187.0	....	.10	....
Rural [2]	94.0	....	N.A.	....
Black	48.0	....	N.A.	....
White [2]	140.0	....	N.A.	....
Seattle-Denver [2]	191.1	....	.09	....
3-year only [2]	152.1	....	.07	....
Males	[5] 241.4	(87.1)	[5] .11	(.04)
3-year only	[4] 213.8	(94.9)	[4] .10	(.04)
Females	[3] 144.9	(84.5)	[3] .07	(.04)
3-year only	95.4	(90.7)	.04	(.05)

[1] Results are for plan with guarantee level equal to 75 percent of the poverty level and tax rate equal to 50 percent.
[2] Significance levels not available (N.A.).
[3] Significant at the 10-percent level.
[4] Significant at the 5-percent level.
[5] Significant at the 1-percent level.

NOTE.—New Jersey estimates are for quarters 3-10 and are from Mallar (1977). Rural estimates are for quarters 1-12 and are from USDHEW (1976). Seattle-Denver estimates are for quarters 8-11 and are from West (1979).

pected, the SIME/DIME estimates are larger than the estimates from the other experiments by a considerable amount, but these responses are consistent with the large differences in the generosity of the treatments.[4] Employment rate reductions range from about 1 percentage point to 10 percentage points, but within groups the variation is usually less. For example, the range of estimates for husbands is between 0 and −.04.

On the basis of these results, we now proceed one step further in an attempt to synthesize the findings from all the experiments. We produce a composite set of estimates that represents an average of the estimates from each experiment, weighted by the number of families enrolled in each.[5]

Table 3.27 presents these weighted estimates across all experiments. For annual hours of work, the reductions range from about 2 weeks of full-time work to about 4 weeks of full-time work. The results are smallest for husbands and largest for youth. The results for wives and single female heads are quite similar. For the employment rate, the same pattern emerges with the effects ranging from −.03 to −.09. Since SIME/DIME has a heavy weight in calculating these estimates, the results may be viewed as responses to a fairly generous NIT program. Rough calculations indicate that the average (weighted) guarantee level in the

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[4] In Chapter 8, we generalize the SIME/DIME findings to nationwide programs that are considerably less generous than the SIME/DIME treatments.

[5] While the sample sizes used to produce the estimates would probably represent a better set of weights, these were not always available from the sources cited. Moreover, the general scope of each experiment, as exemplified by the number of families enrolled, appears to be a reasonable criterion to use for weighting the results.

experiments is about 100 percent of the poverty level and the average (weighted) tax rate is about 50 percent. Hence, the responses in table 3.27 can be viewed as those forthcoming from such a program, although it is important to keep in mind that these responses apply to a specific segment of the low and middle income population. The average responses for the entire U.S. population would be much lower than the average responses reported in table 3.27.[6]

TABLE 3.27.—WEIGHTED AVERAGES OF SINGLE DUMMY VARIABLE ESTIMATES FROM ALL EXPERIMENTS[1]

	Annual hours of work	Employment rate
Husbands	−89	−.03
Wives	−117	−.06
Single female heads	−123	−.07
Youth	−173	−.09

[1] Estimates are weighted average of the total sample results given in tables 3.23 through 3.26, where the weights are the number of families enrolled in each experiment described in table 3.22.

Structural estimates

In one sense, as we have mentioned earlier, structural estimates are easier to compare across experiments than are single dummy variable estimates because unlike single dummy variable estimates, substitution and income effects derived from structural models represent the effects of a specified (usually one dollar) change in the net wage rate and net nonwage income, and are thus directly comparable across experiments. In another sense, however, the methodologies used to obtain structural estimates vary considerably. Thus, identical data sets can produce radically different estimates.

A large number of studies have been conducted using experimental data to estimate structural models of labor supply response. We will not devote space here to discussing these methodologies. Instead, we attempt to synthesize the evidence without making judgments as to their relative merits. Keeley (1981) and Moffitt and Kehrer (forthcoming) discuss the methodological approaches of the various structural models estimated.

Table 3.28 presents averages of the substitution and income effects for family heads estimated from each experiment. The studies upon which these estimates are based include Watts and Horner (1977), Hausman and Wise (1976), Bawden (1977a, 1977b), Ashenfelter (1978), Moffitt (1978), Burtless and Hausman (1978), Cain et al. (1977), Dickinson (1980), Johnson and Pencavel (1979a, 1979b), and the two versions of the ΔW, ΔY structural model presented in Chapter 5. Within each experiment, the substitution and income effects are simple averages.

TABLE 3.28.—ESTIMATED SUBSTITUTION AND INCOME EFFECTS ON ANNUAL HOURS OF WORK[1]

	Substitution effect (per dollar per hour)	Income effect (per thousand dollars)
Husbands:		
New Jersey	102	−12
Rural	123	2
Gary	55	−37
Seattle-Denver	119	−90

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[6] See the discussion in Chapter 8 on generalizing the SIME/DIME estimates.

TABLE 3.28.—ESTIMATED SUBSTITUTION AND INCOME EFFECTS ON ANNUAL HOURS OF WORK [1]—Continued

	Substitution effect (per dollar per hour)	Income effect (per thousand dollars)
Wives:		
New Jersey	−143	−279
Rural	438	9
Gary	592	271
Seattle-Denver	215	−120
Single female heads:		
Gary	288	−303
Seattle-Denver	189	−126

[1] Estimates for New Jersey, Rural, and Gary are sample averages of the undeflated estimates reported in Moffitt and Kehrer (forthcoming, tables 3, 4, and 5) except for results by race which are weighted by the racial composition of the sample. Estimates for Seattle-Denver are simple averages of estimates presented in Dickinson (1980), Johnson and Pencavel (1979a, 1979b), Robins and West (1980a), and the ΔW, ΔY, and Δ_W, Δ_Y models we have presented.

In 80 percent of the cases, the signs of the estimated substitution and income effects are in accordance with theory which, as noted earlier, predicts that the substitution effect will be positive and that the income effect will be negative. Exceptions occur mainly for wives, where one of the substitution effects is negative (New Jersey) and two of the income effects are positive (Rural and Gary).[7] In cases where the signs are consistent with theory, the estimates are also quite similar—very reassuring given the wide variation in the methodological approaches taken. In general, the substitution and income effects tend to be smaller for men.

In table 3.29, we present a composite set of estimated substitution and income effects from all the experiments, based on the results in table 3.28. As before, we weight the estimates by the number of families enrolled in each experiment. These estimates suggest several things. First, in dual-headed families, the wife is much more sensitive to changes in the net wage rate than is the husband, although income effects are remarkably similar. Both findings are consistent with results from the nonexperimental literature.[8] Second, single female heads have a substitution effect that is similar to that for wives, but their income effect is approximately three times as large as the income effect for wives. The apparent implication that single female heads are the most responsive group to a negative income tax is, however, unwarranted. Although single female heads are the most sensitive to a particular change in the net wage rate and net nonwage income, they are currently eligible for benefits under a fairly generous welfare system, and the changes they would experience in these budget variables are quite small relative to the changes experienced by women in dual-headed families. In the next chapter, when we generalize the SIME/DIME estimates to specific NIT programs, we will see that single female heads have the smallest net responses.

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[7] Of course, some of the underlying estimates upon which these averages are based are of the correct theoretical sign. For example, Moffitt and Kehrer (forthcoming) present two sets of estimates for wives in the New Jersey Experiment. One of these sets indicates a negative substitution effect at mean labor supply while the other indicates a positive substitution effect. The positive substitution effect is obtained for a model similar to the ΔW, ΔY model we have presented. The negative substitution effect is obtained for a model in which the treatments are parameterized by the guarantee level and the tax rate (in addition to a treatment dummy).

[8] See Cain and Watts (1973) and Keeley (1981).

TABLE 3.29.—WEIGHTED AVERAGE OF SUBSTITUTION AND INCOME EFFECTS FROM ALL EXPERIMENTS [1]

	Substitution effect (per dollar per hour)	Income effect (per thousand dollars)
Husbands	104	−59
Wives	257	−54
Single female heads	216	−174

[1] Estimates are a weighted average of the results in table 3.28, where the weights are the number of families enrolled in each experiment.

Summary of results across experiments

Despite the wide range of treatments and evaluation methodologies, the labor supply responses from the four NIT experiments conducted in the United States are remarkably consistent. In no case is there evidence of a massive withdrawal from the labor force. On average, husbands reduced labor supply by about the equivalent of 2 weeks of full-time employment. Wives and single female heads reduced labor supply by about the equivalent of 3 weeks of full-time employment. Youth reduced labor supply by about the equivalent of 4 weeks of full-time employment. Because women and youth work fewer hours per year than do husbands, their effects are much larger in percentage terms.

In comparing structural model estimates across experiments, we also find remarkable consistency in the estimated substitution and income effects. When averaged across experiments, the substitution and income effects are similar to those obtained from nonexperimental studies. In general, wives' substitution effects are more than double husbands' substitution effects, and income effects for both spouses are similar. Single female heads have substitution effects of roughly the same order of magnitude as wives, although the income effects are three times as large. Despite the large values of the parameter estimates for single female heads, this group is likely to be the least responsive to a nationwide negative income tax because they already face a fairly generous welfare system that is not universally available to dual-headed families.

CHAPTER VIII

GENERALIZATION OF THE LABOR SUPPLY ESTIMATES

One of the most important objectives of SIME/DIME is to develop a methodology for generalizing the experimental labor supply results to the national population. Such generalization is not straightforward as we have noted throughout, because the experiments were conducted on a specific sample stratified by income, race, and family structure, and tested specific treatments considerably more generous than any program that would be considered for national implementation.[1] Hence, responses observed in SIME/DIME may bear little resemblence to overall responses expected under a nationwide program.

In recent years, microsimulation analysis has been developed for making nationwide predictions of the effects of alternative proposed transfer programs. The microsimulation approach consists of applying social program regulations and estimated behavioral response parameters to a microeconomic data base in order to project program costs and caseloads under varying conditions. Program eligibility requirements and benefit determination schedules are reproduced and the behavior of low-income families regarding welfare participation and various measures of labor supply is estimated. For the NIT, the tax liability, transfer payment, and amount of employment is determined for each family both before and after the NIT is implemented and the results are summed to derive the total change in costs, caseloads, and employment under alternative plans.

In their early developmental stages, microsimulation models did not account for behavioral changes that would occur as a result of new policy initiatives. The first estimates of behavioral response used in microsimulation models were based on nonexperimental data. Subsequently, the models have incorporated estimates of behavioral response based on experimental data.[2] Microsimulation models using experimentally based estimates of behavioral response have proven particularly useful in assessing the impact of welfare reform proposals.[3]

In this chapter, we utilize microsimulation techniques, together with findings from SIME/DIME, to examine how four different NIT plans implemented at the national level would affect work incentives and a variety of other behavioral and programmatic outcomes related to the low-income population.[4]

Labor supply response model used in the simulations

Two types of behavioral response models have been estimated in previous chapters—dummy variable models and structural models. For several reasons, the results from dummy variable models are not of much use for generalization.

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[1] As indicated earlier, the mean SIME/DIME treatment has a guarantee level equal to about 115 percent of the poverty level. Most proposed welfare reform alternatives, such as the Carter Administration's Program for better Jobs and Income, specified a guarantee level of 65 percent of the poverty level.

[2] The first microsimulation analysis using estimates of behavioral response based on nonexperimental data was preformed by Greenberg and Kosters (1973). The first microsimulation analysis based on experimental data was performed by Keeley et al. (1978b).

[3] See Aaron and Todd (1979), for example.

[4] The microsimulation model we use in our analysis is based on a microsimulation model originally developed by Mathematica Policy Research and subsequently extended and refined by SRI. Details of the model are presented in Robins, West, and Lohrer (1980).

First, the mean effects produced by dummy variable models are averages over the several different plans tested in SIME/DIME and are thus not representative of any single plan. Second, mean effects in the experimental sample depend on mean changes in disposable income and the net wage rate, as indicated by the theoretical model discussed in Chapter 2 and these mean changes, which depend on the distribution of income and demographic variables in the experimental sample, are different from the mean changes in a national sample. Finally, the assignment model, which allocated families to treatments on the basis of preexperimental income, must be accounted for in generalizing SIME/DIME results by allowing response to vary with preexperimental income in order to make meaningful comparisons across programs.[5] For all these reasons, a structural model yields vastly superior predictions of the nationwide effects of program alternatives.

The ΔW, ΔY structural model estimated in Chapter 5 is well suited to the task of generalization.[6] Recall that this model has the following general form:

$$\begin{aligned} H_n - H_0 = \Delta H &= \alpha\Delta W + \beta(H_0\Delta W + \Delta N) \\ &= \alpha\Delta W + \beta\Delta Y, \end{aligned} \qquad (1)$$

where ΔH = the difference between labor supply before and after the NIT program is implemented
ΔW = the change in the net wage rate
$H_0\Delta W + \Delta N = \Delta Y$ = the change in disposable income, holding constant labor supply at the preexperimental level
α = the substitution effect
β = the income effect.

The attractive feature of this model is that it differentiates response according to the two main features of an NIT program that are under the control of policymakers, the guarantee level and the tax rate.[7] In addition, response in this model varies with preexperimental labor supply, preexperimental nonwage income, and preexperimental wage rates, the main variables affecting preexperimental income, because the values of ΔW and ΔY depend on these variables. It is important to note, however, that the estimated substitution and income effects are constrained to be the same for each person in the sample.[8] This constraint is imposed to increase the precision of the estimated response. Previous work, cited in Keeley et al. (1978b), suggests that relaxation of these constraints (by allowing the estimated substitution and income effects to vary across certain groups in the sample) does not significantly increase the explanatory power of the model.

The model in equation (1) applies strictly only to persons below the breakeven level before response. However, as the discussion in chapter 5 implies, a small number of persons above the breakeven level before response will reduce their labor supply and receive benefits. In our empirical formulation of this model, we allowed such persons to respond by specifying a dummy variable that takes on the value of one if the person is above the breakeven level before response, and zero otherwise. The estimated coefficient of this dummy variable measures both the probability of going below the breakeven level and the response (both income and substitution) conditional on going below the breakeven level. Although such a specification of the above breakeven response provides useful information about the response of this group to the experiment, it cannot be used to simulate the above breakeven response to a nationwide program because it combines the probability of responding in the experimental sample with the response of those who do respond. An alternative approach, discussed below, uses the income and substitution effects estimated for persons below the breakeven level to simulate the response of persons above the breakeven level.[9]

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[5] For a discussion of this point, see the chapter on the design model and Keeley and Robins (1980).

[6] The alternative structural model discussed in Chapter 5 that utilizes predicted values of ΔW and ΔY could also be used, but the parameters of this model are not estimated very precisely relative to the ΔW, ΔY model.

[7] The guarantee level affects the value of ΔN in equation (1); the tax rate affects the value of ΔW.

[8] Equivalently, they represent the average substitution and income effects in the sample.

Our simulation analysis uses estimates of income and substitution effects for three measures of labor supply—annual hours of work, weeks worked per year, and the probability of working at all during a year. Annual hours of work is used because it is theoretically the most appropriate variable and the most comprehensive measure of labor supply available. Weeks worked is used because our earlier estimates suggest that there are significant effects on this component of annual hours and because the costs of many existing and proposed transfer programs depend on the number of weeks worked. The probability of working is used because our earlier estimates also indicate significant effects on this component of annual hours and because withdrawal from the labor force has important implications for the effectiveness of work requirements in proposed transfer programs.

The estimated income and substitution effects we use in the simulations are presented in table 3.30. The estimates for hours of work are the same as those reported earlier in chapter 5, table 3.11. The estimates for weeks worked and the probability of working are from Robins, West, and Keeley (1980).

TABLE 3.30.—ESTIMATED INCOME AND SUBSTITUTION EFFECTS ON THREE MEASURES OF LABOR SUPPLY FOR FAMILIES BELOW THE BREAKEVEN LEVEL

[Standard errors in parentheses]

Variable	Husbands	Wives	Single female heads
Annual hours of work (Tobit):			
Income effect ($1,000's)	[2]−95.15	[3]−196.75	[2]−147.12
	(41.59)	(63.75)	(70.28)
Substitution effect (dollars per hour)	71.33	151.76	176.37
	(56.31)	(136.33)	(119.57)
Weeks worked (Doubly truncated tobit):			
Income effect ($1,000's)	[1]−3.08	[3]−6.50	−3.22
	(1.77)	(2.37)	(3.05)
Substitution effect (dollars per hour)	3.61	7.35	[1]9.37
	(2.39)	(5.08)	(5.31)
Probability of working (Probit):			
Income effect ($1,000's)	[3]−.346	[3]−.194	−.110
	(.131)	(.075)	(.097)
Substitution effect (dollars per hour)	−.071	.198	.236
	(.170)	(.161)	(.178)
Sample size	2,249	2,332	1,715
Number of experimental families below the breakeven level	912	945	746

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Using the income and substitution effects to make nationwide predictions

The estimated income and substitution effects in table 3.30 can be used as follows to predict the labor supply response to specific NIT plans. First, a family is identified as being above or below the breakeven level. If the family is below the breakeven level, it is assumed individual family members will alter their labor supply in response to the program in the manner implied by the parameter estimates appearing in table 3.30. Some of these families will decrease labor supply if the NIT is more generous than the welfare programs it replaces; the remaining families, who are made worse off by the NIT, will increase their labor supply.

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[9] See Robins and West (1978a, 1980a) for further discussion of the difficulties of estimating income and substitution effects for persons above the breakeven level.

For families below the breakeven level, the changes in disposable income and net wage rate implied by the NIT are then calculated. Because of differences in preprogram income and wage rates, these changes vary considerably across individuals. The changes are based on actual preprogram income, so that persons with temporarily low labor supply are given large values of the change in disposable income, even though they may normally be above the breakeven level.

Once the change in disposable income and the change in the net wage rate are calculated, predictions of the labor supply effects of the NIT program are derived by multiplying these changes by the estimated income and substitution effects in table 3.30 (accounting for truncation at 0 hours of work, 0 weeks worked, and 52 weeks worked) and then summing the results over individuals. Additional information, such as program costs, is calculated using the predicted labor supply response and the hypothesized rules of the program.

Most families below the breakeven level will experience an increase in income as a result of the NIT program; many will move out of poverty. However, some families (particularly those near the breakeven level) will experience a decline in income because the earnings reduction caused by the labor supply response will more than offset the income provided by the program benefit. But although such families may not move out of poverty (some may actually move into poverty) they are certainly better off in a "well-being" sense in that total expenditures on market goods, home goods, and leisure will increase. This increase in well-being (or utility) can be measured using the estimated income and substitution effects and some basic principles of economic theory. By calculating the utility gain from an NIT it is also possible to estimate the potential labor supply response of persons initially above the breakeven level.

The gain in well-being from an NIT

In figure 3.3 the pre-NIT budget line is given by ABD and the NIT budget line by ABC. Prior to the NIT, the individual is in equilibrium at L_0 hours of leisure (on indifference curve I_1); after the NIT, the individual is in equilibrium at L_n hours of leisure (on indifference curve I_2). The NIT program has a guarantee level of S dollars, given by the distance CD. If there were no tax associated with the NIT program, the individual would demand L^{**} hours of leisure and would reach indifference curve I_3. The utility gain from the NIT would simply be equal to S, the amount of the guarantee. Because of the tax, however, the individual is not able to reach indifference curve I_3. Evaluated at pre-NIT prices, the utility gain from the NIT is given by DC−CE=DE, which is less than the support level S.

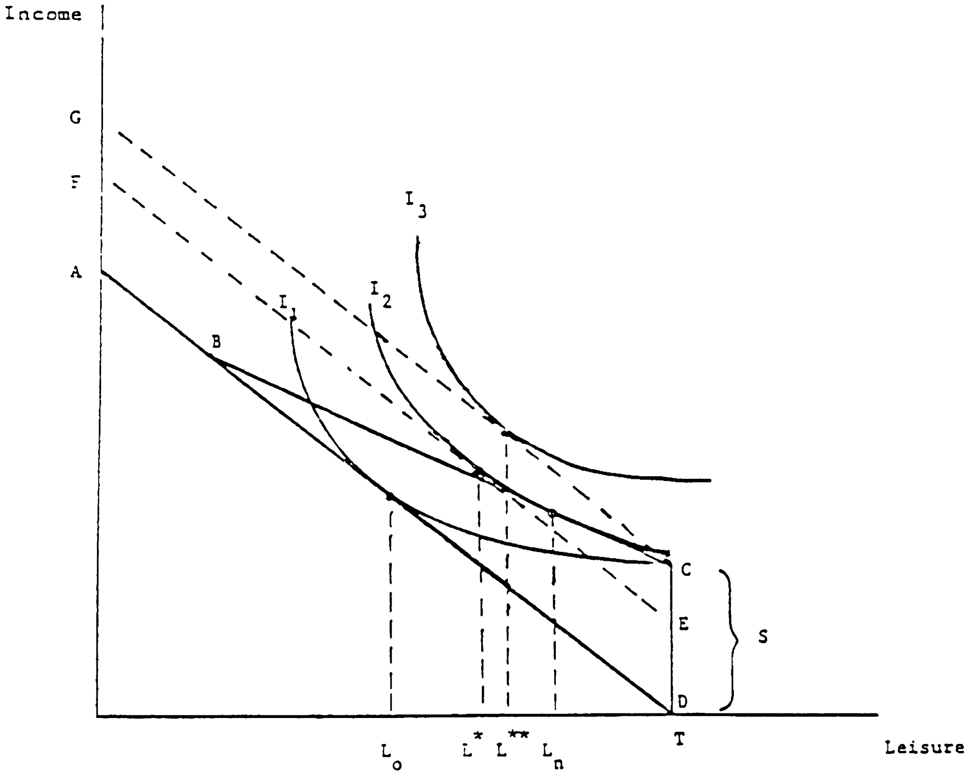

FIGURE 3.3.—Utility gain from an NIT program.

The distance CE is represented by the area of the trapezoid MPQON in figure 3.4. The area of the trapezoid is the sum of the area of the triangle MNO and the area of the rectangle MPQO.[10] The area of the triangle is equal to $\frac{1}{2}(H^* - H_n)tW = \frac{1}{2}\alpha^*(tW)^2$ and the area of the rectangle is equal to tWH_n, where W is the gross wage rate, α^* is the slope of the compensated labor supply curve (the Hicks substitution effect),[11] and $H = T - L$ is hours of work (total time T less leisure time L). The utility gain from the NIT is therefore equal to

$$S - tWH_n - \tfrac{1}{2}\alpha^*(tW)^2. \qquad (2)$$

The first two terms in this expression $(S - tWH_n)$ are equal to the NIT payment after response. The utility gain, therefore, is equal to the payment after response less the deadweight loss $(\frac{1}{2}\alpha^*(tW)^2)$.

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[10] The triangle is sometimes referred to as the welfare or deadweight loss associated with a price change. For an NIT program, it is the value of the reduced output associated with a reduction in the net wage rate that is not distributed as a program benefit (see Harberger, 1964).

[11] The coefficients in table 3.30 are not Hicks substitution effects but rather Slutsky substitution effects. Appendix C of Robins, West, and Lohrer (1980) describes the derivation of the Hicks substitution effects for the model given by equation (1).

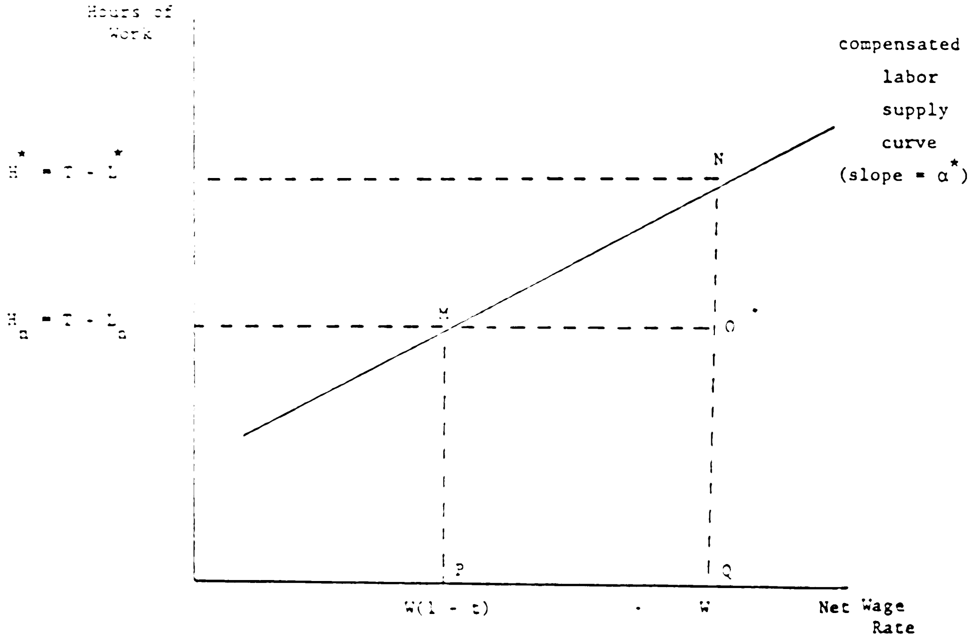

FIGURE 3.4.—Compensated labor supply curve and utility gain from an NIT program.

Response above the breakeven level

The formula given by equation (2) can be used to determine whether an individual initially above the breakeven level will receive positive benefits from the program. It is assumed that an individual will participate in the NIT program if there is an increase in well-being (i.e., if utility is increased). Thus, if the expression given by (2) is positive, the individual is assumed to participate in the program. Labor supply after response, H_n, is calculated in the same way as for persons below the breakeven level

$$H_n = H_o + \alpha \Delta W + \beta \Delta Y, \qquad (1),$$

repeated where H_o is pre-NIT hours of work, α is the substitution effect, β is the income effect, ΔW is the change in the net wage rate, and ΔY is the change in disposable income evaluated at pre-NIT hours of work. It should be noted that the NIT payment (ΔY) is calculated to be negative for persons initially above the breakeven level.[12] Thus, in order for an individual to drop below breakeven, his or her payment after response must be at least as great as the deadweight loss.[13]

The simulation model

As indicated at the beginning of this chapter, nationwide predictions are generated using a microsimulation model. The microsimulation model we use is a computer program called the Tax and Transfer Simulation Model (TATSIM) developed at SRI.[14] TATSIM is based on a microsimulation model developed by Mathematica Policy Research called the Microanalysis of Transfers to Households (MATH) model.[15] The MATH model in turn is a modified version of the Transfer Income Model (TRIM) originally developed at the Urban Institute. The TATSIM model is designed to predict the effects of potential transfer programs, such as an NIT, using as input a microeconomic data base. In this study, we use the 1976 Survey of Income and Education (SIE) to predict the effects of four NIT programs.[16]

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[12] In practice, of course, negative payments are not possible.

[13] This theory assumes that there are no tax or transfer programs prior to the NIT. Since this is not true in the real world, equation (2) must be modified to account for the positive tax system and the various welfare programs. Robins, West, and Loher (1980) discuss these modifications in detail.

[14] Appendix D of Robins, West, and Lohrer (1980) gives a nontechnical description of the TATSIM model and how it simulates labor supply response to a nationwide NIT program. Appendices E and F of the same paper give a complete technical description of the operation of the TATSIM model for the analysis presented here.

Income data from the SIE are annual data for the year 1975. Thus, the calculations represent what the effects of an NIT would have been in 1975. Unfortunately, 1975 is not the best choice of years for performing the simulations because this was a year of substantial unemployment. Nationally, the unemployment rate was 8.5 percent, larger than in any year up to then since the Great Depression. Because of this high unemployment rate relative to other years, our predicted labor supply responses and costs may be overstated because of an abnormal number of persons designated as below the breakeven level. Furthermore, as indicated below, the labor supply response may be overstated if the unemployment was truly involuntary. On the other hand, substantial unemployment insurance benefits could place a significant number of persons above the breakeven level. In any case, no attempt is made to update responses or cost estimates to later years.

The four programs for which predictions are made have constant tax rates of 50 percent and 70 percent on earnings, and guarantee levels of 75 percent and 100 percent of the poverty level ($5,456 for a family of four in 1975). Because the poverty level increases with family size, the support level also increases with family size. The nominal guarantee level is constant across regions.[17] The NIT is assumed to replace the existing AFDC, AFDC-U, and Food Stamp programs, and all other nonlabor income is taxed by the NIT program at a rate of 100 percent. The program reimburses federal and state income taxes and Social Security taxes at the rate of 100 percent.[18]

Limitations of the simulation approach

The simulation model used in this analysis ignores several issues that must be considered in any final evaluation of proposed welfare reform alternatives. We discuss these limitations briefly so that the reader is aware of the shortcomings of this approach and, hence, can assess the findings objectively.

First, a key assumption of the model is that the demand for labor is perfectly elastic. The assumption of perfectly elastic demand implies that gross wage rates are not affected by the NIT. Whether or not this is a good assumption depends in part on the substitutability of the labor of low-income persons who are likely to be participants in an NIT, for labor in general and for capital. If there is substantial substitutability, the effect of an NIT on wage rates is likely to be very small since total labor supply is probably not affected much by an NIT.

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[15] See Beebout (1976) and Maxfield (1977) for a description of the MATH model.

[16] The SIE is a major survey that was conducted by the Bureau of the Census in the spring of 1976 and contains a vast amount of economic, social, and demographic information on approximately 150,000 households. Because the SIE does not contain complete or reliable data on certain types of tax payments and transfer receipts, various methods are used to inpute values to these variables, which are critical inputs into the overall simulation process. Underreporting of welfare benefits is an especially troublesome problem in household surveys and efforts were made to correct for this problem in designing an analysis file from the SIE. For a discussion of the procedures used to impute information on the SIE, and the methods used to correct for underreporting, see Robins, West, and Lohrer (1980) and the references therein, especially the references to work by Mathematica Policy Research, who prepared the basic SIE data file used in the analysis in this chapter.

[17] This implies that the real support level varies across regions because of regional cost of living differences. The calculated variables used in the response function are corrected for regional cost of living differences. Thus, the responses differ across regions for persons who have the same nominal values for all variables.

[18] Full reimbursement of positive taxes was also a feature of SIME/DIME. Many existing programs, such as AFDC and Food Stamps, reimburse positive taxes partially by allowing these taxes to be deducted from the income used to calculate the benefit. The 1977 Carter welfare reform proposal, the Program for Better Jobs and Income, also contained a provision for reimbursing positive taxes partially. Later, we indicate the fraction of program costs that is due to reimbursement of positive taxes.

It is not possible to assess the effects of a downward-sloping demand curve on labor supply a priori, since knowledge of the aggregate supply curve of labor, and how this supply curve shifts (the shift is not a parallel shift), is needed.[19] Actual response may be either larger or smaller than the calculated response, depending on the slope of the aggregate supply curve. If aggregate labor supply is upward-sloping, the aggregate labor supply response would be smaller than is calculated assuming perfectly elastic demand. If aggregate supply is assumed to be backward bending, actual labor supply response would be larger than that calculated assuming perfectly elastic demand. Evidence presented by Greenberg (1980b) suggests that demand responses may be only a minor source of bias in estimating the labor supply responses to a nationwide NIT.

A second problem in calculating cost, but not labor supply response, is due to experimentally induced changes in marital status or fertility. Because the labor supply response function is a reduced-form equation, not conditional on unchanged marital status or fertility, our estimates of the impacts of an NIT on labor supply include any indirect effects caused by changes in marital status or family size. Our calculation of costs, however, assumes unchanged marital status or fertility. Thus, the costs calculated are due only to the labor supply response.[20]

Third, the model assumes that there are no changes in the positive tax system. This assumption implies that no consideration is taken of how the additional revenue is generated to finance the program. The most likely source, an increase in federal income tax rates, could have additional labor supply effects that may be either positive or negative.[21] The approach followed in this study is to analyze the effects of an NIT only on those who are previous recipients of welfare or who are likely to receive positive benefits from the program given their income level.

Fourth, all individuals considered eligible for NIT payments are assumed to receive them. In fact, few programs have 100 percent participation rates among those eligible, and most programs, unlike the NIT, have complex categorical eligibility requirements that are likely to discourage participation. For this reason, the simulations may overestimate the labor supply effects of the NIT.

Fifth, the model assumes that all unemployment prior to the NIT is voluntary. Hence, the possibility that job slots opened up by the NIT (as a result of the reductions in labor supply) would be taken up by unemployed workers is not allowed. Evidence presented by Greenberg (1980b) suggests that if involuntary unemployment is substantial, this effect can significantly offset the first round costs of an NIT. This possibility would have been especially likely for 1975 because, as indicated above, it was a year of abnormally high unemployment.

Sixth, the model assumes that the substitution and income effects estimated in the second year of SIME/DIME represent long term responses to an NIT. The evidence we presented earlier suggests that this is probably a reasonable assumption for husbands and wives, but may not be a reasonable assumption for single female heads who appear to take much longer to respond to the experimental treatments. In any event, the predictions are most appropriately interpreted as responses occurring approximately 2 years after implementation of an NIT.

Seventh, because the experiment affects only a portion of the potentially eligible population, neither the experimental results nor the national predictions reflect any changes in community attitudes that might result from a permanent nationwide program. If the NIT becomes accepted in the community, attitudes toward work may be substantially changed (that is, there may be a substantial reduction in the work ethic) and it is possible that labor supply effects would be larger than those reported here. There is no possible way of determining the extent of such a bias using experimental data.

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[19] Note that the aggregate supply curve is not calculated using the simulation approach; rather, the aggregate quantity of labor supplied is calculated. Thus, our simulation results do not have direct implications for the slope of the aggregate supply curve.

[20] If family stability is reduced by an NIT or fertility is increased, these calculated costs underestimate the total costs of an NIT. There is some evidence that the family stability effects of a nationwide NIT would not be too large (see Part V of this volume by Groeneveld, Hannan, and Tuma).

[21] Essentially, we are assuming that the method used to generate the additional revenue does not affect either supply or demand conditions.

Eighth, because interview data are used to estimate income and substitution effects, the predictions are not corrected for possible income underreporting. In a study of the effects of income underreporting on the size of a nationwide labor supply responses to an NIT, Greenberg (1980b) finds that estimates based on interview data will overstate the true labor supply reductions of an NIT. (Program costs would not be affected if reporting behavior under a national program is the same as reporting behavior under the experiment, because program costs are based on reported, not true, income). As Greenberg points out, however, the data underlying his study are subject to important limitations and, as a consequence, the size of the bias cannot be estimated with much precision.

Finally, the temporary nature of the experiment may cause biases in the predicted effects of a permanent program for a variety of reasons that are described in Robins and Stieger (1980) and Keeley and Wai (1980). These biases may be positive or negative and there is no definitive evidence at the present time on their quantitative importance for specific programs.

Of the nine biases discussed above, it is not clear which are most important, nor is it clear how they would bias the simulation estimates overall.[22] Some of the biases, such as changes in community attitudes, would lead to an underestimate of nationwide impact, whereas others, such as underreporting and the existence of involuntary unemployment, would lead to an overestimate. On balance, there is no convincing evidence that would lead us to conclude that the estimates obtained from the simulation analyses would be substantially different from the responses obtained under a nationwide program, although additional work on the extent of such biases is clearly warranted.

Simulation results

Of the 150,000 households surveyed in the SIE, approximately two-thirds were either husband-wife households or single female-headed households with children in which the household head was between the ages of 16 and 65, was not in the armed forces, and was not an inmate in an institution. To conserve on computational costs, we perform the simulations on random subsamples of these husband-wife and single female-headed families. For husband-wife families, we use about a 12 percent sample, or around 10,000 families. For single female headed families, we use about a 53 percent sample, or around 5,000 families—numbers judged to give sufficiently accurate predictions. All figures are scaled to national estimates using the appropriate sample weights.

For each NIT program, the following effects are calculated separately for husband-wife families and for single female-headed families with children:

1. Labor supply response.
2. Program costs.
3. Changes in income and utility.
4. Changes in poverty status.
5. Changes in well-being for welfare families.

In examining the effects on welfare families, the results are also broken down by regions within the United States.[23] A detailed description of how the variables are constructed is presented in Appendix E of Robins, West, and Lohrer (1980).

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[22] Additionally, factors specific to the experimental design and the evaluation model used in the simulations, such as the limited duration of the treatments, sample attrition, and incomplete adjustment to the treatments, could also bias the simulation estimates.

[23] Robins, West, and Lohrer (1980) present regional breakdowns for all the results presented in this chapter.

Labor supply responses

Predicted labor supply responses to the four nationwide NIT programs are presented in tables 3.31 and 3.32. Table 3.31 reports average responses of participating families for annual hours of work, annual weeks worked, and the fraction of persons working. In table 3.32, the average hours of work responses for the entire U.S. population, including the responses of nonparticipants, are reported for various categories of preprogram income. The nonparticipants who respond are families that previously received welfare benefits and are now earning a sufficient amount of income to place them above the breakeven level of the new NIT program. Because of the lost welfare benefits, these families increase their hours of work when the welfare programs are replaced by the NIT.[24]

Breaking down the response in these two ways gives a fairly comprehensive picture of the nature of the labor supply response to an NIT. Participating families represent those who may be judged poor by social standards because their incomes fall into the category that makes them eligible for government support. Clearly, their response is of policy interest. Examining responses for the entire U.S. population by income category provides an indication of the distributional effects of the programs and the extent of coverage for various socioeconomic groups.

In addition to providing estimates of the size of the response, standard errors of the predicted responses for participating families are also reported in table 3.31. These standard errors give an indication of the degree of confidence one can place in the point estimates. The standard errors account for only the variability in the estimated income and substitution effects and do not account for any variability due to the fact that the simulations are performed on a sample. The method used to calculate the standard errors is presented in appendix G of Robins, West, and Lohrer (1980). For additional details, see Lohrer (1980).

In interpreting the results, it is important to keep in mind that the responses vary not only because of changing guarantee levels and tax rates, but also because the number of participants changes. For example, as the tax rate increases (for a given guarantee), the number of participants decreases; as the guarantee increases (for a given tax rate), the number of participants increases. For the programs simulated, the percentage of families who participate (that is, those who receive benefits) ranges from 10 percent of all families in the United States in the 75/70 program to 42 percent of all families in the 100/50 program.[25]

Individuals participating in the various NIT programs differ considerably in terms of preprogram work effort. Under the least generous (75/70) program (the program with the lowest breakeven level), about 81 percent of the husbands work; under the most generous (100/50) program, 94 percent of the husbands work. For wives, the range is 38 percent to 48 percent. For single female heads it is 39 percent to 60 percent.

For participating husband-wife families (table 3.31), the percentage responses are statistically significant and increase in absolute value with both the guarantee level and the tax rate. All three measures of labor supply indicate fairly sizable reductions in work effort. For hours of work, the responses of husbands range

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[24] Most recent welfare reform proposals contain "hold harmless" provisions that would prevent welfare recipients from being made worse off. It is not clear how such provisions would be implemented over the long run, however. Later, we present some results for situations in which welfare recipients are not assumed to be made worse off.

[25] Throughout the remainder of this chapter, the specific NIT programs are referred to as X/Y, where X is the guarantee level as a percentage of the poverty level and Y is the percentage tax rate.

TABLE 3.31.—AVERAGE LABOR SUPPLY RESPONSE FOR PARTICIPATING FAMILIES

[Estimated standard errors in parentheses]

Program[1]	Hours of work		Weeks worked		Fraction working		Number of participating families (millions)	Fraction of families participating above the breakeven level before response
	Before response	Percentage change due to NIT	Before response	Percentage change due to NIT	Before response	Percentage change due to NIT		
75/50:								
Husbands	1,898	−4.8	42.1	−3.4	.92	−3.4		
		(1.1)		(0.9)				
Wives	467	−13.2	14.0	−13.6	.44	−13.2		
		(2.8)		(2.8)				
Total H&W	2,365	−6.5	56.1	−5.9			7.0	.03
Single female heads	546	7.9	19.5	2.9	.53	−3.2	2.9	.01
		(3.2)		(1.2)				
75/70:								
Husbands	1,434	−5.8	31.4	−7.5	.81	−4.3		
		(2.4)		(1.8)				
Wives	358	−16.8	11.0	−22.1	.38	−18.7		
		(5.2)		(4.8)				
Total H&W	1,792	−8.0	42.4	−11.3			2.4	.06
Single female heads	256	5.2	11.6	−6.3	.39	−5.8	2.2	.01
		(11.6)		(6.4)				
100/50:								
Husbands	1,993	−7.4	44.4	−4.4	.94	−5.5		
		(1.0)		(1.3)				
Wives	569	−18.4	16.8	−17.3	.48	−18.0		
		(3.0)		(4.0)				
Total H&W	2,562	−9.8	61.2	−7.9			15.7	.03
Single female heads	712	−2.2	23.7	−1.2	.60	−5.4	3.6	.01
		(2.7)		(2.5)				
100/70:								
Husbands	1,652	−8.0	36.6	−7.8	.86	−5.8		
		(1.8)		(1.9)				
Wives	426	−21.1	12.7	−24.2	.41	−21.0		
		(4.7)		(5.5)				
Total H&W	2,078	−10.7	49.3	−12.0			5.5	.07
Single female heads	453	−4.4	17.2	−8.3	.49	−8.2	2.9	.01
		(1.0)		(1.6)				

[1] First figure represents the guarantee level expressed as a percentage of the poverty level ($5,456 in 1975 for a family of four). Second figure represents the percentage tax rate.

from −4.8 percent to −8.0 percent, and the responses of wives range from −13.2 percent to −21.1 percent.

For participating single female heads of families, the labor supply responses under the low guarantee programs are generally positive; under the higher guarantee programs they are negative. The hours response varies from +7.9 percent for the 75/50 program to −4.4 percent for the 100/70 program. The same pattern holds with regard to changing guarantee levels and tax rates as holds for husband-wife families; that is, the responses become more negative (less positive) as the guarantee level or the tax rate increases. The fact that the responses are positive under the low guarantee programs suggests that many single female heads are being made worse off by the NIT program in the sense that their lost

TABLE 3.32.—AVERAGE HOURS OF WORK RESPONSES FOR ALL U.S. FAMILIES BY PREPROGRAM INCOME

[illegible]

Program[1]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	Greater than $13,000	Total
75/50:								
Husbands	8.8	5.5	4.6	2.8	1.8	0.9	0.1	0.7
Wives	16.7	14.7	6.2	4.7	2.3	1.2	2	1.0
Total H&W	8.1	8.2	4.2	3.0	1.9	1.0	.2	.8
	[1] (.88)	(.66)	(.55)	(.40)	(.30)	(.23)	(.03)	(.17)
Single female heads	23.8	13.1	8.9	2.8	8.4	8.1	3.0	9.0
	[1] (.97)	(.72)	(.55)	(.37)	(.13)	(.06)	(.01)	(.57)
75/70:								
Husbands	5.6	2.4	2	[illegible]	.5	.3	.1	.04
Wives	13.9	7.1	1.7	2.2	1.3	.8	.3	.1
Total H&W	7.3	4.7	[illegible]	1.0	1.0	.4	.2	.01
	(.86)	(.32)	(.18)	(.10)	(.03)	(.003)	(0.0)	(.06)
Single female heads	16.2	20.2	14.2	15.5	9.9	8.8	3.2	11.5
	(.96)	(.45)	(.18)	(.08)	(.02)	(0.0)	(0.0)	(.43)
100/50:								
Husbands	9.8	10.9	8.9	7.2	6.2	4.4	1.3	2.8
Wives	41.4	29.2	18.9	18.8	12.7	8.4	2.0	−4.8
Total H&W	14.4	16.1	11.4	9.8	7.7	5.3	1.5	−3.4
	(.96)	(.85)	(.75)	(.76)	(.74)	(.61)	(.20)	(.39)
Single female heads	4.8	1.0	1	3.8	2.1	4.2	1.6	1.5
	(.99)	(.92)	(.78)	(.65)	(.53)	(.28)	(.10)	(.72)
100/70:								
Husbands	9.5	8.9	5.6	3.0	1.7	.4	.04	−.8
Wives	26.9	24.0	10.0	5.9	1.7	.3	.2	−1.2
Total H&W	13.0	13.2	6.7	3.6	1.7	.3	.1	−.9
	(.96)	(.83)	(.60)	(.37)	(.25)	(.09)	(.01)	(.14)
Single female heads	12.5	.7	5.9	10.6	8.5	7.9	3.1	5.4
	(.99)	(.87)	(.51)	(.25)	(.10)	(.03)	(.01)	(.56)
Number of husband-wife families (millions)	1.12	1.15	1.97	2.39	3.65	3.60	26.41	40.3
	[2] (.03)	(.03)	(.05)	(.06)	(.09)	(.09)	(.66)	
Number of single female-headed families (millions)	1.86	.56	.63	.58	.50	.30	.63	5.1
	[2] (.37)	(.11)	(.12)	(.11)	(.10)	(.06)	(.12)	

[1] Proportion of families participating in parentheses.
[2] Proportion of families in parentheses.

welfare benefits exceed their NIT benefits.[26] Thus, although single female heads exhibit an overall negative response to the SIME/DIME treatments, the less generous nationwide programs that are simulated induce a positive response for this group.[27]

Average responses of the U.S. population (table 3.32) are generally small relative to the average responses of participants because most families in the United

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[26] In determining whether a family is worse off under the NIT, we use actual dollar receipts of welfare benefits before the NIT. We assume that families treat the value of their food stamp benefits as equivalent to cash income.

[27] Recall that the lowest guarantee level tested in SIME/DIME is 95 percent of the poverty level; the highest is 140 percent of the poverty level.

States do not participate in the programs.[28] The response to the most generous program (100/50) is, however, substantial for husbands and wives, who have hours responses of −2.8 and −4.8, respectively. Their large response occurs primarily because 39 percent of the husband-wife families participate in this program. Although the magnitude of the average responses again increases with the guarantee level (as it does for participants), it decreases with the tax rate for both groups. This inverse relationship between the average U.S. response and the tax rate is a consequence of the fact that the number of participants decreases by an amount large enough to offset the effect of a larger response among participants. Thus, the total disincentive effect of a nationwide NIT program is smaller under higher tax rate programs, despite the fact that the response of participating families is larger. As we shall see, this result also holds for the costs of a nationwide program.

Average responses of the U.S. population generally decrease (in absolute value) with income. This is primarily because the fraction of families participating in each category of income decreases with income, although response among participants also decreases with income. In 1975, the poverty level was about $5,500 for a family of four. As table 3.32 indicates, large fractions of families with incomes below this amount are eligible to participate under all the programs simulated. Under the least generous program (75/70), 58 percent of the husband-wife families and 84 percent of the single female-headed families with incomes under $5,000 in 1975 are eligible to participate. Under the most generous program (100/50), 90 percent of the husband-wife families and 97 percent of the single female-headed families with incomes under $5,000 are eligible to participate. Families in this income range make up 6 percent of the total number of husband-wife families and 48 percent of the total number of single female-headed families in the United States. Hence, with the possible exception of husband-wife families under the least generous program, coverage by an NIT among families officially designated as poor would be almost universal.

The NIT programs would also provide benefits to families not designated as poor in 1975. Under the least generous program (57/70), among those eligible to receive benefits, 39 percent of the husband-wife families and only 7 percent of the single female-headed families have incomes above $5,000. Under the most generous program (100/50), 87 percent of the eligible husband-wife families and 36 percent of the eligible single female-headed families have incomes above $5,000. Thus, under the most generous NIT programs, large numbers of families not officially designated as poor are eligible for benefits. This occurs because the breakeven levels of the programs exceed the poverty level by a considerable amount.

Program costs

Estimated net program costs for the four NIT programs simulated are presented in tables 3.33 through 3.35. Program costs are defined as costs in addition to the current costs of the AFDC, AFDC–U, and Food Stamp programs (about $12 billion in 1976), which are assumed to be replaced by the NIT program. In table 3.33, a comparison is made between program costs before and after the labor supply response. In table 3.34, four sources of program costs are enumerated. In table 3.35, the allocation of program costs to the consumption of leisure and market goods is presented.

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[28] Exceptions occur for single female heads under the low guarantee programs because so many nonparticipants are made worse off by the removal of welfare. Any program that would be considered for actual implementation, however, would probably not allow such families to be made worse off, at least during the early years of the program.

TABLE 3.33.—PROGRAM COSTS BEFORE AND AFTER RESPONSE

Program	Program costs before response (billions)	Change in program costs due to labor supply response (billions)	Program costs after response (billions)	Number of participating families (millions)	Fraction of families participating	Average cost per participating family
75/50:						
Husband-wife families	$4.65	$1.49	$6.14	7.02	0.17	$875
Female-headed families	-1.95	-.57	-2.52	2.87	.57	-878
Total	2.70	.92	3.62	9.89	.22	366
75/70:						
Husband-wife families	.44	.21	.65	2.36	.06	275
Female-headed families	-2.84	-.63	-3.47	2.18	.43	-1,592
Total	-2.40	-.42	-2.82	4.54	.10	-621
100/50:						
Husband-wife families	20.18	7.12	27.29	15.65	.39	1,744
Female-headed families	2.21	-.06	2.15	3.64	.72	591
Total	22.40	7.06	29.45	19.30	.43	1,526
100/70:						
Husband-wife families	5.63	2.21	7.84	5.47	.14	1,433
Female-headed families	.53	-.20	.33	2.86	.56	115
Total	6.16	2.00	8.16	8.33	.18	980

TABLE 3.34.—SOURCES OF PROGRAM COSTS AFTER RESPONSE

[In billions of dollars]

Program	Grants paid after response	Savings in welfare program costs	Positive tax reimbursement	Tax loss due to labor supply response	Program costs after response
75/50:					
Husband-wife families	$6.26	$3.23	$2.63	$0.49	$6.14
Female-headed families	6.27	8.73	.32	.39	-2.52
Total	12.53	11.96	2.95	.10	3.62
75/70:					
Husband-wife families	3.92	3.23	.24	-.28	0.65
Female-headed families	5.71	8.73	.06	-.52	-3.47
Total	9.63	11.96	.30	-.80	-2.82
100/50:					
Husband-wife families	16.42	3.23	10.80	3.31	27.29
Female-headed families	10.26	8.73	.77	-.15	2.15
Total	26.68	11.96	11.57	3.16	29.45
100/70:					
Husband-wife families	9.25	3.23	1.48	.35	7.84
Female-headed families	9.20	8.73	.21	-.36	0.33
Total	18.44	11.96	1.69	-.01	8.16

NOTE.—Program costs = benefits paid − savings in welfare program costs + tax reimbursements + tax loss.

TABLE 3.35.—ALLOCATION OF PROGRAM COSTS INTO CONSUMPTION OF LEISURE AND MARKET GOODS

[Dollars in billions]

Program	Earnings reduction at initial wage rate	Program costs after response	Income change	Percentage of program costs consumed as leisure
75/50:				
Husband-wife families	$2.72	$6.14	$3.43	44.3
Female-headed families	−1.60	−2.52	−.93	....
Total	1.12	3.62	2.50	30.9
75/70:				
Husband-wife families	−.35	0.65	1.01	....
Female-headed families	−1.98	−3.47	−1.50	....
Total	−2.33	−2.82	−.49	....
100/50:				
Husband-wife families	14.16	27.29	13.13	51.9
Female-headed families	−.34	2.15	2.49	....
Total	13.82	29.45	15.63	46.9
100/70:				
Husband-wife families	2.77	7.84	5.07	35.4
Female-headed families	−.95	.33	1.28	....
Total	1.82	8.16	6.34	22.3

Estimated program costs vary widely across programs. Table 3.33 indicates that the most expensive program (100/50) costs $29.5 billion more than the existing welfare system and has 19.3 million participating families, implying a cost of about $1,525 per participating family. Approximately 39 percent of all husband-wife families and 72 percent of all single female-headed families with children participate in the program. Because of the enormous costs and the large number of participants under this program, it is unlikely that an NIT providing a basic benefit for each family equal to the poverty level and a tax rate of 50 percent represents a feasible policy option. The least expensive program (75/70) costs $2.8 billion less than the existing welfare system (a 23 percent savings in welfare program costs) and has about 4.5 million participating families. Only 6 percent of husband-wife families and 43 percent of single female-headed families with children participate in this program. The feasibility of this program depends to a large extent on the number of existing welfare recipients that would be made worse off. Below we present evidence suggesting that close to 80 percent of the existing welfare population would be worse off under this program, probably making it also an infeasible policy option.

The proportion of program costs after response due to the labor supply response is about 25 percent for all the programs with positive costs. The magnitude of this additional cost demonstrates the importance of accounting for adjustments in labor supply when designing a national program. Failure to take these adjustments into account can lead to significant underestimation of total program costs.

Table 3.33 also indicates that the costs of an NIT program are substantially lower under higher tax rate programs than under lower tax rate programs holding the guarantee level constant. For instance, moving from the 100/50 program to the 100/70 program lowers program costs for husband-wife families from $27 billion to $8 billion. Costs are lower because of a substantial reduction in the number of participating families which outweighs the additional costs resulting from the reduced work effort among participants.

Increasing the guarantee level has an enormous effect on program costs. For example, the cost of the 100/50 program ($26 billion) is more than eight times the cost of the 75/50 program. Most of this cost difference is the result of increased coverage, although a significant portion is due to greater benefits received by participating families (as a result of the higher guarantee level and the larger labor supply response).

In table 3.34, four sources of program costs are identified. These include benefits paid, savings in welfare program costs, tax reimbursements, and the tax loss due to the labor supply response. For three of the four programs, savings in welfare costs are smaller than benefits paid under the new program, thus making the NIT more expensive. For the most expensive program (100/50), positive tax reimbursements (which may also be termed tax relief) are about the same as the savings in welfare costs.

As indicated in table 3.35, a significant portion of program costs is consumed as leisure (or results in increased home production).[29] Of the programs with positive costs, the proportion consumed as leisure ranges from 22 percent to 47 percent. This means that measured disposable income rises by 53 percent to 78 percent of program costs. Although participating families are generally made better off in a utility sense by the NIT program (in terms of their command over market goods, home goods, and leisure), they do not all choose to increase their monetary income. Families most likely to experience an increase in monetary income are those farthest below the breakeven level (because of the dominance of the income effect), while the families most likely to experience a decrease in monetary income are those nearest to the breakeven level (because of the dominance of the substitution effect).

Effects on welfare families

The simulations assume that certain welfare programs (AFDC, AFDC-U, and Food Stamps) are replaced by a single NIT program with a uniform benefit structure across the United States. Because of the wide regional variation in benefits under current welfare programs, it is likely that some families in the more generous states would be made worse off by the NIT program. That is, some families who were previously receiving welfare benefits would have those benefits reduced. Table 3.36 presents the percentage of welfare families that are made worse off by the NIT, with and without state supplementation of lost welfare benefits.[30]

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[29] This increased leisure (or home production) has value which should be considered when evaluating alternative programs (see below). From a social point of view, however, increased incomes for the poor may have value to nonparticipating taxpayers, but increased leisure may not.

[30] The figures in table 3.36 abstract from labor supply response to the lost welfare benefits. Since labor supply will generally increase for families made worse off, the NIT benefits will be even smaller and a greater number of families will experience a reduction in total income. The figures in table 3.36 are intended only as a rough comparison of the generosity between the existing welfare system and alternative NIT programs.

TABLE 3.36.—WELFARE FAMILIES MADE WORSE OFF BY THE NIT PROGRAM

[Before response]

Program	Number made worse off (millions)	Amount made worse off per family	Percentage made worse off	Additional program costs to supplement lost welfare benefits (billions)	Total program costs with State supplementation (billions)
75/50:					
Husband-wife families	1.36	$999	44.9	$1.36	$6.01
Female-headed families	2.53	1,189	76.1	3.01	1.06
Total	3.89	1,123	61.2	4.37	7.07
75/70:					
Husband-wife families	2.26	821	74.3	1.86	2.30
Female-headed families	2.78	1,296	83.7	3.60	.76
Total	5.04	1,083	79.2	5.46	3.06
100/50:					
Husband-wife families	.68	1,106	22.4	.75	20.93
Female-headed families	1.21	1,021	36.4	1.24	3.45
Total	1.89	1,051	29.7	1.99	24.39
100/70:					
Husband-wife families	1.25	950	41.0	1.19	6.82
Female-headed families	1.69	1,118	50.7	1.89	2.42
Total	2.93	1,047	46.1	3.08	9.24

As table 3.36 indicates, the percentages of welfare families made worse off by the NIT are quite large, even for the more generous programs. This is an important finding, illustrating the extensive generosity of the existing set of overlapping welfare programs. For example, under the 100/50 program, 30 percent of the welfare families would be worse off (22 percent of the husband-wife families and 36 percent of the single female-headed families). Under the 75/70 program, almost 80 percent of the welfare families would be made worse off. If these families were compensated (through some form of state supplementation that replaced lost welfare benefits) the likely result would be a substantial increase in program costs. An indication of how large these additional costs would be is presented in the last two columns of table 3.36. Whereas the range of program costs (before response) without state supplementation is from —$2.4 billion to $22.4, the range with state supplementation is from $3.1 billion to $24.4 billion. Under certain programs, state supplementation of lost welfare benefits adds significantly to total program costs.

Effects on poverty status

One of the major objectives of a nationwide NIT program is to reduce the incidence of poverty among the U.S. population. In table 3.37, we show how the four simulated NIT programs affect the size of the poverty population and the poverty gap, which measures the total amount of additional income required to eliminate poverty. For purposes of the simulations, poverty is defined in terms of monetary income plus in-kind benefits from the Food Stamp program. Ignored are in-kind benefits, such as Medicaid, public housing, etc., and the value of leisure and home production. These are all important components of a family's well being and should certainly be included in any definition of poverty status; but since our main purpose is to compare effects of alternative NIT programs, we will not concern ourselves with developing the most appropriate measure of poverty status.[31]

TABLE 3.37.—EFFECTS ON MEASURED POVERTY STATUS

	Husband-wife families	Single female-headed families	All families
Before program			
Percentage below poverty	4.4	29.3	7.2
Poverty gap (billions)	$4.01	$1.95	$5.96
After program			
Percentage below poverty			
75/50	3.7	36.9	7.4
75/70	5.0	41.8	9.1
Poverty gap (billions)			
75/50	$1.95	$2.03	$3.98
75/70	$2.29	$2.24	$4.52
Percentage change in the poverty gap			
75/50	−51.3	4.2	−33.2
75/70	−43.0	14.9	−24.2

According to our definition of poverty, 4 percent of the husband-wife families and 29 percent of the single female-headed families are in poverty prior to the NIT. Of those in poverty before the program, the fraction moving out of poverty after the program is implemented varies with both the guarantee level and the tax rate, although the former is clearly the most important determinant of postprogram poverty status. For the low guarantee programs, there is generally an increase in the number of families in poverty, partly because of the labor supply response, partly because the guarantee level is less than the poverty level, and partly because the NIT is less generous than the existing welfare system for many families. For the lower tax rate program (75/50), 31 percent of the husband-wife families and 7 percent of the single female-headed families are able to escape poverty (not shown in table 3.37), although for single female heads this is offset significantly by a large number of families entering poverty. For the higher tax rate program (75/70), virtually all families remain in poverty after the program and many families not previously in poverty become poor. For the high guarantee programs (not shown in table 3.37), poverty is effectively eliminated because the guarantee level (which becomes the income floor under a universal program) is equal to the poverty level.

One frequently cited measure of the extent of poverty is the poverty gap, which measures the total amount of money income required to eliminate poverty. Without the NIT program, the (1975) poverty gap is about $6 billion, or about $1,800 per poor family. Without the NIT program, the poverty gap is eliminated under the high guarantee programs and is reduced by between one-quarter and one-third under the low guarantee programs.

Effects on family well-being

Although evaluating alternative NIT plans on the basis of changes in money income is useful for many purposes, it does not give a complete picture of the change in the well-being of participating families. Since labor supply reductions are voluntary, families are using the NIT payments to purchase additional leisure (at the expense, perhaps, of market goods). Since leisure itself includes time spent in home production and recreational (or consumption) activities, it is important to evaluate these activities in assessing the merits of competing programs since they benefit society as a whole.[32] Even though money income (command over market goods) may decrease, a family may be better off by virtue of increased home production or leisure consumption. Some families, however, will be unambiguously worse off because of the elimination of more generous welfare programs.

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[31] For a discussion of alternative measures of poverty status, see Danziger, Haveman, and Plotnick (1980) and the references therein.

Table 3.38 compares changes in monetary income with changes in utility for the four NIT programs simulated. The figures indicate that each NIT program leads to both increases and decreases in money income of families in the U.S. Money income decreases occur either because of the elimination of more generous welfare programs or because of earnings reductions in excess of NIT payments by families near the breakeven level. Only those families whose welfare income has been eliminated can be worse off in a utility sense. In most cases, the fraction of families with increased income exceeds the fraction of families with decreased income. Exceptions occur for single female heads under the low guarantee programs. Among the various programs, the 100/50 program has the greatest fraction of families with increased income and the greatest ratio of increases to decreases in income. However, the fraction of families with decreased income is about the same under the 100/50 program as under the low guarantee programs. There are two possible explanations for this. First, the labor supply response is greater under the high guarantee program. Second, and perhaps more important, families with decreased incomes are initially very near the breakeven level and there are more such families in a high guarantee program than in a low guarantee program, because the breakeven level is in a "thicker" part of the income distribution. For husband-wife families, the fraction of families with decreased income is greatest for the 100/50 program.

Table 3.38 also indicates that each NIT program leads to both increases and decreases in utility. The decreases in utility arise because certain welfare families are made worse off by the NIT. Although the fraction of families with increased utility generally exceeds the fraction of families with increased income, there are cases where the fraction of families with income gains exceeds the fraction with utility gains. These cases occur under the less generous NIT programs where earnings increase (due to a positive labor supply response) and utility decreases (due to the elimination of more generous welfare programs).

Finally, table 3.38 shows how utility changes for families who experience a decrease in income. This relationship varies considerably across programs. Under the most generous program (100/50), 66 percent of the families that experience a decrease in income also experience an increase in utility, although this effect is almost entirely concentrated among husband-wife families. Under the least generous program (75/70), virtually all families that experience a decrease in income also experience a decrease in utility, reflecting the dominance of welfare over this NIT program.

In the aggregate, the four NIT programs are summarized in table 3.39. with regard to income changes, utility changes, and program costs. For the programs with positive costs, the utility change is approximately equal to program costs, suggesting a fairly small deadweight loss from the programs. The utility changes range from 22 percent to 84 percent more than the income changes. For the program with a negative cost, the utility loss is less than the saving in welfare costs and the utility change is approximately equal to the income change.

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[32] Increases in the consumption of leisure benefit society because they increase the "quality" of life. Of course, it must be remembered that we have ignored the financing of a nationwide NIT. Undoubtedly, increasing taxes to finance the NIT would lower the quality of life for some individuals. A social welfare function is needed to evaluate the benefits to society of such a redistribution of well-being.

TABLE 3.38.—INCOME AND UTILITY CHANGES RESULTING FROM NIT PROGRAMS

Program	Fraction of families with increase in		Fraction of families with decrease in		Fraction of families with decrease in income that have an increase in utility
	Income	Utility	Income	Utility	
75/50					
Husband-wife families	0.14	0.16	0.06	0.03	0.55
Single female headed families	.27	.22	.44	.49	.04
Total	.18	.17	.10	.09	.30
75/70					
Husband-wife families	.06	.05	.05	.06	.13
Single female headed families	.21	.15	.47	.54	.02
Total	.07	.06	.09	.11	.06
100/50					
Husband-wife families	.31	.38	.09	.02	.83
Single female headed families	.56	.54	.22	.24	.14
Total	.34	.40	.10	.04	.66
100/70					
Husband-wife families	.11	.13	.04	.03	.51
Single female headed families	.43	.38	.29	.33	.05
Total	.15	.16	.07	.06	.30

TABLE 3.39.—TOTAL INCOME AND UTILITY CHANGES

[In billions of dollars]

Program	Program costs	Income change	Utility change
75/50	$3.6	$2.5	$3.6
75/70	4.1	−2.5	−2.6
100/50	29.5	15.6	28.7
100/70	8.2	6.3	7.7

Summary of the simulation results

In this chapter we have used the structural model results from SIME/DIME to derive implications about various effects of alternative nationwide NIT programs. Four alternative nationwide NIT programs were examined in an attempt to determine how substitution of a universal NIT program for the existing welfare system would affect work incentives and a variety of other measures related to the low-income population. Several important conclusions emerge from our analysis which we believe provide useful information for the continuing debate on welfare reform.

First, it is evident that a fairly generous welfare system already exists in the United States, especially for single female-headed families. Under all four NIT programs simulated, a substantial number of families already receiving welfare benefits are made worse off. Even under the most generous NIT program (100/50), almost 30 percent of the existing welfare families are made worse off by the NIT. Despite this perhaps undesirable effect, the program would still cost almost

$30 billion more than the existing welfare system, primarily because of the extension of benefits to many working couples that do not currently qualify for welfare benefits.[33] Single female-headed families are made worse off by all the NIT programs simulated and actually increase their labor supply when the existing programs are replaced by the NIT.

Second, under an NIT that replaces the existing welfare system, both labor supply response and program costs vary widely with the guarantee level and the tax rate. As noted above, the most generous program simulated (100/50) would have cost $30 billion more in 1975 than the existing welfare system cost at that time, whereas the least generous program (75/70) would have cost $2.8 billion less. (State supplemenation would add substantially to these cost estimates.) Furthermore, adjustments of hours of work in response to these programs account for a substantial fraction (25 percent) of total costs for those programs with positive costs.

Third, although higher guarantee levels and higher tax rates substantially reduce labor supply for program participants, higher tax rates lead to smaller aggregate reductions in labor supply and smaller program costs than do lower tax rates. This is because, other things equal, higher tax rate programs have lower breakeven levels and hence fewer participants than do lower tax rate programs. Thus, it appears that adequate minimum benefits (a high guarantee level) can only be insured at a feasible cost if the tax rate of the program is high. But, under such a program, participants may decrease their work effort enough to become dependent on the program because of the very low economic return to working.

Fourth, the simulations suggest that a significant portion of the labor supply response takes the form of a reduction in the probability of working. Although SIME/DIME did not include work provisions as part of the eligibility criteria, it is possible that program costs could be reduced if stringent work requirements were attached to receipt of benefits. Hence, an NIT coupled with a rigorously supervised job search assistance program could potentially satisfy three often contradictory goals of welfare reform: adequate minimum benefits, desirable economic incentives, and reasonable cost.

Finally, although a nationwide NIT leads to a reduction in work effort, participating families are made better off because total resources available for expenditures on market goods, home production, and leisure are increased. For the programs simulated, we have found that the total value of this increased well-being is approximately equal to program costs. Although we have not considered the decrease in well-being associated with financing an NIT, it is possible that such a program may have redistributive effects that would be judged desirable from a social point of view.

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[33] Tax relief alone, under the most generous NIT program, amounts to about the same amount of money as the cost of the existing welfare system.

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SIME/DIME FINAL REPORT

VOLUME 1, PART IV

IMPACTS OF COUNSELING AND EDUCATION SUBSIDY PROGRAMS

by

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CHAPTER I

INTRODUCTION

One of the unique features of the Seattle and Denver Income Maintenance Experiment was the simultaneous presence of job counseling and education subsidy options. The rationale for including these counseling and training programs was to determine whether increased labor market information and increased education could offset the decline in work effort that was expected to result from an income maintenance program. All individuals who were eligible could receive job counseling services, and some were also eligible for subsidies of either 50 percent or 100 percent of direct expenses of job-related training and schooling. Both programs were nondirective. Counselors provided clients with labor market information and advice concerning job search methods, but did not attempt to lead them into specific career choices. The subsidy program was essentially a voucher system: all types of schooling and training were subsidized as long as they were job related.

SIME/DIME job counseling and education subsidy programs represent additions to a long series of government sponsored counseling and training programs. Two objectives of the SIME/DIME programs can be distinguished. The first is an antipoverty objective of increasing the long-term economic status of participants, an important objective of many CETA programs as well. The second objective is to reduce public welfare costs by increasing the employability of welfare recipients, a goal of the work incentive program (WIN), which is mandatory for certain AFDC recipients. However, the SIME/DIME counseling and training programs are some of the first to be evaluated experimentally. SIME/DIME families were randomly assigned to be eligible for various treatments or to be in control groups. With the advantage of randomly assigned control groups, the evaluation of SIME/DIME programs can avoid self-selection biases that have plagued the evaluations of most other counseling and training programs. Because of this important advantage, it is desirable to examine whether results of the SIME/DIME program can be generalized to other counseling/training programs.

The explicit purpose of SIME/DIME counseling and training programs was to mitigate the labor supply response to the income maintenance programs by increasing labor market information and labor market skills. It was believed that the counseling program could attain two goals: increased client awareness of individual abilities and personal goals, and increased knowledge of labor market conditions and prospects. By achieving these goals the program could enable a client to make more effective decisions about a career choice, leading to greater earnings in the labor market. The education subsidy program was intended to induce the client to acquire additional schooling or training, to lead to more marketable skills and consequently to greater labor market success.

In summary, the availability of counseling and education subsidy programs was expected to result in the following sequence of events:

1. Participation in the counseling program.
2. Increased schooling and training, particularly among those eligible for a subsidy.

3. A concurrent decrease in work hours and labor force participation in order to make time available for education and training. This decrease might not occur for those initially devoting most of their time to nonmarket activities. (Those who do not seek additional training are not expected to exhibit this response.)
4. A subsequent increase in job search activities, perhaps with an increase in unemployment.
5. An increased wage rate and upgraded occupation.
6. An increase in participants' work hours because of more stable employment.

Overview of Part IV

The analysis presented in part IV indicates that the counseling/training programs did not have training the expected long-run impact on labor market performance for most groups. In the short run, there was substantial participation in the programs resulting in net increases in the amount of counseling and education received. Earnings declined during the eligibility period but, quite unexpectedly, the negative results tended to continue into the postprogram period as well. The programs seem to have eroded rather than improved the labor market positions of those eligible to participate, with the exception of the counseling-only program for single women. These are startling results. The availability of these programs was expected to broaden participants' opportunities and thus increase their economic well-being. That the programs worsened their situation was unanticipated.

To help the reader interpret the unexpected results, chapter 2 presents a detailed description of the counseling and education subsidy programs. Chapter 3 describes the extent of participation in the programs, how participants made use of the programs, and how individuals that chose to participate differed from those who did not. Chapter 4 describes the experimental design and statistical methodology used to estimate the programs' impacts. Chapter 5 presents the impact of the programs, first on the amount of education and counseling received, and then on labor market performance. Because unanticipated results were found, chapter 6 presents an exploratory analysis of the mechanisms by which the programs may have produced those results. The final chapter compares the results of the SIME/DIME programs to those of other programs and draws conclusions about the design of future counseling and training programs.

CHAPTER II

DESCRIPTION OF THE COUNSELING AND EDUCATION SUBSIDY PROGRAMS

Counseling/training options were included in the SIME/DIME experiment to examine whether the decline in work effort expected from income maintenance could be offset by increased labor market information and training. The income maintenance program was designed to minimize interference in recipients' lives. It was noncategorical and based solely on fixed income eligibility rules, in contrast to existing programs (such as AFDC) which apply only to specific categories of low-income families and, in the early 1970s, were subject to substantial administrative discretion in determining grant levels. In keeping with the nonintervention objective of the income maintenance program, the counseling and training programs were designed to enhance the individual's freedom of choice. Although the intent of the counseling program was to increase participants' information about their labor market prospects and opportunities, in designing the actual program the professional counselors involved felt that the mere provision of labor market information, even if accurate and relevant, might not fully meet the clients' needs. It was believed that the counseling process should first prepare the clients to consider making labor market decisions by improving their self-concepts, assessing their past performances, and developing labor market goals. Again, this self-assessment process was designed to be nondirective; the counselors helped clients address specific issues but did not suggest courses of action. The counseling process is described more fully below.

The educational subsidy program was also designed to preserve participants' freedom of choice. The subsidy applied to the direct costs of job-related training (although the amount of the subsidy was limited to the cost of the least expensive local institution that provided the requisite training) and thus, in practice, was limited to classroom and skill training. On-the-job training was not available since it would have entailed substantially different administrative mechanisms to extend the voucher system to subsidize employers.[1]

The remainder of this chapter describes SIME/DIME counseling and training programs in some detail.[2] We first describe the counseling/training delivery system, including the location of the program and characteristics of the staff. We next describe the counseling process and compare it to other job counseling programs. An analogous description of the education subsidy program is then presented.

The counseling/training delivery system

SIME/DIME counseling and training programs were operated in a local community college at each site. When the income maintenance experiment began in Seattle, several alternative intake centers were considered, including the Employment Service, the concentrated employment program, and a local multiservice agency. However, these alternatives were considered less desirable because of bureaucratic and political constraints and, in some cases, poor management of existing programs.

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[1] A vouchered OJT program was used in a WIN program, as reported by Weidman (1979). It is interesting to note that the type of counseling available in that program was similar to that offered through SIME/DIME (including nondirective self-assessment, labor market information, and job search training). Moderate support was found for such a program among employers but no evaluation was conducted of the impact on participants.

[2] See Benus, Halsey, and Spiegelman (1979) for further detail on the counseling program, and Davis and Dickinson (1980) for further detail on the education and training subsidy program.

The Seattle Community College, in contrast, was willing to accommodate the special needs of the experiment and the unknown size of the program. The community college was located in the low-income area and had a good reputation in that community. Further, it offered vocational training in 90 different trades in addition to a full academic program for the first 2 years of college. When the program was expanded to the Denver site, an attempt was made to replicate the Seattle program as closely as possible. Fortunately, the Denver Community College had facilities and capabilities similar to the Seattle Community College.

Directors of the counseling programs were all experienced and highly trained professionals. The Seattle director had about 10 years of experience as a senior counselor and had conducted an extensive study of nondirective counseling. He was, in fact, responsible for establishing the nondirective approach of the SIME/DIME program. The Denver director had a PhD in psychology, had done extensive research in counseling, and had practical experience as well.

Counselors were hired by the community colleges according to criteria established by SRI. These included willingness to accommodate the experimental nature of the program, ability to counsel clients in a nondirective manner, ability to counsel low-income clients without hostility or a patronizing manner, and enough flexibility to learn the counseling techniques required by the program. All but 1 of 28 counselors who were hired for this program had prior counseling experience, the average length of prior counseling experience being 4.5 years. Of the counselors, 71 percent had a postgraduate degree or specific training in counseling beyond a B.A. degree. The racial composition of the staff approximated that of the client population in both sites.[3]

Counselors were trained in the nondirective counseling techniques in group sessions. This included a 3-day workshop at the start of the program, and continuous review training using such techniques as role playing and videotapes of actual sessions. In Seattle, this training was conducted by the project director and an expert in job counseling from the University of Washington. In order to replicate the Seattle process in Denver, the Denver director made two initial visits to Seattle to study the program content, implementation, and counselor training techniques, and to observe the counselors in actual sessions. Later, the Seattle director and head counselor went to Denver to conduct a workshop on self-assessment. Throughout the project, close contact between the two sites was maintained to keep the two projects as similar as possible.

In addition to counselors, the project also employed 8 labor market information specialists who prepared reports on the requirements of and prospects in specific occupations. All specialists had college training, most having done graduate work in social sciences. Their prior work experiences varied, involving one or more of the following: research writing, data gathering and analysis, teaching, and counseling/training services.

The counseling process

Because of its limited duration and constrained eligibility, SIME/DIME contained a large outreach effort not normally found in counseling and training programs. Families were informed of their eligibility at the time of the enrollment interview. Subsequently, a counselor contacted the family at home and explained the program to each eligible individual. Those who did not participate immediately were recontacted in person 6 months to 1 year later, and subsequently by mail, in an effort to keep them continually aware of their eligibility. In a further effort to encourage participation, counseling participants were given $5 for each of the first five sessions attended to cover expenses such as child care and transportation.

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[3] Counselor characteristics are described in more detail in Benus, Halsey, and Spiegelman (1979).

Counseling was offered on an individual and group basis. A majority of participants received individual counseling, but group sessions were also common in the first year in both sites in order to accommodate the large initial inflow of participants. The counseling process was the same in both types of sessions, although in the groups participants often provided "sounding boards" for one other.

The SIME/DIME counseling process was composed of three main features: self-assessment, labor market assessment, and job search assistance. The self-assessment process was designed to encourage participants to assess their own labor market potential by identifying their past strengths and weaknesses as well as their vocational interests. The process was nondirective in that counselors did not suggest specific courses of action, although the self-assessment process was quite structured. Each client was asked the following six questions:

Question 1. Self-evaluation—What things have I done or accomplished to any degree of success?

(a) On jobs for pay
(b) On jobs for which I received no pay
(c) In school
(d) In recreation activities

This question forced the participant to evaluate his or her own past performance by examining his or her work history. Failures were examined as well as successes. The goal of self-assessment was to discover past accomplishments and how they might be built into future labor market successes. The counselor's aim was to help clients build a positive self-image concerning work skills and abilities.

Question 2. Evaluation by others—What things have I done in a manner that was recognized by others, that is, praise, a raise, a compliment, an award, and the like?

Like self-evaluation, this question was used by the counselor to assist the client in probing for facts and reasons for commendation in both work and non-work activities.

Question 3. Evaluation of performance in working situations—What jobs have I held (paid and unpaid), and how did I do on them?

This question served a two-fold purpose: It gave a historical listing of past work experience required for job applications and it served as a way of finding information overlooked in questions (1) and (2), especially data regarding unsuccessful experiences. The historical perspective of work and volunteer activity frequently revealed other jobs or activities to be explored for occupational possibilities.

Question 4. What kind of equipment can I operate or use?

In some cases, this question would suggest alternative occupations in areas not directly related to the particular tool. A machinist, for example, who was able to set shop tools for precision cutting, may have had no desire to work with such tools, but could adapt or transfer his skills in accuracy, measurement, and blueprint-reading to another vocation.

Question 5. What things would I like to do? In what kind of environment would I like to work and live? What kind of people would I like to work around? What kinds of work would I like to do?

Up to this point, the data gathering focused attention on factual work-world accomplishments. In answering this fifth question, the client examined areas of interest, preferences, and tastes. Although many likes or aspirations came from past experiences, the client was encouraged to explore his long-range hopes and unfulfilled expectations. The full range of possibilities was listed and evaluation of practicality postponed.

Question 6. What are the things I do not like to do? In what kind of environment would I not want to live or work? What kind of work or working conditions would I dislike?

This question provided data that were used as part of a reality check as the participant developed a vocational goal. If, for example, a client decided to become a registered nurse, but also stated a dislike for blood or aged people, then these data would have been recalled by the counselor as part of the reality check. The dislike might be so strong as to negate any positive features of the nursing profession.

After the client had completed the self-assessment component and identified potential areas of interest, the labor market assessment component was used. The extent to which information was provided depended on the participant's circumstances. In some cases, the client was well informed about the labor market conditions in his or her desired occupation so that little additional information was required. In many cases, however, labor market information (LMI) specialists prepared detailed descriptions for specific occupations by contacting employers and employees in the desired fields as well as by utilizing resources such as the "Occupational Outlook Handbook" and the "Dictionary of Occupational Titles." [4]

These LMI occupation reports followed a standard format. They provided a detailed description of the tasks required on the job as well as the pay scale, working conditions, and occupational hazards. They described requirements of the job, such as educational level, credentials, and experience, attempting to ascertain both the formal requirements and the criteria actually used in the current, local labor market. Occupational reports also described career ladders in the occupation, including entry level positions, potential for upgrading, and the normal length of time required to achieve the desired occupation. In addition, the specialists attempted to determine through contacts with employers the usual hiring channels for the desired occupation. The reports also listed related occupations that shared characteristics of the desired occupation. Although it was not the purpose of the LMI service to refer clients to specific job openings if, in surveying employers, the LMI specialists found out about openings, these were noted on the reports. The primary purpose of this component was to provide information rather than job referrals.

The culmination of the self assessment and labor market assessment components was the formulation of a plan of action. Such a plan was used to document the client's decision to pursue a new career, to obtain further training, or in some cases, to accept the status quo. Those plans of action that called for new courses of action specified the following:

- The goal.
- Obstacles to achieving the goal.
- The client's assets in overcoming those obstacles.
- The time schedule for obtaining the goal.
- Financial considerations in obtaining the goal.

Counselors encouraged the clients to be as specific as possible in formulating these plans.

The counselor's role was active at this stage—assisting in formulating goals and suggesting new possibilities for the client to consider. Even at this stage, however, the counselor attempted to be nondirective and to avoid assuming a decisionmaking role. For example, a participant might have desired a position that was unrealistic according to labor market analysis because of a glutted market, or wanted a job that normally required a college degree which the participant had not obtained, or a job that normally required certain physical attributes that the participant did not possess. The counselor's responsibility in such cases was to alert the participant to the obstacles involved in such a choice, then help the participant marshal assets and abilities to pursue the selected goal.

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[4] In Denver, 357 separate occupation reports were prepared between October 1971 and April 1975. These occupations covered 70 percent of the U.S. employment at that time. See Benus, Halsey, and Spiegelman, 1979.

The third component to the SIME/DIME counseling program was job search assistance. Again, no direct job placement or referrals were involved. Rather, general assistance in job search strategies was provided. The counselors aided clients in developing résumés that presented their experience and abilities for the desired occupation. Practice job interviews were also used to increase client self-confidence. Many clients had problems in their backgrounds (for example, long periods out of the labor force, being fired from a job, prison records), and the practice interviews helped them develop ways of explaining these problems in an honest but positive manner.

The SIME/DIME job counseling program was designed as a fairly extreme example of nondirective counseling. It does, however, share characteristics with other job counseling programs that are worth noting. In addition to its job referral program, the U.S. Employment Service (ES) offers job counseling to clients who have problems with their occupational choice. The type of counseling offered varies considerably among States, and is constrained by lack of resources in many areas.[5] Nonetheless, the counseling approach advocated by the ES is nondirective. Clients identify their interests and abilities. The counselor assists this process by empathetic listening and some active advice when discrepancies in plans are evident. However, ES counselors are generally instructed to take a more active role than were SIME/DIME counselors, although the difference is a matter of degree. For example, consider this passage from the ES counselor training manual:

ES counseling also offers market assessment assistance so that the client may more realistically choose, change, or adjust to a vocation. General job search assistance is not offered by the ES counseling program although, of course, the regular ES program is a job placement service.

Job-finding clubs have been used in a variety of programs including the WIN program for AFDC recipients. In contrast to SIME/DIME and ES counseling programs, job-finding clubs concentrate primarily on training in job search techniques. These programs view job-seeking as a learning experience itself and participants are encouraged to treat looking for work as a full-time job. Typically, part of the day is spent in contacting prospective employers, discussing the previous day's efforts in groups with other participants, and role-playing interview situations; the other part is spent in actual job interviews. The Intensive Employment Services component of the WIN program also provides job search assistance, administered at the local level and utilizing a variety of job search procedures. Generally, the emphasis of such programs is on developing job search skills, rather than in self-assessment of interests and abilities in specific occupations.

The education subsidy program

The SIME/DIME educational subsidy program operated like a voucher system in that the participant was free to choose the type of education as long as he or she could demonstrate that it was job related. The program provided either

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[5] ES counseling participants receive on average 1.5 sessions, compared to 5 for their counterparts in SIME/DIME.

50 percent or 100 percent (depending on the program to which the family was assigned) reimbursement of direct costs of the schooling or training. There was no maximum amount of subsidy; the only limitation was (as noted) that it not exceed the cost of the least expensive local institution that provided the requisite training. Participants were reimbursed for the costs of tuition, books, materials, and tools required for authorized courses, and paid an allowance for transportation, based on the cost of public transportation, for the days of actual school attendance. Expenditures for child care were authorized if the participant was not eligible for other child care subsidies. In a few cases where individuals could not obtain the required training locally, living expenses (such as housing and long distance transportation) were also subsidized.

In practice, the criterion that the schooling or training be job related was defined very loosely. Participants were required to formulate a plan of action, generally during the counseling process, specifying the employment objective, the educational program needed to meet the objective, and a schedule for carrying out the educational program. However, participants who were unable to formulate an employment objective but who had a specific training objective with vocational relevance were permitted to proceed with their education and to formulate an employment objective at a later date.

The individual could choose to take the desired training in any institution. In practice the community college was often chosen, partly because the amount of subsidy was usually determined by the cost of courses at the community college. But also, since counseling centers were located in the community college, participants were particularly aware of that choice, and the counselors were undoubtedly more familiar with the courses offered there. In Seattle, there was excess capacity at the community college, so that there were few problems in obtaining admission to community college courses.[6] It is also possible that the counselors who were hired for this project through the community college felt some loyalty to that institution, although they were not so instructed in their training as counselors for the SIME/DIME program.

The training provided by the SIME/DIME program was predominantly classroom and skill training. As such, it differed from the WIN and CETA programs, which offer on-the-job training, work experience, and public service employment as well. Further, there has been a shift in both the CETA and WIN programs away from classroom training, with policies aimed toward increasing short-run employability by providing the minimum training necessary to get a job. Nonetheless, both programs offer classroom training, and it is instructive to compare them with SIME/DIME.

It is difficult to characterize both WIN and CETA because they offer opportunities for considerable local initiative. The Manpower Development and Training Act programs that preceded CETA were generally new classroom training programs designed to augment existing programs that were considered inappropriate for disadvantaged workers. With the enactment of CETA and the redirection of WIN toward immediate employability, the emphasis shifted away from separate classes toward referrals to programs that already existed in the community, as in the SIME/DIME program. However, it is likely that SIME/DIME participants, particularly those who had long-run employment goals involving extensive training, had more choice about the type of training they could receive than do CETA or WIN participants.

Another important distinction involves the link between income maintenance and participation in training. The WIN program is mandatory for certain categories of AFDC recipients and failure to participate can result in loss of AFDC eligibility. The CETA program is not mandatory, but is directly linked to income maintenance insofar as participants receive training allowances. There was, of course, no link in SIME/DIME between the education subsidy and the income maintenance programs. Those who were eligible for both could receive income support whether or not they participated in the counseling and training programs; those who were eligible only for counseling and training could not receive any income support whether or not they participated.

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[6] SIME/DIME had an agreement with the college that they would add sections to classes if they were filled and SIME/DIME participants were unable to enroll, with a wait of not more than one semester. SIME/DIME would reimburse the college for the extra classes.

CHAPTER III

UTILIZATION OF THE PROGRAMS

Participation in the counseling and subsidy programs

A large proportion of those eligible took advantage of the counseling and subsidy programs, as shown in table 4.1.[1] Participation was particularly high among single women. The proportion of single women who attended at least one counseling session was 54 percent for those eligible for counseling only (Mls), 64 percent for those eligible for a 50 percent subsidy (M2s), and 72 percent for those also eligible for a 100 percent subsidy (M3s). The proportion of single women who received subsidized training ranged from 35 percent to 47 percent.

TABLE 4.1.—RATES OF PARTICIPATION IN SIME/DIME COUNSELING AND TRAINING PROGRAMS

	Husbands			Wives			Single female heads		
	M1	M2	M3	M1	M2	M3	M1	M2	M3
Counseling program									
Percentage attending at least one session	39.8	51.4	60.6	38.2	52.8	56.6	54.0	64.4	71.9
Of those participating:									
Average number of sessions	4.8	6.6	6.5	4.9	6.4	7.4	5.8	7.3	7.7
Number of sessions attended:									
1	26.9	14.7	11.9	16.0	13.8	10.3	13.7	11.9	12.1
2–3	22.8	22.0	24.0	25.0	20.0	16.2	21.0	24.0	16.0
4–5	16.6	19.7	14.2	24.5	25.8	19.3	23.9	14.2	15.0
6–10	26.0	27.1	33.2	27.5	26.6	28.8	32.7	33.1	32.8
11 or more	7.7	16.5	16.7	7.0	13.8	25.4	8.7	16.8	24.1
Percentage with at least one plan of action	54.8	71.9	79.4	55.5	68.4	83.7	58.5	74.2	78.6
Percentage with at least one training plan of action	28.6	66.6	65.8	27.7	52.8	70.8	33.1	53.5	67.0
Subsidy program									
Percentage receiving some subsidy	—	21.0	36.3	—	21.3	36.5	—	34.9	46.6
Of those receiving subsidy:									
Average amount of subsidy	—	$363	$666	—	$401	$954	—	$650	$857
Average number of academic quarters subsidy received	—	4.0	3.8	—	3.6	3.9	—	4.1	4.1
Number of eligible individuals	510	671	391	510	670	392	374	481	313

Note.—
M1—Eligible for counseling only.
M2—Eligible for counseling and 50 percent subsidy.
M3—Eligible for counseling and 100 percent subsidy.

Participation rates among husbands and wives, although lower, were still substantial. Further, despite the fact that nearly two-thirds of the wives were out of the labor force prior to enrollment, participation rates of wives were similar to those of husbands. For both husbands and wives, the proportion attending counseling ranged from 40 percent to 60 percent and the proportion receiving subsidies ranged from 21 percent to 36 percent.

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[1] Further details on the utilization of the programs are presented in Benus, Halsey, and Spiegelman (1979) and in Davis and Dickinson (1980). Participation rates were somewhat higher than anticipated; the assignment model assumed individual participation rates of between 25 percent and 30 percent for the counseling program and about 10 percent for the subsidy program.

The high take-up rate of the counseling program may have been a result of the large outreach effort. Once individuals attended at least one session, however, they tended to continue with the program: participants attended an average of at least five sessions. With one exception, between 85 percent and 90 percent of participants attended two or more sessions. Among single and married women participating in the M3 program, 25 percent attended more than 10 sessions. The one major exception occurred for husbands participating in the M1 program, 27 percent of whom attended only one session. Even in this group, however, 34 percent of the participants attended more than five sessions. On average, participants attended five sessions, each approximately 40 minutes in length.[2]

Of the participants eligible only for counseling, more than 50 percent devised plans of action. Approximately half of these plans called for additional training, despite the fact that such training was not subsidized. Of participants also eligible for a subsidy, 75 percent specified plans of action in the counseling process, most of which were for training. The outcome for many counseling participants, therefore, was a plan for additional training.

There was considerable divesity in how the education subsidy program was used. On average, participants took 4 academic quarters of training and received between $363 (M2 husbands) and $954 (M3 wives) in subsidies. However, there was considerable variation around those averages. For example, between 25 percent and 30 percent took only 1 quarter of training, while 15 percent took more than 8 quarters. Similarly, 25 percent received $100 or less, while 25 percent of those eligible for 100 percent subsidy received more than $1,000.

Table 4.2 presents the occupational objectives of training plans of action. Many of these goals were fairly ambitious. Nearly 40 percent of all training plans included a professional occupation as a goal. Among women, these goals were

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[2] Halsey (1980) estimates that the cost per individual session was about $70.

TABLE 4.2.—OCCUPATIONAL AND DEGREE OBJECTIVES OF LAST TRAINING PLAN OF ACTION

[Percentages]

	Husbands	Wives	Single female heads
Occupational objective			
Professional:			
Health, social service and teaching professions	13.4	24.9	29.3
Other professional occupations	24.6	11.9	10.0
Managerial	12.1	4.8	4.7
Sales	2.6	1.9	1.8
Clerical	3.4	25.5	28.1
Craftswork	25.5	5.8	1.8
Operative	10.8	3.7	2.9
Labor	1.3	.8	1.1
Servicework	6.3	20.7	20.3
Degree objective			
College, no degree	10.8	11.3	13.5
Associates degree	17.8	22.9	21.4
B.A., B.S. or higher	18.8	11.6	13.0
License or certification (including union certification)	12.1	20.2	16.0
Skill upgrade	26.1	25.2	28.1
Other	14.4	8.8	8.0

concentrated in the areas of health (for example, nurses, health technicians), social services (for example, social workers, recreational workers), and teaching. It should be noted that only 10 percent of those working at enrollment were employed in professional occupations and that the choice of a professional goal was not strongly related to preenrollment occupation or labor force status.

Other training plans called for jobs more closely related to the types of jobs held by sample members at enrollment. Among men, 25 percent of the goals called for craftwork. Such goals were prevalent among those who were already in craftwork or who were in closely related occupations such as operatives and laborers. Among women, over 25 percent of the goals were for clerical jobs and 20 percent were for service work, the two most common occupations of women who were employed at enrollment.

The degree objectives of training plans also reflect some ambitious goals. Nearly half of those specifying a training plan chose to seek some college training and many planned to seek a B.A. degree or equivalent. Those with B.A. objectives had an average of 13 years of education at enrollment so these goals represent plans for considerable additional schooling. However, those seeking a B.A. degree received only 6 quarters of SIME/DIME-subsidized schooling on average. Thus, few participants remained in school long enough to meet their training goals.

Other participants chose more modest goals. For example, approximately one quarter sought training to upgrade their existing skills. A substantial number of others, particularly women, chose training that could lead to a license or certificate.

There is evidence of both ambitious occupational and educational goals, therefore, and to some extent these overlap. Among those seeking a professional goal, two-thirds sought a college degree. The remainder, however, sought substantially less training. It may be that the professional occupation was a long-run goal for these participants and that more limited training was an intermediate step toward that goal. It is also possible that some participants underestimated the educational requirements of their desired occupation.[3]

In summary, there was substantial participation in the counseling and subsidy programs, particularly among single women. Many of those who participated in the counseling program demonstrated interest in obtaining additional training, even those who were not eligible for a training subsidy. There was marked diversity in the goals of those who planned to seek training. A majority chose relatively modest occupational and training objectives, but a substantial minority chose quite ambitious objectives that presumably held lower prospects for successful attainment.

Characteristics of participants

In this section, we examine how individuals who participated in the program differed from those who did not and, specifically, whether participation was high among individuals with severe employment problems or only among those who would have done well on their own.[4] These results are relevant to determining whether the program can serve antipoverty objectives. We also investigate whether eligibility for the income maintenance experiment affects participation in counseling/training services. If income maintenance substantially reduces the demand for these services, voluntary counseling/training would be unlikely to reduce welfare costs.

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[3] There was more congruence between the other occupational goals and the educational goals. Those who sought clerical, sales, craftwork, or operative occupations generally chose to upgrade their skills or to seek other nondegree training. Individuals who sought service work tended to specify associate degrees or licenses as their degree objectives.

This analysis is based on regression models in which four dependent variables were examined: a dummy variable for the receipt of some counseling, the number of counseling sessions attended, a dummy variable for the receipt of some subsidy, and amount of subsidy received. The independent variables included a variety of individual characteristics. Separate models were estimated for persons assigned to the counseling-only treatment and those assigned to one of the counseling-plus-subsidy treatments.

If counseling and training programs are to serve antipoverty objectives, participation must be substantial among the most disadvantaged. With respect to preenrollment employment characteristics, there is some evidence that the more disadvantaged husbands in SIME/DIME were more likely to participate: the wage rate at preenrollment was negatively related to participation (significantly so for counseling), as were preenrollment work hours. Similarly, participation in the counseling program was higher among husbands who were unemployed or who were out of the labor force but had some recent labor market experience.

For wives and single women, the pattern is less clear. Participation in counseling was generally higher among women who were unemployed. Women out of the labor force for 1 year or more were consistently less likely to seek counseling, but even their participation rates ranged from 35 percent (wives eligible only for counseling) to 65 percent (single women eligible for a subsidy). With respect to the subsidy program, single women who were out of the labor force were also significantly less likely to participate, although over one-third did receive some subsidized training. Participation in the subsidy program did vary by wives' employment status. Other preenrollment characteristics had an inconsistent impact on wives' participation. For single women, participation in both programs was higher among those with higher wage rates, perhaps because they had a more serious attachment to the labor market, and thus saw greater benefit to increased labor market information and increased training.[5]

If voluntary counseling and training programs are to reduce welfare costs, then eligibility for welfare must not substantially reduce participation in such programs. In SIME/DIME there is no evidence of such a reduction. The support level significantly increased the likelihood of participation for those eligible for counseling and not training subsidies; there was no relationship between the support level and participation in either program for those eligible for a subsidy as well as counseling. The tax rate had an inconsistent impact on participation in counseling and no impact on participation in the subsidy program. Further, prior receipt of nonexperimental welfare had a weak but generally positive influence on participation in the SIME/DIME training/counseling programs.

The analysis of the characteristics of participants indicates that participation was fairly widespread. Husbands who chose to participate had somewhat poorer employment histories than those who chose not to participate; the pattern was more mixed for women. While the disadvantaged were not consistently more likely to participate, none of the factors considered had such a strong influence as to result in low participation rates among any group. Thus, the pattern of participation does not preclude the attainment of antipoverty objectives. Similarly, eligibility for income maintenance did not inhibit participation in the program so that the goal of reduced welfare costs is also not precluded by the patterns of participation.

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[4] Further analysis of the determinants of participation is presented in Dickinson (1980).

[5] Surprisingly, the age and number of children in the family did not have any systematic influence on the decisions of women to seek job counseling or subsidized training.

CHAPTER IV

EXPERIMENTAL DESIGN AND ANALYSIS METHODOLOGY

Design

As discussed earlier, three counseling/training treatments were tested: counseling only (M1), counseling plus 50 percent subsidy of the direct costs of training and schooling (M2), and counseling plus 100 percent subsidy (M3). Of families assigned to counseling/training treatments, all family members who were over 16 at enrollment and those who turned 16 during the experimental period could receive counseling/training services. We use the word "eligible" here to describe persons who were assigned to any of the three counseling/training treatments.

As with the financial plans, some families were assigned to the counseling/training treatments for 3 years and others for 5 years. However, the 100% subsidy treatment had only a 3-year duration. As noted in part I, durations were used to maintain consistency with the NIT treatments. The designers of the experiment believed it was desirable to avoid any confusion that might arise if a family had both NIT and counseling/training treatments and the programs had different durations. However, there was no interest in how the effects of the counseling and training programs varied specifically with duration.[1]

Individuals were assigned to one of the treatments or to the control group, based on a stratified random procedure.[2] The intention of this stratified procedure was to increase the efficiency of the experiment by accounting for the fact that the cost of an observation varied systematically with both household and treatment characteristics. Costs of both financial and counseling/training treatments were expected to be higher for low-income families. The cost of financial treatment was assumed to be lower among those eligible for the counseling/training treatments. As a result, the eligibility for these treatments was correlated with eligibility for NIT treatments. Approximately 50 percent of the NIT controls and 65 percent of the NIT experimentals were assigned to the counseling and training programs.

Because of random assignment, we are assured that, on average, the only difference between controls and persons assigned a counseling/training treatment is the treatment itself.[3] There are no systematic differences in any other characteristics between experimentals and controls. Consequently, by comparing all persons assigned to each counseling/training treatment with all controls, we obtain an estimate of the treatment effect that is unaffected by selection bias. This absence of selection bias is an analysis advantage not shared by evaluations of regular public programs, which cannot use random assignment and which therefore may be affected by a variety of potential selection biases. Comparisons of program participants to some comparison group will be biased if there are any systematic differences in the processes that generate the two groups. For example, suppose that persons with poor prospects participate disproportionately in a program. In this case, participants have poorer prospects than persons in the comparison group, and estimates of the program's effect will have a negative bias. An opposite bias can occur if persons with better prospects are more likely to participate, as may occur if the persons running the program choose the best applicants.

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[1] The limited duration of the experiment may have some implications for the estimated size of the program's impacts. On the one hand, some individuals may not have participated in the program because they could not free time to take training during the limited eligibility period. On the other hand, participation might have been higher in the eligibility period than it would in a permanent program if individuals shifted any planned future training to the eligibility period. Empirically, we find little difference in program participation by duration of eligibility.

[2] See part I of this volume or Kurz and Spiegelman (1972).

[3] Subsequent analyses that involve the comparison of participants to controls may be subject to self-selection biases. Methods for detecting such a selection bias will be discussed as appropriate.

One important feature of the SIME/DIME counseling and training programs is that participation was voluntary. An eligible person could choose whether to attend counseling sessions or receive a subsidy. As discussed earlier, participation in the program, though substantial, was not universal. Although this fact does not impair the advantages of random assignment to treatments, it has some implications about inferences that can be drawn from the experiment. In particular, the estimated impact here is for a voluntary program. Similar programs with compulsory participation or with different incentives to participate might have different impacts. For example, if those persons who would benefit from the program chose not to participate, the impacts of a compulsory program might be positive, even if a voluntary program had zero or negative impacts.

Analysis methodology

The basic model used in the analyses is an analysis-of-covariance type of regression model. The estimated impacts of the counseling/training treatments represent the difference between the average value of the dependent variable among persons assigned to the treatment and the average of the dependent variable among controls (after accounting for the covariates). The primary model used to estimate the impact of the program on labor market performance has the following form:

$$Y = a + bX + cA + dM + fN + e$$

where Y = dependent variable
X = preenrollment characteristics, including the preenrollment value of the dependent variable
A = assignment model variables, also defined at enrollment
M = vector of counseling/training treatments
N = vector of income maintenance variables.

Models are estimated for several dependent variables: participation in counseling programs other than SIME/DIME, amount of additional schooling, amount of additional training, and a number of variables representing labor market performance—the log wage rate, earnings, annual hours worked, and proportions of the year worked, unemployed, and out of the labor force. In most analyses, the counseling/training treatments are represented by five dummy variables: 3-year counseling only, 5-year counseling only, 3-year counseling plus 50 percent subsidy, 5-year counseling plus 50 percent subsidy, and 3-year counseling plus 100 percent subsidy.

Covariates are included in the equation for two reasons: to account for the assignment process (A), and to increase the efficiency of the estimators of the treatment effects (X). Families were assigned to NIT and counseling/training plans as well as to control status (no plan) randomly within strata determined by normal income level, family structure (husband-wife or single-headed), ethnic group (black, white, Chicano) and site (Seattle, Denver).[4] We account for the assignment by family type by estimating separate models for husbands, wives, and single female heads of families. We account for the assignment by site, ethnic group, and normal income level by including dummy variables for site, ethnic group, and normal income level in the set of nontreatment variables. Other included variables vary somewhat among the analyses presented. However, they generally include the preexperimental value of the dependent variable.[5] The income maintenance variables include whether the family was eligible for the negative income tax treatment and whether they were eligible for 3 or for 5 years.

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[4] See part I of this volume.

Only original enrollees are included in the samples of husbands, wives, and single female heads of families used in this study.[6] Persons were eliminated from samples because of attrition and missing data on independent variables. Samples also changed over time because of the study design. All individuals who were eligible for the 3-year program were scheduled to be followed for at least 4 years. Approximately half of the 3-year individuals were also interviewed in the fifth year. None of the 3-year individuals are included in the 6-year sample. In the analysis of the impacts on earnings and hours of work, the sample is reduced from the original 7,519 persons to 5,167 persons in the 4th year, 3,376 persons in the 5th year, and 1,721 persons in the 6th year.[7] Much of the reduction in the sample between the 4th and 5th years is due to the fact that a 5th year of data was collected for only about half the 3-year families. Most of the reduction between the 5th and 6th years is due to the cessaton of interviewing for 3-year families.

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[5] In the case of the analyses of the impacts on labor market performance, the preexperimental values of the dependent variables are included to account for any accidental differences in the preexperimental circumstances of controls and experimentals. Other included variables are years of schooling, a dummy variable for graduation from high school, a dummy variable for being in school or training, experience (defined as age minus years of schooling minus five), experience squared, a dummy variable for having had training, the log of weeks of training, amount of AFDC received, and a dummy variable for end of year employment status. To ensure that these variables have not been affected by the experiment, we define these variables as of the preexperimental period. Variables included in the analysis of the impacts on education and training are described in Hall and Stieger (1980); those in the analysis of the impacts of other forms of counseling are described in Davis (1980).

[6] However, persons in assignment cells containing only controls were excluded because there are no comparable experimentals (E7's for all groups and E6's for all groups and E6's for female heads).

[7] While the designers of the experiment did not calculate the statistical power of the experiment, ex post calculations demonstrate that it is substantial for husbands, although not for wives and single women. Assuming that the effects of the five counseling/training treatments are evenly spaced from 2 percent to 12 percent, an assumption roughly consistent with those found in Conlisk and Kurz (1972), and using a 10 percent significance level, the actual 4th year sample used to estimate earnings impacts and the unexplained relative variance of earning power is approximately 75 percent for husbands, and 40 percent for wives and single women. The relatively low power for wives and single women implies that findings of no effect should be treated cautiously.

CHAPTER V

IMPACTS OF THE PROGRAMS ON ELIGIBLES

As we hypothesized in the first chapter we expected all three programs to result in a net increase in the amount of job counseling received. We also expected the subsidy program and possibly the counseling program to increase the amount of schooling and training acquired, which would then be followed by improved labor market performance. We now proceed to investigate whether these impacts in fact occurred, using the model described in chapter 4.

Impact on counseling

Although a substantial number of eligible individuals participated in the SIME/DIME programs, it is possible that counseling participants simply reduced their use of other job counseling programs, resulting in little net change in the amount of counseling received. Periodically, throughout the study, respondents were asked about the job counseling services that they had used. Davis (1980) estimated the effects of the counseling/training treatments on participation in counseling other than that provided directly through SIME/DIME using a model similar to that described in chapter 4, except that no distinction was made by program duration and a person is included as an observation each time he or she was asked the questions on use of counseling. The results are presented in table 4.3.

TABLE 4.3.—EFFECTS OF COUNSELING/TRAINING TREATMENTS ON USE OF COUNSELING PROGRAMS OTHER THAN SIME/DIME

[Standard errors in parentheses]

	Husbands	Wives	Single females
Counseling only (M1)	.01	[1].01	−.00
	(.01)	(.01)	(.01)
Counseling plus 50 percent subsidy (M2)	−.01	.00	−.01
	(.01)	(.01)	(.01)
Counseling plus 100 percent subsidy (M3)	−.00	.01	−.01
	(.01)	(.01)	(.01)
Number of cases	7,670	8,170	6,080
Mean of dependent variable	.09	.05	.09

[1] Significant at the 5-percent level.

Eligibility for the SIME/DIME counseling and subsidy programs had no significant impact on the use of other counseling programs for husbands and single women, but significantly increased the use of other programs for wives eligible for counseling only. Thus, the substantial participation in the SIME/DIME counseling program represents an increment to assistance received and not a substitute for other forms of counseling.

As indicated in the table, the mean participation rate in other programs was about 9 percent per year for husbands and single women and 5 percent per year for wives. On a yearly basis, participation in SIME/DIME counseling was considerably higher—about 15 percent for husbands and wives, 24 percent for single women who were eligible only for counseling, and still higher for those eligible for subsidy (see Dickinson, 1980, table 6). Since total counseling is the sum of both types, the SIME/DIME counseling/training programs substantially increased the amount of labor market counseling received.

Impact on schooling and training

Hall and Stieger (1980) estimated the impact of counseling and training programs on schooling and training over the first 3 years of the experiment.[1] They used a statistical technique that accounts for the fact that additional schooling could not be less than zero nor more than 3 years. Separate education equations were estimated for persons under and over 30 years of age. Although younger experimentals were more likely to participate in the programs, younger controls were probably more likely to seek out education and training on their own. It is possible, therefore, that the net impact on education may be larger for older individuals.

In table 4.4, we present predicted changes in years of schooling derived from the estimated coefficients of the statistical model and the means of the independent variables.[2] There is no evidence that the counseling program increased the net amount of schooling of husbands, wives, or single women. Eight of the ten estimated effects are negative.

TABLE 4.4.—PREDICTED AVERAGE EFFECTS OF THE COUNSELING AND EDUCATION AND TRAINING SUBSIDY PROGRAMS ON YEARS OF SCHOOLING UNDERTAKEN DURING THE EXPERIMENT

[Standard errors in parentheses]

	Husbands		Wives		Single female heads	
	Under 30	Over 30	Under 30	Over 30	Under 30	Over 30
Counseling only:						
3-year program	.017	.010	–.041	–.032	–.132	.052
	(.116)	(.055)	(.064)	(.061)	(.099)	(.085)
5-year program	.087	.025	.027	–.021	.078	.030
	(.102)	(.039)	(.053)	(.048)	(.103)	(.064)
Counseling plus 50 percent subsidy:						
3-year program	.032	.068	.009	.067	–.007	[1] .161
	(.104)	(.070)	(.068)	(.096)	(.114)	(.089)
5-year program	.070	.045	.030	.087	[2] .258	[2] .133
	(.087)	(.044)	(.050)	(.097)	(.111)	(.066)
Counseling plus 100 percent subsidy:						
3-year program	.113	[1] .136	[3] .272	.129	.116	[2] .212
	(.124)	(.081)	(.102)	(.140)	(.126)	(.098)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

In the case of the subsidy program the estimated impacts are generally positive. Only 2 of the 18 estimated effects are negative. For husbands, a subsidy program generally increased schooling, but the only significant effect is a 0.14-year increase in education for husbands over 30 who were eligible for the 100 percent subsidy. For wives, there is somewhat more evidence that the subsidy program increased schooling; and the effects tend to increase with the generosity of the program. For single women, the effects of the subsidy program are even more positive. For older women, a subsidy program significantly increased the amount of schooling received, as did the 5-year 50 percent program for younger women.

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[1] Hall and Stieger also estimated the impact over the first 5 years. The pattern of results are similar to those for the first 3 years, although the impacts are somewhat smaller.

[2] The counseling and education and training subsidy treatments were represented by four dummy variables; one for counseling only, one for counseling plus 50 percent subsidy, one for counseling plus 100 percent subsidy, and one for eligibility for only 3 years (rather than 5). Thus, the difference between the effects for 3- and 5-year eligibles is constrained to be the same for the counseling and counseling plus 50 percent subsidy programs (the 100 percent subsidy program had only a 3-year duration). The statistical technique used was doubly truncated tobit. The predicted changes represent the differences between the expected values of additional years of schooling with and without the treatment, holding all other variables constant.

Similar models were estimated for training.[3] The counseling and education and training subsidy programs generally did not have significant effects on training. This is, perhaps, not surprising since counseling centers were located at the local community college. Thus, there was a natural tendency to take subsidized training at the community college, which is likely to have been reported as schooling and not as training.

In summary, we have found that the counseling-only program had no significant effect on schooling and the program of subsidies generally did increase schooling, as was expected. This increase was smallest for husbands and largest for single female heads.

Impact on labor market performance

We turn now to the important issue of whether the programs increased labor market performance. We begin this section by describing the impact of eligibility for the program on several measures of labor market performance, including earnings, annual hours of work, and the wage rates of those who were working.[4] We then examine whether there is variation in impact by experimental and personal characteristics.

Because we are primarily interested in the programs' impacts after program participation, our main focus is on the postprogram periods. We separate the effects of the 3- and 5-year programs because years 4 through 6 are postprogram years for 3-year families, while only year 6 is a postprogram period for 5-year families.

For husbands, eligibility for the counseling-only program had no significant impact on earnings in any year, as shown in table 4.5. The 3-year counseling program had virtually no impact on husbands' earnings; the 5-year counseling program had an uneven but predominantly negative impact, even in the 6th year when experimentals were no longer eligible for counseling.[5] Further, eligibility for the counseling consistently resulted in lower wage rates among husbands who were working.

TABLE 4.5.—ESTIMATED EFFECTS OF ELIGIBILITY FOR COUNSELING/TRAINING TREATMENTS ON ANNUAL EARNINGS, WORK HOURS, AND AVERAGE WAGE RATES OF HUSBANDS

[Standard errors in parentheses]

	Year—					
	1	2	3	4	5	6
Effects on earnings						
Counseling only (M1):						
3-year	6	104	87	−5	9	.....
	(138)	(191)	(227)	(259)	(434)	.....
5-year	−31	203	−253	−185	205	−239
	(172)	(234)	(280)	(320)	(365)	(391)

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[3] Separate training equations were not estimated for the two age groups.

[4] Effects on wage rates are estimated using a sample that includes only persons who worked during the experimental period because rates are not observed for nonworkers. West (1979, appendix A) found similar results after accounting for the possibility of selection bias.

[5] Because of delays between the enrollment date and the date of contact by the counseling/training programs, a few 5-year eligibles were still eligible to receive counseling and/or subsidies in the 6th year.

TABLE 4.5. ESTIMATED EFFECTS OF ELIGIBILITY FOR COUNSELING/TRAINING TREATMENTS ON ANNUAL EARNINGS, WORK HOURS, AND AVERAGE WAGE RATES OF HUSBANDS—Continued

[Standard errors in parentheses]

	Year—					
	1	2	3	4	5	6
Counseling plus 50 percent subsidy (M2):						
3-year	[2] 248	92	108	−101	−58	....
	(123)	(171)	(198)	(227)	(328)	....
5-year	[3] 398	234	−81	−161	−165	−33
	(153)	(208)	(236)	(270)	(307)	(331)
Counseling plus 100 percent subsidy (M3):						
3-year	[2] 317	161	178	−245	−98	....
	(124)	(170)	(197)	(225)	(327)	....
Effects on hours of work						
Counseling only (M1):						
3-year	3.0	34.9	33.1	24.1	−52.5	....
	(37.8)	(49.6)	(57.2)	(61.8)	(99.6)	....
5-year	2.8	32.0	115.3	−80.6	12.8	−115.6
	(47.1)	(60.9)	(70.5)	(76.6)	(83.9)	(87.1)
Counseling plus 50 percent subsidy (M2):						
3-year	[2] 71.7	37.9	62.6	22.4	−16.9	....
	(33.6)	(44.4)	(49.7)	(54.2)	(75.3)	....
5-year	[2] 105.6	[1] 94.2	36.7	−73.7	−45.3	−18.7
	(41.9)	(54.1)	(59.4)	(64.5)	(70.5)	(73.7)
Counseling plus 100 percent subsidy (M3):						
3-year	[1] 88.8	40.3	52.7	−37.9	−43.7	....
	(33.9)	(44.3)	(49.6)	(53.8)	(75.0)	....
Effects on log wage rate of workers						
Counseling only (M1):						
3-year	.021	.006	.024	−.021	−.001	....
	(.019)	(.024)	(.028)	(.030)	(.050)	....
5-year	.006	.017	.027	[2] −.091	−.042	−.030
	(.024)	(.031)	(.035)	(.040)	(.045)	(.047)
Counseling plus 50 percent subsidy (M2):						
3-year	[2] −.038	.022	−.029	[1] −.049	−.021	....
	(.017)	(.022)	(.024)	(.027)	(.038)	....
5-year	.016	.019	.003	−.026	.039	.030
	(.021)	(.027)	(.029)	(.033)	(.038)	(.039)
Counseling plus 100 percent subsidy (M3):						
3-year	[3] −.055	[1] −.038	−.036	−.035	−.025	....
	(.017)	(.022)	(.024)	(.027)	(.039)	....

[1] Significant at the 100-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

The subsidy program significantly reduced husbands' earnings in the first year, due to the experimentals working fewer hours and earning lower wage rates. This early reduction in earnings is consistent with the hypothesis that participants reduced their work effort and perhaps took lower paying part-time jobs while receiving subsidized training. Negative impacts, however, persist throughout the sample period. In the postprogram period the impacts are considerably smaller but still predominantly negative. Thus, neither the counseling nor the subsidy programs had the expected positive impact on postprogram labor market performance for husbands.

A similar pattern is found for wives, with even larger negative effects (see table 4.6). Both the 3- and 5-year counseling programs resulted in lower earnings and work hours for wives in every year, and these effects are generally significant for the 5-year program. In the 6th year, for example, wives eligible for counseling earned $430 less than did comparable controls, a 19 percent reduction.

TABLE 4.6.—ESTIMATED EFFECTS OF ELIGIBILITY FOR COUNSELING/TRAINING TREATMENTS ON ANNUAL EARNINGS, WORK HOURS, AND AVERAGE RATES OF WIVES

[Standard errors in parentheses]

	Year—					
	1	2	3	4	5	6
Effects on earnings						
Counseling only (M1):						
3-year	21	−21	−72	−215	−296	...
	(82)	(117)	(135)	(148)	(259)	...
5-year	[1] −187	[3] −376	[2] −393	[3] −534	−133	[1] −430
	(102)	(143)	(167)	(182)	(223)	(239)
Counseling plus 50 percent subsidy (M2):						
3-year	6	−80	−65	−199	−126	...
	(73)	(104)	(118)	(130)	(208)	...
5-year	[3] −255	[2] −290	[2] −283	−124	−148	−301
	(92)	(129)	(144)	(158)	(193)	(208)
Counseling plus 100 percent subsidy (M3):						
3-year	−37	[2] −210	−81	[2] −222	−95	...
	(74)	(105)	(119)	(132)	(205)	...
Effects on hours of work						
Counseling only (M1):						
3-year	−8.6	−28.6	−24.3	−61.8	−65.3	...
	(34.3)	(46.3)	(54.0)	(57.8)	(96.3)	...
5-year	[1] −77.1	[3] −164.0	[2] −163.9	[3] −197.9	−68.4	−125.8
	(42.6)	(56.7)	(66.9)	(71.2)	(82.9)	(86.2)
Counseling plus 50 percent subsidy (M2):						
3-year	−14.9	−25.3	−5.1	−63.9	−6.2	...
	(30.5)	(41.1)	(47.2)	(50.9)	(77.4)	...
5-year	[3] −107.8	[2] −109.8	−83.2	−43.1	−44.1	−52.1
	(38.5)	(51.0)	(57.8)	(61.8)	(71.7)	(75.1)
Counseling plus 100 percent subsidy (M3):						
3-year	−10.0	[1] −71.1	6.3	[1] −96.1	−64.4	...
	(30.9)	(41.5)	(47.6)	(51.5)	(76.1)	...
Effects on log wage rate of workers						
Counseling only (M1):						
3-year	.020	−.044	.042	−.059	.001	...
	(.043)	(.047)	(.052)	(.051)	(.077)	...
5-year	−.034	−.049	.031	.008	.044	−.040
	(.049)	(.056)	(.064)	(.068)	(.069)	(.064)
Counseling plus 50 percent subsidy (M2):						
3-year	−.018	[3] −.113	.017	−.049	.003	...
	(.037)	(.041)	(.045)	(.044)	(.061)	...
5-year	−.053	−.053	−.075	.007	−.007	−.030
	(.046)	(.048)	(.051)	(.054)	(.058)	(.055)
Counseling plus 100 percent subsidy (M3):						
3-year	−.014	[3] −.110	[1] −.078	−.035	.064	...
	(.038)	(.041)	(.044)	(.044)	(.060)	...

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Wives eligible for the subsidy program also had lower earnings and worked fewer hours than did controls. These effects tend to be significant in the early years but are also large and negative in the later years, particularly for the 50 percent subsidy. The preponderance of evidence indicates that eligibility for the counseling and subsidy programs resulted in reduced earnings for wives.

The pattern for single women is different from that observed for husbands and wives (table 4.7). In particular, the counseling-only program had a consistently positive impact on single women's earnings. Although not statistically significant, differences in earnings are substantial: in the 5th year, single women eligible for the counseling-only program earned between $275 and $300 more per year than did controls, approximately a 10 percent increase. Furthermore, the counseling program did have some significant postprogram impacts on wage rates and work hours of single women. Thus, in contrast to the results for married men and women, the counseling program may have improved the labor market performance of single women.

TABLE 4.7 ESTIMATED EFFECTS OF ELIGIBILITY FOR COUNSELING/TRAINING TREATMENTS ON ANNUAL EARNINGS, WORK HOURS, AND AVERAGE WAGE RATES OF SINGLE FEMALE HEADS

[Standard errors in parentheses]

	Year 1	2	3	4	5	6
Effects on earnings						
Counseling only (M1):						
3-year	148	195	91	144	307	...
	(103)	(144)	(173)	(197)	(316)	...
5-year	123	99	124	184	273	426
	(133)	(184)	(219)	(243)	(277)	(308)
Counseling plus 50 percent subsidy (M2):						
3-year	32	171	203	−207	−108	...
	(96)	(131)	(154)	(172)	(248)	...
5-year	114	229	107	−89	141	−37
	(124)	(172)	(197)	(217)	(246)	(269)
Counseling plus 100 percent subsidy (M3):						
3-year	11	191	[1] 293	−45	−90	...
	(94)	(131)	(154)	(173)	(252)	...
Effects on hours of work						
Counselng only (M1):						
3-year	36.1	47.2	27.8	13.2	106.4	...
	(41.3)	(55.4)	(65.1)	(71.9)	(114.2)	...
5-year	31.8	14.3	51.2	81.1	84.9	[1] 183.4
	(52.9)	(70.9)	(82.2)	(88.6)	(100.2)	(108.1)
Counseling plus 50 percent subsidy (M2):						
3-year	4.6	50.5	52.8	−98.1	−76.5	...
	(38.2)	(50.5)	(57.7)	(62.6)	(89.6)	...
5-year	81.3	[1] 119.2	64.9	−77.9	−1.7	−112.8
	(49.3)	(66.0)	(73.9)	(79.2)	(89.2)	(94.6)
Counseling plus 100 percent subsidy (M3):						
3-year	20.7	[1] 92.6	[1] 111.2	−35.7	35.4	...
	(37.7)	(50.2)	(57.7)	(62.9)	(91.3)	...
Effects on log wage rate of workers						
Counseling only (M1):						
3-year	.041	.060	[2] .086	.028	[2] .141	...
	(.030)	(.037)	(.043)	(.046)	(.062)	...
5-year	.034	.013	.018	.041	.056	.037
	(.038)	(.047)	(.051)	(.057)	(.058)	(.060)
Counseling plus 50 percent subsidy (M2):						
3-year	.007	.031	−.015	−.046	.021	...
	(.028)	(.034)	(.037)	(.039)	(.050)	...
5-year	.005	.048	−.058	−.034	[2] .122	.063
	(.037)	(.046)	(.047)	(.052)	(.054)	(.055)
Counseling plus 100 subsidy (M3):						
3-year	.005	.005	−.014	.045	−.036	...
	(.027)	(.034)	(.038)	(.040)	(.051)	...

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.

The subsidy program, on the other hand, did not produce positive effects for single women. As in the case of husbands and wives, single women eligible for subsidy programs generally had lower earnings and work hours than did comparable controls even in postprogram periods.[6]

West and Dickinson (1980) also examined the impacts of eligibility for the counseling and training programs on labor force status. The results suggest that the early decline in labor market performance of counseling/training eligibles is associated with a withdrawal from the labor market. The evidence for later years does not present a clear picture of why the reduction in employment tended to persist for husbands, wives, and single women who were eligible for subsidies.

Single women who were eligible for the counseling-only program did not reduce their employment. Any withdrawal from the labor force came from those who were unemployed rather than from those who were employed. The fact that this program did not induce single women to reduce their work effort initially offers a potential explanation for why the counseling program was successful for single women but not for husbands and wives.

In interpreting these results it would be useful to be able to distinguish the effects of counseling from the effects of the subsidy program for those who were eligible for both. Unfortunately, the evaluation was not designed to make this distinction and as a result no individuals were assigned eligibility only for the subsidy but not for counseling. Furthermore, virtually all individuals who received subsidized training went through the counseling process. Thus, we cannot test the independent effects of the two programs. If it could be assumed that effects of the two programs were additive, one could subtract the effects of counseling only from the estimated impacts for the M2 and M3 programs to obtain an estimated impact of the education subsidy program alone. This procedure suggests that the subsidy program generally had a small negative effect on husbands' earnings, a small positive effect on wives' earnings, and a large negative effect on single women's earnings. Thus, negative effects of the programs for husbands and wives would be primarily due to counseling.

There are, however, reasons to doubt the additive assumption. As we discuss in chapter 7, it is likely that the choice of the type of education to pursue was influenced by several aspects of the counseling program. It is possible, therefore, that a subsidy program in the absence of the counseling program would have had a different impact than the additive assumption suggests.

Results presented here indicate that only the counseling-only program for single women successfully increased earnings. The counseling program for husbands and for wives and the subsidy program for all three groups did not improve earnings and, in fact, tended to have a negative impact on labor market performance.[7] The negative impact on earnings in the early years is consistent with the hypothesis that participants reduced their labor force attachment while participating in the programs, particularly in the subsidy program. However, negative impacts in the postprogram periods are unexpected and disturbing.[8]

It is worthwhile to reiterate that negative impacts are not due to noncomparability between experimentals and controls. The sample was randomly assigned to experimental and control groups and as a result the only systematic difference between the two groups was the eligibility to participate in counseling and training programs. Furthermore, we have compared the labor force performance of all eligibles, whether or not they participated, to the performance of all controls. This avoids any self-selection bias that might result if, for example, participants had poorer prospects than nonparticipants and only participants were compared to all controls. If that were the comparison being made, negative impacts might be explained by self-selection. The comparison of all experimentals (both good and poor prospects) to all controls (both good and bad prospects) ensures that only the eligibility for the program could have produced the negative impacts.[9]

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[6] There is one exception: in the 5th year, women eligible for the 5-year 50 percent subsidy program had 12 percent higher wage rates and, as a result, somewhat higher earnings than did controls. The wage rate effect was considerably smaller in the next year, however, and because experimentals worked fewer hours than controls, the program had a negative impact on earnings in the 6th year.

[7] It should be noted that the estimated negative impacts are not due to preexperimental differences in the level of earnings between the experimentals and controls. We have controlled for many aspects of the individuals' preenrollment employment situation in our regressions. Further, when we estimate the impact of the programs on change in earnings rather than level, the same pattern of results is found.

[8] Not only did the counseling and training programs impair the labor market success of husbands and wives, but there is some evidence that they also increased marital dissolution rates (see part V, appendix B). The estimated effects of the 3-year programs on separations are positive and the effect of the counseling-only program is significant for blacks and Chicanos. However, the effects of the 5-year programs are generally negative. There are no effects on getting married. The mechanism by which these results might have occurred is unclear. The counseling program was specifically restricted to career-related counseling and did not include any sort of marital counseling, although referrals to other types of counseling were occasionally made when the job counselor felt other problems were impeding employment prospects.

Variation of impact by individual characteristics

Although we have observed that the programs were generally unsuccessful, it is possible that they provided some benefits to certain subgroups. We have argued that if programs are to meet the objectives of reduced poverty and reduced welfare costs, then there should be benefits among the more disadvantaged and among income maintenance recipients. In this section, we disaggregate the impacts by these characteristics to determine whether SIME/DIME programs had a pattern of effects necessary to meet these objectives. Since a number of interactions are included, this analysis is necessarily exploratory.

Because we are presenting a large number of coefficients for each model, we estimate these models only for the 4th and 6th years. The 4th year represents the postprogram period for the 3-year program for which we have the largest sample size. Similarly, the 6th year is a postprogram period for the 5-year program.

The model used for this analysis is somewhat different from the general model described in chapter 4. The interaction model has the following form:

$$Y = a + bX + cA + d_1 Z_1 M + d_2 Z_2 M + fDM_1 + gDM_2 + e,$$

where Y = dependent variable
X = preenrollment characteristics
A = assignment model variables
Z_1 = a dummy variable indicating one level of the interacted characteristic (also a member of X or A)
Z_2 = a dummy variable indicating the other level of the interacted characteristic
M = a vector of counseling/training treatment variables (M_1, M_2, M_3)
D = dummy for 3-year eligibility.

In this model treatment variables are three dummy variables for counseling only, counseling plus 50 percent subsidy, and counseling plus 100 percent subsidy. The coefficients d_1 represent the effects of the counseling/training treatments for persons with one level of the interacted characteristic (Z), and d_2 the effects for persons with the other level of Z.[10]

It should be noted that this model maintains comparability between experimentals and controls. Variables that are interacted with the treatments (Z) are also included as control variables, and all variables are defined at preenrollment. Thus experimentals who were employed at enrollment are compared to controls who were employed at enrollment. Once again the only systematic difference between these two groups is eligibility for the counseling and training programs.

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[9] The self-selection does bear on whether the results of the SIME/DIME programs could be generalized to a mandatory program. The possibility exists that individuals who chose not to participate would in fact have been helped by the program. By observing the impact on only those who chose to participate, we ignore the potential impact on those who chose not to participate.

[10] The coefficients represent the effects of the 5-year counseling-only (M_1) and counseling plus 50 percent subsidy (M_2) programs, and the 3-year counseling plus 100 percent subsidy program.

If the counseling and training programs are to reduce welfare costs, they should have a positive impact on the labor market performance of welfare recipients. To explore whether this was the case in SIME/DIME, we estimate the impact of the programs separately for those who were eligible for the SIME/DIME income maintenance program and for those who were financial controls.

Eligibility for the financial program had no consistent impact on the post-program effectiveness of counseling training programs for husbands, as shown in table 4.6.[11] Both NIT experimentals and NIT controls tended to have lower earnings, work hours, and wage rates as a result of eligibility for the counseling/training. Similarly, there were no consistent variations in counseling/training effects by whether wives were eligible for income maintenance programs (see table 4.7). Thus, the conclusion that counseling and training programs impaired the labor market performance of husbands and wives is not confounded by an interaction between counseling/training and NIT treatments.

For single women, eligibility for income maintenance reduced the effectiveness of the counseling-only program (see table 4.9). For example, in the 6th year among NIT controls, M1s earned $846 more than those who were counseling/training controls; among NIT experimentals, M1s earned only $56 more. This occurred even though female heads with an NIT treatment were more likely to participate in the counseling program than were NIT controls, as discussed in chapter 2.

These results indicate that income maintenance per se inhibits the effectiveness of the counseling program. However, as shown below, the program is most effective for the most disadvantaged single women. Since actual welfare recipients fall into both of these categories, the impact of those receiving welfare depends on the net impact of these two effects. When we further disaggregate the impact of the counseling program by receipt of welfare at preenrollment, we find that the program is more beneficial for AFDC recipients. Thus, the program could potentially reduce welfare costs if it served the current welfare population.

If counseling and training programs are to reduce poverty, then they should have a positive impact on the most disadvantaged. To examine whether this pattern is evident for SIME/DIME programs, we disaggregate impacts in the 4th and 6th years by preenrollment normal income and employment status. There is a difference in the impact of the counseling-only program by normal income for husbands (table 4.6). Those with low normal income generally fared poorly under the counseling-only program, and to a lesser extent under the 50 percent subsidy program. These results suggest that the counseling program was most deleterious for husband with the fewest labor market skills. Difference in program impact by employment status support this hypothesis. Among those not employed at enrollment, husbands eligible for the counseling-only program earned consistently less than comparable controls, particularly in the 6th year. On the other hand, among those who were employed at enrollment, those eligible for counseling only earned approximately the same as controls. A similar, but less pronounced pattern is also evident for the subsidy program.

For wives, there was no variation in impact by normal income. However, this variable is a better measure of the husband's labor market experience than the wife's. With respect to preenrollment employment status, it is working wives who fared poorly under the subsidy program, the opposite of the pattern observed for husbands. Since this differential impact on earnings is primarily due to

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[11] The results presented in tables 4.8, 4.9, and 4.10 are for the postprogram years (4 and 6), the years of interest for the long-term outcome of the counseling/training programs. Part III presents the counseling/training-income maintenance interaction effects for labor supply in year 2. Simultaneous eligibility for the counseling/training programs did not significantly affect the labor supply response to the NIT for any group.

TABLE 48.—VARIATION IN EFFECTS OF COUNSELING/TRAINING ELIGIBILITY BY NIT ELIGIBILITY AND PREEXPERIMENTAL CHARACTERISTICS OF HUSBANDS

[Standard errors in parentheses]

	Earnings		Hours		Log wage rate	
					Workers	
	Year 4	Year 6	Year 4	Year 6	Year 4	Year 6
NIT status						
Counseling only						
NIT experimental	184	311	101.5	122.7	−.082	−.061
	(358)	(512)	(85.6)	(114.3)	(.045)	(.061)
NIT control	298	203	56.4	94.7	[2] −.108	−.013
	(424)	(621)	(101.5)	(138.6)	(.052)	(.078)
Counseling plus 50 percent subsidy						
NIT experimental	285	40	110.3	15.7	−.031	−.030
	(314)	(457)	(75.0)	(102.1)	(.038)	(.055)
NIT control	19	281	25.1	131.4	−.014	.086
	(344)	(472)	(82.2)	(105.5)	(.041)	(.058)
Counseling plus 100 percent subsidy						
NIT experimental	288		54.7	...	−.038	...
	(306)		(73.2)	...	(.037)	...
NIT control	200		19.1	...	−.028	...
	(332)		(79.4)	...	(.040)	...
Normal income						
Counseling only						
Income below $5,000	587	[2] 1,628	140.4	492.3	[3] −.174	−.163
	(485)	(783)	(116.1)	(174.8)	(.065)	(.109)
Income above $5,000	53	112	61.0	14.5	[1] −.072	−.005
	(343)	(439)	(81.9)	(97.9)	(.042)	(.052)
Counseling plus 50 percent subsidy						
Income below $5,000	178	682	91.4	171.4	[1] −.093	−.059
	(424)	(666)	(101.3)	(148.6)	(.055)	(.088)
Income above $5,000	159	6	69.2	17.7	−.008	−.046
	(288)	(370)	(69.0)	(82.5)	(.035)	(.044)
Counseling plus 100 percent subsidy						
Income below $5,000	137	...	0.4	...	−.053	...
	(441)	...	(105.5)	...	(.057)	...
Income above $5,000	277	...	49.5	...	−.028	...
	(258)	...	(61.7)	...	(.030)	...
Preenrollment employment status						
Counseling only:						
Employed	49	131	111.4	−75.6	−.062	−.022
	(343)	(444)	(82.0)	(99.5)	(.041)	(.052)
Not employed	−600	[2] −1,672	18.4	280.8	[3] −.251	[3] −.319
	(506)	(774)	(121.1)	(174.0)	(.070)	(.112)
Counseling plus 50 percent subsidy:						
Employed	−200	45	85.3	−39.0	−.030	−.055
	(285)	(367)	(68.3)	(82.3)	(.034)	(.043)
Not employed	−60	−950	−23.2	−111.8	−.021	−.140
	(452)	(698)	(108.1)	(156.5)	(.061)	(.093)
Counseling plus 100 percentsubsidy:						
Employed	−170	...	−35.3	...	.012	...
	(250)	...	(59.8)	...	(.029)	...
Not employed	−552	...	−49.3	...	[2] −.161	...
	(498)	...	(119.1)	...	(.070)	...

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[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

a differential impact on work hours, it suggests that initially working wives re-

TABLE 4.9.—VARIATION IN EFFECTS OF COUNSELING/TRAINING ELIGIBILITY BY NIT ELIGIBILITY AND PREEXPERIMENTAL CHARACTERISTICS OF WIVES

[Standard errors in parentheses]

	Earnings		Log wage rate			
			Hours		Workers	
	Year 4	Year 6	Year 4	Year 6	Year 4	Year 6
NIT status						
Counseling only:						
NIT experimental	[2]−505	−326	[2]−172.9	−68.7	−.014	.029
	(205)	(319)	(80.3)	(114.8)	(.078)	(.088)
NIT control	[2]−587	−569	[3]−243.4	−191.2	.010	−.142
	(239)	(370)	(93.5)	(133.4)	(.084)	(.101)
Counseling plus 50 percent subsidy:						
NIT experimental	−90	−216	−31.4	10.3	.020	−.041
	(181)	(285)	(70.8)	(102.5)	(.064)	(.076)
NIT control	−187	−346	−62.3	−104.1	−.014	−.035
	(204)	(305)	(79.9)	(109.7)	(.067)	(.084)
Counseling plus 100 percent subsidy:						
NIT experimental	−134	...	−58.4	...	.005	...
	(177)	...	(69.3)	...	(.061)	...
NIT control	-330	...	[1]−142.0	...	−.068	...
	(198)	...	(77.5)	...	(.066)	...
Normal income						
Counseling only:						
Income below $5,000	[2]−614	−665	[2]−220.8	−107.7	−.044	−.085
	(265)	(449)	(103.8)	(161.7)	(.106)	(.143)
Income above $5,000	[2]−502	−352	[2]−188.4	−136.6	.008	.048
	(198)	(278)	(77.4)	(100.1)	(.071)	(.117)
Counseling plus 50 percent subsidy:						
Income below $5,000	−195	134	−43.6	128.4	.001	−.030
	(241)	(120)	(94.6)	(151.3)	(.093)	(.074)
Income above $5,000	−96	[1]−403	−41.2	−98.3	.011	−.071
	(168)	(233)	(65.9)	(84.0)	(.056)	(.062)
Counseling plus 100 percent subsidy:						
Income below $5,000	[1]−445	...	−151.2	...	[1]−.156	...
	(250)	...	(97.8)	...	(.094)	
Income above $5,000	−140	...	−76.1	...	.007	...
	(153)	...	(59.7)	...	(.050)	...
Preenrollment employment status						
Counseling only:						
Employed	[2]−657	−544	[2]−205.9	−177.0	−.033	−.061
	(257)	(417)	(100.6)	(150.2)	(.081)	(.095)
Not employed	[2]−499	−397	[3]−200.2	−108.7	.021	−.027
	(198)	(287)	(77.6)	(103.3)	(.080)	(.090)
Cousneling plus 50 percent subsidy:						
Employed	[2]−521	[2]−779	−139.4	−202.0	−.023	−.104
	(224)	(357)	(87.8)	(128.7)	(.067)	(.084)
Not employed	48	−58	−2.0	24.6	.032	−.004
	(173)	(246)	(68.0)	(88.8)	(.064)	(.073)
Counseling plus 100 percent subsidy:						
Employed	[3]−594	...	[2]−221.1	...	−.061	...
	(229)	...	(89.7)	...	(.065)	...
Not employed	−61	...	−39.7	...	−.002	...
	(158)	...	(61.8)	...	(.059)	...

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

duced work effort in order to pursue education, while nonworking wives reduced time devoted to nonmarket activities.

TABLE 4.10.—VARIATION IN EFFECTS OF COUNSELING/TRAINING ELIGIBILITY BY NIT ELIGIBILITY AND PREEXPERIMENTAL CHARACTERISTICS OF SINGLE FEMALE HEADS

[Standard errors in parenthesis]

	Earnings		Log wage rate			
			Hours		Workers	
	Year 4	Year 6	Year 4	Year 6	Year 4	Year 6
NIT status						
Counseling only						
NIT experimental	38	56	1.2	76.7	−.002	−.028
	(277)	(396)	(101.1)	(139.4)	(.065)	(.077)
NIT control	478	¹846	¹195.3	¹302.3	.096	.066
	(322)	(471)	(171.4)	(165.8)	(.074)	(.095)
Counseling plus 50 percent subsidy						
NIT experimental	182	560	83.8	²−268.1	−.054	−.033
	(246)	(356)	(89.8)	(125.3)	(.061)	(.076)
NIT control	33	583	101.4	84.3	−.019	².162
	(282)	(403)	(102.8)	(141.8)	(.064)	(.080)
Counseling plus 100 percent subsidy						
NIT experimental	96		14.5	....	.051	....
	(217)		(79.2)	....	(.051)	....
NIT control	298		126.0	....	.034	....
	(283)		(103.1)	....	(.064)	....
Normal income						
Counseling only:						
Income below $3,000	¹588	480	²243.0	213.3	¹.130	.071
	(307)	(469)	(111.7)	(164.7)	(.077)	(.103)
Income above $3,000	103	422	35.3	174.2	−.005	−.004
	(277)	(396)	(100.8)	(139.3)	(.062)	(.073)
Counseling plus 50 percent subsidy:						
Income below $3,000	70	50	30.1	−94.0	−.073	.082
	(280)	(463)	(101.9)	(162.6)	(.073)	(.105)
Income above $3,000	196	35	¹150.1	−116.0	−.019	.052
	(243)	(327)	(88.6)	(114.7)	(.056)	(.065)
Counseling plus 100 percent subsidy:						
Income below $3,000	2	....	33.0	....	.059	....
	(260)	....	(94.7)	....	(.071)	....
Income above $3,000	110	....	99.9	....	.037	....
	(223)	....	(81.3)	....	(.047)	....
Preenrollment employment status						
Counseling only:						
Employed	85	216	66.7	156.6	.026	−.049
	(287)	(395)	(104.6)	(138.8)	(.062)	(.073)
Not employed	305	¹834	100.4	260.5	.076	¹.180
	(295)	(468)	(107.6)	(164.6)	(.076)	(.104)
Counseling plus 50 percent subsidy:						
Employed	−33	246	−54.0	17.5	.009	.072
	(259)	(360)	(94.5)	(126.4)	(.057)	(.068)
Not employed	−149	−399	−103.0	262.6	¹−.125	.028
	(259)	(390)	(94.5)	(137.1)	(.069)	(.092)
Counseling plus 100 percent subsidy:						
Employed	14	....	49.8	....	.024	....
	(240)	....	(87.3)	....	(.048)	....
Not employed	−102	....	−30.5	....	.094	....
	(239)	....	(86.9)	....	(.066)	....

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.

For single women there are significant variations in the effectiveness of the counseling-only program by normal income and preenrollment employment status, but the pattern is opposite to that observed for husbands and is more consistent with the objectives of the program. In this case the counseling program was consistently more beneficial for those who had the fewest skills. Among those who had low normal income and who were not working at enrollment, M1s worked more hours, received higher wage rates, and consequently earned more than did controls.

While the counseling program was beneficial for disadvantaged single women, it was most harmful for disadvantaged husbands. Although we cannot be certain why such different patterns of effects are found for husbands and single women, there are some important differences in their preexperimental labor market experiences that suggest possible explanations. Husbands in initially poor labor market situations generally have had substantial labor market experience. Thus, it may be that they have already dealt with any readily solvable problems or defects in their training. Furthermore, those who were not employed at enrollment had a good chance of finding work on their own. Such individuals, therefore, may have had less need for counseling or training, while the opportunity costs of their pursuing any additional training or job search were large. The nondirective counseling program may have led husbands to seek overly ambitious goals, as described in chapter 2, rather than remain in the labor market and gradually find better employment.

On the other hand, single women who were initially in poor labor market circumstances generally had much less labor market experience. Further, those initially not employed were more likely to remain so. Such women, therefore, may have had the greatest need for labor market counseling, and their foregone labor market opportunities were small. Moreover, women with few alternative financial resources, such as NIT controls, may have been less likely to pursue ambitious training plans.

Summary of eligibility results

Up to this point we have found that, as expected, the SIME/DIME counseling and training programs increased the amount of job counseling and the amount of additional schooling received. However, we have determined that they also, quite unexpectedly, reduced the earnings of those eligible to participate, with the exception of the counseling-only program for single women. Further, we have found that these negative impacts are widespread and that the programs, on the whole, were not beneficial even for select subgroups of the population (again, with the exception of counseling only for single women). Since these results are based on a comparison of randomly assigned experiments and controls and thus are not a result of the self-selection and noncomparability problems that plague most other evaluations of employment and training programs, considerable reliance can be placed on these basic findings.

The question remains of how programs that were intended to help individuals' labor market positions actually harmed them. If the results had indicated no effects, one might be able to explain them by hypothesizing that very little actually happened in the counseling and training programs. The finding of negative effects, however, cannot be so dismissed. Something did occur in counseling and in training that actually harmed participants. It is important to investigate what was deleterious in order to learn how to avoid such unintended results in the future.

CHAPTER VI

ANALYSIS OF RESULTS

We turn now to an exploratory analysis of how SIME/DIME counseling and education subsidy programs produced the negative results described in the previous chapter. This is not an easy question to answer since theory is of little guidance. Instead, we take an inductive approach by examining the patterns of results for potential mechanisms that may be responsible for these unanticipated findings. Any resulting hypotheses, of course, cannot be strictly tested with these data, but we do look for confirmation of their validity by comparing SIME/DIME findings to those of other programs. We focus directly on participants here since participation was not universal, and the impact of the programs on participants may, therefore, have been clouded by random variation in the labor market behavior of the large group of nonparticipants. In addition, we examine how participants made use of the programs and how that may have influenced the programs' effectiveness.

In examining participants directly we sacrifice some of the advantages of random assignment. Because participants in the programs are self-selected, the difference between their behavior and the behavior of all controls is not necessarily an unbiased estimate of the programs' impacts. If those with good prospects were more likely to seek counseling, for example, we could observe a positive impact even if the effects were actually negative. The SIME/DIME experiment, however, still has an advantage over nonexperimental evaluations because we can detect whether selection bias exists and, if so, in what direction results are biased. Because the program cannot affect nonparticipants, a comparison of nonparticipants to all controls should yield a zero program effect. Any estimated effect of the programs on nonparticipants, therefore, must be due to selection bias. Further, random assignment ensures that a comparison of all experimentals to all controls is an unbiased estimate of the programs' impacts. Thus, if there is selection bias in the estimated impact on participants it is opposite to the sign of the estimated impact on nonparticipants. To test for the presence of selection bias, therefore, we include in the participation models a dummy variable for eligible persons who did not participate in the counseling and training programs. This relatively simple procedure is feasible only because of the presence of a randomly assigned control group.

We begin by presenting the overall impact on participants after they stopped using the counseling and training programs. Two models are estimated: the counseling participation model and the subsidy participation model. The models are substantially different from the model used in the eligibility chapter. The counseling participation model contains dummy variables for prior and current participation in counseling by those eligible for each of the three programs, plus a dummy for those who did not ever participate in the counseling program. Counseling participation is defined as having at least one formal counseling session.[1] We also include dummy variables for 3 years of eligibility for the counseling-only and 50 percent subsidy programs.

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[1] In earlier work (West, 1979), the impact of the number of counseling sessions was also estimated. The number of sessions had no significant impact on earnings and wage rates for wives or for female heads, and a negative impact for husbands after accounting for attendance at at least one session.

The subsidy participation model contains variables for current and prior but not current receipt of a subsidy for each of the two subsidies available.[2] Among those who never received a subsidy (including M1s) we include dummies for whether they received current counseling, prior but not current counseling, or no counseling. Dummy variables for 3 years of eligibility for the counseling only and 50 percent subsidy programs are also included so that the effects reported are for the 5-year counseling only and 50 percent subsidy programs and the 3-year 100 percent subsidy program. Full results of these models are presented in West and Dickinson (1980). To simplify this discussion we present only selected results here: the impact of prior counseling from the counseling participation model, the impact of prior subsidy receipt for M2s and M3s, and the impact on nonparticipants from the subsidy participation model.

We have hypothesized that a temporary reduction in earnings might occur as participants reduced their labor force attachment while in the programs and then searched for new jobs when they were finished. To determine whether this process is responsible even for the long-run reduction in earnings, we also examine the impact of the programs on participants' labor force status.

Estimated effects of participation in the counseling and subsidy programs for husbands are presented in table 4.11. There does not appear to be a substantial selectivity bias in estimated impacts for husbands. Generally, nonparticipants had somewhat better labor market performance than controls, but these differences are rarely significant. The participation models confirm that the programs had a negative impact for husbands. Those M1s who had previously received counseling earned less, worked less, and had lower wage rates. These negative impacts are larger, and more of the results are significant, than in the eligibility model.

Some insight into the source of the effects on work hours can be gained by examining the impact of prior counseling participation on the proportion of the year individuals were employed, unemployed, or out of the labor force (see second panel, table 4.11). The large and significantly negative impact on employment in the 6th year is primarily a consequence of prior participants being out of the labor force a higher proportion of time than controls.[3] There are two potential explanations for this effect. As a result of counseling, some low income husbands may have decided that their labor market prospects were so poor that they withdrew from the labor market. Alternatively, the counseling program may have induced some husbands to take education that lasted beyond the period of counseling eligibility. Since counseling-only participants did not receive any subsidy, they had no incentive to complete their training during the experimental period. When we examine whether husbands reported that they were in school or training in the 6th year, we find that M1s were more likely to report that they took such training, by about 5 percentage points, although this difference is not significant. Therefore, while there is some evidence to support the latter hypothesis, it is not conclusive.

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[2] In earlier analysis (West, 1979), it was found that the amount of subsidy received did not have a significant impact on earnings or wage rates, after accounting for the receipt of some subsidy.

[3] This increase in the proportion of the year spent out of the labor force is not due to selectivity bias. Those who did not participate in counseling were more likely to be out of the labor force than were controls. Thus, the estimated increase in being out of the labor force for participants is probably an underestimate of the true effect.

TABLE 4.11.—ESTIMATED EFFECTS OF PARTICIPATION IN COUNSELING/TRAINING TREATMENTS OF HUSBANDS

[Standard errors in parentheses]

	Earnings			Hours of work			Log wage rates of workers			Proportion of—								
										Year worked			Year out of the labor force			Year unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
Counseling participation																		
Received prior counseling but not in the current year:																		
M1	[1]−661	−404	[2]−1176	−147.0	−32.1	[3]−314.4	[1]−.083	[3]−.155	−.049	−.052	−.011	[3]−.127	.028	.012	[2].086	.024	−.001	.040
	(377)	(458)	(541)	(90.3)	(105.3)	(121.0)	(.046)	(.055)	(.068)	(.039)	(.045)	(.047)	(.030)	(.035)	(.037)	(.029)	(.034)	(.035)
Subsidy participation																		
Received prior subsidy but not in the current year:																		
M2	[3]−1067	−459	119	[2]−225.7	−118.2	−73.3	[3]−.078	−.011	.096	[3]−.115	−.030	−.051	.007	−.007	.027	[3].108	.038	.024
	(380)	(455)	(541)	(91.2)	(104.6)	(115.7)	(.046)	(.055)	(.061)	(.039)	(.045)	(.045)	(.030)	(.035)	(.035)	(.029)	(.034)	(.033)
M3	65	−346	....	−10.4	−161.6	....	−.008	−.018	....	.011	−.043	....	−.029	.048	....	.017	−.004	....
	(344)	(465)	....	(82.5)	(106.9)	....	(.041)	(.055)	....	(.035)	(.045)	....	(.027)	(.035)	....	(.026)	(.034)	....
Never received counseling or subsidy though eligible	198	[2]631	333	30.3	95.4	12.1	−.020	.034	.005	−.001	.022	−.016	.024	.012	[1].048	−.024	[1]−.034	−.033
	(200)	(268)	(369)	(48.0)	(61.5)	(83.0)	(.025)	(.032)	(.045)	(.021)	(.026)	(.032)	(.016)	(.020)	(.025)	(.015)	(.020)	(.024)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Prior participation in the subsidy program significantly reduced the earnings of M2s. They were more likely to suffer unemployment, particularly in the 4th year, perhaps because they were having trouble finding jobs for which they were trained. There is no consistent pattern for husbands who participated in the M3 program.

The estimated effects of participation in the counseling and subsidy programs for wives are presented in table 4.12. The estimated impact of the programs on participants is biased by self-selection. Those who did not participate in programs earned less than the average of controls,[4] a difference that is significant in the 4th and 6th years. Because of this self-selection, the estimates for participants are biased upward, and should be interpreted with caution.

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[4] This is primarily a result of the fact that nonparticipants were more likely to be out of the labor force, perhaps because wives who did not intend to work were unlikely to choose to participate in the counseling and training programs.

TABLE 4.12.—ESTIMATED EFFECTS OF PARTICIPATION IN COUNSELING-TRAINING TREATMENTS BY WIVES

[Standard errors in parentheses]

	Earnings			Hours of work			Wage rates of workers			Proportion of—								
										Year worked			Year out of the labor force			Year unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
Counseling participation																		
Received prior counseling but not in the current year:																		
M1	−44	91	−385	12.1	79.3	−99.2	.074	−.004	−.039	.015	.060	−.042	−.031	−.078	.040	.016	.018	.002
	(218)	(282)	(340)	(85.1)	(104.7)	(122.5)	(.074)	(.082)	(.093)	(.043)	(.052)	(.061)	(.044)	(.053)	(.061)	(.021)	(.027)	(.027)
Subsidy participation																		
Received prior subsidy but not in the current year:																		
M2	−90	221	−244	−57.2	90.7	−39.4	−.026	.082	.021	−.005	.074	.018	−.013	−.040	−.054	.017	−.034	.036
	(236)	(304)	(341)	(92.2)	(112.8)	(12.3)	(.076)	(.085)	(.090)	(.047)	(.05[illegible]	(.061)	(.048)	(.057)	(.061)	(.023)	(.029)	(.027)
M3	−256	−258	—	−88.1	109.9	—	−.052	.030	—	−.060	−.0[illegible]	—	.029	.009	—	.032	[2] .058	—
	(202)	(304)	—	(78.9)	(109.6)	—	(.065)	(084)	—	(.040)	(.054)	—	(.041)	(.055)	—	(.020)	(.028)	—
Never received counseling or subsidy though eligible	[2] −254	−118	[2] −493	[2] −116.3	−77	[1] −140.9	−.005	[1] .091	−.073	[3] −.067	−.038	−.064	[3] .071	.049	[2] .089	−.004	−.011	−.025
	(116)	(162)	(230)	(45.3)	(60.2)	(83.2)	(.043)	(.052)	(.068)	(.023)	(.030)	(.041)	(.024)	(.030)	(.041)	(.011)	(.015)	(.018)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

NOTE.—No married women eligible for the M3 program received a subsidy in the 5th year.

The effects of prior participation in counseling on earnings and work hours for wives are predominantly negative, despite the upward bias. As was the case for husbands, this effect is largest for prior M1 participants in the 6th year, primarily due to their greater likelihood of being out of the labor force.

Those wives who had previously participated in the subsidy program also tended to earn less despite the upward bias in these estimates. Wives who had previously received a subsidy tended to be unemployed a higher proportion of the year, a pattern similar to that observed for husbands.[5] Thus, there is evidence that wives who took subsidized training had difficulty finding appropriate jobs.

The estimated participation effects for single women, presented in table 4.13, are not substantially biased by self-selection. The nonparticipants earned only slightly more than controls and generally had similar patterns of labor force status.[6] Thus, estimated effects of participation for single women can be viewed with more confidence than the results for wives.

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[5] We have also investigated the impact of participation on whether individuals were unemployed in the middle of the last month of the experimental year, which represents the last postprogram observation for each year. Results of that analysis indicate that wives who were prior subsidy participants were significantly more likely to be unemployed at the end of both the 4th and 6th years.

[6] The one exception occurs with respect to unemployment; nonparticipants consistently experienced less unemployment than average.

TABLE 4.13.—ESTIMATED EFFECTS OF PARTICIPATION IN COUNSELING/TRAINING TREATMENTS BY SINGLE FEMALE HEADS

[Standard errors in parentheses]

	Earnings			Hours of work			Wage rates of workers			Proportion of—								
										Year worked			Year out of the labor force			Year unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
Counseling participation																		
Received prior counseling but not in the current year:																		
M1	[1] 484	[3] 980	[1] 694	[2] 222.6	[3] 355.9	[1] 273.5	.058	.028	.086	[2] .107	[2] .153	[1] .125	[2] −.111	[1] −.095	−.069	.004	−.058	−.056
	(274)	(346)	(400)	(99.8)	(125.3)	(141.3)	(.064)	(.071)	(.077)	(.048)	(.059)	(.066)	(.047)	(.056)	(.062)	(.029)	(.040)	(.038)
Subsidy participation																		
Received prior subsidy but not in the current year:																		
M2	−74	97	235	−81.5	−81.7	−65.7	−.003	.034	.037	−.053	−.036	−.034	.008	−.000	−.028	.045	.036	[1] .062
	(269)	(320)	(390)	(97.9)	(115.4)	(138.0)	(.065)	(.070)	(.075)	(.048)	(.055)	(.065)	(.046)	(.052)	(.060)	(.028)	(.036)	(.037)
M3	112	151	....	26	135.8	....	.087	−.010	....	.010	.085	....	−.035	−.057	....	.025	−.028	....
	(238)	(314)	....	(86.5)	(113.3)	....	(.053)	(.062)	....	(.042)	(.054)	....	(.041)	(.051)	....	(.025)	(.036)	....
Never received counseling or subsidy though eligible	40	59	287	−25.5	−2.6	0.2	.042	.069	.048	−.002	.009	.001	.043	.029	.033	[2] −.042	−.038	−.034
	(172)	(232)	(360)	(62.6)	(83.6)	(127.2)	(.040)	(.050)	(.076)	(.030)	(.040)	(.060)	(.029)	(.038)	(.056)	(.018)	(.026)	(.034)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

NOTE.—No single women eligible for the M3 program received a subsidy in the 5th year.

The counseling-only program significantly increased the earnings of prior participants in every year, due both to more hours worked and higher wage rates. The increase in work is due to a reduction in time spent both out of the labor force and unemployed. These results for participants, therefore, strongly suggest that the counseling-only program was beneficial for single women. Single women who participated in the counseling-only program were more likely to enter the labor market, find better paying jobs, earn more, and suffer less unemployment.

In the previous chapter, it was found that eligibility for the subsidy program generally resulted in a negative impact on earnings of single women, although the effects were not statistically significant. When we distinguish those who are currently receiving a subsidy from those who are no longer taking subsidized training, however, prior participants are estimated to have a small increase in earnings over controls. These differences are very small relative to their standard errors, but results at least indicate that participation in the program did not harm the labor market position of single women.

There is some evidence, however, that single women who participated in the subsidy program had difficulty finding jobs after they left the program, as was the case for husbands and wives. Prior participants in the subsidy program were consistently more likely to be unemployed than controls, and this difference is significant in the 6th year for prior participants in the M2 program.

In this chapter we have focused on the experience of those who actually participated in the counseling and training programs. Our analysis confirms that the counseling-only program was beneficial for single women, and also yields further insights into the source of unexpected negative results. Significant increases in unemployment are observed for all groups—husbands, wives, and single women. Although some increase in unemployment was expected as participants searched for new jobs after leaving the program, the unemployment suffered by prior subsidy participants was substantial and prolonged and did not generally result in eventual increases in employment. It may be that some individuals who sought training had unrealistic expectations about their labor market prospects. As a result, they may have set their reservation wage or occupation too high to ensure finding a job in a short period of time. As noted before, some participants specified very ambitious goals in their training plans of action. Therefore, we next distinguish experiences of individuals who sought different types of training.

Plans of action models

One possible explanation for the general failure of the counseling and training programs is that the nondirective nature of the programs allowed many persons to pursue unrealistic objectives. Although the formal counseling process was structured, it was nondirective in that participants were not guided toward specific occupational choices. The fact that participants in the subsidy program suffered increased unemployment after completing their training is consistent with the hypothesis that some individuals had unrealistic expectations about their labor market prospects. In this section we examine whether the success of the program varied with the type of objectives pursued. In particular, we disaggregate responses by the occupational and degree objectives of participants' training plans of action.[7]

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[7] As discussed above, estimates in this model are subject to selection biases. In the present application, it is likely that the bias of the coefficient of an occupational goal variable is positive when the same occupation is the dependent variable and negative otherwise. This might occur because the occupational goals are to some extent measures of a person's preferences and it is probable that persons will be more likely to enter their preferred occupations even if the counseling and training programs have no effect. Since no similar measure of preferences is available for controls or nonparticipants, the experimental impact on the preferred occupation is likely to be biased upward.

Results for husbands are presented in table 4.14. Because of the large amount of disaggregation, the standard errors of the estimated effects are large and there are relatively few significant effects. A general impression from the results of the occupational objective model is that husbands with a professional or managerial objective did relatively poorly. In contrast, husbands with a clerical or sales (primarily sales) objective did relatively well, increasing earnings and work effort in all 3 years and suffering less unemployment than did controls.

TABLE 4.14.—VARIATION IN THE EFFECTS OF PARTICIPATION IN THE COUNSELING AND TRAINING PROGRAMS ON EARNINGS, HOURS, AND PROPORTION OF YEAR UNEMPLOYED BY TYPES OF TRAINING PLANS OF ACTION FOR HUSBANDS

[Standard errors in parentheses]

	Earnings			Proportion of year					
				Hours of work			Unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
Occupational objective:									
Professional or managerial	434	−586	−554	18.8	[2]−231.7	−223.4	−.032	.001	.004
	(352)	(456)	(638)	(84.3)	(104.8)	(142.1)	(.027)	(.034)	(.042)
Clerical or sales	724	144	1,772	268.2	73.0	[2] 870.0	[2]−.126	−.094	−.021
	(703)	(901)	(1,631)	(168.5)	(207.1)	(363.2)	(.054)	(.067)	(.106)
Operative or laborer	70	−331	−370	−36.2	−218.1	−301.9	.056	.048	.050
	(568)	(709)	(1,028)	(136.1)	(163.1)	(228.9)	(.043)	(.053)	(.067)
Craftswork	518	−1	188	127.0	−7.5	83.9	−.025	−.034	−.067
	(440)	(601)	(857)	(105.4)	(138.1)	(190.9)	(.033)	(.045)	(.056)
Service	−292	−483	1,966	−42.6	−198.9	176.9	.011	.084	−.079
	(711)	(948)	(1,527)	(170.3)	(217.8)	(340.0)	(.054)	(.070)	(.099)
Counseling but no training plan of action occupational goal	86	[1]−760	[1]−911	39.6	−156.2	−127.3	−.004	.024	.018
	(335)	(437)	(519)	(80.3)	(100.4)	(115.7)	(.025)	(.032)	(.034)
No counseling though eligible	147	[2] 567	346	14.4	87.5	9.6	−.024	−.030	−.032
	(210)	(274)	(374)	(50.3)	(62.9)	(83.2)	(.016)	(.020)	(.024)
Degree objective:									
Associates degree	336	−657	[1]−1,969	−5.0	−224.2	[2]−502.8	.027	.018	[1].139
	(499)	(629)	(1,115)	(119.5)	(144.7)	(249.8)	(.038)	(.047)	(.072)
B.A. or higher	350	−759	−1,501	−125.0	[2]−307.9	[2]−448.8	−.035	.026	.064
	(471)	(608)	(961)	(112.9)	(140.0)	(215.2)	(.036)	(.045)	(.062)
License or certificate	896	518	1,618	189.0	−105.1	95.5	[2]−.087	−.036	−.086
	(554)	(765)	(1,159)	(132.6)	(176.1)	(259.7)	(.042)	(.057)	(.075)
Other	259	−324	206	95.1	−80.6	17.4	−.018	−.007	−.043
	(344)	(457)	(582)	(82.3)	(105.1)	(130.4)	(.026)	(.034)	(.038)
Counseling but no training plan of action degree objective	98	[1]−764	−803	45.4	−156.9	−96.6	−.004	.020	.006
	(334)	(434)	(528)	(80.0)	(100.0)	(118.2)	(.025)	(.032)	(.034)

TABLE 4.14.—VARIATION IN THE EFFECTS OF PARTICIPATION IN THE COUNSELING AND TRAINING PROGRAMS ON EARNINGS, HOURS, AND PROPORTION OF YEAR UNEMPLOYED BY TYPES OF TRAINING PLANS OF ACTION FOR HUSBANDS—Continued

[Standard errors in parentheses]

	Earnings			Proportion of year					
				Hours of work			Unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
No counseling though eligible	157 (210)	[2]572 (273)	347 (372)	17.0 (50.2)	89.7 (62.8)	13.7 (83.4)	−.025 (.016)	−.031 (.020)	−.034 (.024)

[1] Significant at the 10-percent level
[2] Significant at the 5-percent level

Results for the degree objective model indicate that those husbands with academic objectives (associate degree, B.A., or higher) also did poorly. Earnings and hours worked were reduced in the 5th and 6th years for them, and several of these reductions are significant. Much of the decline in employment among those with academic objectives was a result of increased unemployment. Husbands who pursued a license or certificate, on the other hand, tended to have higher earnings and hours worked, and had consistently lower unemployment than controls.

Results for wives are presented in table 4.15. The results for the occupational objective model indicate that wives who pursued health, education or social science careers (one of the two most frequent objectives) had relatively poor labor market experiences. The earnings were lower in every year, due to a decrease in hours that is significant in all years.[8]

TABLE 4.15.—VARIATION IN THE EFFECTS OF PARTICIPATION IN THE COUNSELING AND TRAINING PROGRAMS ON EARNINGS, HOURS, AND PROPORTION OF YEAR UNEMPLOYED BY TYPES OF TRAINING PLANS OF ACTION FOR WIVES

[Standard errors in parentheses]

	Earnings			Proportion of year					
				Hours of work			Unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
Occupational objective:									
Health, education, or social services	[1]−453 (268)	−435 (389)	−480 (542)	[3]−280.5 (104.6)	[2]−237.9 (144.6)	[2]−389.8 (194.4)	−.034 (.026)	.014 (.037)	[2].091 (.043)
Other professional or managerial	−302 (296)	−65 (401)	−826 (650)	−175.2 (115.6)	−25.2 (149.0)	−291.5 (233.2)	.039 (.029)	[3].148 (.038)	.029 (.051)

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[8] The source of this decline in employment varies over time, however. In the 4th year wives seeking health, education, or social service professions were significantly more likely to be out of the labor force while in the 6th year they were significantly more likely to be unemployed. This pattern is consistent with a reduction in work effort to pursue training followed by an increase in unemployment while searching for a job.

TABLE 4.15.—VARIATION IN THE EFFECTS OF PARTICIPATION IN THE COUNSELING AND TRAINING PROGRAMS ON EARNINGS, HOURS, AND PROPORTION OF YEAR UNEMPLOYED BY TYPES OF TRAINING PLANS OF ACTION FOR WIVES—Continued

[Standard errors in parentheses]

	Earnings			Proportion of year					
				Hours of work			Unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
Operative, laborer, or craftswork	−303	114	2,629	−111.1	51.7	[2] 1252.5	.002	−.039	−.054
	(384)	(539)	(1,618)	(150.2)	(200.5)	(580.1)	(.038)	(.052)	(.127)
Clerical or sales	−14	117	−34	−14.8	34.9	−7.0	.005	−.010	−.042
	(266)	(374)	(470)	(104.0)	(139.0)	(168.4)	(.026)	(.036)	(.037)
Service	−296	−662	−509	−100.1	−236.7	−225.2	.037	.024	−.019
	(276)	(403)	(541)	(108.1)	(149.7)	(193.8)	(.027)	(.039)	(.043)
Counseling but no training plan of action occupational goal	−58	−206	128	−7.2	−40.2	117.6	−.012	.012	.036
	(206)	(300)	(312)	(80.5)	(111.5)	(111.9)	(.020)	(.029)	(.025)
No counseling though eligible	[2]−256	−104	[2]−538	[2]−118.9	−75.5	[1]−160.8	−.003	−.010	−.027
	(119)	(165)	(232)	(46.7)	(61.2)	(83.1)	(.012)	(.016)	(.018)
Degree objective:									
Associates degree	−318	−157	121	[1]−199.1	−92.2	−17.7	.019	[3]−.120	.021
	(263)	(370)	(618)	(103.1)	(137.6)	(221.6)	(.026)	(.035)	(.049)
B.A. or higher	−37	−94	[1]−1,158	−97.8	−116.5	[3]−702.3	−.040	.030	[3].152
	(381)	(530)	(671)	(149.2)	(197.0)	(240.5)	(.038)	(.051)	(.053)
License or certificate	[1]−463	[1]−744	−778	[1]−176.8	[1]−279.8	−302.9	−.004	.012	.049
	(270)	(386)	(548)	(105.6)	(143.5)	(196.3)	(.027)	(.037)	(.043)
Other	−208	−78	−61	−89.9	−42.0	−23.1	.018	.030	−.031
	(221)	(307)	(406)	(86.5)	(114.0)	(145.6)	(.022)	(.029)	(.032)
Counseling but no training plan of action degree objective	−6	−152	115	5.7	−26.1	111.7	−.008	.025	.033
	(212)	(307)	(316)	(82.9)	(114.0)	(113.4)	(.021)	(.029)	(.025)
No counseling though eligible	[2]−256	−104	[2]−534	[2]−118.5	−76.3	[1]−158.5	−.002	−.007	−.025
	(119)	(163)	(232)	(46.7)	(60.7)	(83.2)	(.012)	(.016)	(.018)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Results for the degree objective model indicate that wives pursuing either a B.A. or higher degree, or a license or certificate, fared poorly. Wives with such objectives tended to earn less and work fewer hours with no change in wage rates. Wives with other objectives, such as upgrading of skills or college courses with no degree, fared relatively well in that there was no deterioration in their labor market circumstances.

Results of both the occupational goal and degree objective models for single women yield little information (table 4.16). The coefficients are generally insig-

TABLE 4.16.—VARIATION IN THE EFFECTS OF PARTICIPATION IN THE COUNSELING AND TRAINING PROGRAMS ON EARNINGS, HOURS, AND PROPORTION OF YEAR UNEMPLOYED BY TYPES OF TRAINING PLANS OF ACTION FOR SINGLE FEMALE HEADS

[Standard errors in parentheses]

	Earnings			Hours of work			Proportion of year Unemployed		
	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6	Year 4	Year 5	Year 6
Occupational objective:									
Health, education, or social services	93 (299)	63 (389)	463 (514)	119.4 (109.1)	5.1 (140.5)	291.0 (181.4)	.006 (.031)	−.008 (.045)	.021 (.049)
Other professional or managerial	−296 (365)	648 (491)	32 (783)	130.5 (133.0)	180.3 (177.3)	25.8 (276.3)	.045 (.038)	.027 (.056)	−.052 (.075)
Operative, laborer, or craftswork	−450 (530)	−495 (617)	1,150 (2,520)	177.4 (193.1)	122.7 (223.2)	907.1 (888.8)	[1] −.107 (.055)	.055 (.071)	−.143 (.241)
Clerical or sales	−239 (277)	307 (372)	145 (509)	78.0 (100.8)	201.7 (134.5)	14.9 (179.5)	.032 (.029)	−.009 (.043)	.024 (.049)
Service	−168 (323)	120 (404)	−216 (761)	77.4 (117.6)	1.5 (146.1)	211.3 (268.5)	.003 (.034)	.016 (.046)	−.069 (.073)
Counseling but no training plan of action occupational	−39 (237)	−11 (316)	93 (393)	16.0 (89.3)	69.7 (114.3)	38.1 (138.6)	.001 (.025)	.029 (.036)	−.005 (.038)
No counseling though eligible	36 (174)	128 (232)	325 (362)	−17.9 (63.3)	23.4 (84.0)	12.6 (127.6)	[2] .043 (.018)	−.036 (.027)	−.032 (.035)
Degree objective:									
Associates degree	−18 (323)	−54 (424)	−937 (695)	−17.4 (117.7)	33.2 (153.1)	382.4 (245.6)	−.012 (.034)	.036 (.048)	.039 (.066)
B.A. or higher	[3] −1,041 (390)	−178 (505)	−174 (708)	[2] −354.3 (142.1)	52.3 (182.4)	209.2 (250.2)	.058 (.041)	.087 (.058)	.109 (.067)
License or certificate	−263 (331)	51 (417)	−79 (638)	−117.9 (120.7)	−25.9 (150.7)	141.6 (225.6)	.042 (.035)	−.000 (.048)	−.083 (.061)
Other	−85 (243)	108 (319)	2 (445)	−69.6 (88.6)	100.8 (115.4)	91.4 (157.4)	−.020 (.026)	−.023 (.036)	−.008 (.042)
Counseling but no training plan of action degree objective	−25 (240)	−31 (321)	263 (404)	−10.6 (87.5)	58.6 (116.1)	1.2 (142.7)	.002 (.025)	.041 (.037)	−.009 (.038)
No counseling though eligible	38 (173)	142 (232)	336 (361)	−17.1 (63.2)	28.8 (84.0)	11.4 (127.6)	[2] −.042 (.018)	−.037 (.027)	−.034 (.034)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

nificant and do not have consistent patterns over time. In general, none of the groups of female heads had the substantial deterioration in labor market experience that occurred for some groups of husbands and wives. This is consistent with their relatively favorable counseling and training experience as evidenced by results for the eligibility and participation models.

Overall, results from the plans of action models for husbands and wives support the hypothesis that those with ambitious goals would have relatively poor labor market experiences. Although some of this reduction in earnings is due to a continued reduction in labor force participation, part is due to increased unemployment in the postprogram period. Thus, it is possible that the nondirective nature of the counseling and training program may have resulted in participants pursuing overly ambitious goals, and forming unrealistic expectations about their labor market prospects.

Occupation

In this section, we summarize the analysis of the extent to which participation in the programs affected the occupation of those who were working.[9] We disaggregate the impact by the occupational goal of participants.[10] The objective is to determine whether there is any evidence that participants' increased job search eventually resulted in finding jobs consistent with their goals.

Results indicate little evidence that husbands with professional goals had a substantially increased probability of being in professional occupations. Although the probability of being in a professional occupation is increased in 2 of 3 years among those with professional goals, the probability of being in some other occupation was increased by a greater amount in every year. On the other hand, husbands with more modest occupational goals were more likely to work in their planned occupation or where their training was likely to be useful. For example, husbands with goals of craftswork were more likely to work as operatives. Although these are generally less desirable occupations, training in craftswork has some payoff in this allied set of occupations.

Results for wives indicate that those whose goals were health, education, or social service professions were substantially more likely to become employed. However, like husbands, wives whose goals were more ambitious managerial or professional positions were not more likely to be employed in these occupations. Wives with less ambitious goals had changes in occupation consistent with their goals. Overall, wives were generally more likely than husbands to get jobs in fields for which they trained; only those wives with the most ambitious goals were unsuccessful.

Single women were generally less successful than wives in obtaining jobs in their planned occupations. Only those whose goals were service work were clearly successful. One result that contrasts greatly with the result for wives is the lack of success for single women who had goals of health, education, or social service professions. These women did, however, have some increase in service occupations, which include closely related occupations such as nurse's aide, or teacher's aide.

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[9] The full details of this analysis are presented in West and Dickinson, 1980.

[10] As discussed above, estimates in this model are subject to selection biases. The same caveats apply as were mentioned earlier with respect to the plan of action analysis.

In general, results of the occupational goal model are consistent. Persons with more ambitious goals were less successful than persons with modest goals in obtaining jobs in occupations for which they planned to train. These results provide further support for the hypothesis that the nondirective nature of the counseling and training programs may have induced participants to set unrealistic goals. The results also indicate that wives who found work were more successful in attaining their goals than were husbands or single women.

CHAPTER VII

SUMMARY AND CONCLUSIONS

As discussed in chapter 1, a successful program was expected to lead to the following sequence of events: participation in counseling, increase in training and schooling (at least among those eligible for subsidy), concurrent reduction in labor supply, subsequent increase in job search activity, upgrading of occupation, and an increase in wage rates, work hours, and earnings. We do find substantial participation in all the programs and an increase in schooling among those in the subsidy programs.

For the counseling-only program we find one positive impact on the last step in this sequence: as a result of the counseling program, single women were more likely to find better-paying jobs, earn more, and suffer less unemployment. There was, however, an important difference between the actual experience of single women and the expected sequence of events. In particular, single women did not reduce their work effort as a result of eligibility for the counseling program. An important element to the success of the counseling-only program for single women may be that it did not induce women to forgo alternative labor market opportunities as a result of the counseling.

Consistent with this hypothesis is the finding that the counseling-only program was more effective for single women with the fewest labor market skills: those with low normal income and those not employed at enrollment. Positive effects of the counseling-only program were also concentrated among single women who were not eligible for the SIME/DIME income maintenance program. Such women may have been less likely to pursue schooling that required substantial reduction in work effort. The subsidy program for single women and both the counseling and subsidy programs for husbands and wives induced individuals to reduce their work effort during the period of program eligibility. This may have contributed to the harmful results these programs had on those eligible to participate.

The second part of the analysis presented explored other aspects of the programs that might explain why they had deleterious effects. This exploratory analysis focused directly on participants and how they used the programs. In contrast to analysis of eligibles, the analysis of participants is subject to selectively bias so that we must interpret results more cautiously. This analysis also indicates that those who participated in the subsidy program tended to experience significantly more and, for women, more prolonged, unemployment than controls after their participation was completed.

Further support for the hypothesis that the nondirective nature of the programs may have led participants to form unrealistic expectations about their labor market prospects is found by disaggregating the impact of the program by participants' degree and occupational objectives. Those who had ambitious goals, such as professional occupations or academic degrees, fared relatively poorly as a result of participation in counseling and training programs. Part of this is due to their remaining out of the labor force in later years but part is also due to their suffering more unemployment. Those with modest goals were more successful in achieving their planned occupation and goals.

Can we use the SIME/DIME experience to draw implications about the ability of counseling and training programs to meet their objectives? Can the results observed for SIME/DIME programs be generalized to other programs? What have we learned from the SIME/DIME experience about design and evaluation of future counseling and training programs?

We have established that the counseling-only program for single women was beneficial, but that the counseling program for husbands and wives, and the subsidy program for all three groups were not beneficial and, in fact, generally hurt the labor market position of participants. The remainder of this chapter explores whether we can go beyond the results for these specific programs to draw conclusions about the efficacy of a broader class of counseling and training programs. We begin by examining specific aspects of SIME/DIME programs that may have been responsible for the negative results. We then compare the outcomes of SIME/DIME programs to outcomes of other programs and attempt to assess what accounts for differences.

Characteristics of SIME/DIME programs

The analyses presented in this report suggest that SIME/DIME counseling and training programs were generally unsuccessful because the nondirective nature of the counseling led to the pursuit of overly ambitious goals by some individuals. As a result, they reduced their employment during the eligibility period and suffered unemployment when they were unable to find jobs for which they trained. The question remains as to which specific aspects of the counseling program led to the formulation of unrealistic goals. Since there was no systematic variation in the type of counseling given and since we can no longer observe the counseling process,[1] we cannot be certain why it failed. Descriptions of the counseling process do, however, suggest some program characteristics that may have generated these results.

Although the counselors were nondirective in the sense that they did not lead clients into specific career choices, they did follow a highly structured counseling process. The self-assessment phase, for example, specifically encouraged clients to explore new vocational possibilities, and to take a broader view of their strengths, weaknesses, and preferences rather than simply to seek jobs in fields they had worked in before. During this process clients were asked to identify what kinds of work they would like. This step is described in Benus, Halsey, and Spiegelman (1979, p. 28) as follows:

> The purpose . . . was to expand vocational possibilities. Hence, it was important for the counselor to remind the participant not to judge the practicality or possibility of the items. Consequently, participants listed things such as working for a "listening" boss, working in a small office, working where there was mountain climbing, or wanting to be a lawyer, police officer, or surgeon.

The labor market assessment and other aspects of the self-assessment process were intended to assist the clients in evaluating the practicality of their goals. Evidently, the overall outcome of counseling was unrealistic expectations for many clients.

The location of the counseling centers in local community colleges may also have contributed to the failure of the programs. Individuals who pursued academic goals generally did less well, and the frequency of such goals may have been influenced by close association between the counseling program and the community college. Because virtually all individuals who received subsidized education also received counseling, we cannot assess whether the subsidy program would have been deleterious by itself. It seems likely, however, that there was an interaction between these two components, with the subsidy program facilitating the pursuit of overly ambitious goals.

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[1] A few tapes of counseling sessions were made for training purposes, but these were not made systematically for research purposes.

Comparison with outcomes of other programs

Counseling programs

Other job counseling programs that have been evaluated include the employment service (ES) counseling program, job-finding clubs, and job search assistance provided to AFDC recipients by the work incentive program (WIN). Of these programs, ES counseling is most similar to SIME/DIME counseling since both contain a nondirective self-assessment component. The job-finding clubs and most WIN job counseling programs, in contrast, provide clients primarily with job search assistance and labor market information.

Several nonexperimental evaluations of ES counseling have found positive effects,[2] but these studies lacked adequate control or comparison groups. An experimental evaluation of the ES counseling program, as reported by Johnson, et al. (1980), found that those who were eligible for counseling earned less than randomly assigned controls 6 months after they applied for counseling. Negative effects were only partly accounted for by counselees seeking out training as a result of the program. Further, negative effects seem to be concentrated among men and married women with the impact for single women positive although differences among these groups are not statistically significant.

The WIN job counseling program offers a range of flexible services that varies considerably among local areas. Schiller (1978) evaluated these programs by comparing participants to a group of nonparticipating AFDC recipients. He corrected statistically for several potential biases, but any residual bias in the estimates probably overstates the program's impact. The WIN counseling programs significantly increased earnings only for women who had no recent employment experience. These programs had a negative, although not significant, impact on men and women who had recent employment experience.

Job-finding clubs, which combine intense periods of actual job search with group sessions to share information and develop job search skills, provide a substantially different type of counseling than did SIME/DIME programs. An experimental evaluation of job-finding clubs for WIN registrants (Azrin, 1978) found that they produced substantial increases in earnings for both men and women.[3]

In summary, all programs had positive impacts for at least some single women; both WIN and SIME/DIME counseling were found to be most effective for single women with no recent employment experience. Further, results for ES, far from conclusive in themselves, confirm the SIME/DIME finding that nondirective counseling is not appropriate for many disadvantaged workers. The evidence from the job-finding clubs indicates that programs focusing exclusively on labor market information and job-search assistance are more likely to raise earnings of low-wage participants. Overall evidence indicates programs with relatively limited goals, such as job search assistance, are more successful than programs like SIME/DIME counseling, which attempt to help individuals make large changes in their economic position.

Training and education programs

A number of other education and training programs have also been evaluated—including WIN education and training programs, MDTA, CETA and Supported Work. Only the supported work evaluation was based on a randomly assigned control group so that the results for other programs are not as reliable as those for SIME/DIME. Schiller's evaluation (1978) of WIN education programs found a negative impact on participants' earnings, although some individuals may have been still in school at the follow-up interview (9 months after WIN registration). He also found that WIN training programs had a significant positive impact on both men and women who did not have recent employment experience, but a negligible effect on those with recent experience.

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[2] See, for example, Brenna (1974) and Jacobsen (1977).

[3] Preliminary analysis of the employment opportunity pilot projects (EOPP) also shows positive impacts for job-finding clubs.

Evaluations of CETA programs and their precursors have found impacts that varied considerably by race and sex. The MDTA programs, which provided both institutional and on-the-job training, were found to have more positive impacts for women than for men. This pattern was found in comparison with both nonparticipating eligibles (Gay and Borus, 1980; Kiefer, 1976) and the Social Security Administration's continuous work history sample, a random sample of workers (Ashenfelter, 1978). Gay and Borus and Keifer also found evidence of negative program impacts, although the program or specific demographic group for which negative effects occurred varied. A study by Westat (1980) of individuals who terminated from CETA in 1975 found that by 1977 the programs significantly increased earnings of those who had relatively low earnings prior to enrollment, but significantly decreased earnings of those with relatively high prior earnings. These results are preliminary, however, and differences by prior earnings level may be due to differences among comparison groups.

A major experimental evaluation of a training program was conducted for Supported Work, a transitional work experience program that provided training in very basic work skills (such as working in crews and taking directions from supervisors). Supported work was found to have a positive postprogram impact on the earnings of single women, in this case long-term AFDC recipients (Masters, 1981). The program was not generally successful for other predominantly male target groups, although there were some positive earnings effects for former drug users who had the least employment opportunities (Dickinson, 1981).

There is also some evidence that supported work raised participants' expectations about their labor market prospects: 9 months after enrollment, the experimentals' reservation wages were significantly higher than those of controls for all target groups. For participating AFDC women, the reservation wages stayed higher in later periods and, indeed, participants earned higher wage rates than did controls. For youths and ex-offenders, the differential in the reservation wage also remained, but no long-term impact of supported work was found. For former drug users, reservation wages of participants eventually declined relative to controls, and a positive postprogram impact on employment is observed for at least some ex-addicts in subsequent periods. Although it is highly unlikely that differences in the reservation wage entirely account for the differential impact across target groups, these results provide some support for the hypothesis that participation in such programs may generate raised expectations which, in some cases, interfere with program effectiveness.

The evaluations of these other education and training programs do not find the pervasive negative impacts we have found for the SIME/DIME subsidy program. Generally, the effects are positive, but there are some negative results. These negative results may be affected by selectivity bias due to the lack of an adequate comparison group or by the use of short-term follow-up periods. Alternatively, negative impacts of both SIME/DIME and other programs suggest the possibility that for some individuals participation in education and training programs involves not only short-term foregone earnings but longer-term opportunity costs as well. Employment in the regular labor market may have a greater long-term payoff than institutional training for some low-income workers. There is evidence that detrimental effects are more likely for those with better alternative labor market opportunities.

Conclusion

Two important issues have emerged from this evaluation. First, the issue of targeting programs has been elevated in importance. The issue is not simply whether transferring resources from nonparticipants to participants will increase social product. In SIME/DIME, the transfer actually harmed some participants. On the other hand, the counseling program was found to be beneficial for single women. Other programs have also had differential impacts for various demographic groups. We have only been able to speculate about why the counseling program worked only for single women. Greater knowledge is required about employment problems of single women and decisions of employers to hire them before we can determine why the SIME/DIME counseling program in particular, and other counseling and training programs in general, are more beneficial for single women than for other groups.

The second issue is the critical nature of program design. The SIME/DIME programs were designed to maximize freedom of choice for participants. They offered nondirective counseling and a wide range of educational opportunities. Evaluation indicates that such programs in general are inappropriate for low-income individuals, causing at least some of them to form unrealistic expectations about their labor market prospects and to pursue overly ambitious goals. SIME/DIME programs may have been a fairly extreme example of nondirective programs; it is, nonetheless, surprising that such seemingly innocuous elements of program design could result in such deleterious effects. We do not know with any certainty whether more directed job search assistance or more narrowly defined training would have worked. Some evidence from other programs suggests that this might be the case.

Intervening in the labor market appears to be a delicate and complex task. Misguided programs aimed at the wrong individuals can actually lead to a deterioration in the labor market position of low-income workers. Yet the prospects for success are substantial if well-designed programs are aimed at the right types of individuals. We have only begun to understand elements of targeting and program design that distinguish failure from success.

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SIME/DIME FINAL REPORT

VOLUME 1, PART V

MARITAL STABILITY

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CHAPTER I

INTRODUCTION

Effects of welfare on marital stability have been a central issue in the welfare reform debate over the last 20 years. The welfare system has been widely cited as responsible for the breakup of low-income families. In a 1969 memorandum prepared for general distribution by the Social Security Administration asserted that "the present AFDC (aid to families with dependent children) system encourages dependency. The preferential treatment of female-headed families has led to increased family breakup" (Social Security Administration, 1973). Both President Nixon's family assistance program and President Carter's better jobs and income program were intended to reduce the incentives toward family breakup in the welfare system.

When experimental research on income maintenance was proposed in the late 1960s it was recognized that such experiments could address the family dissolution incentives of the welfare system. In a background paper on income maintenance research prepared for the Senate Committee on Finance, the Department of Health, Education, and Welfare stated that among the most important questions to be addressed by income maintenance research was the effect on family stability: "Will family stability be enhanced by changes in income maintenance policies and if so what types of plans and variations within them will serve this purpose best? For instance, if male-headed families are included in the program, will this help to reduce family breakups and illegitimacy?" (USDHEW, 1972).

In Part V we present our latest findings on effects of the Seattle and Denver income maintenance experiments (SIME/DIME) on marital dissolution and formation. SIME/DIME is the income maintenance experiment that put the most emphasis on the study of marital stability effects. The first preliminary findings of the effects of SIME/DIME on marital dissolution were reported to the Department of Health, Education, and Welfare in November 1974 (Hannan, Beaver, and Tuma, 1974). A fuller analysis was submitted in December 1975 and subsequently released by SRI International in June 1976 (Hannan, Tuma, and Groeneveld, 1976). An abbreviated version of that report was published in a major social science journal in May 1977 (Hannan, Tuma, and Groeneveld, 1977a). A great deal of additional analysis has been done since those early reports: data files have been corrected, additional interview data have become available, different models have been estimated and different estimation techniques have been used. General findings of those earlier reports have not changed—the negative income tax (NIT) plans tested in SIME/DIME dramatically increased the rates at which marriages dissolved among white and black couples, and decreased the rate at which Chicano women entered marriages—but we have learned more about the way the experimental effects take place.

In chapter 2 we review the literature on the relationship between income and marital stability and on the effects of the present welfare system on marital stability, along with the marital stability analyses conducted in the other NIT experiments. Chapter 3 is a review of the aspects of the SIME/DIME design that are particularly relevant to the marital stability analysis. Chapter 4 is a statement of the transition rate methodology that we use to estimate the experimental effects, and chapter 5 presents the latest estimates of the experimental effects on marital dissolution. We present analyses relevant to the validity of our findings in chapter 6, in which we show that the estimated treatment effects cannot be accounted for by attrition from the sample, by temporary dissolutions followed by reconciliations, or by fraud. Chapters 7 and 8 contain analyses directed at answering the question "How did the effect of the NIT treatments on marital dissolution come about?" Chapter 7 explores the relationship between marital dissolution and the employment status of husband and wife. Chapter 8 explores how the levels of family income and income available outside of marriage (independence income) affect the marital dissolution rate. Chapter 9 analyzes the effect of the NIT treatments on the formation of marriages. Chapter 10 generalizes our findings to determine how a national NIT program might affect marital stability and how changes in marital stability might affect the costs of such a program. Finally, chapter 11 summarizes our findings and discusses some of their implications.

CHAPTER II

INCOME AND MARITAL EVENTS: REVIEW OF PREVIOUS RESEARCH

In this chapter we review theoretical arguments and previous empirical research on the relationship between income and the formation and dissolution of marriages. The results cited provide the substantive motivation for our analyses and a basis for evaluating our findings. We first discuss dissolution of marriages and then formation of marriages.

Nonexperimental studies of income and dissolution

Until recently the sociological literature on dissolution of marriages has focused primarily on the quality of interpersonal relations in the marriage and on the fit among the spouses' attributes (for example statuses, personality factors, and cultural backgrounds). However, one important line of research that dates back at least 25 years examines the relationship between socioeconomic status (SES) and marital dissolution. This research has produced one well-established empirical generalization. There is an inverse relationship between socioeconomic status and the probability of marital dissolution. Any attempt to explain this generalization reveals the complexity of the relationship between SES and dissolution. Consider Goode's widely cited explanations (1966, p. 382) for the lower propensity to divorce among those with high socioeconomic status:

1. The network of social and of kin relations is more extended, more tightly organized, and exercises greater control over the individual.
2. Income differentials between the wife and husband are greater in the upper strata than in the lower strata; consequently, the wife has more reason to maintain the marriage if she can.
3. Toward the upper strata, far more of the husband's income is committed to long-term expenditures, from which he cannot easily withdraw to support an independent existence.
4. The husband in the lower strata can easily escape the child-support payments and other postdivorce expenditures because his life is more anonymous and legal controls are less effective.
5. Strains internal to the marriage are greater in the lower strata: marital satisfaction scores are lower, romantic attachment between spouses is less common, the husband is less willing to share household tasks when the wife is working, and so on.

These propositions intermingle three possible effects. One is a broad social class effect (propositions 1, 3, 4, and 5) that states that patterns of social relations, consumptions habits, values, and orientations toward marriage differ among social classes, and that observed variations in dissolution rates are a function of these differences. A second effect concerns only the financial dependence of spouses on the marriage (proposition 2). It hypothesizes that spouses compare income levels in the marriage with those in singleness (or perhaps in other marriages; see also Becker, Landes, and Michael, 1977). In Goode's discussion, like most others, it is not clear whether these propositions (particularly 3, 4, and 5) involve a third effect that is attributable only to family income. Recently many investigators have argued for pure income effects. For example, Cutright (1971, p. 296) proposes the following:

> It is reasonable to expect that male income will be more important to marital stability than social status derived from years of education or occupational position because income is more directly linked to consumption than either of the other two status indicators. Consumption is a daily activity, and provides the wife with a constant empirical monitoring of how well her husband is doing in his role as breadwinner. A satisfactory level of consumption should help the wife maintain her own feelings of competence in her role of wife and homemaker, and should act to reinforce her positive view of her husband. Of course, the husband's view of himself as an adequate provider may also be directly linked to his evaluation of his current earnings and prospects for future income growth.

Thus, the literature on the relationships of risk of marital dissolution and socioeconomic variables offers three broad propositions:

1. Family social class is inversely related to the rate of marital dissolution (a social class effect).
2. Financial independence (of the lower-income partner) is positively related to the rate of marital dissolution (an independence effect).
3. Family income is inversely related to the rate of marital dissolution (an income effect).

All three hypotheses are plausible. If true, each can explain the typical relationship between income and dissolution rates. To understand the possible effects of income maintenance on dissolution rates, it is essential to separate the hypotheses into two parts: (1) class versus current income (family income and independence) effects, and (2) the two different kinds of income effects.

Much sociological literature suggests that the relationship of income to dissolution reflects only the operation of social class effects. If so, only interventions that alter social class will affect dissolution rates. The stronger version of this argument holds that class-linked behavior patterns are determined relatively early in life. The weaker version holds that major, long-term shifts in resources during adulthood change class-linked behaviors. According to both versions, a permanent, national income maintenance program might affect dissolution rates, but neither condition would be met by a short-term program of modest transfer payments. If the fundamental relationship involves only class-linked patterns of marital behavior and decisionmaking, income maintenance experiments are unlikely to affect dissolution rates.

Research that examines the alternative possibility—that variation in family income affects dissolution rates—focuses on family income and does not distinguish among the male's earnings, the female's earnings, and nonwage income. We expect that the higher a family's income, the greater the benefits received by its members. These greater benefits should tend to lower dissolution rates. We refer to this as "a pure income effect" because this tendency should hold whatever the source of the income.

Increases in family income also change the difference between benefits received within the family and the potential benefits received outside the family. The source of the income is important in making this comparison. If the increase in family income arises from an increase in the husband's earnings, the wife becomes more economically dependent on her husband. Thus, an increase in the husband's earnings should lower the dissolution rate because it increases the wife's dependence as well as total family welfare. On the other hand, if the increase in family income arises from an increase in the wife's earnings or an increase in nonwage income available outside of the marriage (including both AFDC and income maintenance), the wife becomes less dependent upon her husband. Thus, an increase in the wife's earnings or in transferable nonwage income would tend to raise the dissolution rate through its effect on the woman's independence, at the same time as it would tend to lower the dissolution rate because it raises total family welfare. Since income maintenance will increase both income and independence, we cannot predict its net effect. These opposing effects are discussed more fully in chapter 8.

Until recently, most empirical researchers have contrasted the explanatory power of measures of family income with education and other measures of social class. Many published analyses of census materials (Cutright, 1971; Bernard, 1966; Carter and Glick, 1970; Udry, 1966) related family income, education, and other variables to marital status (married, divorced, never married, widowed). Unfortunately, analyses of census data typically involve three complications that greatly diminish their value. First, reliance on current marital status treats divorce as a state occupied by individuals rather than as an event characterizing unions. The measurement of divorce is thereby confounded with rates of remarriage. Second, for at least some variables (notably income), the direction of the causal effect is obscured by the cross-sectional analysis of current marital status. Persons currently divorced may have been in that state for some time and their current levels on socioeconomic variables may be quite different from the levels during their previous marriage. In fact, the dissolution of marriages appears to hamper occupational achievement (and presumably depress income) for males (Duncan, Featherman, and Duncan, 1972). If this is so, the cross-sectional relationship between income and marital status includes both the effects of income on the divorce rate and the effects of being divorced on income. Finally, because of their cross-sectional nature, these studies compare families with different levels of income rather than the impact of changes in income on the dissolution rate.

Several studies have avoided one or more of these methodological problems. Glick and Norton (1971) obtained marital histories and measures of current socioeconomic variables for a sample of 28,000 households from the 1967 Survey of Economic Opportunity. The use of marital histories permits specific rates to be calculated in a way that avoids confounding dissolution and remarriage rates. However, the measurement of income and education at the end of the marital histories (1967) still confounds to some extent causes and consequences of dissolution, as well as prevents inferences based on changes in income. Glick and Norton reported that education and the male's income affected dissolution rates, and concluded that income is more significant than education in determining which men obtain a divorce. The probabilities of divorce by income and education can be properly interpreted as consistent with the hypothesis, particularly during the first 10 years of marriage. But, since income is more likely than education to be causally affected by changes in marital status, this conclusion may be misleading.

Results on changes in marital status during the first 4 years of the Panel Study of Income Dynamics (Morgan et al., 1974), in which approximately 5,000 families were interviewed annually beginning in 1968, partially support Cutright's contention concerning the causal priority of income. A modification of automatic interaction detection (AID) analysis—which divides the sample sequentially into groups, with the largest differences in their distributions being in family composition—yielded a distinctively nonadditive relationship of 1967 income to divorce probability. For those families in which the male head was 18 to 34 years of age and the marriage less than 5 years old in 1968, the probability of divorce decreased with family income; but for marriages of more than 5 years, the probability of divorce increased with family income.[1] This result is unexpected, and it perhaps should not be taken seriously because it does not control education, race, and other variables simultaneously.

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[1] They also report that changes in family income relative to needs (indexed by family size) are related to dissolution rates. The divorce rate for those whose income decreased relative to needs was 13.7 percent between 1968 and 1972 compared to 7.5 percent for those with no change or an increase (see Morgan et al., 1974, table 2.9). Unfortunately, income change is measured after dissolution so their analysis confounds the effects of divorce on income and of income on divorce.

Becker (1973, 1974) has presented the most fully developed economic theory of marriage. Where we have referred to independence and income effects, Becker discusses the effects of changes in full income on relative gains to being married rather than single. Whereas our analysis focuses on comparisons of being married rather than single, he compares the relative gains accruing to a particular marriage to those available in another match. There have been several empirical studies based on Becker's theoretical model. Becker, Landes, and Michael (1977) estimated the effects of earnings on marital dissolution probabilities using data from the 1967 Survey of Economic Opportunity (SEO). Some of Becker's other predictions are supported by the analysis, although the interpretation of the findings must be made with care since the SEO data measured income at the time of the survey and marital dissolutions were reported prior to the survey. Thus, the association between income and marital stability is confounded by any effect of marital dissolution on income.

An interesting implication of Becker's theory is that unexpected changes in income, whether gains or losses, will have a destabilizing effect on marriages. This expectation is partly supported by the analysis of the SEO data. However, since income is measured after marital dissolution has occurred the direction of causality is unclear. Nevertheless, this implication of Becker's theory suggests that an unexpected change in income, such as an experimental NIT program, should increase the rates of maritial instability and the effect should be positively related to the magnitude of the change.

Cherlin (1976) estimated the effects of husband's and wife's earnings on marital dissolution using data from the National Longitudinal Surveys (NLS). He found that husband's earnings in general significantly decreased the probability of marital dissolution. Following Becker, Cherlin argued that the wife's wage rate relative to her husband's, rather than the absolute levels of wage rates or earnings, was the crucial variable. He found that the wife's wage rate relative to her husband's wage rate was a significant predictor of dissolution: the higher the ratio of the wife's wage to the husband's the greater the probablity of a dissolution.

Effects of the current welfare system on marital stability

One of the major criticisms of the current welfare system is that it encourages marital instability by providing more generous benefits to single-parent families than to two-parent families.[2] In many States the earnings of a low-wage worker working full time are less than the benefits from AFDC and Food Stamps available to a single mother and her children. Thus, there is an incentive to dissolve marriages in order to increase income. Bradbury (1978) identifies four categories of concerns regarding welfare programs that encourage marital dissolution. First, many people adhere to moral systems that place a high value on marriage and the traditional two-parent family. Second, children raised in single-parent families may face handicaps not faced by children raised in two-parent families. Third, there are resource costs to any incentive that causes families to break up into smaller units: it is thought to be less expensive per capita for people to live in larger than in smaller groups. The fourth cost is in target efficiency: if a program has nonneutral incentives there may be more people who qualify as needy than there would have been in the absence of the program. Both the Nixon administration's family assistance plan and the Carter administration's program for better jobs and income were designed to reduce the incentives toward marital dissolution by providing more generous benefits for two-parent families than were available previously.

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[2] For a review of the arguments and recent research on this topic, see Moles (1979).

There have been several empirical studies designed to measure the effects of the current welfare system on marital stability. A widely cited study in the literature on welfare reform is Honig's (1973) analysis of the effect of AFDC benefit levels on female headship rates in 44 standard metropolitan statistical areas (SMSAs). Using census data she found that higher welfare payments significantly increased the proportion of females heading families with children for both whites and nonwhites in 1960. The AFDC effect was significant for nonwhites in 1970 but not for whites. Honig interpreted these results as indicating that family dissolution rates rise with the availability of AFDC income. However, as Lehrman (1973) and others have pointed out, the effect of AFDC payment levels on female headship rates does not necessarily indicate that AFDC promotes marital instability. It may mean only that AFDC enables mothers in disrupted marriages to establish separate households rather than live with other relatives. Cutright and Madras (1976) report results that support Lehrman's criticism. Using State census data for 1950, 1960, and 1970, and models similar to those used by Honig, they found no significant effects of generosity of AFDC on the percentage of ever-married women aged 15 to 44 who were divorced or separated, but a positive significant effect on the percentage of women in disrupted marriages who were family heads with children.

Minarik and Goldfarb (1976) conducted an analysis similar to Honig's based on census data for the 50 States, using a more appropriate measure of female headship rates and a different measure of the generosity of State AFDC programs. They found no significant effect of the generosity of AFDC benefits on female headship rates. Bane (1975) used individual level data from the 1970 census 1-in-1,000 sample to measure the effects of AFDC on marital dissolution. Using a sample of ever-married men and women aged 25 to 44 who lived in one of the 123 largest SMSAs, she found no evidence that the average level of AFDC payments in the State of the SMSA was related to the fraction of men or women who had ever been divorced.

Taken together, these analyses of census data do not suggest that levels of welfare payments affect rates of marital dissolution. However, census data are not well suited to the study of such effects for two reasons. First, as we noted above, marital status is measured at only one point in time in the census: there is no measure of the number of persons who have experienced marital disruptions and later remarried. Any association of welfare payments with the number of persons currently divorced or separated confounds effects of welfare payments on dissolution rates with effects on remarriage rates. The second shortcoming of census data is that relevant variables are not always measured or they are measured only at the time of the census. Variables measuring economic status, for example, are measured at the time of the census, not at the time the dissolution occurred. This makes it impossible to separate the effects of economic factors on marital dissolution from the effects of marital dissolution on economic factors.

Studies that follow representative samples of two-parent families over time are better suited to estimating the effects of welfare benefits on marital dissolutions. Such studies allow the observation of marital dissolutions as they occur and place the causes of dissolution in the proper temporal order. Data from the two major panel studies that have been conducted recently—the panel study of income dynamics (PSID) and the national longitudinal surveys (NLS)—have been used to estimate the effect of AFDC benefits on marital dissolution. Sawhill et al. (1975) analyzed the PSID data for the period 1968 to 1972. They conducted their analysis separately for whites and nonwhites and found no significant effect of the average State AFDC payment on the probability that a marriage would dissolve in the 4-year period. On the hypothesis that the effect of the

AFDC payments variable would be stronger for these "poor and near-poor" families than for higher-income families, they performed the same analysis on the subset of the PSID sample whose 1968 family income was less than twice the poverty level. Even with this subsample, they found no significant effect.

Hoffman and Holmes (1976) used 6 years of PSID data (1968–1974) to estimate the effects of AFDC benefit levels on marital dissolution. Rather than using State average payment levels like Sawhill et al., Hoffman and Holmes grouped the families into four categories that reflected whether family income was high or low relative to its needs and whether State welfare benefits were high or low. They found that couples whose incomes were low relative to their needs were significantly more likely to dissolve their marriages if they lived in a high welfare benefit State than if they lived in a low welfare benefit State. In a third study using PSID data, Moore and Waite (1978) found a significant, but weak, effect of 1972 AFDC benefit levels on marital dissolution during the 1972–1976 period.

Cherlin (1976, 1978, and 1979) has estimated the effects of AFDC benefits on marital dissolution rates using several of the NLS samples. He found no effect of AFDC benefit levels on marital dissolution for white, nonfarm women in the 30- to 44-year-old cohort during the 4-year period, 1967–1971 (1979), but did find a significant effect (10 percent level) in an analysis of black women in the same age cohort (1976). He urges caution in interpreting this effect, however, because his measure of AFDC benefit levels (regional benefit levels rather than State benefit levels) is highly correlated with residence in the South, making it impossible to distinguish the AFDC level effect from the effect of living in the South. He did find a significant positive effect of State level AFDC payments on marital dissolution during 1968 to 1972 in an analysis of white women in the NLS 14- to 24-year-old cohort. Moore and Waite (1978) analyzed data for the same cohort using regional AFDC payments and found no effect for whites or blacks. Mott and Moore (1978) also analyzed the NLS young women but used a different measure of AFDC benefit-levels. They categorized the States into three groups: those with low AFDC benefits and low AFDC participation rates, those with high benefits and high participation rates, and the rest. Using multiple classification analysis they found that women in the high benefit, high participation States had significantly higher dissolution rates than women in the low benefit, low participation States. The women in the third, intermediate category of States had the highest dissolution rates of all.

These studies of the effects of AFDC benefit levels on marital dissolution using data from the PSID and the NLS cannot be summarized easily. Given the mixed results it seems unlikely that the effect of AFDC on marital dissolution is strong. The work of Sawhill et al. (1975) is the most convincing of all the studies in showing no effect of AFDC on marital stability, for they found no significant effect even when the analysis was restricted to low-income families in the PSID. However, Hoffman and Holmes (1976) found that the level of AFDC benefits significantly affected marital dissolution rates among low-income families in the PSID. Obviously, further analysis is needed to resolve these inconsistencies. But until such analysis is done, one must be skeptical about assertions that the AFDC incentives toward marital dissolution are important causes of the recent rise in rates of marital dissolution.

Experimental studies of income and dissolution

As we have seen, most nonexperimental research provides shaky support for the hypothesis that variations in current income affect chances of dissolution. Results of three other income maintenance experiments are particularly instructive, since the research designs avoid the methodological difficulties that hamper nonexperimental research. Moreover, the experimental variations in income permit stronger inferences concerning income effects.

The New Jersery experiment

The first study to analyze the effects of an NIT treatment on marital dissolution was Knudsen, Scott, and Shore's contribution (1977) to the final report of the New Jersey experiment. Knudsen et al. did not focus on marital dissolution per se, but on household composition. Although in the final report the terms household and family are used interchangeably (Kershaw and Fair, 1976), Knudsen et al. refer only to households in their analysis and discussion. They identify four household types (p. 254): nuclear, "families consisting of male and female spouses with children" (85.6 percent of the households at preenrollment); extended, "nuclear units plus related individuals living with the family" (8.4 percent); female-headed, "families with children . . . in which there are neither male heads nor extended members present" (4.5 percent), and other (1.5 percent).

Their initial analysis included 1,353 originally enrolled households who completed the first quarterly interview. Treating attrition from the experiment as a category, they cross-tabulated the sample by household type at preenrollment and the 12th quarterly interview. In some of their analyses they focus on 935 households that were either nuclear or extended at enrollment. The percentage of households changing from nuclear to female-headed was greater for experimental than for control households, but the difference was not statistically significant. They also analyzed the preenrollment to 12th quarter transition matrix as a Markov chain. Using techniques developed by Goodman (1970), they decomposed the transition probabilities to estimate the treatment effects. Effects of the NIT treatments on transitions from nuclear to female-headed households were reported separately by guarantee level, and included transitions caused by death or institutionalization of the husband, as well as those caused by marital discord. The low guarantee level (50 percent of poverty level) had the largest effect, while the highest guarantee (125 percent of poverty level) had the smallest effect. Effects of ethnicity, site, family size, preexperimental earnings, age, and education were controlled in the analysis, but tests of significance of individual coefficients were not reported. Knudsen et al. concluded that the New Jersey NIT experiment had no important effects on household composition. However, since they did not intend their analysis to estimate experimental effects on marital dissolution, their findings are only indirectly applicable to that subject.

Sawhill et al. (1975) focused on marital dissolution among 968 families in the New Jersey experiment with a husband and wife present at enrollment. Only families that remained intact and were present at the 12th quarter or that reported a divorce, separation, or desertion on one of the quarterly interviews were included. Marriages that ended for other reasons (e.g., death or institutionalization of a spouse) or where a spouse dropped out of the experiment were excluded. Thirty-eight families that broke up are excluded because the cause was unknown. Twelve percent (116) of the 968 families had a marital dissolution.

The Sawhill study used OLS regression to estimate the effects of the experimental changes in income on marital dissolution (a parallel logit analysis was reported in an appendix). It included a dummy variable for the NIT treatments, along with various representations of family income and NIT payments. The coefficient of the NIT dummy variable was consistently positive and significant, indicating that an NIT treatment increased the probability of a marital dissolution in the sample studied. (The coefficient of the NIT treatments variable was also positive and significant in the logit analysis.) Treatment effects were also estimated separately by guarantee level (the 100 percent and 125 percent of poverty level guarantees were represented by a single high guarantee variable). The three guarantee level coefficients were significant, and the lowest guarantee level (50 percent of poverty level) had the greatest effect. When the analysis was done separately by ethnic group, there were significant positive effects for the black and Spanish subsamples but not for whites.

Sawhill et al. also attempted to model the independence effects of the New Jersey experiment's NIT treatments on marital dissolution, following the earlier work of Hannan, Beaver, and Tuma (1974) on the Seattle-Denver experiment. The wife's independence income was defined as wife's earnings*(1−tax rate)+guarantee. In most of their equations independence income had a significant effect. Even with independence income (as well as the payment to the family) in the equation the NIT treatments variable was significant and positive, suggesting "that it is some feature of the experiment other than the level of payments or guarantee that is responsible for the higher dissolution rate among families receiving payments" (Sawhill et al., 1975, p. 68). It was thought that the apparent effect might be due to attrition bias, but Sawhill and colleagues did not have sufficient data to test this hypothesis.

Wolf (1976) analyzed marital dissolution in the New Jersey experiment. Also, his sample included families with both husband and wife present at enrollment who had completed the 11th quarterly interview at which questions about marital history were asked. He used responses to these questions, rather than the questions asked at each quarterly interview (used by Sawhill et al.), to determine marital status changes. Families having a marital dissolution were included only if at least one spouse was interviewed in the 11th quarter. Temporary dissolutions were ignored if the couple had reconciled before the time of the 11th quarterly interview. Dissolutions followed by attrition before the 11th quarter were not included. Wolf also excluded cases that had missing data on the independent variables included in his equations. His final sample contained only 588 couples with 40 reported dissolutions. He estimated effects on marital dissolution separately for white, black, and Spanish families, representing the experimental treatment with an NIT treatment dummy variable and with guarantee and tax rate variables. (His experimental guarantee and tax rate variables implicitly assume that control families have zero guarantee levels and tax rates.) He found no significant effect of the NIT treatments on marital dissolution.

The Gary experiment

Henry (1975) reported a preliminary analysis of the effects of the Gary experiment on marital dissolution using a sample of families in which the head of the household was married and living with the spouse at the time of the sociology baseline interview, which occurred about 3 months before enrollment. At the time of the baseline there were 674 married couples in the sample. Marital dissolutions were identified from self-reported marital status at the third and sixth interviews, but did not count if the wife had remarried by the time the marital dissolution was first identified. At the third interview, 41 of 566 families responding (excluding families in which a spouse had died) reported a marital status change. At the sixth interview 64 out of 511 families reported a dissolution. (Two families who were dissolved at the third had remarried by the sixth interview. Henry did not indicate whether these were reconciliations or new marriages.)

Henry estimated the effects of the NIT treatments on the probability of being single at the third and sixth interviews, controlling for wife's year of birth and education, preexperimental income, and number of children. He found that the NIT treatment effects were negative but not significant except for the most generous treatment (high guarantee and a low tax rate), which had a slight positive, but insignificant, effect. Henry concluded that the Gary experiment had no significant effect on rates of dissolution.

In his analysis of marital dissolution in the Gary experiment, Wolf (1979) distinguished between those families intact throughout the experiment and those who reported ending their marriage at some time during it. Two definitions of dissolution were used. One, "spouse left," included separations, divorces, and moved out for other reasons. This variable was taken from monthly family composition reports, interviews, and other contacts with the experimental staff. The second variable, "separated or divorced," was derived from the family composition questions in the periodic interviews (the questions used by Henry).

Wolf estimated NIT treatment effects on these variables using both probit and transition rate models. The probit model included intact families only if they completed all interviews. The transition rate analysis included all families. For the spouse-left variable there were 510 cases in the probit analysis and 643 in the transition rate analysis. For the separated or divorced variable there were 467 cases in the probit analysis and 507 in the transition rate analysis. The number of dissolutions was the same for both analytic methods: 64 for the spouse-left variable and 24 for the separated or divorced variable.

Wolf's analysis controlled husband's and wife's age, education, and labor supply, number and age of children, and preexperimental family income. NIT treatments were represented either by a single dummy variable or by a pair of dummy variables for high or low guarantee. Wolf's estimates of the treatment coefficients were not significant for either variable or by either method. All treatment coefficients were negative.

The rural experiment

The rural experiment, which was conducted on a sample of 600 North Carolina and Iowa two-headed households, reported very low dissolution rates over a 3-year period for both experimental and control families: 0.0175 and 0.0181 per year, respectively. Clearly, not enough dissolutions occurred for one to draw dependable inferences about impacts on dissolution rates from this experiment.

Summary of previous experimental studies

None of the previous studies found evidence that the NIT treatments reduced marital dissolution rates significantly. Sawhill et al. (1975) reported that in the New Jersey experiment the NIT treatments significantly increased marital dissolution rates, but Wolf (1976), who analyzed the data differently, did not find significant treatment effects. Neither Henry (1975) nor Wolf (1979) report significant treatment effects in Gary.

Differences in findings among these analyses reflect a number of factors. Different methodologies were used in the various analyses, as well as different treatments, rules of operation, and populations. It is also possible that the marital dissolution process differs across the sites. It is well documented (for example Carter and Glick, 1970) that divorce rates are higher in the Western than in the Eastern United States, for example, and this pattern may have affected the way in which people respond to the experimental treatments.

Remarriage

Virtually all theoretical and empirical work on factors affecting rates of marriage focus on first marriages, particularly on the timing of first marriages. As a result, we know little about the remarriage process. Lacking any better framework, we organize our discussion in terms of the three hypotheses offered earlier in this chapter.

It is not obvious that the probability of remarriage varies with social class. Census materials reveal a U-shaped relationship of women's education to the probability of never marrying. The highest rates of remaining single are for women with the lowest and, particularly, the highest levels of educational attainment. Goode (1966) reports the same pattern for remarriage. These data are now dated, however, and the relationship may have changed in the interim.

Although the existence of a class effect is doubtful, it seems likely that variations in income do affect the rate of remarriage. Consider the effect of a divorced woman's resources (financial and otherwise) on the probability that she will remarry. The larger and the more stable her resources outside marriage, the more likely it is that she can remain unmarried and independent and still satisfy her material needs. However, women with low levels of resources (that is, a combination of low asset levels, low wage rates, and little history of labor force participation) almost certainly have much to gain financially from marriage to wage-earning men. We expect, then, that increases in a woman's wage and non-wage income make her less dependent on marriage and thereby lower the probability of remarriage.

A small amount of empirical research supports the independence hypothesis. Women having children (particularly young children) are more financially dependent on financial support than those who do not. The presence of children in the family lowers the standard of living (per capita income) when income level is controlled. Also, women with young children face more constraints in working (at the least because their wage rates are lowered by the cost of child care). Not surprisingly, therefore, women with children are more likely to remarry. A second finding that supports the independence hypothesis is that divorcees who are employed are less likely to remarry than those without jobs.

Bane (1975), using individual level 1970 census data, found that women who had reported that they had been divorced were less likely to have remarried if they lived in a State with low AFDC benefits than if they lived in a State with high AFDC benefits. Sawhill et al. (1975) and Duncan (1976) analyzed the effects of AFDC on remarriage using PSID data. The Sawhill study found that AFDC recipients were less likely to remarry than nonrecipients, but Duncan found no consistent relationship between AFDC and remarriage. Hutchens (1979) also analyzed the PSID data, using different models than either Sawhill et al. or Duncan. He found consistent evidence that higher AFDC transfers were associated with lower remarriage probabilities. These studies suggest that AFDC affects marital status by discouraging remarriage rather than by encouraging dissolution as is usually argued.

Income variations might also have an effect opposite to that implied by the independence effect. As a woman's resources increase, she might become a more desirable mate (a dowry effect). If so, the number and variety of marriage offers she receives will rise, and the probability that she will receive an acceptable offer will increase. We know of no empirical research on the remarriage process that bears on this hypothesis. There is, however, an ample ethnographic literature on the economics of arranged marriages supporting this view.

To complete the picture of the paucity of evidence on remarriage, we note that none of the other income maintenance experiments has reported any analysis of remarriage rates.

CHAPTER III

EXPERIMENTAL DESIGN

The experimental design and the sample selection procedures for SIME/DIME are described in part I of this volume. In this chapter we discuss only those aspects of the experiment that are particularly relevant to the study of marital stability.

Definitions of marriage and dissolution

As in the New Jersey and Gary experiments, couples were enrolled in SIME/DIME on the basis of self-reported marital status. In New Jersey and Gary, the rules made no provision for couples who lived together as family heads but were not legally married. The SIME/DIME rules of operation dealt explicitly with cohabitation. A cohabiting couple could be enrolled as a married couple only if one of them was eligible as the sole head of a family unit.[1] This meant that there had to be at least one eligible dependent in the household. The SIME/DIME rules state that an individual cohabiting with a head of an eligible family unit is eligible to participate immediately if the cohabitant is the parent of any dependent in the family unit; if the cohabitant is not the parent of any dependent in the family unit, he/she is eligible to participate only after residing with the family unit for three months and submitting an affidavit. If these conditions were met, the cohabiting couple could be enrolled as a family headed by a married couple. Unfortunately, no record of the marital status of the couples who were originally enrolled in SIME/DIME was maintained. Nor, as far as we know, is there any record of the legal marital status of the couples enrolled in Gary and New Jersey. Because SIME/DIME rules dealt explicitly with cohabitation, it seems likely that more cohabiting couples were enrolled in SIME/DIME than in the other two experiments. But the New Jersey and Gary experiments almost certainly included some cohabiting couples as married because marital status was defined simply as what the couples stated the situation to be.

Cohabiting couples are probably less committed to their marriage than legally married couples and thus are more likely to dissolve their union. But it should be noted that few of the SIME/DIME marriages were casual unions: 85 percent of the marriages had existed for 2 years or more at the time of enrollment.[2] Additionally, only 13 percent of the new marriages during the experiment were formed on the basis of cohabitation.

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[1] The following definition of the head of a family unit is from the SIME/DIME rules of operation:

III-C. Head of Family Unit.—Among persons eligible to be members of a family unit, the head, or heads, shall be determined as follows:

1. Where a family unit contains a husband and wife, both between the ages of 18 and 58 at the time of enrollment, then both are heads of the family unit.
2. If the Unit contains a husband and wife and only one of the partners is between the ages of 18 and 58 at the time of enrollment, then he or she is head.
3. If the unit does not contain a husband and wife, then the head is to be an adult between the ages of 18 and 58 at the time of enrollment, who is a parent or guardian of a dependent child in the unit, as defined in rule III-D.
4. Any eligible person between the ages of 18 and 58 who is cohabiting with a head of an eligible family unit will be considered a head of that unit.

[2] Unfortunately, data on the duration of marriages at enrollment for New Jersey and Gary are not available.

A marital dissolution was recorded whenever the couple separated and reported that the separation was permanent. Reports of the separations and of new marriages came from a variety of sources. Those eligible for the NIT treatments could report marital status changes on the monthly income reporting form (IRF). Control families could report using the monthly address postcards. Any family could contact the field offices in person, by mail, or by telephone. Families could also report marital status changes at the time of the periodic interviews. A potential problem arises because NIT families had more frequent contact with the experiment and had more incentives to report changes in family composition than did control families. This could result in undercounting events such as marital status changes among control families. Therefore, we relied on reports of marital status changes taken from the periodic interviews which were administered according to the same schedule for both NIT and control families. Marital status changes reported from other sources were counted only if they were verified by the interviewing process. This minimized any bias in reporting between experimental and control couples due to the experimentals' more frequent contact with the system and their greater incentive to report events.[3]

New families formed by either partner after a marital dissolution were eligible for the treatment of the original family, as long as both spouses had at least one dependent from the original family with them. A single spouse without dependents was eligible for a $1,000 guarantee (1971 dollars), regardless of the guarantee level of the original family. When a single family head married (whether he or she had been enrolled while single or had become single after enrollment), the new spouse was considered an eligible family member.[4] In all cases guarantee levels were adjusted to reflect changes in family size.

Characteristics of the sample

As discussed in part I, the three race-ethnic samples are not representative of the larger populations in Seattle and Denver. The most important differences arise from the income cut-off used in selecting these samples. Because both the black and Chicano populations are considerably poorer than the white populations in these cities, the white sample is least representative of the three. That is, the experiment covered only the bottom of the income distribution for whites but a much larger fraction of this distribution for blacks and Chicanos. With respect to comparisons within these race-ethnic categories, therefore, the white sample was probably the most subjectively disadvantaged.

Inspection of the distributions of various socioeconomic characteristics at the beginning of the experiment reveals other differences among the three samples. Table 5.1 summarizes these distributions for those who were married at the start of the experiment. Table 5.2 gives comparable distributions for women who were not married.

TABLE 5.1.—DISTRIBUTIONS OF ENROLLMENT CHARACTERISTICS OF MARRIED COUPLES BY RACE-ETHNICITY

	Blacks	Whites	Chicanos
Husband's age (years)	26.2	24.1	24.1
25th percentile	33.7	29.3	29.8
75th percentile	43.4	37.9	37.6
Mean	35.6	32.5	32.0

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[3] Some marital status changes were undoubtedly undetected because the participants attrited from the experiment without reporting the change. We deal with this problem in chapter 6.

[4] See parts I and II of this volume for the various rules regarding initial and continuing eligibility of new family heads.

TABLE 5.1.—DISTRIBUTIONS OF ENROLLMENT CHARACTERISTICS OF MARRIED COUPLES BY RACE-ETHNICITY—Continued

	Blacks	Whites	Chicanos
Wife's age (years):			
25th percentile	23.3	22.4	22.0
Median	29.5	26.8	26.1
75th percentile	39.1	34.0	34.2
Mean	32.4	29.8	29.2
Husband's schooling:			
8 years or less (percent)	15.5	10.0	28.4
9–11 years (percent)	31.4	27.3	39.3
12 or more years (percent)	50.4	62.7	32.3
Mean (years)	10.8	11.8	9.7
Wife's schooling:			
8 years or less (percent)	8.7	8.2	23.6
9–11 years (percent)	27.0	30.4	52.1
12 or more years (percent)	54.3	61.4	24.3
Mean (years)	11.2	11.5	9.7
Husband's yearly earnings:			
$0 (percent)	8.7	9.4	6.0
$1–2,999 (percent)	16.0	18.1	14.1
$3,000–4,999 (percent)	15.1	14.0	16.7
$5,000–6,999 (percent)	25.7	19.3	27.9
$7,000 or more (percent)	34.4	39.3	35.2
Mean (dollars)	4,927	5,029	5,142
Wife's yearly earnings:			
$0 (percent)	46.3	59.9	61.2
$1–1,999 (percent)	19.3	20.3	22.2
$2,000–3,999 (percent)	14.7	10.8	11.0
$4,000 or more (percent)	19.6	9.1	5.4
Mean (dollars)	1,385	770	631
Duration of marriage:			
Less than 2 years (percent)	14.7	15.9	12.1
2 to 4 years (percent)	30.3	30.8	24.9
5 to 9 years (percent)	21.0	23.2	26.0
More than 9 years (percent)	34.0	30.1	37.0
Mean (years)	8.4	7.8	9.1
Number of children:			
0 (percent)	10.3	11.9	5.0
1 (percent)	22.5	24.3	20.4
2–3 (percent)	43.4	45.8	48.0
4+ (percent)	17.9	17.9	26.6
Youngest child:			
Less than 5 years old (percent)	51.5	57.8	63.6
Participation in AFDC in year prior to enrollment (percent)	15.1	14.2	24.3

TABLE 5.2.—DISTRIBUTIONS OF ENROLLMENT CHARACTERISTICS OF FEMALE SINGLE HEADS BY RACE-ETHNICITY

	Blacks	Whites	Chicanos
Number of cases	927	778	335
Age (years):			
25th percentile	24.8	25.1	24.3
Median	31.2	32.6	31.0
75th percentile	38.8	42.3	41.3
Mean	33.0	34.7	33.5
Schooling:			
8 years or less (percent)	6.8	8.1	19.4
9–11 years (percent)	34.8	31.7	47.5
12+ years (percent)	58.4	60.2	33.1
Mean (years)	11.4	11.4	10.0
Yearly earnings:			
$0 (percent)	32.0	34.0	34.6

TABLE 5.2.—DISTRIBUTIONS OF ENROLLMENT CHARACTERISTICS OF FEMALE SINGLE HEADS BY RACE-ETHNICITY—Continued

	Blacks	Whites	Chicanos
$1-1,999 (percent)	18.1	19.6	18.0
$2,000-3,999 (percent)	17.4	15.6	21.2
$4,000+ (percent)	32.5	30.7	26.3
Mean (dollars)	2,163	2,044	1,931
Duration of singleness:			
Less than 1 year (percent)	11.9	14.0	13.4
1 To 2 years (percent)	37.2	37.6	37.3
3 to 5 years (percent)	21.1	19.7	24.3
More than 5 years (percent)	29.6	28.5	24.8
Mean (years)	4.6	4.2	4.2
Number of children:			
1 (percent)	27.6	37.8	27.5
2-3 (percent)	50.5	48.2	52.5
4+ (percent)	21.6	13.9	20.0
Mean	2.5	2.1	2.4
Youngest child less than 5 years old	43.6	38.8	40.6
Participation in AFDC in year prior to enrollment	61.1	49.1	59.1

The most systematic difference among the three samples concerns Chicanos (married and single), who had substantially lower educational attainment than either black or white adults in our samples. Although the mean difference is roughly only 1.5 years of schooling, this difference occurs at a critical point in the education ladder. The majority of the Chicano sample did not finish high school (12 years of schooling), whereas the majority of blacks and whites did.

Differences in levels of the preexperimental earnings for husbands are slight, with Chicanos earning more than the blacks and whites (comparisons of hours worked show that Chicano husbands worked an average of 100 hours more than black or white husbands during the preexperimental year). But there is a major difference among wives. Sixty percent of white and Chicano wives were not employed at all during the preexperimental year, but only 46% of black wives had zero earnings. As a consequence, black wives had double the average earnings of wives in the other two groups. The situation is quite different for unmarried women with children; there are very small differences in the female head's earnings among the three race-ethnic samples.

The black married sample is older than the other married samples—both husbands and wives being older than their white and Chicano counterparts. Chicano couples have the longest durations of marriage in our samples and white couples the shortest.

Perhaps the most surprising difference concerns participation in AFDC during the preexperimental year. The ethnographic literature suggests that the Chicano population should be the most resistant to participating in welfare. However, we found participation rates to be substantially higher for Chicano couples (but not for unmarried Chicanos) than for black and white couples.

It is difficult to know how to interpret these differences because they do not form any neat pattern. We think that the most relevant differences are those that involve wives' earnings and levels of educational attainment. Blacks stand out as unusual on the former dimension, while the Chicanos diverge on the latter.

CHAPTER IV

METHODS OF ANALYSIS

Though our methods are closely related to procedures in biostatistics, they have not been used widely in the social sciences and are likely to be unfamiliar to most readers. Consequently, this chapter describes in some detail the models and estimators that we use and compares them with some more familiar alternatives. We study all kinds of marital status changes, including dissolution, reconciliation, and formation of new marriages. For simplicity, we refer primarily to analysis of marital dissolutions—analyses of the other outcomes present similar issues.

The data

SIME/DIME data contain unusual detail on changes in marital status. These data record not only the occurrence of marital changes but also their timing. This means that one can infer each person's marital status at any time during the experiment; call such data event histories (Tuma and Hannan, 1978; Tuma, Hannan, and Groeneveld, 1979). If the information in event histories is used fully, these designs allow powerful analyses.

Censoring is a major methodological problem in analyzing event history data. Data are said to be censored when events that occur outside some time interval are not observed. In SIME/DIME, people were interviewed roughly three times a year for 4 to 6 years. At each interview, changes in marital status since the previous interview were recorded; if a change occurred, the date of the change was noted. In addition, people were asked the date at which they had entered the marital status they held at enrollment. However, there is no information on marital histories prior to that date and data are obviously also unavailable after the date of the last interview. Histories that lack information prior to some date are said to be left-censored, whereas those that lack information after some date are said to be right-censored. It is extremely important to consider censoring in analyzing event history data. Tuma and Hannan (1978) illustrates several ways in which censoring can ruin inference unless proper adjustments are made. Our data do not permit good solutions to the problem of left-censoring. We must assume that adjustments for observed characteristics of sample members pick up the effects of previous marital histories on rates of change during the experiment. We can do better in the case of right-censoring, as we discuss below.

Marital histories record each family head's marital status: married or not married. In our analyses of reconciliations and formation of new marriages, we distinguish between those married to the person who was their spouse at the start of the experiment and those married to some other person. We also include another category, "not observed"; some of those enrolled in the experiment dropped out, could not be located, or died. In these instances we do not observe the person's marital status for the entire experimental period. A change into the "not observed" category may be regarded as an endogenous form of right-censoring. In SIME/DIME 28 percent of the black sample, 19.5 percent of the white sample, and 20.6 percent of the Chicano sample have marital histories that stop before the end of the experimental period.[1]

Figure 5.1 illustrates the event histories of two hypothetical people in SIME/DIME. The experiment begins at the point labelled τ_1. For simplicity, τ_1 is the same for the two hypothetical cases, although in reality it varies from October 1970 to September 1972. The marital status held at enrollment begins at the point denoted by t_M, where M refers to the total number of changes in marital status that the female head has made. The data record t_M for most people,[2] but we do not know M or the dates of any marital status changes before t_M (that is, how many times people have been married or the dates of any prior marriage). In figure 5.1, the person whose history is denoted by a solid line has a marital dissolution at t_{M+2} and then drops out of the experiment at t_{M+2}. The other person, whose history is depicted by a broken line, remains married until the experiment ends at τ_2.

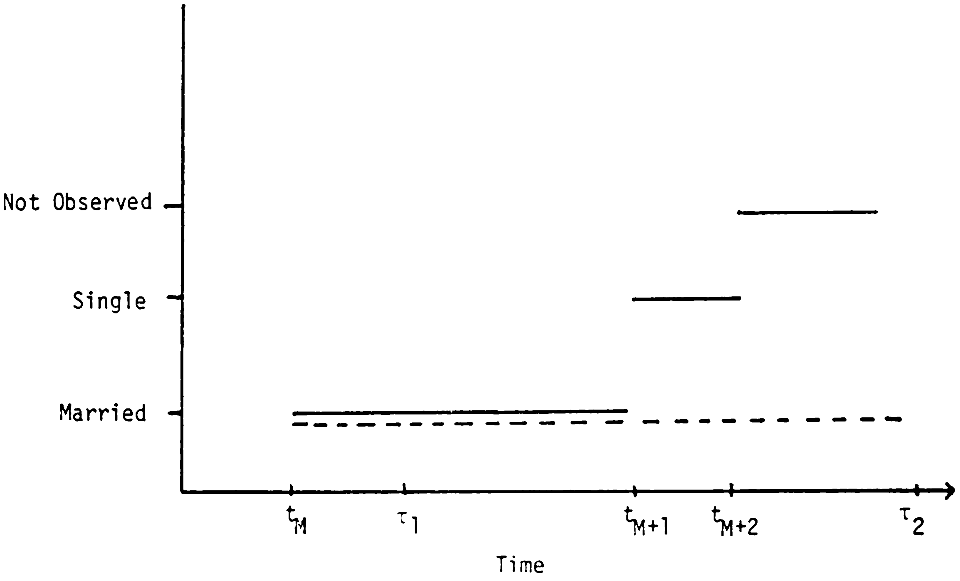

FIGURE 5.1.—Illustration of two types of data records.

SIME/DIME data also include measurements on various individual characteristics that can be used as covariates in analyzing experimental effects. At a minimum, we must take into account the fact that experimental treatments were assigned randomly within categories defined by race-ethnicity, site, marital status, and estimated normal family income. Consequently, one must control for the distribution of these design variables at τ_1, the start of the experiment.

We think that any analysis of the effects of SIME/DIME treatments on marital stability faces three challenges: First, to use fully the information contained in event histories, including information on the timing of events; second, to deal with problems of censoring; and third, to adjust for heterogeneity in rates of marital status change due to variables used in the experimental assignment (and perhaps other covariates).

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[1] The "experimental period" is the time from enrollment until 1 year after eligibility for the NIT treatment ends: that is, 4 years after enrollment for 3-year families and 6 years after enrollment for 5-year families.

[2] Information on t_M is available for 89.5 percent of the SIME/DIME sample. It was predicted for the remainder on the basis of the person's age, education, and other personal characteristics.

Some alternative approaches

It is not practical to discuss all possible ways of analyzing the type of data described above. We discuss some of the most obvious alternatives.

Simple proportions

It is natural to begin by describing these data in terms of the fraction of initially observed marriages that end within the experimental period. For example, one might cross-tabulate the proportion of initial marriages that end before τ_2 by race and by experimental treatment. Such statistics have the advantages of familiarity and apparent ease of interpretation. Unfortunately, they can be misleading. Raw proportions of couples ending their marriages do not adjust for differences in the length of exposure to the risk of a change in marital status. Because some couples drop out in the middle of the experiment and some marriages begin in the middle of the experiment, couples are not exposed equally long to the risk of a dissolution. Such differences are probably not random. For example, couples in the control group are more likely to drop out of the experiment than those with an NIT treatment. Ignoring such differences generally distorts inferences.

Survivor functions

A simple alternative that does adjust for differences in length of exposure is the empirical survivor function, calculated for each group of interest (the control group, treatment groups, etc.). The survivor function, usually denoted by $G(t \mid t_M)$, is the probability that a marriage beginning at t_M survives (has not dissolved) at time t.

Kaplan and Meier (1958) devised a powerful nonparametric method for estimating empirical survivor functions and their standard errors from event history data. As long as one can assume that the process of interest is stationary in calendar time, one can pool observations that enter the state "married" at different times, treating each case's starting time as t_M. This means that we can also include the same person in multiple spells of marriage. Right-censored cases affect the estimated survivor function as long as they are observed. (They give information about the probability that the event has not yet occurred, but they drop out of the calculations when censoring occurs.) Thus an estimated empirical survivor function avoids the major shortcomings of simple proportions.

The Kaplan-Meier product-limit estimator of the survivor function is

$$G(t \mid t_M) = \prod_{t_{(i)}<t} \frac{I-i-c(t_{(i)})}{I-i-c(t_{(i)})+1},$$

where i = a member of the sample of size I initially at risk

$t_{(1)} \leqslant \dots \leqslant t_{(I^*)}$ = ordered times of events

I^* = number in this sample for whom the event is observed

$c(t_{(i)})$ = number of sample members lost from the sample before $t_{(i)}$ because of right-censoring.

The estimated asymptotic sample variance of the Kaplan-Meier estimator is approximately

$$\mathrm{Var}[G(t \mid t_M)] \approx [G(t \mid t_M)]^2 \sum_{t_{(i)}<t} \frac{1}{[I-i-c(t_{(i)})][I-i-c(t_{(i)})+1]}.$$

Survivor functions have an additional advantage. They give information about fundamental parameters of the change process. In particular, the negative of the slope of the natural logarithm of the survivor function is the rate (or intensity) at which the event occurs. If the log of the survivor plot is linear in time, the rate is time-invariant.

Indeed, the use of empirical survivor plots is so attractive that if we had enough data we would be tempted to base our entire analysis on this approach. Unfortunately, the sample size is not large enough. At the start of the experiment (τ_1) there are 2,776 couples and 1,974 single women, but the design is complex. If we partition the data by all assignment and treatment categories and estimate survivor functions within each category, the data are spread too thinly to provide estimates in which one can have confidence. (Each empirical survivor function has a very large standard error.) Thus we need to supplement analyses of survivor functions by some other method that lets us parameterize the effects of the assignment and treatment variables (and other covariates) and pool all observations.

Regression strategies

In the social sciences, the most common approach to analyzing data like ours uses some sort of linear regression analysis. Some analyses of earlier income maintenance experiments relied on this approach. For example, Middleton and Haas (1976) regressed a dummy variable indicating whether a couple was still married at the end of the rural experiment on a set of experimental treatments and selected covariates. Henry (1975) used a similar procedure in analyzing the effects of the Gary income maintenance experiment. These so-called linear probability models have undesirable statistical properties in general and have been largely replaced by regression analysis of transformations of the binary dependent variable.

Typical procedures analyze either a logit (the logarithm of the odds of an outcome) or a probit (a transformation of the cumulative normal distribution) function. For example, Sawhill et al. (1975) used logit analysis to study the effects of the New Jersey Experiment on marital dissolution. Wolf (1976) followed their lead in reanalyzing the data from the Gary experiment.

These methods allow the analyst to parameterize the effects of design variables and covariates. But they share all the other fundamental defects of analyzing simple proportions. They do not deal with the problems of right-censoring, differential variation in the length of exposure to risk of the event, or the occurrence of multiple events to the same individual.

These methods have another less obvious problem: all implicity use a comparative-static logic, assuming that the process of marital formation and dissolution has reached a steady state by the end of the observation period τ_2. Based on analyses of SIME/DIME data using a model that does not assume a steady state has been reached, we estimate that nearly 20 years must elapse before the NIT treatments' effects on the proportion married is reasonably close to its steady-state value. If equilibrium is not reached before τ_2, comparative static methods of analysis give inaccurate estimates of casual effects. Valid inference necessitates procedures that do not asume that a new equilibrium is achieved within a short time after the initiation of the NIT treatments.

Transition-rate analysis

We analyze effects of the experimental treatments on instantaneous rates of transition from one state to another, for example the instantaneous rate of a change from married to not married, which we call the marital dissolution rate. Roughly speaking (see formal definition below), an instantaneous transition rate is the probability of making a particular transition or change per unit of time, given that a change has not yet occurred. A typical value for a marital dissolution rate would be .05 per year, which implies that roughly .05 (or 5 percent) of the marriages existing at the beginning of a year dissolve before 1 year elapses.

Our strategy consists of parameterizing the effects of experimental treatments, design variables, and other covariates on various instantaneous transition rates of special interest: the marital dissolution rate, the marriage rate (sometimes decomposed into a reconciliation rate and a rate of forming a new marriage), and (to a lesser extent) attrition rates (rates of dropping out of the experiment).

Transition rates are fundamental parameters in dynamic models of changes between states of a categorical variable, such as marital status. All other quantities used to summarize event histories can be derived from them, including such outcomes as transition probabilities (used in comparative-static analyses), state probabilities (the unconditional probability of occupying a selected state at a particular time), the expected number of particular transitions during some time interval, and equilibrium probabilities (see Tuma, Hannan, and Groeneveld, 1979, for details).

In some of our analyses of marital histories from SIME/DIME, we analyze data only until the first change in marital status (or until some exogenously determined end of an observational record). For short, we term these "original-spell analyses." In other parts we analyze data on all observed spells of marriage (until some exogenously determined end of observational records). In the latter analyses the same person may be married several times and have more than one marital dissolution. We call these "all-spell analyses."

We find both original-spell and all-spell analyses useful because each has particular advantages and disadvantages not shared by the other. Original-spell analyses have the advantage of simplifying comparison of the treatment and control groups. Treatments were assigned randomly within categories defined by marital status at enrollment (as well as by race and normal income). Consequently, randomization helps rule out alternative hypotheses only when we analyze a sample that is homogeneous on marital status at enrollment. For example, persons with an NIT treatment who marry during the experiment (and thereby become at risk of a marital dissolution) are likely to differ systematically from those in the control group who marry during the experiment. We can avoid dealing with such complications by analyzing only the first change in marital status.

But original-spell analysis has an important disadvantage when we wish to examine time dependence in responses to the treatments, for example, delayed responses, bursts of dissolutions following enrollment, reduced responses due to anticipation of the experiment's end. Original-spell analyses tend to confound time dependence with unobserved heterogeneity. As the experimental clock goes on, the average dissolution rate in the sample of originally enrolled marriages tends to fall simply because couples with the highest dissolution rates tend to have the earliest events and are therefore no longer at risk of a first dissolution. By continually replenishing the sample with newly formed marriages, as we do in all-spell analyses, we minimize temporal variation in the average dissolution rate in the sample due to unobserved heterogeneity. This enables us to investigate variation in response over the experimental period more effectively.

Transition rate models

Our fundamental objective is to understand the effects of SIME/DIME's treatments on the dynamic process generating a person's marital status. Before we can model this phenomenon, we must define various technical terms on which our models are built. Readers who wish only a general view of our methods may skip the following section.

Basic definitions

Since few unmarried men were enrolled in SIME/DIME, our analyses concentrate on female heads. Let Y(t) denote the random variable denoting a person's marital status at time t, and let y(t) denote a particular realization or outcome of the random variable for a given case or person. At t_0, the start of the process (the person's date of birth), the person is unmarried. The person may marry for the first time at t_1, later end this marriage at some time t_2, marry again for the second time at t_3, and so on; eventually the person dies at some time t_d. Clearly the state space of Y(t)—the set of all possible values Y(t) can take—must contain at least three values, say, 1 = unmarried, 2 = married, and 3 = dead.

A person's entire life history is the set of all values of y(t) for all times in the interval t_0 to t_d (inclusive). We denote this history by

$$\omega[t_0,t_d] = \{y(\tau)\colon \tau_0 \leq t \leq t_d\}.$$

Letting y_n represent the value of $y(t_n + \epsilon)$ where ϵ is infinitesimal, we can write this as

$$\omega[t_0,t_d] = \{\tau_0,y_0,t_1,y_1, \ldots ,t_N,y_N, t_d,y(t_d)\},$$

where N is the total number of marital events in a person's lifetime. It is convenient to have a simpler way of writing a person's history from t_0 (the start of the process) to t_n (the time of the n^{th} event), so we let

$$\omega_n = [\omega t_0,t_n] = \{t_0,y_0,t_1,y_1, \ldots t_n,y_n\}.$$

In SIME/DIME we have data on only a portion of a person's complete life history—from t_M to τ_2.

We have already mentioned analysis of survivor functions, and described Kaplan and Meier's nonparametric procedure for estimating them. Here we define the survivor function formally since it plays a central role in models and methods for analyzing transition rates. The survivor function for the n^{th} event occurs after time, t, given a certain history up through time t_{n-1}, the time of the previous marital event:

$$Gy_{n-1}(t\,|\,t_{n-1},\omega_{n-2}) = \Pr[T_n \geq t\,|\,\omega_{n-1}] = \Pr[T_n \geq t\,|\,t_{n-1},y_{n-1},\omega_{n-2}].$$

Note that, in general, the probability that the n^{th} event occurs after some time t depends on the time and outcome of not only the $(n-1)^{th}$ event—t_{n-1} and $y_n - 1$—but also on the time and outcome of all previous marital events, $\omega_n - 2$. Unfortunately, in SIME/DIME only one piece of ω_{n-2} is known: t_0, the date of birth. Even the number of marital events that the person had when the experiment began is unknown. As we discuss below, this missing information necessitates simplifing assumptions. To keep the discussion of basic terms general, however, we continue to define terms conditional on history through time t_{n-1}, i.e., on ω_{n-1}.

The hazard function for the n^{th} event is defined as follows:

$$h_j(t\,|\,\omega_{n-1}) = \lim_{\Delta t \downarrow 0} \frac{G_j(t\,|\,\omega_{n-1}) - G_j(t+\Delta t\,|\,\omega_{n-1})}{G_j(t\,|\,\omega_{n-1})\Delta t},$$

where $j = y_n - 1$, the outcome of the most recent marital change. A hazard function is analogous to the probability of the n^{th} event occurring between t and $t + \Delta t$, given that this event has not occurred before t and given previous history $\omega_n - 1$. Equivalent expressions for the hazard function include

$$h_j(t\ \omega_{n-1}) = -\frac{d \ln G_j(t \mid \omega_{n-1})}{dt}$$

$$= \frac{f_j(t \mid \omega_{n-1})}{G_j(t \mid \omega_{n-1})}, \tag{1}$$

where

$$f_j(t \mid \omega_{n-1}) = -\frac{d\, G_j(t \mid \omega_{n-1})}{dt}$$

is the probability density for the n^{th} event at time t, given ω_{n-1}.

Note that the hazard function at time t is the slope of the natural logarithm of the survivor function at time t plotted against t. Thus, empirical survivor functions, estimated by the procedure developed by Kaplan and Meier, provide a way of estimating hazard functions. On the other hand, if we know the hazard function (empirically or by assumption), the survivor function is determined because (1) implies that

$$G_j(t \mid \omega_{n-1}) = \exp\left[-\int_{t_{n-1}}^{t} h_j(s \mid \omega_{n-1})ds\right]. \tag{2}$$

We are interested in the outcome of the n^{th} event, y_n, as well as its time of occurrence. For example, we certainly do not wish to treat marital dissolution and dropping out of the experiment as equivalent outcomes. For distinguishing outcomes of an event, it is useful to define the conditional transition probability:

$$m_{jk}(t \mid \omega_{n-1}) = \Pr[Y_n = k \mid \tau_n = t,\ \omega_{n-1}].$$

This is just the probability that the n^{th} event consists of a move to state $k = y_n$, given that the event occurs at time t and given previous history ω_{n-1} (including t_{n-1} and $j = y_{n-1}$). It is important to stress that these are conditional transition probabilities. In particular, they "condition" on the occurrence of the n^{th} event at time t. They have no direct relationship to the transition probabilities that analysts attempt to estimate from state histories—the states occupied by members of a sample at a series of discrete points in time.

Finally, we define the instantaneous transition rate as the product of the hazard function and the conditional transition probability:

$$r_{jk}(t \mid \omega_{n-1}) = h_j(t \mid \omega_{n-1})\ m_{jk}(t \mid \omega_{n-1}).$$

An instantaneous transition rate (or simply transition rate for short) is analogous to the probability of a transition from state j to state k at time t, given that the n^{th} event has not occurred before time t and given the person's history up to the $(n-1)^{th}$ event, ω_{n-1}.

Since clearly

$$\sum_k m_{jk}(t|\omega_{n-1}) = 1,$$

it must be the case that

$$h_j(t|\omega_{n-1}) = \sum_k r_{jk}(t|\omega_{n-1}).$$

Consequently,

$$G_j(t|\omega_{n-1}) = \exp\left[-\int_{t_{n-1}}^{t} \sum_k r_{jk}(s|\omega_{n-1})ds\right].$$

Thus, specifying all possible transition rates from state j also defines the survivor function for j.

We choose to model dynamics of marital status in terms of transition rates rather than in terms of hazard functions and conditional transition probabilities. We find this the most natural approach for several reasons. First, separate modeling of hazard functions and conditional transition probabilities seems more appropriate when the choice of changing status precedes the choice of an outcome. In the present case we believe that people do not first decide to have some change in status and then choose whether to break up their marriage or to drop out of the experiment. Second, the process of observing family heads in SIME/DIME introduces a possible outcome (namely, dropping out of the experiment) that does not occur in the natural environment. If we estimate hazard functions during the experiment but wish to predict the survivor function in the natural environment, we must eliminate that part of the hazard function due to dropping out of the experiment before making such projections. By focusing directly on transition rates, i.e., separating dissolution rates from attrition rates, we avoid having to make such a post hoc adjustment. This does not imply that no adustment occurs, merely that it occurs as the result of a decision made before analysis rather than after estimation.

Log-linear models of transition rates

We construct models in which experimental treatments, design variables, and other covariates affect the rate of dissolution. In most of the work reported below, we specify a log-linear relationship between the explanatory variables and the rate. We use a log-linear specification because it is the simplest function that imposes a nonnegativity constraint on the rate. Thus, in the simplest models, which treat the explanatory variables as fixed at their levels at the time of enrollment, we specify models with the form

$$r_{jk}(t|t_M) = \exp[\acute{\alpha}_{jk}\underset{\sim}{x}(\tau_1)],$$

where $\underset{\sim}{x}(\tau_1)$ = vector of covariates measured at τ_1 (start of the experiment)
t_M = date that enrollment marital status was entered.

The vector of covariates includes three kinds of variables: experimental treatments, design variables, and nonexperimental covariates. We use a variety of specifications of the experimental treatments. In most cases these consist of sets of dummy variables for various treatments—in the simplest case, a single dummy variable that distinguishes those with NIT treatments from those in the control group.

The coefficients associated with dummy variables have a simple interpretation in a log-linear specification. For cases with a value of unity on a variable, say x_1, the variable's effect on the rate is $\exp(\alpha_1)$. For those with the value of zero, the effect is just unity. Thus $\exp(\alpha_1)$ is the ratio of the rate for those with the value of unity on the variable to those with the value of zero (assuming that the cases are alike in all other respects). That is, it gives the multiplier of the rate due to the dummy variable (e.g., the experimental treatment). For example, suppose that x_1 is a dummy variable indicating eligibility for the NIT treatment and the estimated value of α_1 is .34. Then $\exp(\alpha_1)=1.40$, indicating that the dissolution rate is 1.4 times greater among those eligible for the NIT than among controls. A multiplier of unity means that the treatment has no effect; a multiplier less than unity means that the treatment lowers the dissolution rate; a multiplier greater than unity implies that the treatment increases the rate.

We control the effects of some design variables by conducting separate analyses within categories. For example, when we analyze only original spells, we implicity control for marital status at enrollment. We also analyze the three race-ethnic groups separately because our early work showed significant differences in the causal structure among these groups. We found that we could pool observations across sites with no significant loss of information, We do, however, always include a dummy variable for site. We control for the effects of income level in a variety of ways. In the simplest analyses, we include a set of dummy variables for the normal income categories used in the assignment of treatments (see part I of this volume for details of the assignment).

Given stratified random assignment to treatments, it is not necessary to include nonexperimental covariates in analyzing effects of treatments. But we do include additional covariates for two reasons. First, as usual, we are concerned with improving precision of estimates. Second, as mentioned above, we are concerned with problems of left-censoring in these data. We cannot control for previous marital histories because they were not recorded. As a surrogate for such controls, we adjust for a variety of current characteristics of the spouses. These include ages and years of schooling of the spouses, the number of children living in the family, a dummy variable that indicates the presence of a child less than 5 years old, and a dummy variable that indicates whether the family received AFDC payments in the year prior to the beginning of the experiment. In analyzing first spells we also control for the duration of the marriage. It may seem strange to use duration at enrollment as a covariate and duration during the experiment as an implicit part of the outcome, but previous duration is an important predictor of dissolution rates. We include it to control for differences due to unobserved previous marital history.

In most of our previously published analyses, covariates are fixed at their enrollment levels. Below we refer to such analyses as ones with "enrollment covariates." Estimates from such analyses give proper reduced-form estimates of experimental effects. If we want to know whether the NIT treatments have any effects on dissolution rates, we should not control for other behavioral responses to the treatments. On the other hand, if we wish to understand how observed effects occur, we decompose the total effect into a series of effects, including direct effects and indirect effects mediated by other behavioral responses. Then it is appropriate to use current levels of covariates, as we describe in chapter 8. We call such analyses ones with "current covariates."

Time dependence

There are several reasons for expecting responses to NIT treatments to vary over time. Some arguments suggest a delay before the full response is realized. For instance, it might take some time before participants fully understand the options open to them. Other arguments suggest that the initial response exceeds the long-run response. For example, the treatments may be a shock that upsets unstable equilibria in some marriages, inducing a burst of dissolutions at the outset. (For a version of this argument, see Becker, Landes, and Michael, 1977.) Since we are interested in learning about the possible long-range effects of NIT treatments, we need to investigate the pattern of time dependence in the response during the 3 to 5 years of the experiment.

If one single argument for time dependence was compelling, we could specify the functional form of the time variation in rates and use a parametric approach to studying time dependence. However, the various arguments do not agree even about the gross qualitative features of the expected time variation. Therefore, we use a simple nonparametric approach to time dependence. We break the experimental period into segments of varying lengths (for example, 6 months), estimate experimental effects within each segment, and test null hypotheses that the experimental effects are the same in each segment. As mentioned earlier, we typically use all spells in estimating models with time dependence.

In one case we do use a parametric specification for time variation in rates. In analyzing possible NIT effects on the rate of reconciliation (remarriage of a husband and wife following a dissolution but before any intervening marriage), we noticed that the rate drops roughly exponentially with time elapsed since the dissolution. Therefore, we fitted a so-called Gompertz model to these data, allowing the rate to decline exponentially with time in the state (single but not yet remarried). Details of this analysis are reported in Groeneveld, Hannan, and Tuma (1980).

We have also used an approach developed by Cox (1972) that permits unobserved, unspecified time variation in the rate for all couples at risk. Cox's partial likelihood estimator adjusts for such time dependence as long as it is the same for all cases at risk (see Tuma and Hannan, forthcoming, and Hannan and Carroll, 1981).

Time-varying covariates

When we wish to use contemporaneous measures of covariates such as income and family size, we use a modification of the time-dependent model discussed above. We break the experimental period into segments a month long and update the levels of all covariates, including duration and age, at the beginning of each month. However, we asume that the coefficients associated with the covariates are time-invariate. This means that we approximate changes in the levels of all metric covariates with step functions that can change on the first day of each month. Given that the observation period is 36 months, this approximation allows considerable variability in the covariates over the period analyzed.

Some other covariates are qualitative states that change at known dates. The most relevant of these is employment status, for which we have complete histories of changes during the experiment. We use a similar procedure to incorporate contemporaneous measures of employment status in analyzing marital dissolutions. Consider the situation drawn in figure 5.2, which we used in our analysis of the interactions of marital status and employment status responses to NIT treatments. The four states on the borders of the figures are defined in terms of husbands' and wives' employment statuses. The state in the center is single. Couples can move to the state single from any one of the four other states. We want to test hypotheses about differences in NIT effects on the rates of the four types of moves. Couples can move among the four employment statuses—that is, they can move around the borders of the figure (assuming that both husband and wife do not change employment status at the same instant).

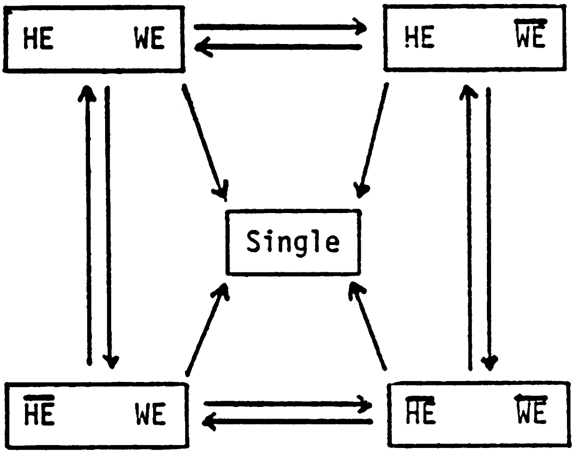

FIGURE 5.2.—Illustration of the state space for a model of differential rates of dissolution by spouses' employment statuses.

An important question arises: How does the rate of moving from any of the four states to the single state depend on the history of moves among the four states? In general, we might expect some complicated form of dependence of rates on the history of changes in employment status. Because we want to analyze experimental effects in the multistate case, we must use some form of simplifying assumption about the effects of history. We use the Markov assumption, namely, that the history of past moves does not affect transition rates once the effects of current status (and all covariates) are taken into acount. The Markov assumption lets us pool all spells into one analysis, ignoring the differences in previous history. We comment further on the suitability of the Markov assumption in discussing of the results of our research.

Estimation and testing

We rely almost exclusively on maximum likelihood (ML) estimators. These estimators have excellent large sample properties, as is well known. Our samples are almost surely large enough for these asymptotic properties to hold fairly well. Moreover, ML estimators are well suited to censored data.

The likelihood function $\mathcal{L}$ is defined as the joint probability (or joint density) of the sample observations. If observations on the different sample members are independent, as is assumed here, the likelihood is just the product of the contributions of the sample members. To find ML estimates of some model's parameters, one first expresses each sample member's contribution, $\mathcal{L}_i$, in terms of the parameters of the model and the sample data on i. Then one selects estimates of parameters that make the sample data most likely. For the models that we study, the parameters cannot be written as simple functions of the observations. Therefore we use the numerical optimization (based on the Gill-Murray (1972a, b) procedure to find estimates.

In addition to finding estimates of parameters, ML estimation produces an estimate of the lower bound of the variance-covariance matrix of parameters (the Fisher information matrix)

$$\left[\frac{\partial^2 \log \mathcal{L}}{\partial \alpha^2}\right]^{-1},$$

where α denotes the parameters that are estimated. Among other things, empirical estimates of this matrix yield the estimated standard error of each parameter. We use the asymptotic standard errors to test hypotheses about specific parameters. In particular, we test the null hypothesis that each parameter (especially those associated with experimental treatments) is zero in the population. We use two-sided tests and the 10-percent level of significance. Since many social scientists prefer lower significance levels, we also indicate whether the estimates are significant at the 5- and 1-percent levels.

We test hypotheses about sets of parameters (for example, a set of experimental treatments) with a likelihood ratio test. Let $\mathcal{L}_1$ and $\mathcal{L}_2$ represent the likelihoods for two different models, one with no constraints on its parameters (the alternative hypothesis) and the other with k constraints (the null hypothesis), respectively. For example, the alternative hypothesis might be that k experimental treatments affect the dissolution rate while the null hypothesis assumes that each of the k treatments has no effect on this rate. The likelihood ratio λ is defined as $\max(\mathcal{L}_2)/\max(\mathcal{L}_1)$. For large samples, $-2 \log \lambda$ has a χ^2 distribution with k degrees of freedom under the null hypothesis.

CHAPTER V

EFFECTS OF NIT TREATMENTS ON DISSOLUTION RATES

This chapter summarizes our basic findings on the effects of NIT treatments on rates of marital dissolution. The analyses presented here were designed to answer the question "Do the treatments have any effect on marital dissolution?" The analyses in Chapters 7 and 8 attempt to answer the question of how the treatments' effects on marital dissolution came about.

Proportions of marriages that dissolved

The most straightforward way of estimating the effects of the treatments on marital dissolution compares the proportions of marriages that ended during the experimental period for couples with NIT treatments and for those in the control group. Table 53 reports these figures at the end of the 36th month of the experiment, at which point treatments end for those on the 3-year plan.[1] The probability of dissolution was much higher for black and white couples with NIT treatments—the fractions of marriages that ended within 36 months were roughly 40 percent higher for couples with NIT treatments than for controls. For the Chicano sample, the effect of the treatment was also positive but much smaller. Black and white couples assigned to the 5-year NIT treatments were more likely to report a dissolution than couples assigned to the 3-year treatments. There was almost no difference between the 3- and 5-year treatments for Chicanos. For all three samples the NIT treatments increased the probability of a marital dissolution.

TABLE 5.3.—PROPORTIONS OF ORIGINAL MARRIAGES OBSERVED TO END DURING THE FIRST 36 MONTHS OF THE EXPERIMENT

[Number of couples in parentheses]

	Blacks	Whites	Chicanos
Original marriages only:			
Controls	.205	.145	.185
	(435)	(608)	(200)
NIT treatment	.278	.203	.222
	(504)	(691)	(338)
3-year treatment	.270	.198	.223
	(333)	(479)	(238)
5-year treatment	.292	.212	.220
	(171)	(212)	(100)

---

[1] Throughout our analyses we have omitted the marital histories of the 20-year families after they were assigned to the 20-year treatment. Their marital histories prior to that time (about 2 years after enrollment) are included. In our analyses all experimental families who become 20-year families are classified as 5-year experimental families until the length of treatments is changed.

Empirical survivor functions

Next we used the Kaplan-Meier estimator of the empirical survivor function described in chapter 4. Figures 5.3 to 5.5 present empirical log-survivor plots for original marriages. The plots are done separately by race-ethnicity. These plots show that adjusting for the length of exposure to risk does not eliminate the NIT effect for black and white couples; the probability of remaining married for a specific period dropped sharply under the NIT treatments. The pattern is less clear for the Chicano sample. The survivor function for those with NIT treatments tends to lie beneath that for the control group, but the differences are slight and inconsistent over time. Inspection of the confidence intervals shows that the curves for the experimental and control groups differ significantly at the 5 percent level over most of the 36-month period for the black and white samples, but do not differ significantly for the Chicano sample.

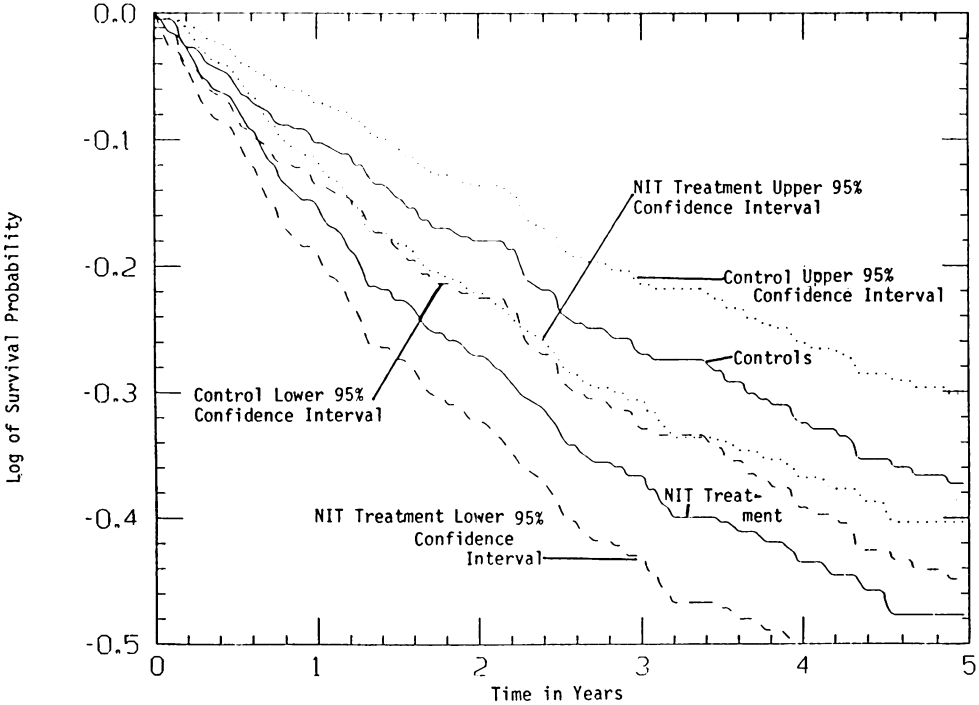

FIGURE 5.3.—Estimated survivor plots: Originally married blacks.

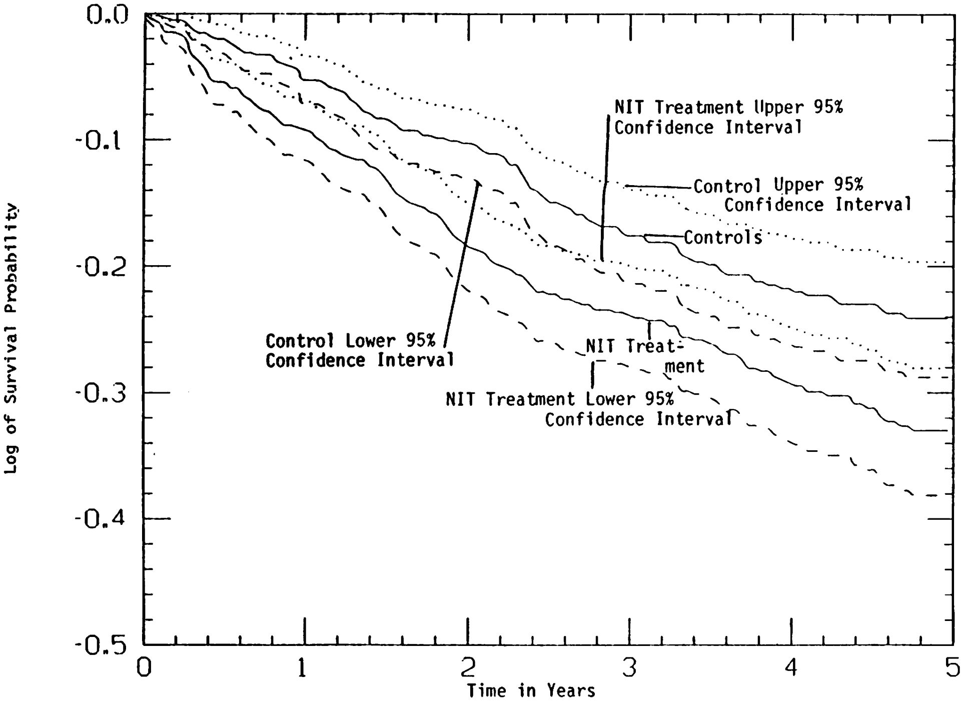

FIGURE 5.4.—Estimated survivor plots: Originally married whites.

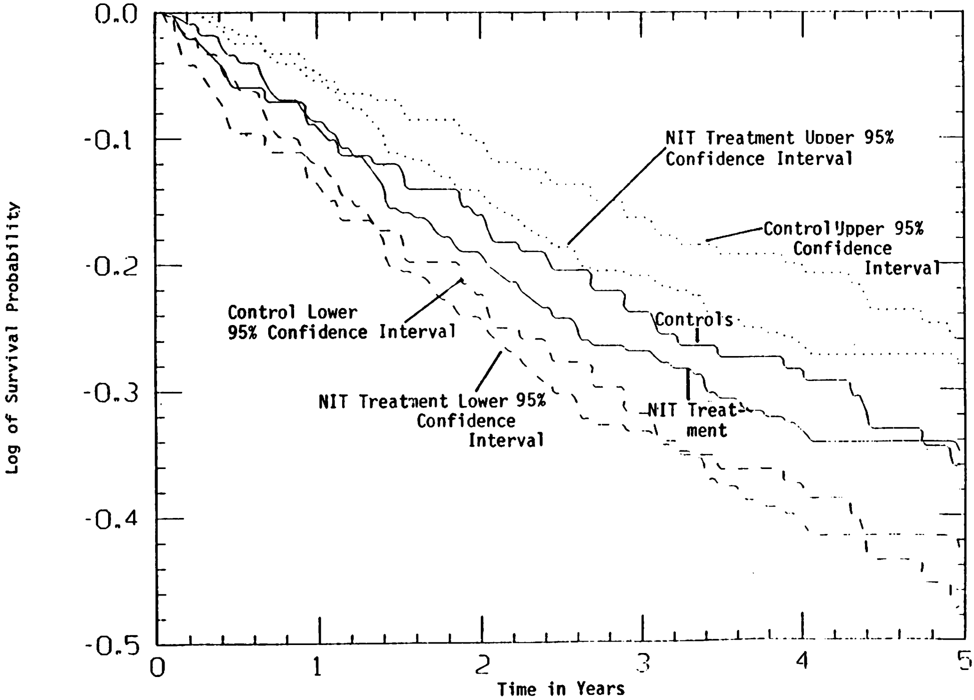

FIGURE 5.5.—Estimated survivor plots: Originally married Chicanos.

We do not include separate plots for the different lengths of treatment. Our study of those plots revealed that the dissolution rates for both whites and blacks were greater under the 5-year treatment than under the 3-year treatment, but the two treatments did not differ significantly. The plots for the 3- and 5-year groups of Chicanos are almost indistinguishable.

One final aspect of these empirical survivor functions deserves mention. In chapter 4 we noted that linearity of the log of the survivor function implies that the rate is time-invariant. The plots in figures 5.3 to 5.5 are essentially linear for all three race-ethnic groups, suggesting that the rate of dissolution did not vary over the 36-month period. These plots are consistent with the view that the NIT treatments shifted the rates upward at the start of the treatment but that the rates remained constant at these elevated levels for the rest of the study.

Had assignment to treatments been completely random, we could now answer the main question affirmatively: NIT treatments significantly increased dissolution rates for black and white sample but not for the Chicano sample. However, assignment was random only within categories of the design variables and some of these design variables, especially normal income, are likely to be correlated with a couple's initial (preexperimental) rate of dissolution. Therefore, we must adjust for differences in the initial distributions on all design variables. In addition, we want to adjust for some nonexperimental covariates in the interest of obtaining more precise estimates of the treatments' effects.

Effects on marital dissolution rates

We now report results of the analyses that give the clearest picture of the total effects of the NIT treatments: maximum likelihood estimates of the effects of NIT treatments on dissolution rates, adjusted for design variables and other covariates. We use the log-linear specification discussed in Chapter 4:

$$r_D(u) = \exp[x'\alpha + z'\beta], \tag{1}$$

where

$r_D(u)$ = dissolution rate time u (where $u = t - \tau_1$)
x = vector of experimental treatments
z = vector of assignment variables and nonexperimental covariates.

The various analyses reported include different treatment variables in *x*. The vector *z* always includes the same set of variables: a dummy variable for site, six dummy variables for the normal earnings categories used in assignment, age and years of education of the wife, number of children in the family, a dummy variable for the presence of a child less than 5 years old, and a dummy variable that indicates whether the family received any AFDC payments in the year prior to enrollment. In analyses of original marriages, we also include age and years of education of the husband and the duration of marriage at enrollment in years. Although all of these variables may change over the course of the experiment (and age and duration must change), we keep them fixed at their enrollment values. (In chapter 7 we report estimates from models that use current levels of the covariates).

Because our samples are stratified and generally not representative of all married couples, the main effects of the covariates in *x* are not particularly interesting. Nonetheless, it is reassuring to learn that their effects are similar to those reported from studies of national samples. We have included estimated effects of these covariates, which we sometimes call the "background function," in appendix A.

There are many ways of representing the experimental treatments in estimation equations. The model that we stress most contains two variables for the NIT treatment: a dummy variable that indicates whether the couple received any NIT treatment and a dummy variable that indicates that the couple received the NIT treatment that lasted for 3 years.[2] In our tables and discussion we refer to the variables as the "5-year NIT treatment" and the "3-year treatment adjustment." We can express this model using the notation of equation (1):[3]

$$r_d(u)=\exp[\alpha_1\text{NIT}+\alpha_2 3\text{YRAD}+z'\beta]. \tag{2}$$

In this model, α_1 is the estimated effect of the 5-year NIT treatment on the log of the dissolution rate. The antilog of α_1 (that is, $\exp(\alpha_1)$) is the dissolution rate for a couple in the experimental group expressed as a multiple of the rate of a control couple with comparable values on z; α_2 is the estimated difference between the 5-year and the 3-year NIT treatments. The effect for a couple assigned to a 3-year NIT treatment is the sum of the two estimates, $\alpha_1+\alpha_2$. Exp $(\alpha_1+\alpha_2)$ gives the dissolution rate for a couple on a 3-year NIT treatment expressed as a multiple of the rate of a control couple with comparable values on z.

Readers should note that this specification, and most specifications used in this report, stresses the effects of the 5-year NIT treatment because the longer treatment is more like a permanent program than the 3-year treatment. Our intent was to design our analysis so as to determine what the effect would be of a permanent NIT program. Other specifications, such as a model using a single NIT dummy, give a better indication of the average experimental effect within the sample but do not give a better indication of the effect of a permanent program.

Table 5.4 reports maximum likelihood estimates of these two treatment parameters for both original marriages and all marriages observed during the first 36 months of the experiment.[4] We find that the 5-year NIT treatment significantly increased the dissolution rates for black and white couples. Adjusting for design variables gives larger estimated 5-year NIT treatment effects than are apparent in the simple proportions. The multiplier is 1.57 for black original marriages and 1.53 for white original marriages. This means that the 5-year NIT treatment increased the rate 57 percent for blacks and 53 percent for whites. The multipliers in the all-marriage analysis are slightly larger.[5] In both analyses these effects are not only large, but are also statistically significant.

TABLE 5.4.—ESTIMATED EFFECTS OF THE NIT TREATMENTS ON DISSOLUTION RATES: ORIGINAL MARRIAGES AND ALL SPELLS OVER FIRST 36 MONTHS OF THE EXPERIMENT

[Standard errors in parentheses]

	Blacks			Whites			Chicanos		
	$\hat{\alpha}$		exp $(\hat{\alpha})$	$\hat{\alpha}$		exp $(\hat{\alpha})$	$\hat{\alpha}$		exp $(\hat{\alpha})$
Original marriages only:									
5-year NIT treatment	[2].45		1.57	[1].43		1.53	.01		1.01
	(.21)		....	(.22)		....	(.34)		....
3-year treatment adjustment	−.30		.74	−.33		.72	−.00		1.00
	(.23)		....	(.24)		....	(.37)		....
Number of marriages	....	939	....	....	1,299	....	....	538	....

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[2] Approximately 32 percent of the experimental couples were assigned to the 5-year plans and 68 percent to the 3-year plans.

[3] For convenience we include the variables representing the counseling/training treatments in the vector *z*.

[4] "All marriages" includes marriages that began during the experiment.

[5] The equations estimated also included variables for the SIME/DIME counseling/training treatments. We have investigated whether the NIT effects reported here may be due to the interaction of the counseling/training and NIT treatments. There is no evidence that this is the case for blacks. The interaction of the NIT and counseling/training treatments is statistically significant for whites. We conclude that the interaction is probably due to sampling variability. The estimates are presented and discussed in appendix B.

TABLE 5.4.—ESTIMATED EFFECTS OF THE NIT TREATMENTS ON DISSOLUTION RATES: ORIGINAL MARRIAGES AND ALL SPELLS OVER FIRST 36 MONTHS OF THE EXPERIMENT—Continued

[Standard errors in parentheses]

	Blacks			Whites			Chicanos		
	$\hat{a}$		exp ($\hat{a}$)	$\hat{a}$		exp ($\hat{a}$)	$\hat{a}$		exp ($\hat{a}$)
All marriages									
5-year NIT treatment	[3].48		1.61	[2].45		1.56	.01		1.01
	(.18)			(.19)			(.28)		
3-year treatment adjustment	.31		.73	−.20		.82	.12		1.13
	(.21)			(.21)			(.31)		
Number of marriages		1,118			1,554			648	

[1] Significant at the 10-percent level

[2] Significant at the 5-percent level

[3] Significant at the 1-percent level

NOTE.—All equations contain the design variables, counseling/training treatment variables, and nonexperimental covariates discussed in the test. The estimated coefficients of these background variables for the original marriages analysis are reported in appendix A.

The effect of the 3-year treatment is roughly three quarters as large as the 5-year effect—the multiplier for the 3-year treatment dummy is close to .75 for both blacks and whites. The difference between 3-year and 5-year effects is not significant, however. The fact that the estimated effects of the 3-year treatments are smaller than the 5-year treatment effects has two implications. First, in a specification of the NIT treatments that used a single dummy variable, the estimated experimental-control difference would be smaller than the coefficients for the NIT variable in table 5.4. Second, if the longer treatment more closely approximated the effects of a permanent program, a permanent program would have even larger effects than the 5-year program. We return to this issue in chapter 11 when we discuss the problem of extrapolating from a limited duration experiment.

Findings for the Chicano sample differ considerably. Adjustments for initial distributions on design variables and demographic characteristics eliminate any suggestion of an experimental effect. The multiplier for the 3-year treatment is essentially unity.

Results in table 5.4 cover only the first 3 years of the experiment and ignore both the remaining 2 years of the experiment for the 5-year families and the post-experimental period for all families. By allowing the experimental effects to vary from period to period, we can estimate the experimental effects over the entire period of the experiment.[6] That is, we can estimate NIT treatment effects over 3

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[6] We have extensively analyzed the variation in the NIT treatment effects on marital dissolution over time (Groeneveld, Hannan, and Tuma, 1980; Tuma, Hannan, and Groeneveld, 1979, 1977). Two conclusions can be drawn from those studies. First, the experimental effect differed significantly between the experimental and postexperimental period for blacks and whites. Second, the magnitude of the NIT treatment effects generally declines over time for whites and blacks. The time dependence analysis is complicated by the time variation in the control dissolution rate. The statistical significance of the time dependence of the NIT treatment effects during the treatment period depends on how the control rates are parameterized (specifically, how the constant is allowed to vary over time in the equations) and on how the experimental period is subdivided.

There are several reasons why the treatment effects might vary over time. First, there may be an initial period of learning during which treatment effects are small as people become acquainted with the rules of the experiment and learn to "trust" the treatments. This suggests that the effects would be small initially and then increase. This was found only for black couples eligible for the 5-year treatment and, less clearly, for white 3-year couples. Second, the treatment effects may vary over time if people anticipate the ending of the experiment. One would expect that the effect of the NIT treatment on marital dissolution would be greater when several years of the treatment are available than when the treatment is expected to end in a few months. The decline in the magnitude of the treatment effects reported in Groeneveld, Hannan, and Tuma (1980) is consistent with this expectation. A third reason for time variation in the NIT treatment effects is unobserved differences in marital dissolution rates. Initially some marriages are more likely to end in dissolution than others. As time passes the more risky marriages are selected out through dissolutions and the overall level of risk in the sample declines. We have attempted to control for

Continued

years for couples eligible for the 3-year treatment and 5 years for 5-year couples. In addition, we can compare the dissolution rates during the experiment to those in the postexperimental period. If the dissolution rates for experimental and control families differ during the postexperimental period, then the NIT effects we have estimated may be due to preexperimental differences between the control and experimental groups rather than to the treatments. If, however, the experimental dissolution rates return to the level of the dissolution rates in the control group during the postexperimental period, then we have more confidence that the differences during the experimental period were caused by the NIT treatments.

We analyzed data for the entire observation period no matter how long it was. The only exception is that we censored the event histories of the 3-year families at the end of 5 years. Only a few 3-year families had more than 5 years of data, and by censoring their event histories in this way we avoided having to distinguish between 3- and 5-year families during the period beginning 5 years after enrollment. Records for both experimental and control 3-year families were censored in this way.

We used three time periods for this analysis: enrollment to 3 years later (period 1), 3 years after enrollment to 5 years after (period 2), and more than 5 years after enrollment (period 3). We estimated the following model:

$$r_p = \exp(z'\beta + \alpha_{3p}YR3 + \alpha_{5p}YR5) p = 1,2,3, \tag{3}$$

where r_p = period-specific marital dissolution rate
z = same vector of assignment and background variables included in the equations reported in Table 5.4
β = vector of time invariant coefficients
YR3, YR5 = dummy variables indicating eligibility for 3- and 5-year NIT treatments[7]
α_{3p}, α_{5p} = period-specific NIT treatment effect for 3- and 5-year couples.

Because we truncated the event histories for all 3-year couples at 5 years, $\alpha_{33} = 0$. In order to estimate the NIT treatment effect for 5-year couples over the entire period of their eligibility, we constrained α_{51} and α_{52} to be equal. Thus, the NIT effect for 3-year couples during the period of eligibility is α_{31}, and α_{32} is the effect during the posttreatment period. Similarly, α_{51} (which equals α_{52}) is the effect for 5-year couples while they are eligible for the NIT treatment, and α_{53} is the posttreatment effect for 5-year couples.

Because of the length of time covered by this analysis (up to 5 years for 3-year families and 7 years for 5-year families), we used all marital spells. The estimated treatment effects for the model defined in equation (2) are reported in table 5.5. The value of the chi-square statistic reported at the bottom of table 5.5 is for a comparison of the model in equation (2) with a model in which the ef-

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this heterogeneity in two ways: by including variables known to be related to marital dissolution rates in the equations, and by adding new marriages to the sample as they are formed. Even with these measures it is possible that dissolution rates would decline over time due to this heterogeneity. Our analysis produced mixed results in this regard. The dissolution rates for control couples decline over time for blacks (as expected), but for white controls the rates increase during the first 3 years and then decline.

The fourth reason for time variation in the NIT treatment effects is perhaps the most serious. The treatment may induce couples to split earlier than they would have in the absence of the experiment (economists refer to this as an intertemporal substitution effect). There is no change in the number of dissolutions that occur, only in the timing. This effect is present in a limited duration experiment but would not be present in a permanent program. The expected pattern of time dependence from this effect is an increase in marital dissolution rates among experimental couples during the treatment period, followed by a decrease to a level below that of comparable controls during the posttreatment period. Our analysis did not reveal such a pattern: the experimental marital dissolution rates returned to the control level during the posttreatment period but did not decline below the control rates.

[7] Note that the NIT treatment variables in equation (3) are different from those in equation (2). In particular, the coefficient of 3YRAD in equation (2) estimates the difference between the 5- and 3-year treatments while the coefficient of YR3 in equation (3) is the effect of the 3-year NIT treatment compared to controls.

fects of the 3- and 5-year treatments are constant over the entire observation period (that is, no distinction is made with respect to the period of eligibility for the treatments). The chi-square values are statistically significant for all three race-ethnic groups, indicating that the effects differed according to current eligibility for the treatment.

TABLE 5.5.—EFFECTS OF NIT TREATMENTS ON MARITAL DISSOLUTION RATES DURING THE EXPERIMENTAL AND POSTEXPERIMENTAL PERIODS: ALL MARRIAGES

	Blacks			Whites			Chicanos		
	α		exp(α)	α		exp(α)	α		exp(α)
3-year treatment:									
Experimental period	[2].38		[2]1.47	[2].34		[2]1.41	.18		1.19
	(.13)		....	(.12)		....	(.17)		....
Postexperimental period	.00		1.00	−.15		.86	−.25		.78
	(.19)		....	(.18)		....	(.25)		....
5-year treatment:									
Experimental period	[2].37		[2]1.44	[2].36		[2]1.43	.02		1.02
	(.14)		....	(.14)		....	(.20)		....
Postexperimental period	.05		1.05	.12		1.12	−.76		.47
	(.31)		(.26)	(.51)		....	....		....
χ^2 test of model versus constant treatment effects for across all three periods (2 df)		[1]5.14	....	....	[1]8.84	....	....	[1]5.76	....
Number of marriages		1,203	....	....	1,714	....	....	698	....

[1] Significant at the 10-percent level

[2] Significant at the 1-percent level

NOTE.—All equations contain the design variables, counseling/training treatment variables, and nonexperimental covariates discussed in the text.

For whites the NIT effect during the treatment period is positive and significant for both 3- and 5-year groups. During the postexperimental period the NIT effect does not differ significantly from the rate for comparable controls.[8]

For blacks the NIT effect during the treatment period is also positive and significant for both 3- and 5-year groups. The magnitude of the coefficients is similar to that of other estimates of the NIT effects. During the postexperimental period the estimated multipliers for both 3- and 5-year couples are close to 1.0, which indicates no NIT-control group difference in the posttreatment period. There are no significant experimental effects for Chicanos, in agreement with all previous findings. Estimated coefficients are close to 1.0 except for the 5-year posttreatment effect.

It is important to note the magnitude of the NIT effect for black and white 5-year families. For both whites and blacks the coefficients, which are estimated using 5 years of data, indicate that the NIT treatments increased the dissolution rates by over 40 percent. The magnitude of these coefficients, estimated over such a long time period, is perhaps the strongest evidence yet that the treatments increased marital dissolution rates. The fact that the posttreatment effects for both whites and blacks are close to 1.0 and not significant is strong evidence that the effects during the treatment period result from the treatment and are not artifacts of preexperimental differences between the NIT and control groups. If our findings were due to differences between the experimental and control group, these differences would have persisted over time and not have disappeared when eligibility for the experiment ends.

The fact that the posttreatment effects for whites and blacks are close to 1.0 is also evidence against a hypothesis that the observed treatment effect is due to a change in the timing of marriages and not to an increase in the number of dissolutions (an intertemporal substitution effect). Such a hypothesis holds that the treatment effect arises because couples who would eventually dissolve their marriage do so earlier to take advantage of the more generous income guarantees in order to finance a period of singleness and perhaps the search for a new spouse (this effect might be termed a "sale on marriages"). We discuss this hypothesis and the evidence against it at greater length in chapter 11.

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[8] The estimates in table 5.5 are not directly comparable to the estimates in table 5.4 because the effects for controls are averaged over 3 years in table 5.4 and over 5 or more years in table 5.5.

The analysis to this point reveals strong ethnic differences in responses. The NIT treatments significantly increased dissolution rates of black and white couples but did not affect the rate of Chicano couples. In the remainder of this chapter, we concentrate on results for the two samples in which we detect systematic effects.

What about differences among the treatments? In SIME/DIME there were 11 distinct combinations of the three guarantee levels and the four tax rates.[9] To investigate the pattern of effects across the 11 treatments, we replaced the NIT dummy variable used in the model reported in table 5.4 with 11 dummy variables, one for each combination of the guarantee levels and tax rates. For simplicity, we return to analyzing only the first 3 years of data. Note that the counseling/training treatment is still represented by a set of three dummy variables, as before.

Table 5.6 reports the estimated effects of the 11 guarantee/tax rate combinations. The multipliers of the specific treatments generally exceed unity, as expected. The effects are largest for treatments with the $4,800 guarantee in the black sample and for the $3,800 guarantee in the white sample. There seems to be a pattern of effects by guarantee levels, but there is no apparent pattern in the effects of the tax rates. This is especially notable in the case of the lowest tax rate, the 50 percent constant tax. This tax rate has the strongest destabilizing effect for blacks and whites when combined with the low and medium guarantee levels. However, when the 50 percent tax is combined with the high guarantee, it actually lowers the dissolution rate. Indeed, this is the only specific treatment that significantly decreased the dissolution rate for either blacks or whites.

What do we gain from considering such fine detail? From a statistical perspective a likelihood ratio test of the model with 11 NIT effects against the model with one NIT effect is insignificant at the 10-percent level for whites. That is, we cannot reject the null hypothesis that the effects of all 11 treatments are the same for whites. However, we do reject the hypothesis that all the effects are the same for blacks; the likelihood ratio test of the hypothesis is significant at the 1-percent level. Much of the improvement in fit seems to stem from the significant negative effect of the high guarantee 50-percent tax treatment (given that all the other significant effects are positive). But using 11 effects makes it difficult to understand the response—there is too much detail. Therefore, we shifted to a level of resolution that falls between the two extremes. We have examined effects of both guarantee levels and tax rates separately. We have always learned more from the former. Tax rates do not seem to have strong main effects. Thus we concentrate on the effects of guarantee levels.

Table 5.7 reports estimates of a model that replaces the 11 separate effects with three guarantee level effects. For blacks, significant effects occur for the $3,800 and $4,800 guarantee levels, with implied increases in the rate of 60 percent and 91 percent, respectively. Only the $3,800 guarantee has a significant effect for whites. The most striking feature of table 5.7 concerns the estimated effect of the most generous treatment. The $5,600 treatment has by far the smallest effect of the three for both blacks and whites. None of the guarantee level

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[9] Actually, there are 22 distinct treatments if we consider the length of the treatment. However, we have found that treatment length does not have any strong effect on the response, nor does it interact with the combinations of tax rates and guarantee levels.

TABLE 5.6 ESTIMATED EFFECTS OF THE 11 GUARANTEE/TAX RATE TREATMENTS ON DISSOLUTION RATES: ORIGINAL MARRIAGES OVER FIRST 36 MONTHS OF THE EXPERIMENT

[Standard errors in parentheses]

	Blacks		Whites		Chicanos	
	a	e^a	a	e^a	a	e^a
5-year treatments						
$3,800 50 percent	².655	1.92	³.735	2.09	.393	1.48
	(.314)		(.284)		(.477)	
$3,800 70 percent	.535	1.71	¹.644	1.90	².900	2.46
	(.374)		(.381)		(.439)	
$3,800 70 percent d [4]	.144	1.15	¹.610	1.84	.132	1.14
	(.369)		(.334)		(.510)	
$3,800 80 percent d [4]	¹.663	1.94	.468	1.60	–.551	.576
	(.401)		(.389)		(.598)	
$4,800 50 percent	¹.597	1.82	.5620	1.68	–.271	.763
	(.324)		(.327)	(.506)		
$4,800 70 percent	².616	1.85	¹.570	1.77	–.079	.924
	(.313)		(.311)		(.573)	
$4,800 70 percent d [4]	².613	1.85	¹.592	1.81	.208	1.23
	(.349)		(.341)		(.527)	
$4,800 80 percent d [4]	².776	2.17	–.283	.754	.122	1.13
	(.324)		(.486)		(.464)	
$5,600 50 percent	²–2.22	.109	–.604	.547	–.875	.417
	(1.02)		(.735)		(.801)	
$5,600 70 percent d [4]	³.933	2.54	.382	1.46	–.288	.749
	(.352)		(.375)		(.576)	
$5,600 80 percent d [4]	.323	1.38	.231	1.26	–.169	.844
	(.329)		(.396)		(.504)	
3-year treatment adjustment	.323	.724	¹–.408	.665	–.023	.977
	(.237)		(.244)		(.382)	
Number of spells	939		1,299		538	
χ^2	116.51 (31 df)		150.28 (31 df)		57.07 (30 df)	
Likehood ratio χ^2 test versus model in table 5.4	³25.11 (10 df)		9.8 (10 df)		134.7 (10 df)	

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.
[4] d = declining rate.

effects is significant for Chicanos. The low guarantee level effect increases the dissolution rate and the middle and high guarantee level effects decrease the rate.[10]

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[10] Our findings about the effects of program generosity are fairly robust, especially for whites and Chicanos. In the black sample it appears that stratified assignment to treatments contributed to the pattern of responses by guarantee level (Groeneveld, Tuma, and Hannan, 1980b).

TABLE 5.7.—ESTIMATED EFFECTS OF THE GUARANTEE LEVELS ON DISSOLUTION RATES: ORIGINAL MARRIAGES OVER FIRST 36 MONTHS OF THE EXPERIMENT

[Standard errors in parentheses]

	Blacks			Whites			Chicanos		
	$\hat{\alpha}$		$\exp(\hat{\alpha})$	$\hat{\alpha}$		$\exp(\hat{\alpha})$	$\hat{\alpha}$		$\exp(\hat{\alpha})$
$3,800 guarantee	[1].47		[1] 1.60	[2].60		[2] 1.82	.26		1.30
	(.25)			(.24)			(.37)		
$4,800 guarantee	[3].65		[3] 1.91	.40		1.49	−.05		.95
	(.23)			(.24)			(.37)		
$5,600 guarantee	.18		1.20	.13		1.14	−.42		.66
	(.27)			(.30)			(.43)		
3-year treatment adjustment	−.34		.71	−.33		.72	.05		1.05
	(.23)			(.24)			(.37)		
Number of spells		939			1,299			538	
χ^2 test of model versus model in table 5.4 (2 d.f.)		4.04			3.41			4.26	

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

We think this pattern presents an interesting paradox. The 3,800 guarantee is close to the poverty line and to the financial guarantees in the welfare systems available to families in the control group (AFDC and Food Stamps). Nonetheless, this guarantee level significantly increased the rate for both groups. At the same time, the treatment that offers the highest level of income support has much smaller effects. Indeed, we cannot reject the null hypothesis that the $5,600 treatment has no effect at the 10-percent level. We have paid a great deal of attention to this nonmonotonic pattern of effects, which we sometimes have called the "paradox of the guarantee level effects."

Although we attach considerable importance to these estimates, the model with three guarantee levels does not improve significantly over the model with a single NIT effect. According to the likelihood ratio tests at the bottom of table 5.7, we cannot reject the hypothesis that the effects of all three guarantee levels are the same. Moreover, for blacks the model with three guarantee level effects fits significantly less well than the model with 11 financial treatments. Statistical significance and possible substantive significance appear to disagree. From a strict hypothesis-testing perspective, we should work henceforth with the simple one parameter model for whites and the 11 parameter model for blacks. But the pattern in the estimates suggests that adopting this strategy might foreclose an opportunity to learn about the forces producing the overall effect. Therefore, we have chosen to accept the guarantee level effects as a meaningful baseline for further analysis.

Results reported in this section are our best estimates of the total effects of the 5-year NIT treatments on dissolution rates. Our answer to the question "Did the treatments affect the rate of dissolution?" is clear. The 5-year treatments significantly increased rates of marital dissolution among whites and blacks, but did not affect the dissolution rate among Chicanos.

Family composition and the NIT treatment effects

Results presented in the previous sections have relied heavily upon the logic of the experimental design: we estimated experimental-control differences while controlling for the assignment to the treatments and other factors known to affect marital dissolution rates. The estimated effects have been the effects in the entire experimental sample which included a variety of family types. About 90 percent of the couples had one or more children at enrollment (see chapter 3, table 5.1) and the majority had one or more children less than 5 years old (52 percent of the blacks, 58 percent of the whites, and 64 percent of the Chicanos).

There are several reasons for considering how the NIT treatment effects on marital dissolution vary by family composition. First, the process of marital dissolution may vary by family composition. The presence of children has been shown to be a stabilizing influence on marriages (Bumpass and Sweet, 1972), although Cherlin (1977) found that the stabilizing influence occurred only when children were of preschool age. If the presence and the age of children affects the stability of marriages, it is possible that the effect of the treatments may depend on the presence or age of children.

The second reason for considering how the treatment effects may vary with family composition is that the policy implications of the results may differ depending on family composition. The current welfare system treats childless couples differently than couples with children. The implications of the effects on families with young children may be different than the effects on families with school-aged children. Therefore, we investigated how the experimental effects differed according to whether children were present and whether there were preschool-aged children in the family.

Presence of children and the treatment effects

To determine whether the treatment effects we observed differed for childless couples and couples with children, we interacted a dummy variable indicating childlessness with the treatment variable (or variables). The analysis focuses on the first 3 years of the experiment and is limited to couples who were married at enrollment. The equations estimated are similar to those reported in tables 5.4 and 5.7 except for the childlessness variables. The NIT treatment is represented in two ways: with a 5-year NIT treatment variable and a 3-year treatment adjustment variable, and with three guarantee level variables (5-year treatment) and a 3-year treatment adjustment variable. The results are presented in table 5.8.

TABLE 5.8.—ESTIMATED EFFECTS OF NIT TREATMENTS ON MARITAL DISSOLUTION AMONG COUPLES WITH CHILDREN AND CHILDLESS COUPLES: ORIGINAL MARRIAGES OVER FIRST 36 MONTHS OF THE EXPERIMENT

[Standard errors in parentheses]

	Blacks			Whites			Chicanos		
	$\hat{\alpha}$		exp $(\hat{\alpha})$	$\hat{\alpha}$		exp $(\hat{\alpha})$	$\hat{\alpha}$		exp $(\hat{\alpha})$
5-year NIT treatment	[2].51		1.67	.30		1.36	.06		1.06
	(.21)		....	(.23)		....	(.35)		....
5-year NIT treatment[1] no children	−.72		.49	[1].89		2.44	−.19		.82
	(.48)		....	(.47)		....	(1.25)		....
No children	.57		1.77	[2] −.96		.38	−.76		.47
	(.38)		....	(.45)		....	(1.09)		....
3-year treatment adjustment	−.30		.74	−.30		.74	−.05		.95
	(.23)		....	(.24)		....	(.38)		....
χ^2 test for no children interaction (1 d.f.)	2.22		....	[2] 3.95		....	.02		....
$3,800 guarantee	[2].57		1.76	[1].46		1.59	.27		1.31
	(.25)		....	(.25)		....	(.38)		....
$3,800 guarantee[1] no children	−.15		.22	.86		2.36	.26		1.29
	(1.07)		....	(.55)		....	(1.46)		....
$4,800 guarantee	[3].68		1.98	.33		1.39	−.00		.99
	(.24)		....	(.25)		....	(.38)		....
$4,800 guarantee[1] no children	−.30		.74	.22		1.25	−.23		.80
	(.55)		....	(.66)		....	(1.45)		....
$5,600 guarantee	.28		1.32	−.25		.78	−.37		.69
	(.28)		....	(.34)		....	(.43)		....

TABLE 5.8.—ESTIMATED EFFECTS OF NIT TREATMENTS ON MARITAL DISSOLUTION AMONG COUPLES WITH CHILDREN AND CHILDLESS COUPLES: ORIGINAL MARRIAGES OVER FIRST 36 MONTHS OF THE EXPERIMENT—Continued

[Standard errors in parentheses]

	Blacks			Whites			Chicanos		
	$\hat{a}$		exp $(\hat{a})$	$\hat{a}$		exp $(\hat{a})$	$\hat{a}$		exp $(\hat{a})$
$5,600 guarantee[1] no children	−1.08		.34	[3] 1.92		6.81	−8.20		.00
	(.82)		....	(.62)		....	(88.8)		....
No children	.59		1.81	[2] −1.00		.37	−.70		.49
	(.38)		....	(.45)		....	(1.09)		....
3-year treatment adjustment	−.36		.70	−.27		.76	.03		1.03
	(.24)		....	(.24)		....	(.38)		....
χ^2 test for no children interactions (3 d.f.)	4.00		....	[2] 10.16		....	.93		....
Number of cases	....	939	....	....	1,299	....	....	538	....

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

The interaction of the childless variable with the NIT 5-year treatment is not significant for blacks whether the single treatment variable is interacted or the guarantee level variables. The interaction terms are all negative, indicating that the experiment stabilized marriages of childless couples relative to couples with children. Notice that for control couples being childless destabilized marriages (as expected). Thus the interaction indicates a stabilizing effect of the NIT for childless black couples relative to experimentals with children and childless controls. However since the interaction coefficients are not significant these conclusions can only be suggestive.

The interaction of childlessness with the treatment variables has a different pattern for whites.[11] The chi-square tests for the interactions are significant indicating that the NIT treatment effects differ for childless couples and couples with children. When childlessness is interacted with the single NIT treatment variable, the coefficient of the treatment variable is positive but not significant.[12] The coefficient of the interaction term is positive and significant. The NIT treatment increased the dissolution rate much more among white childless couples than among white couples with children. This finding is of great potential importance: if the treatment effect observed for whites is mainly an effect on marital dissolution of childless couples, one cannot generalize from SIME/DIME to a national program that excludes childless couples without taking this interaction into account.

We can see how this interaction between childlessness and the NIT treatment varies with the generosity of treatment by examining the guarantee level interactions. The three interaction terms are positive for white couples, indicating that NIT treatments have greater effects for white childless couples than for white couples with children across the three guarantee levels. However, the difference between childless couples and couples with children is significant only for the $5,600 guarantee level. For this highest guarantee level the NIT treatment was slightly stabilizing for couples with children and destabilizing for couples without children (the multiplier of 6.81 indicates an increase in the dissolution rate almost sevenfold compared to childless control couples)—an effect that is only slightly greater for white childless couples receiving the $4,800 guarantee level treatments than for white couples with children.

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[11] The childlessness coefficient is negative for whites when the treatment interactions are included in the equations. However, the coefficient has the expected positive sign and is not significant in equations without the interactions.

[12] In chapter VIII, where this analysis is replicated using a measure of "current" childlessness, the estimated effect is significant.

Although the effect of the $3,800 guarantee level treatments is also greater for white childless couples, the treatment effect for couples with children is also significantly different from zero. This is particularly important because the low guarantee level treatments are the most like the support levels of current welfare system and proposed program reforms. This suggests that a program similar to SIME/DIME with guarantee levels similar to those recently proposed would increase marital dissolution among white couples with children. We discuss this implication at greater length in Chapter 11. The NIT treatments had no significant effects among Chicano couples whether they are childless or have children.

The presence of young children and the treatment effects

Another important aspect of family composition is the age of the youngest child in the family. We interacted the NIT treatments with a dummy variable indicating that the family had one or more children under age 5 living with them at enrollment. These interactions indicate how the treatment effect depends on whether young children are present or not. The results are reported in table 5.9.

TABLE 5.9.—ESTIMATED EFFECTS OF NIT TREATMENTS ON MARITAL DISSOLUTION AMONG COUPLES WITH AND WITHOUT YOUNG CHILDREN: ORIGINAL MARRIAGES OVER FIRST 36 MONTHS OF THE EXPERIMENT

[Standard errors in parentheses]

	Blacks		Whites		Chicanos	
	$\hat{a}$	exp ($\hat{a}$)	$\hat{a}$	exp ($\hat{a}$)	$\hat{a}$	exp ($\hat{a}$)
A. 5-year NIT treatment	.07	1.07	[2].59	1.80	.22	1.25
	(.26)		(.28)		(.49)	
5-year NIT treatment [1] young children	[2].63	1.88	−.29	.75	−.23	.79
	(.28)		(.29)		(.46)	
Young children	[3] −.63	.53	−.24	.78	−.49	.63
	(.24)		(.26)		(.45)	
3-year treatment adjustment	−.29	.75	−.30	.74	−.06	.94
	(.23)		(.24)		(.38)	
χ^2 test for young children interactions (1 d.f.)	[2] 4.95		1.01		.25	
B. $3,800 guarantee	.18	1.19	[1].63	1.88	.20	1.22
	(.37)		(.33)		(.57)	
$3,800 guarantee [1] young children	.51	1.66	−.09	.91	.09	1.10
	(.40)		(.35)		(.56)	
$4,800 guarantee	.31	1.36	.48	1.62	−.14	.87
	(.31)		(.34)		(.64)	
$4,800 guarantee [1] young children	[1].59	1.80	−.17	.84	.17	1.18
	(.34)		(.37)		(.63)	
$5,600 guarantee	−.36	.70	[1].70	2.01	.48	1.61
	(.40)		(.37)		(.59)	
$5,600 guarantee [1] young children	[1].89	2.44	[2] − 1.11	.33	[2] − 1.32	.27
	(.47)		(.50)		(.66)	
Young children	[2] −.61	.54	−.25	.78	−.44	.65
	(.24)		(.26)		(.45)	
3-year treatment adjustment	−.35	.71	−.29	.75	.00	1.00
	(.23)		(.24)		(.38)	
χ^2 test for young children interactions (3 d.f.)	5.44		5.25		5.38	
Number of cases		939		1,299		538

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

The NIT effect differs for blacks significantly between couples with young children and other couples.[13] The treatment has no significant effect among black couples except for those with young children. The treatment multiplier for black couples with young children is 2.01 (1.07 times 1.88). This pattern holds in the equation using the three guarantee level variables, although the interaction variables are not jointly significant. The difference between couples with young children and other couples is significant for the $4,800 and $5,600 guarantee levels. The guarantee level multipliers for black families with young children are 1.97 (1.19 times 1.66) for the $3,800 guarantee, 2.45 for the $4,800 guarantee, and 1.71 for the $5,600 guarantee. Thus, both models in table 5.9 indicate that the increase in marital dissolution rates caused by the NIT treatment among black couples is primarily an effect on couples with young children.

Interactions show a different pattern for white couples. The NIT effect does not differ significantly between couples with young children and other couples when the single NIT interaction variable is used. The multiplier for couples with young children is 1.35 compared to the multiplier of 1.80 for other couples. When the guarantee levels and interactions are used, the interactions differ by guarantee level. The $3,800 guarantee level significantly increases the dissolution rates for all couples and the effect is slightly, but not significantly, smaller for couples with young children. The multiplier for couples with young children is 1.71. Effects for the $4,800 guarantee level are similar to the $3,800 guarantee level effects but smaller, and neither coefficient is significant. The multiplier for couples with young children is 1.36. The effect of the $5,600 guarantee level is significant and positive for couples without young children. But this treatment actually stabilizes marriages of white couples with young children. The NIT effect at the high support level on the marital dissolution rate is a multiplier of 2.01 (a 101-percent increase) for couples without young children and .66 (a 34-percent decrease) for other couples.

Comparison with the results of the New Jersey NIT experiment

There have been two other major urban negative income tax (NIT) experiments—in New Jersey and Gary—in which efforts were made to determine the effects of the NIT treatments on marital dissolution. (See Chapter 2 for a review of the findings of the other experiments.) Sawhill et al. (1975) found that the NIT treatments significantly increased marital dissolution rates in New Jersey, whereas Wolf (1976) concluded that there was no effect. No significant effects were found in Gary (Henry, 1975; Wolf, 1979).

In this section we compare the effects of the New Jersey Experiment on marital dissolution with our findings from SIME/DIME. In our analysis of the Gary data prepared by Mathematica Policy Research (Citro, 1980), we found no significant experimental efect (Groeneveld et al., 1980). But in that analysis we found significant problems in the way marital events were defined in the public use file, which led us to be skeptical about the results. The public use data file had fewer dissolutions than either Henry or Wolf found and actually had more marriages end in death than in dissolution. Therefore we ignore Gary in this report.

New Jersey had less generous treatments and lower tax rates than SIME/DIME. However, the treatments tested in the two experiments overlap considerable. New Jersey used four guarantee levels: 50 percent, 75 percent, 100 percent, and 125 percent of the poverty level. The three SIME/DIME guarantee levels were approximately 90 percent, 110 percent, and 125 percent of the poverty level. (See Kershaw and Fair, 1976, for a comparison of the experimental treatments.) In addition, the New Jersey Experiment has no counseling/training treatment. Appendix C contains a comparison of the experiments and a description of the data used in the anlaysis.

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[13] Other couples includes childless couples in order to maintain comparability with the effects reported in tables 5.4 and 5.7.

Effects of the New Jersey NIT experiment on marital dissolution

We began by estimating the New Jersey NIT treatments' effects on marital dissolution and comparing these results to the findings of earlier investigators and to our findings for SIME/DIME. In our analysis of the New Jersey data, we included variables for the age and education of both the husband and the wife, the number of children in the family, the presence of any child younger than 5 years, and the family's report of any welfare income in the year before the experiment. We also included the variables used to assign families to the treatments, and variables for the specific New Jersey city and for the poverty level. We represented the NIT treatments in two ways: model 1 includes a single dummy variable for the NIT treatment; model 2 includes separate dummy variables for the guarantee levels.

Table 5.10 presents the estimated NIT effects in New Jersey by race-ethnicity. The NIT variable is significant for black and Spanish-speaking couples. The NIT treatments thus increased the marital dissolution rate by 52 percent for blacks and by 116 percent for Spanish-speaking couples. The effect for whites is also positive, but it is not significant.

TABLE 5.10.—ESTIMATED EFFECTS OF THE NEW JERSEY NIT EXPERIMENT ON MARITAL DISSOLUTION

	Blacks	Whites	Spanish-speaking
Model 1: NIT treatment	[1] 1.52	1.22	[3] 2.16
Model 2:			
50 percent guarantee	[1] 1.80	[2] 2.71	[2] 2.63
75 percent guarantee	1.06	1.29	[3] 2.55
100 percent guarantee	[1] 1.79	.83	1.67
125 percent guarantee	[2] 2.20	.84	1.38
Likelihood ratio χ^2 test for guarantee levels—Model 2 versus Model 1 (3 d.f.)	4.07	4.83	2.15
Number of cases	443	415	392
Number of events	84	52	60

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

We also estimated the effects of the guarantee levels (model 2). For whites the lowest guarantee level has a large significant effect, increasing that dissolution rate by 171 percent. All four guarantee levels have positive effects for blacks, and all but the 75 percent guarantee level have significant effects. The greatest effect for blacks is for the highest guarantee level, but the likelihood ratio test for the differences in the guarantee level variables is not significant. The guarantee level effects for Spanish-speaking couples are all positive, and the effects of the 50 percent and 75 percent guarantee levels are significant.

Our findings agree with those of Sawhill et al. (1975) that the New Jersey NIT treatments increased marital dissolution rates for black and Spanish-speaking couples but not for white couples. The results in table 5.10 are also consistent with our findings that SIME/DIME increased dissolution rates; however in SIME/DIME the rates increased for whites and blacks but not for Chicanos.

Combining the New Jersey and SIME/DIME samples

Our next step was to combine the samples from the two experiments. In this combined analysis, we represent the experimental treatment in two ways: as a single dummy variable that indicates eligibility for the NIT treatment in either experiment (model 1), and as two dummy variables, one for the NIT treatment in New Jersey and one for the NIT treatment in Seattle/Denver (Model 2).[14] Thus, the representation of the SIME/DIME NIT treatment combines the 3- and 5-year treatments rather than representing the treatments separately as was done in our other analyses. We report the estimated NIT treatment effects on the marital dissolution rates in table 5.11. In the combined sample the NIT treatments significantly increased marital dissolution rates for all three race-ethnic groups. For blacks the separate NIT variables for the two experiments are both significant, but for whites the effect is significant only in Seattle/Denver. The likelihood ratio test indicates that the treatment effect do not differ significantly in the two sites for either blacks or whites. The NIT also significantly increased the dissolution rate in the combined sample for Hispanic couples. When we allow separate effects for the two experiments in the combined sample, only the New Jersey treatment variable is significant. The likelihood ratio test indicates that the NIT treatment effect differs significantly between the two experiments.[15]

TABLE 5.11.—ESTIMATED EFFECTS OF THE NEW JERSEY AND SEATTLE/DENVER EXPERIMENTS ON MARITAL DISSOLUTION: COMBINED SAMPLES

[Standard errors in parentheses]

	Blacks		Whites		Hispanics	
	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$
Model 1:						
NIT treatment	[3].32	[3] 1.37	[1].22	[1] 1.24	[1].29	[1] 1.33
	(.12)		(.13)		(.17)	
Model 2:						
New Jersey NIT treatment	[1].39	[1] 1.48	.14	1.15	[3].78	[3] 2.17
	(.23)		(.29)		(.29)	
SIME/DIME NIT treatment	[2].29	[2] 1.34	[1].24	[1] 1.27	.00	1.00
	(.14)		(.14)		(.21)	
Likelihood ratio tests:						
For separate treatment variables in combined sample (1 d.f.)		.13		.08		[2] 4.80
For combining samples (6 d.f.)		10.53		9.25		8.36
Number of cases		1,382		1,714		930
Number of events		313		280		172

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Results reported in table 5.11 strengthen our confidence that the NIT treatments increased the marital dissolution rates in both the New Jersey and the Seattle/Denver Experiments. While the pattern of effects by ethnic group differs between the two experiments, the process appears to be similar in the two experiments since the effects of the background variables are similar. Differences between the designs of the experiments do not seem to have led to differences in the experimental effects on dissolution rates.

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[14] The background variables are the same as in the separate analyses: husband's and wife's age and education, number of children, and dummy variables for the presence of children 5 or younger, and for receipt of welfare payments in the year before the experiment. The assignment variables for both experiments are also included: New Jersey poverty level variables, Seattle/Denver normal income variables, and dummy variables for the individual cities.

[15] The second likelihood ratio test in Table 5.11 tests differences in the effects of the background variables in the equations. The test is not significant for any of the race-ethnic groups, indicating that effects of the background variables in the two experiments were similar.

CHAPTER VI

POTENTIAL SOURCES OF BIAS

Analyses presented in chapter 5 indicate that the SIME/DIME NIT treatments induced significantly higher rates of marital dissolution among black and white couples. Three major factors, however, may bias these estimates: attrition, reconciliations, and fraud. In this chapter we describe how these three factors might bias our estimates and present analyses that address the amount of bias. In no case do we find evidence that qualitatively alters our conclusions.

Sensitivity of SIME/DIME dissolution rates to attrition

Some women originally enrolled in the experiment did not participate for the complete experimental period. In this section, we investigate the potential bias in our results that may arise if attrition from the experiment is related both to our dependent variable (marital dissolution) and the independent variables (experimental treatments). First, we describe the amount of attrition and the relationship of attrition to the experimental treatments. Second, we discuss the possible consequences for our results if attrition is related to both experimental treatments and changes in marital status. Third, we present an analysis designed to test the sensitivity of our results to several hypotheses about the associations among attrition, marital status change, and experimental treatment.

Attrition from the sample

Throughout our analyses we have defined attrition as a premature termination of a scheduled interview sequence.[1] Every control and experimental family was assigned a length of treatment of either 3 or 5 years. Families assigned to the 3-year treatment were to be interviewed for at least 4 years and families assigned to the 5-year treatment were to be interviewed for at least 6 years. We always take the date of the last completed interview as the end of the period of observation of an individual. An individual is considered an attrition if he/she was not observed for at least 4 years if assigned to a 3-year treatment or 6 years if assigned to a 5-year treatment.[2]

Table 5.12 gives the percentage of originally enrolled wives who attrited before the end of the postexperimental year. The percentages are given separately by race-ethnicity and length of treatment. In general, blacks attrited more frequently than whites or Chicanos, and controls attrited more frequently than women assigned to one of the financial treatments. The only exception is for blacks assigned to the experiment for 5 years: black women eligible for the 5-year NIT treatment attrited more than the controls. Table 5.12 does not allow a comparison of attrition rates for 3- and 5-year treatments, since the observation time differs for the two lengths of treatment groups. We have examined attrition during the first 3 years of the experiment, however, and found that 5-year women attrited less frequently than 3-year women.

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[1] We do not treat death of heads of households as attrition. When the male head dies, we end the record for the marriage without recording a dissolution and place the female head in the single state. When the female head dies, the entire observation is ended.

[2] Control couples were assigned for 3 to 5 years. This assignment affected only the length of the interviewing period and was not a "treatment."

TABLE 5.12.—PERCENTAGE OF ORIGINALLY ENROLLED WIVES WHO ATTRITED BEFORE THE END OF THE POSTEXPERIMENTAL YEAR

[Number of cases in parentheses]

	2-year NIT treatment	3-year controls	5-year NIT treatment	5-year controls	Total
Blacks	21.6	31.2	35.7	27.6	27.9
	(333)	(250)	(171)	(185)	(939)
Whites	16.3	24.9	19.8	17.0	19.5
	(479)	(373)	(212)	(235)	(1,299)
Chicanos	16.4	26.1	20.0	25.8	20.6
	(238)	(111)	(100)	(89)	(538)

NOTE.—Attrition is defined as having less than 4 years of data for women assigned to 3-year treatments or less than 6 years of data for women assigned to 5-year treatments. The 31 women who died during the experiment are not counted as attritions but are included in the total number of cases.

We conducted an analysis to determine whether the attrition rates for experimentals and controls were significantly different when we controlled for the background variables included in our marital dissolution equations. Details of that analysis are reported in Groeneveld, Hannan, and Tuma (1980). The results of that analysis confirm the findings in table 5.12 that attrition rates are lower for experimentals than for controls. The experimental-control difference was significant only for black and Chicano 3-year couples. It was not significant for white 3-year couples or for the 5-year couples of any of the three race-ethnic groups.

The problem of attrition

There are reasons to suspect that attrition is affected by marital status changes. Reports from our interviewers indicate that refusals to continue in the study were sometimes associated with a recent marital status change.[3] That is, female heads who married and couples that separated resisted our attempts to complete interviews more than others. Also, marital status changes may lead to residential moving that makes it difficult to locate the individuals involved. Unfortunately, we cannot observe the effects of marital status changes on attrition. Precisely those marital status changes that lead to attrition before the next interview are unobserved by definition.

TABLE 5.13.—HYPOTHETICAL EXAMPLE OF BIAS DUE TO NONRANDOM ATTRITION

	No marital dissolution		Marital dissolution		Total
	Remain	Dropout	Remain	Dropout	
Experiments	855	45	95	5	1,000
Controls	810	90	60	40	1,000
Total	1,665	135	155	45	2,000

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[3] The marital status changes were observed in these cases, hence attrition is not a problem. However, the fact that some marital status changes preceded attrition suggests that marital status changes may also have been the reason for attrition in some cases where the reason for attrition was not determined. It is these cases that pose the problem.

	True proportion having a marital status change	Observed proportion having a marital dissolution
Experimentals	100/1,000=.10	95/950=.10
Controls	100/1,000=.10	60/870=.07

Whenever attrition is related to both independent and dependent variables, analysis of the surviving sample gives misleading results. Consider the hypothetical (intentionally exaggerated) example in table 5.13. The two samples—experimental and control—have identical proportions experiencing a marital dissolution: 0.10. However, attrition is related additively to both experimental status (controls have higher rates) and to marital status changes (those who change have higher rates). Attrition is also related nonadditively to the same two variables: it is disproportionately high for controls who change their marital status. Although the true proportions are the same in the two populations, the observed proportions in the remaining samples differ substantially. The observed proportion for the controls is 0.03 less than that for experimentals. Even in less extreme cases, the differences between controls and experimentals can be substantial. If everything were the same as the hypothetical example in table 5.13, except that 30 instead of 40 of the controls who had a marital dissolution dropped out, the observed rate of marital status change for controls would be 0.08. If only 20 of the controls who change their marital status dropped out the observed rate would be 0.09 instead of the true rate of 0.10.

Any nonrandom attrition of this type biases the estimates of experimental impacts on marital dissolution rates. If the relationship between marital dissolution and attrition in our sample is similar to this example, we have underestimated the true rate of marital status change among the controls. Thus, the experimental effects reported may be due to underestimating the rates for controls because of attrition rather than to a true experimental impact on marital dissolution.[4]

Sensitivity Analysis

Our primary concern is to determine how attrition from SIME/DIME might bias our estimates of the experimental effects on marital dissolution. To do this we conducted a sensitivity analysis in which we reestimated our marital dissolution equations under several hypotheses about the relationship between attrition and marital dissolution. Our analysis is similar to that reported in an earlier report of results (Hannan, Tuma, and Groeneveld, 1976), but we use a different and statistically more defensible methodology.

We constructed artificial marital histories for all cases that ended in attrition before the end of 3 years. The assumptions about the relationship between attrition and marital dissolution rates depended on whether the woman was an experimental or a control. In constructing these marital histories for experimentals we assumed that the rate of marital status change during the period following attrition was the same as the rate estimated during the observation period. For controls we assumed that the marital dissolution rate increased by a constant multiplier, α, in the period following attrition. We reconstructed the data for values of α equal to 1, 2, 3, 4, 5, and 10. We did not change the data for cases that did not end in attrition.

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[4] Evidence from another of our studies suggests that inflation of the relative dissolution rates among experimental couples due to attrition is not a great problem. We searched court records in both Seattle and Denver for records of divorces for all couples who dropped out during the first 3 years of the experiment. Fifteen percent of the control couples who dropped out and 14 percent of the financial couples had records of divorce proceedings (Groeneveld, 1979). Attriting couples eligible for the high guarantee level programs, who had the most to gain from remaining in the experiment, had more divorces recorded than controls.

Our method is a severe test of attrition bias in that the multipliers we use in reconstructing the data are large. The least extreme hypothesis, that the dissolution rate among controls who drop out is double the rate of controls who remain ($\alpha = 2$), is probably unrealistically high. Of course it is impossible to know precisely what the appropriate value of α is.[5]

For each spell that ended in attrition we computed the estimated dissolution rate using the estimates from the following model:

$$r = BX + \gamma F + \delta Y,$$

where

$F, Y =$ dummy variables for NIT treatment and 3-year treatment
$Y =$ vector of background variables
$B, \gamma, \delta =$ estimated coefficients.

We then randomly selected a time of the next marital status change for that spell from a distribution of ending times using the technique described by Tuma and Hannan (1978). The time to the next marital dissolution is defined by the following formula:

$$u_i = \frac{-\ln v_i}{\alpha r_i},$$

where
$u_i =$ time to next marital event
$v_i =$ random number from uniform (0,1) distribution
$r_i =$ estimated rate
$\alpha =$ multiplier of rate used to implement our hypotheses about relationship between attrition and dissolution rates.

We computed u_i for every spell ending in attrition, and added that to the observed time of attrition to obtain a new ending time for the spell. For experimentals α always equals 1, for controls α took on values of 1, 2, 3, 4, 5, and 10. If the new ending time was less than 3 years after the start of the experiment, we recorded a marital dissolution for that spell. If the new ending time was greater than 3 years after enrollment, we censored the observation period at 3 years for that spell, just as all nonattrition spells are censored. This was done separately for each of the six values of α. We reestimated our marital dissolution equations using the reconstructed spells for attriters.

Estimates are reported in table 5.14 for two representations of the NIT treatment. Model 1, which uses a dummy variable for the NIT treatment and another dummy variable for the 3-year treatment, is the same as the model reported in table 5.4. Model 2 uses separate dummy variables for the three NIT guarantee levels and is the same as the model reported in table 5.7. All spells of marriage were included in this sensitivity analysis, whether the marriage existed at enrollment or was formed during the experiment. This was done to include the largest number of attritions possible in the analysis. The first column of estimates in table 5.14 are for the observed data: the ending time of attrition spells are unchanged. The next six columns report the estimates using the reconstructed marital histories for attriters. Each column is for a different value of α (which only affects control attriters), and as α increases the hypothesized relationship between attrition and marital dissolution for controls becomes stronger. The number of dissolutions in the first column is the number actually observed. The number of dissolutions in the next six columns is the number observed plus the number of dissolutions created in our reconstruction of postattrition marital histories.

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[5] A more severe test could have been used in which attrition was assumed to have opposite effects among experimentals and controls. The argument would be that while control attriters are *more* likely to have dissolutions than control stayers, experimental attriters are *less* likely to have dissolutions than experimental stayers. This would occur because experimentals who have a marital dissolution may gain economically by remaining in the experiment. We did not run such a test. However, the implication of this argument is the same as that of the test we did run: marital dissolutions are undercounted for controls relative to experimentals. The test reported here does measure the sensitivity of the results to such an undercount.

TABLE 5.14.—SENSITIVITY ANALYSIS OF EXPERIMENTAL EFFECTS ON MARITAL DISSOLUTION DURING THE FIRST 3 YEARS: ALL SPELLS OF MARRIAGE

[Entries are multipliers of the control rate]

	Observed data	Values of α					
		1	2	3	4	5	10
Blacks (1,118 spells)							
Model 1:							
Financial treatment	[3]1.61	[2]1.54	[1]1.39	[1]1.33	1.28	1.24	1.10
3-year treatment adjustment	.73	.78	.79	.76	.73	.73	.73
Model 2:							
$3,800 guarantee	[1]1.52	[1]1.46	1.33	1.29	1.23	1.19	1.06
$4,800 guarantee	[3]1.90	[3]1.72	[2]1.55	[2]1.47	[1]1.42	[1]1.37	1.22
$5,600 guarantee	1.36	1.38	1.23	1.67	1.13	1.10	.97
3-year treatment adjustment	.72	.78	.79	.76	.73	.73	.72
Number of dissolutions	292	320	331	339	347	351	367
Whites (1,554 spells)							
Model 1:							
Financial treatment	[1]1.56	[2]1.61	[2]1.44	1.33	1.29	1.28	1.14
3-year treatment adjustment	.82	.84	.85	.86	.85	.85	.84
Model 2:							
$3,800 guarantee	[3]1.80	[3]1.82	[3]1.60	[1]1.47	[1]1.43	[1]1.43	1.26
$4,800 guarantee	[2]1.50	[2]1.57	[1]1.41	1.30	1.27	1.26	1.12
$5,600 guarantee	1.25	1.31	1.19	1.11	1.07	1.06	.94
3-year treatment adjustment	.82	.84	.85	.86	.86	.85	.85
Number of dissolutions	306	323	335	343	347	349	366
Chicanos (648 spells)							
Model 1:							
Financial treatment	1.01	.90	.86	.85	.83	.78	[2].57
3-year treatment adjustment	1.13	1.21	1.20	1.19	1.17	1.19	1.47
Model 2:							
$3,800 guarantee	1.25	1.05	1.00	1.00	.98	.91	.69
$4,800 guarantee	1.02	.92	.87	.85	.83	.78	[2].57
$5,600 guarantee	.62	.63	.60	.59	.57	[1].53	[3].39
3-year treatment adjustment	1.16	1.24	1.23	1.21	1.19	1.22	1.50
Number of dissolutions	152	162	164	165	167	170	181

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Comparing the first two columns of table 5.14 indicates how the estimated NIT treatment effects would change if we had observed all couples for the entire 3-year period and attrition was unrelated to dissolution. When α equals 1 the hypothesis is that marital dissolutions continue to occur during the period after attrition at the same rate as they are observed to occur before attrition for both experimentals and controls. Thus, the changes in the estimated experimental effects from the first to the second column reflect only the relationship between attrition and the experimental treatment. The estimated effects become smaller for blacks and larger for whites, but the changes are slight and would not affect any substantive conclusions. This suggests that the differential attrition between experimental and control couples observed in table 5.12 does not by itself affect the estimated treatment effects. Attrition bias is only a problem if the relationship between marital dissolution and attrition differs between experimentals and controls. The remaining columns in the table test this hypothesis.

The remaining columns indicate that our estimates of the experimental effects of SIME/DIME on marital dissolution are robust with respect to attrition bias. The experimental effect using Model 1 is significant for blacks even when the control marital dissolution rate after attrition is three times the rate before attrition ($\alpha = 3$). The effects for blacks using Model 2 are equally robust, especially the medium guarantee effect, which is significant even when α equals 5. The low guarantee effect for blacks is not significant when α is 2 or greater but it remains large in magnitude.

Results for whites indicate that our estimates are robust for them as well. The financial treatment effect is positive for all values of α and is significant when α equals 2. The low guarantee effect for whites in Model 2 is significant even when we hypothesize that the control dissolution rate after attrition is five times the rate before attrition ($\alpha = 5$). The medium guarantee effect is significant for α equal to 2 and is large even when α is 5.

We have found no significant experimental effects on marital dissolution for Chicanos. The sensitivity analysis indicates that for the most extreme hypothesis ($\alpha = 10$), the experimental effects are significant and indicate large decreases in marital dissolution rates. This reflects our method of reconstructing postattrition marital histories which, beginning with no difference between experimental and control rates, increased the control dissolution rate as α increased.

Results of our sensitivity analysis reconfirm our conclusion from our earlier sensitivity analysis (Hannan, Tuma, and Groeneveld, 1976) that our estimates of the experimental effects on attrition for blacks and whites are relatively insensitive to attrition. If we wished to adjust the estimated treatment effects reported in Chapter 5 for attrition bias, the necessary adjustment would be small. The observed treatment effect is reduced by 14 percent for blacks and 8 percent for whites when α equals 2. We think that even this least extreme hypothesis overstates attrition bias and that reasonable adjustments for attrition bias are on the order of 10 percent for blacks and 5 percent for whites.

Reconciliations and new marriages to initially married females

We have paid close attention to the problem of defining marital dissolutions and the associated problem of handling reconciliations.[6] The SIME/DIME rules treat a marriage as having ended when the partners establish separate residences and report that the separation is intended to be permanent. We treat the date at which the separation is reported to have occurred as the end of the marriage. But what about those cases in which the couple later reconciles and reestablishes a marital relationship? Should these be considered marital dissolutions followed by spells of singleness ending in new spells of marriage or as continuous spells of marriage? We have consistently followed the former rule, treating all endings reported by the couple as permanent endings of marriage. This procedure possibly inflates estimates of marital status change.

An alternative procedure is to impose a waiting period so that a marriage would not be counted as having ended until some minimal period of separation had passed. This approach has three important defects. Any waiting period is arbitrary and is likely to be less meaningful than the reports of the couple about the intended permanency of the separation. Second, both AFDC and food stamp programs use the same criterion as SIME/DIME. Under each of these programs, a woman and her children become eligible for increased payments immediately after a separation. We expect that any national NIT program would have similar rules. If so, introducing a waiting period in estimating impacts of programs would tend to underestimate costs due to marital change because it would ignore payments made during waiting periods. Finally, use of a waiting period compromises any attempt to conduct logically consistent analyses of impacts on remarriage. Are individuals at risk of making a new marriage during the waiting period? If they are, in what sense are they still married? If they are not, what do we do with the new marriages that occur shortly after marital dissolutions?

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[6] Our analyses of the problems raised by reconciliations are reported in Hannan, Tuma, and Groeneveld (1976), and Groeneveld, Hannan, and Tuma (1980). The empirical analyses referred to here are reported in detail in the latter paper.

Nonetheless, it is important to know whether experimental effects on dissolution rates are offset by effects on reconciliation rates. One would draw different policy implications about an NIT program that raised the dissolution rate but also greatly increased the rate of reconciliation than we would about a program that raised the dissolution rate but did not alter the rate of reconciliation. The latter program would surely increase the time spent single; the former might not.

Incidence of dissolutions, reconciliations, and marriages

We begin by reporting the frequency of various sequences of events during the SIME/DIME observation period. Figure 5.6 is a tree diagram that shows the paths taken by the originally enrolled wives during the first 5 years of the experiment for the first three events (frequencies are very small thereafter). Several facts bear mentioning. First, most women who left their initial marriages remained single (only 267 out of 713 reconciled or married a new spouse). Second, reconciliation was more than twice as common as a new marriage. Third, reconciliations faced a high risk of failure. Almost half (43 percent) of all reconciliations were followed by another separation within our observation period. Moreover, reconciliations appear to be riskier than new marriages. This fact suggests that it would be a mistake to discount completely dissolutions that are followed by a reconciliation. The marriages we count as having dissolutions were indeed unstable.

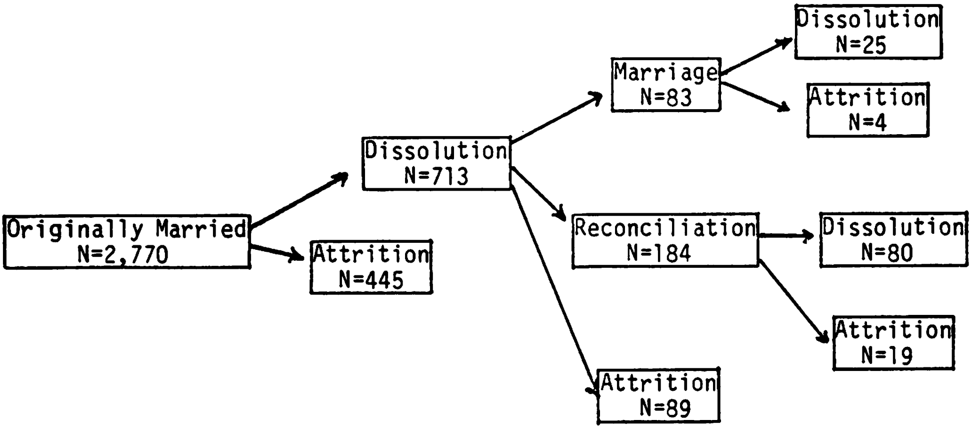

FIGURE 5.6.—Tree diagram for marital status for the first 60 months of SIME/DIME—all originally enrolled married female heads.

Impacts on rates of reconciliation and new marriage

Raw frequencies do not give a good picture of the underlying structure when the data are censored and transition rates are unlikely to be constant over individuals or over time for the same individual. Moreover, we want to see whether the treatments affect these rates. Thus, we use the transition rate models described in chapter 4 to consider the causal effects on the rates of reconciliation and new marriage. Although our primary interest is in reconciliation, the effects on new marriage help to put the reconciliation analysis in perspective. Because the number of cases is small we pool the three race-ethnic samples. We investigated the differences among the three samples in the effects of the treatments on reconciliation and new marriage and found no significant differences.

We are interested in reconciliations during the experimental and postexperimental periods. The model we use permits the treatment effects to shift at the end of the third year while constraining the effects of background variables to be the same for both periods. Table 5.15 gives estimates for this two-period model. There is at best slight evidence of impacts on rates of reconciliation: none of the treatment coefficients is significant. Note especially that the multipliers of both the 3-year and 5-year treatments are slightly lower in the second period. There is no rush to report reconciliations after the treatments end.

TABLE 5.15.—ESTIMATED NIT TREATMENT EFFECTS ON RECONCILIATION AND REMARRIAGE RATES IN A TWO-PERIOD MODEL

	Reconciliation			New marriage		
	α	Standard error	exp (α)	α	Standard error	exp (α)
Period 1: 0 to 35 months:						
3-year treatment	.103	.212	1.11	.030	.332	1.03
5-year treatment	.307	.238	1.36	−.276	.436	.758
Period 2: 36 to 60 months:						
3-year treatment	.066	.311	1.07	.047	.347	1.05
5-year treatment	.209	.326	1.23	.056	.375	1.06

There are also no significant effects of the treatments on the rates of forming new marriages. The only suggestion of an impact is a roughly 25 percent lowering of the rate for the 5-year treatment during the first period. However, the estimated standard error of this effect is much larger than the estimated effect. Thus, we conclude that the treatments did not affect the rate at which initially married women enter new marriages following a dissolution of the initial marriage.

Discussion

The possibility of reconciliation complicates attempts at modeling the impacts of the experiment treatments on marital status. However, we continue to believe that all endings reported to be permanent should be counted as real dissolutions. The fact of reconciliation does not rob the earlier dissolution of its meaning. Marriages formed by a reconciliation appear to be fragile. Moreover, periods of singleness increase the costs of an NIT whether the spell ends in a reconciliation or in a new marriage.

Of course the lack of statistical significance in the reconciliation analysis reflects the small sample sizes and should not necessarily be interpreted as evidence of a lack of practical significance. The estimated treatment effects on reconciliation rates are sizable, especially for the 5-year treatment.[7] This can be interpreted as indicating that the marital dissolutions of experimental couples are "less permanent" than those of controls. Our procedure for defining the endings of marriages inflates the estimated treatment effect to the extent that the dissolutions of experimental couples are less permanent than those of control couples. It is difficult, however, to make an adjustment for such a reconciliation "bias" since many of these reconciled unions are observed to dissolve again. We believe that our procedures for defining the endings of marriages, which closely approximate those used by existing welfare programs, offer a substantial advantage to any ad hoc recodings of the data and we conclude that our earlier estimates regarding the overall NIT effect on marital dissolution give a good picture of the impact on time spent outside marriage.

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[7] Attrition may be inflating the treatment effects on reconciliation for the same reasons that it inflates the effects on dissolution rates.

Fraud

Another possible source of bias in estimating NIT treatment effects on marital dissolution is fraud. The SIME/DIME design may create an incentive for those on NIT plans to lie about their marital status. Whenever one spouse earns more than another, the couple will gain financially by reporting a dissolution, thereby causing the high-earner's income not to be taxed in computing the payment for the low earner and the children. Couples on NIT treatments have an incentive to report fraudulently that they have dissolved their marriages while actually continuing to live together. This reasoning suggests that we may have overestimated the true effect of the experiment. The evidence, however, suggests that any such overestimate is slight.

First, the incentive to commit fraudulent behavior is not special to the NIT treatments. Families in the control group also gain financially if they report a false dissolution and the low-earner applies for AFDC and Food Stamps. So the question is whether the incentive for fraud is greater in the NIT treatments. It is not clear that it is. Indeed, we find significant increases in the dissolution rate on low guarantee plans that differ little in financial terms from welfare. Presumably there are approximately equal expected gains from such fraud on these two alternative forms of income support. In fact, the risks of detection of fraud (and of eventual prosecution) were higher for the SIME/DIME families since we interviewed families in their residences three times a year. Thus we have never been convinced that the incentive to commit such fraud was higher on the NIT plans.

Second, the pattern of guarantee level effects (chapter 5, table 5.7) indicates that the couples assigned to the most generous guarantee level have lower dissolution rates than the other experimental couples. This is inconsistent with the assertion that the observed experimental response is due to fraudulent reporting, since the high guarantee level couples have the most to gain from such fraudulent behavior.

Third, the analyses of reconciliations in the previous section provides another opportunity to address fraud problems. We pay special attention to the period after persons lose their eligibility for NIT payments. Once the financial guarantees end, the incentive to continue to lie about marital status presumably ends also. If fraud has been extensive, high levels of "reconciliation" should occur in the postexperimental period. In particular, the rate of reconciliation in the postexperimental period should rise for the groups on NIT plans. We addressed this question by focusing on the fourth and fifth years of observations on families eligible for a 3-year financial treatment. Our analysis of reconciliations (table 5.15) showed no evidence that reconciliations increased among experimentals in the period after eligibility ended. The rate of reconciliation did not rise in the 2 years after the 3-year subjects were disenrolled from the financial treatments.

Finally, there was little evidence of fraud. The field staff suspected fraudulent reports of marital status changes in four or five cases and such fraud was never proven. The evidence does not support the hypothesis that the observed experimental effects are the result of fraudulent reports of marital dissolutions. While some fraud undoubtedly occurred it does not appear to have been widespread enough to account for the large treatment effects observed.

CHAPTER VII

EFFECTS OF CURRENT EMPLOYMENT STATUS ON MARITAL DISSOLUTION

Introduction

This chapter and the next (chapter 8) explore how intervening changes in aspects of labor supply alter our conclusions about the magnitude of effects of NIT treatments on dissolution rates. We attempt to estimate the direct effects of the treatments after labor supply adjustments are taken into account. Since we are concerned with marital stability, we do not attempt the more difficult task of providing a complete picture of the joint labor supply/marital status response.[1] Our approach is partial; we ask only how observed labor supply changes alter the NIT effect on the dissolution rate.

We focus on the effects of changes in employment statuses, distinguishing employment from nonemployment (unemployment and nonparticipation in the labor force). At any point in time, each spouse may be employed or not. Cross-classifying their two possible employment statuses gives four states: (1) the husband may be employed and the wife nonemployed (labeled HEWU), (2) both may be employed (HEWE), (3) both may be nonemployed (HUWU), and (4) the husband may be nonemployed and the wife employed (HUWE). These states are listed in the order of their relative frequency among the couples enrolled in SIME/DIME at the time of their enrollment (see table 5.16).

TABLE 5.16.—HUSBAND'S AND WIFE'S JOINT EMPLOYMENT STATUS AT ENROLLMENT BY RACE-ETHNIC GROUP

	Blacks	Percent	Whites	Percent	Chicanos	Percent
HEWU	389	43.8	710	56.7	325	63.0
HEWE	308	34.7	281	22.4	113	21.9
HUWU	122	13.7	187	14.9	67	13.0
HUWE	69	7.8	75	6.0	11	2.1
Number of couples	888		1,253		516	

According to the findings of Tuma and Robins, (1980), Robins Tuma, and Yaeger (1980), and Tuma and Smith-Donals (1980), much of the labor supply effect measured in terms of annual hours of work comes about (largely for wives and to a lesser—but still considerable—extent for husbands) because those with NIT treatments entered employment at a lower rate than those in the control group, causing both husbands and wives to spend greater fractions of time out of work.

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[1] See Tuma (1980) for estimates of effects on rates of employment status changes within this framework (that is, within the first spell of marriage) and Keeley (1980) for estimates of a structural model of hours of work and marital status.

In the analyses reported below, we address two related questions: (1) Does the impact of NIT treatments on rates of dissolution depend on the current employment status of husbands and wives? (2) Do estimates of NIT impacts on dissolution rates that adjust for current employment status differ from estimates that adjust only for preexperimental labor supply? Analyses designed to answer the first question help us make more complete models of the joint response of labor supply and marital status. Answering the second question tells us whether conclusions based on previous work should be revised.

It may seem that the first question does not deal directly with policy concerns, that it should be enough to provide a reliable measure of the total effect of NIT treatments on rates of marital dissolution. However, estimates of the underlying structure are needed for two kinds of policy applications. The first concerns forecasting responses in populations other than those that we studied in SIME/DIME. If, as seems likely, the distribution of spouses' employment statuses differs across important segments of the population, it can be highly misleading to generalize responses from Seattle and Denver to other populations. One should know whether the response to an NIT program varies with employment status and adjust forecasted responses for local variations in the distributions of such statuses. The second way in which knowledge of the causal structure plays a role in policy analyses involves attempts to design alternative forms of an NIT program that might lessen the impact on the marital dissolution rate. If the total response reflects mainly labor supply adjustments to NIT treatments, attempts at redesigning the program should presumably focus on ways to minimize labor supply responses. If, however, NIT treatments affect dissolution rates directly, such redesigns could fail completely to lessen the destabilizing impact of an NIT program.

Models estimated

The model used in chapter 5 had the following form:

$$\ln r_D(t) = \sum_{m=1}^{M} \alpha_m x_m,$$

where

$r_D(t)$ = dissolution rate
x_m = background variables and experimental treatments.

The data used to analyze this model consisted of observations on the timing of the dissolution of a spell in the state "married"—see the upper panel of figure 5.7.

In the analyses reported in this section, each spell of marriage is partitioned into a series of spells defined by the husband's and wife's joint employment status as illustrated in the lower panel of figure 5.7. Note that only the last spell in the series may end with a marital dissolution; the others end when either the husband or wife changes his/her employment status.

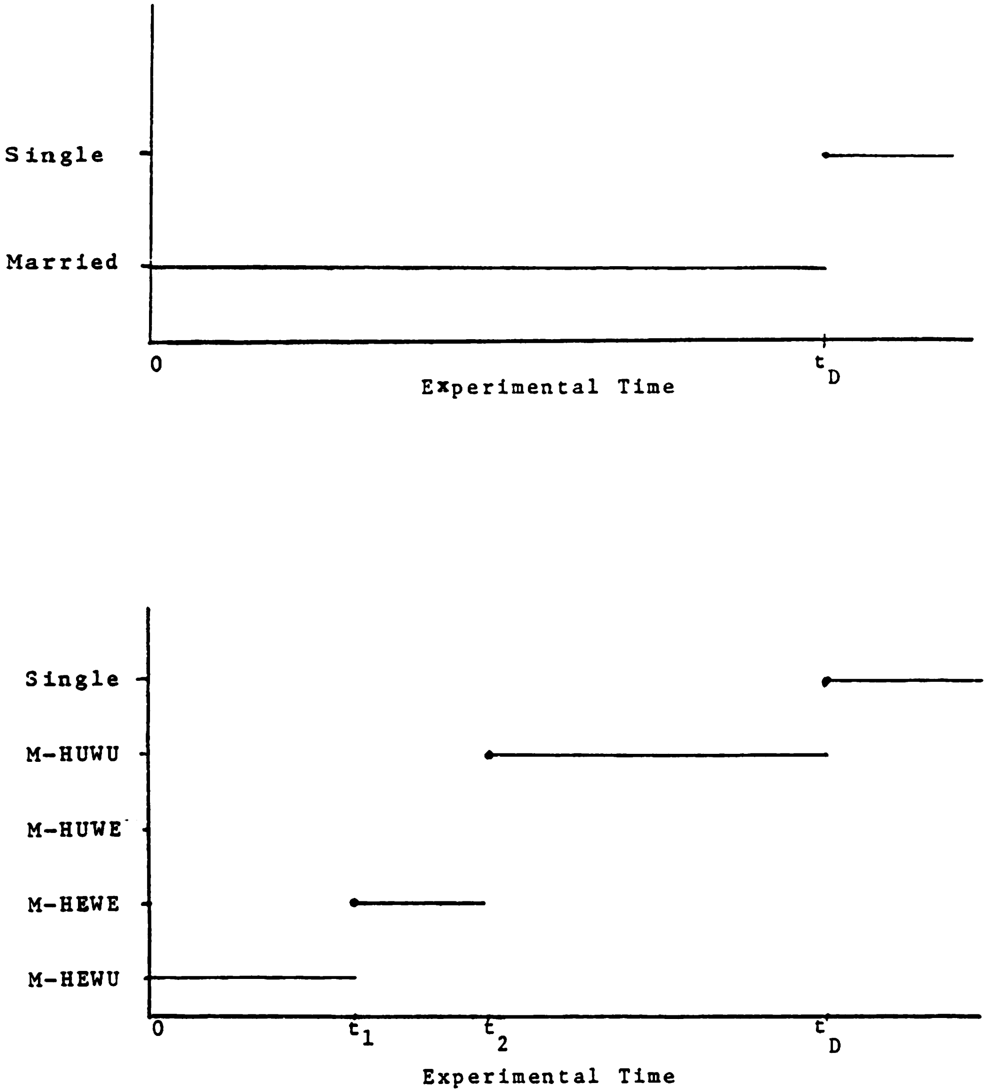

FIGURE 5.7.—A hypothetical example of the partitioning of spells of marriage into spells defined by marital status and spouses' joint employment status.

First, analogous to our previous analyses, we estimate models of the form

$$\ln r_{nD} = \beta_1 HE(0) + \beta_2 WE(0) + \sum_{m=1}^{M} a_{mD} x_m,$$

where HE(O) and WE(O) are dummy variables equal to 1 if the person (H=husband and W=wife) is employed at the beginning of the experiment, and n is the number of changes in husband's and wife's joint employment status observed since the experiment began. (With this definition of n, husband's employment status during the first spell equals HE(O), husband's employment status at enrollment.) More generally, we estimate models with the following form:

$$\ln r_{nD} = \sum_{j=1}^{4} \sum_{m=1}^{M} a_{jmD} x_m z_j.$$

where z_j, j=1, . . ., 4, denotes a dummy variable indicating one of the four possible combinations of husband's and wife's current employment status. This model lets the current employment status of the husband and wife have both main effects on the marital dissolution rate and effects that interact with all other variables.

We also estimate models in which the effects of the background variables and husband's and wife's employment status are constrained to be the same for all spells, and only the effects of the NIT treatments are allowed to depend on husband's and wife's current employment status:

$$\ln r_{nD} = \beta_1 HE(n) + \beta_2 WE(n) + \sum_{m=1}^{M^*} a_{mD} x_m + \sum_{j=1}^{4} \sum_{m=M^*+1}^{M} a_{jmD} x_m z_j.$$

Equation (3) constrains the effects of M* covariates to be the same across all four types of spells.

Finally, we estimate models that constrain all background and experimental treatments to have effects that are independent of z_j, but still allow main effects of husband's and wife's current employment status:

$$\ln r_{nD} = \beta_1 HE(n) + \beta_2 WE(n) + \sum_{m=1}^{M} a_{mD} x_m,$$

Comparison of a model with the form in equation (3) versus a model with the more constrained form in equation (4) lets us test whether spouses' employment status interacts with the NIT treatments.

Consistent estimation of these models requires certain assumptions. Some are fairly obvious and others are not. First, we must assume that the likelihood of the data on all spells through the ending of a marriage factors into a product of the likelihoods for each spell. Thus, we must assume that there is no dependence across spells for the same couple, i.e., that unobservable heterogeneity does not persist across spells (such as person-specific and couple-specific effects that operate throughout the observation period), and that rates do not depend on previous history. Second, we must assume that the coupling between labor supply and marital status change has a special form: state-dependence (each employment status affects rates) rather than rate-dependence (one rate affects the other rates).

Undoubtedly these assumptions do not hold exactly. To deal with the first problem (nonindependence of spells), we also tried to conduct analyses that use only the first observed spell for each couple. Unfortunately, because employment statuses change much more frequently than marital statuses, original spells give almost no information about marital dissolution; almost all original spells end in an employment status change. We can only caution the reader that our testing procedures assume independent spells. If the spells contain dependencies, the estimated standard errors of parameters understate the true standard errors. There may, of course, be other consequences that are difficult to foresee.

The second problem (misspecification) is potentially more damaging to correct inference. But, we are not convinced that rate-dependence is more realistic than state-dependence for this problem. We think that the specification we use is superior to any used previously to study the effects of labor supply on marital decisions. Though we do not claim perfection, we believe that the present specification adds insight and represents a useful step forward in understanding how NIT treatments affect the marital dissolution rate.

Results

We conducted an analysis of NIT effects using the various models described above.[2] We began with the most complex model, the one in equation (2), which allows effects of all variables to depend on combinations of wives' and husbands' employment statuses. These models have four times as many parameters as those that ignore changing employment statuses completely. Using so many parameters strains the capacity of our data. The number of marital dissolutions within any combination of statuses is small for each race-ethnic group. Consequently, estimated standard errors of all effects are large, making it difficult to detect substantively important effects. For this reason we do not emphasize the point estimates of parameters for this "full interaction" specification. The estimated treatment effects and joint tests on treatment effects are reported in Tuma, Hannan, and Groeneveld (1980). The model of full interactions of current employment statuses with all other effects serves mainly as a baseline against which to evaluate the fits of simpler, more constrained models.

Next we tested whether the data could be pooled across the four categories by testing the model in equation (3) against the model in equation (2). We calculated likelihood ratio test statistics of a model that incorporates main effects of HE(n) and WE(n) but constrains all other parameters to be the same across the four combinations of spouses' employment statuses. The test statistics are reported in Tuma, Hannan, and Groeneveld (1980). We found that the data may indeed be pooled in this way for blacks and whites; in neither sample is the likelihood ratio test statistic significant at the 0.10 level. However, for the Chicano sample, the pooled model fit significantly less well than the model that permits all parameters to vary with husband's and wife's combined employment status. Inspection of the point estimates for the various parameters revealed that for Chicanos the differences across joint employment status categories involve effects of background variables and not experimental treatments. Since we have found no significant effects on marital dissolution for Chicanos, we direct most of our attention to estimates for the black and white samples. For purposes of comparison we report estimates from analyzing the pooled data on Chicanos, ignoring the fact that pooling disregards significant differences in the effects of background variables.

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[2] The sample used in this analysis differs in one respect from that used in analyses reported in previous chapters (and in the next chapter). In order to maintain comparability with Tuma's analysis (1980) of impacts on rates of entering and leaving employment, we exclude cases with the highest level of preexperimental family income (all families in that category were placed in the control group).

The test of the model with only main effects of HE(n) and WE(n) against the fully interactive model is too broad for our purposes because the test statistic may be dominated by many insignificant coefficients of the background variables and of the counseling and training programs. Since we want to focus on the effects of the NIT treatments, we investigated possible interactions of spouses' employment statuses with the NIT treatments and with the income guarantee levels. We constrained the effects of all background variables, the counseling/training treatments, and the length of the NIT treatment to be constant across the husband's and wife's employment status but permitted the effects of the various financial treatments to depend on HE(n) and WE(n). The results are reported in table 5.17.[3] For black couples the NIT effect is substantially stronger when the wife is currently employed. Indeed, the effect of the NIT treatment is negligible for those couples in which the wife is not employed. We also find that the NIT treatment has a large but insignificant stabilizing effect for Chicano couples when the wife works, essentially neutralizing the destabilizing main effect of wife's employment. For whites, the NIT effects are insignificant though both husband's and wife's employment status do affect the dissolution rate significantly.

TABLE 5.17.—ESTIMATED EFFECTS ON THE MARITAL DISSOLUTION RATE OF HUSBAND'S AND WIFE'S CURRENT EMPLOYMENT STATUS, THE NIT TREATMENTS, AND THE INTERACTIVE EFFECTS OF THE NIT TREATMENTS AND THEIR CURRENT EMPLOYMENT STATUSES

[Entries are multipliers of the rate for the control group]

Explanatory variable	Blacks	Whites	Chicanos
HE(n)	[1].60	[2].55	.55
WE(n)	.92	[1] 1.55	[2] 2.40
5-year NIT treatment	.84	1.12	1.25
3-year treatment adjustment	.88	.88	1.07
HE(n)*NIT	1.29	1.19	.89
WE(n)*NIT	[2] 1.77	1.26	.58
Number of spells	2,872	4,755	1,975

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.

We specified two models of how husband's and wife's employment status influences the effect of the NIT treatments on marital dissolution.[4] In the first model, which fits best for whites, the NIT treatment effect does not depend on the employment status of either the husband or the wife. In the second model, which fits best for blacks and Chicanos, the NIT effect depends on the wife's current employment status but not on the husband's. The estimates for both models are reported in table 5.18. For both whites and Chicanos the effects are substantial and nearly symmetric: husband's employment stabilizes marriage nearly as much as the wife's employment destabilizes it. The NIT effect for whites in Model 1 is not significant but has the same magnitude as in the earlier analyses (chapter 5, table 5.4). In model 2 the NIT treatment effect for working wives is larger than the effect for nonworking wives but the difference is not significant. As in table 5.17, the NIT effect on dissolution does not depend on wife's employment status for whites. The NIT treatment effect is insignificant for Chicanos in both models, consistent with our earlier findings.

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[3] The results reported here differ slightly from those reported in Tuma, Hannan, and Groeneveld (1980) due to updating of the data.

[4] We also estimated models that contained effects of enrollment employment statues—both main effects and interactions with the NIT treatment variable. In only one of the three samples did these terms significantly improve the fit of the model. Moreover, the estimated effects of preexperimental employment status was not significant at the .10 level. Therefore, we concentrate on the effects of current employment statuses in what follows.

TABLE 5.18.—ESTIMATES OF EFFECTS OF THE NIT TREATMENTS AND OF EMPLOYMENT STATUSES ON DISSOLUTION RATES—REVISED MODELS

	Blacks	Whites	Chicanos
Model 1			
HE(n)	[1].72	[3].62	[3].51
WE(n)	[1] 1.30	[3] 1.79	[2] 1.72
NIT treatment	1.38	1.40	.95
3-year treatment adjustment	.87	.88	1.07
Model 2			
HE(n)	[1].71	[3].62	[3].51
WE(n)	.91	[1] 1.55	[2] 2.43
NIT treatment	1.01	1.27	1.15
NIT*WE(n)	[2] 1.82	1.28	.57
3-year treatment adjustment	.88	.88	1.08
Number of spells	2,872	4,755	1,975

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level

The estimates for black couples are puzzling. The NIT effect in model 1 is insignificant but has about the same magnitude as the effect estimated with preexperimental variables (again, see chapter 5). In model 2, as in table 5.17, the NIT treatments affect the dissolution rate for blacks only when the wife is employed, but the wife's employment does not affect the dissolution rate for black couples in the control group. While the husband's employment lowers the dissolution rate in the control group it does not appear to alter the effect of the NIT treatment on the marital dissolution rate.

We sought to gain a better understanding of the interaction between the NIT treatment and the wife's employment status for blacks by estimating models that let current employment status and the difference in employment status relative to enrollment affect the dissolution rate. We constructed two additional variables. The first, denoted by ΔWE(n), equals one only for those women who are currently employed but who were not employed at enrollment. The second variable, denoted by ΔWU(n), equals one only for those women who are currently not employed but who were employed at enrollment. We let these variables have additive effects on the log of the dissolution rate; we also interacted them with the dummy variable indicating the NIT treatment. The results are reported in table 5.19.[5] Clearly, it is the interaction of the wife's current employment status with the NIT treatment that matters. Neither of the two new variables has any impact on the dissolution rate, either additively or interactively with the NIT treatment. Thus the puzzle cannot be easily resolved.

TABLE 5.19.—ESTIMATED EFFECTS OF WIFE'S CHANGES IN EMPLOYMENT ON DISSOLUTION RATES FOR BLACKS

Independent variable	Multiplier of the rate
WE(n)	1.02
ΔWE(n)	.93

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[5] We also estimated this model for whites and Chicanos. The new variables—ΔWE(n) and ΔWU(n)—did not significantly affect the dissolution rate.

TABLE 5.19.—ESTIMATED EFFECTS OF WIFE'S CHANGES IN EMPLOYMENT ON DISSOLUTION RATES FOR BLACKS—Continued

Independent variable	Multiplier of the rate
ΔWU(n)	1.58
NIT	1.05
NIT*WE(n)	[1] 1.74
NIT*ΔWE(n)	1.03
NIT*ΔWU(n)	.79
3-year treatment adjustment	.88

[1] Significant at the 1-percent level

NOTE - Likelihood ratio test of the exclusion of the four effects involving change in wife's employment status is 1.78, not significant at the 10-percent level

We also investigated whether the effect of both spouses being employed may differ from what one would predict from the separate effects of wives' and husbands' employment. To explore this possibility, we estimated a model that added a dummy variable indicating that both spouses were employed and that let this variable interact with the NIT treatment. We found that both this variable and its interaction with the NIT variable had insignificant and small effects on the dissolution rate.

How does adding employment statuses change estimated NIT effects?

Finally, we address the question that motivated this analysis. The analyses reported in Chapter 5 used only preexperimental information in analyzing the impacts of NIT treatments on the rate of marital dissolution; they ignored actual labor supply responses to the NIT treatments. Such analyses do not tell how much of the observed impact results from the direct effect of an NIT program on marital decisions and how much results from the NIT's effects on each spouse's labor supply, which in turn affects marital decisions. By including information on spouses' current employment status responses the present analyses pick up spouses' employment status responses. If the NIT treatments still affect the dissolution rate when spouses' current employment statuses are taken into account, we can be more confident that the NIT program affects marital decisions directly.

We approach this issue by comparing estimates from otherwise identical models that use information on employment status in two different ways. One uses information on employment status in the month prior to enrollment; that is, it uses equation (1). This model gives what we call the total effect. The second uses current employment status, HE(n) and WE(n); that is, it uses equation (3). This gives the direct effects of the NIT. For the black sample we include either the NIT*WE(−1) or NIT*WE(n) interaction. Table 5.20 presents the relevant reports for the two models. We concentrate on the black and white samples since the NIT treatments apparently have neither a total nor a direct effect on the dissolution rate in the Chicano sample.

We find that for whites the estimated total effects of the NIT treatments and of the income guarantee levels are virtually the same as the estimated direct effects. This should not be viewed as a paradoxical result. We know that NIT effects on rates of change in the wife's employment status are stronger for whites than for blacks; the effect of wife's employment status on the marital dissolution rate is also stronger for whites than for blacks. Thus, adding current employment status to the model should improve the precision of estimates more for whites than for blacks. But it is not obvious that the improvement should be directional, i.e., that the total effect either understates or overstates the direct effect.

TABLE 5.20.—COMPARISON OF ESTIMATED IMPACTS ON MARITAL DISSOLUTION RATES FOR MODELS THAT IGNORE CURRENT EMPLOYMENT STATUS (THE TOTAL EFFECT) WITH THOSE THAT CONTAIN EFFECTS OF CURRENT EMPLOYMENT STATUS (THE DIRECT EFFECT)

[Entries are multipliers of the rates for controls]

	Total effects			Direct effects		
	Blacks	Whites	Chicanos	Blacks	Whites	Chicanos
A. Effects of the 5-year NIT treatment:						
NIT	1.22	[1] 1.41	.93	1.01	1.40	.95
NIT*WE(−1)	1.29					
WE(−1)	1.14	[1] 1.33	[2] 1.73			
NIT*WE(n)				[2] 1.82		
WE(n)				.91	[3] 1.79	[2] 1.72
3-year treatment adjustment	.86	.86	1.16	.88	.89	1.07
B. Effects of the guarantee levels (5-year treatment):						
Low guarantee	1.37	[2] 1.70	1.23	1.04	[2] 1.67	1.27
Medium guarantee	1.22	1.42	.85	1.19	1.41	.87
High guarantee	1.04	.98	.62	.90	.98	.61
WE(−1)* Low guarantee	.84					
WE(−1)* Medium guarantee	[1] 1.85					
WE(−1)* High guarantee	1.09					
WE(−1)	1.13	[1] 1.37	[2] 1.74			
WE(n)* Low guarantee				1.79		
WE(n)* Medium guarantee				[2] 2.08		
WE(n)* High guarantee				1.45		
WE(n)				.90	[3] 1.79	[2] 1.74
3-year NIT treatment adjustment	.84	.86	1.22	.81	.88	1.14
Number of spells	2,873	4,755	1,975	2,873	4,755	1.975

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

The difference between the two sets of estimates for blacks is harder to summarize. In the total effects model the NIT effect is somewhat stronger for initially working than for nonworking wives, but the difference is not significant except for the medium support effect. In the direct effects model the NIT treatments have a negligible effect for currently nonworking wives and a large effect for the roughly 60 percent of the black couples with employed wives. If we exclude the wife working interaction in the model for blacks, the total and direct effects model look the same: the NIT effect in both cases is 1.38. If we had not explored the effect of the wife working, both the black and white samples would have essentially the same total and direct effects.[6]

For Chicanos, as for whites, the total effects are almost identical to the net effects. In neither case is the NIT effect significant. The effect of the wife being employed for Chicanos is the same in both models.

Discussion

The work reported in this chapter takes the first step toward estimating a model of the NIT effects on marital dissolution that adjusts for labor supply responses to the treatments. We have focused on the distinction between employment and nonemployment because much of the NIT treatments' effects on annual hours of work comes from longer spells of nonemployment. We asked first whether effects of the NIT treatments on the marital dissolution rate depend on spouses' current (that is, experimental) employment statuses. We then asked whether this change in our model alters our previous conclusions about the impact of an NIT program on the marital dissolution rate.

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[6] Including the wife working interaction in the model for whites changes the difference between the direct and total effects models only slightly: the NIT effect for working wives becomes somewhat stronger from the total to the direct models, while the NIT effect for nonworking wives becomes weaker.

We found that spouses' employment statuses do indeed affect rates of marital dissolution. The dissolution rate rises when husbands are not employed and when wives are employed. These effects are quite substantial for all three race-ethnic groups.

Since the effects of the NIT treatments on labor supply cause both husbands and wives to spend longer periods out of work, the indirect effects of an NIT program on marital dissolution rates are unclear. By inducing longer spells out of work for husbands, an NIT program tends to increase dissolution rates. But, by inducing wives to spend longer periods out of work, it tends to lower dissolution rates. Thus the indirect effects of an NIT program can either stablilize or destabilize marriages. The stabilizing effect seems likely to predominate in SIME/DIME, because the labor supply response is much larger for wives than for husbands (Tuma, 1980), and because the main effects of husbands' nonemployment and wives' employment are almost equal in magnitude.

Unfortunately, the situation is more complicated for the black sample. We found that for blacks the effect of the NIT treatments on rates of dissolution was stronger for couples in which the wife was employed. For black couples in our sample, the indirect effect of the NIT tends to stabilize marriage by moving couples to the combinations of employment statuses in which the NIT impacts on marital stability are weaker.

This complication notwithstanding, we answer the first question in the affirmative: the NIT treatments do affect rates of dissolution indirectly via the labor supply response. For white couples the indirect effect is quite simple—the NIT simply shifts couples to employment status positions that differ in their "main effects" on rates of dissolution. For black couples the indirect effect apparently has an additional component. Current employment status of black wives affects the direct effect of the NIT treatments on dissolution rates.

Having established that NIT treatments do have indirect effects on rates of dissolution, we asked whether our previous broad conclusions need to be revised. On the whole we conclude that they do not. When we adjust for the possible indirect effects of the NIT treatments via changes in employment status, we find essentially the same pattern of direct effects of the NIT treatments on dissolution rates.

CHAPTER VIII

EFFECTS OF FAMILY INCOME AND WIFE'S INDEPENDENCE ON MARITAL DISSOLUTION

Introduction

According to the estimates for the black and white samples in chapter 5, table 5.7, the treatment-control difference in the dissolution rate tends to decline as the level of income guarantee increases. Those on the program with the lowest guarantee have a significantly higher dissolution rate than comparable controls. So do those with a medium guarantee. But the dissolution rate of those with the highest guarantee level is not significantly higher than the rate for comparable controls.

As we noted in chapter 5, this pattern may seem paradoxical: improvements in economic circumstances appear to affect the dissolution rate more when they are smaller. We cannot easily dismiss the genuineness of this pattern. It holds for a variety of specifications of the other variables in the model. Moreover, it is robust with respect to possible biases due to sample attrition (see discussion in chapter 6).

How can we explain this pattern? We think that an NIT program (like many other public transfer programs) has two opposing effects on the marital dissolution rate. It tends to raise family income, thereby stabilizing marriages. (See the discussion of the relationship between income and marital stability in chapter 2.) We refer to this stabilizing effect as the income effect. But it also decreases the dependence of spouses on the marriage by providing improved financial alternatives to marriage, and thereby tends to destabilize marriages. We call this destabilizing effect the independence effect. Depending on the strength and forms of the two effects, an NIT program may increase the marital dissolution rate, decrease it, or leave it unchanged.

Nonpecuniary differences between the NIT treatments and the current welfare system are important in understanding this pattern. For families with a given income, benefits received under the low guarantee treatment are financially similar to benefits available under current welfare programs (i.e., AFDC and Food Stamps). Only nonpecuniary differences can explain our finding that those with the low guarantee have a substantially higher dissolution rate than comparable controls.

We think that there are at least three important nonpecuniary differences between the NIT treatments and current welfare programs. First, knowledge of the financial benefits available to persons who end their marriage is likely to be greater for those with NIT treatments than for those in the control group. Both rules of eligibility and the method of calculating payments were carefully explained to those with NIT treatments at the start of SIME/DIME. Although public knowledge about AFDC and Food Stamp programs has risen in recent years, neither program advertises its benefits or methods of obtaining them. Couples married at enrollment, who have low rates of preexperimental participation in these welfare programs, probably know much less about them than unmarried women with dependent children who have high participation rates among those eligible.

Second, the cost of acquiring the financial benefits for which a family is eligible appears to be much lower for the NIT treatments than for current welfare programs. NIT benefits are adjusted monthly as family members mail in information on their earnings, income, and family composition. On the other hand, applying for benefits from AFDC and Food Stamps requires trips to local welfare offices, lengthy (and for some people, confusing) forms to fill out, and documented evidence of family composition and other sources of support. Thus, applying for benefits is likely to be much more costly (in terms of both money and time) for current welfare programs than for the NIT programs in SIME/DIME.

Third, the social and psychological costs of receiving benefits differ for the NIT treatments and current welfare programs. Applying for NIT payments is largely private; recipients need not face the general public's disapproval of those on relief. The use of Food Stamps, on the other hand, puts recipients in the position of publicly and frequently acknowledging their dependency on welfare. Though AFDC does not involve the use of special coupons, it does expose applicants and recipients to face-to-face interaction with welfare workers who may treat clients in a condescending fashion. In short, receiving aid is stigmatizing for current welfare programs, but not for the NIT treatments.

All three reasons suggest that income from the current welfare system is discounted in its behavioral consequences. That is, we think that a dollar from current welfare programs affects marital stability less than a dollar of earned income or a dollar from an NIT treatment.

To test these arguments, we must translate them into a particular model that we can estimate. We could choose a variety of functional forms to represent the hypothesis that the marital dissolution rate falls as family income increases. Similarly, we could select a variety of forms to capture the ideal that the dissolution rate rises as independence increases. To choose among the various possibilities, we must introduce additional considerations about the effect of family income and independence income on the dissolution rate. The critical issue is the way that we expect the NIT-introduced changes in family income and independence to affect the marital dissolution rate.

It has been widely observed that marital dissolution rates decrease as family income increases (see chapter 2). Therefore, in modeling the effects of the NIT treatments on dissolution we want a function that decreases as income increases. We expect a given NIT-induced change in family income, ΔI_1, to have the strongest effect on the dissolution rate at intermediate levels of preexperimental family income I_1. When preexperimental family income is high, a typical change, ΔI_1, has little impact on the family's manner of living and should have a weaker effect on the rate than the same change at lower levels of preexperimental family income. At the other extreme, when preexperimental family income is very low, a typical change, ΔI_1, may still leave the family destitute and do little to make married life more tolerable. Consequently, the effect of ΔI_1 on the rate when I_1 is very low may be less than its effect at higher levels of family income. In other words, we think that the relationship between family income and the marital dissolution rate has a backwards S shape. A simple decreasing function behaving in this fashion is

$$r_D = \exp[\alpha(I_1 + I_1)^2] r^{\dagger}, \tag{1}$$

where $\alpha < 0$ and $r^{\dagger}$ summarizes the effects of all causal variables other than $I^{\dagger}$ and ΔI_1.

We expect that the marital dissolution rate increases as independence increases. Therefore we seek a model in which the dissolution rate is an increasing function of independence. We also expect a given NIT-induced change in independence, ΔI_2, to have the strongest effect on the dissolution rate at intermediate levels of preexperimental independence, I_2. When preexperimental independence is very low, a typical change in independence, I_2, may still leave one or both partners very dependent on marriage. When preexperimental independence is high, a typical increase in independence, ΔI_2, may not greatly augment the partners' already satisfactory financial opportunities outside marriage and so may have a weaker effect on the rate than at lower levels of preexperimental independence. A simple increasing function known to us that has a shape consistent with this pattern is

$$r_D = \exp[\beta(I_2 + \Delta I_2)^{-1}]r^*, \tag{2}$$

where $\beta < 0$ and r^* denotes the combined effects of all explanatory variables other than I_2 and ΔI_2.

The forms in equations (1) and (2) are very flexible and place few constraints on results a priori, except that the dissolution rate be monotonically related to family income and independence. Depending on the estimate of α and β, the effects on the dissolution rate of the experimentally-induced changes in family income and independence may be small or large, and greatest near the lower or upper end of the ranges of income and independence in the sample. Moreover, if the estimate of α or β is positive rather than negative as posited, the entire shape of the relationship between the dissolution rate, family income, and independence will be contrary to our hypotheses. Thus, there are ample possibilities for the findings to be inconsistent with our arguments.

Our strategy for evaluating this model is as follows. We first estimate the model using measures of family income and independence income that ignore NIT payments. In this stage we estimate effects of the financial treatment and of the three guarantee levels. These are the baseline findings. Then we incorporate NIT payments into measures of family income and independence and reestimate the model. We conclude that the model accounts for the experimental effect if the estimated NIT effects are substantially smaller in the second analysis than in the first analysis. Ideally the NIT effects should be reduced to zero (multipliers of unity).

We use both enrollment and current measures of income and independence in our analyses. By using current levels of income and independence we are able to incorporate the effects of labor supply changes on the dissolution rate. By allowing the levels of income and independence to vary we believe that we can better estimate their effects on marital dissolution.

Data

We analyze only original marriages, following such spells for 3 years or until a dissolution or attrition, whichever comes first. We use the final data base, which differs somewhat from those used in our earlier published reports.[1] We measure preexperimental family income, I_1, as the total income reported by the family, and we compute the change in family income, ΔI_1, as the payment that a family will receive under SIME/DIME's rules when its income level is I_1.[2] The mean change in family income is $1,530.

Measurement of independence, I_1, is a little complicated. Because financial dependence on marriage is usually more critical for wives than husbands, we measure independence by the wife's expected income if she were single. We calculate the latter in the following way. Let E_2 denote a woman's earnings if single

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[1] It corrects some misclassified events and various other data errors.

[2] We might have used either of two other measures of the NIT payments: the family's reported payment from the interviews and the amount of the check the family received. We investigated and rejected both measures because much of the data were either missing or obviously inaccurate.

(which we assume is the same as her earnings while married[3]) and N_2 her nonwage income (excluding welfare benefits) if single. Based on the values of E_2, N_2, and the number of dependent children in the family, we calculate W_2, the value of the AFDC and Food Stamp grants that a woman could receive if single. The mean value of W_2 is \$2,320. To incorporate a welfare discount, we treat the discounted welfare benefit as 50 percent of W_2.[4] Then independence income, I_1, is $(E_2+N_2+0.5W_2)$. For those with an NIT treatment, we also calculate the predicted NIT payment if single, P_2, on the basis of E_2 and N_2. The mean of P_2 is \$3,390. The change in independence due to the NIT treatments, ΔI_2, is measured as $(P_2-0.5W_2)$ or zero, whichever is greater.

We measured the level of family income and wife's independence both at enrollment and during the experiment. The current data were measured over intervals of 1 month or less. We define the current observation periods as illustrated in the following example.

Figure 5.8 shows a 4-month segment containing two events. There is an employment change at t_a and a marital dissolution at t_b. To distinguish spells defined in terms of employment status, we must break this record into at least two observations, one ending at t_a, and one beginning at t_a and ending at t_b (as was done in chapter 7). In addition, we want to allow levels of earnings and nonwage income to change monthly. Therefore we break the record into a finer set of observations—the five numbered observations indicated above the time line in figure 5.8. This shows that observations are created both by changes in the employment statuses of husbands and wives and by the beginnings and ends of months.

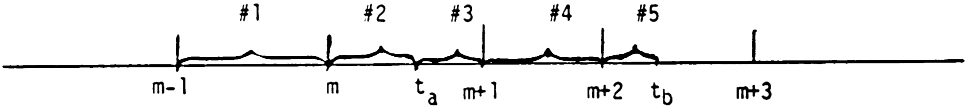

FIGURE 5.8.—Illustration of the way observation periods are defined.

We must decide how to associate monthly data with the observations. A problem arises in treating the month in which a marital dissolution occurs because the totals of earnings and nonwage income for this month combine both pre- and postdissolution levels. Since we do not want to let postdissolution outcomes affect the rate of dissolution, we lag earnings and nonwage income data by 1 month. Even though the earnings data are lagged 1 month, for simplicity we still refer to these as contemporaneous or current data.[5] We predict the NIT payments (both the payment to the intact family and the payment the wife would receive if single) for each month using the observed monthly earnings and nonwage income data. Using these monthly measures we calculate monthly values of I_1, ΔI_1, I_2, and ΔI_2 as described above.

Besides updating information on labor supply and income, we also continually update several other variables at the beginning of each month. Since family size affects marital dissolution, we use a monthly record of family composition, lagging family size in the same way as earnings. We also update age and marital duration so that these variables have the values that apply at the beginning of the relevant month.

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[3] We conducted a similar analysis using a predicted value for wife's earnings if single. We estimated the prediction equation over single female household heads. Like the prediction equations of other analysts, our equations had low predictive power. Using the predicted values had little effect in the income and independence equation.

[4] We have experimented with a variety of welfare discounts. It turns out that the qualitative pattern we report with the 50 percent discounts holds for discounts ranging from 25 percent to 75 percent. Using data from the first 24 months of the experiment, we managed to estimate the welfare discount for the black sample—52 percent (see Hannan, 1979). However, we never obtained a stable convergent solution for the white sample, which is not surprising since we were using a more complex model for whites. The results that we did obtain suggested that the welfare discount was quite large in the white sample, much higher than 50 percent. Since we have not had the time and resources to modify the software for estimating the discount using current data, we have not replicated these analyses in the framework used here.

[5] Based on this procedure, we should use the month prior to enrollment to measure earnings data associated with the first month on the experiment. Since we do not have monthly data for the preexperimental period, we used the first month's data on earnings and nonwage income.

When we use monthly observations on family composition and income variables, fewer cases have complete data than in analyses that use only preexperimental values of covariates. Since we want to maximize the amount of usable data, the values of income and family composition variables are taken from the most recent month in which they were observed whenever observations on family composition or income variables are missing. We do lose some cases that have no monthly observations—mainly cases who dropped out of the experiment early in the interviewing cycle (generally before the first periodic interview).

Results

Effects of using contemporaneous measures, of age, duration of marriage, and family composition

We begin by comparing estimates of NIT treatments from analyses using enrollment characteristics with those using current characteristics.[6] The effect of this change can be seen in table 5.21. The first column gives the estimated NIT effects when all covariates are measured at enrollment; the second column gives the estimated effects when these covariates are measured currently. Results for the black and Chicano samples are similar whether covariates are measured at enrollment or currently. But the estimated effect of the NIT treatment for the white sample is considerably stronger in the analysis that uses current measures of the covariates.

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[6] We also add a variable indicating whether the husband is away from the home at the start of the month (all husbands are present at enrollment). Husbands may be absent because they have taken a job in another location, or been jailed or hospitalized.

TABLE 5.21.—ESTIMATES OF NIT EFFECT ON DISSOLUTION RATES USING COVARIATES FIXED AT ENROLLMENT OR MEASURED FOR EACH MONTH: ORIGINAL MARRIAGES OVER FIRST 36 MONTHS OF THE EXPERIMENT

[Standard errors in parentheses]

	Whites				Blacks				Chicanos			
	Enrollment	Current	Enrollment	Current	Enrollment	Current	Enrollment	Current	Enrollment	Current	Enrollment	Current
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
NIT treatment (α)	[1].383	[3].557	.245	[1].382	[1].319	[1].351	[1].361	[1].334	−.164	−.124	−.142	−.117
	(.193)	(.198)	(.202)	(.203)	(.184)	(.185)	(.191)	(.191)	(.288)	(.294)	(.291)	(.298)
Antilog (α): the multiplier	[1] 1.47	[3] 1.74	1.28	[1] 1.46	[1] 1.37	[1] 1.42	[1] 1.43	[1] 1.40	.848	.884	.868	.889
NIT treatment*no children (O)			[2] 1.07				−.392				−.980	
			(.469)				(.500)				(1.44)	
Multiplier			[2] 2.90				.676				.375	
NIT treatment*no children (n)				[3] 1.61				.173				−.160
				(.466)				(.513)				(1.13)
Multiplier				[3] 5.01				1.19				.852
Number of cases			1,280				904				527	
Number of observations			42,631				26,297				17,233	

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

NOTE.—All equations contain additional covariates listed in Appendix A. For odd-numbered columns, these covariates are set at their values at the time of enrollment. Even-numbered columns use current observations (as of the beginning of each month). All equations also contain the 3-year adjustment variable.

Our previous empirical research indicated that the NIT effects on the dissolution rate of childless couples differ qualitatively from the NIT effects on the breakup rate of couples with children. Since families gained and lost children during the experiment (mainly as older children left to set up their own households), it is important to learn the consequences of using current rather than enrollment measures of family size and composition. The third and fourth columns of table 5.21 show the consequences of allowing NIT effects to depend on enrollment and current measure indicators of childlessness (defined as having no children living with the couple). Only 31 couples became childless during the first 3 years of the experiment.[7] For the black and Chicano samples, the effects of the NIT treatments do not differ significantly whether or not the couple has children. This is not the case for the white sample, however. The effect of the NIT treatments is significantly greater when the couple is childless, especially when the current measure is used. The effect for white couples with children is substantial, indicating a 46 percent increase in the dissolution rate when the current data are used. Because of the difference in the effect on childless white couples we concentrate on the subsample of couples with children at enrollment.

Estimates for income and independence models

In our earlier estimates of models of the effects of family income and wife's independence (Hannan, Tuma, and Groeneveld, 1978), we used preexperimental measures of all variables and analyzed data from only the first 2 years of SIME/DIME. We found that this model explained the NIT effects on the dissolution rate reasonably well for black couples, but much less well for white couples, even when we let the effects of income and independence depend on the number of children and let the effect of the payment to the family (ΔI_l) depend on the preexperimental level of family income (I_l).

As mentioned earlier, we now analyze data from the first 3 years of the experiment and use contemporaneous rather than preexperimental measures of the income and independence variables. We would like to know the consequences of these changes. We would also like to know whether the interactions with number of children and the interaction of I_l with ΔI_l are still needed. Table 5.22 provides results relevant to these issues. There are five columns of estimates for each race in the table. All of the estimated equations contain current levels of the background variables. The first two columns contain estimates using enrollment levels of the income and independence variables, the first column without NIT payments and the second column with them. The third and fourth columns contain estimates based on the current levels of the variables, without and with payments, respectively. The equations for each column contain an NIT dummy variable. By comparing the first column to the second and the third to the fourth, we see how the addition of the NIT payments to the equation reduces the estimated NIT effect. The fifth column also contains estimates of a model that uses current data, but which excludes the NIT treatment dummy variable. In the lower panel of the table, the estimated guarantee level effects are reported from equations in which the NIT treatment dummy is replaced by guarantee level dummies.[8]

The model of the income and independence effects stated in equations (1) and (2) is used for the three samples. The more complicated model used in our earlier work (Hannan, Tuma, and Groeneveld, 1978; Groeneveld, Hannan, and Tuma, 1980) is no longer necessary for the white sample. It does not either significantly improve the fit of the model or further reduce the estimated NIT effect.

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[7] Further investigation indicated that the effect of childlessness on the NIT treatment effect for couples who became childless during the experiment was not significantly different from the effect for couples who were childless at enrollment.

[8] We include a 3-year adjustment variable in all equations containing the income and independence variables. The coefficient of that variable is never significant and always indicates that the 3-year treatment has a slightly smaller effect than the 5-year treatment.

Table 5.22 shows that using current income data improves the ability of the model to explain the treatment effects for whites. In columns 1 and 2 the addition of the NIT payments to the equation reduces the estimated NIT effect by almost one-half, but the effect remains substantial. When current data are used, the NIT effect is explained completely when NIT payments are added to the equation (column 3 versus 4). With the payments in the equation using current data (columns 4 and 5), both the income and the independence effects are significant at the 1-percent level.

TABLE 5.22.—ESTIMATES OF FINANCIAL TREATMENT EFFECTS AND OF INCOME AND INDEPENDENCE EFFECTS FROM ALTERNATIVE SPECIFICATIONS OF EQUATION (1): ANALYSIS OF ORIGINAL MARRIAGES OVER THE FIRST 36 MONTHS EXCLUDING OBSERVATIONS ON COUPLES WITH NO CHILDREN, 5-YEAR TREATMENT

[Standard erro s in parentheses]

Covariates	Whites					Blacks						Chicanos			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Financial treatment	.466	.279	.319	.013	...	[3].539	.029	[3].539	[1].396	...	−.197	−.398	−.271	−.488	...
	(.232)	(.290)	(.232)	(.282)	...	(.207)	(.243)	(.207)	(.240)	...	(.307)	(.415)	(.310)	(.404)	...
Antilog (the multiplier)	[1]1.59	1.32	1.38	1.01	...	[2]1.71	.972	[3]1.71	[1]1.49	...	.822	.672	.762	.614	...
Income effects:															
Enrollment levels without payments: $Y(O)^2$	−.0033	...	...	...	...	[1]−.0042	...	...	...	...	.0006	...	...	...	...
	(.0030)	...	...	...	...	(.0025)	...	...	...	...	(.3071)	...	...	...	...
Enrollment levels with payments: $[Y(O)+P(O)]^2$	...	−.0041	...	...	...	...	[2]−.0051	...	...	...	...	−.0038	...	...	...
	...	(.0029)	...	...	...	...	(.0024)	...	...	...	...	(.0037)	...	...	...
Current without payments: $Y(m-1)^2$	...	...	[3]−.0110	...	...	...	...	[3]−.0077	...	...	...	...	[2]−.0066	...	...
	...	...	(.0026)	...	...	...	...	(.0022)	...	...	...	...	(.0030)	...	...
Current with payments: $[Y(m-1)+P(m-1)]^2$	...	...	...	−.0119	[3]−.0119	...	...	...	[3]−.0066	[3]−.0065	...	...	...	[2]−.0081	[2]−.0079
	...	...	...	(.0026)	(.0026)	...	...	...	(.0021)	(.0021)	...	...	...	(.0032)	(.0032)

TABLE 5.22.—ESTIMATES OF FINANCIAL TREATMENT EFFECTS AND OF INCOME AND INDEPENDENCE EFFECTS FROM ALTERNATIVE SPECIFICATIONS OF EQUATION (1): ANALYSIS OF ORIGINAL MARRIAGES OVER THE FIRST 36 MONTHS EXCLUDING OBSERVATIONS ON COUPLES WITH NO CHILDREN, 5-YEAR TREATMENT—Continued

[Standard errors in parentheses]

Covariates	Whites					Blacks						Chicanos			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Independence effects: Enrollment levels without payments: $[Y_s(O)]^{-1}$	[1]−.871					[3]−193					−.709				
	(.466)					(.471)					(.634)				
Enrollment levels with payments: $[Y_s(O)+Y_s(O)]^{-1}$		−.804					[3]−291								
		(.671)					(.698)					−.656			
Current without payments: $[Y_s(m-1)]^{-1}$			[3]−172					[3]−139				(.986)	[1]−126		
			(.511)					(.459)					(.65)		
Current with payments: $[Y_s(m-1)+Y_s(m-1)]^{-1}$				[3]−2.36	[3]−2.38				[2]−133	[3]−179				−1.57	−.799
				(.717)	(.593)				(.634)	(.565)				(.967)	(.732)

$3,800 guarantee	[3].723	.480	[2].540	.158		[2].558	.020	[1].475	.317		.163				
	(.266)	(.302)	(.263)	(.296)		(.243)	(.267)	(.244)	(.268)		(.337)	−.242	.065	−.340	
Antilog	[3]2.06	1.62	[2]1.72	1.17		[2]1.75	1.02	[1]1.61	1.37		1.18	(.403)	(.335)	(.396)	
												.785	1.07	.712	
$4,800 guarantee	[1].486	.206	.342	−.021		[3].771	.122	[3].793	[2].616		−.271				
	(.259)	(.315)	(.262)	(.309)		(.249)	(.279)	(.248)	(.277)		(.347)	−.718	−.372	−.720	
Antilog	[1]1.63	1.23	1.41	.979		[3]2.16	1.13	[3]2.21	[2]1.85		.762	(.450)	(.351)	(.439)	
												.488	.689	.486	
$5.600 guarantee	−.121	−.424	−.225	−.568		.290	−.322	.374	.276		−.679				
	(.360)	(.412)	(.358)	(.405)		(.278)	(.318)	(.277)	(.316)		(.416)	[2]−112	[1]−.792	[2]−.110	
Antilog	.886	.654	.798	.567		1.34	.724	1.45	1.32		.507	(.530)	(.421)	(.515)	
												.325	.453	.334	
Number of cases			1,087					767					484		
Number of observations			33,895					21,833					14,846		

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[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

NOTE.—See text for definitions of variables. All equations also contain the covariates listed in Appendix A and all equations include a 3-year adjustment variable.

We surmise that our earlier difficulties in modeling the response for whites were due to the confounding effects of labor supply adjustments. As long as we confined our attention to preexperimental measures of labor supply we could not understand the full response. Once we took current behavior into account, we uncovered an orderly structure. As originally conjectured, NIT treatments have both income and independence effects. The lowest guarantee level has the strongest effect, at least partly because its independence effect is not compensated by an equally large income effect. The highest guarantee program has both a large independence effect and a large income effect, leading to an overall null effect.

Using the current data for blacks does not improve the fit of the model, however. A comparison of columns 6 and 7 indicates that adding the NIT payment to the equation when enrollment data are used explains the NIT effect quite well. However, when current data are used, the NIT payments account for only a third of the NIT treatment effect (columns 8 and 9) and the NIT dummy is still significant. The guarantee level effects at the bottom of the table show that the $4,800 guarantee effect remains significant and substantial.

Notice that the income effect for blacks is only half as large as for the white sample. This difference is consistent with the much higher rate of dissolution in our black sample (both control and experimental). Increases in family income in the range in which our sample is concentrated stabilize white marriages much more than otherwise similar black marriages.

Overall, we conclude that our earlier success in modeling the response for black couples was partly spurious, due to the confounding effects of labor supply adjustment. Once labor supply adjustments are considered explicitly, we find that our parameterization has not explained all of the NIT effect. Actually, given the interaction of wife's current employment status and the NIT treatment (chapter 7), we are not surprised at this outcome. It is clear that something more complicated than just the income and independence effects is at work in the black response.

Since we observe no NIT treatment effect on dissolution for Chicanos we cannot assess our model in the same way as we have for blacks and whites. The independence effect is significant only when current data are used and NIT payments are excluded (column 13). Notice, however, that when current data are used for Chicanos the income effects are significant and are similar to those we found for whites and blacks.

Effects of wife's employment

In chapter 7, we analyzed the effects of NIT treatments on dissolution rates when we controlled for husbands' and wives' current employment statuses. Recall that the NIT effect depended on the wife's employment status for the black sample, but not for the white and Chicano samples. When black wives were employed, the NIT treatment greatly increased the rate of dissolution. For black couples with nonworking wives, there was no NIT effect.

We now return to this issue. We continue with our model of income and independence but add a dummy variable indicating whether the wife is currently employed. This revised specification allows the wife's employment to affect the dissolution rate in three ways: directly, through the independence effect, and through the income effect. We use the current levels of income for this analysis and report the results in table 5.23.

TABLE 5.23—ESTIMATES OF MODELS IN TABLE 5.22 BUT WITH TERMS INVOLVING WIFE'S CURRENT EMPLOYMENT STATUS, 5-YEAR TREATMENT

[Standard errors in parentheses]

Covariates	Whites				Blacks				Chicanos			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Financial treatment	^{2}.466	^{1}.511	^{2}.542	.552	^{3}.557	^{2}.562	.353	.032	−.117	.021	.179	.807
	(.234)	(.309)	(.275)	(.410)	(.208)	(.262)	(.273)	(.405)	(.316)	(.440)	(.361)	(.583)
Multiplier	^{2} 1.59	^{1} 1.67	^{2} 1.72	1.74	^{3} 1.74	^{2} 1.76	1.42	1.03	.900	1.02	1.20	2.24
Income effect	^{3}−.0123		^{3}−.0123									
Current without payments:												
$Y(m-l)^2$	(.0026)		(.0026)		^{3}−.0079		^{3}−.0081		^{2}−.0079		^{2}−.0076	
		^{3}−.0130		^{3}−.0129	(.0022)		(.0022)		(.0033)		(.0032)	
Current with payments:												
$[Y(m-l)+P(m-l)]^2$		(.0026)		(.0026)		^{3}−.0072		^{3}−.0077		^{3}−.0096		^{2}−.0084
						(.0022)		(.0022)		(.0033)		(.0033)
Independence effect	−.440		−.402									
Current without payments:												
$[Y_s(m-l)]^{-1}$	(.621)		(.623)		^{1}−1.11		^{2}−1.24		−.317		−.099	
		−.846		−.779	(.607)		(.62)		(.780)		(.776)	
Current with payments:												
$[Y_s(m-l)+Y_s(m-l)]^{-1}$		(.810)		(.922)		−.726		^{1}−1.61		−.281		1.16
						(.732)		(.906)		(1.06)		(1.23)
	^{2}.836	^{2}.827	^{3}.951	^{2}.870								
Wife working (m)	(.233)	(.216)	(.314)	(.353)	.161	.319	−.098	−.211	^{2}.650	^{3}.720	^{3} 1.21	^{3} 1.59
	^{2} 2.31	^{3} 2.29	^{3} 2.59	^{2} 2.39	(.230)	(.200)	(.32)	(.374)	(.298)	(.272)	(.416)	(.554)
Multiplier					1.17	1.38	.91	.810	^{2} 1.92	^{3} 2.05	3.37	^{3} 4.90
			−.190	−.061								
Wife working*financial treatment			(.351)	(.397)			.373	^{1}.662			^{1}−.891	^{2}−1.18
			.83	.941			(.329)	(.396)			(.473)	(.554)
Multiplier							1.45	^{1} 1.94			.41	^{2}.307
Number of cases			1,087			767				484		
Number of observations			33,895			21,833				14,846		

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

Note.—All equations included a 3-year adjustment variable.

When the wife working variable is in the equation, we find that for whites the addition of the NIT payments to the income and independence model does not account for the NIT effect (columns 1 and 2). The estimated independence effect is much smaller in this model than in the model without the wife working variable (columns 3 and 4 of table 5.22). This is to be expected since the wife's employment is an important component of her independence income. For whites, we conclude that the model in table 5.22, without the wife working variable, is a better representation of the process underlying the NIT treatment effects than the model with the wife working variable. Interacting the NIT treatment with wife working does not alter the estimated income and independence effects or the NIT treatment effect for whites.

Adding the wife working variable also reduces the estimated independence effect for blacks (columns 5 and 6) compared to the estimates in table 5.22 (columns 6 and 7). When we interact the NIT treatment with the wife working variable we find, as in chapter 7, that the NIT effect is larger for working wives (columns 7). The NIT treatment multiplier for nonworking wives is 1.42 and for working wives is 2.06 (1.42 times 1.45). When we add the NIT payment to the equation with the wife working treatment interaction, the NIT effect for nonworking wives falls to 1.03. However, the effect for working wives remains almost the same, 2.00 (1.03 times 1.94). Our income and independence model does a good job of explaining the NIT effect for nonworking wives but, as we found in chapter 7, it is the black couples with working wives who respond most to the NIT treatments. Thus, adding the wife working variable to the income and independence model does not improve the model over the one reported in the previous table.

We still do not understand why the model does not work as well for blacks as it does for whites, nor how the interaction between the wife working and the NIT treatments comes about for blacks. To learn more about it, we estimated various models that interacted wife's employment status with income and independence effects. We did not find that wife's employment status altered income and independence effects. Nor did we find that including interactions of wife's employment status with income and independence effects eliminated the interaction of employment status with the NIT treatment effect.

It is worth noting that inclusion of wife's employment status does not eliminate the independence effect for blacks, although it reduces it. The obvious collinearity between wife's employment status and independence increases the standard error of the independence effect, but the effect is still large and significant. Actually, it differs only slightly in magnitude from the effect reported in table 5.22 which ignores employment status. Quantitative variations in independence do seem to matter for the black sample. Wife's employment status has a very small main effect—the qualitative difference between employment and nonemployment does not greatly affect the dissolution rate among couples in the control group. Only on the NIT program does wife's employment status make a difference—it greatly increases the rate of dissolution. It appears for this sample that the combination of employment and the NIT guarantee has a joint effect although the components do not have separate effects.

Parallel results for Chicanos are reported in columns 9–12 of table 5.23. The results of column 9 show that wife's employment has a strong destabilizing effect when one controls current levels of income and independence. Adding NIT payments to the income and independence measures (column 10) gives a slightly larger estimate of the effect of wife's employment. There is still no main effect of the NIT treatment. Columns 11 and 12 show the consequences of adding an interaction between the NIT treatment and wife's employment status. This interaction has a strong effect. In the control environment, wife's employment status increases the dissolution rate almost fivefold (a multiplier of 4.9). But the NIT treatment almost completely counteracts this effect. The effect of wife's employment status for those within an NIT treatment is 1.50 (4.9 times .307). So wife's employment increases the dissolution rate "only" 50 percent when the couple is on an NIT program. The NIT treatment stabilizes marriages of Chicano couples when the wife is working.

The interaction of wife working and the NIT treatment is significant in table 5.23 for Chicanos but the same effect was insignificant in chapter 7, table 5.18, which used a different methodology. The point estimates in the two analyses are similar, however. Given the results in table 5.23 we revise our conclusion in chapter 7 and suggest that the NIT effect may differ for working and nonworking Chicano wives. We hold this conclusion tentatively, in that it is not supported by both analyses. Notice that the NIT wife working interaction has opposite effects for blacks and Chicanos: It is destabilizing for blacks and stabilizing for Chicanos.

Conclusions

Several broad conclusions apply to all three race-ethnic samples. Insofar as current labor supply behavior affects family income and wives' independence, it affects rates of marital dissolution and gives altered estimates of NIT effects. However, much of the NIT impact on dissolution rates does not reflect simple labor supply adjustments. Even when we take into account current family composition, earnings, and nonwage income, the NIT impacts are still substantial. Finally, we have seen that our model of income and independence effects describes marital dissolution in all three samples. We have found consistent evidence that income and wife's independence affect dissolution rates in the predicted direction for whites, blacks, and Chicanos.

The differences in results among the three samples are as important as the similarities. The main story breaks in two parts according to whether wife's employment status affects the process. For black and Chicano samples, couples with employed wives respond differently to an NIT program than do couples in which the wife is not currently employed. Surprisingly, the effect differs in the two samples. For blacks, the NIT treatment appears to be especially destabilizing when the wife is employed, even when we take income and independence effects into account. For Chicanos, the NIT treatment largely offsets the strong destabilizing effect of wives' employment. That is, the NIT program stabilizes marriage of Chicano couples with employed wives but does not affect the dissolution rate of couples with nonemployed wives.

Our attempt to learn more about the processes that underlie the interaction of wife's employment with the effects of NIT treatments has been unsuccessful. Something more than variations in income and independence seems to be at work. Letting income and independence effects vary according to wife's employment status does not help matters. Thus, the deeper structure of the effects of NIT treatments on black and Chicano marriages remains obscure.

We have been much more successful in understanding the response of the white sample. We learned that the estimated response for whites is sensitive to the timing of measurements of earnings and nonwage income. Apparently our earlier problems in modeling the white response arose from our use of preexperimental measurements of income and independence. When we use current data, a simple model fits the data well. Not only do we find evidence of strong income and independence effects, but our explanation accounts for almost the whole NIT effect.

CHAPTER IX

EFFECTS OF THE EXPERIMENT ON THE FORMATION OF MARRIAGES

One of the major implications of an increase in marital dissolution rates is that the proportion of families headed by single women increases. The overall effect of the experiment on the fraction married in a population also depends on the marital formation rate. A sufficient increase in the marital formation rate could offset the increased dissolution rate brought about by the NIT treatments for whites and blacks. On the other hand, if the NIT treatments reduced the marital formation rate, the proportion of families headed by single females would increase even more than if only the dissolution rate increased. Thus, it is important to consider the NIT treatments' effects on marital formation in order to understand the total effect of the treatments upon the fraction married in the population.

The major difficulty with studying marital formation in SIME/DIME arises because the experiment was not a universal program. The formation of a new marriage involves decisions by both parties. However, except for the rare case when two eligibles marry, the potential spouses in SIME/DIME were affected differently by the treatments. Of particular importance in this regard was the provision allowing people to "marry into" the experiment. Whenever a single family head married, the new spouse became eligible for the treatment. The guarantee level was adjusted to reflect the change in family size, and the new spouse's income was considered in computing NIT payments. After a 90-day waiting period, the new spouse retained eligibility even if the marriage dissolved. This possibility of marrying into the experiment introduces a complexity to analyses of marriage that does not occur in analyses of marital dissolution. It creates an incentive for marital formation that would not be present in a universal NIT program. Because of this incentive, sometimes called a dowry effect, an experimental NIT program is expected to increase the marital formation rate more than a universal program.

Proportions of single women marrying

As described above, families headed by single women were enrolled in SIME/DIME.[1] In addition, some women became single during the experiment through marital dissolution or the death of a husband. Table 5.24 indicates the proportion of originally single women who married during the first 3 years of SIME/DIME. For blacks and whites, women eligible for the financial treatment were more likely to remarry than control women. For Chicanos, experimentals were less likely to marry. As in the analysis of the NIT treatments' effects on dissolution, comparisons of raw proportions are not adequate to determine the NIT effects on marital formation.

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[1] In addition, 64 families headed by single males were enrolled. Because of the small number of unmarried males in the original sample, we ignore them in this discussion of marital formation. The experimental effects on marital formation among eligible males (both males who were single at enrollment and those who became single during the experiment) are discussed in Groeneveld (1977).

TABLE 5.24.—PROPORTION OF ORIGINALLY SINGLE WOMEN OBSERVED TO MARRY DURING THE FIRST 36 MONTHS OF THE EXPERIMENT

[Number of women in parentheses]

	Blacks	Whites	Chicanos
Controls	.104	.203	.230
	(364)	(310)	(113)
NIT treatment	.122	.222	.180
	(540)	(441)	(206)
3-year NIT treatment	.117	.234	.203
	(403)	(316)	(143)
5-year NIT treatment	.139	.192	.127
	(137)	(125)	(63)

Effects on marriage rates

In table 5.25 we report the estimated NIT effects on the marital formation rate.[2] The NIT treatment does not have a significant effect for blacks or for whites but it significantly reduces the rate at which Chicanos form marriages. In the original-spells analysis, the NIT reduced the marital formation rate to about one-third of the rate of comparable controls among Chicanos. This large decrease in their formation rate has persisted throughout our analyses.

TABLE 5.25.—ESTIMATED EFFECTS OF NIT TREATMENTS ON MARITAL FORMATION

[Standard errors in parentheses]

	Blacks		Whites		Chicanas	
	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$
Original spells:						
5-year NIT treatment	.32	1.37	−.20	.82	[1]−.94	.39
	(.34)		(.29)		(.50)	
3-year treatment adjustment	−.29	.74	.04	1.05	.60	1.82
	(.37)		(.31)		(.54)	
Number of spells	904		751		319	
χ^2 for equation	59.47		122.49		36.16	
	(19 df)		(19 df)		(17 df)	
Likelihood ratio χ^2 test for treatment variables	.88		.97		4.01	
	(2 df)		(2 df)		(2 df)	
All spells:						
5-year NIT treatment	.35	1.42	−.12	.88	[2]−1.12	[2].32
	(.30)		(.24)		(.47)	
3-year NIT treatment	−.34	.71	−.00	1.00	.62	1.87
	(.34)		(.27)		(.52)	
Number of spells	1,206		1,071		479	
χ^2 for equation	32.01		101.67		29.17	
	(18 df)		(18 df)		(18 df)	
Likelihood ratio χ^2 test for treatment variables	1.35		.64		[2]7.93	
	(2 df)		(2 df)		(2 df)	

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.

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[2] The assignment variables included in the equations are normal income level and site. The background variables are age, education, number of children at enrollment, and dummy variables for having a child younger than 5 and for participating in AFDC in the year before the experiment. Variables for the counseling/training treatment are included in all equations. In the original-spells analysis, the length of time single is also included. The effects of the assignment and background variables for the original spells analysis are reported in appendix A. In appendix B we discuss the effects of the counseling and training program on marital stability.

Table 5.26 reports the NIT effects on the marital formation rate for each of the 11 financial treatments for originally single women. This representation of the NIT treatment does not improve on the single variable model in Table 5.25. For Chicanos 10 of the 11 coefficients are negative. Table 5.27 presents the estimated NIT effects on marital formation rates by guarantee level. For Chicanos, the low and high levels have significant effects in the original-spells analysis and all three guarantee levels significantly reduced the marital formation rates in the all-spells analysis. The chi-square test indicates that the guarantee level effects are not significantly different from one another.

TABLE 5.26.—ESTIMATED NIT TREATMENT EFFECTS ON MARITAL FORMATION RATES: ORIGINALLY SINGLE WOMEN OVER FIRST 36 MONTHS OF THE ENROLLMENT

[Standard errors in parentheses]

NIT treatment	Blacks		Whites		Chicanos	
	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)
$3,800/50 percent	.30	1.35	.12	1.13	[1]−1.19	.30
	(.43)		(.36)		(.70)	
$3,800/70 percent	−.06	.95	−.12	.88	[1]−1.27	.28
	(.58)		(.45)		(.72)	
$3,800/70 percent d [2]	.66	1.93	−.26	.77	−.47	.62
	(.47)		(.47)		(.66)	
$3,800/80 percent d [2]	.23	1.26	−.34	.71	−.52	.59
	(.60)		(.52)		(.70)	
$4,800/50 percent	[1].51	1.66	.51		−.55	.58
	(.52)		(.44)	(.60)	(.84)	
$4,800/70 percent	.55	1.73	−.18	.84	−1.19	.30
	(.45)		(.37)		(.74)	
$4,800/70 percent d [2]	.65	1.92	[1]−.49	.61	−.52	.59
	(.51)		(.47)		(.92)	
$4,800/80 percent d [2]	.35	1.42	.13	1.14	−.29	.75
	(.56)		(.40)		(.63)	
$5,600/50 percent	−.58	1.78	−.14	.87	−9.60	$7*10^{-5}$
	(.84)		(.72)		(69.83)	
$5,600/70 percent d [2]	−1.10	.33	[1]−1.08	.34	−1.33	.26
	(1.04)		(.65)		(.90)	
$5,600/80 percent d [2]	.65	1.92	−.14	.87	−1.59	.20
	(.52)		(.45)		(.85)	
3-year treatment adjustment	−.40	.67	.05	1.05	−.58	1.78
	(.37)		(.32)		(.55)	
Number of spells		904		751		319
χ^2 for equation	66.09		130.42		45.65	
	(29 df)		(29 df)		(27 df)	
Likehihood ratio χ^2 test versus model in table 5.6	6.62		7.93		9.46	
	(10 df)		(10 df)		(10 df)	

[1] Significant at the 10-percent level.
[2] d=declining rate.

TABLE 5.27.—ESTIMATED GUARANTEE LEVEL EFFECTS ON MARITAL FORMATION

[Standard errors in parentheses]

	Blacks		Whites		Chicanas	
	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)
Original spells (5-year treatment):						
Low guarantee	.27	1.31	−.06	.94	[1] −.95	.39
	(.36)		(.32)		(.54)	
Medium guarantee	.47	1.60	−.23	.79	−.68	.51
	(.37)		(.31)		(.54)	

TABLE 5.27.—ESTIMATED GUARANTEE LEVEL EFFECTS ON MARITAL FORMATION—Continued

[Standard errors in parentheses]

	Blacks		Whites		Chicanas	
	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$
High guarantee	.15	1.16	−.43	.65	[2] −1.74	[2] 1.75
	(.46)		(.39)		(.71)	
3-year treatment adjustment	−.32	.73	.04	1.04	.64	1.89
	(.37)		(.31)		(.54)	
Likelihood ratio χ^2 test versus model in table 5.6	.90		1.44		4.17	
	(2 df)		(2 df)		(2 df)	
All spells (5-year treatment):						
Low guarantee	.29	1.34	−.07	.94	[2]−1.15	[2].32
	(.34)		(.27)		(.50)	
Medium guarantee	.42	1.52	−.15	.86	[1]−.96	[1].38
	(.34)		(.26)		(.51)	
High guarantee	.37	1.45	−.19	.82	[2]−1.57	[2].21
	(.40)		(.33)		(.64)	
3-year treatment adjustment	−.34	.70	−.00	1.00	.66	1.93
	(.34)		(.26)		(.52)	
Likelihood ratio χ^2 test for guarantee level effects	.22		2.49		1.50	
	(2 df)		(2 df)		(2 df)	

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.

In an earlier report (Tuma, Groeneveld, and Hannan, 1976), we reported results that shed further light on these effects. In that analysis, we let the effects of the guarantee level depend on the length of time single. The likelihood ratio tests indicated that for all three race-ethnic groups, allowing the guarantee levels to vary with duration single fit the data significantly better. The NIT treatments lowered the marital formation rates for black and white women who had been single for less than 4 years at enrollment and increased the rate among women who had been single for longer periods.[3] It is possible that some women who have recently become single enter suboptimal marriages for reasons of economic security. Given a nonstigmatizing alternative to welfare, some of these women may delay marriage while searching for a better match.

The findings presented here do not support the hypothesis of a dowry effect. For no experimental group did the NIT treatment significantly increase the marital formation rate. If a dowry effect exists, it must be offset by a negative effect of the NIT treatments—a hypothesis that cannot be tested with the available data.

The effects of the NIT treatments on the observed marital dissolution and formation rates imply that overall the NIT increased the proportion of families headed by single females. For blacks and whites, the increase was due to the increase in dissolution rates; for Chicanos, the increase was due to the decrease in the marital formation rate. For all three race-ethnic groups, the net effect is increased welfare costs because the proportion of the population most likely to depend upon welfare rises.

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[3] This analysis covered only the first 24 months of the experiment.

CHAPTER X

GENERALIZATION OF THE SIME/DIME EFFECTS ON MARITAL STABILITY TO A NATIONAL SAMPLE

Introduction

We have presented two types of estimated effects of SIME/DIME on marital stability. The first type estimated the experimental effect as differences between dissolution and marriage rates of the experimental and control groups (see chapters 5 and 9). The experimental effects in these models were reported as multipliers of the rates for comparable controls. These analyses clearly demonstrate that the NIT treatments substantially increased the rate at which marriages dissolved for white and black couples and decreased the rate at which Chicano single female heads of families formed marriages. These effects imply an increase in the proportion of families headed by single females for all three race-ethnic groups.

Our examination of the experimental effects by guarantee level indicated that the magnitude of the effect was not a simple function of a program's generosity. For all three race-ethnic groups, programs with the most generous guarantee level had the smallest effect on the dissolution rate. Among Chicanos programs with the highest guarantee level had the largest effect on marriage rates. We can conclude from these analyses that a national NIT program similar to SIME/DIME would affect marital stability. The consequences would be in the direction of increasing the proportion of families headed by single females. However, the magnitude of the effects would not be a simple linear function of program generosity.

In the second stage of our analysis we modeled the NIT treatments' effects in terms of the changes in income that this generated. As we discussed in chapter 8, the effects on dissolution could operate both through a change in the income available to the couple (the income effect) and through a change in the income available to the wife (and/or husband) if the marriage dissolves (the independence effect). These analyses showed that the magnitude of the NIT effect on marital dissolution rates depended not only on the amount of change in income due to the NIT program, but also on the level of income prior to the NIT program. In these models we also incorporated nonpecuniary differences between the NIT program and the existing welfare system in the form of a welfare discount (see chapter 8).

However, these two sets of analyses have not provided information about the magnitude of the effects of a national NIT program on marital stability or about differences in the effects of different national NIT plans. If the experimental effects depend on levels of income, the effects of national programs will depend upon the income distribution of the population as well as the program's generosity. Also, none of these analyses considered how changes in marital formation and dissolution rates due to an NIT program affect the costs of a national program.

Methodology

The general approach

We applied the estimated effects of SIME/DIME to a representative sample of a national population to estimate the effects of a national program. The aggregate effect is obtained by summing over all the family units in the nationally representative sample using the appropriate sampling weights. We used the Current Population Survey (CPS) from March 1975 in generalizing the SIME/DIME results. It is an annual household survey that contains data on the same variables used in analyzing SIME/DIME data and includes detailed information about family income, labor supply, and family composition. All variables necessary for the generalization were measured in the CPS or could be approximated by reasonable surrogates.

We identified a subsample of family units in the CPS sample that would be categorically eligible for an NIT program using the SIME/DIME initial eligibility rules. We call this subsample the generalization sample; it includes all husband-wife families in which the husband is between the ages of 18 and 58 and is able to work, and all female-headed families in which the female head is between the ages of 18 and 58, able to work, and has at least one dependent in her household.[1] The sample contained 24,381 married couples and 3,196 single female household heads. Using the CPS sample weights, the generalization sample represents 39.3 million married couples and 5.2 million single females.

In designing the generalization, we employed some well-known implications of transition rate models.[2] If we assume that the transition rates do not depend on the history of transitions among previous states (technically, if we assume the Markov property), we can predict a number of useful properties of event histories as functions of the transition rates. In particular, given the estimated dissolution and marriage rates for an individual, we can derive algebraically the expected number of marriages and dissolutions, the probability of being single at any point in time, and the expected fraction of any time interval in a particular marital status.

We first calculated the estimated dissolution and remarriage rates for each case in the generalization sample.[3] Using the estimated rates, we then computed for each case the probability of being single after 2 years and the expected number of marriages and dissolutions within 2 years.[4] We also calculated the estimated cost during the first 2 years of supporting each unit under the current welfare program and under several NIT programs, given the expected fraction of the year in each marital status. The aggregate effects were estimated by summing over all units in the generalization sample using the CPS sampling weights.

Comparison with the microsimulation approach

Our approach differs from that utilized in other generalizations of marital stability analyses. The approach employed in the past, which we refer to as the microsimulation approach, uses the predicted probability of an event (a dissolution or marriage) to simulate the events that would occur in the population (see

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[1] Female family heads are regarded as married if they were separated from their spouse for reasons other than marital discord. If the husband was absent because of military service, the family unit was not included at all.

[2] The methodology is described fully in Groeneveld, Tuma, and Hannan (1980a). Readers interested in the operating characteristics on WYKNOT, the computer program that generated the generalization estimates, should consult Moeller (1981).

[3] The marriage rate for a married unit was calculated using the wife's characteristics and her estimated independence income. The dissolution rate for a single female was calculated using her characteristics and the characteristics of the husband of a married woman in the sample who was of similar age, race, and employment status and had the same number of children. See the discussion of "synthetic marriages and divorces" in Moeller (1981) for details.

[4] We used a 2-year period because only data over the first 24 months of the experiment had been analyzed at the time the generalization was done.

Orcutt, Caldwell, and Wertheimer, 1976). For example, to stimulate marital dissolution, the probability of having a dissolution is computed for each case at risk (only married persons may dissolve marriages). This probability is compared to a random number drawn from a uniform (0,1) distribution. If the probability is less than the random number, no change is simulated and the welfare cost is computed assuming no change. If the probability is greater than the random number, a dissolution is simulated, and a date for the dissolution is determined. The welfare cost for that case is then computed, taking into account the marital status change and the length of time in each marital state. The total number of events occurring and the total welfare cost is obtained by summing over all cases, with each case weighted by its sampling weight.

Our objection to using a microsimulation approach has to do with its "all or none" character. A decision is made for each case, to simulate a marital status change or not. If a change is simulated, the number of changes predicted in the national population is the weight of that case. Take, for example, a case with a weight of 1,600 (the average weight of a unit on the CPS) and the probability of an event in the first year of .05. With a microsimulation approach, it is decided whether that case has an event or not. If an event occurs, we predict 1,600 events in the population. If not, we predict no events. If the CPS contains a large number of cases similar to this case, or if a large number of simulations are run, the predicted number of events in the population will be .05 x 1,600 = 80. But usually only one simulation is run, and the number of cases with a particular set of characteristics is often small. In this situation, the predicted number of events and predicted costs may vary considerably from one simulation to another.

It might be argued that such variation will "wash out" in a single simulation of a national program, but there is no evidence to support this claim. We are aware of no testing of the sensitivity of the microsimulation approach to this kind of variation. This problem becomes more acute when subsets of the population are examined. The smaller the subset selected, the greater the variation from one simulation to another.

Our approach, which we call an expectations approach, avoids this problem by directly computing the expected numbers of events and the expected costs. These expectations are the mean numbers of events and the mean costs that would be obtained if a large number of microsimulations were run. Since the results of any single microsimulation may differ substantially from these expectations, our approach produces better estimates of average outcomes than can be obtained with the microsimulation approach.

Equations for rates of marital formation and dissolution

The equations for the marital formation and dissolution rates that we used for the generalization differ from those discussed in chapter 5. The major difference is that the generalization equations average over employment statuses and use only preenrollment data. Spells are defined by marital events without regard to employment status or changes in employment status. The equations are similar to those reported in our earlier work (Hannan, Tuma, and Groeneveld, 1978). Equations for the dissolution rates of blacks and whites differ.[5] The same marriage rate equation is used for both racial groups, but the coefficients are estimated separately.[6] The marital dissolution equation for blacks is

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[5] Hispanic persons are not easily identifiable in the CPS. Moreover, we have evidence that the Denver Chicano population is atypical of Hispanic Americans generally (Groeneveld, 1979). Thus, we did not attempt to generalize the experimental effects for Chicanos. Any Hispanics in the CPS are treated as whites.

[6] The equations used to estimate these coefficients contained a 3-year treatment adjustment variable. The coefficients used in the generalization are estimates of the effects of a 5-year treatment which is a better approximation of a permanent program than the 3-year treatment.

$$r=\exp[\alpha(I_1+\Delta I_1)^2+\delta(I_2+\Delta I_2)^{-1}+X\theta], \qquad (1)$$

where I_1 = couple's disposable income
ΔI_1 = change in couple's disposable income due to NIT
I_2 = wife's independence income
ΔI_2 = change in wife's disposable income due to NIT
X = vector of variables known to be related to marital dissolution
α, δ, θ = estimated coefficients.

The marital dissolution equation for whites is more complicated:

$$r=\exp[\alpha^* I^1{}_1+\beta^*\Delta I^2{}_1+\delta^* I_1\Delta I_1+\delta^*(I_2+\Delta I_2)^{-1}+X\Theta], \qquad (2)$$

where
$\alpha^*=\alpha_0+\alpha_1 C, \beta^*=\beta_0+\beta_1 C, \gamma^*=\gamma_0+\gamma_1 C, \delta^*\delta=\delta_0+\delta_1 C,$

and C denotes the number of children in the family.[7]

The marriage rate equation for both races is

$$r=\exp[\alpha(I_2+\Delta I_2)+Z\Phi]. \qquad (3)$$

In the marriage equation I_2 and ΔI_2 refer to the income and NIT-induced change in income of the single female; Z is a vector of variables known to affect marriage rates. The definitions of all variables in the equations and the estimated coefficients are given in Groeneveld, Tuma, and Hannan (1980a).

An important aspect of our methodology is that the dissolution and marriage equations predict marital status changes both under an NIT program and for the present system. The estimates for the current system are obtained by setting ΔI_1 and ΔI_2 to zero in the dissolution and marriage equations. Thus, in order to evaluate the effects of the NIT plans, we must consider the marital status changes that would have occurred under the current welfare system.[8]

Results

We simulated three NIT programs similar to those in SIME/DIME except for their tax rates and guarantee levels. The programs were a 70 percent tax rate with a guarantee level of 75 percent of the poverty line (70/75), a 50 percent tax rate with a guarantee level of 75 percent of the poverty line (50/75), and a 50 percent tax rate with a guarantee level of 100 percent of the poverty line (50/100). The 50/100 program is somewhat more generous than the least generous SIME/DIME program. The other two programs are less generous than any in SIME/DIME, and are also less generous than current welfare (AFDC and Food Stamps) programs in some states.

Annual costs and other outcomes are calculated using a "hold harmless" provision. This provision means that no family unit was allowed to be worse off under an NIT program than under the current welfare system. The cost estimates include only payments to families (no administrative costs). All costs are given in 1974 dollars.

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[7] In chapters 5 and 8 we noted that the NIT treatment effect was greater for white childless couples than for white couples with children. The form of the dissolution equation used for whites captures that difference by allowing the coefficients to vary with family size.

[8] Including marital status changes under the present system is a major methodological difference between the marital stability generalization reported here and the labor supply generalization reported in part III, chapter 1. The labor supply generalization used only the coefficients of the experimental variables (analogous to the ΔI_1 and ΔI_2 variables in our equations). That was possible only because they used a linear additive model. In our model, the effect of a change in any variable on the rate is a function of the level of the rate at the time of the change. To maintain consistency between our estimation and generalization methodologies, we must assume that both the estimated coefficients of the experimentally changed variables and the other variables from SIME/DIME apply to a national population.

Estimated marital status changes

We estimated the proportion of families headed by single women, the number of dissolutions and marriages, and the number of children in families headed by single women for both the current system and the three NIT programs. These estimates are reported in table 5.28. The top panel reports the estimates for all families in the generalization sample. The middle and bottom panels report the outcomes for upper- and lower-income families, respectively. The dividing point between the two income groups is annual family income (excluding welfare payments) of $12,501. The outcomes reported for the upper-income families need to be interpreted cautiously. The SIME/DIME sample contained few families with income that high. We must extrapolate beyond the region of our observations to obtain the upper income results.

TABLE 5.28.—SUMMARY OF ESTIMATED MARITAL STATUS CHANGES OVER 2 YEARS [1]

	No marital status changes	Current system	NIT program (tax rate/guarantee as percent of poverty level)		
			70/75	50/75	50/100
All families:					
Proportion single after 2 years	.122	.152	.154	.156	.158
			(1.3)	(2.6)	(3.9)
Number of dissolutions in 2 years (millions)		2.23	2.30	2.37	2.42
			(3.5)	(6.3)	(8.9)
Number of remarriages in 2 years (millions)		.94	.92	.92	.88
			(−2.0)	(−2.4)	(−6.2)
Number of children in families headed by single women after 2 years (millions)	11.19	12.20	12.48	12.64	12.76
			(2.3)	(3.6)	(4.6)
Upper-income families:					
Proportion single after 2 years	.022	.056	.056	.056	.056
			(0)	(0)	(0)
Number of dissolutions in 2 years (millions)		.90	.91	.91	.91
			(1.1)	(1.1)	(1.1)
Number of remarriages in 2 years (millions)		.11	.10	.11	.10
			(0)	(0)	(0)
Number of children in families headed by single women after 2 years (millions)	.85	2.04	2.05	2.05	2.07
			(0)	(0)	(0)
Lower-income families:					
Proportion single after 2 years	.246	.271	.276	.280	.285
			(1.8)	(3.3)	(5.2)
Number of dissolutions in 2 years (millions)		1.32	1.40	1.46	1.52
			(6.1)	(10.6)	(15.2)
Number of remarriages in 2 years (millions)		.83	.82	.81	.78
			(−1.2)	(−2.4)	(−6.0)
Number of children in families headed by single women after 2 years (millions)	10.34	10.16	10.43	10.59	10.69
			(2.7)	(4.2)	(5.2)

[1] Numbers in parentheses are percentage changes from the respective current system entry.

Initially, 12 percent of the family units in the generalization population were headed by single women. The model predicts that under the current welfare system 15 percent of the family units would be headed by single women after 2 years. The change in 2 years under the current system is proportionately much greater among the upper-income families than the lower-income families. The proportion increases from .022 to .056 among the upper income families, an increase of 155 percent, whereas among the lower-income families the proportion increases from .246 to .271, a 10 percent increase.[9] These increases are due to behavioral responses to the current system.

The differences between the current system and the NIT plans after 2 years are small. Under the NIT plans there would be somewhat more dissolutions and fewer marriages; the population would have more families headed by single women and more children living in these families. There is no change for the upper-income families. The proportion of families headed by single women under the most generous NIT plan is only 5.2 percent greater than what is predicted under the current system.

The predicted number of dissolutions during the 2-year period increases from 2.23 million for the current system to 2.43 million under the 50/100 NIT plan. Almost all the increase occurs among the lower-income families. The number of marriages decreases from the current system to the NIT plans; most of the change occurs among lower-income families.

The number of children in families headed by single women increases by about 1 million children from the start of the period until the end of the 2 years under the current system. This increase is concentrated among the upper-income families. In fact, the number of children in families headed by single women decreases in the lower income group. The number of children in families headed by single women after 2 years under the NIT plans is fairly close to the number under the current system. There is no difference between the current system and the NIT plans in the upper-income group and only a slight increase in the lower-income group.

The largest changes from the current system are for the most generous NIT program, 50/100. This appears to contradict our estimates in chapter 5, table 5.7 which indicated that the most generous SIME/DIME treatments had the least effect on marital dissolution rates. But the NIT programs in the generalization are not as generous as the SIME/DIME treatments. In fact, the 50/100 program is about as generous as the least generous SIME/DIME treatment. The NIT programs used in the generalization do not appear to be generous enough for income effects to offset independence effects. If more generous programs were simulated, the stabilizing income effects would begin to offset the independence effects and the pattern observed across the SIME/DIME treatments would appear.

Cost estimates

In comparing the costs of the three NIT programs to the costs of the present welfare system, we must distinguish between costs due to changes in the coverage of the programs and costs due to changes in the rates of marital formation and dissolution.[10] The former costs occur because the payments to eligible family units are more generous than the current system and because, under a more generous program, more families are eligible to participate. In particular, many two-parent families who are categorically ineligible for payments under the present system would be eligible for NIT payments. The latter costs result from changes in marital stability rates induced by the new programs. In presenting the cost estimates, we have been careful to distinguish these sources of differences in costs.

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[9] In Groeneveld, Tuma, and Hannan (1980a) we evaluated the marital stability generalization model by comparing its predictions under the current system to what was observed in the June 1975 CPS marital history survey. While differences in the data available prevented an exact test, it appeared that the generalization model overpredicted the percentage single at the end of 2 years for originally single women and underpredicted the percentage for originally married women. However, because of the way marital histories are reported in the CPS, women with frequent marital status changes did not have complete marital histories and could not be included in the analysis. This caused the June 1975 CPS marital history data to have somewhat lower dissolution estimates of marital status changes than actually occurred.

[10] Another source of difference in the cost estimates is participation. The cost estimates for the current system do not assume 100 percent participation. Current welfare costs are estimated by first estimating a probability of participating and then estimating the cost conditional on participating. The NIT costs assume 100 percent participation. Thus, some of the cost increases are due to increases in participation. The computer program used for the generalization did not allow us to determine what part of the cost increases are due to increased participation.

The cost estimates for the current welfare system and the three NIT plans are reported in table 5.29. The first row gives the annual cost of each program, assuming no change in the marital composition of the population. The costs range from $9.65 billion for the current system to $39.92 billion for the most generous NIT plan.[11]

TABLE 5.29.—SUMMARY OF INITIAL MARITAL STABILITY GENERALIZATION COST ESTIMATES

	Estimated cost (dollars in billions)			
	Current system	70/75	50/75	50/100
Annual cost with no marital status changes: all families	9.65	14.27	20.12	39.93
Increase in cost over current system		4.62	10.47	30.28
Annual cost with no marital status changes: married families only	1.84	5.06	10.27	27.15
Increase in cost over current system		3.22	8.43	25.31
Percentage of increase in cost due to extension of coverage to married families		69.7	80.5	83.6
Cost including changes in marital status: all families:				
First year	9.96	14.64	20.50	40.35
Increase in cost over no marital status change cost	0.31	0.37	0.38	0.42
Second year	11.16	16.06	21.93	41.97
Increase in cost over no marital status change cost	1.51	1.79	1.81	2.04
Estimated number of families initially included (millions)				
All families	5.47	7.65	13.29	22.11
Married families	2.20	4.01	9.43	17.87

The second row gives the difference between the NIT plans and the current system. Each entry is the increase in cost that results from switching from the current system to an NIT plan, assuming no changes in either marital status or labor supply are induced by such a switch. The next three rows show that the increase in cost mainly comes from including more two-parent families under the NIT plans. Two-parent families receive 1.84 billion dollars annually under the current welfare system, about 19 percent of the total payments. Under the NIT plans, the number of eligible two-parent families increases greatly (see bottom panel)—much more than the number of one-parent families. The more generous the NIT plan, the greater the extent to which the increase in cost is accounted for by the two-parent families. For the 50/100 program 83.6 percent of the cost increase goes to two-parent families who would receive 68.0 percent of the NIT payments.

The bottom panel of table 5.29 shows the estimated increase in the number of families who would receive benefits under each plan. As in the case of the cost increases, the increase in coverage under the NIT plans is concentrated in the increase in the number of two-parent families.

The next four rows give the costs of the four programs assuming marital status changes. Since the underlying model of marital stability is dynamic, the costs vary over time. The costs for each of the first 2 years are reported. The change in costs due to changes in marital status are small in the first year and larger in the second year. Under the current system the increase in costs due to marital status changes is 0.31 billion dollars in the first year and 1.50 billion in the second year. The model predicts about a 16 percent increase in the annual cost of the current welfare system due to changes in marital status. The increases in cost due to changes in the distribution of marital status are slightly larger under the NIT plans than under the current system but actually represent a smaller percentage of the total cost.

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[11] The cost estimates we have reported are higher than those reported in the labor supply generalizations (part III) because the sample selection criteria used in the two generalizations were somewhat different, and because the marital stability generalization included a "hold harmless" provision not present in the labor supply generalization. The result of the hold harmless provision is to increase the costs of the NIT program relative to the current system.

Table 5.29 indicates that the cost increases of national NIT plans like SIME/DIME would be substantial, even for the less generous plans. Cost increases are mainly due to the extension of coverage to two-parent families, not to changes in the fraction married in the population.

The costs in table 5.29 need to be interpreted with some caution. Part of the costs are payments to families whose incomes are higher than those in the SIME/DIME sample. In table 5.30 the costs are reported separately for upper and lower income families, with an annual nonwelfare income of $12,500, dividing the two groups as in table 5.28. Table 5.30 shows that the increase in cost due to changes in marital status occurs disproportionately among the upper income families. This is consistent with the results in table 5.28, which indicated large percentage increases in the proportion single in the upper income group. The changes in costs due to marital status changes in the upper income group must be interpreted cautiously because they extrapolate beyond the income range of the SIME/DIME sample.

TABLE 5.30.—MARITAL STABILITY GENERALIZATION COST ESTIMATES BY INCOME LEVEL

[In billions of dollars]

	Current system	70/75	50/75	50/100
Upper Income Families				
Annual payments with no marital status changes	0.15	0.15	0.55	4.93
Annual payments with predicted marital status changes:				
First year	.40	.43	.85	5.31
Second year	1.26	1.44	1.93	6.69
Lower-Income Families				
Annual payments with no marital status changes	9.50	14.12	19.58	35.00
Annual payments with predicted marital status changes:				
First year	9.57	14.21	19.65	35.04
Second year	9.89	14.62	20.00	35.28

Comparison with the estimated experimental effects

The results we reported in the preceding section indicate that the short-run marital stability effects of the national NIT plans are modest. This finding may seem inconsistent with the estimated effects reported in Chapter 5 which indicated 50 percent increases in dissolution rates for blacks and whites. How can these predictions be reconciled with the small increases in the number of divorces and in the proportion single estimated for the 50/100 NIT plan which is close to the average low guarantee plan in SIME/DIME?

There are two reasons why we observed large changes in the dissolution rates in SIME/DIME and yet obtained small differences between the current welfare system and the NIT plans. The first reason is that the generalization covers only the first 2 years. In Tuma, Hannan, and Groeneveld (1977) we explored the time-path from the time an NIT plan is initiated until a steady state is reached. We found that, given changes in the rates of the magnitude we observe in SIME/DIME, the proportion single was only halfway to the steady-state value. Thus, the differences between the current system and the NIT plans in the first 2 years are smaller than the differences that would occur later.

The second reason that we observed large changes in the dissolution rates in SIME/DIME and yet obtained small differences between the current welfare system and the NIT plans in the generalization is that the income distributions of the two populations are very different. Our income and independence model demonstrated that marital dissolution rates varied with family income and with the income the wife could expect outside the marriage (her independence income). The change in the dissolution rates in response to the SIME/DIME NIT plans depended on both initial levels of income and on the magnitudes of the changes in income. The SIME/DIME sample was concentrated in the lower-income range where dissolution rates are most sensitive to changes in income. The NIT treatment effects we estimated (e.g., tables 5.4 and 5.5 in chapter 5) represent the average change in marital dissolution rates from the control environment to the NIT treatments among low-income couples, not among a representative sample of the U.S. population. The purpose of the generalization is to extrapolate from the SIME/DIME sample to the U.S. population. The results of the generalization reflect the fact that the majority of the U.S. population is not concentrated at the income levels where the greatest responses to the NIT plans occur.

This can be seen more clearly by comparing the income distributions in SIME/DIME and in the generalization and by examining how the marital dissolution rate varies with income. Our model specifies that the dissolution rates depend on both family income and the wife's independence income, so we should look at joint income distributions. However a table cross-tabulating family income by independence income would have too many entries to be easily interpretable. In table 5.31 we present the population distributions for married couples from the CPS used in the generalization and from the sample of SIME/DIME couples eligible for the NIT treatment. Table 5.31 also includes the mean dissolution rate for each income level computed in the generalization for both the current system and the 50/100 NIT. The 50/100 NIT is closest to the treatments in SIME/DIME.

TABLE 5.31.—POPULATION DISTRIBUTIONS AND MEAN DISSOLUTION RATES FOR MARRIED WHITES AND BLACKS, BY FAMILY INCOME

	Percentage of married population		Mean dissolution rate		Percentage change in mean rate
	CPS	(SIME/DIME) experimentals	Current system	50/100 NIT	
Whites:					
$0	0.6	12.4	.048	.077	60.4
$1 to $1,000	0.5	4.9	.053	.089	67.9
$1,001 to $3,000	1.5	15.6	.056	.091	62.5
$3,001 to $6,000	5.6	43.5	.052	.066	26.9
$6,001 to $10,000	14.6	21.0	.042	.044	4.8
$10,001 to $12,500	14.0	2.0	.030	.031	3.3
$12,501 to $15,000	12.9	0.4	.024	.024	0
$15,001 to $20,000	22.4	0	.019	.019	0
$20,001 or more	27.9	0.1	.011	.011	0
Weighted average dissolution rate using the CPS distribution			.025	.027	8.0
Weighted average dissolution rate using the SIME/DIME distribution			.049	.067	36.1
Blacks:					
$0	0.5	9.1	.053	.083	56.6
$1 to $1,000	1.2	4.4	.086	.145	68.6

TABLE 5.31.—POPULATION DISTRIBUTIONS AND MEAN DISSOLUTION RATES FOR MARRIED WHITES AND BLACKS, BY FAMILY INCOME—Continued

	Percentage of married population		Mean dissolution rate		Percentage change in mean rate
	CPS	(SIME/DIME) experimentals	Current system	50/100 NIT	
$1,001 to $3,000	4.6	11.7	.068	.147	116.2
$3,001 to $6,000	12.4	39.3	.091	.137	50.5
$6,001 to $10,000	24.2	32.7	.093	.125	34.4
$10,001 to $12,500	14.5	2.2	.095	.107	12.6
$12,501 to $15,000	12.8	0.6	.100	.104	4.0
$15,001 to $20,000	16.2	0	.103	.102	−0.1
$20,001 or more	13.6	0	.077	.077	0
Weighted average dissolution rate using the CPS distribution			.092	.112	21.7
Weighted average dissolution rate using the SIME/DIME distribution			.085	.129	51.6

The family income variable is total family income (excluding welfare) in 1974 dollars. The income distribution for the CPS is for the population of married couples included in the generalization sample. The SIME/DIME income distribution is preexperimental family income adjusted for inflation using the CPI. It is similar to the CPS income variable. Notice how the income distributions for the generalization and SIME/DIME differ. The SIME/DIME sample is concentrated in the range of $0 to $6,000. In the national population only 8.2 percent of the married whites and 18.7 percent of the married blacks have income in that range.

Table 5.31 also includes the mean dissolution rates from the generalization for each income level. Rates for both the current system and the 50/100 NIT are reported. These are the rates used to produce the generalization outcomes. For whites the mean dissolution rates generally decrease with increases in family income, both under the current system and under the 50/100 NIT plan. For blacks, the mean dissolution rates do not decrease with increasing family income. This is due to the effects of the wife's independence income on the rates. In similar tables where independence income is controlled the mean rates for blacks decrease monotonically as family income increases.

The columns labeled "percentage change in mean rate" are simply the 50/100 NIT mean rate divided by the current system mean rate times 100 for each income level. This is a rough approximation of the change in the rates for families in that income group.[12] Table 5.31 indicates that the greatest changes in the mean dissolution rates occur at the lowest income levels where the SIME/DIME sample is concentrated and the generalization population is thinnest. This indicates how averaging over the SIME/DIME sample would produce much higher changes in the dissolution rates than are produced by averaging over the generalization population.

For both races we calculated the weighted average dissolution rate first for the CPS sample and then for the SIME/DIME sample. Again, these are not the mean rates but rather the weighted averages calculated from the table. The true means would be somewhat different because of the nonlinear dependence of the rates on family income. Nevertheless, they are sufficiently close to the true means to illustrate the point. Using the CPS distributions increases the mean dissolution rate 8.0 percent for whites and 21.7 percent for blacks from the current system to the 50/100 NIT. Using the SIME/DIME distributions increases the rates 36.1 percent for whites and 51.6 percent for blacks—increases that are much closer to those found in our analysis of the NIT treatment effects.

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[12] It is not, however, the mean change in the rate. Since the generalization output did not include mean changes in rates, we will use this surrogate.

Discussion

The generalization of the SIME/DIME marital stability results indicates that a national NIT plan similar to the plans tested in SIME/DIME would have small effects on marital stability after 2 years. The number of dissolutions and marriages predicted by the model under the NIT plans does not differ greatly from the numbers predicted under the current system. The increases in cost of the NIT plans are due primarily to the inclusion of more two-parent families than under the current system. The cost increases due to increased dissolution rates are small in comparison.

These findings need to be interpreted cautiously, however, for several reasons. First, to generalize we must extrapolate our findings beyond the range of the SIME/DIME sample. We do not know how well the model fits populations that have incomes much higher than those observed in the SIME/DIME sample. Note especially the predictions for the current welfare system. The model predicts a large increase in the proportion of the families headed by single women in the income range above the SIME/DIME sample. We have no way at this time to test whether these predictions are realistic.

Second, these results apply only to NIT plans with rules of operation similar to those in SIME/DIME. We cannot say how differences in the administration of the programs or in rules of eligibility would affect the outcomes. In particular, we do not know how work requirements or job guarantees would affect a generalization. Throughout our work we have argued that the way welfare programs are administered is an important factor in determining a program's effects on marital stability. The impact of the administrative aspects of NIT plans on marital stability should not be ignored in policy discussions. But SIME/DIME does not provide data that give quantitative answers about such impacts.

Third, the cost estimates do not include any administrative costs. In 1974 (the year on which the generalization is based), State and local administrative costs accounted for about 11 percent of the costs of the AFDC program. A national NIT program with a simpler administrative procedure than the current system might offset some of the increased costs through administrative savings.

Further work is necessary before marital stability generalizations such as the one presented here can be used confidently as a policy tool. First, work should be done to determine the reliability of the predictions for the current welfare system. Second, repeated generalizations should be done to determine the sensitivity of the predictions to the various assumptions in the model. Third, more work needs to be done on modeling the rate of marital formation. The SIME/DIME sample and the generalization population are not representative of the population of women at risk of marrying; in particular, both exclude childless single women. Fourth, research is needed on both marriage and dissolution rates in families with higher incomes than in the SIME/DIME sample. Given the U.S. income distribution, costs of a national income program could be altered drastically if marital formation and dissolution in such groups were sensitive to such a program. The designers of SIME/DIME assumed such affects would not occur, but this assumption has no empirical basis.

We believe that the results of this generalization are primarily useful for comparing different NIT plans. They provide an indication of the relative size of the various plans' short-run effects on marital stability. Until further work is done on the methodology of generalizations, we urge that these findings be interpreted cautiously.

CHAPTER XI

DISCUSSION AND CONCLUSIONS

Summary of the findings

We have discussed the effects of the NIT treatments on marital dissolution and marriage rates at length. The main findings are clear: the NIT treatments significantly increased marital dissolution rates among whites and blacks (by 40 to 60 percent—see tables 5.4 and 5.5 in chapter 5) but did not affect remarriage rates. The treatments did not affect dissolution rates of Chicano couples but did reduce remarriage rates by over 60 percent for single Chicano women (see table 5.25 in chapter 9). If we wish to adjust the dissolution effects to account for attrition bias we should reduce these effects by about 10 percent for blacks and 5 percent for whites (see chapter 6).

However, these estimates are unsatisfactory as a summary of the experimental effects because they obscure much of what we have learned about them. First, the experimental effects vary by the generosity of the treatments. The most generous NIT treatments has the least effect on dissolution rates in all three race-ethnic groups. In some cases (for example, the combination of the highest guarantee level and the lowest tax rate), generous programs actually lowered dissolution rates slightly. The $5,600 guarantee level treatments increased dissolution rates by 20 percent for blacks and 14 percent for whites (neither increase was significant). The less generous $3,800 guarantee level treatments significantly increased the dissolution rate 60 percent for blacks and 82 percent for whites.

The treatment effects also vary by the composition of the families. The treatment increased dissolution rates more among white childless couples than among those with children. The treatment effect is greater among black couples with young children than among other couples (see tables 5.8 and 5.9 in chapter 5).

If one wishes a single set of numbers to summarize our findings one might choose the effects of the $3,800 guarantee level treatments because it is closest to the current system and to likely welfare reforms. The results should emphasize couples with children since it is likely that any NIT program implemented nationally would differentiate childless couples as the present system does. Taking the estimates from chapter 5's table 5.8 and adjusting for attrition bias, one obtains estimates of 58 percent increases in dissolution for blacks and 51 percent increases for whites. These are estimates of the experimental-control differences in the SIME/DIME population for the most feasible programs tested.

We caution the reader against uncritically extrapolating from these or any other summary measures of the effects of SIME/DIME on marital dissolution. A 51 percent increase in a marital dissolution rate in the experimental sample cannot be interpreted as predicting a 51 percent increase in marital dissolution rates if a national NIT program is implemented. Our generalization results (chapter 10) show that the effects in a national population would be much smaller because the magnitude of the effect depends on the level of income and on the independence of wives. Had the income distribution of the SIME/DIME sample been different the average effects would be different.

There are several other factors that need to be considered in interpreting our findings. First, the relationship of the effects of a limited duration experiment to a permanent national program are ambiguous. While we are convinced that a short-term experiment understates the effects of a national program, some analysts will argue the opposite. Second, nonpecuniary differences among programs may have important effects. Third, the social context of the experiments and of a national program may differ in important respects. Fourth, the effects of a national program on remarriage may differ from the effects of the experimental treatments because of the incentive to marry into the experiment. Last, the differences among the race-ethnic groups in the treatment effects indicate to us that we do not yet fully understand the experimental effects. We address each of these points in the following sections.

Nonpecuniary differences in welfare programs

Any explanation of the pattern of guarantee level effects must deal with the fact that the lowest guarantee level significantly increased dissolution rates, while the highest guarantee level had the smallest impact (or, in some cases, actually stabilized marriages). Benefits under the lowest guarantee hardly differ in financial terms from the combined benefits of AFDC-U and food stamps, which are available to all controls. Why then did it affect behavior? The income transfer programs available to the control and treatment groups differed in important nonpecuniary ways which are likely to affect behavior. We think that welfare benefits available to controls are "discounted" in their behavioral effects.

Nonpecuniary differences between programs may explain why the low guarantee treatment affected marital dissolution rates. They do not, however, explain why the pattern of effects reverses. Our survey of the previous theoretical and empirical literature (see chapter 2) suggested that there must be at least two opposing effects, which we labeled income and independence effects. We constructed a mathematical model in which the rate of marital dissolution depends nonlinearly on family income and the income available to the female head and the children if the marriage were to break up. According to this model, both wife's earnings and nonwage income transfers like NIT payments have income effects as well as independence effects. On the one hand, they stabilize marriages, improving the family's ability to purchase essential goods and services. On the other hand, they destabilize them by improving the economic quality of alternatives to marriage. Since the hypothesized effects are nonlinear, the effect of any particular change in nonwage income (or in the earnings of either spouse) depends on the family's current income. A given NIT treatment will stabilize some marriages and destabilize others. NIT treatments with a low guarantee level will have a reasonably strong independence effect but a negligible income effect. As the guarantee level rises, the strength of the income effect rises relative to the independence effect. If the guarantee level is sufficiently high, the income effect will dominate for many families, leading to a decrease in the overall dissolution rate. Thus, this model can explain why a program with a low guarantee level increases the rate while a program with a high guarantee level decreases it.

This model has been reasonably successful in explaining the pattern of results in SIME/DIME. We find that the income and independence effects operate as expected for all three race-ethnic groups. The estimated effects have the predicted signs and have steep slopes within the observed range of income. The two effects are statistically significant for blacks and whites even when we include the NIT payments. We interpret this result as indicating that NIT guarantees and payments have both income and independence effects.

This model explains almost the entire observed response in the white sample. Allowing NIT payments to have income and independence effects reduces the multiplier for the pure NIT treatment almost to unity. However, the model does not explain the complete response in the black sample. Even with the NIT payments included in the model, black couples in which wives are working were still much more likely than comparable controls to end their marriages. We discuss possible implications of this finding below.

Extrapolating from a limited duration experiment

Another important set of questions about these findings focuses on the importance of the limited duration of the experiment. Did the fact that income guarantees expired after 3 or 5 years affect dissolution rates? When our research began, most social scientists interested in these issues assumed that a 3- or 5-year period was too short to detect effects on marital decisions. They assumed that evidence from a short-term program would understate the effect of a permanent program. Comparing the effects of the 3- and 5-year treatments suggests that the limited duration experiment did understate the effects of a permanent program since the longer program had stronger effects. However, once it became clear that SIME/DIME treatments affected marital dissolution rates, intuitions shifted.

Questions began to focus on the possibility that a short-term income guarantee might overstate the long-term effect of NIT programs. We have encountered two serious arguments on this matter. One suggests that a short-term program creates a "sale" on dissolutions that leads people to shift events forward in time, creating an initial surge of dissolutions. The second argument contends that any new income supplement program creates a temporary disequilibrium in marriage markets. Due to the disequilibrium, the rate of dissolution rises as the market reequilibrates, at which time the dissolution rate should fall to its old (preintervention) level.

When some commodity is offered at an exceptionally low price for a brief period, sales rise. Buyers who had planned to purchase the commodity tend to shift their purchases forward in time. But, even though sales rise in the short run, overall consumption of the commodity might not increase. We have argued that the NIT plans lowered the costs of marital dissolution (the independence effect). Were dissolutions that would have occurred anyway moved forward into the "sale" period? If so, a rise in the dissolution rate over the short term need not imply a long-run increase in the dissolution rate under a permanent NIT.

The sale theory implies that the increase in the rates during the experiment would be offset by lower rates after the experiment since the number of dissolutions does not increase in the long run. The evidence does not support this expectation: the dissolution rates for experimental couples during the posttreatment period are the same as the rates of comparable control couples (see table 5.5 in chapter 5).

The second hypothesis about timing has more substantive appeal. Becker's economic theory (1973 and 1974) of marriage implies that the environment affects the kinds of marriages that occur. Suppose that the environment has been stable long enough (relative to rates of marriage and dissolution) for the marriage market to have reached equilibrium. If the environment changes unexpectedly (as when an NIT program suddenly appears), some persons will find themselves in the wrong marital status and others will find that they no longer have an optimal spouse. If they act rationally and the transaction costs are small, those persons will alter their circumstances. The marriage market will undergo a period of intense reshuffling, implying that rates of dissolution will rise temporarily. Eventually, according to this theory, the marriage market will hit a new Pareto-optimal equilibrium. At this point, the dissolution rate will fall to some low level that reflects localized uncertainties and random events.

This theory implies that the dissolution rate will rise in the short run in both a limited experiment and a national program. However, the rise in the rate reflects only the mobility required to reequilibrate the marriage market. The program may not, in other words, change the long-run rate. Evidence from the first few years of an experiment or a national program might overstate the long-term effects of an NIT on marital stability.

The empirical work of Becker, Landes, and Michael (1977) offers one point of contact with our work. They propose that unexpected changes in income (either positive or negative) destabilize marriages. The strength of this effect supposedly increases monotonically with the size of the change in income. Applying this logic to the NIT experiment (which yields an unexpected increase in nonwage income) leads to the prediction that dissolution rates will rise most sharply for those families who receive the most generous payments. But, as we reported, the most generous program (the high support/low tax combination) actually decreased dissolution rates for black and white couples. At the same time, couples assigned to less generous programs had sharp increases in rates. There is no evidence that the largest changes in income produce the largest marital stability response in this experiment. Indeed, our specification of income and independence effects permits payments to families to have a stabilizing effect. The empirical estimates agree.

Keeley (1980) has modified the Becker model to take account of effects of an NIT program on the tradeoffs between marriage and singleness. After assuming signs for certain theoretically indeterminate effects (involving cross substitutions), he shows that the model is consistent with the experimental results. However, his final specification allows the NIT treatment to either increase or decrease the dissolution rate; in this respect it mirrors our arguments about income and independence effects. Keeley has changed the model from one that contains only a short-term shock effect into one that specifies the ways in which changes in tax rates and nonwage income flows (and guarantees for single persons) affect dissolution rates. The revised model implies that an NIT could affect long-run rates.

Earlier (Tuma, Hannan, and Groeneveld, 1979), we proposed a variety of reasons why the response to an NIT program might vary over time. Some of the arguments suggest that the initial effect might understate the long-term effect (as people learn about the system and gradually become accustomed to receiving payments). Arguments, such as those discussed above, suggest that the initial response might overstate the long-term effect. We showed that the initial response was very strong for whites, but small for the other two race-ethnic groups. However, further inquiry revealed that the fluctuations over time reflected variation in the rate in the control groups over the experimental period; the rate for those on NIT plans was quite stable over time. We examined the time-dependence of the response with the final data set and found no significant variation over time in the effects of NIT treatments, except that the effects disappear when the treatments end, as they should. Therefore, we conclude that our findings do not represent a strong initial response that will decline markedly over the next few years. Still, we cannot be confident that the long-term effect (say 20 years after implementation of an NIT program) will not differ markedly from the effect in the first few years.

It is important to repeat a warning made earlier (Tuma, Hannan, and Groeneveld, 1979) that even short-term increases in dissolution rates have an impact on the numbers of marriages for a reasonably long period. For example, an effect that increases dissolution rates for only 2 or 3 years may take as much as 20 years to disappear (for the proportion of people who are married to return close to the initial level). This fact is an implication of the low rate of remarriage—once a woman becomes single, she typically remains single for some time. Stocks of marriages do not respond quickly to changes in the underlying rates. This is one reason why the predicted effect in the generalization over 2 years is so small. This fact should alter the ways in which we think about short-run and longrun effects. Short-run increases in dissolution rates imply that the number of families headed by females will remain elevated for some time.

Effects of social factors

Economists tend to view the treatments as combinations of guarantee levels, tax rates, periods of eligibility, and accounting periods. Indeed, the first three of these are the parameters that were manipulated in the experiment. It is natural to confine attention to the formal (designed) properties of the treatments. But from a sociological perspective, the treatments are more complex. Beginning with our first research report on the marital dissolution effects of SIME/DIME, we have argued that the treatments mix designed variations in financial parameters with a social context. The financial treatments were embedded in an administrative structure and delivered to families by a social structure. The SIME/DIME treatments included a rationale for participation, a structure for collecting information, and a full-fledged welfare bureaucracy that determined eligibility, delivered checks, and handled questions and complaints. Families with NIT treatments faced different social structures from those faced by control group families, which may account partially for the rise in the dissolution rate among those on NIT plans. To learn exactly how the experimental effect came about, we must understand how the two structures differed. Unfortunately, the experiment did not contain designed variations in the structure that delivered the treatments. Therefore, our arguments and inferences about these important matters must be indirect.

Recognition that social factors played a role suggests to some people that the SIME/DIME findings do not apply to other situations—that the results reflect an "administrative artifact." This phrase connotes that the SIME/DIME treatments contained special incentives for marital dissolution that are not present in other income support programs. This characterization may be partly true, as we indicate below. However, we think that it is a mistake to conclude that the effects are artifactual. All transfer programs use specific social structures to determine eligibility and distribute benefits. We think that the research on marital stability in SIME/DIME has made an important contribution to policy analysis by focusing attention on the nonpecuniary features of income support programs. We have shown that marital decisions in low-income populations are sensitive to them.

We emphasized three of the many possible differences between NIT treatments in SIME/DIME and existing welfare programs: transaction costs associated with participation, information about eligibility, and personal and social stigma associated with participation. We think that participation in SIME/DIME was less costly in terms of time and inconvenience (transaction costs) and in terms of social acceptability (stigma) than participation in AFDC. In addition, we think that knowledge about eligibility was more widespread among those on NIT programs than among the control group.

It is not clear exactly how the SIME/DIME treatments and a national NIT program compare on these dimensions. Not the least of the problems in making such comparisons is ignorance about the likely structure of a national program. For a baseline comparison, we might consider Milton Friedman's original proposal for an NIT program. He proposed using the existing positive tax system to determine eligibility and to disburse benefits. Such an arrangement would have minimal bureaucratic overhead. Since most families already file income tax forms, the additional transaction costs associated with participating in this kind of structure would be small—presumably smaller than in SIME/DIME (which required thrice-yearly interviews and monthly income reports). Moreover, knowledge about eligibility and benefits would be widespread because participation would be tied to the nearly universal obligation of filing income tax returns. It is not obvious that the receipt of NIT payments in such a system would involve stigma. We assume that the stigma associated with participation in an income support program declines as the program becomes more universal in coverage and more formalistic in the calculation of eligibility and benefits. If the NIT payments were viewed in the same light as refunds on overpayments on positive taxes, personal or social costs involved in receiving payments would be minimal. Considering all three factors, it seems likely that an NIT scheme based on the current positive tax system would have a more powerful effect on marital stability than the schemes used in SIME/DIME.

Alternatively, a national NIT program might build on existing State and local bureaucracies. If so, the transaction costs of participation might be considerably higher than in the scheme discussed above. But, it is hard to imagine that the existing bureaucracies are capable of processing the millions of additional participants using current procedures. The additional caseload would probably push the system toward more formal and limited connections between clients and bureaucracies. Such shifts would lower the costs of participation relative to the present system. If the system were administered locally, involving registration with an existing welfare bureacracy, the social and psychological costs of participation might differ little from the present situation. Considering all of these possibilities together, we surmise that such a system would fall between the SIME/DIME structure and today's welfare system in terms of impacts on marital stability. In other words, we speculate that shifting to such a system would increase the dissolution rate but not as much as in SIME/DIME.

Finally, a national NIT program might impose much higher costs on participants. These costs might include extensive or demeaning procedures for enrolling and reporting, stigma arising from some form of public identification as a member of the undeserving poor (perhaps in the form of required participation in visible "make work" jobs), and so forth. Clearly, such programs would have smaller impacts on marital stability than those in SIME/DIME. In other words, we are suggesting that higher rates of marital dissolution may be one of the social costs associated with a universal program of income support that treats recipients with dignity.

Considering these three nonpecuniary characteristics, we think that the structure implemented in SIME/DIME approximates reasonably well the national programs that have been proposed in recent years. Still, the fact that families participated voluntarily has no counterpart in a national program. That the program had no clear public identity also means there was no generalized negative connotation associated with receiving payments. This might not be true in a national program. In these two respects, we judge that the results from SIME/DIME overstate somewhat the likely increase in dissolution rates in similar populations under a national program with similar financial characteristics.

Effects on remarriage rates

Another important question about the general implications of our findings focuses on remarriage rates. Our experiment did not include all members of the marriage market in either Seattle or Denver. When experimental subjects entered a marriage, they almost always married someone who was not a subject in the experiment. The SIME/DIME rules of operation extended eligibility to new spouses (immediately if the marriage was official and after a 3-month waiting period otherwise). The purpose of this rule was to eliminate a possible disincentive for single adults to marry under an experimental NIT program. However, this rule may have improved the status of single adults in the general marriage market—they controlled eligibility to the NIT program, providing a kind of dowry which may have made them more attractive as potential spouses. The dowry effect should increase the rate of marriage for single adults because it improves the quality of marriage offers. However, such an effect is an artifact of the experimental design and rules of operation. It would not exist in a national program.

It turned out, as we reported in chapter 9, that the remarriage rates of black and white women with NIT treatments did not differ from those of comparable controls. Chicanos on NIT treatments had significantly lower rates. If there was a positive dowry effect, there must have been an offsetting negative NIT effect on marriage rates for black and white women. Since the dowry effect would not exist in the national program, the marriage rate for unmarried women with children would presumably decline. If this reasoning is sound, its implications are disturbing. The combination of rising dissolution rates and falling remarriage rates could imply a substantial increase in the proportion of single-parent families.

We do not give much credence to the hypothesized dowry effect. There is no independent evidence that such an effect operated in SIME/DIME. It seems much more likely that the NIT treatments did not affect remarriage rates of black and white women and lowered the remarriage rates of Chicanos.

After considering each of the questions that have been raised about the general importance of our findings, we take a fairly positive position. We agree that the social structural (nonpecuniary) features of SIME/DIME's negative income tax treatments may not be duplicated in other income support programs. The SIME/DIME program was unusual in that it sought to enroll families and tell them about their rights. It was also unusual in that it did not have a cultural definition and therefore did not stigmatize participants. For these reasons, we conclude that the effects in most feasible national programs would be smaller than those we estimated. Unfortunately, we cannot tell how much smaller they would be. We do not understand the relative importance of information, transaction costs, and stigma. Neither do we know how a national program would be structured on these dimensions. Still, we think that general income support programs will affect rates of dissolution if they are structured at all like SIME/DIME.

Race-ethnic group differences

Finally, there is the question of the differences in response among the three ethnic groups. Because the literature emphasizes cultural differences in marriage patterns and processes among such groups, we have always allowed full interactions between ethnicity and all other variables of interest. When we have tested explicitly whether the data could be pooled across the three samples, we have found that the differences in response were statistically significant. Both substantive and statistical considerations have argued in favor of analyzing each of these groups separately. Unfortunately the sample sizes for blacks and whites appear just barely large enough to detect substantively important changes in rates over the period of the study, while the Chicano sample is too small. This means that ordinary sample fluctuations generate seemingly large differences among the three groups. There is a real danger of overinterpreting the data when we try to reconcile all of the observed differences. It is unreasonable to expect estimated effects to be very similar in three random samples ranging in size from 500–1,000 with rates as small as those we have estimated.[1] Furthermore, our testing indicates that these three samples are not from the same population; however, it does not tell us how the populations differ. It is here that the danger of overinterpretation arises.

The problem of sample size is most severe for the Chicano sample, which is roughly half as large as the other two. The small size of this sample made it very difficult to obtain precise estimates on effects. This difficulty is compounded by the fact that we know so little about Chicano family structure. The research literature has paid little attention to Chicano families. Moreover, as noted, there is some evidence (Groeneveld, 1979) that the Denver Chicano population differs in important ways from the larger Southwest and West Coast populations.

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[1] The annual marital dissolution and remarriage rates were between 0.05 and 0.12 in the SIME/DIME samples.

Given these uncertainties, we hesitate to make strong claims about the likely effects of NIT programs on Hispanic families generally. Our data suggest that NIT programs may have stronger effects on their rate of remarriage than on their dissolution rate. As we have shown in chapter 9, the remarriage rate for Chicano women in the control group was roughly four times as high as the rates for black and white women. But for Chicanos receiving NIT treatments the remarriage rate was close to those of the black and white women. The experimental treatment surely has an effect. In this case it seems that the presence of an effect for Chicanos but not for black and white women reflects an interaction of a cultural pattern with the NIT. Apparently, the existing social and economic supports for a single life for Chicanos with children are slight, leading them to remarry at high rates. Providing only modest financial supports for the single status strongly affects behavior. If such an effect existed in a national program, it would speed the convergence of Chicano family demography toward the black-white pattern.

Because we have larger samples of black and white families, we have focused more attention on them. We have tried to reconcile differences in response between blacks and whites. In this section we comment further on such differences. However, it is important not to lose sight of the similarities in response. The total effects of the NIT treatment are similar for the black and white samples. In each sample, the NIT treatment increased the dissolution rate significantly, about 40 percent. In broad terms, we hardly need to discriminate between the two groups.

The differences in response between blacks and whites concern the details of the process. We found that among black couples the NIT treatment effect is concentrated among families with young children, whereas for whites the treatment effect is strongest among childless couples. Thus, family composition must be considered in any attempt to understand the race-ethnic differences.

The response of black (but not white) families to NIT treatments depends strongly on the wife's current employment status—the NIT treatments increased the rate of dissolution for blacks only when wives were employed. This difference between races persists through our analysis of income and independent effects. Income and independence effects must be supplemented by some other process(es) for blacks.

We are left with the puzzle of why the effects of NIT treatments on the marital dissolution rate are much stronger for blacks when the wife is employed. This finding is especially puzzling because wife's employment status does not affect the dissolution rate in the black control group. In a sense, the NIT treatment sensitized black marriages to the wife's employment status. Put this way, the finding is all the more curious, since wife's employment is much more common in the black sample than in the white or Chicano samples. Why should a common status become more salient under an NIT program?

Perhaps black family structures have been adapted to high levels of wives' financial independence. After all, black women have for a long time contributed significantly to family earnings. Suppose that the eligibility for NIT payments upon leaving a marriage is functionally equivalent to wife's employment in generating financial independence. Given this historical context, a dose of independence of this magnitude does not markedly alter the relative advantages of being married or single. For couples in which the wife is not employed, the introduction of the NIT does not make much difference. But employed wives receive a double dose—they receive an income guarantee in addition to their own demonstrated earning power. Perhaps the double dose has much more than twice the behavioral impact of the single dose for blacks simply because the black population has by its history become inured to the latter.

Conclusion

We have reported the results of many analyses that were carried out to answer the questions of whether the NIT treatments affected marital dissolution and remarriage rates and how those affects came about. We are convinced that marital dissolution rates among whites and blacks were higher and remarriage rates among Chicanos were lower as a result of the experimental NIT treatments. The magnitudes of these effects are substantial: increases in the dissolution rates of 40 percent to 60 percent among black and white couples eligible for the 5-year treatment, and a decrease of over 60 percent in the remarriage rate of Chicano women.

We have discussed a number of factors that complicate attempts to extrapolate from SIME/DIME to any other NIT program. None of these factors alter our qualitative conclusions. Given the magnitude of these findings, it is unlikely that any national NIT program would be neutral with respect to marital stability. Although the effects of a national NIT program are unlikely to be as dramatic as the experimental effects, the potential for such effects must not be ignored.

APPENDIX A

ESTIMATED EFFECTS OF ASSIGNMENT AND CONTROL VARIABLES ON MARITAL FORMATION AND DISSOLUTION RATES

TABLE 5.A.1.—ESTIMATED EFFECTS ON MARITAL DISSOLUTION RATES DURING THE FIRST 3 YEARS FOR ORIGINAL MARRIEDS[1]

	Blacks		Whites		Chicanos	
	Parameter	Standard error	Parameter	Standard error	Parameter	Standard error
Normal earnings:						
$0 to $999	.40	.36	[3] 1.01	.44	−.02	.80
$1,000 to $2,999	−.11	.35	[4].89	.31	−.19	.44
$3,000 to $4,999	−.10	.22	.66	.27	.14	.37
$5,000 to $6,999	−.28	.20	[3].52	.26	.06	.35
$7,000 to $8,999	[2]−.34	.20	.23	.27	−.01	.37
$9,000 to $12,999	...	...	...	...	...	...
Unclassified	−.59	1.02	1.37	1.03	7.52	70.22
Site (Denver=1)	[3].28	.14	−.20	.14	...	...
Duration married	[4]−.05	.01	[3]−.10	.03	−.03	.03
Wife's age	−.01	.02	.01	.02	−.06	.03
Wife's education	.01	.05	[3]−.08	.04	−.03	.05
Husband's age	[3]−.03	.01	−.02	.02	−.00	.02
Husband's education	[3]−.08	.03	.02	.03	.03	.04
Number of children	.08	.05	.05	.06	.13	.09
One if child younger than 6	[2]−.29	.16	[2]−.29	.16	−.43	.28
One if preexperimental AFDC	.05	.19	[4].50	.19	[4].67	.24
Manpower treatment:						
Counseling only	[2].45	.20	.32	.21	.52	.27
50% subsidy	.30	.20	.14	.20	.13	.29
100% subsidy	.26	.21	[2].33	.20	.18	.32
5-year manpower treatment	−.38	.26	−.29	.27	−.04	.40
NIT treatment	[3].45	.21	.43	.22	.01	.34
3-year NIT treatment	−.30	.23	−.33	.24	−.00	.37
Constant	.07	.73	[2]−1.59	.69	−1.01	1.06
Number of cases	939	...	1,299	...	538	...

[1] The estimates reported here are for the equation reported in Table 5.2 for original marriages.
[2] Significant at the 10-percent level.
[3] Significant at the 5-percent level.
[4] Significant at the 1-percent level.

TABLE 5.A.2.—ESTIMATED EFFECTS ON MARITAL FORMATION RATES DURING THE FIRST 3 YEARS FOR ORIGINALLY SINGLE WOMEN[1]

	Blacks		Whites		Chicanos	
	Parameter	Standard error	Parameter	Standard error	Parameter	Standard error
Normal earnings:						
$0 to $999	−.69	.51	−.65	.50	−.61	.88
$1,000 to $2,999	[3] −1.14	.50	−.73	.48	−.86	.85
$3,000 to $4,999	−1.01	.48	−.58	.46	−.55	.83
$5,000 to $6,999	−.86	.50	−.28	.46	−.47	.82
$7,000 to $8,999	−.82	.53	−.59	.50	−.82	.90
$9,000 to $12,000	....	....	....	....	....	....
Unclassified	−13.21	1136.0	−1.16	1.11	....	....
Site (Denver = 1)	.06	.21	−.11	.17	....	....
Duration married	[4] −.15	.04	[4] −.12	.03	−.04	.04
Wife's age	[3] −.04	.07	[4] −.07	.02	[4] −.08	.02
Wife's education	−.09	.05	−.01	.05	.04	.07
Number of children	−.01	.08	.12	.08	.05	.11
One if child younger than 6	.16	.26	.08	.21	.09	.35
One if preexperimental AFDC	−.10	.25	.08	.20	−.12	.31
Manpower treatment:						
Counseling only	.17	.32	−.21	.25	.52	.40
50 percent subsidy	.38	.28	−.12	.24	.37	.36
100 percent subsidy	.18	.31	−.26	.26	−.31	.39
5-year manpower treatment	−.31	.40	−.23	.34	−.16	.57
NIT treatment	.32	.34	−.20	.29	[2] −.94	.50
3-year NIT treatment	−.29	.37	.04	.31	.60	.54
Constant	.29	1.07	.82	.90	−1.60	1.24
Number of cases	904	....	751	....	319	....

[1] The estimates reported here are for the equation reported in table 9.2 for original spells.
[2] Significant at the 10-percent level.
[3] Significant at the 5-percent level.
[4] Significant at the 1-percent level.

APPENDIX B

EFFECTS OF COUNSELING/TRAINING TREATMENTS ON MARITAL STABILITY

There are several reasons why one might expect a counseling and training program to have an effect on marital stability. One is that it can alter the economic opportunities available to participants. In our discussions of the NIT treatment effects on marital stability we introduced the idea of the income and independence effects on marital stability (see chapter II). Both effects operate through experimentally induced changes in economic opportunity. Increases in family income are expected to lower the marital dissolution rate by lessening strains on the marriage that are due to inadequate income and by easing quarrels over finances.

The independence effect operates through the change that an NIT introduces in the economic opportunities outside marriage, particularly for the partner most economically dependent upon the marriage (usually the wife). In most cases, the NIT benefits for a single mother were greater than the benefits available from the current welfare system. The difference is even greater when the stigma associated with the current system and the transaction costs are considered. We hypothesized that an NIT program that increased income available outside the marriage would increase marital dissolution. The total effect of an NIT program on marital dissolution is the combination of the income and the independence effects and may be positive or negative, depending on which effect is greatest.

A counseling and training program may also affect marital stability independently of its effects on incomes. If it enables persons to find more satisfying jobs, marital strains may be reduced and consequently dissolution rates decreased. Similarly, additional training may increase self-esteem or increase the spouse's evaluation of each other, which could decrease pressures toward dissolution. Of course, training or counseling may increase one's self-confidence and result in one being more willing to risk leaving a marriage.

The effects of a counseling and training program are not immediately apparent. Increasing market skills by training or education takes time. Additionally, once training is completed the benefit may not be immediate. If training or education enables the individual to change occupations there may be an initial loss of income upon entry into the new occupation even though income may rise considerably after a time in the new occupation. Thus, the full effects of the SIME/DIME programs on marital stability may not have been manifested in the short time we observed participants.

In several of our earlier reports on the NIT effects on marital stability the effects of the counseling and training programs were reported, but with little discussion. For the most part the estimated effects were not significant. The few coefficients that were significant were not consistent from analysis to analysis.

The first step in our present analysis is to look at the percentage of originally enrolled women who changed their marital status during the first 3 years of the experiment. These percentages are given in table B1 separately by counseling/training treatment and race-ethnic groups. For single women there does not appear to be any difference in the percentage marrying across the treatments for any of the three race-ethnic groups. For married women the counseling-only treatment has a higher percentage of dissolution for blacks and Chicanos. Of course, in table B1 we have not controlled for any independent variables that might affect marital stability.

TABLE 5.B.1.—PERCENTAGE OF ORIGINALLY ENROLLED WOMEN CHANGING MARITAL STATUS WITHIN 3 YEARS BY COUNSELING/TRAINING TREATMENT, RACE, AND MARITAL STATUS AT ENROLLMENT

[Number of cases in parentheses]

	Counseling/ training controls	Counseling only	50 percent subsidy	100 percent subsidy
Married at enrollment:				
Blacks	22.2	28.3	24.6	25.6
	(397)	(173)	(236)	(133)
Whites	15.9	19.3	17.0	21.2
	(561)	(238)	(311)	(189)
Chicanos	17.6	29.6	19.0	19.7
	(210)	(115)	(137)	(76)
Single at enrollment:				
Blacks	10.4	10.6	14.0	11.7
	(376)	(161)	(222)	(145)
Whites	22.5	20.8	21.5	19.5
	(307)	(149)	(177)	(118)
Chicanos	20.2	18.5	21.2	18.2
	(114)	(65)	(85)	(55)

To determine the effects of the counseling and training programs on marital stability controlling for the effects of other variables, we estimate the parameters of log-linear models for the rates of dissolution and remarriage. All the results presented below pertain to models that include measures of experimental treatments and preexperimental characteristics of the husband, the wife, and their family. The counseling/training treatments are represented by a series of dummy (0,1) variables. These treatment variables are defined so that those with the particular treatment indicated by the variable label have the value 1 and those without this treatment have the value 0.

We only analyze the first 3 years of the experiment to avoid the problems associated with the ending of the NIT and counseling/training treatments. We focus our analysis on the women who were enrolled as wives or single female family heads at the start of the experiment and estimate the effects on dissolution and remarriage rates separately. In the marital dissolution equations we include both marriages that existed at enrollment and those that were formed during the first 3 years of the experiment—that is, we conduct an all-spells analysis.

We use two models to estimate the effects. In the first, each of the three counseling/training treatments is represented by a dummy variable. A fourth dummy variable takes the value of 1 if the woman was originally assigned to a counseling/training treatment for 5 years and 0 otherwise.[1] In the second model three additional variables are included that indicate the woman is eligible for both a counseling/training treatment and an NIT treatment. The three variables are "counseling-only and NIT treatments," "50 percent subsidy and NIT treatment," and "100 percent subsidy and NIT treatment." For each model we included two variables for the NIT treatments: an NIT dummy and a 3-year NIT treatment dummy. The equations estimated also include assignment variables and other background variables. Separate equations are estimated for each ethnic group.[2]

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[1] We chose a dummy variable for the 5-year counseling/training treatment because no persons were to be assigned to the 100 percent subsidy treatment for 5 years. The length of treatment for the counseling/training and the NIT treatments is the same for any person eligible for both.

Table B2 presents the estimated effects of counseling/training treatments. The top panel contains estimates of only the main effects of the three treatments and the duration of the treatments. All nine main effects are positive. Thus, it appears that the treatments increase dissolution rates. However, the treatments are not jointly significant for any race-ethnic group (that is, we cannot reject the null hypothesis that all treatment effects are zero at the 10 percent level). Three of the treatment effects are individually significant at the 10 percent level—the counseling-only treatments for blacks and Chicanos and the 100 percent subsidy for whites. Surprisingly, the counseling-only treatment has a larger effect than the combination of counseling and a 50 percent training subsidy for all three race-ethnic groups. The significant effect of the counseling-only option is puzzling since it was such a minimal treatment (see part IV of this volume for a description of the counseling-only treatment).

TABLE 5.B.2.—EFFECTS OF COUNSELING/TRAINING TREATMENTS ON MARITAL DISSOLUTION RATES: ORIGINAL SPELLS OVER FIRST 36 MONTHS

[Standard errors in parentheses]

	Blacks		Whites		Chicanos	
	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$
Model 1:						
Counseling only	[2].453	1.57	.316	1.37	[1].516	1.67
	(.202)		(.208)		(.269)	
50 percent subsidy	.298	1.35	.142	1.15	.134	1.14
	(.199)		(.204)		(.290)	
100 percent subsidy	.258	1.29	[1].333	1.40	1.84	1.20
	(.215)		(.200)		(.317)	
5-year counseling/training treatment	−.385	.68	−.286	.75	−.039	.96
	(.260)		(.269)		(.391)	
NIT treatment	[2].45	1.57	[1].43	1.53	.01	1.01
	(.21)		(.22)		(.34)	
3-year NIT treatment	.30	.74	−.33	.72	−.00	1.00
	(.23)		(.24)		(.37)	
Likelihood ratio χ^2 for counseling/training treatments (4 df)	5.43		4.14		4.43	
Model 2:						
Counseling only	[1].477	1.61	.227	1.02	.419	1.50
	(.283)		(.343)		(.429)	
Counseling only and NIT	.017	1.02	.410	1.50	.164	1.18
	(.373)		(.400)		(.512)	
50 percent subsidy	.078	1.08	−.454	.64	−.186	.83
	(.291)		(.359)		(.503)	
50 percent subsidy and NIT treatment	.404	1.50	[2].819	2.27	.467	1.60
	(.356)		(.402)		(.580)	
100 percent subsidy	−.154	.86	[1].528	1.70	.147	1.16
	(.369)		(.272)		(.564)	
100 percent subsidy and NIT treatment	.683	1.98	−.385	.68	.087	1.09
	(.462)		(.398)		(.691)	
5-year counseling/training treatment	−.425	.65	−.198	.82	−.012	.99
	(.271)		(.294)		(.407)	
NIT treatment	.306	1.36	.173	1.19	−.166	.85
	(.248)		(.275)		(.438)	
3-year NIT treatment	−.38	.68	−.21	.81	.03	1.03
	(.24)		(.27)		(.39)	

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[2] The assignment variables were a site dummy and normal income level dummies. The background variables were women's age and education, number of children, presence of a child aged 5 or younger, and a dummy variable for previous AFDC experience.

TABLE 5.B.2.—EFFECTS OF COUNSELING/TRAINING TREATMENTS ON MARITAL DISSOLUTION RATES: ORIGINAL SPELLS OVER FIRST 36 MONTHS—Continued

[Standard errors in parentheses]

	Blacks		Whites		Chicanos	
	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)
Likelihood ratio χ^2 for interactions (3 df)	3.25	....	[2] 7.97	....	.70	....
Number of marriages	....	939	....	1,299	....	538

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.

NOTE.—Equations for both models include background and assignment variables. The estiamtes are from the same equations as the original marriage results in chapter V, table 5.4. The estimated effects of the other variables in the equations for Model 1 are reported in table A1.

The bottom panel of table B2 presents estimates of the combined effects of counseling/training and NIT treatments (we add three interactions to the models reported in the top panel). The set of interactions are jointly significant for whites but not for blacks or Chicanos. For blacks the counseling-only effect is significant, but there is no significant difference in the counseling-only effect between couples eligible only for the counseling treatment and those eligible for both counseling and NIT treatments. The same is true for the counseling-only effect among Chicanos. The interaction between the NIT treatments and the counseling/training treatments for whites is confusing. There is no significant effect of the counseling only option, whether the couples are eligible for a NIT treatment or not: the effect of the 100 percent subsidy for whites holds whether the couples are eligible or not; the 50 percent subsidy has no effect by itself, but increases the dissolution rate when it is combined with an NIT treatment. The most plausible explanation of this unsystematic interaction between the NIT and counseling/training treatments is that it is due to sampling variability. The number of cases in any of the combinations of both types of treatment is small.

Table B3 presents the counseling/training and NIT treatment effects on remarriage rates. There are no significant effects for blacks or whites, although the NIT treatment significantly reduces remarriage rates for Chicanos in model 1, and in model 2 the counseling only and NIT treatments effect is significant. We interpret table B3 as supportive of the hypothesis that the counseling/training treatments had no effect on remarriage.

TABLE 5.B.3.—EFFECTS OF COUNSELING/TRAINING ON MARITAL FORMATION RATES: ORIGINAL SPELLS OVER FIRST 36 MONTHS

[Standard errors in parentheses]

	Blacks		Whites		Chicanos	
	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)	$\hat{\alpha}$	exp ($\hat{\alpha}$)
Model 1:						
Counseling only	.166 (.318)	1.18	−.208 (.249)	.81	.052 (.396)	1.05
50 percent subsidy	.384 (.282)	1.47	−.116 (.236)	.89	.367 (.357)	1.44
100 percent subsidy	.177 (.314)	1.19	−.264 (.256)	.77	−.031 (.392)	.97
5-year counseling/training treatment	−.306 (.399)	.74	−.230 (.343)	.80	−.159 (.571)	.85
5-year NIT treatment	.316 (.337)	1.37	−.202 (.288)	.82	[1]−.938 (.503)	[1].39
3-year NIT treatment	−.29 (.37)	.74	.04 (.31)	1.05	.60 (.54)	1.82

TABLE 5.B.3.—EFFECTS OF COUNSELING/TRAINING ON MARITAL FORMATION RATES: ORIGINAL SPELLS OVER FIRST 36 MONTHS—Continued

[Standard errors in parentheses]

	Blacks		Whites		Chicanos	
	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$	$\hat{\alpha}$	exp $(\hat{\alpha})$
Likelihood ratio X^2 for counseling/training treatment (4 df)	.88		2.79		1.37	
Model 2:						
Counseling only	.530	1.70	.009	1.01	.657	1.93
	(.439)		(.386)		(.543)	
Counseling only and 5-year NIT	−.598	.55	−.339	.71	−1.34	.32
	(.590)		(.459)		(.745)	
50 percent subsidy	.198	1.22	−.076	.93	.426	1.53
	(.434)		(.371)		(.599)	
50 percent subsidy and 5-year NIT treatment	.277	1.32	−.089	.91	−.161	.85
	(.514)		(.439)		(.708)	
100 percent subsidy	−.173	.84	−.147	.86	.219	1.24
	(.631)		(.399)		(.617)	
100 percent subsidy and 5-year NIT treatment	.461	1.59	−.209	.81	−.425	.65
	(.733)		(.514)		(.804)	
5-year counseling/training treatment	−.309	.73	−.234	.79	−.147	.86
	(.403)		(.340)		(.544)	
5-year NIT treatment	.293	1.34	−.100	.90	−.580	.56
	(.406)		(.332)		(.594)	
3-year NIT treatment	−.31	.73	.05	1.06	.60	1.82
	(.38)		(.31)		(.51)	
Likelihood ratio X^2 for interactions (3 df)	3.38	....	.059	....	2.53	....
Number of single women	904	....	751	....	319	....

[1] Significant at the 10-percent level.

NOTE.—Equations for both models include background and assignment variables. The estimated effects of the other variables in the equations for model 1 are reported in table A2.

APPENDIX C

COMPARISON OF THE NEW JERSEY AND SEATTLE-DENVER EXPERIMENTS

Comparison of benefits

There were important differences in the level of benefits available to families from AFDC and Food Stamps between the two experimental sites. In both Seattle and Denver the NIT guarantee levels exceeded those available from AFDC and Food Stamps for the high and medium guarantee levels. The low guarantee level was approximately equal to the AFDC and Food Stamps support level (see Hall 1976). In New Jersey, the NIT treatments initially were more generous than AFDC and Food Stamps but in January 1969, shortly after enrollment began, New Jersey significantly increased the level of AFDC benefits and extended the AFDC program to two-parent families where the father was unemployed. The AFDC benefit levels in Pennsylvania were comparable to the increased New Jersey benefit levels. Thus, for many of the NIT treatments the level of benefits from AFDC and Food Stamps was higher than the benefits available from the NIT treatments. While couples in the New Jersey Experiment would not be eligible for AFDC-U benefits when the husband worked more than the maximum number of hours allowed, wives would be eligible for higher benefits from AFDC, should the marriage dissolve, than they could obtain from the NIT treatments (see Rossi and Lyall, 1976, for further details). These differences mean that the NIT treatments in Seattle and Denver represented a greater change from the prevailing welfare system than the NIT treatments in New Jersey and Pennsylvania.

Although the three experiments were similar in design, there were differences in the rules of operation that could cause them to have different effects on marital dissolution. The two most important differences are the definition of marriage and the eligibility of family heads after a marital dissolution.[1] Both experiments enrolled couples as "married" on the basis of their self-reported marital status. In New Jersey the rules made no explicit provision for couples who lived together as family heads but were not legally married. The SIME/DIME rules of operation, on the other hand, explicitly dealt with cohabitation (see part V, chapter 2). Consequently, more cohabiting couples were probably enrolled in SIME/DIME than in New Jersey, although the New Jersey Experiment may well have included some cohabiting couples as married.

As we pointed out in Chapter 3, including cohabiting couples in the married sample results in higher dissolution rates if those couples are less committed to their unions than legally married couples. However, since over 85 percent of the couples in the SIME/DIME sample had been cohabiting more than 2 years at enrollment, the number of casual unions included in the sample is likely to be small. We do not believe that the explicit provision for cohabitation caused the SIME/DIME sample to differ greatly from the sample in New Jersey.

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[1] Many other differences in the rules of operation of the three experiments are discussed in Kershaw and Fair (1976). The effects of these other differences on marital dissolution seem inconsequential.

The second relevant difference in the experiment's rules of operation concerns a spouse's eligibility after a marital dissolution. In both experiments both families formed after a marital dissolution were eligible for the treatment of the original family as long as both spouses kept at least one dependent from the original family. The rules differed, however, when one spouse left and formed a new family without any dependents. In New Jersey the single spouse was eligible for a guarantee equal to 50 percent of the guarantee for a two-spouse family with no dependents; but if the spouse remarried, the new spouse did not become eligible. In SIME/DIME a single spouse without dependents was eligible for a $1,000 guarantee (constant 1971 dollars), regardless of the guarantee level of the original family. If that spouse married, the new spouse became eligible after a 1-month waiting period. In all cases guarantee levels were adjusted to reflect the changes in family size. Thus, the economic incentives for a single spouse leaving a marriage differ among the experiments. Kershaw and Fair (1976) concluded that while there were differences among the treatments, the rules for adjusting guarantee level after marital dissolutions were an incentive against dissolution in both the New Jersey Experiment and SIME/DIME.

The samples

Data for our analysis of the New Jersey Experiment were taken from the Cross-Site Analysis File (CSA) described in detail in Citro (1980). The CSA file was constructed from the data bases of the three urban NIT experiments to facilitate comparative analysis of NIT effects on both marital stability and labor supply.

Using the CSA data, we constructed variables for New Jersey that paralleled those used in our SIME/DIME analyses. These included treatment and assignment variables, demographic variables, preexperimental income variables, and marital history variables. We selected all families in which both spouses were present at enrollment. The definition of a marriage was the one used by both experiments: Any family with two heads present at enrollment was considered married.

Several differences among the samples should be noted. (See Groeneveld et al., 1980, for a more complete comparison.) The proportion of husbands and wives in Seattle-Denver who finished high school was almost double the proportion in New Jersey. The New Jersey sample had both larger families and more families with preschool age children than SIME/DIME. The number of children in a family, especially young children, is usually thought to have a stabilizing effect: the more children, the lower the dissolution rate. If so, the dissolution rate of controls should be higher in Seattle-Denver than in New Jersey.

SIME/DIME had fewer husbands who were not working in the year before enrollment and fewer husbands who employed full time. Many more wives worked in Seattle-Denver than in New Jersey, and family income was higher in Seattle-Denver than in New Jersey. New Jersey has the lowest proportion of families reporting welfare income for the year before the experiment, largely because there was no AFDC-U program in New Jersey at the time the experiment began. Few couples in either experiment received more than $1,000 from welfare in the year before the experiment.

Differences in family income are important to the study of the effects of the NIT treatments on marital dissolution. Our analysis of SIME/DIME has shown that the effect of a NIT treatment depends not only on the magnitude of the NIT payment but also on the initial level of the family's income. Thus, similar NIT treatments may have different effects, depending on the levels of family income in the different sites.

Couples are included in the analysis even if they dropped out before the end of the experiment. The period of observation for a couple is from the time of enrollment until the date of marital dissolution (if one occurs), the date they dropped out of the experiment, or 3 years after enrollment (the end of the experiment for all except the 5- and 20-year samples in Seattle and Denver). The attrition rate for both control and experimental couples was higher in New Jersey than in Seattle/Denver.

Definitions of marriage and dissolution

As stated above, the same definition of marriage was used in both experiments. Any family identified as having two family heads present at enrollment was considered married. A marital dissolution occurred when a couple reported that they were no longer living together.

For about three-fourths of the recorded marital status changes in New Jersey, the CSA file does not indicate whether the change was a marital dissolution (divorce or separation) or some other change (such as the spouse living away from home for work or being institutionalized). Based on our understanding of the way in which dissolutions were identified in New Jersey and on the work of other investigators (Sawhill et al., 1975; Wolf, 1976), we believe that counting the cases for which the reason that the marriage ended is unknown as dissolutions gives the best estimate of the number of dissolutions.[2]

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[2] For a more complete discussion of this problem, see Groeneveld et al. (1980).

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SIME/DIME FINAL REPORT

VOLUME 1, PART VI

OVERVIEW OF RESEARCH ON HEALTH, CONSUMPTION, AND SOCIAL BEHAVIOR

by

Margaret R. Davis and Kenneth Kehrer

CHAPTER I

INTRODUCTION

The central purpose of SIME/DIME was to estimate the effect of alternative NIT plans on labor supply and program costs. Another important objective was to examine the effects on marital stability. There were, however, a number of other studies that examined the effects of SIME/DIME on a broad range of social and economic outcomes.

Some of these outcomes were of interest because they were expected to have indirect effects on the labor supply response (that is, migration, child care, residential mobility, household production, fertility) or to be long-run determinants of labor supply and earnings (health, medical expenditures, children's school performance, and infant health). Other outcomes were examined because of their relationship to marital stability (marital conflict and attitudes toward family roles). Some of the research was initiated with the objective of replicating the findings on consumption, well-being, family change, and the quality of life in other NIT experiments.

In Part VI, we discuss several of these studies. The studies are heterogeneous in that they were performed by different researchers, using somewhat different analysis samples from the SIME/DIME data base, measuring experimental treatment in different ways, and with different levels of resources devoted to each. Resources available were considerably lower than those devoted to the central topics of interest in SIME/DIME covered in Parts III, IV and V.

For these reasons, most of the studies that we discuss should be regarded as tentative explorations of the issues, in no way comparable to the exhaustive and large-scale investigations reported in preceding chapters. The discussion of each study or topic is, therefore, brief. Although we have attempted to report central theoretical and methodological concerns in each case, together with the most important findings, of necessity much detail is omitted and many smaller analyses are referred to only in passing or not at all. Readers interested in a full exposition of a particular subject are urged to consult the uncondensed reports or papers cited.

Much of the research reported here attempts to capture the effects of the experiment by the use of a dummy variable to represent eligibility for the NIT, with an additional dummy variable to represent three- or five-year treatment. The argument, explicit or otherwise, put forward by the researchers for expecting an experimental effect is based on the notion that on average the experiment improves the financial situation of the experimental group. For this premise to hold, it is not necessary that the experiment improve the financial situation of all participants. Certain combinations of family size and experimental tax rate did result, for those at the lowest guarantee level, in some experimental families being in no better financial situation than control families. However, even these families were better off in the sense that the additional option of the NIT gave them greater security. In addition, the relative ease of applying for and receiving the NIT as compared with, for instance, applying for and receiving AFDC, can be regarded as making experimental families better off (see part V).

One problem arises with the use of the simple dummy variable treatment specification adopted by most studies reviewed here. Although, in the absence of nonlinearities in the experimental response, the treatment dummy provides an unbiased estimate of the average treatment effect in the study population, it may obscure some experimental effects. The lack of statistical significance of the dummy treatment variable may be the result of a general lack of experimental effects across experimental subgroups, or it may be the result of averaging the response of small intervention (and no response) groups with larger intervention (and significant response) groups. It is also possible that the statistical tests used in many of these studies do not have sufficient power to detect experimental effects.[1]

We would clearly be able to assess with more confidence the impact of the experiment on these dependent variables if substantially greater resources had been devoted to these studies. For example, it might have been possible to develop models of response and to test alternative specifications, as was done in the labor supply and marital stability studies. As already noted, however, more limited resources were allocated to these ancillary studies. Simple dummy variable models used to test experimental impact are economical, and in the main do capture overall experimental-control differences.

The following sections are organized by topic. Chapter 2 presents the effects of the experiments on health and fertility, including infant health and psychological distress. Next, chapter 3 summarizes the findings on the effects of SIME/DIME on consumption, including housing consumption, expenditures on nondurable goods, and demand for asset stocks. Chapter 4 examines the effects on mobility, including intracity mobility and migration out of the area. Chapter 5 reports on two child-related issues: the demand for child care and its effects on labor supply, and children's school performance. Effects on family life, including household production, marital conflict and attitudes toward family roles, are presented in chapter 6. Finally, in chapter 7, we assess what we have learned from this body of research about the effects of an NIT program on labor supply, marital stability, and program costs.

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[1] This issue has generally not been addressed by the researchers involved.

CHAPTER II

EFFECTS ON HEALTH AND FERTILITY

The health and fertility of the poor are two public policy concerns that continue to influence the redesign of income maintenance and other social welfare programs. Concern about adequate health care for the poor has led to programs both to reduce the cost of health care (that is, Medicaid and proposals for national health insurance) and to increase the supply of health services in underserved areas (that is, the Neighborhood Health Centers, the National Health Service Corps). Concern over the effect of income support on fertility has been one reason why children's allowances have not been adopted in this country.

In this chapter we review the research of SIME/DIME on health and fertility. The first two sections report the findings on the effects of SIME/DIME on health status and psychological distress. The third section assesses SIME/DIME effects on infant health at birth. The final section discusses the findings on fertility.

Health status

Health is desired both for its own sake (that is, consumption) and as an investment. Thus, the demand for good health is in part a derived demand for earnings capacity. An NIT can be expected to increase the demand for health as consumption, due to the income effect.[1] However, if the NIT increases the marginal tax rate on earnings, the rate of return on health is reduced, which may discourage investment in health status. Nonetheless, an NIT would also be expected to have indirect beneficial effects on health, through its impact on nutrition, housing, and other aspects of well-being.

Stieger and Wai (1980) investigated the effect of SIME/DIME on health status in the context of a simultaneous equations model of medical care utilization and health status. Because their interest was in estimating the gross (direct and indirect) experimental effect on health, they analyzed a reduced form specification.

Several alternative measures of health status were examined: (1) number of work days lost due to illnesses, (2) number of hospital stays and (3) number of days hospitalized in the last 2 years, (4) number of work days missed in the last 6 months, (5) presence of a functional limitation on doing household tasks, (6) presence of a chronic condition that limits activities of daily living or (7) market work, (8) the duration of the chronic condition, (9) a mental health index, and (10) self perception of overall health.

Stieger and Wai regressed all these measures of health status against preexperimental medical care expenditures (to partially control for preexperimental health status, which was not ascertained in baseline interviews), a treatment dummy, an interaction term for treatment and duration of experimental plan, assignment, and control variables.[2] Separate regressions were estimated for husbands, wives, and female heads. Findings were inconclusive as few were statistically significant and the effects could have occurred by chance. For husbands, there was a beneficial effect on number of days hospitalized. For wives, there was significant improvement in the mental health index and, for female heads, a reduction in the duration of chronic illness. There were not appreciable differences in estimates for 3- and 5-year duration families.

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[1] Although there is evidence that the income elasticity of demand for health is negative among the affluent (HHS 1980, p. 159).

Thus, we must conclude that SIME/DIME probably did not affect the health of participants, although Stieger and Wai caution that the many measures used in their study may not adequately assess health.

Stieger and Wai, in an exploratory analysis, also investigated the effect of SIME/DIME on medical care expenditures, which reflect a derived demand for health status. Their interest was in analyzing medical care utilization, but they did not have data on the large class of medical services available free to the study population under Medicaid, free clinics, etc. They found that SIME/DIME did result in increased medical care expenses: average experimental effects in the third year of the experiment were about $10 a year per family member for husbands and wives and $6 for single family heads. While small in dollar terms, these effects are large in percentage terms: the husband-wife (female-headed) families receiving experimental payments increased their medical expenditures by 28 (40) percent compared to control families.

Psychological distress

Considerable literature over the past 50 years has demonstrated that psychological distress varies inversely with level of income: the greater the individual's income, the less will be his or her reported psychological distress (Parker and Kleiner, 1966; Bradburn and Caplovitz, 1965; Bradburn, 1969; Phillips, 1966; Gurin et al., 1960; Srole et al., 1962; Dohrenwend and Dohrenwend, 1969).[3] However, little is known about the effect of *changes* in income on psychological state.

Thoits (1979a, 1979b) offered two alternative theoretical explanations for the relationship between income and psychological distress that lead to different predictions concerning the effects of an NIT on psychological distress. First, according to the social causation hypothesis, psychological disturbance is a result of social status (Dohrenwend and Dohrenwend, 1969; Kohn, 1968): because the lower income individual has fewer resources with which to resolve day-to-day crises—accidents, job layoffs, unexpected car repairs, etc.—these events are more likely to produce psychological distress than they are for higher-income individuals. This social causation, or "resource," hypothesis does not address the question of how a change in income affects psychological distress. However, it suggests that NIT payments, by increasing or stabilizing income, will make it easier for the individual to cope with everyday crises, and thus lead to an overall decrease in psychological distress.

The second explanation is what Thoits and Hannan (1978) called the "life events" hypothesis. Life events are defined as "objective events that disrupt or threaten to disrupt the individual's usual activities" (Dohrenwend and Dohrenwend, 1969). These events require the individual to readjust his or her behavior substantially, which causes stress in the individual. Intense or prolonged stress in turn may result in physical or psychological disorder. Several studies demonstrate a positive relationship between life events and distress: the more life events a person experiences, the more psychological distress he or she exhibits or reports (Brown and Birley, 1968; Paykel, 1974; Myers, et al., 1971, 1972, 1974).

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[2] These were age, age squared, education, number of children less than five years old, dummy variables for eligibility for Medicaid and presence of health insurance (to account for the consumption of medical services with no out-of-pocket costs), a job satisfaction index (to control for any tendency to become "sick" more often), and preexperimental per capita medical expenditures. Most of the independent variables were based on preexperimental data, to avoid endogeneity with experimental treatment, although job satisfaction was measured only during the experiment.

[3] "Psychological distress" as used here means a general state of unpleasant arousal which is accompanied by physiological symptoms such as nervousness, upset stomach and dizziness. Such arousal, if intense, is expected to impair daily functioning.

The experiment may cause changes in the individual's life that could be termed life events. For example, divorce, joblessness, and migration are significant life events and it has been shown that SIME/DIME significantly increased the breakup of experimental families (part V), their rate of not working (part III), and their rate of geographic mobility (later in part VI).

There are two variants of life events theory. One variant suggests that only undesirable life events cause distress.[4] The second variant argues that it is the frequency of events, regardless of their desirability, that causes distress.[5]

Two previous NIT experiments have reported social-psychological outcomes. In the New Jersey Experiment (Middleton and Allen, 1977), individuals eligible for experimental payments reported somewhat higher rates of psychosomatic symptoms overall than did controls. Further, distress was weakly but positively related to the levels of actual payments received by the experimental group. However, none of these experimental effects was significant. In the rural experiment (Middleton, 1976), individuals eligible for experimental payments had significantly higher distress scores than did controls, overall. Also, amount of payment received had a significant direct effect on distress.

Thoits and Hannan (1978, 1979, 1980) and Thoits (1979a, 1979b) studied the effect of SIME/DIME on psychological distress at two points in time early in the experiment, using a series of psychophysiological symptom items that are a close variant of the Macmillan Health Opinion Survey index (MacMillan, 1957). The items were administered to both experimental and control groups in similar ways, as part of the periodic interviews. Separate analyses were conducted for married men, married women, and divorced or separated women using ordinary least squares regression. Four dummy variables were used to indicate experimental treatment and duration of treatment. Each equation contained all variables used in assignment, as well as several other variables: normal preexperimental family income, parental status, education, age, occupational status, and employment status. Since it was found that treatment duration interacted with NIT effects and with race-ethnicity, whites, blacks, and Chicanos were analyzed separately.

The experiment had a few significant effects scattered across subgroups. Two points in time were studied, at both of which white husbands and white wives receiving the 5-year treatment in Denver and black husbands with the 3-year treatment in Seattle reported significantly more distress than controls. At the second point in time several subgroups reported significantly higher distress than controls: black wives on the 5-year treatment in Seattle, black single women on the 3-year treatment in both Seattle and Denver, black and white single women on the 5-year treatment in Denver. There were no significant effects among the other groups of married or single black and white experimentals, and no significant experimental effects among any group of Chicanos. While the experimental effects are scattered and appear to reveal no pattern, the statistically significant effects observed indicate that experimentals reported more psychological distress than did controls. Eleven of the 72 tests for experimental effects reported in the text are statistically significant.

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[4] Baswoitz, Persky, Korchin, and Grinker (1955); Birley and Brown (1970); Bremer (1951); Brown and Birley (1968); Clayton, Halikas, and Maurice (1972); Fried (1963); Grinker and Spiegel (1945); Hudgens, Morrison, and Pepper (1971); Paykel (1974); Rogler and Hollingshead (1965); Schmale (1958); Star (1959).

[5] Coates, Moyer, and Wellman (1969); Dohrenwend and Dohrenwend (1970); Froberg, Karlsson, Levi, and Lindberg (1971); Markush and Favero (1974); Myers et al. (1971, 1972, 1974); Paykel, Myers, Dienelt, Klerman, Lindenthal, and Pepper (1969); Paykel, Prusoff, and Myers (1974); Uhlenhuth and Paykel (1973).

Thoits and Hannan interpreted this pattern of effects as offering some support for the life events hypothesis and none for the resource hypothesis that an increase in resources, such as income, should lower psychological distress. In one analysis (Thoits and Hannan, 1978), a variable measuring the magnitude of the change in disposable income a family experienced on being assigned to an NIT treatment was added to the equation. There were almost no significant effects for this variable—the one exception being that for single women in Seattle at the time of the first observation, the level of distress reported rose with level of payment received. The authors concluded that it appears to be the fact of experiencing the experimental treatment, rather than the amount of the payment received, that affects levels of distress. On the other hand, the change-in-income variable may be a better way of characterizing the magnitude of the experiment's intrusion in the lives of these families. In that light, the experimental effects virtually disappear.

Thoits (1979a, 1979b) pursued other approaches to understanding these findings, including determining what kinds of life events were related to psychological distress in this sample. One of the strongest and most consistent findings for all subgroups was that physical health problems raised the level of psychological distress. This raises the still unresolved issue of whether, in fact, the psychological distress index tends to confound physical illness and mental distress.

The clearest overall finding was that for most groups there is no experimental effect on psychological distress. Most of the observed statistically significant effects indicate that the experiment increased distress levels; however, significant effects are scattered among subgroups in such a way that it is difficult to derive a consistent and convincing explanation for them. It is equally clear that there is no support whatever for the "resource hypothesis"; that is, one apparently cannot increase the psychological well-being of the poor merely by providing income guarantees and supplemental payments of the magnitudes or over the periods provided in SIME/DIME.

Infant health status

Because low birth weight and infant mortality are inversely related to socioeconomic status, an NIT might be able to improve infant health status. Mechanisms through which improved health at birth might be achieved include improved nutrition of the mother and fetus during pregnancy, increased and earlier use of prenatal care, reduction of work at strenuous jobs during pregnancy, and an increase in the quality of life (that is, better housing, reduced stress) associated with higher income.

Kehrer and Wolin (1977) examined the effect of the Gary Experiment on the birth weight of children born during the experiment. Beneficial effects were found for children born to high risk mothers—those women who had any two of the following characteristics:

- age less than 18 or over 34
- short interval between pregnancies, or
- being a smoker.

The beneficial effects range as high as an additional pound.

While infant health was not studied in any of the other experiments, nutrition—a potentially important mechanism for improving health status—was investigated. O'Connor, Madden, and Prindle (1976) found no effects of the Rural Experiment on nutrition in the Iowa sample, but did observe that North Carolina families with an inadequate diet used the experimental payments to improve the quality of their nutrition. No nutritional effects were found among those members of the North Carolina sample who had nutritional intake as high as the average level of nutrition observed in the Iowa sample.

Kehrer and Wolin (1980) attempted to replicate and extend their Gary findings on birth weight in a study of the effect of SIME/DIME on low birth weight and infant mortality in the Denver sample. They derived a reduced form birth weight response equation from a structural model. Because they defined low birth weight as deviation of actual birth weight below either 2,501 or 3,001 grams (two limits suggested by the public health and pediatrics literature), they used a tobit procedure to estimate experimental effects. Their explanatory variables include infant's sex and race/ethnicity, mother's age, parity, whether there was a short interval since the mother's last delivery, whether the mother smoked, mother's educational attainment and health problems, whether the infant was legitimate, mother's predicted wage rate, whether the mother was the sole family head, the usual assignment variables, and a dummy variable for experimental treatment.

Defining low birth weight to be below 2,501 grams was a more severe specification of low infant health status than that used in Gary. Contrary to the Gary findings, no significant experimental effects were observed for that measure of low birth weight. When the same specification of low birth weight used in the Gary research—under 3,001 grams—was used in SIME/DIME, only one of the 12 cohorts studies was observed to have an experimental effect. Chicano infants whose mothers did not have a short interval between pregnancies and who did not smoke had a small (70 grams or 0.15 pound) but statistically significant increase in birth weight.

Kehrer and Wolin also specified the dependent variable as birth weight, a continuous variable, but found no significant experimental effects. They concluded that SIME/DIME, in contrast to the Gary experiment, does not appear to have had an effect on the infant health status, at least of the Denver sample. The authors point out, however, that they could not control for medicaid eligibility or insurance coverage in their analysis, and speculate that effects of these programs could have swamped any treatment effect that might have occurred.

NIT effects on fertility

There is interest in effects of an NIT on fertility for two principal reasons. First, because income support programs generally pay larger benefits to larger families, an important component of long-run programs costs is determined by fertility. A related issue is the extent to which the program benefit schedule encourages fertility.

An NIT program could affect fertility in several ways, but the net effect is uncertain. The NIT provides a subsidy for additional children, because payment increases with family size, just as in most other income support programs such as AFDC. In addition, to the extent that an NIT increases the tax rate on earnings, a family would forgo less income if a woman reduced her labor supply to have a child. The net effect of an NIT is uncertain because the effect of increasing income on fertility is unknown.

Previous research on fertility in the other experiments was concerned with methods to solve difficult analytic problems arising from the short duration of the experiments: (1) a limited duration experiment may encourage earlier timing of childbirth, because additional children are subsidized only during the experiment, and (2) the duration of the experiments are short in relation to the time it takes to conceive, bear, and give birth to a child. Cain (1974) examined fertility in the New Jersey Experiment using three measurees of fertility: (a) being pregnant by the end of the second quarter, (b) being pregnant or having a child less than 1 year old by the end of the 6th quarter, and (c) being pregnant or having a child less than age 2 by the end of the 10th quarter of the experiment. His results suggest that the New Jersey Experiment increased fertility among blacks, decreased fertility among whites, and had no effect on Spanish-speaking families. Wolin (1981) found that the NIT reduced fertility among unmarried women in Gary.

Keeley (1978, 1979) examined the effects of SIME/DIME on fertility during two different periods: during the first year after enrollment and during a period that covered the second and third years. He excluded women who were over 40 at enrollment, and analyzed two groups: women who were married at enrollment and women single at enrollment. To account for the censored sample generated by the fertility process, he used the rate model from the marital stability analysis (see part V).

Keeley controlled for stratification in the experimental design by estimating equations for husband-wife and female-headed families separately by ethnic group. Dummy variables for the normal income groups were used as control variables for the calendar quarter of the observation, and dummy variables for age, training/counseling treatment and site, and other economic and education variables.

Keeley's estimates of the effect of SIME/DIME on fertility, using a dummy variable to capture experimental effects, are presented in table 6.1. He found effects of SIME/DIME on fertility of women who were married at enrollment, but not on women who were single at enrollment. However, the effects for married women differ sharply by ethnic group: SIME/DIME increased the fertility of Chicano married women (for women on 5-year plans fertility occurred at twice the rate of their control group). On the other hand, the fertility rate of white married women in 5-year plans was substantially lower (57 percent) than similar controls, but 7 percent higher than controls among married white women on 3-year plans. No statistically significant effects were found for black married women.

TABLE 6.1.—EFFECTS OF SIME/DIME ON FERTILITY BY DURATION OF EXPERIMENTAL PLAN

[Annual birth rates relative to controls during second and third years]

	Married females		Single females	
	3 years	5 years	3 years	5 years
Blacks	0.93	1.00	0.71	1.36
Whites	[3] 1.07	[2] .57	.42	1.07
Chicanos	[4] 1.20	[1] 2.23	1.36	.91

[1] Significant at the 5-percent level.
[2] Significant at 10-percent level.
[3] Additive 3-year effect significant at 5-percent level.
[4] Additive 3-year effect significant at 10-percent level.

Keeley investigated whether his estimates changed if he interacted his measure of treatment with the probability of giving birth, or if he controlled for whether a family was actually receiving any NIT payments. He found that, for black women, the higher the predicted probability of giving birth, the smaller the experimental effect on fertility. Estimates controlling for actual receipt of NIT payments (as opposed to experimental status) are generally of similar magnitude to those presented in table 6.1, although statistical significance generally decreases for married women and increased for single women.

Keeley's efforts to estimate separate income and price effects of the experiment were generally not successful. With the exception of Chicanos, none of the estimated coefficients was significant.

Reviewing SIME/DIME fertility findings along with results of the other NIT experiments, one is left with the view that an NIT appears to affect fertility, but the findings suggest different directions of the response across ethnic groups, and these responses are not always consistent across experiments. Both the New Jersey and Seattle-Denver Experiments found evidence that whites receiving an NIT reduced their fertility. On the other hand, the New Jersey findings suggest that for blacks the NIT increased fertility, but there were no effects for Spanish families. SIME/DIME increased the fertility of Chicanos, but had no effect on black fertility. In Gary, the NIT appears to have reduced fertility among unmarried women, but not among married women. The cumulative evidence on fertility effects summarized here is somewhat bewildering and raises questions about both the theory and empirical evidence on the relationship between income maintenance and fertility. There may be regional differences in the fertility responses to an NIT among black and Spanish-speaking families, but the evidence from the experiments is too limited for us to be able to test such a hypothesis.

CHAPTER III

CONSUMPTION

There is considerable public interest in expenditure patterns of the poor and how welfare recipients spend their transfer payments. Much of this interest is normative—public concern over whether public funds are being used to pay for the basic human needs of the poor or whether taxes are being used to finance activities that are deemed frivolous or immoral. In addition, better knowledge about expenditure patterns of the poor facilitates the prediction of the impact of government policies and exogenous events, such as the energy crisis, on their well-being. Understanding how the poor choose to allocate their expenditures between consumption and investment also helps us to understand the process of escaping poverty. These two concerns—normative and behavioral—come together in the design of public transfer programs. Better understanding of the consumption choices of the poor can inform decisions about whether public transfer programs should be designed to restrict the use of transfer funds to certain types of expenditures (for example, Food Stamps, housing allowances).

The increase in income generated by an NIT can be expected to increase the consumption of most goods, but the relative effect of an NIT across types of expenditures is unclear on theoretical grounds. If marginal tax rates are increased under an NIT, there will be an incentive to shift away from market-produced goods toward home-produced goods and goods that are complementary with leisure. If the NIT taxes some assets, as SIME/DIME taxed the asset value of a home, there will be disincentives to hold some forms of wealth rather than others. Moreover, economic theory indicates that how much of an increase in income is consumed varies depending on whether the increase is considered permanent or transitory. Families tend to use transitory income increases in ways that increase their net worth (for example, paying off debts, saving and purchasing durable goods), while permanent income increases tend to be spent on nondurables (for example, food, entertainment). Income from transfer programs is generally regarded as transitory in the literature, although empirical problems and the lack of adequate data have hindered testing this hypothesis. Some attempts were made to infer from families' expenditures whether the income support payments under SIME/DIME were regarded as transitory or permanent. These attempts were unsuccessful, perhaps because this issue is bound up (and probably confounded) with the limited duration of the experiment.

In this chapter we summarize the effects of SIME/DIME on consumption. First, we examine the impact of SIME/DIME on nondurable consumption. Then we discuss the effects on housing consumption. Finally, we examine findings on asset demand.

Effects of NIT on nondurable consumption

Johnson, Pozdena, and Stieger (1979) investigated the effects of SIME/DIME on nondurable consumption in the context of a static model of household consumption where the household maximizes within a single period. They derive an empirical specification that expresses the change in consumption as a function of the change in wage rates of each household head and the change in household income. This specification requires baseline consumption data, but preexperimental observations on nondurable consumption were not obtained in SIME/DIME. The authors attempted to overcome this problem by predicting consumption for all households through regression of the consumption of control families on a set of exogenous variables, using data from the first year of the experiment.[6]

Their empirical specification also included a dummy variable to capture those families who were above the tax breakeven level before the experiment, assignment variables, dummy variables for the training/counseling treatments, and other control variables, including earnings, nonwage income, homeownership and vehicle ownership (all measured before the experiment), a set of age and education variables, and variables describing the ages of the children in the household.

They estimated experimental effects both with a treatment dummy specification and their structural model, and estimated effects separately for dual-headed and female-headed families.

Because the SIME/DIME expenditure data are not sufficiently comprehensive to permit analysis of the utilization of total disposal income, the authors estimated separate effects for several items noting that although it would be preferable to use a tobit estimation procedure in this situation, their limited resources would only permit estimation with ordinary least squares (OLS).

Table 6.2 presents experimental effects estimated using a treatment dummy. Overall, these estimates indicate that the experiment had some effect on consumption patterns of the SIME/DIME families, but the effects are not consistent across family types. In dual-headed households, the experiment led to a 15 percent reduction in expenditures on meals, but the (statistically insignificant) effect in female-headed households is an increase of almost 14 percent. Female-headed households used their experimental payments to increase their consumption of clothing by about 20 percent and records by 74 percent. The experiment also appears to have increased expenditures on clothing in dual-headed households. None of the other percentage effects in table 6.2 border on statistical significance.

TABLE 6.2.—ESTIMATED AVERAGE PERCENTAGE EFFECTS OF SIME/DIME ON CONSUMPTION

	Dual headed households expenditures	Single female headed households expenditures
Clothing	8.8	19.7
Meals	[1] −15.3	13.9
Trips	−18.0	−7.9
Records	−0.4	[1] 73.9
Events	−3.9	−2.7
Gifts	12.5	33.5
Housewares	−0.8	103.2
Beauty	16.8	−39.5

[1] Based on estimated coefficients significant at the 5-percent level.

Thus, Johnson et al. found that families used their experimental payments to increase the share of their income spent on clothing. Female-headed families, who tend to be substantially poorer than dual-headed families, may have increased their relative expenditures on food.

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[6] Results of these regressions are not reported in their paper. As in the labor supply work reported in Part III, Johnson et al. also estimate wages for nonworkers on data for workers.

Effects on housing

Because housing costs account for such a large share of total expenditures by low-income families, estimating the effect of an NIT on housing expenditures answers an important component of the question of how families would spend increased income support payments. But interest in the effect of SIME/DIME on housing expenditures also arises because of the debate over alternative programs to insure adequate housing for the poor: housing allowances versus income support.

Previous research on other NIT experiments indicates that families receiving experimental payments increased their rental payments and were more likely to purchase homes than control families. The Gary Experiment increased average monthly rental expenses by about 4 percent (Kaluzny 1980), and similar experimental effects were found in New Jersey (Wooldridge 1977).[7] The Gary and New Jersey Experiments also had the effect of increasing the probability of homeownership of black families by around 9 percent (Kaluzny 1980; Wooldridge 1978). The Gary Experiment studied only black families; in the New Jersey Experiment the effect on purchasing a home was found for black but not white or Spanish-speaking families.

Ohls (1980) estimated the impact of SIME/DIME on housing consumption using housing demand functions. He examined several measures of housing consumption:

- monthly rent (for renters)
- estimated current market price (for owners)
- hedonic quality indices, and
- whether dwelling units met housing standards.

The hedonic quality indices are calculated by regressing rents and housing prices on dwelling and neighborhood characteristics. They provide a means of translating dwelling characteristics into the amounts of housing services the dwellings provide in a way that is independent of rents and prices paid.

Ohls regressed these various measures of housing consumption on 3-year average household disposable income, a vector of demographic variables (stratification variables for ethnicity and number of household heads, and a series of dummy variables for different family sizes), and alternative measures of experimental treatment. For most of his analyses, he used average experimental payments received by families over the 3 years prior to the collection of housing data, but he also computed average experimental effects using a dummy variable for treatment. OLS techniques were used for analysis of continuous dependent variables, but probit techniques were employed for dependent variables that are dichotomous.

Ohls found that SIME/DIME increased housing consumption in the rental sector, whether he measured consumption as market rent or as a hedonic index. However, the effects, though statistically significant, are small; the elasticity of housing demand with respect to NIT payments is between 0.2 and 0.6. However, when he accounted for two other indirect effects of the experiment on housing consumption—effects generated by the impact of SIME/DIME on the reduction of nonexperimental transfer income and labor supply—estimated experimental effects were less than half as large and no longer statistically significant.

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[7] The Housing Allowance Demand Experiment increased rental expenditures by 19 percent (HUD 1978). These results are not strictly comparable to the NIT experiment, however, because the benefits in the housing experiment were restricted to purchasing housing—and for some of the experimental treatments, there were also minimum housing standards specified in order to be eligible for the benefits. Two types of housing allowances were tested: a "percent of rent" allowance (where experimental group families were reimbursed for a specified fraction of their gross rent) and a "housing gap" allowance (where families receiving this experimental treatment obtained an allowance based on a specified fraction of the difference between its income and the cost of adequate housing). See Kennedy (1980) and Rosen (1981).

In other analyses, Ohls found that SIME/DIME increased the probability of meeting minimum housing standards in Denver, but not in Seattle. Separate analysis of the housing consumption of homeowners indicated no experimental effect, but combined analysis of the rental and ownership sectors revealed larger experimental effects than when each sector was analyzed separately. Ohls interpreted these findings as suggesting that switching from rental housing to owner-occupied housing was one of the major ways in which families increased their housing consumption in response to the experiment. For example, movers had somewhat higher experimental effects than nonmovers.

Ohls compared experimental effects for 3- and 5-year duration families and concluded that his housing consumption findings are not sensitive to the duration of the experiment.

An important finding that emerges from the social experiments is the differences in income elasticities of rental expenditures between the NIT experiments and the Housing Allowance Experiment. The NIT experiments found that 6 percent or less of net income increases were spent on housing, which is somewhat less than observed in the general population (Carliner 1973). Constrained payments under the Housing Allowance Experiment, however, resulted in 36 percent of net income increases being devoted to increased housing consumption. One reason for low income elasticities in the NIT experiments, suggested by Philip Robins, is that the experiments were undertaken in low-income neighborhoods with relatively poor quality housing, so that families in the study samples may have less preference for quality housing than the general population. Of course, the differences in the share of net income spent on housing between the income maintenance experiments and the housing allowance experiment may be due to supply differences or price differences across the items studied. There may also have been a Hawthorne effect in the housing experiments. Nonetheless, findings from the social experiments provide evidence that low income families will spend strikingly less on housing under an NIT than under a housing allowance program, in which benefits are tied to housing and minimum housing standards are specified.

Effects on asset stocks

Little is known about asset stocks of the poor, or about how an NIT would affect their demand for asset stocks. As with other aspects of consumption discussed above, if NIT payments are regarded as transitory income, they will tend to result in an increase in assets. To the extent that an NIT program taxes the return on assets (or imputes a return where the asset does not generate taxable income), as was the case for selected assets in SIME/DIME, an NIT would discourage asset acquisition. Because an NIT changes the value of time by changing the net wage rate, asset stocks become more or less attractive depending on whether they require or save time. Finally, an NIT creates a floor below which income cannot fall, thereby reducing one motive for saving and building up asset stocks. Thus, it is not at all clear a priori how an NIT would affect asset demand.

In the other experiments, there was some evidence that the NIT increased net savings. Findings from the New Jersey (Nicholson 1978), rural (Michael 1976) and Gary (Hotz, Metcalf, and Minchella 1978) Experiments all indicate that debt was reduced by one or another subgroup. The Rural Experiment also resulted in an increase in savings. In Gary, there is evidence that the experimental payments enabled husband-wife families to increase their financial assets and shift their debts away from high interest lending situations such as stores and credit cards to more traditional lending institutions such as banks, credit unions, and savings and loan associations. Thus, higher incomes and more stable income streams provided by the experiment may have induced an increase in the supply of credit from traditional sources. Although no effects on the assets or debt of female-headed families were observed in Gary, both husband-wife and female-headed families tended to use the experimental payments to acquire home appliances, but not automobiles.

Pozdena and Johnson (1979) examined the effect of SIME/DIME on the acquisition of durables, vehicles, liquid assets, nonmortgage debts, home value, and mortgage debt. Previous research on asset demand in the experiments used a portfolio adjustment model, using predicted values of a stock demand model as estimates of desired stocks. Pozdena and Johnson adopted an alternative approach that postulates a partial adjustment process for each asset stock. Their empirical specification included a vector of variables that capture cross-section variation in user costs, wealth, and tastes affecting the desired stock (net wage prior to the experiment, predicted preenrollment income, net worth, family size, age of the head, and a dummy variable for nonwhites). The NIT treatment was estimated both by (1) a set of dummy variables and (2) predicted changes in the tax rate and disposable income. Normal income (part of the assignment variables) and counseling/training treatment dummies were also included in the equations.

In an attempt to estimate effects of the income security (reduced risk) facet of an NIT, separate effects were estimated by Pozdena and Johnson for families that were either below or above a predicted NIT breakeven level; they interpreted below breakeven estimates as payments effects and above breakeven estimates as income security effects.

NIT effects were estimated for the 6th experimental quarter. The nature of the dependent variable suggested a tobit technique; but no differences were found between OLS and tobit estimates, so Pozdena and Johnson reported only OLS results.

Their findings for below predicted breakeven level families are presented in table 6.3. SIME/DIME appears to have little effect on asset demand. Husband-wife families below predicted breakeven significantly increased their nonmortgage debt, while there appear to be some small (and not statistically significant) effects on increasing acquisition of durables and mortgage debt for female-headed families below the predicted breakeven level.

TABLE 6.3.—LONG-RUN EFFECTS OF SIME/DIME ON ASSET DEMAND

	Husband-wife families	Female-headed families
Durable	−49.25	133.49
	(107.22)	(97.24)
Vehicles	153.99	−119.34
	(219.74)	(143.07)
Liquid assets	20.21	31.02
	(56.64)	(79.13)
Nonmortgage debt	[1] 407.96	−27.94
	(177.17)	(119.47)
Home value	−560.54	−153.13
	(1354.88)	(2092.73)
Mortgage debt	35.16	592.53
	(676.88)	(578.44)

[1] Significant at the 1-percent level.

For families above predicted breakeven, SIME/DIME has an effect on debt acquisition: husband-wives families above predicted breakeven acquired significantly more mortgage debt and similar female-headed families acquired more nonmortgage debt.

Pozdena and Johnson tested for differential response by race and duration of the experiment, but found no statistically significant effects.

The SIME/DIME findings on asset demand contrast sharply with the findings from other NIT experiments. While other experiments observed an increase in saving and reduction of debt, SIME/DIME resulted in increased debt. The different models used in SIME/DIME might account for the different findings. The differences could also be the consequence of supply side differences across the experimental sites, or differences in the relative generosity of the treatment across the experiments.

CHAPTER IV

MOBILITY

There are several reasons why it is important to have knowledge concerning the impact of a negative income tax on geographical mobility. Migration can be seen as an equilibrating mechanism in a changing economy in that it tends to reduce interregional differences in real wage rates, property values, and unemployment rates. Any substantial redistribution of the low-income population among different parts of the country has important consequences for regional economic growth and development; and, to the extent that tax revenues and expenditures are affected, it can also have important fiscal effects on local government. Similarly, population shifts within a city can result in institutional change and fiscal strain in the neighborhoods affected, and may even alter the extent of ethnic-racial integration. Finally, the effect of welfare reform on the pattern of migration to and from depressed regions or neighborhoods with high unemployment rates is an important policy concern.

Quite apart from macroeconomic considerations, however, migratory behavior has potentially important economic and sociological consequences for the individual in affecting his or her level of well-being.

The SIME/DIME data base provides a valuable opportunity to examine the impact of a nationwide negative income tax in the United States on the tendency to move and on destination choices and earnings of movers. Because all SIME/DIME enrollees who moved within the continental United States continued to receive the NIT treatments for which they were eligible and continued to be interviewed, the benefit structure can be considered to be independent of region.

This chapter first summarizes research by Keeley on the impacts of an NIT on geographical mobility and on destination choices and earnings of individuals who migrated out of the Denver and Seattle areas. Next, there is a summary of work by Ohls and Thomas on intracity mobility and by Avrin on the use of publicly subsidized housing.

Migration

The decision to migrate, like many choices, is expected to be influenced by the costs and benefits—both pecuniary and nonpecuniary—expected to result from the decision. Keeley proposed two models of human behavior to explain migration, one focusing on pecuniary benefits and the other on nonpecuniary benefits.[1] The two models are not mutually exclusive.

The first is the human capital model. There is considerable empirical evidence that wage differentials importantly motivate migration (DaVanzo, 1976). Viewed as a response to wage differentials, migration may thus be regarded as an investment in human capital—similar to investments in schooling, work experience, and health. According to the human capital model, an NIT influences the decision to migrate because it affects the costs and benefits of an investment in migration. Assuming that hours of work or labor supply are fixed, if the monetary costs of migration are to be borne by the individual (as was the case in SIME/DIME), then the tax rate of the NIT acts to reduce the pecuniary gain from any increase in wages brought about by the move.[2] We know, of course, that the assumption of fixed labor supply is unrealistic: both the guarantee and tax components of an NIT actually *reduce* desired hours of work. Under these conditions, the profitability of migration is further reduced.

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[1] The discussion in this section is a nontechnical exposition of Keeley's theoretical discussion. For a detailed, technical exposition, refer to Keeley (1977) on geographical mobility and to Keeley (1979) on destination choices.

There are, nevertheless, two facets of the NIT that would lead us to expect a higher rate of migration. First, difficulty in raising money for migration presumably deters many low-income people from moving, and the NIT will facilitate self-financing of migration. Second, the NIT tends to reduce the financial risk of migration because both the guarantee and tax components reduce the variability of disposable income.

The second explanatory model of migration focuses on nonpecuniary consumption. For example, the quality of the environment is better in some areas of the country than others. Holding other factors equal, there tends to be a negative tradeoff between the level of a region's real wage level and the region's environment. That is, locations with more desirable environments have lower real wage rates. This is because a region with a superior environment [3] and equal or better wage rates continues to attract new population until either real wages fall or congestion reduces the quality of the environment. Thus, migration is one important way individuals can substitute nonmonetary consumption (the environment) for income or market consumption.

An NIT increases the desirability of environmental consumption relative to market consumption, because the high tax rate of the NIT reduces the return from market work but does not tax consumption of the environment. Thus, this model of migration predicts that an NIT will lead to a redistribution of the population to regions with better environments and lower wages.

NIT effects on rates of migration

The sample used for the analysis of experimental effects on the rate of migration consisted of all originally enrolled black and white family heads (Keeley, 1977; 1980).[4] The unit of analysis was the individual rather than the family.[5] A person was considered to have moved out of the area if he or she moved away from the greater Seattle or Denver metropolitan area and remained away long enough to miss a periodic interview. The rate of mobility was estimated with data from the first nine experimental quarters after enrollment, using Tuma's log linear stochastic rate model (Tuma, 1976). This uses a maximum likelihood technique to estimate effects of the experiment on the duration of time elapsing from enrollment until a move occurs.[6] Estimating equations included a set of control variables which were: E-level (dummy variables representing normal preexperimental income levels used in assigning families to treatment), annual income prior to enrollment, site (Denver-Seattle), counseling/training treatment dummy variables, employment status (a dummy variable that equals one if the person was working preexperimentally), number of family members at enrollment, age, and education. In the first equation, experimental treatment was represented by two dummy variables: one indicating assignment to 5-year treatment and the other indicating assignment to 3-year treatment. Results are shown in table 6.4.[7]

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[2] If monetary costs are deductible from income, the tax rate reduces gains and costs proportionately and thus would not affect the rate of return to migration.

[3] Keeley did not specifically define "superior" environment. It is clear however, that he considered climate to be a key factor.

[4] Data for Chicanos were not available at the time of this analysis.

[5] This is because there was considerable family change in SIME/DIME due to divorce, marriage, and remarriage. If the family were used as the unit of analysis, response would have to be measured conditional on the family remaining intact. There is evidence that experimental treatments affect dissolution rates. Thus, if mobility were related to family status, such an analysis might lead to biased estimates of treatment effects on mobility.

[6] There is a complete technical exposition of this model in Part V.

TABLE 6.4.—EXPERIMENTAL EFFECTS ON RATE OF MIGRATION [2]

	Average rate [3]	5-year NIT effect	Differential effect for 3-year treatment
White:			
Married males	.072	[1] 1.495	.889
Married females	.069	[1] 1.450	.904
Single females	.077	.510	3.177
Black:			
Married males	.027	.817	1.573
Married females	.023	.916	1.260
Single females	.070	.850	1.150

[1] Significant at the 10-percent level.

[2] Multipliers of the rate are presented. These give the percentage effect of the NIT treatment on the rate of mobility. For example, for white married males, the ratio of 1.495 indicates that experimentals in the 5-year sample moved out of the area at a rate 1.495 times greater than controls. The differential 3-year effect of 0.889 indicates that white married males in the 3-year sample moved out of the area at a rate 1.329 (1.495 × .889) times greater than controls.

[3] This is the average rate per year of moving out of the area of both experimentals and controls.

Over the period of observation of approximately 2.5 years the average rate of mobility for white husbands and wives, both experimentals and controls, was approximately 7 percent per year. White married males receiving the 5-year NIT treatments had a rate of mobility 49 percent greater than controls and white married females in the treatment group moved at a rate 45 percent greater than controls ($p<.10$). For married black males and females, the average rate of mobility was only about 2.5 percent per year and the NIT treatment had no statistically significant effect. No statistically significant effects were found for single females. Nor were there statistically significant differences between effects of 3-year and 5-year NIT treatments on mobility rates for any married subgroup, although single women receiving the 3-year treatment had a greater rate of migration than single women receiving the 5-year treatment.

In a second equation, treatments were parameterized in terms of the following three variables: whether or not someone was above the breakeven level; and, if the person was below, the percentage change in disposable income and the percentage change in the net wage. (Results do not appear in tabular form.) For white married males, there were statistically significant effects of income and tax changes on the rate of mobility. A decrease in the net wage resulted in a statistically significant increase ($p<.01$) in the rate of mobility. An increase in disposable income led to a decrease in the rate of mobility ($p<.10$). Similar results were found for white married females although the coefficients were smaller and the standard errors were larger. No significant effects for other subgroups were detected; however, in all cases, an increase in income resulted in a lower rate of mobility.

The finding that higher tax rates lead to increased migration supports the notion that people move in response to the NIT for environmental reasons. This is because the wage-differential model predicts that higher tax rates would make people less likely to move, a prediction opposite to that of the environmental model. The negative income effect, however, is not consistent with either model.

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[7] While the rate model is estimated in the log linear form

$$\ln R = b_0 + b_1x_1 + b_2x_2 + \ldots + b_jx_j,$$

results are easier to interpret if reported in the equivalent multiplicative form,

$$r = e^{b_0} . e^{b_1x_1} . e^{b_2x_2} \ldots . e^{b_jx_j}.$$

When dummy variables are used in the analysis, the interpretation of e^{bx} is straightforward. It is the ratio of the rate of mobility for experimental and control subjects with similar characteristics.

Finally, there were no statistically significant effects of site and NIT interaction on the rates of mobility. This finding suggests that the experimental effects do not depend on the unemployment rate, which was much higher in Seattle than in Denver.

It is possible that estimated rates of mobility were biased by selective attrition from the experiment. This could be the case, for example, if people receiving low payments and with high probabilities of moving tended to drop out at a greater rate than other people. Keeley estimated the probability of attrition for each of the two equations discussed above. In the first case, where treatment was represented by two dummy variables, there were no significant differences in attrition rates between experimentals and controls. In the second case, however, when treatment was parameterized in terms of changes in income and the net wage, results indicate that for all subgroups except white married males at least one of the treatment coefficients was significant at the 10-percent level, indicating a possible biasing effect of selective attrition. Attrition is negatively and significantly related to the percentage change in income for white married females ($p<.10$) and white single females ($p<.05$).

Keeley (1979) also investigated the effects of the NIT on the destination choices and labor market behavior of migrants. The sample consisted of black, white, and Chicano individuals who migrated out of either Seattle or Denver during the first nine quarters of the experiment, and who completed a post-move interview.[8] Approximately 70 percent of all migrants completed at least one post-move interview, resulting in a sample of 181 married males and 121 single females.[9]

Using OLS regression, and controlling for assignment variables, Keeley found that men receiving the NIT treatments chose regions with higher January temperatures and smaller temperature variations; also, experimentals were more likely than controls to indicate that climate was an important reason for their choice of destination. This finding is consistent with the environmental model of migration discussed earlier. However, there was no NIT effect on climate among destination choices of single females.

In further analyses, Keely found that, for married males, the distance moved (for the white subgroup only) and the costs of moving (for all married males and the white subgroup) were higher for those eligible for the NIT than for controls. There were no significant differences between experimental and control migrants concerning their intention to work at their destination, whether they had a job lined up, hours of work per week expected, or expected earnings per week. Again, for single females, there were no experimental-control differences in these job-related variables.

Effects on labor market behavior and experience of migrants

For this analysis, data pertaining to the year subsequent to migration for all persons who moved during the first nine quarters of the experiment were used—including married males, married females, and single female heads. Persons who moved out of the area and then returned to the sites within 1 year were excluded from the sample.

Table 6.5 presents experimental treatment coefficients for (a) married males, (b) married females, and (c) single females. For married males, there were large, statistically significant negative effects on total annual hours worked and days worked. In addition, there was a significant negative effect on earnings. Since there was a larger treatment effect on earnings than hours, experimentals who moved must have received lower wages than controls. There were no significant effects on unemployment or liquid assets.

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[8] No destination information is known about migrants who did not complete postmove interviews.

[9] Responses for married women were not analyzed separately as their responses are the same as their husbands'.

TABLE 6.5.—EXPERIMENTAL EFFECTS ON LABOR SUPPLY VARIABLES OF MOVERS, FOR MARRIED MALES, MARRIED FEMALES, AND SINGLE FEMALES

[Standard errors in parentheses]

Dependent variable	Married males	Married females	Single females
Number in sample	181	201	121
Total days worked:			
Mean	247.69	90.00	83.93
NIT effect	[3]−74.32	−23.17	[2]−82.61
	(17.44)	(18.51)	(34.89)
Total hours worked:			
Mean	1,547.47	473.70	426.75
NIT effect	[3]−514.80	−97.54	[2]−452.39
	(133.42)	(103.77)	(199.45)
Days involuntarily unemployed:			
Mean	72.02	40.83	81.99
NIT effect	26.99	[1]−24.80	30.36
	(17.13)	(13.51)	(32.97)
Total earnings:			
Mean	5,038.84	985.24	1,218.83
NIT effect	[3]−2,014.00	−169.70	[2]−1,355.05
	(505.51)	(239.40)	(675.69)
Liquid assets:			
Mean	1,378.07	1,351.47	1,204.17
NIT effect	−430.56	−146.06	[1]−1,931.83
	(506.16)	(563.92)	(1,019.40)

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

For married females, all treatment effects were negative, but the only statistically significant effect was on unemployment. Since many women are "tied" movers (that is, they move with their husbands), there is less reason to expect a treatment effect.

For single females, the NIT treatments had a significant negative impact on all variables except days unemployed. As is true for married males, the treatments had a larger proportional effect on earnings than hours, suggesting that single female experimentals who moved obtained lower wage jobs than similar controls.

To summarize, the results for both married males and single female heads are consistent with the environmental consumption model of migration. That is, the experiment induced persons to move to locations with lower wages and also to reduce hours of work. One additional important implication of these findings is that since the SIME/DIME treatment caused migrants to have large reductions in labor supply, results from the NIT experiments that excluded migrants may seriously underestimate the labor supply response to an actual program.

Residential mobility

As with migration, intracity mobility is a way families can adjust their circumstances. Families may change residence to upgrade the quality of their housing or neighborhood, to move closer to jobs or to other family members, to reduce housing costs, or to leave a personally troublesome situation.

The effects of income maintenance on residential mobility have been examined in other income maintenance experiments. Using data from the rural income maintenance experiment, Johnson (1976) found that the experiment increased the probability of changing residence by 5 percentage points. No statistically significant effects on housing mobility were found in the New Jersey (Wooldridge 1977) and Gary (Kaluzny 1980) Experiments. However, families eligible for experimental payments in New Jersey (Wooldridge 1977) and Gary (Kaluzny 1976) did move out of public housing into private housing at significantly greater rates than control families.[10]

In contrast to the findings from other NIT experiments, the housing allowance payments in the Housing Allowance Experiment had little effect on residential mobility. Housing allowance recipients did not move significantly more than controls (MacMillan 1978). Nor did the housing allowances result in increased integration—minority families did not leave neighborhoods with concentrations of minorities. For those families that moved during the housing allowance experiment, there was also no experimental effect on upgrading of dwellings (Vidal, 1978).

We next summarize work by Ohls and Thomas (1979) on the effect of SIME/DIME on the probability of an intracity move and on relocation patterns of movers. This is followed by a discussion of Avrin's (1978) investigation of the effect of an NIT on the utilization of subsidized housing.[11]

NIT effects on the probability of moving

About half the families in SIME/DIME moved at least once during the experiment. The effect of the experiment on the probability of moving was estimated in a regression context using a probit model. In addition to a measure of experimental status, the model included average nonexperimental income, measures of changes in family circumstances that are typical life cycle milestones, characteristics of the preexperimental dwelling unit, and demographic characteristics.

Households were classified into two groups for separate analysis: intact households—where the same single adult or pair of adults was head of household throughout the experiment, and nonintact households—where individual heads either left the household or joined it during the experiment. In addition, the two sites were analyzed separately.

The experimental variable for the analysis of intact households was average experimental income (calculated over 3 years prior to the 10th periodic interview). The data tape available for this study did not permit creation of this variable for nonintact households, so an experimental dummy was used for analysis of nonintact households.

For the most part, these analyses found that SIME/DIME had little effect on mobility. There were no significant effects on the probability of moving in Seattle, and nonintact households in neither Seattle nor Denver appeared to move as a result of the experiment. However, for intact households in Denver, the experiment did result in greater residential mobility; families who received $2,000 a year in experimental payments were 9 percent more likely to move than control families.

Thus, the mixed mobility response found in the other income maintenance experiments was repeated in SIME/DIME. We conclude that, while an NIT can be expected to increase the residential mobility of some types of households in some sites, it will not result in a general increase in mobility.

Effects on relocation patterns

Ohls and Thomas (1979) also examined whether SIME/DIME had impacts on where families moved. In particular, they investigated whether SIME/DIME households moved to areas with higher mean incomes, to areas with smaller proportions of racial minorities, or to locations further from the center of the city. Movement of households was plotted by census tracts on maps, and regression analysis was used to examine whether the experiment affected types of neighborhoods selected. Regressions included preexperimental housing status, nonexperimental income, and demographic variables as well as experimental income. Separate analyses were conducted for each site, and all households in a site were analyzed both in the aggregate and separately by ethnic group.

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[10] The probability of leaving subsidized housing was not studied in the Rural Experiment.

[11] These researchers examined other related issues also, such as housing search strategies employed by SIME/DIME experimentals; see Ohls and Thomas (1979).

SIME/DIME did have some significant effects on relocation decisions, although these effects vary by site and are concentrated in particular ethnic groups. Families eligible for experimental NIT payments tended to move to higher income neighborhoods in Denver but not in Seattle. In Seattle, the experiment resulted in families moving to neighborhoods with lower proportions of blacks, but this movement may have been concentrated among white families in the experiment since, overall, black families in Seattle appear to have moved somewhat in the direction of greater segregation. In Denver, the experiment had no effect on integration, while the study sample generally was becoming more segregated as the experiment progressed.

The experiment influenced the relocation of black and Chicano households in Denver to neighborhoods further from the center of the city, but no such effects were observed in Seattle nor for whites in Denver.

Effects on utilization of subsidized housing

Avrin (1978) investigated the effect of SIME/DIME on the probability (1) of living in subsidized housing, (2) of leaving subsidized housing, and (3) of moving into subsidized housing.

As in other NIT experiments, in SIME/DIME the public transfer represented by a housing subsidy was not taxed in the payment calculation, even though other public transfers were generally taxed. Also, as in other experiments, arrangements were made with the local housing authorities to ensure that receipt of experimental payments did not cause participants to be forced to leave public housing.

Avrin posited both an income effect and a type of substitution effect of the NIT on the use of subsidized housing. An increase in family disposable income will increase the quantity of all goods, including housing, that a family can consume. At the same time, increased income raises the price of subsidized housing relative to nonsubsidized housing since the price of subsidized housing is a constant proportion of income. The theory implies that the direct effect of an NIT-caused increase in disposable income, given no labor supply or family structure changes, would be to decrease the utilization of subsidized housing. However, the model predicts increased use of subsidized housing as the experimental tax rate increases. This is because labor supply and income are expected to decline as the tax rate increases, thus reducing the price of subsidized housing.

Avrin estimated housing demand equations similar to those described earlier. The effects of the experiment were measured by first, an estimated experimental payment based on a quarter of preenrollment earnings; second, a variable measuring the change in the marginal tax rate (to capture the effect of the NIT on labor supply, income, and hence the relative price of subsidized housing); and third, a dummy variable representing families above the breakeven level of income at enrollment. Also included as explanatory variables were family size, normal income category, race, site, and number of family heads (the assignment variables), family size interacted with treatment, age, and whether a family was on welfare or lived in subsidized housing prior to the experiment. Separate estimates for sites were computed, although samples were pooled by race and marital status.[12]

For female heads in Seattle and Denver, SIME/DIME reduced their probability of residing in subsidized housing. However, this NIT effect diminished the larger the family size. In addition, in Denver, female heads responded to the tax rate of the NIT by increasing the likelihood of residing in subsidized housing. (A doubling of the tax rate resulted in female heads in Denver being about 9 percent more likely to reside in subsidized housing.) The families with female heads in Denver were also 13 percent less likely to live in subsidized housing if they received NIT payments and lived in subsidized housing prior to the experiment. Table 6.6 summarizes the findings for female heads. For a family of four, the net effect of a $1,000 increase in disposable income and a 25 percent increase in the tax rate decreases the probability of living in subsidized housing by 4.3 percent in Seattle and 13.9 percent in Denver, if the families resided in public housing prior to the experiment.

TABLE 6.6.—PROBABILITY OF RESIDING IN SUBSIDIZED HOUSING BY FAMILY SIZE BASED ON AN INCREASE OF $1,000 IN DISPOSABLE INCOME AND A 25 PERCENT INCREASE IN TAX RATE

Family size at enrollment	Male Heads		Female Heads		Intact families
	Denver	Seattle	Denver	Seattle	
2	.31	.51	.53	.55	.70
4	.31	.56	.56	.59	.72
6	.32	.60	.58	.64	.74
8	.33	.66	.61	.68	.76

For male heads in Seattle, a similar picture emerges, except that estimated coefficients are not as highly statistically significant as in the case of female heads. For male heads in Denver, the signs of coefficients on the change in disposable income and tax rate variables are opposite to those of other groups; an increase in disposable income reduces the probability of residing in subsidized housing, and an increase in the tax rate increases the probability. Strikingly, a family with a male head receiving the NIT is 26 percent less likely to reside in subsidized housing if they resided there prior to the experiment. Net effects of an increase of $1,000 in disposable income and 25 percent in the tax rate, if the male-headed family resided in subsidized housing prior to the experiment, was zero in Seattle and 27.2 percent less in Denver. (See table 6.7).

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[12] OLS estimates are reported. Avrin notes that she tested logit models, but the alternative procedure did not change her findings.

TABLE 6.7.—PROBABILITY OF RESIDING IN SUBSIDIZED HOUSING FOR A FAMILY OF FOUR WITH AN INCREASE OF $1,000 IN DISPOSABLE INCOME AND A 25 PERCENT INCREASE IN TAX RATE

	Lived in subsidized housing prior to experiment		Did not live in subsidized housing prior to experiment	
	Control probability	Change	Control probability	Change
Female heads:				
Seattle	.706	−.043	.089	−.007
Denver	.503	.000	(¹)	(¹)
Male heads:				
Seattle	.731	−.139	.022	−.006
Denver	.627	−.272	.046	−.003

¹ Not applicable.

Avrin also examined the probability of living in subsidized housing for a sample of intact families, and the probability of leaving and entering subsidized housing for all three samples—female heads, male heads, and intact families. The results amplify the findings summarized above, and generally confirm that SIME/DIME provided incentive for families to leave subsidized housing.

Across all samples, the magnitude of NIT effects diminishes as family size increases. Avrin suggests that this may be the result of three factors. First, housing subsidies are greater for larger families with a given income than for smaller families. Second, a given change in disposable income induced by the NIT is less important to a larger family, so that a larger family would require a larger income increase to abandon subsidized housing. Third, larger families may have more difficulty than smaller ones in obtaining suitable housing in the private rental market.

Conclusions

First, there is some basis for the tentative conclusion that one result of a nationwide NIT program would be a redistribution of the low-income population to regions with better environments and lower wage rates. This is based on Keeley's findings that white migrants receiving experimental treatments were more likely than controls to move to destinations with a better climate, and (for males) to move further, and spend more on the move. Moreover, the move is likely to result in lower earnings and lower hours of work for experimentals than for controls.

Second, as with the other income maintenance experiments, SIME/DIME has little effect on intracity residential mobility and no effect on integration. The only significant finding was that intact households in Denver who received $1,000 a year in experimental payments were 9 percent more likely to move than control families. Also, in Denver, movers receiving experimental payments tended to move to higher-income neighborhoods than movers who were controls.

Finally, as with the other income maintenance experiments, SIME/DIME resulted in a decrease in the utilization of subsidized housing. However, the latter effect would not necessarily materialize under a national program unless there was a supply response in the private housing market.

CHAPTER V

CHILDREN

In this section, we investigate two issues, each of which relates to labor market impacts of SIME/DIME. The first is child care for working parents. The availability of good child care undoubtedly enhances the employment potential of many mothers. Child care subsidies of several forms were available to all low-income families in Seattle and Denver (including, of course, SIME/DIME control families); families receiving experimental treatments were eligible for somewhat more generous subsidies than those otherwise available. In the first part of this chapter, we discuss work by Munson et al. (1980) that examines the impact of the SIME/DIME program on both the demand for child care and the mother's labor supply. Results of other studies by Kurz, Robins and Spiegelman (1975) and by Robins and Spiegelman (1978, 1979) on the demand for child care are also briefly summarized.[1]

The second issue dealt with in this chapter is the effects of the NIT on children's school performance. The studies reviewed below investigated NIT impact on school performance and NIT effects on home environment (which is held to affect school performance). Research here is motivated by the assumption that there is a relationship between a child's scholastic achievement and his or her future employability.

Child care

The issue of child care for working parents has become increasingly important over recent years with growing proportions of mothers entering the labor force. The availability of publicly subsidized child care may have a strong impact on labor supply decisions of mothers. Moreover, the issue is critical for public policy because of both its impact on labor supply and the costs involved in government-sponsored child care programs.

Forms of child care subsidies

Several forms of child care subsidies were available to families in Seattle and Denver. For one-parent families and two-parent families on AFDC and AFDC–U, child care expenses were fully reimbursed up to a limit specified in the regulations through title IV–A of the Social Security Act. The SIME/DIME program offered subsidies that were essentially identical with AFDC to eligible one-parent families, the major difference being that SIME/DIME subsidies were available to experimental families regardless of income whereas under AFDC the subsidy is income-conditioned. For two-parent families receiving an NIT treatment, child care was treated as a work expense that was reimbursed fractionally depending on income and the tax schedule of the experimental treatment.[2] One-parent families not on AFDC or SIME/DIME (that is, non-AFDC control families) were eligible for subsidy under title XX of the Social Security Act, provided their incomes were sufficiently low.

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[1] In addition to the work reviewed here, studies were conducted on the supply of day care available in Denver and Seattle during the experiment and estimation of costs of child care (Hall and Weiner, 1977; Weiner, 1978; Hall, 1978).

Typology of child care

For policy purposes, it is important to distinguish among formal and informal market and nonmarket care. Formal market care includes modes of child care licensed by the state, such as day care centers and licensed family day care homes. Informal market care consists of all arrangements that are not licensed by the state, but for which a regular fee is charged. It includes sitter caring for children in sitter's home (equivalent to unlicensed family day care), and sitter caring for children in user's home. Nonmarket care is that for which no fee is charged and includes care by persons in the family other than the mother, and self-care by the child. Half the working women in the Seattle and Denver samples used nonmarket forms of care in 1972 (Kurz et al. 1975). This is consistent with findings of other studies (see Hall and Weiner, 1977).

It is reasonable to believe that subsidies available for child care will influence the decisions of working women concerning the mode of care selected. Kurz et al. (1975) examined the effects of a child care subsidy, as measured by a dummy variable representing eligibility for either AFDC or SIME/DIME on the child care mode selected by working mothers (both experimental and control subjects) in both one-parent and two-parent families. Robins and Spiegelman (1978; 1979) focused on working mothers in two-parent families, and compared the effects of the SIME/DIME subsidy with those available to two-parent families under the Federal income tax.

In brief, results point to the conclusion that a child care subsidy (through either AFDC or SIME/DIME in the Kurz et al. study and through SIME/DIME in the Robins and Spiegelman studies) will lead to greater use of market rather than nonmarket care.[3] In addition, the researchers' models predict that the demand for market care varies with the worker's wage rate and with family structure. For families with only children under 5, the utilization rate of market care was about 70 percent; for families with only children between 6 and 12, the utilization rate was about 50 percent; for families with at least one teenager or other adult present, the rate dropped to 20 percent.

On the basis of these findings, Kurz et al. conclude that a national program of child care subsidies, such as that available under the Federal income tax system, is likely to increase the usage of market rather than nonmarket forms of care. Moreover, because of its greater sensitivity to price and wage rate changes, they suggest formal market care will exhibit relatively greater changes in use.

The above studies treat the choice of child care mode as conditional on the decision to work. In fact, the child care decision and the decision to work are

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[2] The formula is as follows:

$$G = S - t_2 T_B = S - t_2 Y + t_2 C,$$

where G = the NIT grant
S = the NIT guarantee
t_2 = the NIT rate
$T_B = Y - C$ = the tax base (income Y less child care expenses C)

This indicates that for families receiving negative income tax benefits, the grant is increased by an amount equal to $t_2 C$.

[3] In both Seattle and Denver, a large number of day care centers were funded through nonwelfare government programs (for example, WIN, Head Start, and Model Cities). Because many persons being classified in these analyses as ineligible for subsidization were actually able to receive child care subsidies through these other programs, subsidy variables used tend to underestimate the true effect of subsidization for the use of formal care. Robins and Spiegelman (1979, footnote 22) point out that it is also possible effects are overestimated to the extent the price variable may be measuring effects of being in the experiment rather than changes in the price of child care.

likely to be made simultaneously. Munson et al. (1980) investigated the simultaneous effects of the SIME/DIME child care subsidy component on child care demand and on labor supply. Since the availability of subsidies for child care raises net wage rates of users of subsidized modes of care, this subsidy effect may to some extent offset the disincentive effects of the NIT program. The sample used in Munson's analysis consisted of 818 originally enrolled single women with at least one child under the age of 13 at the time the child care survey was administered.[4] Results must be interpreted conditional on the subjects' remaining single, having young children, and having remained in the experiment through the time of the child care survey. There were 406 working women and 412 nonworking women. Women were defined to be working if they had any hours of market work during the week of the child care interview or during any of the 3 weeks immediately before the interview.[5]

The demand for child care and mother's labor supply was estimated simultaneously using a multinomial logit model. The model is based on their assumption that an individual chooses one of four mutually exclusive and exhaustive combinations of work behavior and child care mode: work and market care, work and nonmarket care, no work and market care, and no work and nonmarket care.

Table 6.8 shows the estimated probabilities for some selected characteristics of one-parent families.[6] The first column shows the effects of independent variables on the probability of working. The second column shows effects on the probability of using market care given that the mother works. The third column shows the joint probability of market care and work. To determine the effect of a given independent variable, compare the rows under each major subheading. For example, a comparison of probabilities in the first column for the first and second rows of the table indicates the difference in the probability of the mother working without and with a full child care subsidy.

TABLE 6.8.—PREDICTED PROBABILITIES FOR SELECTED CHARACTERISTICS OF ONE-PARENT FAMILIES

[Asymptotic standard errors in parentheses]

	Probability of—		Joint probability of market care and work
	Working	Market care given work	
Child care subsidy effects SIME/DIME:			
No subsidy[2]	.461	.415	.191
	(.086)	(.093)	(.055)
Full subsidy	.410	.443	.182
	(.047)	(.057)	(.031)
SIME/DIME manpower program:			
No subsidy:[2]	.455	.428	.194
	(.024)	(.036)	(.019)
Full subsidy	[1] .298	.462	.138
	(.048)	(.086)	(.034)
Experimental treatment effects change in net hourly wage rate:			
–$0.30	.432	.424	.183
	(.023)	(.035)	(.018)
–$0.40	.427	.434	.185
	(.022)	(.035)	(.017)

[1] Indicates that the difference in predicted probabilities associated with the change in the specified variable is greater than 1.96 times the standard error of the difference in predicted probabilities.

[2] Might receive subsidies from sources other than SIME/DIME.

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[4] Data for this analysis were obtained during the 8th periodic interview in Seattle and the 9th in Denver. This is later than the period analyzed in the Kurz et al. and Robins and Spiegelman studies which used data from the 4th periodic interview in Seattle and the Denver post enrollment interview.

[5] The authors warn that their sample may be biased through attrition and marriage, and because of experimental effects on fertility. However, they did not investigate these potential sources of bias.

[6] These probabilities are computed for the given value of each specified variable and mean values of all other variables in order to depict a "typical" response.

The findings indicate that there is no support for the hypothesis that SIME/DIME subsidies for work-related child care increase work effort. In fact, for women eligible for reimbursement of training and education-related child care expenses through the SIME/DIME counseling/training treatments, there was a significant reduction of work effort. (See first column of the table and compare the third and fourth rows.) This is presumably because the subsidy had a strong effect on the probability of enrolling in education or training programs, which usually entails a reduction in work effort during program participation. Consistent with the earlier findings, this study also found that family structure has a significant impact both on the decision to work and the choice of child care mode.

Eligibility for the subsidy, either through the NIT treatment or through the training/counselor treatment, did increase the probability of using market care given that the mother works;[7] however, effects are not statistically significant. The authors suggest that the lack of a strong effect is due to the fact that so many families in the sample were eligible for full reimbursement of some form of work-related subsidy, either through SIME/DIME or AFDC.

Recall that the earlier analysis by Kurz et al. and Robins and Spiegelman reported much stronger effects of subsidization on the use of market care for female-headed families than did the Munson et al. study. There are several factors that might account for this difference. The later analyses (by Munson et al.) used a sample consisting of all single women, both working and nonworking; the early analyses focused on working women only. The later analysis used data from later administrations of the child care module than did the early analyses. In the later analysis, child care was specified as either market or nonmarket; the earlier analysis, in contrast, split market care into formal and informal care. There were also several differences in the way independent variables were specified. In the early analyses, eligibility for either SIME/DIME or AFDC was specified by one dummy variable; in the later analysis, there were separate dummy variables to represent eligibility for SIME/DIME and for AFDC. Mothers' hours of work and earnings were also differently specified: the former two used data from the month before the child care module was administered; the latter used preexperimental data.

It seems likely that in both the earlier studies and in the later study, findings concerning effects of SIME/DIME subsidies for child care are confounded by the fact that control families were also eligible for various forms of subsidy. In any event, the strong effects of the subsidy on the use of market care reported in the earlier studies were not found in the later study using a different sample and somewhat different measures.

Children's school performance

The question to be investigated in this section is whether income transfers have the potential of raising the human capital of children by improving their performance in school, thereby leading to the possibility of increased long-term earning power. This issue was investigated with data from the rural experiment using measures of attendance, comportment, academic grades, and standardized achievement scores. The findings from that experiment indicated that elementary school children in North Carolina whose families received the NIT demonstrated significant improvement in their school performance as compared with controls. However, there were negligible effects of the NIT on the performance of older

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[7] The formula for estimating the probability of using market care given that the mother works is:

$$\text{Pr(Market Care Given Work)} = \frac{\text{Pr(Market Care and Work)}}{\text{Pr(Market Care and Work)} + \text{Pr(Nonmarket Care and Work)}}.$$

That is, the numerator is the joint probability of market care and work (third column of the table) and the denominator is the probability of working.

children in North Carolina and no effects on children of any age in Iowa.[8] The authors whose work is reviewed here base their arguments on a model that predicts a negative income tax will affect certain aspects of home environment which, in turn, will affect school performance. We first discuss the theoretical model and then present findings from five studies investigating different aspects of the model.

Knickman (1979) suggested that a complete model of determinants of school performance would specify that the child's school achievement in any one year is related to characteristics of the home environment, the school environment, the child's peers, and the child's genetic endowment.

There are three ways in which an NIT might alter the home environment: through an income effect, through a labor supply effect, and through a marital stability effect. The income effect results from the family's ability to purchase goods and services that would alter the quality of the environment: for example, better housing. The labor supply effect results from the well-documented tendency of the NIT to reduce the labor supply of parents due to income and tax effects of the NIT, an effect that is particularly strong for married women. This implies that parents in such households, and especially mothers, will have more time to spend with their children, possibly having a positive effect on student achievement. Also, to the extent that the NIT reduces employment time for children and youths, they will have more time for study and school-related activities.[9]

In summary, through its income and labor supply effects, the experiment is expected to improve the home environment of school children and thereby to have a beneficial effect on school achievement.

Our strategy in discussing studies on school achievement in SIME/DIME is as follows. First, we discuss research on the effect of an NIT on school achievement (Manheim and Minchella, 1978a). This study shows that an NIT essentially had no impact on school achievement. We then turn to a study of effects of the NIT on home environment (Manheim and Minchella, 1978b), and again—perhaps not surprisingly in view of results from the first study—there is basically no effect.[10]

Finally, we briefly discuss some methodological issues that arose in the course of this research together with an attempt by Hanushek and Knickman (1980) to explore an alternative methodological approach to studying school performance.

NIT effects on school performance

Manheim and Minchella (1978a) investigated the effect of experimental treatment on three indicators of school performance: standardized achievement scores, academic grades, and absence rates. As they point out, each indicator has both advantages and drawbacks. Standardized scores provide a relatively objective way of measuring achievement but will be relatively insensitive to learning gains in specific fields; they may also be fairly slow to change in response to a specific stimulus like the NIT. Academic grades are more likely to reflect achievement in specific subjects, but they are subject to teacher biases and may be normalized by immediate peer group performance. Finally, absence rates are on the one hand likely to reflect a commitment to schooling, but on the other hand will include absence due to student illness (not measured in SIME/DIME).

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[8] The explanation offered by the authors to account for differences between North Carolina and Iowa revolved around the socioeconomic differences between the two States. The Iowa sample had substantially higher socioeconomic status than the North Carolina sample; thus, a given level of support was of relatively greater importance in North Carolina. Also, there were differences in the data bases used. When the North Carolina analysis was redone using only data similar to those used in Iowa, the experimental response diminished.

[9] It is not clear that the NIT does reduce employment hours of young people in school. West (1979) found that the experiment reduced the labor supply of youths aged 16 to 21 at enrollment, and effects were strongest for those youths who remained in the home rather than moving out on their own. However, when he interacted NIT treatment variables with variables indicating whether or not the youth attended school, he found that the experimental effect was strongest for those not in school; for those in school, the experiment had a negative but not a statistically significant effect on labor supply (West, 1980). As we will report later in this chapter, Manheim and Minchella (1978b) found no NIT effects on students' outside hours of employment. It is also possible that much of the apparent labor supply effect for youths is due to underreporting bias.

[10] There was another group of studies (Knickman, 1979; Knickman and Phillips, 1980; Knickman and Wolf, 1980) that attempted to determine the relationship between home environment and school achievement. Findings were complex in that home environment variables, as a group, did seem to explain variance in school achievement, yet the research failed to identify conclusively which aspects of home environment are most critical in this regard.

There were several differences between Seattle and Denver in forms of data available; thus separate analyses were conducted for the two sites.[11] Sample size also varied widely according to site and the indicator of school achievement used.

Measures of school performance used by Manheim and Minchella for most of their analyses were obtained approximately 2½ years after the start of the experiment. Experimental treatment was specified in two ways: first, as a dummy variable indicating whether respondent is enrolled as experimental (=1) or control (=0), and second, the dollar increase in income that would occur under the experiment.

With only one exception, there were no significant effects of experimental treatment or any indicators of school achievement for either Denver of Seattle. The single exception is that in Seattle the experimental treatment dummy was positively related to grade point average ($p<.10$).[12] Since nonsignificant effects were small and not consistent in direction, the only safe conclusion seems to be that there was no experimental effect on school achievement.

NIT effects on home environment

Manheim and Minchella (1978b) investigated the possibility that the lack of an NIT impact on school achievement was due to a lack of NIT impact on home environment. Their dependent variable fell into three groups: (a) those related to time allocation, (b) those related to goods, and (c) those related to aspirations. Time allocation measures included hours spent on homework, hours spent on after-school activities, time spent talking to parents about books, T.V., school work, and future occupation; whether parents help with homework, days absent from school. Goods input measures were whether a desk or table were available for studying, whether an encyclopedia had been purchased in the last 4 years, availability of a quiet place to study. Aspirations were measured by asking whether the student planned to attend college, and by asking parents what education they wanted for their child. Data from Denver and Seattle were pooled, and the sample divided into two groups by grade level: those in grades 5 through 8 and those in grades 9 through 12. Results indicated that there was no experimental impact on these measures of home environment.[13]

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[11] For Denver, composite academic grade point averages were available for grade levels 1 through 12 (N=711). Attendance rates were available only for grade levels K through 6 (N=335). Standardized test data were available for grades K through 8 (N=94 for math scores and N=85 for reading scores). However, the type of test administered differed both by grade and year. In preexperimental years, the Stanford Achievement Test (SAT) was administered, while in subsequent years the Metropolitan Achievement Test (MAT) and Iowa Test of Basic Skills (ITBS) were given. Percentiles, grade equivalents and raw scores were available for each test. In Seattle, both pre- and during-experiment standardized test data were available for grades 2 through 8 (N=234 for math scores and N=261 for reading scores). In all cases the same test—the Metropolitan Achievement Test—was administered, and percentile scores recorded. On the other hand, composite academic grade point averages were collected only for Seattle high school students (N=453). Attendance data were again available for only grades 1 through 8 (N=210).

[12] The authors report that further analysis revealed that this finding occurred only for those whose preexperimental adjusted income was greater than $4,800 (those who would receive the lowest experimental transfer), and for those male-headed families where the wife did not work in the preexperimental year. Thus, the effect seems random—it cannot, for example, be explained because wives reduced their labor supply to spend more time with their children.

[13] The only two statistically significant effects appear to be random: for grades 5 through 8 children in experimental families were more likely to have parental help with homework ($p <.10$), but they were less likely than controls to talk to their parents about their plans and recreational activities. ($p <.05$).

It is somewhat surprising that the experiment had so little effect on the home environment of students. However, it is possible that the time students spend doing homework and the availability of study aids in the home are less reflective of income than of such factors as family attitudes toward work and study and their understanding of what is conducive to good study habits.[14]

Discussion

There is no doubt that the school performance studies were hampered by a paucity of good data. There were three data sources available, but few students for whom data were available from all three sources. The first consisted of the periodic questionnaires administered to family heads. The home environment questionnaire was the one most heavily utilized in the studies reported here, and this was not administered until very late in the experiment—after many 3-year families had already been disenrolled. A second data source consisted of school records obtained retrospectively from the schools; this source was flawed because of a large amount of missing data. Furthermore, even the kinds of tests administered were different in the two sites, and in Denver several different kinds of tests were administered. The third source consisted of interviews with a subset of students.

Virtually all estimates were obtained using OLS regression.[15] Knickman suggests this may have been too insensitive a technique to isolate effects of specific independent variables, and in a separate study Hanushek and Knickman (1980) developed a structural equations model as an alternative method of estimation. In this model, which was based on the LISREL model developed by Jöreskog (1973), math and reading scores were treated as indicators of the unobservable underlying dimension of school achievement. This approach permitted explicit investigation of the effects of measurement errors on estimation results. As this was primarily an exploration of an alternative methodology, only a limited number of variables were used and results do not add substantively to findings already reported. However, Hanushek and Knickman did find fairly substantial differences between models that assume no error in test score measurement and those that incorporate measurement errors, and concluded that the structural model would be a valuable method for investigating changes in family factors over time.

The only conclusion we can draw at this point is that, in SIME/DIME at least, the NIT treatment did not influence school performance (Manheim and Minchella, 1978a). Its failure to affect home environment (Manheim and Minchella, 1978b) may be irrelevant in light of the fact that other research not reported here indicates that home environment variables available in SIME/DIME appear to be largely unrelated to school performance (Knickman, 1979; Knickman and Phillips, 1980; Knickman and Wolf, 1980). The theoretical causal model underlying these studies seems reasonable, and the lack of significant effects probably is due either to the poor measures available, or to the fact that much larger increases in income or more time would be required for the predicted processes to occur.

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[14] The lack of experimental effect on parental aspirations for children is generally consistent with findings in the study by Davis (1980a). The few significant effects in Davis' study indicated that, if anything, mothers in the experimental group had somewhat lower expectations and aspirations than their controls.

[15] Manheim and Minchella (1978b) reported probit partial derivatives in one analysis.

CHAPTER VI

FAMILY LIFE

This section covers two studies dealing with quite different aspects of family life: household production, and marital conflict and individual attitudes toward family roles. The first study posits a relationship between time spent on market work and on household work, arguing that to the extent the NIT affects the former it will affect the latter also. The second study examines effects of marital conflict and individual attitudes toward family roles on marital stability and labor supply, together with the extent to which conflict and attitudinal variables interact with experimental treatment to affect these outcomes.

Household production

Household production here refers to all work performed in the home: meal preparation, housecleaning, yard work, helping children with homework, and so on. On the assumption that household production enhances family well-being, it is of interest to investigate whether the NIT has an impact on time spent by family members on household work and the purchase of services for the home. Our discussion is based primarily on the report by Yaeger (1980) which revises and expands earlier work by Keeley and Yaeger (1979).

Theoretical framework

Keeley and Yaeger used the model proposed by Gronau (1977) as their theoretical basis for predicting effects of an NIT on household production. In Gronau's model, household activities are classified as either work or leisure. Work includes activities that one can pay others to do or that have close substitutes in the market (such as laundry, cleaning, yard work); leisure includes activities that give direct satisfaction and that therefore cannot be performed by someone else (such as reading or watching a movie).

A utility maximization framework is used to develop predictions of NIT impacts on household production. This assumes that the household will attempt to maximize utility which depends on goods and services, purchased either in the market or produced at home, and on the consumption of leisure time of household heads.

The NIT increases unearned income through the guarantee level and reduces the net wage rates of both family heads through the NIT tax rate. The theoretical model predicts that the change in the tax rate will have a pure substitution effect—that is, as a tax is applied on the time spent in market work, the individual will substitute leisure or household production for market work. However, if one spouse is working and the other is not, because the tax will encourage the working spouse to spend more time on household work, the model predicts that the nonworking spouse will spend less time.

The guarantee level is expected to have a pure income effect. The model predicts that for an employed individual there will be no change in time spent on household production; any increase in unearned income will result in an increase in leisure time and an increase in household expenditures. One could argue, incidentally, that this is a serious flaw in the model: an increase in household expenditures is likely to include expenditures on household services (such as eating out, housecleaning services, etc.) that will in turn reduce time spent on household production.

For a nonemployed head, the model predicts that an increase in unearned income will lead to less household production. This prediction is based on the assumption that an individual will decide to participate in the labor market if and only if his or her market wage rate exceeds the shadow wage, which is a function of nonlabor income. An increase in unearned income will increase the shadow wage (assuming that leisure is a normal good) and thereby reduce time spent on household production.

The sample used to estimate this model consisted of individuals who responded to the SIME/DIME household chores module (module 57). This was administered approximately 4 years after enrollment as part of the twelfth periodic interview, when many families on the 3-year financial treatments had already been disenrolled.[1] Both dual-head and single-head families (based on status at time of enrollment) were analyzed.

Estimates were obtained for each family head on hours spent in the previous week on tasks related to housecleaning, cooking, laundry, home repairs and improvements, helping children with homework.

In the first analysis, experimental treatment was represented by a dummy variable that equaled one if the individual had been assigned to an experimental program. The dependent variable was the sum of eight specific tasks. Results indicated that for single female heads, the NIT induced a significant increase in housework time of 2.5 hours per week. There were no significant treatment effects for husbands and wives.

Next, the NIT treatment was parameterized in terms of three variables: first, the change in disposable income caused by the NIT, ΔY, calculated at the preexperimental budget position (defined to be zero for those above breakeven and for controls); second, the linearized change in tax rate, Δt, calculated at the preexperimental budget position (defined as zero for those above the breakeven and for controls); and third, a dummy variable equal to one for experimentals above the breakeven level at enrollment (FABOVE).[2] (These variables are described in detail in part I.) Results are in table 6.9.

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[1] Although data for these disenrolled families could not be included in the main analysis, they were analyzed separately in a study of postexperimental effects.

[2] Yaeger's analysis included a fourth parameter, a dummy variable that equaled one for families assigned to the declining tax progams. Effects were not significant, so for ease of presentation, we omit these coefficient estimates here.

TABLE 6.9.—ESTIMATED EFFECTS OF TREATMENT PARAMETERS ON TOTAL WEEKLY HOURS OF NONMARKET WORK

[Standard errors in parentheses]

	Husbands (N=1084)			Wives (N=1043)			Single female heads (N=791)		
	Δ Y	Δ t	FABOVE	Δ Y	Δ t	FABOVE	Δ Y	Δ t	FABOVE
Basic parameter effect	.000	−1.817	.512	−.001	8.384	9.891	[3].004	−5.412	.907
	(.001)	(2.726)	(1.331)	(.001)	(5.734)	(2.522)	(.001)	(5.083)	(2.169)
Treatment interaction with:									
(a) Program length:									
3-year financials	.003	[3]−29.027	−4.287	.012	−13.187	−4.479	.006	4.764	5.908
	(.005)	(10.714)	(11.640)	(.015)	(23.770)	(9.778)	(.010)	(21.889)	(11.895)
5-year financials	.000	3.253	.588	−.002	11.431	[1]4.478	[3].004	−2.783	−.497
	(.001)	(3.423)	(1.360)	(.001)	(7.154)	(2.593)	(.002)	(5.954)	(2.240)
20-year financials	[1].009	−1.173	6.923	−.009	10.352	—	−.002	[1]19.597	[1]11.470
	(.005)	(5.134)	(12.786)	(.146)	(11.457)	—	(.010)	(11.727)	(6.640)
F-test	[1]F=2.713			F=.411			F=1.340		
(b) Own employment in preexperimental year:									
Employed financials	.000	−1.644	.660	−.003	[1]15.210	[3]9.457	.003	−4.566	.571
	(.003)	(2.863)	(1.397)	(.003)	(8.922)	(3.480)	(.004)	(6.620)	(2.377)
Nonemployed financials	[2].005	−5.352	−1.780	.0001	3.426	−1.882	[3].007	−8.971	1.932
	(.002)	(9.069)	(4.300)	(.002)	(7.346)	(3.588)	(.002)	(8.250)	(5.056)
F-test	F=1.428			F=.174			F=.425		

[1] Significant at the 10-percent level.
[2] Significant at the 5-percent level.
[3] Significant at the 1-percent level.

We turn first to the effects of a change in disposable income induced by the NIT (Δ Y). It may be recalled that the theoretical model predicted that in both single- and dual-headed familes, there would be no change in housework hours if the head was employed and a decrease if the head was unemployed. The results are contrary to prediction in that husbands not employed preexperimentally and husbands on the 20-year program showed increased housework hours relative to controls. Similarly, single female heads who were unemployed preexperimentally showed an increase in housework hours relative to controls.

The predicted positive effects concerning the impact of a change in tax rate were found for wives employed in the preexperimental year and for Chicano wives. Most of the remaining tax rate effects were not significant. However, housework hours of husbands on the 3-year NIT treatment and single female heads on the 20-year treatment were lower than controls.

There were some positive significant effects of being above breakeven level. Wives in the NIT group who were employed preexperimentally or were on a 5-year program and single female heads on the 20-year program performed more housework than controls in the equivalent subgroups. Yaeger and Keeley suggest we may interpret these findings as evidence that employed wives reduce their hours of market work as a result of the treatment (thereby bringing them below the breakeven level) and significantly increase their hours of nonmarket work.

Keeley and Yaeger also compared the overall experimental effects on nonmarket and on market hours of work. They reported that for some subgroups, the NIT treatment led to the predicted shift from market to nonmarket work. For husbands on 3-year programs, the average decrease in hours of market work (3 hours per week) was counterbalanced by the average increase in nonmarket work (5.9 hours per week). Single female heads on the 5-year program significantly increased nonmarket work by 3 hours per week and reduced market work by 6.3 hours per week.[3]

Conclusions

We can draw two overall conclusions from these findings. First, there seems to be a negative relationship between time spent on market work and time spent on household work. Second, where experimental effects are significant, they suggest primarily a positive relationship between the NIT and household production.

The researchers (Keeley and Yaeger, 1979; Yaeger, 1980) comment that their findings do not consistently support the predictions they derive from Gronau's theory of the allocation of time. That is, they had predicted that the change in tax rate induced by the NIT would increase housework time, but that the change in unearned income due to the NIT would make no difference or would decrease housework time. Leaving aside Gronau's model, however, their findings are clearly consistent both with the findings in the socioeconomic literature and with SIME/DIME findings on labor supply. The negative relationship between time spent on market work and on household work has been well-documented, especially in the extensive social science literature on the effects of maternal employment (that is, see Walker and Gauger, 1973; Davis, 1979; Bahr, 1974). Moreover, we know that the NIT induces a reduction in labor supply. These two factors make a positive relationship between experimental treatment and household production quite reasonable.

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[3] Results presented in the first column of table 6.9 are based on data from a somewhat larger sample than the results in the second and third columns. This is because only families responding to the Household Chores Module were included in analyses of time spent on nonmarket work.

Marital conflict and attitudes toward family roles

The NIT treatment in SIME/DIME reduced the labor supply of husbands and wives. It also, in conjunction with the counseling/training treatment, increased family break up. Davis (1980b) explored the effect of including variables related to marital conflict and normative attitudes toward sex roles in models estimating NIT impacts on rates of marital dissolution and employment status changes. The primary purpose of her study was to gain a greater understanding of processes that lead people to dissolve their marriages and to change employment status. In addition, she wished to determine whether the impact of experimental treatment varies according to the level of marital conflict or type of normative attitude expressed by family members—for example, whether the NIT destabilizes initially high conflict marriages but stabilizes initially low conflict marriages, or whether the NIT is more likely to reduce the labor supply of women with traditional normative attitudes and less likely to reduce the labor supply of traditional men.

Marital conflict

Previous research on marital satisfaction concludes that there is higher marital conflict and instability at lower income levels in the population than at higher levels (Glick and Norton, 1971; Hicks and Platt, 1970; Cutright, 1971; Renne, 1970). One hypothesis advanced to account for this phenomenon, which is known as the economic strain theory of family relations, is that economic problems generate tensions that strain interpersonal relations (Middleton and Haas, 1976). This theory would therefore predict that the NIT would reduce marital conflict because of its potential for alleviating financial uncertainty.

As a first step, Davis (1980a) used OLS regression to estimate experimental impacts on several indicators of marital satisfaction and conflict, including husbands' and wives' reports of the frequency of arguments over money and the use of free time, and wives' reports of number of topics argued about, whether she has refused to speak to her spouse after an argument, whether she or her spouse have left home, whether she has considered separation in the previous year. The sample consisted of couples married at enrollment who were still married by the time the measures were obtained (the 4th periodic interview for husbands and 5th for wives). Equations included background and assignment variables (race, site, preexperimental normal income levels, whether family received AFDC preexperimentally, number of children, presence of a child under 5 years of age, age and education of husband and wife.).

Although most of the NIT treatment coefficients were very small, the signs were consistently negative. This suggests that intact families in the experimental group experienced less conflict than control families. Specifically, married women in the experimental group were less likely than controls to report ever arguing over income spending, were less likely to argue over two or more different topics, and were less likely to report having become sufficiently upset after an argument to leave home.[4]

These findings seem supportive of economic strain therory. However, recall that the measures used in the above analysis were obtained relatively late in the experiment (about the middle of the second year) and the questions were asked only of respondents still married at that time. Thus, it is possible that many high conflict couples had already split, leaving predominantly low-conflict couples still intact. It is also possible that the NIT has a different effect on initially high-conflict and initially low-conflict families. For example, if the initial level of conflict is high, the experimental income may destabilize the marriage either by providing each spouse with an independent source of support that facilitates any existing desire to separate, or perhaps by providing them with yet another topic of conflict. On the other hand, for couples with initially low levels of conflict, the NIT may stabilize marriage by easing everyday financial burdens. This explanation was explored in a second study by Davis (1980b), who first estimated the direct effects of conflict on the rate of marital dissolution and the rates of leaving and entering employment, and then tested for interaction effects to determine if the NIT had a different impact on the outcomes depending on the initial levels of conflict reported.

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[4] The effects of training counseling eligibility, although again chiefly small, were interestingly in the opposite direction where significant: for example, women eligible for the most generous training subsidy were more likely than controls to report arguing over the use of free time, child care, and household chores.

No preexperimental measures of family conflict were obtained in SIME/DIME. To ascertain the impact of initial levels of conflict, therefore, Davis (1980b) used the earliest available measures based on responses from Denver wives obtained during the first and second periodic interviews. These women were followed from the date of enrollment until either the end of the experiment (5 or 7 years), the date of attrition from the experiment, or the date of the first marital status change—whichever date was earliest.

The method of analysis used was the rate model developed by Tuma (1976). Both the overall rate of dissolution was estimated, using the entire sample of women, and the separate rates conditional on the initial joint employment state of husband and wife. There were 4 states possible: husband employed/wife not employed (HEWU), husband employed/wife employed (HEWE), husband not employed/wife not employed (HUWU), and husband not employed/wife employed (HUWE).[5] In addition to measures of conflict, each rate equation contained the following treatment, background and assignment variables: variables representing experimental treatment, 3-year treatment, manpower treatments, race, preexperimental normal income, husband's and wife's hours worked, receipt of AFDC in preexperimental year, duration of marriage, family's net worth, number of children, presence of child under 5 years, husband's and wife's age and education.

Three separate analyses were conducted using measures of conflict obtained during the first periodic interviews. The three conflict variables were: frequency of arguing over money decisions, how to spend free time, and (for parents only) bringing up the children.[6] Responses were coded on an ordinal scale with values from 1 to 4, ranging from "never" to "all the time."

A fourth analysis used data obtained from the 2nd periodic interview. Five dummy variables, each representing a dimension of conflict were entered simultaneously into the rate equation. The variables were:

(a) Respondent reports arguing with spouse about two or more topics.

(b) Respondent and spouse have been so mad at each other they would not speak for a while.

(c) Respondent has become so upset she left home.

(d) Respondent's spouse has become so upset that he left home.

(e) Respondent has considered separating from spouse during the past year.

Table 6.10 presents the effects of experimental treatment and each conflict measure on the rate of marital dissolution.[7] Because the effect of a variable on the rate itself is of more interest and simpler to interpret than the effect on the log of the rate, the table reports the multipliers (antilogs). (Appendix A, part VI, discusses the interpretation of effects using the rate model.) Both conflict over money decisions and conflict over the use of free time significantly increased the rate of dissolution in the pooled sample (p <.001). In the separate analyses of families in each of the four initial joint employment states, all three forms of conflict significantly raised the dissolution rate in the HEWU group. Other effects were trivial, except that in the smallest group (HUWE) conflict over childrearing significantly lowered the dissolution rate.

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[5] The numbers of families in each group were, respectably 515, 317, 80, 45.

[6] It was only in the case of conflict over childrearing that the sample was limited to parents. In analyses using all other conflict indicators, respondents included both parents and nonparents.

[7] The only tables included in this chapter are those showing the main effects of conflict variables on marital dissolution and employment status changes. Refer to Davis (1980b) for tables showing results of models that include interaction effects, and all tabular results of effects of normative attitudes on marital stability and employment status changes.

TABLE 6.10.—EFFECTS OF MARITAL CONFLICT ON RATES OF MARITAL DISSOLUTION OF DENVER BLACKS AND WHITES (WIVES' RESPONSES)

	All spells	Joint Employment Status			
		HEWU	HEWE	HUWU	HUWE
First Periodic Interviews[1]					
A. Conflict over money decisions:					
Number of spells	4,467	1,701	1,451	686	498
Number of dissolutions	213	71	93	26	20
NIT	1.31	.92	1.35	2.14	[3] 4.03
Conflict over money decisions	[6] 1.36	[4] 1.37	1.23	1.56	.57
B. Conflict over use of free time:					
Number of spells	4,447	1,700	1,449	678	489
Number of dissolutions	212	71	92	26	20
NIT	1.26	.91	1.33	1.82	[3] 3.41
Conflict over use of free time	[6] 1.39	[4] 1.42	1.23	1.31	1.22
C. Conflict over childrearing:					
Number of spells	3,947	1,529	1,286	591	416
Number of dissolutions	190	64	81	24	18
NIT	1.14	.98	1.11	1.17	[4] 8.26
Conflict over childrearing	1.08	[3] 1.39	1.01	.99	[4] .28
Second Periodic Interviews[2]					
Number of spells	4,398	1,677	1,444	657	489
Number of dissolutions	195	68	86	22	16
NIT	1.25	.85	[3] 1.67	1.60	3.71
Argued about more than 2 topics	1.19	1.36	1.25	.46	.82
Refused to speak to spouse	1.07	.75	1.25	.45	[3] 5.73
Left home after argument	1.17	1.33	1.15	1.41	.74
Spouse left home after argument	1.20	.76	1.36	[4] 6.60	2.34
Considered separation	[6] 2.56	[6] 3.60	[5] 2.60	1.80	.26

[1] Conflict variables are coded from 1–4 (with 4 representing the highest level of conflict).
[2] Conflict variables are dummy (0–1) variables.
[3] Significant at the 10-percent level.
[4] Significant at the 5-percent level.
[5] Significant at the 1-percent level.
[6] Significant at the 0.1-percent level.

NOTE.—Multipliers of rates are presented. Equations contain all background and assignment variables. Each analysis uses only cases that have no missing data for the relevant conflict variable.

Using indicators from the second periodic interviews, the strongest predictor of dissolution was having considered a separation. This factor significantly raised the dissolution rate for the pooled sample (p<.001) and for couples in the HEWU (p<.001) and HEWE (p<.01) categories. Other dimensions of conflict that raised the dissolution rate were the husband leaving home after an argument (HUWU), and refusing to speak to spouse (HUWE).

When interaction with NIT treatment was added to the model, it was found that among HEWU couples, those who argued about childrearing and were in the NIT group, were significantly more likely to split than other couples (p<.05). (Recall that conflict over childrearing had no direct effect on the rate of dissolution.) This suggests that some unemployed women may maintain unsatisfactory marriages when they have dependent children and a husband to support them. Some external source of support, like an NIT, might tip the balance in favor of dissolution.

There were only a few significant effects in the two largest groups with employed husbands (HEWU and HEWE) when the NIT was interacted with conflict measures from the 2nd periodic interviews. Among HEWU families, the dissolution rate was higher for experimentals who argued about two or more topics ($p<.05$) and where the spouse had left home ($p<.05$). Among HEWE families, results for one conflict indicator—wife refusing to speak to her spouse—suggest that the NIT increases the dissolution rate more for the low conflict than high conflict families.

In summary, we have some evidence here that marital conflict is associated with rates of marital dissolution. Using measures from the 1st periodic interviews (which occurred typically a few months after the start of the experiment), there were no interaction effects of experimental treatment with conflict over money or use of free time. However, families who argued over childrearing and received an NIT were significantly more likely to split than other families. For conflict measures from the 2nd periodic interviews, most interaction effects were trivial. The few significant effects occur for families where the husband was employed and vary by wife's employment status. If the wife was unemployed, the NIT increased dissolution more among high conflict than among low conflict families; if the wife was employed, the NIT increased the dissolution rate more among low conflict than high conflict families.

Marital conflict and employment status of husbands and wives

Previous research indicates that the husband's role as economic provider is an important component of marital satisfaction (Cutright, 1971; Hicks and Platt, 1970). On the other hand, there is evidence that the wife's employment and earnings are positively related to marital dissolution (Carter and Glick, 1970; Havens, 1973).

Studies dealing with the relative power of husband and wife typically explain the distribution of power in terms of the relative resources each spouse controls. Money is a key resource in U.S. society, and thus we can predict that where husband and wife are relatively equal regarding the amount of money each provides to the household (both employed or both unemployed), the balance of power between them will be more equal than in the case where one spouse is employed and the other not. Such an argument has been supported by findings of earlier studies (Blood and Wolfe, 1960; Scanzoni, 1970).

For wives, it would seem that—other things being equal—marital conflict will have the general effect of inducing unemployed women to seek employment and of reducing the tendency of employed women to leave work, so that they can improve or maintain their power position relative to the husband. To the extent that an NIT is accepted as a substitute for the independence that employment brings, it may mediate the impact of conflict on women's work efforts. The same argument would presumably apply to husbands also. To test these ideas, Davis used the rate model to estimate husbands' and wives' rates of entering and leaving employment, conditional on their initial joint employment status.[8]

Table 6.11 shows results for wives. Most conflict variables did not affect the rate at which wives change employment status; however, significant findings were frequently supportive of the argument above. Conflict over money reduced wives' rates of leaving employment, regardless of husband's work status ($p<.10$). Women with employed husbands (HEWU and HEWE) who had considered separation entered employment at a significantly greater rate ($p<.10$ for HEWU) and left at a significantly reduced rate ($p<.01$ for HEWE) than women who had not considered separation. On the other hand, considering separation reduced the rate at which women with unemployed husbands entered employment ($p<.01$). Also, working women with husbands (HEWE) were more likely to leave their jobs when they reported having left home after an argument ($p<.01$).

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[8] During each periodic interview, data were gathered concerning dates of changes in employment status. This information was used to define spells of employment and nonemployment for the entire observation period which, to reiterate, extends up to but not beyond the date of the first marital status change. Conflict and attitudinal measures were not obtained until the 1st and 2nd periodic interviews, and it is possible that some employment status changes occurred prior to this time.

TABLE 6.11.—EFFECTS OF MARITAL CONFLICT ON WIVES' RATES OF ENTERING AND LEAVING EMPLOYMENT, DENVER BLACKS AND WHITES (WIVES' RESPONSES)

	Wives' rates of—			
	Entering employment		Leaving employment	
	HE	HU	HE	HU
First Periodic Interviews [1]				
A. Conflict over money decisions:				
Number of spells	1,701	686	1,451	498
Number of changes	818	136	725	115
NIT	.86	[5].47	.94	1.06
Conflict over money decisions	.99	[4].76	[3].89	[3].76
B. Conflict over free time:				
Number of spells	1,700	678	1,449	489
Number of changes	818	130	725	110
NIT	.86	[5].52	.95	.96
Conflict over free time	1.03	.84	.96	1.19
C. Conflict over childbearing:				
Number of spells	1,529	591	1,286	416
Number of changes	740	114	649	89
NIT	.81	[6].39	1.01	1.06
Conflict over childbearing	1.05	1.13	.93	.99
Second Periodic Interviews [2]				
Number of spells	1,677	657	1,444	489
Number of changes	820	130	726	110
NIT	.88	[6].36	.98	1.36
Argued over more than 2 topics	1.03	1.00	1.13	1.25
Refused to speak to spouse	.98	1.08	.92	1.22
Left home	1.10	1.01	[5] 1.45	.72
Spouse left home	.82	1.27	1.17	1.06
Considered separation	[3] 1.23	[5].38	[5].68	.95

[1] Conflict variables are coded from 1–4 (with 4 representing the highest level of conflict).
[2] Conflict variables are dummy (0–1) variables.
[3] Significant at the 10-percent level.
[4] Significant at the 5-percent level.
[5] Significant at the 1-percent level.
[6] Significant at the 0.1-percent level.

NOTE.—Mutipliers of rates are presented. Equations contain all background and assignment variables. Each analysis uses only cases that have no missing data for the revelant conflict variables.

When NIT treatment was interacted with conflict measures (refer to Davis, 1980b, for tables presenting results for interaction models), there were few significant effects. Among women with employed husbands, experimentals who argued about money were less likely to enter employment than were controls or low-conflict experimentals. In this same group, the interaction term indicates that experimentals reporting high conflict over childrearing entered employment at a lower rate than low-conflict experimentals; high-conflict controls entered employment at a higher rate than low-conflict controls.

Also in the HEWU group, there was some tendency for the NIT to interact with the conflict measures of arguing over more than two topics and refusing to speak to spouse. High-conflict controls were more likely to enter employment than low-conflict controls. However, the effect of the NIT on rates of entering and leaving employment was not altered by the level of conflict. Another indication that some women with employed husbands may regard the NIT as a substitute source of independent income is the finding from the HEWE families that controls were far less likely to leave employment if they had considered separation than were the NIT group. One contrary finding is that for unemployed women with employed husbands (column 1), experimentals were more likely to enter employment if they reported their spouse had left home, whereas controls were significantly less likely to do so. Also, in the HEWE group, for controls, when the women reported having left home, she was significantly more likely to leave her job, whereas this measure of conflict did not much affect the rate at which NIT wives left work.

Table 6.12 shows findings with respect to husbands' rates of entering and leving employment. The overall effect of the NIT treatment was to reduce significantly the husbands' rates of entering employment, whether or not the wife was employed, and increase rates of leaving employment if the wife was unemployed. However, if the wife was employed, the NIT significantly reduced the rate at which the husband leaves employment. There was little evidence that marital conflict increased husbands' labor supply—in fact, the reverse appears to be true. Conflict over money increased the rate at which husbands left employment if the wife was employed ($p<0.01$), and reduced the rate of entering employment if the wife was unemployed ($p<0.10$). Conflict over free time also reduced entry into employment for men with unemployed wives ($p<0.05$). Effects of conflict measures from the 2nd periodic interviews were mainly nonsignificant, but significant effects again point to a pattern of marital conflict reducing the work efforts of husbands. Men entered employment at a lower rate if their wives had left home (HUWE), if they had argued over more than two topics (HUWU); and they left their jobs at a greater rate if the wife had left home (HEWE). The one contrary finding is that arguing over more than two topics was associated with greater rates of entering employment for men in HUWE families.

TABLE 6.12.—EFFECTS OF CONFLICT ON HUSBANDS' RATES OF ENTERING AND LEAVING EMPLOYMENT, DENVER BLACKS AND WHITES (WIVES' RESPONSES)

	Husbands' rate of—			
	Entering employment		Leaving employment	
	WE	WU	WE	WU
First Periodic Interviews [1]				
A. Conflict over money decisions:				
Number of spells	498	686	1,451	1,701
Number of changes	327	461	321	501
NIT	[6].54	[5].71	[4].69	[5] 1.43
Conflict over money decisions	1.07	[3].89	[5] 1.27	1.11
B. Conflict over free time:				
Number of spells	489	678	1,449	1,700
Number of changes	325	460	318	500
NIT	[6].53	[4].72	[4].67	[5] 1.42
Conflict over free time	.91	[4].87	.96	1.00
C. Conflict over childrearing:				
Number of spells	416	591	1,286	1,529
Number of changes	281	406	271	441
NIT	[6].45	[5].65	[5].58	[5] 1.43
Conflict over childrearing	.93	.96	1.11	1.04
Second Periodic Interviews [2]				
Number of spells	489	657	1,444	1,677
Number of changes	327	442	317	481
NIT	.80	[4].74	[4].68	[5] 1.41
Argued over more than 2 topics	[4] 1.32	[6].59	.96	1.12
Refused to speak to spouse	1.14	1.09	[4] 1.32	1.11

TABLE 6.12.—EFFECTS OF CONFLICT ON HUSBANDS' RATES OF ENTERING AND LEAVING EMPLOYMENT, DENVER BLACKS AND WHITES (WIVES' RESPONSES)—Continued

	Husbands' rate of—			
	Entering employment		Leaving employment	
	WE	WU	WE	WU
Left home	[5].62	1.12	[4] 1.47	1.02
Spouse left home	.98	1.16	1.25	[5] 1.56
Considered separation	1.12	1.01	.91	.97

[1] Conflict variables are coded from 1–4 (with 4 representing the highest level of conflict).
[2] Conflict variables are dummy (0–1) variables.
[3] Significant at the 10-percent level.
[4] Significant at the 5-percent level.
[5] Significant at the 1-percent level.
[6] Significant at the 0.1-percent level.

NOTE.—Multipliers of rates are presented. Equations contain all background and assignment variables. This analysis uses only cases that have no missing values for the conflict variables shown.

There were few significant interaction effects. In general, where the main effect of the NIT was to reduce labor supply the effect was stronger among high-conflict than among low-conflict families.

Summarizing the effects of marital conflict on husbands' and wives' rates of entering and leaving employment, we find some evidence that wives in high-conflict marriages tend to try to establish some degree of financial independence by maintaining or increasing their work effort. This is in keeping with a resource theory of marital power which would predict that individuals will try to maximize personal resources. Results concerning the way the NIT treatment interacts with conflict to affect wives' work effort are mixed. In some instances, conflict was associated with higher rates of entry into employment for controls but not NIT families, suggesting that if some women seek employment as a source of security in an unhappy marriage, the income guaranteed by the NIT may act as an alternative source of security. Among husbands, it seems that conflict is associated overall with reduced work effort. Nearly all statistically significant findings indicate that conflict reduced husbands' rates of entering employment and increased rates of leaving employment. In general, the NIT effect does not appear to depend on the level of conflict. The few significant interaction effects indicate that where the main effect of the NIT was to reduce labor supply, the effect was stronger among high-conflict than among low-conflict families.

Normative attitudes toward family roles

There is no general consensus concerning the causal relationship between an individual's attitudes and his or her behavior. "Normative" attitudes refer to the individual's beliefs concerning the way that behavior ought to be. In the first study in which Davis investigated normative attitudes of SIME/DIME subjects toward sex-linked family roles (1980a), she did not expect to find significant experimental effects. But, there were some interesting differences in attitudes among ethnic subgroups, and also some NIT effects that suggested further study would be fruitful.[9] Specifically, blacks and Chicanos, both male and female, were more likely than whites to prefer traditional roles inside the home. But black males and females and Chicano males were less traditional than whites in that they were more likely to approve of market work for wives. (Chicano women did not differ from whites in this respect.) Females receiving experimental treatment were more traditional in their attitudes toward both domestic and market roles than were controls. Davis suggested that possibly attitudes largely reflect what individuals perceive to be appropriate for someone in their own particular economic and social situation. For example, if it is necessary for the wife to work, her expressed attitudes may be relatively nontraditional. If she is able to stop work, and the NIT did lead to many wives doing this, her attitudes may become more traditional to match her change to more traditional behavior. A recent study by Molm (1978) confirms this argument. Molm found that, although there was little relationship between a woman's sex-role attitudes and her employment status, what relationship there was resulted from employment modifying attitudes rather than vice versa. In her second study, Davis (1980b) examined the effect of normative attitudes on the rate of marital dissolution and rates of entering and leaving employment of husbands and wives.

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[9] The two measures of normative attitudes focused on here refer to the degree of preference for internal differentiation of family roles and for external differentiation of family roles. "Differentiation" refers to the traditional sex-linked allocation of tasks and responsibilities with, for example, females responsible for housework and males responsible for earning the family income. These measures were obtained using the model of normative differentiation developed by Tuma et al. (1974). Refer to Davis (1980a) for a fuller discussion.

We have no reason for believing, a priori, that a preference either for differentiation or nondifferentiation of family roles will directly affect the rate of marital dissolution. However, it does seem plausible that families where expressed normative beliefs conflict with the current employment status of family heads will be less stable than those marriages where beliefs are congruent with employment status. For example, an employed wife who approves of wives working is probably less likely, other things being equal, to dissolve her marriage than an employed wife who believes wives should not work. This hypothesis was tested by examining the effects of normative attitudes on the rate of dissolution for the pooled sample and separately for couples in each of the four joint employment states.

Results indicated that normative attitudes had little effect on dissolution rates for either the pooled sample or for the separate joint employment groups. However, when the NIT treatment was interacted with the attitudinal variables, results for both HEWE and HUWU families indicated that the dissolution rate was significantly lower for experimentals who had traditional attitudes toward market work. There are many employed wives who would prefer not to work if there were no financial necessity to do so. It is possible that NIT payments enable some of these women in the HEWE group to leave employment, thus bringing their behavior into line with their preferences—with an eventual positive impact on marital stability. Similarly, NIT payments might soften stresses that presumably accompany a situation where an unemployed husband has a wife who believes in external differentiation of roles, again resulting in a lower dissolution rate among experimentals with traditional beliefs.

In the next part of her analysis, Davis examined the impact of normative attitudes on husbands' and wives' rates of entering and leaving employment. Traditional normative attitudes would support employment for the husband and nonemployment for the wife. The NIT should be less likely to reduce the labor supply of males and more likely to reduce the labor supply of females in families where attitudes are traditional than in nontraditional families.[10] For wives, results indicated that attitudes neither toward domestic nor toward market roles had a direct impact on rates of changing employment status. For husbands, traditional attitudes concerning market roles were associated with increased rates of entering employment and decreased rates of leaving employment. For the two largest groups there were no significant effects when NIT treatment was interacted with attitudinal variables.[11]

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[10] However, other research has found that economic need is the most important reason that wives work and that their normative attitudes toward sex-roles are only weakly related to their employment behavior (Gordon and Kammeyer, 1980; Molm, 1978).

[11] The only significant effects occurred in the HUWU group where the experimentals who believed in traditional domestic or market roles were less likely to enter employment than nontraditional experimentals or controls.

In summary, normative attitudes do not generally mediate NIT effects on either marital stability or labor supply, although in two subgroups (HEWE and HUWU) dissolution rates were lower for experimentals with traditional attitudes. It is also possible that the NIT may indirectly affect normative attitudes through its effects on behavior. That is, if these women do attempt to make attitudes consistent with behavior by reducing their labor supply and thereby making behavior more traditional, the NIT may ultimately be responsible for the shift toward more traditional attitudes reported by Davis (1980a). This question can only be answered by further analysis of the effects of employment changes on changes in normative attitudes.

CHAPTER VII

CONCLUSIONS

In this chapter we have discussed several sets of studies that examine the impact of SIME/DIME on a wide range of economic and social behaviors. We now briefly summarize these findings in terms of how they might affect our interpretation of the impact of SIME/DIME on labor supply, marital stability, and program costs. Among the factors that we would expect to affect short-run labor supply are fertility, household production, migration, residential mobility, child care, marital conflict, and attitudes toward family roles.

Labor supply

Fertility can affect short-run labor supply by interrupting a woman's labor force participation and increasing the cost of working (e.g., increasing the cost of child care). SIME/DIME appears to have resulted in a substantial increase in fertility among Chicano women, particularly those in the 5-year group. On the other hand, white women in the 5-year group experienced a decline in fertility relative to similar control women. It is thus possible that some portion of the labor supply reduction of Chicano women is the indirect effect of the direct experimental impact on their fertility. It is also possible that the reduced fertility among white women in response to the experiment had a moderating effect on their labor supply reductions.

The findings on household production are consistent with the overall labor supply findings: the experiment appears to have reduced labor supply and increased time spent in household production.

SIME/DIME increased the rate of interarea migration (by just under 50 percent) for white married couples. Although these findings may be biased upwards (control families that moved had little incentive to continue to report and may be overrepresented among the families that quit the experiment), they provide some insight into the question of whether these moves improved the distribution of the population vis-à-vis available jobs. SIME/DIME appears to have financed longer moves than were undertaken by control families; however, we conclude that the findings do not indicate that these moves were made to take better jobs. Male and female heads of family worked fewer hours in their new locations and worked for lower wage rates than control group individuals who moved.[1]

It is difficult to assess the impact of subsidization of child care on labor supply from the SIME/DIME analyses of the issue. This is because these studies simply compared families eligible for SIME/DIME child care subsidies with the control population who, in many cases, were also eligible for child care subsidy through AFDC and other programs. Thus, the lack of significant findings reported does not necessarily imply that child care subsidies have no effect on labor supply.

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[1] SIME/DIME also increased the probability of intracity moves for some subgroups. While the question of whether these moves resulted in greater access to employment was not directly addressed, conversations with project staff in Denver revealed that blacks and Chicanos in Denver tended to move away from, rather than closer to, areas where the suburban job market was expanding.

Marital conflict has some direct effects on labor supply in that married women in high-conflict homes tended to increase their labor supply, and men in high-conflict homes tended to reduce it. In those cases where conflict interacts significantly with experimental treatment, the effect was to weaken the main effect of conflict for women and to strengthen it for men. That is, women receiving the NIT were less likely to increase their labor supply where there was marital conflict, whereas men were more likely to decrease their labor supply when there was conflict.

There is no evidence that the labor supply response to the NIT was affected by the normative attitudes of respondents. Men and women with traditional attitudes toward male and female roles were not differently affected from nontraditional men and women.

Among the outcomes studied, some possible long-run determinants of labor supply include health status and children's school performance. In general, there were no effects of SIME/DIME on these outcomes. There was a statistically significant but small experimental effect on increased out-of-pocket medical expenses, suggesting that an NIT increases investment in one's health. Although there was no systematic pattern of findings with respect to psychological distress, the scattered significant effects indicate that experimental subjects in SIME/DIME experienced higher levels of stress than did controls. This finding is consistent with those of the New Jersey and Gary Experiments.

Marital stability

There is some evidence that SIME/DIME resulted in less marital conflict among married women who remained with their husbands than was experienced by similar control women. On the other hand, there is also evidence that SIME/DIME may have been destabilizing for families who reported conflict over childrearing. SIME/DIME may have encouraged families with this kind of conflict to split early in the experiment, leaving the experimental couples with less conflict than observed among control couples.

Whether or not individuals hold traditional views about marital roles generally did not appear to influence the effect of SIME/DIME on marital stability, although two subgroups with traditional attitudes experienced lower experimentally-induced dissolution rates than the rest of the experimental group.

Program costs

While the findings on labor supply reductions, marital dissolution, and underreporting of income provide a basis for estimating the effect of a program such as SIME/DIME on national welfare costs, several of the studies we have reviewed also have implications for program costs. These cost impacts are largely the result of how the families spent their NIT payments.

The expenditure pattern revealed by consumption studies reviewed here indicates that families in SIME/DIME, as in the other income maintenance experiments, spent their added income largely on food, clothing, and housing. As in the other income maintenance experiments, the share of increased income spent on housing was lower than would have been expected from previous studies of the income elasticity of housing expenditures.

SIME/DIME had the effect of upgrading the status of some families from renters to owners, and encouraged families to move out of public housing into private dwellings. These findings are consistent with those of the other income maintenance experiments. These effects suggest long-run decreases in welfare costs, as families reap the financial benefits of home ownership and as the population in public housing is reduced. Of course, such responses might not materialize under a national program similar to SIME/DIME, given housing supply constraints.

The kind of child care subsidy offered under SIME/DIME had a tendency to increase total welfare costs, because these subsidies led to more use of higher-priced market care rather than informal arrangements. Although the services provided under market care may be superior to informal care, a shift from informal care to subsidized market care increases welfare program costs.

Concluding comments

The findings we have reported do not have as strong and clear-cut an impact either on the lives of the families participating or in terms of social costs as is the case with the labor supply, training/counseling and marital stability results reported earlier in the volume.

It seems reasonable to conclude that where significant effects were detected, the overall impact of SIME/DIME on health, consumption, and social behavior was beneficial. Some of the positive findings from the other income maintenance experiments were not repeated—for example, SIME/DIME did not appear to improve school performance and attendance, nor infant birth weight. However, based on findings from SIME/DIME and from the other income maintenance experiments, we can be reasonably confident that an NIT along the lines tested would increase expenditures on food, clothing, and housing, and stimulate home ownership and movement out of public housing—outcomes that imply an increase in the quality of life for the families concerned.

APPENDIX A

INTERPRETING EFFECTS IN THE RATE MODEL

The following discussion is an attempt to simplify the interpretation of the rate coefficients presented in the tables in part VI, chapter 6. For a complete technical exposition of Tuma's log-linear stochastic rate model, please refer to part V.

The mathematical model used in this analysis can be expressed in two ways. The log-linear version (which is actually estimated) is:

$$\ln \mathrm{R} = AX + \mathrm{bF} + \mathrm{cC}, \tag{1}$$

where R is the rate being estimated, X is a vector of assignment and background variables, F is a dummy variable for the financial treatment, and C is a measure of the level of conflict, A is a vector of coefficients of the assignment and background variables, b and c are the estimated effects of the NIT treatment and conflict, respectively. Alternatively, the model can be expressed in a multiplicative form:

$$\mathrm{R} = \exp(AX)\exp(\mathrm{bF})\exp(\mathrm{cC}). \tag{2}$$

Because the effect of a variable on the rate itself is of more interest than the effect on the log of the rate, we report the antilog of estimated parameters. That is, when reporting the experimental effect we report exp(b) rather than b from equations (1) and (2). When the variable of interest is a dummy variable, such as the NIT treatment variable, the interpretation of the coefficient is straightforward: exp(b) is the ratio of the rate for an experimental to the rate for a control who is equivalent on all other variables in the equation. Another way of stating it is that exp(b) is the multiplier of the control rate.

When the variable is continuous or treated as continuous, as some of the measures of conflict and normative attitudes are, the interpretation of the effect is somewhat more complicated. In these cases the antilog of the estimated parameter is the multiplier of the rate for a unit increase in the variable. The total effect of the variable then is $(\exp c)^{\mathrm{C}}$. An example will help to clarify this.

Refer to table 6.11, first column. For conflicts over money decisions, exp (bF) is .86 and exp(cC) is .99. This indicates that the rate of entering employment for an experimental is .86 times the rate for an otherwise equivalent control, or the rate for experimentals is 14 percent $((.86-1.00)\times 100)$ less than the rate for controls. Conflict over money is measured on an ordinary scale with numeral values from 1 to 4. Thus, a change of one unit of conflict multiplies the rate by .99 or alternatively, decreases the rate by 1 percent. The effect at the highest level of conflict is $(.99)^4$ or .96 times the rate.

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